Goldbach Programme

Master File

Research knowledge base for future models

Factorization-graph programme for the binary Goldbach conjecture: the canonical selector $p_0(N) = \min\{p \in \mathbb{P} : p \geq N/2\}$, the stopped traversal, Canonical Root Dynamics (CRD), the complete theorem registry, the current proof frontier, and the full legacy failure log plus the seven post-v4.81 additions, with the exact post-v4.81 raw checkpoint preserved losslessly in an appendix.

This is not a publication draft. It is a project bible. It contains no marketing language and no suggestion that Goldbach is proved anywhere. Binary Goldbach is **OPEN**. Universal success of the canonical root is **OPEN**. The *old finite-selected-transcript* p_0 architecture is **SUPERSEDED**; complete-support and multigeneration p_0 reachability remain live.

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Chapter 1

AI Navigation Map

HOW TO USE THIS DOCUMENT

If you are a model receiving only this file, read in this order.

1. **Chapter 2** (three pages) — what is true right now. Do not skip it; several beliefs that were live in earlier versions of the corpus are dead.
2. **Chapter 3** — the notation is fixed. One symbol has one meaning throughout. Do not silently re-define h, m, Q, R, T, U, J, η, ν, γ.
3. **Chapter 4** — the theorem registry. Before using any result, look up its status and dependencies here.
4. **Chapter 9** — the only place where new work should start.
5. **Chapters 12 and 13** — read *before* proposing anything. The legacy log has 239 entries and the post-v4.81 continuation adds seven binding failure/scope-correction records.

Three rules that are violated most often.

1. Failure of the canonical traversal is *not* global Goldbach failure. Never invoke a theorem whose hypothesis is $\mathrm{Good}_N = \varnothing$ under p_0-failure alone. `FAIL-UPPERHALF-SCOPE-001`
2. A successful-world witness does not refute a statement conditioned on a closed failed world; a noncanonical root does not refute a canonical statement. `FAIL-F1-DOMAIN-MIX-001`
3. Never quote a count without its domain. The unit-complement row $N - r = 1$ is a separate class and is excluded everywhere. `FAIL-UNIT-COMPLEMENT-001`, `FAIL-CENSUS-DOMAIN-001`

Never raise a status. OPEN, CONJECTURE, EMPIRICAL PATTERN, VERIFIED COMPUTATION and PROVED are different things and stay different. A finite scan with zero counterexamples is never promoted to a theorem. If you prove something, add a new registry row; do not edit an old one.

IF YOU NEED X $\longrightarrow$ GO TO SECTION Y

If you need ...	Go to ...
definitions (all of them)	Chapter 3: notation registry §3.1, objects §3.2, valuation layer §3.7
p_0 — definition, geometry, why it matters	§3.6 (definition), Chapter 6 (whole programme), §6.5 (what is and is not special)
BFS / stopped traversal — exact algorithm	§3.3 (definition), §3.4 (runnable reference implementation), §3.5 (historical variants)

If you need ...	Go to ...
EAC — exceptional autonomous component	§3.8; orientation warning in §3.8; prior-art boundary §15.2
canonical failed world	§3.9 (definition $R_N(r)$, closure, unique source); fixed point B_N^{low} in §3.10
CRD — Canonical Root Dynamics	Chapter 8; epistemic convention §8.1; architecture wall §8.2
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C0b — canonical reflection-defect ledger	§8.4; branch targets TI/TD/TS/TF in §8.5
P1 / P1BR — branchwise production upgrade	§9.4
L1 — actual layer realisation	§9.5
E1 — escape or coupling	§9.6
W1 — simultaneous clean witnesses	§9.7
K1 — source-sensitive capacity	§9.8
X1 — canonical failure exclusion	§9.9
RPD — repeated prime-power redundancy	§9.10
computational evidence — all of it, with ranges	Chapter 11; reproducible generator in Appendix A
failed approaches — curated narrative	Chapter 12
full failure log — 239 legacy entries + 7 post-v4.81 additions	Chapter 13; index §13.1
literature	Chapter 15; what each external theorem does and does not give
current frontier	**§10.8 first** (binding 2026-09-07C override); then Chapter 10 and the historical v4.81 frontier in Chapter 9. The handoff chapter has been updated accordingly.
exact product locks (T_S, T_R, overlap, orthogonality, full-Q)	§5.3
residue no-gos ($q \geq 5$ saturation, mod 3, topology)	§5.4
FRESH/SMOOTH, smoothness tax, first-front conservation and split	§7.3
lossless prime-power root ledger, collision budget, weighted cut	§7.4
Exact-D pair compensation, primitive normal form, gap filters, branch $k = 1$	§7.5
sparse geometric core lock (FFS layer)	§7.6

If you need ...	Go to ...
surviving lemmas of killed theories (CMFT, CGST, CSD, CAPT, RSD, rough-P_2, H1-FP)	§12.3
rejected literature — what was tried and why it failed	§15.3; the 52-source audit §15.3.2; the frontier salvage audit §15.4.1; named missing lemmas §15.5
scope classes A / B / C / D — is this really about p_0?	§5.1; full classification table there
what to work on next — triage rules and tiers	§14.1; the five best moves in §17.4
Euler defect — and why its normalisation matters	§5.7
shadow / retention arithmetic	§5.5
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$T = UJ$, **Euler defect,** η, ν	§3.7, §5.7
D_q, M_q, **FRESH/SMOOTH**	§7.1
B_{abs}, B_{re}, **incidence ledger**	§7.2
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parity barrier / why it is the real wall	§12.4
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what not to try again	Chapter 17

Status vocabulary

Status	Meaning
PROVED	A proof is reproduced here or was independently reconstructed from definitions in the latest hostile audit, and no later correction is known.
PROVED WITH REPAIR	Proved after an explicit repair to hypotheses or to a step. The repair is always stated.

Status	Meaning
AUDITED IDENTITY	An exact identity that survived independent reconstruction on a declared domain, but has not been promoted to canonical PROVED.
VERIFIED COMPUTATION	An exact finite computation with a stated range and provenance. *Never* evidence for a universal statement.
EMPIRICAL PATTERN	A pattern observed in data, with no causal isolation. Not a theorem, not evidence for one.
CONJECTURE	A precisely stated conjecture, usually with computational support.
OPEN	A precisely stated target for which no proof is known.
KILLED	Killed by a proof, a domain correction, or an exact counterexample. Not usable as a lemma.
SUPERSEDED	Retained historically; not a live target. Its correct lemmas may still be used, its programme may not be restarted.
SCOPE-CORRECTED	The statement survives on a narrower domain; what died is a use outside that domain. Used inside the imported failure log, where it is the entry-level analogue of PROVED WITH REPAIR, and for imported results whose domain was narrowed.
Two statuses carried over verbatim from the sources, so that no import is silently promoted:	
PROVED – PENDING RED TEAM	The source states a complete proof but records that the result has not been adversarially audited ([BIB] `PROVED-PENDING-REDTEAM`). Stronger than CONJECTURE, weaker than PROVED. **Do not upgrade it by citing it.**
NEEDS VERIFICATION	The source states the result but flags an unverified step or an unverified uniform dependence ([BIB] `NEEDS VERIFICATION`). Treat as CONJECTURE until the flagged step is discharged.

Status mapping used when importing from the sources. [BIB]/[CORE] `PROVED` → PROVED; `PROVED-PENDING-REDTEAM` → PROVED – PENDING RED TEAM; `PROVED-PENDING-INDEPENDENT-AUDIT` → PROVED – PENDING RED TEAM; `NEEDS VERIFICATION` → NEEDS VERIFICATION; `TRANSFERRED-PROVED` → PROVED; `SOURCE CLAIM: NEW-PROVED` and `SOURCE: PROVED-PENDING` → PROVED – PENDING RED TEAM; `CONDITIONAL` → CONJECTURE with the condition stated; `COMPUTATIONAL/EXPERIMENT` → VERIFIED COMPUTATION; `HEURISTIC` → EMPIRICAL PATTERN; `WORKING` → OPEN; `REJECTED/DISPROVED/ABANDONED` → KILLED; `CORRECTED` → SCOPE-CORRECTED; `RETIRED` → SUPERSEDED. The mapping never raises a status.

Chapter 2

Current State of the Project

Binding state after the 2026-09-07C continuation. The project studies binary Goldbach through the factorization graph $p \to q$ when $q \mid N - p$ and through the canonical selector $p_0(N) = \min\{p \in \mathbb{P} : p \geq N/2\}$. The exact boundary, path, valuation and closed-world theory remains available, but the previous blanket "finite bad-tree / finite-local universality" wall was **scope-corrected**: CRT programming controls prescribed selected edges and valuations, not every residual cofactor of a complete BFS frontier. The live obstruction is therefore not "all local work is impossible" but the possibility of a large, moving, completely closed BAD reachable support; no proof of binary Goldbach or universal p_0 reachability is known.

2.1 Top-level status

Statement	Status	Where
Binary Goldbach: every even $N \geq 4$ is a sum of two primes	**OPEN**	external; §15.2
Every even $N \geq 6$ with $N/2$ composite has $\mathrm{Good}_N \neq \varnothing$ where the canonical traversal from p_0 terminates in SUCCESS	**OPEN**	Chapter 9
The old p_0 finite-selected-transcript architecture	**SUPERSEDED**	§8.2; complete-support reachability remains open
Canonical failure has an exact fixed-point characterisation	**PROVED**	§3.10
Exactly six $N \leq 6 \cdot 10^4$ have $B_N^{\mathrm{low}} \neq \varnothing$	**VERIFIED COMPUTATION**	§6.6
p_0 is a genuinely distinguished vertex (boundary, genealogy, uniqueness)	**PROVED**	Chapter 6
A universal dynamic explanation of the empirical reliability of p_0	**OPEN**	§10.9

2.2 What is actually proved

The following are theorems with reproduced or reconstructed proofs. They are safe to use as lemmas; their exact statements and hypotheses are in Chapter 4 and their proofs in Chapters 5–8.

1. **Canonical gap equivalence** (§3.6). For an upper root $r = x + h_r$: $r = p_0 \iff [x, r) \cap \mathbb{P} =$

$\varnothing \iff h_r < G_r$. This is the whole content of "p_0 is special" at the level of the boundary, and it is categorical, not quantitative.

2. **Strict non-reuse and the reflection gate** (§3.6). Every prime $\ell \geq h$ divides at most one element of the strict reflected block $\mathfrak{B}(p_0)$; every upper mirror c_j is composite. The gate gives compositeness only — *not* activation ($N = 242$).
3. **Genealogical defect rank** (§6.1). $\kappa_{\text{gen}}(p_k) = k$ on the non-diagonal domain, the defect ledger is a lossless code of local prime history, and p_0 is the unique reflection-complete upper root.
4. **Boundary filtration** (§6.2). The reflected blocks of consecutive upper roots nest with a shift, and the gcd-coupling matrix of level k embeds as a submatrix of level $k+1$.
5. **Phase trichotomy** (§6.3). $m_0 - p_- = G - 2h$ splits the canonical cell into Type I / II / III with exact consequences for immediate success.
6. **Shadow arithmetic** (§5.5). The prime-gap shadow identity $r \mid N - p \iff r \mid \delta - p$, retention–escape $\mathrm{PF}(N-q) \cap Q = \mathrm{PF}(d_q)$, minimum-factor escape, shadow cycle localisation.
7. **Path invariants** (§5.6). The full path Bézout invariant $K_t p_t - C_t x = (-1)^t h$, the winding normal form, the $\leq 2\tau(h)$ reduced-state bound, no-bad-return-to-p_0, and the global two-sheet collision budget.
8. **Valuation layer** (§5.7). $T_S = U_S J_S$ and the layered Euler defect $\eta(S) + \nu(S) \geq |S| + |\operatorname{Src}(S)|$, canonically $\eta(R) + \nu(R) \geq |R| + 1$.
9. **Post-root factorization ledger** (§7.1, §7.2). $D_q = \gcd(m_0, N-q) = \gcd(B_q, |2h - q|)$, the FRESH/SMOOTH split, and the incidence identities IL1–IL6.
10. **CRD fixed point** (§3.10). $B_N^{\text{low}} = \bigcap_i B_i$ and p_0 fails $\iff$ $N - p_0$ is composite and $\mathrm{PF}(N-p_0) \subseteq B_N^{\text{low}}$.
11. **C0a** (§8.3). The small-gap canonical obligation, via the residual argument.
12. **Exact product locks** (§5.3). The multi-source lock $T_S \equiv (-1)^{|S|} \prod_{\mathrm{Src}} d \pmod N$, the rooted full-front lock $T_R \equiv (-1)^{|R|} p_0$, the source-overlap identity, the orthogonality of disjoint child-closed worlds, and the full-Q ledger and production bounds.
13. **Residue no-gos** (§5.4). The residue-only route saturates for every $q \geq 5$; modulus 3 is the sole exception; and topology alone provably cannot supply incremental J.
14. **The rooted first-front layer** (§7.3–§7.5). Conservation and split identities (source status PROVED – PENDING RED TEAM), and — fully PROVED — the actual prime-power collision budget, the exact weighted cut identity, joint D_q, D_r rigidity, Exact-D pair compensation with its primitive normal form, and the two canonical gap filters.
15. **Scope-corrected partial transcript programming** (§10.2). Prescribed finite divisibilities/-valuations can be programmed, but this does *not* control every unprescribed prime factor of a complete BFS row; the old blanket FBTU/finite-local wall is not a theorem.
16. **Closed-support counting and height** (Chapter 10). A fixed prime alphabet supports only finitely many closed active BAD worlds; Matveev gives an unrestricted closed-support height inequality, an effective bounded-core threshold, and pointwise growth of the largest core label.
17. **Shallow asymptotic structure** (Chapter 10). In the $h = 1$ family, complete radius-two BADness holds for almost every prime start, simultaneously for a growing same-N prefix of nearby roots; short successful paths obey simultaneous label- and cofactor-product barriers.
18. **Exact $h = 1$ conjugacy and canonical embedding** (§10.5). For $N = 2(P-1)$ the complete non-direct canonical BFS from P is exactly the BFS from root 2 after the initial-root replacement. Any finite hereditary first-support graph/valuation obstruction seen at an arbitrary upper BAD root can be embedded, at the same gap, into infinitely many canonical worlds; therefore such an invariant cannot explain universal p_0 reliability.
19. **Low-gap complete-support restrictions** (§10.6). If $0 < h < q = \min \mathrm{PF}(N-r)$ and no GOOD vertex appears through depth 2, at least three distinct nonroot labels occur by depth 3. Reciprocal returns satisfy $b \geq 3q + 2h$; monomial returns force an exponent drop; the remaining three-label obstruction is reduced to a closed three-prime world or an explicit prime-power Diophantine system.
20. **The canonical threshold engine** (§9.2). The container identity, its slack bound, the charge-free overlap normal form, and the trichotomy of surviving routes remain historically correct, but the

binding frontier is now §10.8.

2.3 What is only computational

Everything in Chapter 11 is **VERIFIED COMPUTATION** or **EMPIRICAL PATTERN** and never more. The load-bearing numbers are: the exhaustive canonical scan to $N \leq 10^7$ with zero canonical failures (inclusive tested-vertex convention audited on 2026-09-07); the six values of $N \leq 6 \cdot 10^4$ with $B_N^{\text{low}} \neq \varnothing$, namely 128, 1718, 1862, 1928, 2200, 6142; the ten $N \leq 2 \cdot 10^5$ with escape margin one; and the 301 failing non-canonical upper roots for $N \leq 2 \cdot 10^4$ concentrated on exactly those six N. External verification of Goldbach itself reaches $4 \cdot 10^{18}$ [12]; the project has never verified anything at that scale and must not cite it as its own.

2.4 The current frontier in one paragraph

Current (binding 2026-09-07C). The principal unresolved object is a *moving closed BAD support*. In the canonical $h = 1$ subfamily $N = 2(P-1)$, P prime, the full non-direct BFS from $p_0 = P$ is exactly conjugate to the full BFS from the fixed root 2. Thus the clean global question is whether there can exist a finite active prime set C containing $\mathrm{PF}(P-2)$ such that every $q \in C$ has $2(P-1)-q$ composite with its *entire* prime support still in C. No such failed canonical world is known; no theorem excludes it. The exact local subproblem is the three-label obstruction split of §10.6; the exact asymptotic subproblem is the uncontrolled radius-three region where both the label product and cofactor product are large. See §10.8.

What changed from v4.81. The old finite-local wall was too broad. CRT/Dirichlet programming proves selected finite edge and valuation patterns, but does not suppress unprescribed cofactors, so it cannot be promoted to a theorem about complete bounded-depth BFS neighbourhoods. Conversely, the new canonical valuation-embedding theorem gives a rigorous no-go for a precise class of hereditary first-support invariants. This narrows, rather than closes, the search: any successful explanation of p_0 must use complete support, external factors, absolute size/root location, or genuine multigeneration reachability.

Historical v4.79–v4.81 frontier. The container/population/residue programme in §9.2 remains part of the theorem corpus, but it is no longer the sole or binding description of the frontier.

2.5 The architecture wall

SCOPE-CORRECTED — **what the finite-programming wall actually says.** [2026-09-05 hostile audit; see §10.2]

Finite Affine Canonical Programmability survives: compatible prescribed congruence, valuation and compositeness constraints on finitely many affine forms can be realised in infinitely many canonical $h = 1$ worlds. What does *not* survive is the quantifier jump from selected prescribed edges to a complete BAD BFS neighbourhood. Unspecified cofactors may introduce additional children, including GOOD children. The exact $N = 92$ witness contains an arbitrarily reusable selected BAD cycle while the complete canonical BFS succeeds at depth 1.

Consequently:

1. selected finite BAD transcripts are not contradiction-generating by themselves;
2. a theorem about *complete* bounded-depth frontiers is **not** ruled out by FACP/CRT programming;
3. any proposed complete-failure theorem must explicitly control every residual cofactor in every expanded row;

4. the later hereditary first-support embedding theorem supplies a separate rigorous no-go, but only for its stated hereditary graph/valuation class and not for complete support or eventual reachability.

The old blanket "Finite Bad-Tree Universality" and "Finite-local universality wall" are retained historically only in their scope-corrected form.

2.6 What a new model should not do

Short list; the full list is Chapter 17 and the log itself.

1. Do not restart the local certificate programme in a new coordinate system. Five successor theories already did; all five hit the same wall.
2. Do not treat canonical failure as Goldbach failure. Almost every scope error in the log is a version of this.
3. Do not promote a finite scan. The scan to 10^7 says nothing about 10^8.
4. Do not re-derive proved theorems. Use them by ID.
5. Do not bridge two different functionals with an inequality that holds "morally". That is precisely what $P1^{br}$ and K1 need and precisely what has never been produced.

Chapter 3

Canonical Notation and Definitions

One symbol, one meaning, for the whole file. If a future revision needs a new object, give it a new symbol; do not overload h, m, Q, R, T, U, J, η, ν, γ, κ. Several failure-log entries are nothing but collisions of notation (`FAIL-NOTATION-KAPPA-001`, `FAIL-SYMBOL-R-COLLISION-001`).

3.1 Symbol registry

Symbol	Meaning (fixed throughout)
N	the even target, $N \geq 6$; always $N = 2x$
x	the midpoint $N/2$
$\mathbb{P}$	the primes
p_0	the canonical selector $\min\{p \in \mathbb{P} : p \geq x\}$
p_k	the k-th upper root: $p_0 < p_1 < p_2 < \cdots$, all primes $\geq x$
r	a generic upper root (any prime $\geq x$), *not necessarily* p_0
h, h_r	the offset $p_0 - x$, resp. $r - x$
m_0, m_r	the complement $N - p_0 = x - h$, resp. $N - r = x - h_r$
δ	$2h$; note $N = 2m_0 + \delta$
p_-	the prime predecessor of p_0
G, G_r	the gap $p_0 - p_-$, resp. $r - r^-$
θ	the phase $h/G \in (0, 1)$
Q	$\mathrm{PF}(m_0)$, the canonical prime set
$\mathrm{PF}(n)$	the set of distinct prime factors of n; $\mathrm{PF}(1) = \varnothing$
$P^+(n)$, $P^-(n)$	largest, resp. smallest, prime factor of n
$\Omega(n)$	number of prime factors of n with multiplicity
$\tau(n)$	number of divisors of n
$\mathcal{A}_N$	the active primes: $\{p \in \mathbb{P} : p < N,\ p \nmid N\}$
Good_N	$\{p \in \mathcal{A}_N : N - p \in \mathbb{P}\}$ — the Goldbach witnesses
Bad_N	$\mathcal{A}_N \setminus \mathrm{Good}_N$
$\mathfrak{B}(r)$	the strict reflected block of r
c_j, a_j	upper mirror $r - 2j$, lower mirror $N - c_j = m_r + 2j$
t_r	$\lfloor (h_r - 1)/2 \rfloor$, the block half-length
$\kappa_{\mathrm{gen}}(r)$	genealogical defect rank of the upper root r
ρ_0	bidirectional prime rank, $\#\{s \geq 1 : h < D_s < 2h\}$

Symbol	Meaning (fixed throughout)
R, $R_N(r)$	the stopped failed world of the root r
$\mathrm{Src}(S)$	the sources of S: vertices of S with no parent in S
Reach	reachability closure under $p \to q$
B_i, B_N^{low}	the CRD filtration and its fixed point
G_{esc}	escape capacity (number of escape edges)
D_q, M_q, B_q	$\gcd(m_0, N-q)$, $(N-q)/D_q$, m_0/q
T_S, U_S, J_S	total valuation mass, unit part, layer part on S
η, ν	Euler defect terms: excess valuation, resp. layer count
γ	the coupling/capacity functional in $\mathrm{P1}^{\mathrm{br}}$ and K1
K_t, C_t, W_t	path Bézout and winding coefficients

Warning 3.1 (Reserved letters). R is *always* the failed world, never a ring, a remainder, or a generic set. κ with no subscript is never used; only κ_{gen} and $\kappa^{\pm}$ appear. Q is $\mathrm{PF}(m_0)$ and nothing else — the smooth domain in the incidence ledger is written Q_{smooth}.

3.2 The active prime graph

Definition 3.2 (Activity, edges, GOOD/BAD)**.** Fix an even $N \geq 6$.

- p is *active* iff $p \in \mathbb{P}$, $p < N$ and $p \nmid N$. Write $\mathcal{A}_N$ for the set of active primes.
- The *row* of p is the integer $N - p$ together with its factorization.
- There is a directed edge $p \to q$ iff $q \mid N - p$ and $q \nmid N$, i.e. $q \in \mathrm{PF}(N-p) \cap \mathcal{A}_N$. Write $\Gamma_N(p) = \mathrm{PF}(N-p) \cap \mathcal{A}_N$ for the children of p.
- p is *GOOD* iff $N - p \in \mathbb{P}$ (equivalently $(p, N-p)$ is a Goldbach pair), and *BAD* otherwise.

The condition $q \nmid N$ is not cosmetic: without it the graph acquires absorbing cycles through the divisors of N and every statement about sources fails. [CORE, ARS-ACTIVITY]

Remark 3.3 (Unit complement). $N - p = 1$, i.e. $p = N - 1$, is neither GOOD nor a normal BAD row: its row has empty factorization. It is excluded from every count in this file. Failing to exclude it is `FAIL-UNIT-COMPLEMENT-001`.

Definition 3.4 (Sources)**.** For $S \subseteq \mathcal{A}_N$, $\mathrm{Src}(S) = \{p \in S : \nexists q \in S,\ q \to p\}$. A *closed* set is one with $\Gamma_N(S) \subseteq S$.

3.3 The stopped traversal

Definition 3.5 (Stopped traversal / factor-flow)**.** Given N and an active root r, run breadth-first search from r in the graph of Definition 3.2 with one modification: a popped GOOD vertex is *recorded and not expanded*, and a GOOD child is recorded as an *escape edge* and not enqueued. The traversal returns SUCCESS at depth d if a GOOD vertex is first recorded at depth d, and FAILURE otherwise. The set of BAD vertices actually expanded is the *stopped failed world* $R_N(r)$.

If $2r = N$ (the diagonal case, $x \in \mathbb{P}$) the traversal returns SUCCESS at depth 0 with $R = \varnothing$.

Definition 3.6 (Two modes)**.** Two implementations appear in the corpus and are both correct.

`census` stops at the first popped GOOD vertex. This reproduces the historical node and depth statistics.
`world` continues to exhaustion, building the full $R_N(r)$ and the full escape-edge set.

They agree on the verdict and on the success depth; they differ only in how much of the world is materialised. [CORE, ARS-TRAVERSAL-MODES]

Lemma 3.7 (Mode equivalence). ***PROVED***. *For every N and every active r,* `census` *and* `world` *return the same verdict, and when the verdict is* SUCCESS *the same depth.*

Proof. BFS pops vertices in non-decreasing depth order, and neither mode expands a GOOD vertex, so the multiset of vertices popped strictly before the first GOOD pop is identical in the two modes. The first GOOD pop is therefore the same vertex at the same depth, and the verdict is FAILURE in one mode iff no GOOD vertex is ever popped, iff the same in the other. □

3.4 Reference implementation

The following is the canonical executable semantics of Definitions 3.2–3.6. It is self-contained (standard library only), deterministic (fixed Pollard-rho seed), and is the file `p0_algorithm.py` of the project repository with its comments translated into English; the executable code is unchanged. A future model that is unsure what a definition means should run this rather than guess.

Listing 3.1: `p0_algorithm.py` — canonical selector and stopped traversal, runnable as given.

```python
#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""
p0_algorithm.py -- minimal, fully self-contained implementation of the
canonical selector p_0 and of the historical stopped traversal.

No imports outside the standard library. Copy and run:

    python3 p0_algorithm.py 1928 3 # N = 1928, roots p_0, p_1, p_2
    python3 p0_algorithm.py # demonstration on the six EAC
"""
import sys
from collections import deque
from math import gcd
from random import Random

# --- 1. primality: deterministic Miller-Rabin for n < 3.3 * 10^24 ----------
BASES = (2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37)


def is_prime(n):
    if n < 2:
        return False
    for p in BASES:
        if n % p == 0:
            return n == p
    d, s = n - 1, 0
    while d % 2 == 0:
        d //= 2
        s += 1
    for a in BASES:
        x = pow(a, d, n)
        if x in (1, n - 1):
            continue
        for _ in range(s - 1):
            x = x * x % n
            if x == n - 1:
                break
        else:
            return False
    return True


def next_prime(n):
    m = n + 1
    while not is_prime(m):
        m += 1
    return m


def prev_prime(n):
```

```
    m = n - 1
    while m >= 2 and not is_prime(m):
        m -= 1
    return m if m >= 2 else None


# --- 2. factorization: trial division + Pollard rho (deterministic seed) ---
def _rho(n, rng):
    if n % 2 == 0:
        return 2
    while True:
        c, y = rng.randrange(1, n), rng.randrange(0, n)
        g, r, q, x, ys = 1, 1, 1, 0, 0
        while g == 1:
            x = y
            for _ in range(r):
                y = (y * y + c) % n
            k = 0
            while k < r and g == 1:
                ys = y
                for _ in range(min(128, r - k)):
                    y = (y * y + c) % n
                    q = q * abs(x - y) % n
                g, k = gcd(q, n), k + 128
            r *= 2
        if g == n:
            g = 1
            while g == 1:
                ys = (ys * ys + c) % n
                g = gcd(abs(x - ys), n)
        if g != n:
            return g


def factorize(n):
    """Returns {p: v_p(n)}; factorize(1) = {}."""
    f, rng = {}, Random(20260904)
    for p in range(2, 100000):
        if p * p > n:
            break
        while n % p == 0:
            f[p] = f.get(p, 0) + 1
            n //= p
    stack = [n] if n > 1 else []
    while stack:
        m = stack.pop()
        if m == 1:
            continue
        if is_prime(m):
            f[m] = f.get(m, 0) + 1
            continue
        d = _rho(m, rng)
        stack += [d, m // d]
    return dict(sorted(f.items()))


# --- 3. canonical selector and the successive upper roots ------------------
def p0(N):
    """p_0(N) = min{ p prime : p >= N/2 }."""
    if N % 2 or N <= 2:
        raise ValueError("N must be even and greater than 2")
    x = N // 2
    return x if is_prime(x) else next_prime(x)


def upper_roots(N, count):
    """[p_0, p_1, ..., p_{count-1}] -- consecutive primes >= N/2."""
    out, p = [], p0(N)
    while len(out) < count and p < N:
        out.append(p)
        p = next_prime(p)
    return out


# --- 4. graph: edge, GOOD vertex, activity --------------------------------
```

```
def is_active(N, p):
    """p active for N <=> p prime, p < N, p does not divide N."""
    return is_prime(p) and p < N and N % p != 0


def is_good(N, p):
    """GOOD <=> N - p prime, i.e. N = p + (N-p) is a Goldbach pair."""
    return N - p >= 2 and is_prime(N - p)


def children(N, p):
    """Active children of the row N - p: {q prime : q | N-p, q does not divide N}."""
    return sorted(q for q in factorize(N - p) if N % q != 0)


# --- 5. HISTORICAL STOPPED TRAVERSAL / FACTOR-FLOW -------------------------
def stopped_traversal(N, root):
    """Full BFS with stopping at GOOD vertices.

    Returns a dict:
      verdict 'SUCCESS' or 'FAILURE' of the root,
      depth depth of the first GOOD (None on failure),
      witness Goldbach pair (p, N-p) found by the traversal,
      R the stopped bad world (BAD vertices that were expanded),
      escapes escape edges (p, q): p in R, q GOOD,
      rows factorizations of N - p for p in R.
    """
    x = N // 2
    if 2 * root == N: # diagonal case
        return dict(verdict="SUCCESS", depth=0, witness=(root, root),
                    R=set(), escapes=set(), rows={}, diagonal=True)
    if not is_active(N, root):
        raise ValueError("root must be an active prime")

    queue, queued = deque([(root, 0)]), {root}
    R, rows, escapes = set(), {}, set()
    verdict, depth, witness = "FAILURE", None, None

    while queue:
        p, d = queue.popleft()
        if p in R:
            continue
        if is_good(N, p): # GOOD: record, do NOT expand
            if witness is None:
                verdict, depth, witness = "SUCCESS", d, (p, N - p)
            continue
        R.add(p) # BAD: expand the whole row
        rows[p] = factorize(N - p)
        for q in sorted(q for q in rows[p] if N % q != 0):
            if is_good(N, q):
                escapes.add((p, q))
                if witness is None:
                    verdict, depth, witness = "SUCCESS", d + 1, (q, N - q)
                continue # a GOOD child is not enqueued
            if q not in R and q not in queued:
                queued.add(q)
                queue.append((q, d + 1))

    return dict(verdict=verdict, depth=depth, witness=witness, R=R,
                escapes=escapes, rows=rows, diagonal=False)


# --- 6. report ---------------------------------------------------------------
def show(N, k=1):
    x, r0 = N // 2, p0(N)
    pm = prev_prime(r0)
    h, m = r0 - x, N - r0
    fm = factorize(m)
    print(f"N = {N}, x = N/2 = {x}")
    print(f"p_0 = {r0}, h = {h}, m_0 = {m} = "
          + (" * ".join(f"{p}^{e}" if e > 1 else str(p) for p, e in fm.items())
             or "1"))
    print(f"previous prime = {pm}, G = {r0 - pm}, h < G : {h < r0 - pm}")
    print(f"delta = 2h = {2*h}, Q = PF(m_0) = {sorted(fm)}")
    for i, r in enumerate(upper_roots(N, k)):
```

```python
        t = stopped_traversal(N, r)
        hr = r - x
        tt = (hr - 1) // 2
        kap = sum(1 for j in range(1, tt + 1) if is_prime(r - 2 * j))
        print(f" p_{i} = {r:<10d} {t['verdict']:<8s} depth={t['depth']}"
              f" |R|={len(t['R']):<4d} escapes={len(t['escapes']):<3d}"
              f" kappa_gen={kap} witness={t['witness']}")


if __name__ == "__main__":
    if len(sys.argv) > 1:
        show(int(sys.argv[1]), int(sys.argv[2]) if len(sys.argv) > 2 else 1)
    else:
        for N in (128, 1718, 1862, 1928, 2200, 6142, 11822, 14198):
            show(N, 3)
            print()
```

Remark 3.8 (Reading the code as a definition)*.* Three lines carry all the semantics that prose gets wrong. `if is_good(N, p): ...continue` is the stopping rule (a GOOD vertex is never expanded). `escapes.add((p, q)); ...continue` is the escape rule (a GOOD child is recorded, not enqueued). `if N % q != 0` in `children` is activity. Every historical variant that differs from the corpus differs in exactly one of these three lines.

3.5 Historical variants and how they differ

Variant	Difference	Consequence
Canonical (audited)	As Definition 3.5	The reference. All statistics in this file use it.
Expanding GOOD	Expands GOOD vertices too	Depth statistics change; the world is no longer the failed world; several early tables in the corpus were produced this way and do not match later ones. `FAIL-TRAVERSAL-EXPANDGOOD-001`
Ignoring activity	Allows $q \mid N$ as a child	Creates absorbing cycles; "source" becomes ill-defined; kills the Euler defect bound.
DFS instead of BFS	Depth-first order	Verdict unchanged, depth statistic meaningless. Depth is only defined relative to BFS.
Unit row admitted	$p = N - 1$ counted	Off-by-one in every census; see Remark 3.3.
Non-canonical root	Starts at p_k, $k \geq 1$, or below x	Legitimate, but its results belong to scope class B (root-generic) and must never be quoted as canonical. `FAIL-F1-DOMAIN-MIX-001`

3.6 The canonical selector and midpoint geometry

Definition 3.9 (Normal form)**.** Write

$$N = 2x, \qquad p_0 = x + h, \qquad m_0 := N - p_0 = x - h, \qquad \delta := 2h,$$

$$p_- := \max\{p \in \mathbb{P} : p < p_0\}, \qquad G := p_0 - p_-, \qquad Q := \mathrm{PF}(m_0), \qquad \theta := \frac{h}{G}.$$

If $h = 0$ then $x \in \mathbb{P}$ and $p_0 = x$: the *diagonal* case. Unless stated otherwise the *non-diagonal* case is assumed: x composite, $h \geq 1$. The exact identity

$$\boxed{N = 2m_0 + \delta}$$

holds always. The same symbols with subscript r denote the analogous quantities for an arbitrary upper root $r = x + h_r \geq x$.

Theorem 3.10 (Canonical gap equivalence). **PROVED**. *Let $r = x + h_r \geq x$ be prime, r^- its predecessor and $G_r = r - r^-$. Then*

$$\boxed{r = p_0 \iff [x, r) \cap \mathbb{P} = \varnothing \iff h_r < G_r.}$$

Proof. If $r = p_0$ then $r^- < x$, so $G_r = r - r^- > r - x = h_r$. Otherwise some prime lies in $[x, r)$, hence $r^- \geq p_0 \geq x$ and $G_r \leq h_r$. [CORE, C-GAP-EQUIVALENCE] □

Warning 3.11 (How *not* to read $h < G$). The inequality $h < G$ looks like a statement about "a large prime gap before p_0". That reading is wrong and was a recurring source of error. Theorem 3.10 is *categorical*, not quantitative: no earlier upper prime lies inside the reflected interval. The correct formulation [CORE, V71.43] is

$$\boxed{p_0 \text{ is a } \textit{defect-free boundary condition selected by midpoint minimality.}}$$

Equivalently [CORE, V71.42]: p_0 is the unique upper root whose predecessor lies geometrically *outside* its reflected block. Nothing about the *size* of G is asserted, and no bound of Baker–Harman–Pintz type is used or needed here.

Definition 3.12 (Strict reflected block). For an upper root $r = x + h_r$ with $h_r \geq 3$ put

$$t_r = \left\lfloor \tfrac{h_r - 1}{2} \right\rfloor, \quad \mathcal{J}_{h_r} = \{1, \dots, t_r\}, \quad c_j = r - 2j, \quad a_j = N - c_j = m_r + 2j,$$

$$\mathfrak{B}(r) = \{a_1, \dots, a_{t_r}\} \cup \{c_1, \dots, c_{t_r}\}.$$

Then $m_r < a_j < x < c_j < r$ and $a_j + c_j = N$. The endpoints $u = 0$ and $u = h_r$, i.e. m_r and r themselves, are *not* in $\mathfrak{B}(r)$. Call c_j an *upper mirror* and a_j a *lower mirror*.

Lemma 3.13 (Strict non-reuse of large primes). **PROVED**. *If an odd prime ℓ divides two distinct elements of $\mathfrak{B}(r)$ then $\ell < h_r$. Equivalently, every prime $\ell \geq h_r$ occurs in exactly one element of $\mathfrak{B}(r)$.*

Proof. For $i \neq j$, $|a_i - a_j| = |c_i - c_j| = 2|i - j|$, and for any i, j, $c_j - a_i = 2(h_r - i - j)$. Since $i + j \leq 2t_r < h_r$ the last quantity is positive, so all three differences are nonzero. In each case half the difference is an integer in $\{1, \dots, h_r - 1\}$; an odd prime dividing twice that half divides the half itself, hence is smaller than h_r. [CORE, C-STRICT-NONREUSE] □

Warning 3.14 (No widening of the block). Lemma 3.13 may *not* be applied to the widened block $\{m_r + 2u : 0 \leq u \leq h_r\}$. The endpoint pair $(u, v) = (0, h_r)$ has difference exactly $2h_r$ and is not covered by the proof; it would require a separate canonical input $\gcd(h, m_0) = 1$ and is nowhere used that way below. [FAIL, FAIL-BROADBLOCK-ENDPOINT]

Lemma 3.15 (Reflection gate). **PROVED**. *Let $r = p_0$ and $0 < 2j < h$. Then $c_j \in (x, p_0)$ and c_j is composite. Its full factorization is a graph row with parent a_j* if and only if *a_j is an active prime; in that case $\Gamma_N(\{a_j\}) = \mathrm{PF}(c_j)$.*

Proof. Compositeness of c_j is immediate from $[x, p_0) \cap \mathbb{P} = \varnothing$; the rest is the definition of an edge. [CORE, ARS-REFLECTION-GATE] □

Warning 3.16 (The gate gives compositeness, not activation). **KILLED** is every strengthening of the gate to activation. The gate gives *only* compositeness of each c_j: it does not give primality of a_j, does not give incidence coupling, does not give basin coupling and does not give capacity. Exact counterexample: $N = 242$, $x = 121$, $p_0 = 127$, $p_- = 113$, so $h = 6 < G = 14$; both reflected values $a_1 = 117 = 3^2 \cdot 13$ and $a_2 = 119 = 7 \cdot 17$ are composite, so *not a single active reflected row exists.* [FAIL, FAIL-REFLECTION-ACTIVATION-001]

This is not an accident: $N = 242$ is in phase Type I, and a guaranteed active lower mirror exists exactly in phase Type III (§6.3).

Scope note 3.17 (Diagonal correction to the genealogical layer). **SCOPE-CORRECTED**. The corpus states the genealogy and reflection-completeness uniqueness theorems for "every prime $r \geq x$" [CORE, V71.21, V71.31], while the proof uses the strict inequality $p_i > x$. The difference is real.

If $x = N/2$ is *prime* then $p_0 = x$, $h = 0$, and the hole corresponding to p_0 sits at index $h_k/2 > t_k$, i.e. *outside* the strict reflected block. Then $\kappa_{\text{gen}}(p_k) = k - 1$ for $k \geq 1$, and the root p_1 is also reflection-complete — violating literal uniqueness.

VERIFIED COMPUTATION: the smallest witness is $N = 14$, $x = 7 \in \mathbb{P}$, $p_0 = 7$, $p_1 = 11$, block $\{c_1 = 9 = 3^2\}$, $\kappa_{\text{gen}}(p_1) = 0$ although $p_1 \neq p_0$. For $N \leq 6 \cdot 10^4$, $k \leq 5$, *every* violation has x prime, and for x composite there is not a single violation.

What survives. Theorem 3.10 is unaffected in all cases. Only the equivalence with "every $r - 2j$ is composite" fails, because the strict reflected block covers the *open* interval (x, r) while the canonical condition concerns the *closed* interval $[x, r)$. Correct repair: add the hypothesis "x composite", or replace $[x, r)$ by (x, r).

Why this does not disturb the rest of the file. All failure analysis assumes $h \geq 1$: for x prime the canonical root wins immediately and there is no closed world. The domain of every p_0-specific theorem is therefore non-diagonal by definition.

3.7 The valuation layer

Definition 3.18 (Valuation mass, unit part, layer part)**.** For a finite set S of active rows put, for each active prime q,

$$d_S(q) = \sum_{p \in S} v_q(N - p), \qquad \kappa_S(q) = \#\{p \in S : \ q \mid N - p\},$$

and define, the products running over $q \in \Gamma_N(S)$,

$$T_S = \prod_q q^{d_S(q) - 1}, \qquad U_S = \prod_{p \in S} \frac{N - p}{\operatorname{rad}(N - p)}, \qquad J_S = \prod_q q^{\kappa_S(q) - 1}.$$

U_S is the *unit part* (excess valuation, i.e. repeated primes inside one row), J_S the *layer part* (support reuse, i.e. the same prime appearing in several rows).

Theorem 3.19 (Exact minimal-collision factorization)**.** ***PROVED***. $T_S = U_S J_S$.

Proof. Fix $q \in \Gamma_N(S)$ with positive valuations $e_1, \ldots, e_k$ ($k = \kappa_S(q)$, $\sum e_i = d_S(q)$). Its exponent in T_S is $\sum e_i - 1$, in U_S it is $\sum e_i - k$, and in J_S it is $k - 1$; these add up. [CORE, G-TUJ] □

Remark 3.20 (Reading). The radical defect of the world splits *exactly* into excess valuation and support reuse. This is completely generic (scope class B): it uses nothing about p_0.

The Euler-defect terms η and ν built on top of this, and the inequality $\eta(S) + \nu(S) \geq |S| + |\operatorname{Src}(S)|$ that the corpus records, are discussed with their exact status in §5.7. Read that section before using them: the normalisation matters and is easy to get wrong.

Warning 3.21 (η is not a capacity). η measures valuation leaving the set. It is *not* the functional γ appearing in $\mathrm{P1}^{\mathrm{br}}$ and K1, and the repeated attempt to identify them is the single most common dead end in the log (`FAIL-ETA-GAMMA-IDENTIFY-001` and its successors).

3.8 Exceptional autonomous components

Definition 3.22 (EAC)**.** An *exceptional autonomous component* for N is a nonempty set $C \subseteq \mathrm{Bad}_N$ that is closed under Γ_N ($\Gamma_N(C) \subseteq C$) and contains no GOOD vertex. Equivalently: a nonempty closed bad set. Bożek [8] introduced this notion for the Goldbach factorization graph.

Warning 3.23 (Orientation). Two orientations of the Goldbach graph exist in the literature and in this project. The corpus uses $p \to q$ iff $q \mid N - p$ ("the row of p points to its prime factors"). Reversing the arrows turns closed sets into reachable sets and *silently changes what "autonomous" means.* Several early comparisons with [8] were wrong for this reason (`FAIL-EAC-ORIENTATION-001`). Always state the orientation.

Remark 3.24 (EAC and canonical failure). The canonical root p_0 fails exactly when it lies in an EAC. This is the qualitative form of the fixed-point characterisation in §3.10; the quantitative form is the statement that the relevant EAC is B_N^{low} restricted to the reachable set of p_0.

3.9 The canonical failed world

Definition 3.25 (Stopped failed world)**.** $R_N(r)$ is the set of BAD vertices expanded by the stopped traversal from r (Definition 3.5). An *escape edge* is a pair (v, q) with $v \in R_N(r)$, $q \in \Gamma_N(v)$ and q GOOD. The traversal fails iff there is no escape edge and r itself is BAD.

Lemma 3.26 (Closure and unique source)**.** **PROVED**. *If the traversal from r fails then $R_N(r)$ is closed ($\Gamma_N(R_N(r)) \subseteq R_N(r)$), consists entirely of BAD vertices, and $\mathrm{Src}(R_N(r)) = \{r\}$ provided no $v \in R_N(r)$ has $r \in \Gamma_N(v)$. For $r = p_0$ the proviso holds unconditionally (no-bad-return-to-p_0, §5.6), so*

$$\mathrm{Src}(R_N(p_0)) = \{p_0\}.$$

Proof. Closure and badness are immediate from the traversal rules (a BAD vertex is expanded and all its active children enqueued; failure means no child is GOOD). r is a source by construction. Any other source would be a vertex enqueued with no parent in the world, impossible since every enqueued vertex other than r was enqueued from its parent. The canonical case uses Theorem 5.33. [CORE, ARS-CLOSED-WORLD] ☐

Scope note 3.27 (Failure of the canonical traversal is not Goldbach failure). $R_N(p_0) \neq \varnothing$ and no escape means only that *this traversal from this root* found nothing. Good_N may still be large. The six values of N with $B_N^{\mathrm{low}} \neq \varnothing$ all satisfy Goldbach comfortably. Every theorem whose hypothesis is $\mathrm{Good}_N = \varnothing$ is a *class-D* statement (Goldbach-counterexample-conditioned) and may never be invoked under canonical failure alone. `FAIL-UPPERHALF-SCOPE-001`

3.10 The CRD fixed point B_N^{low}

Definition 3.28 (CRD filtration)**.** Put

$$B_0 = \{p \in \mathcal{A}_N :\ p < N/3,\ N - p \text{ composite}\}, \qquad B_{i+1} = \{p \in B_i :\ \mathrm{PF}(N-p) \subseteq B_i\},$$

and $B_N^{\mathrm{low}} = \bigcap_{i \geq 0} B_i$. The sequence is decreasing and finite, so the intersection is reached in finitely many steps.

Theorem 3.29 (Fixed-point characterisation of canonical failure). ***PROVED***. *B_N^{low} is the largest Γ_N-closed subset of $\{p < N/3 : \ N-p$ composite$\}$, and*

$$p_0 \textit{ fails} \iff N - p_0 \textit{ is composite and } \mathrm{PF}(N-p_0) \subseteq B_N^{\mathrm{low}}.$$

Proof. B_{i+1} removes from B_i exactly the elements with a child outside B_i, so B_N^{low} is closed and contains every closed subset. For the equivalence: if p_0 fails then $R_N(p_0) \setminus \{p_0\}$ is closed, bad, and contained in the $N/3$ range (a child $q \mid N-p$ with $q \geq N/3$ would force $N-p \in \{q, 2q\}$, both excluded for a BAD row with $p > N/3$), hence contained in B_N^{low}; and $\mathrm{PF}(N-p_0) \subseteq R_N(p_0)$. Conversely if $\mathrm{PF}(N-p_0) \subseteq B_N^{\mathrm{low}}$ then the traversal never leaves B_N^{low} after the first step and B_N^{low} contains no GOOD vertex. [PCRD, CRD fixed point] □

Remark 3.30 (Why the $N/3$ threshold). For $p > N/3$ and $N-p$ composite, every prime factor q of $N-p$ satisfies $q \leq (N-p)/2 < N/3$. So the interesting closed sets live below $N/3$ and the threshold costs nothing. Using $N/2$ instead re-admits p_0 itself and makes the filtration non-monotone in N; this is `FAIL-BLOW-THRESHOLD-001`.

Scope note 3.31 (B_N^{low} is a computation, not a mechanism). Theorem 3.29 converts "does the canonical traversal fail" into "is a certain explicitly computable set nonempty". That is an exact reformulation and it made the $6 \cdot 10^4$ census possible. It is *not* progress towards a proof: deciding $B_N^{\mathrm{low}} = \varnothing$ for all N is at least as hard as the original question, and the finite-local universality wall (§2.5) says no bounded certificate decides it.

Chapter 4

Theorem Registry

How to use this chapter. Every named result of the project has one row. Before using a result anywhere, read its row: the *status* tells you whether it is a theorem, the *dependencies* tell you what breaks if a dependency is later corrected, and the *location* points at the proof or at the failure-log entry that killed it.
Rules. (i) A row is never edited to a higher status. If you prove something that was CONJECTURE, add a new row and mark the old one SUPERSEDED. (ii) A VERIFIED COMPUTATION row always carries its range. A range is part of the statement. (iii) A KILLED row is not a lemma. It may not be used "in the direction that still works" without a new row stating precisely that direction. (iv) Dependencies are transitive; the registry lists only immediate ones.

4.1 Foundational and boundary layer

ID	Statement / short name	Status	Dependencies	Location
`C-GAP-EQ`	Canonical gap equivalence: $r = p_0 \iff [x,r)\cap\mathbb{P} = \varnothing \iff h_r < G_r$	PROVED	—	Thm. 3.10
`C-BOUNDARY-READING`	p_0 is a defect-free boundary condition selected by midpoint minimality; *not* a large-gap statement	PROVED	`C-GAP-EQ`	Warn. 3.11
`C-STRICT-NONREUSE`	Every prime $\ell \geq h_r$ divides at most one element of $\mathfrak{B}(r)$	PROVED	Def. 3.12	Lem. 3.13
`C-BROADBLOCK-FAIL`	The widened block $\{m_r+2u : 0 \leq u \leq h_r\}$ does *not* satisfy strict non-reuse (endpoint pair)	KILLED	`C-STRICT-NONREUSE`	Warn. 3.14
`ARS-REFLECTION-GATE`	Every upper mirror c_j of p_0 is composite; its row has parent a_j iff a_j is an active prime	PROVED	`C-GAP-EQ`	Lem. 3.15
`C-GATE-ACTIVATION`	The gate also gives activation of some a_j	KILLED	—	Warn. 3.16; $N = 242$
`C-DIAGONAL-SCOPE`	Genealogy and uniqueness hold on the non-diagonal domain (x composite); the diagonal case is a genuine exception	SCOPE-CORRECTED	`V71.21`, `V71.31`	Scope 3.17; $N = 14$

ID	Statement / short name	Status	Dependencies	Location
C-NORMALFORM	$N = 2m_0 + \delta$, $\delta = 2h$, $\theta = h/G$	AUDITED IDENTITY		Def. 3.9

4.2 Graph, traversal and world layer

ID	Statement / short name	Status	Dependencies	Location
ARS-ACTIVITY	Activity ($q \nmid N$) is required for the source theory; without it the graph has absorbing cycles	PROVED	—	Def. 3.2
ARS-TRAVERSAL-MODES	`census` and `world` agree on verdict and success depth	PROVED	Def. 3.6	Lem. 3.7
ARS-CLOSED-WORLD	On failure, $R_N(r)$ is closed and all-BAD; $\mathrm{Src}(R_N(p_0)) = \{p_0\}$	PROVED	ARS-NORETURN	Lem. 3.26
ARS-NORETURN	No-bad-return-to-p_0: no $v \in R_N(p_0)$ has $p_0 \in \Gamma_N(v)$	PROVED	C-GAP-EQ	Thm. 5.33
CRD-FIXPOINT	B_N^{low} is the largest closed bad set below $N/3$; p_0 fails iff $N - p_0$ composite and $\mathrm{PF}(N - p_0) \subseteq B_N^{\mathrm{low}}$	PROVED	ARS-CLOSED-WORLD	Thm. 3.29
CRD-THRESHOLD	The $N/3$ threshold is the correct one; $N/2$ breaks monotonicity	PROVED	CRD-FIXPOINT	Rem. 3.30
EAC-DEF	EAC = nonempty closed bad set; p_0 fails iff p_0 lies in an EAC	AUDITED IDENTITY	CRD-FIXPOINT	§3.8
EAC-ORIENT	The orientation convention is $p \to q \iff q \mid N - p$; reversing it changes "autonomous"	PROVED	—	Warn. 3.23
C-SCOPE-D	Canonical failure does *not* imply $\mathrm{Good}_N = \varnothing$	PROVED	—	Scope 3.27

4.3 Genealogy, filtration and phase

ID	Statement / short name	Status	Dependencies	Location
V71.21	Genealogical defect rank: $\kappa_{\mathrm{gen}}(p_k) = k$	PROVED	C-DIAGONAL-SCOPE	Thm. 6.2
V71.31	p_0 is the unique reflection-complete upper root ($\kappa_{\mathrm{gen}} = 0$)	PROVED	V71.21	Cor. 6.5
C-LEDGER-LOSSLESS	The defect ledger is a lossless code of local prime history: hole positions recover the prime gaps	PROVED	V71.21	Thm. 6.3
C-LEDGER-EXPLAINS	The lossless ledger *explains* why p_0 succeeds	KILLED	—	Warn. 6.7
C0b-RD	The canonical ledger partitions as $\mathcal{J}_h = \mathcal{I} \uplus \mathcal{D} \uplus \mathcal{S} \uplus \mathcal{F}$ with no fifth class	PROVED	V71.21	Thm. 8.6

ID	Statement / short name	Status	Dependencies	Location
C-FILTRATION	Boundary filtration: $c^{(k+1)}_{i+s} = c^{(k)}_i$, $a^{(k+1)}_{j+s} = a^{(k)}_j$, $t_{k+1} = t_k + s$, $s = (p_{k+1} - p_k)/2$	PROVED	Def. 3.12	Thm. 6.8
C-GCD-EMBED	The gcd-coupling matrix $M^{(k)}_{ij} = \gcd(c_i, a_j)$ embeds as a submatrix of level $k+1$	PROVED	C-FILTRATION	Cor. 6.9
C-PHASE	Phase trichotomy from $m_0 - p_- = G - 2h$: Type I ($2h < G$), Type II ($2h = G$), Type III ($2h > G$)	PROVED	C-GAP-EQ	Thm. 6.12
C-PHASE-I	Type I ⇒ m_0 composite ⇒ immediate success impossible	PROVED	C-PHASE	Thm. 6.12
C-PHASE-II	Type II ⇒ $m_0 = p_-$ ⇒ immediate success automatic	PROVED	C-PHASE	Thm. 6.12
C-PHASE-III	Type III ⇒ $p_- = a_{j^*}$ with $j^* = h - G/2$	PROVED	C-PHASE	Thm. 6.12
C-PHASE-CELL	Cell decomposition $1+(G/2-1)+1+(G/2-1) = G$; θ symmetric under $\theta \mapsto 1-\theta$	AUDITED IDENTITY	C-PHASE	§6.3
C-RHO	$\rho_0 = \#\{s \geq 1 : h < D_s < 2h\}$, $D_s = p_0 - p_{-s}$; and $\rho_0 \geq 1 \iff$ Type III	PROVED	C-PHASE	Thm. 6.17
C-RHO-USERFORM	The originally proposed form of ρ_0 (without the lower bound $D_s > h$)	KILLED	C-RHO	Warn. 6.18
C-SIGNATURE	Bidirectional signature $(\kappa^+, \kappa^-) = (\kappa_{\text{gen}}, \rho_0)$	AUDITED IDENTITY	T7.21, C-RHO	§6.4

4.4 Shadow, path and valuation layers

ID	Statement / short name	Status	Dependencies	Location
S-SHADOW-ID	Prime-gap shadow identity: $r \mid N - p \iff r \mid \delta - p$ for $r \mid m_0$	PROVED	C-NORMALFORM	Thm. 5.22
S-RETENTION	Retention–escape: $\mathrm{PF}(N-q) \cap Q = \mathrm{PF}(d_q)$ with $d_q = \gcd(m_0, \lvert\delta - q\rvert)$	PROVED	S-SHADOW-ID	Thm. 5.24
S-MINFACTOR	Minimum-factor escape: if $P^-(N-q) \notin Q$ the traversal escapes Q at q	PROVED	S-RETENTION	Cor. 5.25
S-CYCLE-LOC	Shadow cycle localisation: a cycle inside Q forces all its vertices into an explicit residue window	PROVED	S-SHADOW-ID	Thm. 5.27
S-TWOCYCLE	Shadow two-cycle equation $\delta = q + s + tqs$	PROVED	S-CYCLE-LOC	Prop. 5.28
S-PPWITNESS	Prime-power witness: a prime power row forces an explicit escape or an explicit divisibility	PROVED	S-RETENTION	Prop. 5.29

ID	Statement / short name	Status	Dependencies	Location
`P-BEZOUT`	Full path Bézout invariant $K_t p_t - C_t x = (-1)^t h$	PROVED	—	Thm. 5.30
`P-WINDING`	Winding normal form $K_t p_t = W_t N + (-1)^t p_0$	PROVED	`P-BEZOUT`	Cor. 5.31
`P-STATES`	At most $2\tau(h)$ reduced states per vertex	PROVED	`P-WINDING`	Thm. 5.32
`P-COLLISION`	Global two-sheet collision budget: at most two collision endpoints	PROVED	`P-STATES`	Thm. 5.34
`P-COLLISION-CONTRA`	The collision budget yields a contradiction from canonical failure	KILLED	`P-COLLISION`	§5.6
`V-TUJ`	$T_S = U_S J_S$ (excess valuation $\times$ support reuse)	PROVED	Def. 3.18	Thm. 3.19
`V-EULER`	Layered Euler defect $\eta(S) + \nu(S) \geq \lvert S\rvert + \lvert \mathrm{Src}(S)\rvert$; canonically $\eta(R) + \nu(R) \geq \lvert R\rvert + 1$ — *normalisation of η, ν not reconstructed here*	CONJECTURE	`V-TUJ`, `ARS-CLOSED-WORLD`	§5.7
`V-EULER-A`	Active-mass form $\sum_q d_R(q) \geq \lvert R\rvert + 1$	VERIFIED COMPUTATION	`V-TUJ`	§5.7
`V-EULER-B`	Trivial form $\sum_{p \in R}(\Omega(N-p)-1) + \nu(R) \geq \lvert R\rvert + 1$	PROVED	—	§5.7
`V-ETA-GAMMA`	η is the capacity functional γ	KILLED	—	`FAIL-ETA-GAMMA-IDENTIF`

4.5 Post-root factorization and incidence

ID	Statement / short name	Status	Dependencies	Location
`PR-DQ`	$D_q = \gcd(m_0, N - q) = \gcd(B_q, \lvert 2h - q\rvert)$ for $q \mid m_0$	PROVED	`S-SHADOW-ID`	Thm. 7.1
`PR-MQ`	$M_q = (N-q)/D_q$ is coprime to Q	PROVED	`PR-DQ`	Cor. 7.3
`PR-FRESH`	FRESH/SMOOTH split: q is FRESH iff $P^+(N-q) \geq B_q$	AUDITED IDENTITY	`PR-MQ`	Def. 7.4
`IL1-IL6`	Incidence ledger $\prod_q M_q = U_F\,(J_F B_{\mathrm{re}})\,B_{\mathrm{abs}}$ on $S = Q_{\mathrm{smooth}}$	AUDITED IDENTITY	`PR-MQ`	Thm. 7.6
`IL-DOMAIN`	IL1–IL6 hold on Q_{smooth} only; the FRESH part is excluded	SCOPE-CORRECTED	`IL1-IL6`	Warn. 7.7
`PR-BABS`	$B_{\mathrm{abs}}, B_{\mathrm{re}}, B_{\mathrm{abs,int}}$: definitions and the exact relation between them	AUDITED IDENTITY	`IL1-IL6`	§7.2
`PR-HIGHGAP`	High-gap amplification $h^2 > 4m_0$	SCOPE-CORRECTED	`PR-DQ`	Warn. 7.5

4.6 CRD obligations and open targets

ID	Statement / short name	Status	Dependencies	Location
C0a	Small-gap canonical obligation (residual argument)	PROVED	C-GAP-EQ, C0b-RD	§8.3
TI	Injective branch obligation of C0b	OPEN	C0b-RD	§8.5
TD	Root-anchored dominance obligation of C0b	OPEN	C0b-RD	§8.5
TS	Stability branch obligation of C0b	OPEN	C0b-RD	§8.5
TF	Fresh-layer branch obligation of C0b	OPEN	C0b-RD	§8.5
TD-DENSITY	Density form of TD	SUPERSEDED	TD	§8.5
P1BR	Branchwise production upgrade: realised production bounded below by the *same* functional as capacity	OPEN	TD	§9.4
L1	Actual layer realisation	OPEN	V-EULER	§9.5
E1	Escape or coupling	OPEN	S-RETENTION	§9.6
W1	Simultaneous clean witnesses	OPEN	L1	§9.7
K1	Source-sensitive capacity $K_\times(R,\mathcal{P}) + K(\mathcal{P}) \le \mathcal{C}_N(R)$	OPEN	P1BR	§9.8
X1	Canonical failure exclusion	OPEN	K1	§9.9
RPD	Repeated prime-power redundancy: $D_{\mathrm{high}}(H_Q) \le O(\log\log N)\,\eta_{\mathrm{high}} + o(M)$	OPEN	V-EULER	§9.10
C-SAMEFUNC-BRIDGE	Global same-functional margin bridge	OPEN	P1BR, K1	§9.11
C-GENEALFREE-DYN	Genealogy-free boundary-to-dynamic escape for p_0	OPEN	V71.31	§9.11
C-POP1-SEP	One-defect dynamic separation between p_0 and p_1	OPEN	V71.21	§9.11

4.7 Computational rows

ID	Statement / short name	Status	Dependencies	Location
EXP1-SCAN	No canonical failure for even $6 \le N \le 10^7$	VERIFIED COMPUTATION	—	§11.2
EXP7-BLOW	$B_N^{\mathrm{low}} \neq \varnothing$ for exactly six $N \le 6 \cdot 10^4$: $128, 1718, 1862, 1928, 2200, 6142$	VERIFIED COMPUTATION	CRD-FIXPOINT	§6.6
EXP3-CENSUS	301 failing non-canonical upper roots for $N \le 2 \cdot 10^4$, all at those six N; 2261 unit rows excluded	VERIFIED COMPUTATION	Rem. 3.3	§11.3
EXP4-MARGIN	Ten $N \le 2 \cdot 10^5$ with escape margin $G_{\mathrm{esc}} = 1$; per-dyadic-block minimum rises $1 \to 2 \to 3$	VERIFIED COMPUTATION	Def. 3.25	§6.7
EXP5-PHASE	Phase-type frequencies for $N \le 2 \cdot 10^5$	VERIFIED COMPUTATION	C-PHASE	§6.3

ID	Statement / short name	Status	Dependencies	Location
EXP6-FILT	Filtration embedding verified for $N \le 2 \cdot 10^4$, $k \le 5$	VERIFIED COMPUTATION	C-FILTRATION	§6.2
CERT-34	All 34 hostile certificates of the corpus reproduced (regressions 11822, 3542948; counterexamples 242, 128, 322, 858, 4736, 6142, 14198)	VERIFIED COMPUTATION	—	§11.4
EMP-DEPTH	Success depth of the canonical traversal is almost always ≤ 2	EMPIRICAL PATTERN	—	§11.5
EMP-P0FIRST	p_0 is not measurably better than p_1 at escaping	EMPIRICAL PATTERN	EXP3-CENSUS	§6.5

4.8 Post-v4.81 continuation registry (2026-09-05–2026-09-07C)

These rows are imported from the final 6420-line checkpoint. Their source statuses are preserved; no novelty claim is added. Exact statements, proofs, hostile checks, experiments and later corrections are preserved verbatim in Appendix B. The 2026-09-07C tail is binding when an earlier frontier summary conflicts with it.

ID	Short name	Status	Location
AUDIT-PREFIX-TRANSFER-20260905	Audit Prefix Transfer	PROVED	Chapter 10
CRT-TRANSCRIPT-EXACT-SCOPE-20260905	Crt Transcript Exact Scope	PROVED	Chapter 10
COPRIME-SUPPORTED-AFFINE-COUNT-20260905	Coprime Supported Affine Count	PROVED	Chapter 10
COMPLETE-H1-TWO-ROW-FINITENESS-20260905	Complete H1 Two Row Finiteness	PROVED	Chapter 10
CLOSED-ALPHABET-COUNT-20260905	Closed Alphabet Count	PROVED	Chapter 10
SMALL-BAD-CHILDREN-PREFIX-20260905	Small Bad Children Prefix	PROVED	Chapter 10
UNRESTRICTED-CLOSED-SUPPORT-HEIGHT-20260905	Unrestricted Closed Support Height	PROVED WITH REPAIR	Chapter 10
EFFECTIVE-BOUNDED-CORE-THRESHOLD-20260905	Effective Bounded Core Threshold	PROVED	Chapter 10
POINTWISE-CORE-LABEL-GROWTH-20260905	Pointwise Core Label Growth	PROVED	Chapter 10
H1-TWO-GENERATION-ENVELOPE-ESCAPE-20260905	H1 Two Generation Envelope Escape	PROVED	Chapter 10
H1-ALL-BRANCH-INJECTIVE-ESCAPE-20260905	H1 All Branch Injective Escape	PROVED	Chapter 10
TWO-SMALLEST-ROWS-HEIGHT-20260906	Two Smallest Rows Height	PROVED	Chapter 10
UNIFORM-FEW-FORMS-SIEVE-CONSUMER-20260906	Uniform Few Forms Sieve Consumer	PROVED	Chapter 10
COMPLETE-H1-FIRST-FRONT-RARE-SUCCESS-20260906	Complete H1 First Front Rare Success	PROVED	Chapter 10
SAME-N-GROWING-ROOTS-COMPLETE-FIRST-FRONT-20260906	Same N Growing Roots Complete First Front	PROVED	Chapter 10
SAME-N-NEARBY-ROOT-FIRST-FRONT-SIEVE-20260906	Same N Nearby Root First Front Sieve	PROVED	Chapter 10
COMPLETE-FIRST-FRONT-NO-GO-GROWING-SAME-N-PREFIX-20260906	Complete First Front No Go Growing Same N Prefix	PROVED	Chapter 10
COMPLETE-H1-RADIUS-TWO-RARE-SUCCESS-20260906	Complete H1 Radius Two Rare Success	PROVED	Chapter 10
MULTIGENERATION-SUCCESS-PATH-PRODUCT-BARRIER-20260906	Multigeneration Success Path Product Barrier	PROVED	Chapter 10
GOLDBACH-TRUE-WITH-COMPLETE-SHALLOW-BAD-20260906	Goldbach True With Complete Shallow Bad	PROVED	Chapter 10
AUDIT-RADIUS-TWO-INPUT-20260906B	Audit Radius Two Input	AUDITED IDENTITY	Chapter 10
WEIGHTED-COMPATIBLE-GCD-AVERAGE-20260906B	Weighted Compatible Gcd Average	PROVED	Chapter 10
DISTINCT-PRIME-FORMS-DEGENERACY-FILTER-20260906B	Distinct Prime Forms Degeneracy Filter	PROVED	Chapter 10
SAME-N-RADIUS-TWO-PAIR-COUNT-20260906B	Same N Radius Two Pair Count	PROVED	Chapter 10
SAME-N-GROWING-ROOTS-COMPLETE-RADIUS-TWO-20260906B	Same N Growing Roots Complete Radius Two	PROVED	Chapter 10
SAME-N-RADIUS-TWO-ADVERSARIAL-SCAN-20260906B	Same N Radius Two Adversarial Scan	VERIFIED COMPUTATION	Chapter 10
COFACTOR-CHAIN-EXACT-MODULUS-20260906B	Cofactor Chain Exact Modulus	PROVED	Chapter 10
ORDERED-COFACTOR-HARMONIC-SIMPLEX-20260906B	Ordered Cofactor Harmonic Simplex	PROVED	Chapter 10

ID	Short name	Status	Location
GROWING-FORMS-UNIFORM-SIEVE-CONSUMER-20260906B	Growing Forms Uniform Sieve Consumer	PROVED	Chapter 10
MULTIGENERATION-COFACTOR-PRODUCT-BARRIER-20260906B	Multigeneration Cofactor Product Barrier	PROVED	Chapter 10
MULTIGENERATION-LABEL-PRODUCT-PAIR-COUNT-20260906B	Multigeneration Label Product Pair Count	PROVED	Chapter 10
SAME-N-GROWING-DEPTH-DUAL-PRODUCT-BARRIER-20260906B	Same N Growing Depth Dual Product Barrier	PROVED	Chapter 10
MULTIGENERATION-COFACTOR-ADVERSARIAL-SCAN-20260906B	Multigeneration Cofactor Adversarial Scan	VERIFIED COMPUTATION	Chapter 10
SAME-N-SUPPORT-WITNESS-RANK-CORRECTION-20260907	Same N Support Witness Rank Correction	SCOPE-CORRECTED	Chapter 10
EXACT-SCAN-WORK-CONVENTION-AUDIT-20260907	Exact Scan Work Convention Audit	SCOPE-CORRECTED	Chapter 10
PRIME-LIFT-FULL-BFS-CONJUGACY-20260907	Prime Lift Full Bfs Conjugacy	PROVED	Chapter 10
LOW-GAP-COMPLETE-THREE-LABEL-ALTERNATIVE-20260907	Low Gap Complete Three Label Alternative	PROVED	Chapter 10
PRIME-LIFT-AND-LOW-GAP-EXHAUSTIVE-AUDIT-20260907	Prime Lift And Low Gap Exhaustive Audit	VERIFIED COMPUTATION	Chapter 10
CANONICAL-FIRST-SUPPORT-CYCLE-REALIZATION-20260907	Canonical First Support Cycle Realization	PROVED	Chapter 10
SAME-N-FIRST-SUPPORT-CYCLE-NON-DOMINANCE-20260907	Same N First Support Cycle Non Dominance	VERIFIED COMPUTATION	Chapter 10
FIXED-GAP-CANONICAL-VALUATION-EMBEDDING-20260907	Fixed Gap Canonical Valuation Embedding	PROVED	Chapter 10
HEREDITARY-FIRST-SUPPORT-P0-NO-GO-20260907	Hereditary First Support P0 No Go	PROVED	Chapter 10
CANONICAL-VALUATION-EMBEDDING-EXACT-CERTIFICATE-20260907	Canonical Valuation Embedding Exact Certificate	VERIFIED COMPUTATION	Chapter 10
LOW-GAP-SHARP-TWO-STEP-RETURN-BARRIER-20260907C	Low Gap Sharp Two Step Return Barrier	PROVED	Chapter 10
LOW-GAP-SQUARE-ROW-NO-RETURN-20260907C	Low Gap Square Row No Return	PROVED	Chapter 10
LOW-GAP-MONOMIAL-RETURN-EXPONENT-DROP-20260907C	Low Gap Monomial Return Exponent Drop	PROVED	Chapter 10
H1-SINGLETON-SQUARE-ROW-RESIDUE-FILTER-20260907C	H1 Singleton Square Row Residue Filter	PROVED	Chapter 10
CLOSED-SUPPORT-MULTIPLICATIVE-SUBGROUP-TEST-20260907C	Closed Support Multiplicative Subgroup Test	PROVED	Chapter 10
LOW-GAP-THREE-LABEL-EXACT-OBSTRUCTION-SPLIT-20260907C	Low Gap Three Label Exact Obstruction Split	PROVED	Chapter 10

4.9 Killed and superseded programmes

ID	Statement / short name	Status	Dependencies	Location
P0-ENGINE	Old finite-selected-transcript p_0 proof architecture	SUPERSEDED	FACP, AUDIT-FBTU-PARTIAL-VERSUS-COMPLETE-20	§2.5
FACP	Finite Affine Canonical Programmability: every prime is $p_0(2(P-1))$, so no finite affine canonical certificate is universal	PROVED	—	§8.2
FBTU	Partial BAD-transcript programmability; blanket complete-BFS reading withdrawn	SCOPE-CORRECTED	CRT-TRANSCRIPT-EXACT-SCOPE-20260905	§8.2
FLUW	Blanket finite-local complete-BFS universality wall	SCOPE-CORRECTED	AUDIT-FBTU-PARTIAL-VERSUS-COMPLETE-20	§8.2
CPCT	Canonical Prime Cell Theorem	PROVED	FACP	§8.2
CMFT	Canonical Multiplicative Flow Theory	KILLED	FLUW	§12.2
CGST	Canonical Graph Spectral Theory	KILLED	FLUW	§12.2
CSD	Canonical Shadow Descent	KILLED	FLUW	§12.2
CAPT	Canonical Affine Programmability Transfer	KILLED	FACP	§12.2
RSD	Root Structure Descent	KILLED	FLUW	§12.2
ROUGHP2	Global rough-P_2 matrix	KILLED	parity	§12.4
H1-FP	H1 root-factor descent with shift-2 orthogonality	KILLED	Pillai/Bennett	§12.2

ID	Statement / short name	Status	Dependencies	Location
TWOCORE	Two-core wall $a^\alpha - a = b^\beta - b$	PROVED	[79]	§12.2

4.10 Imported results: exact product locks and no-gos (CRD Core)

ID	Statement / short name	Status	Dependencies	Location
G-MULTISOURCE-LOCK	$T_S \equiv (-1)^{\lvert S\rvert} \prod_{\mathrm{Src}} d \pmod N$ for child-closed S	PROVED	V-TUJ	Thm. 5.7
G-ROOT-LOCK	$T_R \equiv (-1)^{\lvert R\rvert} p_0 \pmod N$; $T_R = (2a+1)x + (-1)^{\lvert R\rvert} h$, $a \geq 3$; $\max(U_R, J_R) > \sqrt{N}$	PROVED	G-MULTISOURCE-LOCK, ARS-CLOSED-WORLD	Thm. 5.8
G-CHILD-CLOSED-OVERLAP-IDENTITY	Exact source-overlap identity (OV1), (OV2)	PROVED	V-TUJ	Thm. 5.9
G-DISJOINT-CHILD-CLOSED-ORTHOGONALITY	Disjoint child-closed worlds are arithmetically orthogonal (OB1)–(OB3)	PROVED	ARS-ACTIVITY, G-COLLISION-CUT	Thm. 5.10
G-FULLQ-LEDGER	$\prod_{q\in Q_r} M_q = U_F(J_F B_{\mathrm{re}}) B_{\mathrm{abs}}$ on the *whole* Q_r under failure	PROVED	ARS-CLOSED-WORLD, V-TUJ	Thm. 5.11
G-FULLQ-PRODUCTION	$\mathfrak{N}_{Q_r} > (5/3)^{\lvert Q_r\rvert} \operatorname{rad}(m_r) > 1$	PROVED	—	Thm. 5.13
G-COLLISION-CUT	$K(S) = \sum \log\gcd$; $C_{\mathrm{cut}} = (\lvert S\rvert - 1)L - 2K$; every $\gcd \mid \lvert s - s'\rvert$	PROVED	—	Thm. 5.14
G-CLEAN-POLYLOG-PARENT-SELECTION	Clean polylogarithmic parent selection; support load $\leq \rho(\mathcal{L})$	PROVED	Siegel–Walfisz, lower sieve, C-GAP-EQ	Thm. 5.15
W1-MISATTRIBUTION	Earlier listing of G-CLEAN-POLYLOG-PARENT-SELECTION as W1's open target	SCOPE-CORRECTED	G-CLEAN-POLYLOG-PARENT-SELECTION	Warn. 5.16
C-RESIDUE-ONLY-RETURN-NOGO	$(G \setminus \{u\})^2 = G$ for $q \geq 5$: no residue-only forced return	PROVED	—	Thm. 5.17
C-MOD3-TAGGED-FORCED-RETURN	$N \equiv 2 \pmod 3$ forces a sibling $r \equiv 2 \pmod 3$ with $r \to 3$	PROVED	—	Thm. 5.18
C-TOPOLOGY-NOGO	Topology alone cannot force $\Delta \log J \gg W(E)$	PROVED	—	Thm. 5.20

4.11 Imported results: the rooted first-front layer (Biblia v7.3–v7.5)

ID	Statement / short name	Status	Dependencies	Location
B-FRESH-PACKING	(FP): $\prod_{q\in E} s_q \geq m_0^{\lvert E\rvert} / \prod q$	PROVED – PENDING RED TEAM	P-COLLISION	Prop. 7.10
B-SMOOTH-TAX	(ST): $B_q^{e_q} > q$, $e_q = j_q + u_q$	PROVED – PENDING RED TEAM	PR-FRESH	Prop. 7.11
B-SMOOTH-EXCESS	$\Omega(U_R) + \Omega(J_R) - 1 \geq \sum e_q$	PROVED – PENDING RED TEAM	B-SMOOTH-TAX	Cor. 7.12

ID	Statement / short name	Status	Dependencies	Location
B-FFC	(FFC), (FFC-Ω): root first-front conservation	PROVED – PENDING RED TEAM	V-TUJ	Thm. 7.14
B-FFRE	(FFRE): $\prod \mathrm{rad}(N-q) = J_{S_1} R_\partial(S_1)$	PROVED – PENDING RED TEAM	B-FFC	Cor. 7.15
B-UJD-SPLIT	(U-split), (JR-split): exact first-front $U/J/\partial$ split	PROVED – PENDING RED TEAM	B-FFRE	Thm. 7.16
B-GCD-PACK	$\nu_\ell \le \min(\lvert Q\rvert, 1+\lfloor m_0/\ell\rfloor)$	PROVED – PENDING RED TEAM	—	Lem. 7.18
B-POWERFUL-TAIL	Powerful-integer density is too weak for the adaptive first front	KILLED	—	Scope 7.19
B-RSCP	Root Smoothness Conservation Problem	OPEN	B-UJD-SPLIT, B-SMOOTH-TAX	Target 7.20
B-PP-LEDGER	(PP-I), (PP-D), (PP): lossless prime-power root ledger	PROVED – PENDING RED TEAM	PR-DQ	Thm. 7.25
B-POSTROOT-DEMAND	$D_q = \gcd(B_q, \lvert 2h-q\rvert)$; (Demand); (Post-root tax)	PROVED – PENDING RED TEAM	S-SHADOW-ED	Thm. 7.26
B-ROOT-VAL-RIGID	$v_r(N-q) > a \Rightarrow v_r(\lvert 2h-q\rvert) = a$	PROVED – PENDING RED TEAM	S-SHADOW-ED	Lem. 7.27
B-SHARP-PP-PACK	$\ell^k \mid n_q, n_r \Rightarrow \ell^k \mid q-r$; the diameter bound on $\nu_{\ell,k}$	PROVED – PENDING RED TEAM	—	Lem. 7.28
B-VPC	(VPC): actual prime-power collision budget	PROVED	PR-MQ	Thm. 7.30
B-SHARED-MASS	$L_{\mathrm{shared}} \le 2\log \Delta_{\mathrm{odd}}(S)$	PROVED	B-VPC	Cor. 7.31
B-PAIR-FLOOR	$L_{\mathrm{pair}}(q,r) \ge \log 5$	PROVED	PR-FRESH	Lem. 7.32
B-WEIGHTED-CUT	(Cut): $\sum L_{\mathrm{pair}} = \sum_\lambda \mu_\lambda(s-\mu_\lambda)w_\lambda$; $C_{\mathrm{cut}} \ge \binom{s}{2}\log 5$	PROVED	B-PAIR-FLOOR	Thm. 7.33
B-LOW-MULT	Low-multiplicity layer lower bound $W_{\le u}$, W_1	PROVED	B-WEIGHTED-CUT, B-VPC	Cor. 7.34
B-EXACTD-OUTSIDE	$t < r \Rightarrow M_q M_r > \delta^2$; failure regime (ED-res)	PROVED	PR-DQ	Thm. 7.36
B-GAP-FILTER-1	$9p_0 \ge 18G + 5G^2 \Rightarrow M_q M_r > \delta^2$	PROVED	B-EXACTD-OUTSIDE, C-GAP-EQ	Cor. 7.37
B-JOINT-D	(Joint-D): $D_q D_r \mid m_0 \gcd(c, r-q)$	PROVED	PR-DQ	Thm. 7.38
B-PNF	(PNF): primitive residual normal form	PROVED	B-EXACTD-OUTSIDE	Thm. 7.39
B-HIGH-GAP	(HG-r), (HG): $h^2 > 4m_0$ under *pair* failure	PROVED	B-PNF	Thm. 7.40
B-GAP-FILTER-2	$p_0 < 2G + G^2/4$	PROVED	B-HIGH-GAP	Cor. 7.42
B-K1-GAP	$r > 4^{d+1}qu^2$; $p_0 < 2G + G^2/12$	PROVED	B-HIGH-GAP	Thm. 7.43
B-K1-QUOTIENT	$k = 1$ quotient normal form in (A, B)	PROVED	B-PNF	Thm. 7.44
B-K1-COVERAGE	Structured composite coverage; $\gcd(B, qu-1) = 1$	PROVED	B-K1-QUOTIENT	Prop. 7.46
B-K1-BARRIER	Canonical $k = 1$ gap/covering barrier	OPEN	B-K1-COVERAGE	Target 7.47
B-MULTIPLEX-RIGID	$D \ge \delta m^{\kappa_{\mathrm{col}}}$	PROVED – PENDING RED TEAM	V-TUJ	Thm. 7.49
B-COMP-BARRIER	$m < (N/6)^{\chi/\eta}$; $\eta \ge t\chi \Rightarrow m < (N/6)^{1/t}$	PROVED – PENDING RED TEAM	B-MULTIPLEX-RIGID	Cor. 7.50
B-SPARSE-CORE-LOCK	$X - 1 \le U_C J_C \le U_C (D/2)^{\chi_\tau}$	PROVED – PENDING RED TEAM	B-MULTIPLEX-RIGID	Cor. 7.51

ID	Statement / short name	Status	Dependencies	Location
B-SUPPORT-EXPLOSION	$\lvert S\rvert \geq \lceil \log A_S / \log B_S \rceil$	PROVED – PENDING RED TEAM		Thm. 7.53

4.12 Imported results: the canonical threshold programme (v4.79–v4.81)

ID	Statement / short name	Status	Dependencies	Location
V79-NAGURA	$p_0 < 3N/5$, $m > 2N/5$ for $N \geq 50$	PROVED	[1], C-GAP-EQ	Prop. 9.1
V79-FLOOR	(V79.4): $U_R^2 J_R B_{\mathrm{abs}} > m(5/3)^{\lvert Q\rvert} > 2N/3$	PROVED	G-FULLQ-PRODUCTION	Thm. 9.2
V79-COEFF-DICH	$F_Q^{\mathrm{coeff}} = 1 \iff$ primitive; $F_Q^{\mathrm{coeff}} > 1 \Rightarrow T_R > \frac{5}{2}N$	PROVED	G-ROOT-LOCK	Thm. 9.3
V80-LOADONE	Primitive $\Rightarrow D_q = 1$; $\prod_{q\in Q}(N-q) > \frac{25}{36}N^2$	PROVED	V79-COEFF-DICH	Thm. 9.5
C-P0-PRIMITIVE-OWNERPRESERVING-QUADRATIC-UJD-CHAR	$\mathscr{C}_Q \mid U_R J_R D_{\lambda_N}$	PROVED	V80-LOADONE	Thm. 9.6
C-P0-PRIMITIVE-FIRSTFRONT-UNITMARGIN-PRODUCTION	$U_R(J_R B_{\mathrm{re}}) B_{\mathrm{abs}} \geq 10N/9$	PROVED	V79-FLOOR	§9.2.1
C-P0-PRIMITIVE-BASEABS-ESCAPE-OR-REUSE-CHARGE	External escape/reuse recharge to charge base absorption	SUPERSEDED	C-P0-PRIMITIVE-OWNERPRESERVING-QUADRA	§9.2.1
V81-CONTAINER	(V81.1): $U_R J_R D_{\lambda_N}$ is the row product of the world	PROVED	V80-LOADONE	Thm. 9.7
V81-SLACK	(V81.2): container slack $> 5N/6$ for $N \geq 500$	PROVED	V81-CONTAINER	Thm. 9.8
V81-SIZE-DEATH	The archimedean size route is closed	PROVED	V81-SLACK	Cor. 9.9
V81-OVERLAP	(V81.3): reflected/selected overlaps are charge-free	PROVED	ARS-REFLECTION-GATE	Thm. 9.10
C-P0-PRIMITIVE-FIRSTFRONT-REFLECTION-DISJOINT-CAP	Previous pre-reflection dichotomy; its alternatives (1) and (2)	KILLED	V81-SLACK, V81-OVERLAP	§9.2.2
V81-C1-SHADOW	(V81.11): primitive load-one $\iff$ $\gcd(m, 2h-u) = 1$ on $R \setminus \{p_0\}$	PROVED	S-SHADOW-ID	Thm. 9.12
V81-C2-RESIDUAL	R' child-closed, nonempty, $\min R' < \sqrt{N}$	PROVED	ARS-CLOSED-WORLD	Thm. 9.14
V81-C3-FACTOR	$T_R\Sigma' = (\prod_{q\in Q}(N-q))T_{R'}$	PROVED	V81-C2-RESIDUAL	Thm. 9.15
V81-C4-RESLOCK	(V81.13): selected residue lock	PROVED	G-MULTISOURCE-LOCK	Thm. 9.16
V81-C5-ALPHABET	Adaptive fringe cofactor alphabet $\mathcal{L}_N \subseteq [\lambda, 2\lambda)$	PROVED	V80-LOADONE	Thm. 9.17
V81-IMMUNITY	Product-comparison immunity of the minimal survivor	PROVED	V81-SLACK	Thm. 9.19
V81-TRICHOTOMY	Only routes (A) population, (B) residue, (C) GOOD remain	PROVED	V81-IMMUNITY	Thm. 9.20
C-P0-PRIMITIVE-LOWCORE-SMOOTHROW-POPULATION-CAPAC	Effective upper bound $\lvert\mathscr{C}_\lambda\rvert \lesssim \Psi^*(N, \lambda_N)$	OPEN	V81-TRICHOTOMY, V81-C5-ALPHABET	§9.2.4
C-P0-PRIMITIVE-SELECTED-RESIDUE-LOCK-CAPACITY	Independent second determination of $\mathscr{C}_Q$ mod N	OPEN	V81-C4-RESLOCK, G-ROOT-LOCK	§9.2.4

ID	Statement / short name	Status	Dependencies	Location
C-PO-PRIMITIVE-SELECTED-UID-CAPACITY	Capacity for the owner-preserving selected charge	OPEN	C-PO-PRIMITIVE-OWNERPRESERVING-QUADRA	§9.2.1
G-CONDITIONAL-EFFECTIVE-PO-N0	Conditional effective $N_0^{p_0}$	PROVED (conditional)	V79-FLOOR	§9.2.1
FAIL-SUNIT-COUNT-BOUNDS-SHELL-LOWCORE-LOG	S-unit counting bounds $\lvert C_\lambda \rvert$	KILLED	[18]	§9.2.4

4.13 Imported results: surviving lemmas of killed programmes

ID	Statement / short name	Status	Parent (killed)	Location
PCRD-CMFT-1	Lossless anchor filtration; $\mathscr{H}_{j+1} = \mathscr{H}_j \sqcup \mathscr{A}_j$	PROVED	CMFT	Thm. 12.2
PCRD-CMFT-2	Tail hard-edge universality: local tail geometry does not distinguish p_0	PROVED	CMFT	Thm. 12.3
PCRD-CMFT-3	Canonical least-factor split	PROVED	CMFT	Thm. 12.4
PCRD-CMFT-4	All-root collision identity	PROVED	CMFT	Thm. 12.5
PCRD-CMFT-5	Prime-pressure failure ceiling: $\mathcal{P}_N > \sqrt{x}\log x \Rightarrow$ Goldbach	PROVED	CMFT	Thm. 12.6
PCRD-CGST-1	Canonical straddle theorem	PROVED	CGST	Thm. 12.7
PCRD-CGST-2	Arbitrarily deep finite reflection obstruction	PROVED	CGST	Thm. 12.8
PCRD-CGST-3	Diagonal collapse: the cell dimension is false	PROVED	CGST	Thm. 12.9
PCRD-CSD-1	Canonical conductor–gap compatibility	PROVED	CSD	Thm. 12.11
PCRD-CAPT-1	Fixed-(r,s) compression to two linear prime forms	PROVED	CAPT	Thm. 12.12
PCRD-RSD-1	Exact rough signed identity	PROVED	RSD	Thm. 12.14
PCRD-RSD-2	Distribution–roughness phase law $\theta > 2/(k+1)$	PROVED	RSD	Thm. 12.15
PCRD-M-1	$K_{3,3}$ explosion at the natural scales	CONJECTURE (conditional)	ROUGHP2	Thm. 12.17
PCRD-M-2	Boundary blindness of generic operator norms	PROVED	ROUGHP2	Thm. 12.18
PCRD-M-3	$h = 1$ matrix invisibility	PROVED	ROUGHP2	Thm. 12.19
PCRD-M-4	Fixed-modulus spectrum recovers only the singular series	PROVED	ROUGHP2	Thm. 12.20
PCRD-P0-3	Gap scalar freedom: h, G, h/G have no closing regime	PROVED	P0-ENGINE	Thm. 12.22
PCRD-H1-1	H1 root-factor descent; $\mathrm{PF}(N - q_*) \cap \mathrm{PF}(m_0) = \varnothing$	PROVED	H1-FP	Thm. 12.24
PCRD-H1-2	H1 shift-2 orthogonality $\gcd(N - u, u + 2) = 1$	PROVED	H1-FP	Thm. 12.25

ID	Statement / short name	Status	Parent (killed)	Location
PCRD-H1-3	3-core normal form; no 3-core found in finite boxes	VERIFIED COMPUTATION	H1-FP	Rem. 12.26
MATCHED-EUCLID-GLUING	Matched Euclidean gluing lemma (named missing theorem)	OPEN	—	§15.5

Completeness statement. This registry is *not* an exhaustive index of the sources. The CRD Theory Core alone contains 801 live titled results and the Biblia 673; what is imported here is the load-bearing subset — exact identities, no-go theorems, frontier reductions, and the lemmas that survive their killed parent programmes. If a needed result is not here, it may still exist in the sources; search [CORE] by THEOREM-ID and [BIB] by theorem title before concluding that something is unproved.

Chapter 5

Core Results (the Biblia layer)

This chapter holds the exact machinery that the project actually owns: shadow arithmetic, path invariants, and the valuation ledger. Almost all of it is *root-generic* — it works from any upper root and never uses minimality. Reading it with that in mind is the fastest cure for the most common error in the project's history, which is to quote a generic lemma as a canonical one.

5.1 Scope classes A / B / C / D

Definition 5.1 (The four scope classes)**.** A result T is

class A — genuine p_0-specific if there is an *explicit proof step* that fails when p_0 is replaced by another upper root, *and* the content of the conclusion actually changes;
class B — root-generic if every step goes through verbatim for $r = x + h_r$ with h_r, m_r in place of h, m_0;
class C — failure-conditioned if its hypothesis is that the traversal from the root fails;
class D — Goldbach-counterexample-conditioned if its hypothesis is $\mathrm{Good}_N = \varnothing$.

Separately there is the *vacuity-canonical* case: the construction goes through for p_k on those coordinates where it is defined at all, and canonicity only guarantees that it is defined everywhere. This is **not** class A. The corpus repeatedly confused the two.

Object / result	Class	What happens under $p_0 \to p_k$
Relation $p \to q$, $\mathcal{A}_N$, valuations, components	B	nothing; this is prior art §15.2
$T_S = U_S J_S$	B	nothing
High-root compression, unique source	B + C	nothing; $r > N/3$ and $N - r$ composite suffice
$q \le m_r/3 < N/6$	B + C	nothing
$\gcd(h_r, m_r) = 1$, q^e congruences	B	nothing; root algebra
Shadow identity, retention, D_q, M_q	B	nothing `FAIL-MR2-CANONICAL-ONLY-001`
Bézout / winding / state count	B	nothing; $h \mapsto h_r$
Shadow cycles, prime-power witness	B + C	nothing
Strict non-reuse of the block	B	nothing; the proof uses index differences only
No-bad-return, two-sheet budget	C	nothing beyond the failure hypothesis

Object / result	Class	What happens under $p_0 \to p_k$
Fixed point B_N^{low} and the failure characterisation	C	nothing; the object depends on N alone
Reflection gate (*activation*)	vacuity	for p_k exactly k coordinates are prime and there is nothing there for the gate to open
Exact bundle lift	vacuity	the clauses go through verbatim on every row with composite c'_j; 25.2% of rows are lost in the recorded scan [CORE, P0-CANONICITY-ABLATION]
Zero hole count as *mass*	KILLED	loss is $O_k(\log N)$; not a mechanism
$h < G$ / empty interval	A	everything: this is the definition and the only canonical input
$\mathcal{P}_{\text{def}} = \varnothing$	A	exactly k prime atoms appear
$\kappa_{\text{gen}} = 0$	A	the rank rises to k; the equivalence is exact in both directions
Reflection-completeness	A	uniqueness is lost
Ledger completeness $\mathcal{I} \uplus \mathcal{D} \uplus \mathcal{S} \uplus \mathcal{F}$	A	a fifth class $\mathcal{P}_{\text{def}}$ appears
Lossless code of prime history	A	p_k encodes k earlier roots; p_0 encodes none
Phase trichotomy and $j^* = h - G/2$	A	the notion "predecessor outside the block" exists only for the canonical root
$\rho_0 \geq 1 \iff \theta > 1/2$	A	for p_k the predecessor is p_{k-1}, so $D_1 \leq h_k$ and the whole characterisation collapses
Exact-D filter $p_0 < 2G + G^2/4$	A + C	without $h < G$ the filter does not exist; non-canonical countermodels exist
Exact-D model elimination	A + C	the models return `FAIL-EXACTD-NONCANONICAL-001`
Upper-half compression	D	requires $\text{Good}_N = \varnothing$ `FAIL-UPPERHALF-SCOPE-001`
Symmetric fracture, forbidden offset band	D	require the minimal-counterexample hypothesis

The separation result in three sentences.

1. Almost the whole local arithmetic of p_0 — shadow, D_q, M_q, Bézout, winding, cycles, the U/J ledger — is *generic upper-root algebra*. Minimality is used nowhere in it.
2. The only genuine canonical input is $h < G$, equivalently emptiness of $[x, p_0) \cap \mathbb{P}$. It enters proofs in exactly one place: it makes *every* c_j composite.
3. The consequences of that single input — $\kappa_{\text{gen}} = 0$, reflection-completeness, ledger completeness, the phase trichotomy, the ρ_0 formula — are the only known properties of p_0 that have an *exact converse*. Every other candidate either has no converse (least-upper-parent) or was refuted (mass, density, local algebra, statistics).

5.2 The generic layer

Theorem 5.2 (High-root compression and unique source)**.** ***PROVED***, *class $B + C$. Let $r > N/3$ be an active odd prime, $N - r$ composite, and suppose the traversal from r fails. Then every non-root state lies below $N/3$ and*

$$\boxed{\Gamma_N(R_N(r)) = R_N(r) \setminus \{r\}.}$$

Proof. For a bad row, $N - p$ is odd composite, so every prime factor q has an odd cofactor ≥ 3, whence $q \le (N-p)/3 < N/3$. If r had a parent $p \in R$ then $N - p = kr$ with k odd and $k < 3$, so $k = 1$ and $p = N - r$ would be prime, contradicting compositeness of the complement. [CORE, G-HIGH-ROOT-SOURCE; canonically ARS-UPPER-EXCLUSION] ☐

Reading. The bad world has one source and the whole upper half apart from the root is external to it. **This is not a property of p_0** — it is a property of being a high root with composite complement. VERIFIED COMPUTATION on 301 real closed worlds ($N \le 2 \cdot 10^4$), zero violations.

Theorem 5.3 (Canonical prime-free geometry)**.** ***PROVED***. Hypotheses: *canonical failure, non-diagonal case.* Conclusion:

$$[x, p_0) \cap \mathbb{P} = \varnothing; \qquad m_0 \textit{ odd composite}, \ m_0 \ge 9; \qquad \gcd(m_0, N) = \gcd(h, m_0) = 1;$$

$$q \in Q \ \Rightarrow\ q \le \frac{m_0}{3} < \frac{N}{6}; \qquad q^e \| m_0 \ \Rightarrow\ x \equiv h, \quad p_0 \equiv 2h = \delta \pmod{q^e}.$$

Classes: *emptiness of the interval — A;* $\gcd(h, m_0) = 1$ *and the* q^e *congruences — B (identical for any upper root with* h_r, m_r*); the inequality* $q \le m_r/3$ *—* $B + C$*.*

Proof. Emptiness is minimality. m_0 is odd as the complement of an odd prime, and composite, else the root is GOOD. Next $\gcd(m_0, N) = \gcd(p_0, N)$, and $p_0 \mid N$ with $p_0 \ge N/2$ would force $N = 2p_0$, the diagonal case. If $d \mid h$ and $d \mid m_0 = x - h$ then $d \mid x$, hence $d \mid x + h = p_0$; from $0 < d \le h < p_0$ and primality of p_0, $d = 1$. The inequality: the cofactor m_0/q is odd and ≥ 3. The congruences are $m_0 = x - h$ and $p_0 = x + h$. [BIB, normal form; CORE, ARS-CANON-GEOMETRY] ☐

VERIFIED COMPUTATION: 4 915 canonical bad roots for $N \le 1.5 \cdot 10^4$, zero violations.

Scope note 5.4 (Inequality repair). SCOPE-CORRECTED. Some earlier notes wrote $q < m_0/3$. The correct inequality is $q \le m_0/3$; equality occurs e.g. for $N = 32$, $p_0 = 17$, $m_0 = 15$, $q = 5$. The consequence $q < N/6$ remains strict. [BIB, minor inequality repair]

Theorem 5.5 (Least upper parent)**.** ***PROVED***, *class A. If* $r = p_0$ *then for every prime power* $q^e \mid N - r$

$$\boxed{r = \min(\mathcal{P}_N(q^e) \cap [x, N)), \qquad \mathcal{P}_N(q^e) = \{p \in \mathcal{A}_N : q^e \mid N - p\}.}$$

[CORE, ARS-LEAST-UPPER-PARENT]

Warning 5.6 (No converse). KILLED as a selector. In the recorded scan of 301 failing roots, eleven non-canonical roots satisfy the same first-front minimality, among them $(128, 73)$, $(128, 79)$, $(1718, 877)$, $(1862, 941)$, $(1928, 971)$, $(6142, 3089)$; exactly one has the stronger world-level property, $(1928, 971)$. So *first-front parent minimality does not by itself explain* p_0. [CORE, V71.41]

5.3 Exact product locks and world identities

These are the exact multiplicative identities of the CRD Core. All are PROVED with short proofs, all are safe to use as lemmas, and none of them was in earlier compressions of this project. They are the algebraic backbone that every capacity argument silently uses.

Theorem 5.7 (Multi-source product lock, `G-MULTISOURCE-LOCK`)**.** ***PROVED***, *class B. If a finite* $S \subseteq \mathcal{A}_N$ *is child-closed and* $D_S^{\mathrm{src}} = S \setminus \Gamma_N(S)$*, then*

$$\boxed{T_S \equiv (-1)^{|S|} \prod_{d \in D_S^{\mathrm{src}}} d \pmod N.}$$

Proof. $T_S = \prod_{s\in S}(N-s)/\prod_{q\in\Gamma_N(S)} q$. Modulo N replace each $N-s$ by $-s$ and cancel the invertible active labels of $\Gamma_N(S)$; exactly the sources survive. [CORE, G-MULTISOURCE-LOCK] ☐

Limitation. A rooted one-source lock cannot upper-bound an enlarged ambient multi-source system; adjoining external parents changes the source term.

Theorem 5.8 (Rooted full-front lock, `G-ROOT-LOCK`). ***PROVED***, *class C. If canonical p_0 fails, $R = R_N(p_0)$, $r = |R|$, then*

$$T_R = U_R J_R \equiv (-1)^r p_0 \pmod N, \qquad \textit{and for an integer } a \ge 3, \qquad \boxed{T_R = (2a+1)x + (-1)^r h.}$$

In particular $\max(U_R, J_R) > \sqrt{N}$.

Proof sketch. Unique-source (Lemma 3.26) gives $T_R = m_0 \prod_{p\in R\setminus\{p_0\}}(N-p)/p$, and the congruence follows from Theorem 5.7. Every non-root row lies below x, so each ratio exceeds one. Writing $q = P^-(m_0)$, $B = m_0/q$: if $m_0 \ne 9$ then $B \ge 5$, so $T_R \ge B(N-q) \ge 5N - m_0$, exceeding every affine value with $a \le 2$. If $m_0 = 9$ then $3 \nmid N$, row 3 is bad, and Theorem 5.2 places its non-root child s below $N/3$, giving $T_R \ge 9 \cdot \frac{N-3}{3} \cdot \frac{N-s}{s} > 3N$ for $N \ge 6$, while every affine value with $a \le 2$ is at most $5x + h < 3N$. [CORE, G-ROOT-LOCK] ☐

Reading. This is the sharpest *unconditional* statement the project has about a canonical failed world: its total valuation mass is pinned to p_0 modulo N and forced into an explicit affine form. It is the container of route (B) in the surviving-route trichotomy (§9.2).

Theorem 5.9 (Exact source-overlap identity, `G-CHILD-CLOSED-OVERLAP-IDENTITY`). ***PROVED***, *class B. Let $A, B \subseteq \mathcal{A}_N$ be finite child-closed sets with $C = A \cap B \ne \varnothing$, $U = A \cup B$, and write $\Sigma(S) = \prod_{d\in D_S^{\mathrm{src}}} d$. Then*

$$\boxed{T_A T_B\, \Sigma(U)\Sigma(C) = T_U T_C\, \Sigma(A)\Sigma(B).} \tag{OV1}$$

If moreover $D_U^{\mathrm{src}} = D_A^{\mathrm{src}} \uplus D_B^{\mathrm{src}}$, then

$$\boxed{T_A T_B\, \Sigma(C) = T_U T_C.} \tag{OV2}$$

Proof. For finite child-closed S, $\Gamma_N(S) = S \setminus D_S^{\mathrm{src}}$, so $T_S = \Sigma(S)\prod_{s\in S}(N-s)/s$. The multiplicative row ratio satisfies exact inclusion–exclusion over A, B, U, C; substituting gives (OV1), and the disjoint-source hypothesis cancels $\Sigma(U) = \Sigma(A)\Sigma(B)$. [CORE, G-CHILD-CLOSED-OVERLAP-IDENTITY] ☐

Theorem 5.10 (Disjoint child-closed worlds are arithmetically orthogonal, `G-DISJOINT-CHILD-CLOSED-ORTHOGONALIT`
PROVED, *class B. Let $S_1, \dots, S_b \subseteq \mathcal{A}_N$ be finite, pairwise disjoint and child-closed, $U = \bigcup_i S_i$. Then for $i \ne j$,*

$$\boxed{\gcd(N-u, N-v) = 1 \qquad (u \in S_i,\ v \in S_j),} \tag{OB1}$$

the supports $\Gamma_N(S_i)$ are pairwise disjoint, and

$$\boxed{T_U = \prod_i T_{S_i}, \qquad K(U) = \sum_i K(S_i), \qquad \mathrm{Src}(U) = \biguplus_i \mathrm{Src}(S_i),} \tag{OB2}$$

$$\boxed{\eta(U) = \sum_i \eta(S_i), \qquad \nu(U) = \sum_i \nu(S_i).} \tag{OB3}$$

Proof. A prime q dividing both $N-u$ and $N-v$ is active (Definition 3.2), so child-closure puts it in $S_i \cap S_j$ — impossible. Hence no prime-power layer occurs in two sets, and the primewise definitions of T, K, η, ν are additive; absence of cross edges splits the source sets. [CORE, G-DISJOINT-CHILD-CLOSED-ORTHOGONALITY] ☐

Reading. Disjoint failed packets carry *no* shared arithmetic whatsoever. This is why "combine several failed worlds and count" arguments gain nothing: the combination is a product, not an interaction.

Theorem 5.11 (Full-Q failure ledger, `G-FULLQ-LEDGER`). ***PROVED***, *class C. If the stopped traversal from $r > N/2$ fails and $N - r$ is composite, then with $S = Q_r$*

$$\boxed{\prod_{q \in Q_r} M_q = U_F(J_F B_{\mathrm{re}})B_{\mathrm{abs}}, \qquad U_F \mid U_R, \quad J_F B_{\mathrm{re}} \mid J_R.}$$

Every residual prime lies in R, and the same formulas hold for any subfamily of Q_r.

Proof. Failure puts every first-front row q in R, so $\kappa_R(\ell) \geq m_\ell$, and closure puts every residual factor in R. For $\ell \notin Q_r$ split $E_\ell = (E_\ell - m_\ell) + (m_\ell - 1) + 1$: the final copy is reused when $\kappa_R > m_\ell$ and absorbed otherwise. For $\ell \in Q_r$ the root row already supplies the base copy and all residual copies enter U_F. The same primewise comparison gives the two divisibilities. [CORE, G-FULLQ-LEDGER] ☐

Warning 5.12 (This is the correct domain of the incidence ledger). Theorem 5.11 *supersedes* the reading of IL1–IL6 as a Q_{smooth}-only statement: on the failed branch the identity holds on the whole of Q_r. The Q_{smooth} version (Theorem 7.6) is the restriction. The unconditioned extension to arbitrary Q_0 still fails at $N = 20$ (`FAIL-Q0-LEDGER-001`), because a successful first-front row can give $\kappa_R < m_\ell$.

Theorem 5.13 (Full-Q production, `G-FULLQ-PRODUCTION`). ***PROVED***, *class B. For an active prime $r > N/2$ with odd composite $m_r = N - r$,*

$$\boxed{\mathfrak{F}_{Q_r} := \prod_{q \in Q_r} M_q > \left(\tfrac{5}{3}\right)^{|Q_r|} \operatorname{rad}(m_r) > 1.}$$

Proof. For $q \mid m_r$, activity gives $q \nmid N$, hence $q \nmid D_q$ and $D_q \leq m_r/q$. An odd composite has $q \leq m_r/3$, and $m_r < N/2$, so $M_q \geq q(N-q)/m_r > \frac{5}{3}q$. Multiply over distinct $q \in Q_r$. [CORE, G-FULLQ-PRODUCTION] ☐

Limitation. The unit complement has $Q = \varnothing$ and is excluded (Remark 3.3, `FAIL-UNIT-COMPLEMENT-001`).

Theorem 5.14 (Exact collision and cut identities, `G-COLLISION-CUT`). ***PROVED***, *class B. For odd row values Y_s ($s \in S$), with $L(S) = \sum_{s \in S} \log Y_s$ the total row mass, $K(S)$ the collision mass and $C_{\mathrm{cut}}(S)$ the cut functional,*

$$\boxed{K(S) = \sum_{s<s'} \log \gcd(Y_s, Y_{s'}), \qquad C_{\mathrm{cut}}(S) = (|S|-1)L(S) - 2K(S).}$$

If $Y_s = N - s$ then every such gcd *divides $|s - s'|$.*

Proof. Use $v_\ell(\gcd(Y_s, Y_{s'})) = \sum_{k \geq 1} \mathbf{1}_{\ell^k | Y_s} \mathbf{1}_{\ell^k | Y_{s'}}$ and sum; the cut identity is $(n-1)\mu - 2\binom{\mu}{2} = \mu(n-\mu)$. A common divisor of $N - s$ and $N - s'$ divides $s - s'$. [CORE, G-COLLISION-CUT] ☐

Limitation. A singleton layer contributes nothing ($\binom{1}{2} = 0$); the identity is generic and cannot by itself supply a canonical absorption trichotomy (`FAIL-GENERIC-ABSORPTION-001`).

Theorem 5.15 (Clean polylogarithmic parent selection, `G-CLEAN-POLYLOG-PARENT-SELECTION`). ***PROVED***. *Fix $A > 0$. For all sufficiently large even N, let canonical p_0 fail, $R = R_N(p_0)$, and let $\mathcal{L} \subseteq \{(\ell, k) : \ell \in R,\ \ell^k \leq (\log N)^A\}$ be finite and nonempty, with $\rho_\ell = \#\{k : (\ell, k) \in \mathcal{L}\}$ and $\rho(\mathcal{L}) = \max_\ell \rho_\ell$. Then there exist* distinct *upper-half external active primes $p_{\ell,k} > p_0$, one per $(\ell, k) \in \mathcal{L}$, with*

$v_\ell(N - p_{\ell,k}) = k$ and the clean-support property: if a prime r divides $N - p_{\ell,k}$ and $N - p_{\ell',k'}$ for two distinct selected parents, then

$$\boxed{r = \ell = \ell'.}$$

Consequently $\max_r \#\{p \in \mathcal{P} : r \mid N - p\} \le \rho(\mathcal{L})$, and if the layers have pairwise distinct base primes the selected complements are pairwise coprime and the global support load is one.

Hypotheses beyond canonical failure: *Siegel–Walfisz, a standard lower-bound sieve, Mertens, Theorem 5.2 and Theorem 3.10. [CORE, G-CLEAN-POLYLOG-PARENT-SELECTION]*

Warning 5.16 (Correction to an earlier reading of W1). **SCOPE-CORRECTED**. Earlier compressions of this project listed `G-CLEAN-POLYLOG-PARENT-SELECTION` as the *open target* of W1. It is not: it is a **PROVED** theorem of the CRD Core. What remains open in W1 is the passage from the *polylogarithmic* layer family $\ell^k \le (\log N)^A$ to the *actual* layers of the world — i.e. W1 is blocked on L1 (§9.5), not on the selection lemma. See §9.7 as corrected.

5.4 Residue no-go theorems

Theorem 5.17 (Residue-only return no-go for $q \ge 5$, `C-RESIDUE-ONLY-RETURN-NOGO`). ***PROVED***. *Let $q \ge 5$ be prime, $G = (\mathbb{Z}/q\mathbb{Z})^\times$, $a = N \bmod q$. Then*

$$\boxed{(G \setminus \{a\})^2 = G.}$$

Hence every possible tagged quotient residue is a product of two sibling residues neither congruent to $N \pmod q$. Consequently the single residue $b \bmod q$, even together with $\Omega(b) \le 3$, cannot *force a direct sibling return to q. [CORE, C-RESIDUE-ONLY-RETURN-NOGO]*

Theorem 5.18 (Exceptional forced return at modulus 3, `C-MOD3-TAGGED-FORCED-RETURN`). ***PROVED***. *Assume $N \equiv 2 \pmod 3$ and $N - p = 3^e b$ with $3 \nmid b$, $b \equiv 2 \pmod 3$. Then some sibling prime $r \mid b$ satisfies $r \equiv 2 \pmod 3$, hence $3 \mid N - r$ and $r \to 3$. [CORE, C-MOD3-TAGGED-FORCED-RETURN]*

Warning 5.19 (Modulus 3 is an exception, not a mechanism). Theorems 5.17 and 5.18 together say exactly this: the pure local-residue route *saturates* for every $q \ge 5$, and 3 is the single exception. Any programme that hopes to iterate the mod-3 forced return into a general mechanism is dead before it starts; this is one of the convergent no-gos that killed CRD1 (§8.1).

Theorem 5.20 (Topology alone cannot convert external seed weight into ambient incremental J, `C-TOPOLOGY-NOGO`). ***PROVED***. *No theorem using only finite child closure, source counts, support-graph topology, edge multiplicities and the reverse-extension ledger can force $\Delta \log J \gg W(E)$ for arbitrarily large external source weights $W(E)$.*

Proof. A finite realisation permits a family with one small receiving target q and arbitrarily many external components feeding it. Topology and attachment count grow linearly, while the horizontal weight at the receiving label grows only like $e \log q$; external source labels may carry arbitrarily larger evaluation weights without altering the recorded topology. The missing ambient factor must therefore come from arithmetic/order/root information, not from the listed ledgers. [CORE, C-TOPOLOGY-NOGO] □

Reading. This is the precise reason the "η is the capacity" family of attempts is hopeless (Warning 3.21): the quantity that would have to do the work is provably not a function of the topology.

Remark 5.21 (First-occurrence subtlety). Fresh support is *not* incremental J: if $\kappa_R(\ell) = 0$ then the first occurrence of ℓ belongs to the fresh-support factor $F_R(A)$, and only a *later* occurrence contributes a full $\log \ell$ to J_S/J_R. Confusing the two is a recurring source of phantom capacity.

5.5 Shadow and retention arithmetic

Theorem 5.22 (Prime-gap shadow identity). **PROVED**, *class B. For $r \in Q$ and any prime p,*

$$\boxed{r \mid (N-p) \iff r \mid (\delta - p),}$$

and the same equivalence holds modulo r^e when $r^e \mid m_0$.

Proof. $N - p = 2m_0 + \delta - p \equiv \delta - p \pmod{r^e}$. [BIB, thm:shadow; CORE, ARS-SHADOW] □

Reading. The whole part of the support that returns to the first alphabet Q is governed by the *single* number $\delta = 2h$. This is the first point at which the gap parameter enters the multiplicative dynamics. *Limitation:* the theorem does not replace the full factorization of $N - p$; a factor outside Q may lead to success, to a new transient component, or to another sink.

Warning 5.23 (This is *not* a p_0 selector). **KILLED** as a canonical statement. For *any* upper root $r = x+h_r$ with $m_r = x-h_r$ we have $r = m_r+2h_r$, so for every odd $q \mid m_r$, $q \mid (r-2j) \iff q \mid (h_r-j)$. Minimality is not used. [FAIL, FAIL-MR2-CANONICAL-ONLY-001]

Theorem 5.24 (Retention–escape). **PROVED**, *class B. For $q \in Q$ and $d_q = \gcd(m_0, |\delta - q|)$,*

$$\boxed{\mathrm{PF}(N-q) \cap Q = \mathrm{PF}(d_q).}$$

In particular $d_q = 1$ means every active child of the row $N - q$ is new relative to Q.

Proof. For $r \in Q$: $r \mid N - q \iff r \mid \delta - q \iff r \mid \gcd(m_0, |\delta - q|)$; collecting over all r gives the claim. [BIB, thm:retention] □

VERIFIED COMPUTATION, $N \le 10^4$, zero violations.

Corollary 5.25 (Minimum-factor escape). **PROVED**, *class B. Let $a = P^-(m_0)$. If $a > h$ then $d_a = 1$.*

Proof. Every prime divisor of d_a lies in Q, hence is $\ge a$; but $a > h$ gives $0 < |2h - a| < a$, and $2h = a$ is impossible by parity. [BIB, cor:min-factor-escape] □

Warning 5.26. A child outside Q does *not* mean success; it means only that the first alphabet has been left.

Theorem 5.27 (Shadow cycle localisation and normal form). **PROVED**, *class $B + C$. On Q consider the auxiliary edge $q \rightsquigarrow s$ whenever $s \mid \delta - q$. Then every directed cycle lies in $(0, \delta)$; a simple cycle of length ≥ 3 has all vertices $< \delta/3 = 2h/3$; and a two-cycle $\{q, s\}$ satisfies*

$$\boxed{\delta = q + s + tqs, \qquad t \ge 0 \text{ even},}$$

with $t \ge 2$ inside a larger terminal core; hence a terminal component $C \subseteq Q$ with $|C| \ge 3$ satisfies $C \subset (0, 2h/3)$. [BIB, lem:qcycle-below-delta, lem:shadow-two-cycle, cor:qcore-compression]

Reading. The scale $\delta = 2h$ controls the *size* of the whole recursive part of the first front — the cleanest example of a mechanism in which the gap steers multiplicatively. But δ is a function of the root, not of its minimality.

Proposition 5.28 (Shadow two-cycle). **PROVED**, *class B. Contained in Theorem 5.27: a two-cycle $\{q, s\}$ in the shadow graph is exactly a solution of $\delta = q + s + tqs$ with $t \ge 0$ even.*

Proposition 5.29 (Prime-power witness). ***PROVED***, *class C. Let $q \in Q$ with $\mathrm{PF}(N-q) \subseteq Q$. Then there is $r \in Q$ with*

$$\boxed{v_r(m_0) = v_r(\delta - q) > 0.}$$

Proof. Put $s = \delta - q$, so $N - q = 2m_0 + s$. If for every $r \mid N - q$ the valuations $v_r(m_0)$ and $v_r(s)$ differed, then for odd r we would have $v_r(N-q) = \min\{v_r(m_0), v_r(s)\} \le v_r(s)$; since all factors of $N-q$ lie in Q, we would get $N - q \mid |s|$, impossible because $N - q > |\delta - q|$. [BIB, thm:witness] ☐

5.6 Path invariants and winding

Theorem 5.30 (Full path Bézout invariant). ***PROVED***, *class B. For a path $p_0 \to p_1 \to \cdots \to p_t$ with $N - p_i = k_i p_{i+1}$, and $K_0 = C_0 = 1$, $K_{t+1} = K_t k_t$, $C_{t+1} = 2K_t - C_t$:*

$$\boxed{K_t p_t - C_t x = (-1)^t h, \qquad \gcd(K_t, C_t) = \gcd(K_t, h).} \tag{PBI}$$

Proof. For $t = 0$ the left side is $p_0 - x = h$. Induction: from $p_t = 2x - k_t p_{t+1}$ we get $K_t k_t p_{t+1} - (2K_t - C_t)x = (-1)^{t+1}h$. Each k_i divides $N - p_i$ and is coprime to N, so $\gcd(K_t, x) = 1$; the second identity follows. [BIB, thm:bezout-path; CORE, ARS-BEZOUT] ☐

Reading. Every path from the root remembers h as an exact Bézout invariant. This is the strongest sense in which "gap geometry" propagates dynamically — but h is replaced by h_r for any root, so the class is B. **VERIFIED COMPUTATION** on real paths, $N \le 5 \cdot 10^3$, zero violations.

Corollary 5.31 (Irreducible winding normal form). ***PROVED***, *class B. With $W_t = (C_t - (-1)^t)/2$,*

$$\boxed{K_t p_t = W_t N + (-1)^t p_0, \qquad \gcd(W_t, K_t) = 1,} \tag{WN}$$

where $W_0 = 0$, $W_{t+1} = K_t - W_t$. On a bad path $W_t > 0$, $W_t \equiv t \pmod 2$, and for $t \ge 2$, $K_t p_t > 2N$. [BIB, thm:winding-normal-form]

Reading. The root enters the equation multiplicatively and modulo N, not merely as a starting point.

Theorem 5.32 (Finitely many states per vertex). ***PROVED***, *class B. After reduction by $d = \gcd(K_t, h)$, with $B = K_t/d$, $A = C_t/d$, $H = h/d$, the state has primitive form $Bp - Ax = \pm H$ with $\gcd(A, B) = \gcd(B, H) = 1$; for fixed p, fixed divisor $H \mid h$, fixed sign and $0 < B < x$ the number B is determined uniquely modulo x, so a single vertex carries at most $2\tau(h)$ reduced states. [BIB, cor:path-state-count]*

Reading. The divisor count of the gap bounds the richness of the path state — the only place where the *size* of h (rather than its relation to G) enters.

Theorem 5.33 (No-bad-return-to-p_0). ***PROVED***, *class C. Under failure no bad path contains an edge $u \to p_0$. Consequently a non-root endpoint of a bad path satisfies $r < N/3$.*

Proof. Under failure $p_0 > N/2$; if $p_0 \mid N - u$ then from $0 < N - u < N < 2p_0$ we get $N - u = p_0$, so $N - u$ is prime and u is GOOD. [BIB, lem:v71-no-return-p0; I160] ☐

VERIFIED COMPUTATION: all canonical worlds $N \le 10^4$, zero violations.

Theorem 5.34 (Global two-sheet collision budget). ***PROVED***, *class C. In the whole bad world reachable from the canonical p_0 there are at most* two *distinct endpoints r reached by two distinct bad paths whose label products $K_1, K_2 < N$ have opposite winding parities.*

Sketch. The two-sheets base gives $K_1 + K_2 = N$ and $m_0 = q_1q_2$; comparing two collisions gives $m_0 \mid K - K'$, and oddness of the products gives $2m_0 \mid K - K'$. After cutting off small N (Nagura) we have $m_0 > 2N/5$, so $2m_0 > 4N/5$: three distinct numbers in one class modulo $2m_0$ cannot lie in $(0, N)$. For fixed K the winding form gives $Kr \equiv \pm p_0 \pmod{N}$, and the endpoint satisfies $r < N/3$ (Theorem 5.33), so each K encodes at most one endpoint. [BIB, thm:v71-global-two-sheet; l160] ☐

Reading. The first *global* result over the whole p_0 world, independent of component size and topology. *Limitation:* it is not an escape theorem and does not exclude a closed bad world. The attempt to turn it into a contradiction is **KILLED**.

5.7 The Euler-defect layer

Read this box before using η or ν. The corpus records a "layered Euler defect" in the form

$$\eta(S) + \nu(S) \;\geq\; |S| + |\,\mathrm{Src}(S)|, \qquad \text{canonically} \qquad \eta(R) + \nu(R) \;\geq\; |R| + 1,$$

but the material available to this file does not fix the normalisation of η and ν unambiguously. Two natural readings were tested here on all canonical failed worlds with $50 \leq N \leq 6 \cdot 10^4$ (21 093 worlds), and one obvious reading is *false*. Do not use the inequality without saying which normalisation you mean.

Definition 5.35 (Candidate normalisations)**.** With $d_S(q)$, $\kappa_S(q)$ as in Definition 3.18, and $\nu(S)$ the number of BFS layers of S:

(N1) row-excess form $\eta_1(S) = \sum_{p\in S}(\Omega(N-p) - |\,\mathrm{PF}(N-p) \cap S|)$.
(N2) valuation/reuse split $\eta_2(S) = \sum_q(d_S(q) - \kappa_S(q))$, $\nu_2(S) = \sum_q(\kappa_S(q) - 1)$, over $q \in \Gamma_N(S)$; these are the logarithmic masses of U_S and J_S counted in prime factors.
(N3) active-mass form $\sum_q d_S(q)$, the total active valuation mass of the world.
(N4) trivial form $\eta_4(S) = \sum_{p\in S}(\Omega(N-p) - 1)$ with ν the layer count.

VERIFIED COMPUTATION — canonical failed worlds, $50 \leq N \leq 6 \cdot 10^4$, 21 093 worlds, all with $|\,\mathrm{Src}\,| = 1$:

(N1)	$\eta_1(R) + \nu(R) \geq \lvert R\rvert + 1$	**6 044 violations**; smallest $N = 556$ ($\lvert R\rvert = 17,\ \eta_1 = 10,\ \nu = 6$)
(N2)	$\eta_2(R) + \nu_2(R) \geq \lvert R\rvert + 1$	**3 804 violations**; smallest $N = 50$
(N3)	$\sum_q d_R(q) \;\geq\; \lvert R\rvert + 1$	0 violations
(N4)	$\eta_4(R) + \nu(R) \;\geq\; \lvert R\rvert + 1$	0 violations

Proposition 5.36 (The trivial form is a theorem)**.** ***PROVED***. *For a canonical failed world R, $\eta_4(R) + \nu(R) \geq |R| + 1$.*

Proof. Every $p \in R$ is BAD, so $N - p$ is composite and $\Omega(N-p) \geq 2$, whence $\eta_4(R) \geq |R|$; and $\nu(R) \geq 1$. ☐

Remark 5.37 (What this means for the programme). (N4) is a theorem but carries no information — it is $\Omega \geq 2$ in disguise. (N3) is the only nontrivial normalisation that survived; it says the world carries at least $|R| + 1$ units of active valuation mass, i.e. strictly more mass than vertices. It is here **VERIFIED COMPUTATION** only, and a proof is a reasonable small target (see §14.1). (N1) and (N2) are **KILLED** as stated. Any argument in the corpus that used the Euler defect in the form (N1) or (N2) must be re-derived.

Warning 5.38 (Do not conflate the layers). $T_S = U_S J_S$ (Theorem 3.19) is exact and safe. The Euler defect built on it is not. Warning 3.21 remains in force: none of these quantities is the capacity functional γ of $\mathrm{P1}^{\mathrm{br}}/\mathrm{K1}$.

Chapter 6

The Canonical p_0 Programme

This chapter is where p_0 genuinely differs from every other upper root, and also where the project has most often overclaimed. Sections 6.1–6.4 are the positive results; §6.5 is the mandatory negative section and is, for a future model, the most valuable part of the chapter.

6.1 Genealogical defect rank

Definition 6.1 (Prime-defect set and genealogical rank). For the upper root $p_k = x + h_k$ (the k-th prime $\geq x$) put

$$\mathcal{P}_{\text{def}}(p_k) = \{j \in [1, t_k] : c_j^{(k)} = p_k - 2j \in \mathbb{P}\}, \qquad \kappa_{\text{gen}}(p_k) = |\mathcal{P}_{\text{def}}(p_k)|.$$

Theorem 6.2 (Genealogy of prime holes). **PROVED**, *class A*. Domain hypothesis: *non-diagonal case (x composite; see Scope note 3.17). For every $k \geq 0$,*

$$\boxed{\{c_j^{(k)} : j \in \mathcal{P}_{\text{def}}(p_k)\} = \{p_0, \ldots, p_{k-1}\}, \qquad \kappa_{\text{gen}}(p_k) = k,}$$

the unique hole carrying p_i sits at index

$$\boxed{j_i^{(k)} = \frac{p_k - p_i}{2}}$$

and its lower mirror is

$$\boxed{a_{j_i^{(k)}}^{(k)} = N - c_{j_i^{(k)}}^{(k)} = N - p_i.}$$

Proof. Every prime reflected value lies in (x, p_k), and the primes in that interval are exactly $p_0, \ldots, p_{k-1}$. Conversely, for $i < k$ the number $p_k - p_i$ is positive and even, and $p_i > x$ gives $0 < p_k - p_i < h_k$, so $1 \leq (p_k - p_i)/2 \leq \lfloor (h_k - 1)/2 \rfloor = t_k$; at that index $c_j^{(k)} = p_k - (p_k - p_i) = p_i$. Uniqueness follows from injectivity of $j \mapsto p_k - 2j$. The mirror is $a_j = N - c_j$. [CORE, G-UPPERROOT-PRIMEHOLE-GENEALOGY, V71.21–V71.24] □

VERIFIED COMPUTATION: all $N \leq 10^3$ with composite x and $k \leq 5$, zero violations; the diagonal case is isolated separately (Scope note 3.17).

Theorem 6.3 (The defect ledger is a lossless code of prime history). **PROVED**, *class A. Knowing p_k and the index set $\mathcal{P}_{\text{def}}(p_k)$ recovers* all *earlier upper primes,*

$$\boxed{p_i = p_k - 2j_i^{(k)},}$$

and the differences of consecutive hole positions recover the consecutive prime gaps:

$$\boxed{j_i^{(k)} - j_{i+1}^{(k)} = \frac{p_{i+1} - p_i}{2} = \frac{g_{i+1}}{2}, \qquad 0 \le i < k-1.}$$

Proof. The first identity is a rewriting of $j_i^{(k)} = (p_k - p_i)/2$; the second is its difference. □

Reading. $\mathcal{P}_{\mathrm{def}}(p_k)$ is not a "counter of missing rows". It is a *complete, lossless record of the local history of the primes above the midpoint*, gaps included. In exactly that sense p_k "remembers" all its predecessors, and p_0 remembers nobody.

Theorem 6.4 (Defect-rank-zero characterisation). **PROVED**, *class A (non-diagonal). For a prime $r = x + h_r \ge x$ the following are equivalent:*

(i) $r = p_0$;
(ii) $[x, r) \cap \mathbb{P} = \varnothing$;
(iii) $h_r < G_r$;
(iv) $\mathcal{P}_{\mathrm{def}}(r) = \varnothing$;
(v) $\kappa_{\mathrm{gen}}(r) = 0$;
(vi) every strictly reflected value $r - 2j$, $1 \le j \le \lfloor (h_r - 1)/2 \rfloor$, is composite.

[CORE, C-P0-DEFECTRANK-ZERO-EQUIVALENCE]

Proof. (i)–(iii) is Theorem 3.10. Items (ii), (iv), (vi) are equivalent because the strictly reflected values enumerate exactly the numbers of the same parity as r lying *strictly* between x and r — this is where non-diagonality enters. (v) is the cardinality form of (iv). □

Corollary 6.5 (Uniqueness of the reflection-complete root). **PROVED**, *class A (non-diagonal). Call an upper root r* reflection-complete *if every strictly reflected value $r - 2j$ is composite. Then*

$$\boxed{r \textit{ is reflection-complete} \iff r = p_0.}$$

[CORE, C-P0-UNIQUE-REFLECTION-COMPLETE-ROOT]

VERIFIED COMPUTATION: $N \le 10^3$, zero violations for composite x; the diagonal exception is located exactly ($N = 14$, $r = 11$).

Theorem 6.6 (Reflection ledger completeness). **PROVED**, *class A. Among all upper prime roots, p_0 is the* only *one whose strictly reflected ledger is exhausted by four multiplicative-dynamical classes with no prime-defect class:*

$$\mathcal{J}_h = \mathcal{I} \uplus \mathcal{D} \uplus \mathcal{S} \uplus \mathcal{F} \qquad \textit{versus} \qquad \mathcal{J}_{h_k} = \mathcal{P}_{\mathrm{def}} \uplus \mathcal{I} \uplus \mathcal{D} \uplus \mathcal{S} \uplus \mathcal{F}, \quad |\mathcal{P}_{\mathrm{def}}| = k.$$

[CORE, C-P0-REFLECTION-LEDGER-COMPLETENESS, V71.32–V71.34]

Warning 6.7 (What these theorems do *not* say). **Do not sell this as an automatic explanation of the success/failure anomaly.** Theorems 6.2–6.6 are an *exact structural exceptionality*, but:

1. there is no proved implication $\kappa_{\mathrm{gen}} = 0 \Rightarrow$ traversal success; the target `C-P0-GENEALOGYFREE-BOUNDARY-TO-DYNAMIC` is **OPEN**;
2. the property may be *partly equivalent to the ordering of the primes itself*: it says that p_0 is the first element of the sequence $p_0 < p_1 < \cdots$, translated into the language of the reflected block. Theorem 6.3 cuts both ways here: since the defect ledger is a *lossless code* of prime history, the statement "$\kappa_{\mathrm{gen}}(r) = 0$" carries exactly as much information as the statement "r is the first element of that sequence" — no more, no less;
3. the logarithmic mass of the holes is negligible: passing from p_0 to p_k the loss is $\sum_{i<k} \log p_i = O_k(\log N)$, which formally *kills* the mass interpretation (§6.5, `FAIL-P0-ZEROHOLE-ESSENTIAL-001`).

Honest formulation: *the type of relation available on the boundary coordinate changes, not the amount of information.* A prime hole $c_j = p_i$ has no nontrivial factorization $c_j = s\ell$ with $s, \ell > 1$, so on that coordinate there is no multiplicative carrier of the kind that modern factor-flow machinery requires [CORE, V71.44]. That is *eligibility*, not automatic activation — Warning 3.16 remains binding.

6.2 Boundary filtration

Theorem 6.8 (Embedding of reflected blocks)**.** **PROVED**, *class B (the filtration) + A (its initial object). Let $p_k < p_{k+1}$ be consecutive upper primes with $p_{k+1} - p_k = 2s$. Then*

$$\boxed{t_{k+1} = t_k + s, \qquad c^{(k+1)}_{i+s} = c^{(k)}_i, \qquad a^{(k+1)}_{j+s} = a^{(k)}_j \qquad (1 \le i, j \le t_k),}$$

so $\mathfrak{B}(p_k) \subseteq \mathfrak{B}(p_{k+1})$ and the new coordinates are exactly the indices $1, \dots, s$ of the block of p_{k+1}. In particular there is a filtration

$$\boxed{\mathfrak{B}_0 \subset \mathfrak{B}_1 \subset \mathfrak{B}_2 \subset \cdots, \qquad \mathfrak{B}_k := \mathfrak{B}(p_k),}$$

whose unique initial object is p_0.

Proof. Since $h_{k+1} = h_k + 2s$, $t_{k+1} = \lfloor (h_k + 2s - 1)/2 \rfloor = \lfloor (h_k - 1)/2 \rfloor + s = t_k + s$. Further $c^{(k+1)}_{i+s} = p_{k+1} - 2(i + s) = p_k + 2s - 2i - 2s = p_k - 2i = c^{(k)}_i$, and since $a_j = N - c_j$ in both blocks the same index shift gives $a^{(k+1)}_{j+s} = a^{(k)}_j$. Index ranges agree: $i \le t_k$ gives $i + s \le t_{k+1}$. □

VERIFIED COMPUTATION: 731 446 checked indices ($N \le 2 \cdot 10^4$, $k \le 6$), zero violations; EXP6.

Corollary 6.9 (Embedding of the coupling matrix)**.** **PROVED**, *class B. With $M^{(k)}_{ij} = \gcd(c^{(k)}_i, a^{(k)}_j)$, $1 \le i, j \le t_k$,*

$$\boxed{M^{(k)}_{ij} = M^{(k+1)}_{i+s,\, j+s},}$$

i.e. $M^{(k)}$ is exactly the submatrix of $M^{(k+1)}$ cut out by shifting both indices by s. The entire coupling geometry of the earlier root is inherited unchanged.

Proof. Immediate from Theorem 6.8: both gcd arguments are identical. □

Corollary 6.10 (New strip and new hole)**.** **PROVED**, *class A. The new coordinates of the block of p_{k+1} are exactly $j = 1, \dots, s$ with $s = (p_{k+1} - p_k)/2$; among them exactly one, $j = s$, is a prime hole and carries the root pair $(p_k, N - p_k)$. The other $s - 1$ new coordinates are composite.*

Proof. New indices are $1 \le j \le s$, since indices $> s$ come from the embedding. For $j < s$ the value $c^{(k+1)}_j = p_{k+1} - 2j$ lies strictly between p_k and p_{k+1}, hence is composite; for $j = s$ it equals p_k. The lower mirror is $N - p_k$. □

Reading the filtration.
- p_0 is the rank-zero initial object: $\mathfrak{B}_0$ has no hole.
- p_1 contains the *whole* boundary of p_0 plus one new strip of width $s = (p_1 - p_0)/2$ containing exactly one new hole.
- Every further root inherits the entire earlier geometry (coupling matrix included) and adds exactly one new genealogical hole.

This is interesting independently of Goldbach: the reflected boundaries of the upper primes form a natural ascending chain in which the defect rank is the height.

6.3 The canonical phase diagram $\theta = h/G$

Theorem 6.11 (Canonical prime cell). ***PROVED***, *class A. Let P be an odd prime, P_- its predecessor and $G = P - P_-$. Then for every $0 \le h < G$,*

$$\boxed{p_0(N_h) = P, \qquad N_h := 2(P-h).}$$

In particular every *odd prime is a canonical root: for $N = 2(P-1)$ we get $p_0(N) = P$ and $h = 1$.*

Proof. $x = P - h > P - G = P_-$, so $[x, P)$ contains no prime and P is the least prime $\ge x$. [PCRD, Canonical Prime Cell Theorem] ☐

VERIFIED COMPUTATION: full cells for $P \in \{67, 127, 1009, 104729\}$, agreement at every h. *Reading.* The label "p_0" does *not* single out any class of primes. All canonical information sits in the *pair* (P, h), and the natural parameter is $\theta = h/G \in [0,1)$. This theorem is also the seed of the architecture wall (§8.2).

Theorem 6.12 (Gap-position identity and trichotomy). ***PROVED***, *class A. For canonical $p_0 = x + h$ with predecessor p_- and $G = p_0 - p_-$,*

$$\boxed{m_0 - p_- = G - 2h,}$$

and in the non-diagonal case ($h \ge 1$, so G is even) exactly one of the following holds.

Type I ($2h < G$, i.e. $\theta < 1/2$). *Then $p_- < m_0 < x < p_0$, and since p_- and p_0 are consecutive primes and $m_0 \ne p_-$, the number m_0 is* composite. *Hence the canonical root is BAD and* immediate success is impossible.

Type II ($2h = G$, i.e. $\theta = 1/2$). *Then* $\boxed{m_0 = p_-}$*, so m_0 is prime and $N = p_0 + p_-$ is a Goldbach representation:* immediate success is automatic.

Type III ($2h > G$, i.e. $\theta > 1/2$). *Then $m_0 < p_- < x < p_0$ and the previous prime appears as a* lower mirror *of the reflected block at index*

$$\boxed{j^* = h - \frac{G}{2} \in \{1, \dots, t\}, \qquad a_{j^*} = p_-.}$$

Proof. $m_0 - p_- = (x-h) - (x+h-G) = G - 2h$; its sign gives the trichotomy. *Type I:* $G - 2h > 0$ gives $m_0 > p_-$; also $m_0 < x < p_0$, and (p_-, p_0) contains no prime, so m_0 is composite (for N large enough that $m_0 > 1$). *Type II:* $G = 2h$ gives $m_0 = p_-$ directly. *Type III:* setting $a_j = m_0 + 2j = p_-$ gives $2j = p_- - m_0 = 2h - G$, i.e. $j^* = h - G/2$, an integer because G is even. Range: $j^* \ge 1 \iff 2h \ge G + 2$, which is Type III; and $j^* \le t = \lfloor (h-1)/2 \rfloor$ follows from $2j^* = 2h - G \le h - 1$, i.e. $h \le G - 1$, which always holds since $h < G$ (Theorem 3.10). ☐

VERIFIED COMPUTATION: all even $N \le 2 \cdot 10^5$: zero Type I cases with m_0 prime; zero Type II cases with $m_0 \ne p_-$; zero index violations in Type III.

Corollary 6.13 (Cell decomposition). ***PROVED***, *class A. The cell of a prime P with even gap G splits into exactly*

$$\underbrace{1}_{\text{diagonal } (h=0)} + \underbrace{\left(\tfrac{G}{2} - 1\right)}_{\text{Type I}} + \underbrace{1}_{\text{Type II } (h=G/2)} + \underbrace{\left(\tfrac{G}{2} - 1\right)}_{\text{Type III}} = G$$

cases. In particular the number of diagonal cases equals the number of Type II cases, and the distribution of θ is symmetric under $\theta \mapsto 1 - \theta$.

Proof. h runs over $0, \dots, G-1$; $h = 0$ is diagonal, $h = G/2$ is Type II, and the rest split symmetrically about $G/2$ via the bijection $h \mapsto G - h$. ☐

EXP5 — phase census. VERIFIED COMPUTATION, all even $N \le 2 \cdot 10^5$:

	diagonal	Type I	Type II	Type III
count	9 591	40 407	9 591	40 409
with immediate success	9 591	**0**	9 591	7 422

The second row is exactly the content of Theorem 6.12: in Type I immediate success is *impossible*, in Type II and the diagonal case it is *automatic*, and in Type III it happens (here in 18.4% of cases). The difference of 2 between Type I and Type III counts comes from the incomplete cell at the end of the range.

Corollary 6.14 (Reflection-gate phase transition)**.** ***PROVED***, *class A. A guaranteed* active prime *lower-mirror coordinate in the strictly reflected block exists exactly in phase Type III:*

$$\theta > \tfrac{1}{2} \implies a_{j^*} = p_- \in \mathbb{P}, \qquad j^* = h - \tfrac{G}{2}.$$

If in addition $p_- \nmid N$ (***VERIFIED COMPUTATION*** *: true in* all 40 409 *Type III cases with $N \le 2 \cdot 10^5$) then a_{j^*} is an active row and the reflection gate (Lemma 3.15) gives the edge $a_{j^*} \to \mathrm{PF}(c_{j^*})$. In Types I and II no such guarantee exists.*

Remark 6.15 (This explains the counterexample $N = 242$). Warning 3.16 cites $N = 242$ as proof that an empty interval does *not* activate any reflected row. The phase diagram shows this is not an accident: $N = 242$ has $h = 6$, $G = 14$, so $\theta = 3/7 < 1/2$ — Type I, where the guaranteed-mirror theorem does not apply. By contrast $N = 240$ (same $p_0 = 127$, $h = 7$) is Type II and the canonical root wins immediately, and $N = 124$ ($h = 5$, $G = 6$, $\theta = 5/6$) is Type III with two prime lower mirrors, 59 and $61 = p_-$.

6.4 Bidirectional prime rank ρ_0

Definition 6.16 (Lower rank)**.** For canonical $p_0 = P$ let $P_{-1} > P_{-2} > \cdots$ be the consecutive earlier primes and $D_s = P - P_{-s}$ (so $D_1 = G$). Put

$$\rho_0 := \#\{j \in [1, t] : \ a_j = m_0 + 2j \in \mathbb{P}\},$$

the number of *prime lower mirrors* in the strictly reflected block.

Theorem 6.17 (Lower-rank formula)**.** ***PROVED***, *class A. For $h \ge 3$,*

$$\boxed{\rho_0 = \#\{s \ge 1 : \ h < D_s < 2h\}.}$$

Proof. A lower mirror $a_j = m_0 + 2j = P - 2h + 2j$ equals $P_{-s} = P - D_s$ iff $2j = 2h - D_s$, i.e. $j = h - D_s/2$ (D_s is even for $P > 3$). The condition $j \ge 1$ gives $D_s \le 2h - 2$, and $j \le t = \lfloor (h-1)/2 \rfloor$ gives $2h - D_s \le h - 1$, i.e. $D_s \ge h + 1$. Since D_s is even, $D_s \le 2h - 2 \iff D_s < 2h$. The map $s \mapsto j$ is injective because D_s is strictly increasing. □

Warning 6.18 (Correction to the originally proposed formula). **KILLED** as stated. The version proposed in the input material had only the upper bound, $\rho_0 = \#\{s \ge 1 : D_s < 2h\}$. It is missing the lower condition $D_s > h$, which is exactly what confines the mirror to the *strictly* reflected block (index $j \le t$, not $j \le h - 1$). Without it the formula also counts mirrors lying outside the strict block. VERIFIED COMPUTATION: all even $N \le 2 \cdot 10^5$ with $h \ge 3$ (72 451 cases): the corrected formula agrees with direct counting in *every* case, zero violations.

Corollary 6.19 (The lower rank detects the phase exactly)**.** ***PROVED***, *class A.*

$$\boxed{\rho_0 \ge 1 \iff G < 2h \iff \theta > \tfrac{1}{2} \iff \textit{Type III.}}$$

Proof. ($\Leftarrow$) In Type III, $h < G = D_1 < 2h$, so $s = 1$ satisfies the condition of Theorem 6.17. ($\Rightarrow$) D_s is strictly increasing, so if $D_1 = G \geq 2h$ then $D_s \geq 2h$ for all s and no s qualifies. □

VERIFIED COMPUTATION: $N \leq 2 \cdot 10^5$: the number of cases with $\rho_0 \geq 1$ is 40 409 — *exactly* the number of Type III cases. Distribution of ρ_0 over the 72 451 cases with $h \geq 3$:

ρ_0	0	1	2	3	4	5	6	7	8	9
$\#N$	32 042	24 288	10 200	3 936	1 331	404	170	57	16	7

Definition 6.20 (Bidirectional signature)**.**

$$(\kappa^+, \kappa^-) := (\kappa_{\text{gen}},\ \rho_0).$$

The upper coordinate counts primes *above* the midpoint recorded in the reflected blocks, the lower one counts primes *below*.

Why the signature is a better descriptor than κ_{gen} alone. For the canonical root $\kappa^+ = 0$ always, so κ^+ separates no two canonical worlds — it is a constant. But $\kappa^- = \rho_0$ is non-constant and, by Corollary 6.19, encodes exactly the phase:

$$\kappa^- = 0 \iff \theta \leq \tfrac{1}{2}, \qquad \kappa^- \geq 1 \iff \theta > \tfrac{1}{2}.$$

The signature $(0, \rho_0)$ is therefore the first quantity in this file that is simultaneously (i) an invariant of the boundary geometry, (ii) p_0-specific in its first coordinate, and (iii) *nontrivially varying* between canonical worlds.

6.5 What does *not* make p_0 exceptional

Mandatory section. For a future model this is the most valuable part of the chapter: "this does not work, and we know why" is worth as much here as a theorem.

6.5.1 Refutations bearing directly on p_0-exceptionality

Mass of the reflected boundary. KILLED. Hypothesis: the absence of prime holes in the canonical reflected block is itself a significant mass mechanism. Counterexample: for p_1 the block has exactly *one* hole, for p_k exactly k; iterating the pointwise short-interval bound gives $p_k - x = O_k(x^{0.525})$, so removing the holes costs only $\sum_{i<k} \log p_i = O_k(\log N)$ against a full block mass of order $t_k \log(x/h_k)$. Hence hole mass/block mass $\to 0$. [FAIL, FAIL-P0-ZEROHOLE-ESSENTIAL-001; CORE, V71.35]

p_0 as an information selector. SUPERSEDED (not refuted, but **not established**). The ablation `P0-CANONICITY-ABLATION` classifies every clause of the bundle-lift construction as "fixed N" or "arbitrary reflected pair $a + c = N$". Replacing p_0 by p_1 leaves the whole construction *word for word* on every row with composite c'_j. Corpus verdict: *p_0 is quantitatively useful and structurally inessential* for that construction; a non-vacuity guarantee is not a principle of information selection. [FAIL, FAIL-P0-INFORMATION-SELECTOR-001]

Shadow / MR2 congruences. KILLED as a canonical statement (Warning 5.23).

First-front parent minimality. KILLED as a selector (Warning 5.6).

Activation of reflection by the prime-free boundary. KILLED (Warning 3.16, $N = 242$). Structurally explained by the phase diagram (Remark 6.15).

Statistical exceptionality after controlling for row size. KILLED. The median rank percentile of p_0 within fixed N, over all reported metrics, is 0.20982; row-size-aware ranks do not support a single universal claim that p_0 dominates on every axis. Comparing canonical and matched non-canonical B_{abs} on 119 reproducible pairs gives a median difference of $4.6127 \cdot 10^{-5}$ — no clean causal separation. [D4, questions 5, 12] — *reported computational result, not reproduced in this document.*

Minimality does not determine local geometry. KILLED. At fixed $P = 927\,961$ ($N = 1\,855\,760$) a single change of h gives $\Delta \log B_{\text{abs}} = 14.434$, $\Delta G_{\text{esc}} = -230$, $\Delta|W| = -240$; full sweeps at fixed p_0 find 23 canonical transitions $B_{\text{abs}} \to B_{\text{re}}$ and 15 in the reverse direction. [D4, questions 6, 17] — *reported, not reproduced here.*

Largest first-front factor as a heuristic. KILLED. The variant "follow the largest factor of m_0" fails in two steps where the smallest factor wins (Example 6.21). [PCRD, FAIL-P0-LARGEST-FACTOR-TWOSTEP]

Example 6.21 ($N = 124$)**.** VERIFIED COMPUTATION. $x = 62$, $p_0 = 67$, $h = 5$, $G = 6$ (Type III), $m_0 = 57 = 3 \cdot 19$. First-front rows:

$$124 - 3 = 121 = 11^2, \qquad 124 - 19 = 105 = 3 \cdot 5 \cdot 7.$$

Following the *largest* factor 19, the children are $\{3, 5, 7\}$ and none is GOOD ($124 - 5 = 119 = 7 \cdot 17$, $124 - 7 = 117 = 3^2 \cdot 13$). Following the *smallest* factor 3, the only child is 11, and $124 - 11 = 113$ is prime. The full traversal succeeds at depth 2 with two escape edges, $3 \to 11$ and $5 \to 17$.

6.5.2 The ablation $p_0 \to p_1$

The ablation removes *only* minimality and keeps everything else. This table is the central empirical result about p_0.

EXP8 — observables at p_0 and at p_1. VERIFIED COMPUTATION. All even $50 \le N \le 2 \cdot 10^5$; 99 976 cases per root; `world` mode.

observable	p_0	p_1	change
root FAILURES	**0**	**2**	$0 \to 2$
success depth, mean	1.3501	1.5234	+12.8%
world size $\lvert R\rvert$, mean	52.71	59.38	+12.7%
margin G_{esc}, mean	24.51	26.96	+10.0%
$\Omega(m)$, mean	2.373	2.521	+6.2%
shadow retention, mean	0.5545	0.5128	−7.5%
$P^+(m)/m$, mean	0.31455	0.23966	−23.8%
κ_{gen}, mean	**0.0000**	**0.9041**	$\mathbf{0 \to\approx 1}$
κ_{gen}, maximum	**0**	**1**	categorical

Definitions. $\Omega(m)$ — number of prime factors of the complement with multiplicity; shadow retention — $\sum_{q \in Q} |\mathrm{PF}(N - q) \cap Q|$ normalised; $P^+(m)/m$ — normalised largest prime factor; κ_{gen} — genealogical rank (Definition 6.1).

Remark 6.22 (Where 0.9041 comes from, rather than 1). The mean $\kappa_{\text{gen}}(p_1)$ falls short of 1 by exactly $9\,591/99\,976 = 9.59\%$ — exactly the fraction occupied by the diagonal cases (Scope note 3.17 and Corollary 6.13). This is an independent confirmation of the domain repair: for prime x the hole of p_0 falls outside the strict block.

Distribution of Ω (a negative result). VERIFIED COMPUTATION, even $N \le 2 \cdot 10^5$:

Ω	1	2	3	4	5	6	7	8
$m_0 = N - p_0$	26 585	32 687	24 592	10 967	3 699	1 085	272	71
$m_1 = N - p_1$	18 534	36 385	27 198	12 249	4 048	1 161	307	78

Outside the bucket $\Omega = 1$ (which corresponds to immediate success, hence to a *definition*, not to structure) the two distributions are nearly identical. The local factorization of the canonical complement does *not* look unusual. This is one of the main negative results of the project.

Conclusion of the ablation. All aggregate observables — depth, world size, margin, Ω, retention, normalised largest factor — are *nearly interchangeable* between p_0 and p_1. The only quantity that changes categorically is the defect rank. This is exactly the content of [CORE, V71.37]: *p_0-specificity is invisible to many aggregate observables and visible in the exact structure of relations.*

6.5.3 Is the exceptionality local, multiplicative, dynamic, or just the empty interval?

1. **Not local-arithmetic.** Exact-D countermodels satisfy every local condition — normal form, smoothness, pair failure, endpoint primality — and die *only* on canonicity. Smallest recorded model: $q = 3$, $r = 5\,413\,103$, $m_0 = 81\,196\,545$, $N = 297\,720\,668$, endpoint $p = 216\,524\,123$ prime, but $x + 9$ is prime and lies much earlier. [FAIL, FAIL-EXACTD-NONCANONICAL-001]
2. **Not purely multiplicative.** D_q, M_q and the whole prime-power ledger are generic; p_0 enters them only through the value $\delta = 2h$.
3. **Not (yet) dynamic.** There is no proved passage from boundary geometry to traversal behaviour; the target `C-P0-GENEALOGYFREE-BOUNDARY-TO-DYNAMIC-ESCAPE` is OPEN.
4. **But not "just" an empty interval either.** The empty interval is *equivalent* to something substantially richer: the boundary is multiplicatively complete, the ledger has no fifth class, the predecessor lies outside the block, and ρ_0 detects the phase. It is a property of the *type of relation* available on each coordinate, not of the length of an interval.
5. **The reservation in the other direction.** Theorem 6.3 shows the defect ledger is a lossless code of prime history. So "$\kappa_{\text{gen}}(r) = 0$" may be *informationally equivalent* to "r is the first element of the sequence of upper primes". We do not claim that the exceptionality of p_0 goes beyond the ordering of the primes; we claim it is an *exact* translation of that ordering into the multiplicative structure of the boundary.

The honest one-sentence summary.

The exceptionality of p_0 is an *exact boundary-closure property*, not a density effect; it is invisible to aggregate observables and visible in the structure of relations; and no theorem is known leading from it to dynamic behaviour.

6.6 The six special N

EXP7 — nonempty fixed points. VERIFIED COMPUTATION. For all even $N \le 6 \cdot 10^4$ the set B_N^{low} is nonempty for *exactly* six values:

N	$\lvert B_N^{\mathrm{low}}\rvert$	B_N^{low} (initial segment)
128	10	$\{3, 5, 7, 11, 13, 17, 23, 29, 37, 41\}$
1718	55	$\{3, 5, 7, 11, 13, 17, 29, 31, 37, 41, 43, \dots, 571\}$
1862	55	$\{3, 5, 11, 13, 17, 29, 41, 43, 47, 53, 59, \dots, 619\}$
1928	41	$\{3, 5, 7, 11, 13, 17, 29, 43, 53, 71, 73, \dots, 641\}$
2200	2	$\{3, 13\}$
6142	66	$\{3, 5, 7, 13, 17, 19, 31, 43, 67, 71, 73, \dots, 2011\}$

This replicates exactly the list reported in [PCRD] for the same range.

EXP3 — full census of failing upper roots. VERIFIED COMPUTATION. Searching *all* upper roots $p \in (N/2, N)$ with composite complement, for all even $N \le 2 \cdot 10^4$:

- exactly **301** roots whose traversal fails;
- they sit at exactly *six* values of N, distributed as 128 : 8, 1718 : 68, 1862 : 63, 1928 : 57, 2200 : 4, 6142 : 101;
- **not one** of those 301 roots is canonical;
- unit rows $N - p = 1$ excluded separately: 2 261.

The number 301 is the one the corpus carries as "301 composite-complement failed upper roots" `FAIL-CENSUS-DOMAIN-001`; the computation above fixes its exact domain and structure, which the corpus did not record. The canonical root wins in *all six* cases: $p_0 = 67, 859, 937, 967, 1103, 3079$ at depths $0, 0, 2, 2, 0, 2$. These six N are exactly the six numbers with an exceptional autonomous component reported up to 10^8 by Bożek [8].

Remark 6.23 (The two censuses agree although they measure different things). EXP3 asks whether some failing upper root exists; EXP7 asks whether a closed bad set exists. The lists coincide on $N \le 2 \cdot 10^4$. That is expected in one direction (a failing root forces a closed bad set) but the converse is not a theorem, and the agreement is data, not proof.

Example 6.24 (Failure structure at $N = 128$)**.** VERIFIED COMPUTATION. For $N = 128$ ($x = 64$, $p_0 = 67$):

k	p_k	$N - p_k$	verdict
0	67	61 (prime)	SUCCESS, $d = 0$
1	71	$57 = 3 \cdot 19$	SUCCESS, $d = 1$ ($19 \to 109 \in \mathbb{P}$)
2	73	$55 = 5 \cdot 11$	FAILURE
3	79	$49 = 7^2$	FAILURE
4	83	$45 = 3^2 \cdot 5$	FAILURE
5	89	$39 = 3 \cdot 13$	FAILURE
6	97	31 (prime)	SUCCESS, $d = 0$
7	101	$27 = 3^3$	FAILURE
8	103	$25 = 5^2$	FAILURE
9	107	$21 = 3 \cdot 7$	FAILURE
10	109	19 (prime)	SUCCESS, $d = 0$
11	113	$15 = 3 \cdot 5$	FAILURE

All eight failures share the *same* bad world $\{3, 5, 7, 11, 13, 23, 29, 41\}$. The set of failing roots is neither an initial nor a final segment: $p_1 = 71$ wins, $p_2 = 73$ loses. *Minimality alone does not order* *the outcome* — the first signal that "p_0 is simply the smallest" is not an explanation.

Remark 6.25 (The witness $N = 2200$ and the two-core wall). VERIFIED COMPUTATION. The core $B_N^{\text{low}} = \{3, 13\}$ at $N = 2200$ is an exact witness that a source-free set is *not* by itself contradictory:

$$2200 - 3 = 2197 = 13^3, \qquad 2200 - 13 = 2187 = 3^7, \qquad 3^7 - 3 = 13^3 - 13 = 2184.$$

At the same time $p_0(2200) = 1103$ and $m_0 = 1097$ is prime, so the canonical root wins immediately and never enters that core. Excluding such cores in general leads to a Pillai/Bennett-type equation $a^\alpha - a = b^\beta - b$, i.e. to an external Diophantine wall. [PCRD, 2-core wall]

6.7 Escape margin

Success depth turned out to be a *bad* measure of difficulty. The right measure is the margin: how many ways out exist at all.

Definition 6.26 (Escape capacity)**.** For even N with BAD p_0, let $W = R_N(p_0)$ and

$$G_{\text{esc}}(N) = \#\{(v, q) : \ v \in W, \ q \mid N - v, \ q \in \mathcal{A}_N, \ N - q \in \mathbb{P}\}.$$

Then Goldbach failure $\Rightarrow G_{\text{esc}}(N) = 0$, and $G_{\text{esc}}(N) > 0 \Rightarrow$ Goldbach for N. The converse implication is *not* proved. [BIB, escape capacity; repair relative to v5.8]

EXP4 — escape margin. VERIFIED COMPUTATION. All even $N \le 2 \cdot 10^5$:

- 73 394 numbers have BAD p_0; $G_{\text{esc}} = 0$ *never* occurs;
- $G_{\text{esc}} = 1$ for 52 numbers, of which exactly **10** have $|W| \ge 3$:

$$368, \ 632, \ 1258, \ 1598, \ 1862, \ 1928, \ 3692, \ 4582, \ 6142, \ 14198,$$

  with world sizes $12, 14, 25, 22, 23, 16, 21, 22, 12, 64$ — *exactly* the list and sizes of [BIB, ten numbers with margin one], obtained there by a scan to $3 \cdot 10^6$;
- distribution of small margins at $|W| \ge 3$: $G_{\text{esc}} = 1{:}10,\ 2{:}38,\ 3{:}122,\ 4{:}477,\ 5{:}571,\ 6{:}656$;
- largest canonical bad world: $|W| = 204$ at $N = 152\,288$.

The minimum margin grows — answer to an open point of the corpus. VERIFIED COMPUTATION. The corpus left as a task the question "does the $G_{\text{esc}} = 1$ list really end at 14198" [BIB, next move]. Minimum G_{esc} over dyadic blocks $[2^{k-1}, 2^k)$, with $|W| \ge 3$ and $N \le 2 \cdot 10^5$:

k	7	8	9	10	11	12	13	14	15	16	17	18
$\min G_{\text{esc}}$	2	2	1	1	1	1	**1**	2	2	3	3	3

In this range the $G_{\text{esc}} = 1$ list at $|W| \ge 3$ does end at $14198 \in [2^{13}, 2^{14})$, and the minimum margin in higher blocks rises to 2 and 3. This is a *signal*, not a proof: the range is finite and an order of magnitude smaller than the scan in [BIB].

Channel substitution. EMPIRICAL PATTERN, not a theorem. For $N = 14198$ a world of 64 bad vertices hangs on a *single* edge $71 \to 277$. Preserving the congruence behind that edge at $N \pm 554$ loses the canonical p_0; preserving p_0 in the nearest tested perturbations destroys that edge but creates 10–26 replacement edges. Across the whole campaign 459 compensation candidates were recorded with *zero* true orphans [D4, q. 18; ORPH, q. 9–10]. Reading: closing one channel opens another. Neighbouring sweeps do not isolate causality, so this remains a pattern, not a law.

Chapter 7

Factorization and Reachability Machinery

7.1 Post-root factorization

Theorem 7.1 (The post-root gcd)**.** ***PROVED***, *class B. Let $q \in Q$ (so q is an odd prime dividing m_0) and put $B_q = m_0/q$. Then*

$$\boxed{D_q := \gcd(m_0,\, N-q) = \gcd(m_0,\, |\delta - q|) = \gcd(B_q,\ |2h-q|), \qquad q \nmid D_q.}$$

Proof. The first equality is the shadow identity (Theorem 5.22) applied to $N - q = 2m_0 + (\delta - q)$. For the rest: $\gcd(h, m_0) = 1$ (Theorem 5.3) and $q \mid m_0$ give $q \nmid h$, and q is odd, so $q \nmid \delta = 2h$; hence $q \nmid (\delta - q)$ and $q \nmid D_q$. Writing $m_0 = q^a m'$ with $q \nmid m'$ we get $\gcd(m_0, \delta - q) = \gcd(m', \delta - q) = \gcd(q^{a-1}m', \delta - q) = \gcd(B_q, \delta - q)$, and $\gcd(\cdot, y) = \gcd(\cdot, |y|)$. [LOG; FND] □

Remark 7.2 (The identity is root-generic). Every step uses only $m_r = x - h_r$ and $\gcd(h_r, m_r) = 1$, so Theorem 7.1 holds verbatim for any upper root. It is class B and must not be quoted as canonical.

Corollary 7.3 (The cofactor is Q-free)**.** ***PROVED***, *class B. With $M_q = (N - q)/D_q$ we have $\mathrm{PF}(M_q) \cap Q = \varnothing$: all of the first alphabet that survives in the row $N - q$ is concentrated in D_q.*

Proof. By Theorem 5.24, $\mathrm{PF}(N - q) \cap Q = \mathrm{PF}(d_q)$ and $d_q \mid D_q$; dividing out D_q removes every prime of Q from the row. □

Definition 7.4 (FRESH / SMOOTH split)**.** $q \in Q$ is *FRESH* if $P^+(N - q) \geq B_q$, and *SMOOTH* otherwise. Write Q_{smooth} for the smooth part. The split is the domain boundary of the incidence ledger below.

Warning 7.5 (High-gap amplification is conditional). **SCOPE-CORRECTED**. The inequality $h^2 > 4m_0$ ("high-gap amplification") is *not* a consequence of CRD failure. Its actual hypothesis is failure of the Exact-D pair, and the corpus states it under that hypothesis [CORE, ARS-EXACTD]. Using it under canonical failure alone is an error; this is the **SCOPE-CORRECTED** noted against [BIB].

7.2 The incidence ledger

Theorem 7.6 (Incidence ledger IL1–IL6)**.** ***AUDITED IDENTITY*** *on the domain $S = Q_{\mathrm{smooth}}$. With U_F, J_F the unit and layer parts of the ledger restricted to S, and B_{re}, B_{abs} the reused and absorbed base parts,*

$$\boxed{\prod_{q \in S} M_q \;=\; U_F\,(J_F\,B_{\mathrm{re}})\,B_{\mathrm{abs}}.}$$

Two independent reconstruction routes, 1 000 *worlds,* 0 *errors* [ILA].

Warning 7.7 (Domain). SCOPE-CORRECTED. IL1–IL6 hold on $S = Q_{\text{smooth}}$ *only.* The FRESH part is excluded. Quoting the identity on all of Q is a recorded error.

Definition 7.8 (Base atoms)**.** B_{abs} is the absorbed base (atoms consumed inside the world), B_{re} the reused base (atoms appearing in more than one row), and $B_{\text{abs,int}}$ the internal absorbed base. The corpus proves $B_{\text{abs,int}} = B_{\text{abs}}$ without vacuity [CORE, ARS-BABS-INTERNAL].

Killed here. *Unconditional CAB*: refuted by $N = 322$ ($U_R = J_R = 1$, $B_{\text{abs}} = 11 \cdot 29$) and, in the internal version, by $N = 858$ `FAIL-CAB-001`, `FAIL-CAB-INTERNAL-001`. *$\mathcal{C}_S < \mathcal{D}_S$ from the first front*: refuted by $N = 4736$ ($\mathcal{C}_S \approx 22.60 > \mathcal{D}_S \approx 20.16$) `FAIL-POTENTIAL-CAPACITY-001`. *Failure always produces nontrivial B_{abs}*: refuted `FAIL-BABS-PRODUCTION-001`. *Four-channel ledger on all of Q_0*: refuted by $N = 20$ (a GOOD row outside R gives $\kappa_R < m_\ell$) [ILA]. *Constant $C = 1$ in charging*: 53 violations, maximum 9.42243 `FAIL-CHILD-C1-001`. *Naive support expansion of all sources*: Hall compression $456 \to 17$ at $N = 51\,032$ `FAIL-HALL-SUPPORT-001`.

Escape-or-reuse (OPEN as a theorem, VERIFIED COMPUTATION as data). Every internal base atom $\ell \in B_{\text{abs,int}}$ has either a GOOD edge reachable inside one of its first-front branches, or an active parent outside R. Both alternatives are necessary: $(N, \ell, p) = (6142, 877, 3511)$ shows the second is needed. Bounded charging multiplicity: observed maximum 1, boundedness unproved. Data: 61 452 instances, 0 orphans; independent red team 60 413 atoms, 0 orphans [ORPH, RT]. Known scope corrections around it: external reuse is *not* local ($(N, \ell) = (1\,825\,372, 52\,153)$, distance 53 750, `FAIL-LOCAL-REUSE-001`); the external parent is *not* necessarily a success root ($(6142, 877, 3511)$: 3511 is active, above p_0, and its own traversal fails, `FAIL-EXTERNAL-SUCCESS-001`); escape need not be directly adjacent ($N = 14198$: atoms $19, 167$ at distances 8 and 7, `FAIL-DIRECT-ADJACENCY-001`).

7.3 The rooted first-front layer

This section and the two that follow are the Biblia's v7.3–v7.5 rooted first-front theory. It is the most quantitatively developed part of the whole corpus and it is where the exact ledgers live. Read the statuses carefully: the *conservation and split identities* carry the source's PROVED – PENDING RED TEAM label, while the *collision budget and Exact-D geometry* of §7.4–§7.5 are fully PROVED.

Throughout, $q \in Q = \mathrm{PF}(m_0)$, and

$$n_q := N - q, \quad B_q := \frac{m_0}{q}, \quad D_q := \gcd(m_0, n_q), \quad I_q := \operatorname{rad} D_q, \quad M_q := \frac{n_q}{D_q},$$

$$U_q^{\text{row}} := \frac{n_q}{\operatorname{rad}(n_q)}, \qquad E_q := \frac{\operatorname{rad}(n_q)}{I_q}, \qquad j_q := \omega(n_q) - 1, \qquad u_q := \Omega(n_q) - \omega(n_q),$$

so that $e_q := \Omega(n_q) - 1 = j_q + u_q$ splits the row cost exactly into a *collision/branching channel* j_q and a *squarefull channel* u_q.

7.3.1 FRESH / SMOOTH dichotomy

Definition 7.9 (FRESH and SMOOTH, restated)**.** For $q \in Q$: the row is **FRESH** if $P^+(n_q) \geq B_q$, in which case it has a subthreshold prime child $s_q \geq B_q$; it is **SMOOTH** if $P^+(n_q) < B_q$, in which case every child of the row exceeds the threshold already at the second step.

Proposition 7.10 (Fresh prime packing, `B-FRESH-PACKING`). **PROVED – PENDING RED TEAM**. *For distinct first-front primes q_i choose FRESH children s_i with two-step cost $< N$. The paths have the same parity, so the two-sheet budget (Theorem 5.34) forbids their merging; hence the s_i are pairwise distinct and for every $E \subseteq Q_{\text{fresh}}$*

$$\boxed{\prod_{q\in E} s_q \;\geq\; \frac{m_0^{|E|}}{\prod_{q\in E} q}.} \tag{FP}$$

[BIB, prop:v73-fresh-packing]

Proposition 7.11 (Smoothness Complexity Tax, `B-SMOOTH-TAX`). **PROVED – PENDING RED TEAM**. *On the SMOOTH branch, with $e_q = \Omega(n_q) - 1$,*

$$\boxed{B_q^{e_q} > q.} \tag{ST}$$

Moreover $e_q = (\omega(n_q) - 1) + (\Omega(n_q) - \omega(n_q)) = j_q + u_q$, so the cost splits exactly *into the collision/branching channel J and the squarefull channel U. [BIB, prop:v73-smooth-tax]*

Corollary 7.12 (Root smoothness excess tax). **PROVED – PENDING RED TEAM**. *For the full closed world,*

$$\boxed{\Omega(U_R) + \Omega(J_R) - 1 \;\geq\; \sum_{q\in Q_{\text{smooth}}} e_q \;>\; \sum_{q\in Q_{\text{smooth}}} \frac{\log q}{\log(m_0/q)}.}$$

Warning 7.13 (The two-channel wall, and what it forbids). The external argument “many SMOOTH rows ⇒ many powerful integers” is **KILLED**. A SMOOTH row can pay almost its whole cost through support *diversity* — the J channel — with no high prime multiplicity at all. This is exactly why Proposition 7.11 must keep both j_q and u_q. The correct picture is two-channel:

$$\text{SMOOTH} \longrightarrow \begin{cases} j_q \text{ large} & \longrightarrow \; J/R_\partial \text{ load} \\ u_q \text{ large} & \longrightarrow \; U \text{ load} \end{cases}$$

and what is needed is a theorem bounding the *joint* capacity of both channels while keeping the root geometry of p_0. That is the *Root Smoothness Conservation Problem* (§7.3.4).

7.3.2 Root First-Front Conservation

Put $S_1 = \{p_0\} \cup Q$ and $b_1 := |\Gamma_N(S_1) \setminus S_1|$. The root row contains every $q \in Q$, so p_0 is the unique source of S_1.

Theorem 7.14 (Root First-Front Conservation, `B-FFC`). **PROVED – PENDING RED TEAM**. *The fundamental factor-flow decomposition gives*

$$\boxed{m_0 \frac{\prod_{q\in Q}(N-q)}{\prod_{q\in Q} q} = U_{S_1} J_{S_1} R_\partial(S_1).} \tag{FFC}$$

Applying Ω,

$$\boxed{\Omega(U_{S_1}) + \Omega(J_{S_1}) + |\Gamma(S_1) \setminus S_1| = \Omega(m_0) + \sum_{q\in Q} (\Omega(N-q) - 1).} \tag{FFC-Ω}$$

[BIB, thm:v73-ff-conservation]

Corollary 7.15 (Exact recycle–escape identity, `B-FFRE`). **PROVED – PENDING RED TEAM**. *Since $U_{S_1} = \frac{m_0}{\operatorname{rad}(m_0)} \prod_{q\in Q} \frac{N-q}{\operatorname{rad}(N-q)}$ and $\prod_{q\in Q} q = \operatorname{rad}(m_0)$, the squarefull factors cancel and*

$$\boxed{\prod_{q\in Q} \operatorname{rad}(N-q) = J_{S_1} R_\partial(S_1).} \tag{FFRE}$$

The first front therefore has an exact *decomposition of its support into recycle/collision and true boundary escape. [BIB, cor:v73-recycle-escape]*

7.3.3 The exact first-front split ledger

Theorem 7.16 (Exact First-Front $U/J/\partial$ Split, `B-UJD-SPLIT`). **PROVED – PENDING RED TEAM**. *With* $S_1 = \{p_0\} \cup Q$ *and* $b_1 = |\Gamma(S_1) \setminus S_1|$,

$$\boxed{\Omega(U_{S_1}) = \Omega(m_0) - |Q| + \sum_{q \in Q} u_q,} \tag{U-split}$$

$$\boxed{\Omega(J_{S_1}) + b_1 = |Q| + \sum_{q \in Q} j_q.} \tag{JR-split}$$

Summing recovers Root First-Front Conservation: $\Omega(U_{S_1}) + \Omega(J_{S_1}) + b_1 = \Omega(m_0) + \sum_{q \in Q} e_q$.

Proof. By definition $U_{S_1} = \frac{m_0}{\mathrm{rad}(m_0)} \prod_{q \in Q} \frac{N-q}{\mathrm{rad}(N-q)}$; applying Ω and $\omega(m_0) = |Q|$ gives (U-split). From Corollary 7.15, $\prod_{q \in Q} \mathrm{rad}(N-q) = J_{S_1} R_\partial(S_1)$, and the boundary radical is squarefree with b_1 prime atoms, so $\sum_{q \in Q} \omega(N-q) = \Omega(J_{S_1}) + b_1$; since $\omega(N-q) = 1 + j_q$ this is (JR-split). [BIB, thm:v74-exact-split] □

Corollary 7.17 (Smoothness tax in the separated channels). **PROVED – PENDING RED TEAM**. *For* $q \in Q_{\mathrm{smooth}}$ *one has canonically* $B_q^{e_q} > q$ *with* $e_q = j_q + u_q$. *Hence every SMOOTH row must pay its required cost through exactly one of three routes: larger* u_q *feeding* U, *larger* j_q *feeding* J *or the boundary support, or a mixture splitting the cost between the channels.* ***There is no "free" SMOOTH variant*** *— but at the present stage there is still no global upper capacity bound producing a contradiction.*

Lemma 7.18 (First-front GCD packing, `B-GCD-PACK`). **PROVED – PENDING RED TEAM**. *For a prime* ℓ *put* $\nu_\ell := \#\{q \in Q : \ell \mid N - q\}$. *All* q *counted by* ν_ℓ *lie in one class modulo* ℓ, *so*

$$\boxed{\nu_\ell \le \min\Bigl(|Q|,\ 1 + \Bigl\lfloor \frac{m_0}{\ell} \Bigr\rfloor\Bigr).}$$

[BIB, lem:v74-gcd-pack]

Limitation. The lemma is correct but is *not* a knockout: it bounds incidence multiplicity, not total capacity.

Scope note 7.19 (Why the classical powerful-tail bound is too weak). **KILLED** as a route. The global sparsity of powerful integers gives $O(N^{(1+\alpha)/2})$ exceptions up to N, whereas the first front requires control of only $|Q| = \omega(m_0) \le \log_2 m_0$ special points. Global density of powerful integers is therefore far too weak to force even one non-powerful row in the *adaptive* set $\{N - q : q \mid m_0\}$. [BIB, v7.4 powerful-tail remark]

7.3.4 The Root Smoothness Conservation Problem

Open target 7.20 (Root Smoothness Conservation Problem, `B-RSCP`). **OPEN**. For a hypothetical failure, find a uniform, deterministic capacity theorem linking

$$B_q^{j_q+u_q} > q \qquad (q \in Q_{\mathrm{smooth}})$$

to the exact ledger

$$\Omega(U_{S_1}) = \Omega(m_0) - |Q| + \sum u_q, \qquad \Omega(J_{S_1}) + b_1 = |Q| + \sum j_q,$$

together with source attenuation and terminal absorption taxes, and leading to a contradiction or a forced escape. [BIB, def:v74-rscp]

Warning 7.21. This target is the Biblia's own statement of the *same* obstruction that Chapter 9 records as the same-functional bridge (§9.11). It is not an independent problem: solving one solves the other. What the Biblia adds is the exact two-channel bookkeeping the bridge has to respect.

7.4 The lossless prime-power root ledger

Proposition 7.22 (Root-recycle-free row identity). **PROVED – PENDING RED TEAM**. *For every $q \in Q$,*

$$\boxed{I_q = \operatorname{rad}\gcd(B_q, |2h-q|), \qquad U_q^{\mathrm{row}} E_q = \frac{N-q}{I_q}.}$$

In particular the first copy of every prime recycled from the root alphabet is removed exactly from the weighted tax. [BIB, prop:v75-root-recycle-free]

Proposition 7.23 (Full first-front recycle split). **PROVED – PENDING RED TEAM**. *For the full first front $S_1 = \{p_0\} \cup Q$ the recycling factor splits as*

$$\boxed{J_{S_1} = J^{\mathrm{rec}} J_Q^{\mathrm{ext}}, \qquad J^{\mathrm{rec}} = \prod_{q \in Q} I_q,}$$

and after removing the root recycle the remaining lower demand is

$$\boxed{U_{S_1} J_Q^{\mathrm{ext}} R_\partial(S_1) > \left(\tfrac{9}{5}\right)^{|Q|} m_0.}$$

This is a lower*-capacity statement; it does not yet give an upper bound closing SMOOTH.*

Warning 7.24 (Two sharp interpretive countermodels). SMOOTH does *not* force $E_q > 1$: one can have $E_q = 1$ with the whole cost staying in the valuations. And the same external prime may occur in several rows. Externalisation must therefore never be identified with pairwise-disjoint support, nor with automatic escape.

Let $\varnothing \neq S \subseteq Q_{\mathrm{smooth}}$, $\nu_{\ell,k}(S) := \#\{q \in S : \ell^k \mid n_q\}$ and $a_r := v_r(m_0)$ for $r \in Q$.

Theorem 7.25 (Lossless Prime-Power Root Ledger, `B-PP-LEDGER`). **PROVED – PENDING RED TEAM**. *Three exact decompositions hold:*

$$\boxed{\log \prod_{q \in S} \frac{n_q}{I_q} = \sum_{\ell \notin Q} \sum_{k \geq 1} \nu_{\ell,k} \log \ell + \sum_{r \in Q} \sum_{k \geq 2} \nu_{r,k} \log r,} \tag{PP-I}$$

$$\boxed{\log \prod_{q \in S} \frac{D_q}{I_q} = \sum_{r \in Q} \sum_{k=2}^{a_r} \nu_{r,k} \log r,} \tag{PP-D}$$

$$\boxed{\log \prod_{q \in S} \frac{n_q}{D_q} = \sum_{\ell \notin Q} \sum_{k \geq 1} \nu_{\ell,k} \log \ell + \sum_{r \in Q} \sum_{k \geq a_r+1} \nu_{r,k} \log r.} \tag{PP}$$

Proof. For each prime ℓ the layer-cake identity $\sum_{q \in S} v_\ell(n_q) = \sum_{k \geq 1} \nu_{\ell,k}(S)$ holds. Dividing by I_q subtracts one copy of every root prime present in the row, so the root sum starts at $k = 2$. Dividing by the full D_q subtracts $\min\{a_r, v_r(n_q)\}$, leaving exactly the layers $k \geq a_r + 1$. External primes do not divide m_0, so their layers start at $k = 1$. [BIB, thm:v75-pp-root-ledger] □

Reading. This is the exact sense in which the root-supported valuation mass can be subtracted *without losing demand*: (PP-D) isolates precisely what the root alphabet already paid for.

Theorem 7.26 (Exact post-root demand). **PROVED – PENDING RED TEAM**. *For $q \in Q$,*

$$\boxed{D_q = \gcd(B_q, |2h-q|)} \quad \text{and} \quad \boxed{\frac{n_q}{D_q} \geq \frac{N-q}{\min\{B_q, |2h-q|\}}.} \tag{Demand}$$

Setting $w_q := \sum_{\ell \notin Q} v_\ell(n_q) + \sum_{r \in Q} (v_r(n_q) - a_r)_+$, every SMOOTH row satisfies

$$\boxed{B_q^{w_q} > \frac{n_q}{D_q} \geq \frac{N-q}{\min\{B_q, |2h-q|\}}.} \tag{Post-root tax}$$

Proof. Since $q \mid m_0$ and $\gcd(h, m_0) = 1$ we get $q \nmid 2h - q$, so $\gcd(m_0, n_q) = \gcd(m_0, 2h - q) = \gcd(B_q, |2h - q|)$. Hence $D_q \le B_q$ and $D_q \le |2h - q|$, which is (Demand). After dividing by D_q all remaining prime factors of a SMOOTH row are still smaller than B_q, and their number with multiplicity is exactly w_q. [BIB, thm:v75-post-root-demand] □

Lemma 7.27 (Root valuation rigidity). **PROVED – PENDING RED TEAM**. *Let $r^a \| m_0$. Then*

$$\boxed{v_r(N-q) > a \implies v_r(|2h-q|) = a,} \quad \text{and hence} \quad \boxed{r^a > |2h-q| \implies v_r(N-q) \le a.}$$

Proof. Write $m_0 = r^a M$ with $r \nmid M$ and $b = v_r(|2h - q|)$. From $n_q = 2m_0 + (2h - q)$,

$$v_r(n_q) = \begin{cases} b, & b < a, \\ a, & b > a, \\ a + v_r\left(2M + \frac{2h-q}{r^a}\right), & b = a. \end{cases}$$

Excess above a can occur only in the case $b = a$; and if $r^a > |2h - q|$ then $v_r(|2h - q|) = a$ is impossible. [BIB, lem:v75-root-valuation-rigidity] □

Lemma 7.28 (Sharp Prime-Power Packing, `B-SHARP-PP-PACK`). **PROVED – PENDING RED TEAM**. *If $\ell^k \mid n_q$ and $\ell^k \mid n_r$ then*

$$\boxed{\ell^k \mid q - r.}$$

All incident roots therefore lie in one class modulo ℓ^k, and for any eligible set $\mathcal{E}_{\ell,k}$ containing all actually incident roots,

$$\boxed{\nu_{\ell,k} \le \min\left(|\mathcal{E}_{\ell,k}|,\ 1 + \left\lfloor \frac{\operatorname{diam} \mathcal{E}_{\ell,k}}{\ell^k} \right\rfloor\right).}$$

In particular $\ell^k > \operatorname{diam} \mathcal{E}_{\ell,k}$ forces $\nu_{\ell,k} \le 1$. [BIB, lem:v75-pp-pack]

Warning 7.29 (Potential capacity is not actual capacity). **KILLED**. One may *not* assume $\mathcal{C}_S < \mathcal{D}_S$ uniformly on the basis of the first front alone. The canonical model

$$N = 4736, \quad p_0 = 2371, \quad h = 3, \quad m_0 = 2365 = 5 \cdot 11 \cdot 43, \quad Q = S = \{5, 11, 43\}$$

has all three rows SMOOTH,

$$N - 5 = 3 \cdot 19 \cdot 83, \qquad N - 11 = 3^3 \cdot 5^2 \cdot 7, \qquad N - 43 = 13 \cdot 19^2,$$

and for the potential-capacity definition used, $\mathcal{C}_S \approx 22.60 > 20.16 \approx \mathcal{D}_S$. This is not a closed bad world, so it does *not* refute a future theorem with a closure hypothesis; it does refute deriving the needed inequality from local eligibility geometry alone. `FAIL-POTENTIAL-CAPACITY-001`

7.4.1 Actual prime-power collision budget

For residual rows $M_q = (N-q)/D_q$ set $\mu_{\ell,k}(S) := \#\{q \in S : \ell^k \mid M_q\}$ and $\Delta_{\text{odd}}(S) := \prod_{q<r,\ q,r \in S} |q - r|_{\text{odd}}$.

Theorem 7.30 (Actual Prime-Power Collision Budget, `B-VPC`). **PROVED**. *Exactly,*

$$\boxed{\sum_{\ell} \sum_{k \ge 1} \binom{\mu_{\ell,k}}{2} \log \ell = \sum_{q<r} \log \gcd(M_q, M_r) \ \le\ \log \Delta_{\text{odd}}(S).} \tag{VPC}$$

Proof. For a fixed pair $q < r$, $\sum_{\ell,k} \mathbf{1}_{\ell^k | M_q, M_r} \log \ell = \log \gcd(M_q, M_r)$. Summing over pairs, the number of pairs sharing layer (ℓ, k) is $\binom{\mu_{\ell,k}}{2}$. Moreover $\gcd(M_q, M_r) \mid \gcd(N - q, N - r) \mid r - q$, and the M_q are odd, so the common divisor divides the odd part $|r - q|_{\text{odd}}$. Multiplying over pairs gives the last inequality. [BIB, thm:v75-actual-pp-collision] □

Corollary 7.31 (Actual shared-mass bound)**.** PROVED*. With* $L_{\text{shared}} := \sum_{\mu_{\ell,k}\geq 2} \mu_{\ell,k} \log \ell$*,*

$$\boxed{L_{\text{shared}} \leq 2 \log \Delta_{\text{odd}}(S).}$$

Proof. For $\mu \geq 2$ one has $\mu \leq 2\binom{\mu}{2}$; sum over the layers that actually occur and apply Theorem 7.30. ☐

For a pair $q < r$ put $g_{q,r} := \gcd(M_q, M_r)$ and $L_{\text{pair}}(q,r) := \log \frac{M_q M_r}{g_{q,r}^2}$.

Lemma 7.32 (Pair isolation floor)**.** PROVED*. For two distinct SMOOTH roots,* $\boxed{L_{\text{pair}}(q,r) \geq \log 5.}$

Proof. $g_{q,r}$ is odd and divides $|r-q|_{\text{odd}} < r/2$. For the larger SMOOTH root r, $M_r > \frac{9}{5}r$, so $M_r/g_{q,r} > 18/5$. That quotient is a positive odd integer, hence at least 5; and $M_q/g_{q,r} \geq 1$. [BIB, lem:v75-pair-log5] ☐

Theorem 7.33 (Exact weighted cut identity, `B-WEIGHTED-CUT`)**.** PROVED*. Let* $s = |S|$ *and let the layer* $\lambda = (\ell, k)$ *carry weight* $w_\lambda = \log \ell$ *and multiplicity* μ_λ*. Then*

$$\boxed{\sum_{q<r} L_{\text{pair}}(q,r) = \sum_{\lambda} \mu_\lambda (s - \mu_\lambda) w_\lambda,} \tag{Cut}$$

and consequently $C_{\text{cut}} := \sum_\lambda \mu_\lambda (s-\mu_\lambda) w_\lambda \geq \binom{s}{2} \log 5$*.*

Proof. Layer λ contributes w_λ to $L_{\text{pair}}(q,r)$ exactly when it belongs to precisely one of the two rows; the number of such unordered pairs is $\mu_\lambda(s-\mu_\lambda)$. Summing over layers gives the identity, and Lemma 7.32 gives the lower bound. [BIB, thm:v75-weighted-cut] ☐

Corollary 7.34 (Low-multiplicity layer lower bound)**.** PROVED*. For* $1 \leq u \leq s-1$ *put* $W_{\leq u} := \sum_{1 \leq \mu_\lambda \leq u} w_\lambda$*,* $R_u := \frac{2(s-u-1)}{u}$ *and* $H_u := \frac{s-1}{u}\lfloor \frac{(u+1)^2}{4} \rfloor$*. Then*

$$\boxed{W_{\leq u} \;\geq\; \frac{\left[\binom{s}{2}\log 5 - R_u \log \Delta_{\text{odd}}(S)\right]_+}{H_u},}$$

and in particular

$$\boxed{W_1 \;\geq\; \frac{\left[\binom{s}{2}\log 5 - 2(s-2)\log \Delta_{\text{odd}}(S)\right]_+}{s-1}.}$$

Proof. Subtract a multiple of the collision budget from (Cut). For $\mu \geq u+1$, $\frac{\mu(s-\mu)}{\binom{\mu}{2}} = \frac{2(s-\mu)}{\mu-1} \leq R_u$; for $1 \leq \mu \leq u$ the coefficient $\mu(s-\mu) - R_u\binom{\mu}{2} = \frac{s-1}{u}\mu(u+1-\mu)$ does not exceed H_u. Apply $K \leq \log \Delta_{\text{odd}}(S)$ and take positive parts. ☐

Warning 7.35 (The sharp abstract obstruction). Cut geometry alone does *not* force singletons once the right-hand side of Corollary 7.34 vanishes. Abstract systems in which every layer has multiplicity exactly 2 (or generally $u+1$) realise the obstruction. Globalisation needs congruences, source structure or absorption data in addition.

7.5 Exact-D pair compensation and canonical pair geometry

This section reduces a hypothetical Exact-D failure to an explicit rigid regime, then to an explicit normal form, and finally to a canonical gap filter. Everything here is PROVED. The branch $k = 1$ ends at a stated **OPEN** barrier. Notation: $t := 2h$, $m_0 = qrc$ with $q < r$ distinct in Q_{smooth}, and $\delta := |r - q|_{\text{odd}}$.

Theorem 7.36 (Exact-D pair compensation outside the regime $r < t$, `B-EXACTD-OUTSIDE`)**.** ***PROVED***. *If $t < r$ then $\boxed{M_q M_r > \delta^2}$; more precisely $\boxed{\frac{M_q M_r}{\delta^2} > \frac{36q}{5}}$. Consequently every potential Exact-D failure satisfies*

$$\boxed{q < r < t, \qquad r \mid D_q, \qquad t \equiv q \pmod{r}.} \tag{ED-res}$$

Proof. Suppose $M_q M_r \le \delta^2$. Since r is SMOOTH, $M_r > \frac{9}{5}r$, so

$$D_q = \frac{N-q}{M_q} > \frac{9r(N-q)}{5\delta^2} > \frac{9qr^2c}{5\delta^2} > \frac{36}{5}qc, \tag{ED1}$$

using $\delta < r/2$. If $t < r$ then $|t - q| < r$; from $D_q \mid rc$, $D_q \mid |t - q|$ and $\gcd(t, m_0) = 1$, D_q cannot contain r, so $D_q \mid c$, contradicting (ED1). Hence failure forces $r < t$ (equality is excluded by parity). By (ED1), $D_q > c$, so every divisor $D_q \mid rc$ must contain r, i.e. $r \mid D_q \mid (t - q)$. [BIB, thm:v75-exactd-pair-outside] □

Corollary 7.37 (First canonical gap filter)**.** ***PROVED***. *Writing $G = p_0 - p_-$ (so canonicity gives $h < G$, hence $t < 2G$), every Exact-D pair failure satisfies*

$$p_0 < 2G + \frac{5(2G-q)\delta^2}{9r} < 2G + \frac{5}{9}G^2,$$

and in particular

$$\boxed{9p_0 \ge 18G + 5G^2 \implies M_q M_r > \delta^2 \quad \textit{for all SMOOTH pairs.}}$$

Theorem 7.38 (Joint D_q, D_r rigidity, `B-JOINT-D`)**.** ***PROVED***. *If $s^a \| m_0$ and $\alpha = v_s(D_q)$, $\beta = v_s(D_r)$, then*

$$\boxed{\alpha + \beta \le a + \min\{a,\ v_s(r-q)\},}$$

and globally

$$\boxed{D_q D_r \;\Big|\; m_0 \gcd(m_0, r-q) = m_0 \gcd(c, r-q).} \tag{Joint-D}$$

Proof. $\max(\alpha, \beta) \le a$, while $s^{\min(\alpha,\beta)} \mid \gcd(D_q, D_r) \mid \gcd(N-q, N-r) \mid r - q$. Adding exponents gives the first bound; the primewise product gives (Joint-D). Since neither q nor r divides $r - q$, the common part $\gcd(m_0, r-q)$ comes from c. [BIB, thm:v75-joint-D] □

Theorem 7.39 (Primitive residual normal form, `B-PNF`)**.** ***PROVED***. *Assume the regime (ED-res) and write $t - q = rk$, $m_0 = qrc$ with k positive and odd; put $g := \gcd(c, k)$, $c = gu$, $k = gv$, $\gcd(u, v) = 1$. Then exactly*

$$\boxed{\begin{array}{rlr} t = q + rgv, & m_0 = qrgu, & D_q = rg, \\ N - q = rg(2qu + v), & M_q = 2qu + v, & p_0 = q + rg(qu + v), \\ h = \frac{q+rgv}{2}, & D_r = \gcd\big(qgu,\ q + (gv-1)r\big), & M_r = \frac{2qrgu + q + (gv-1)r}{D_r}. \end{array}} \tag{PNF}$$

The formulas remain correct when q or r occur to higher powers in m_0; the extra copies are contained in $c = gu$.

Proof. The key identity is $D_q = \gcd(rc, rk) = r\gcd(c, k) = rg$; the rest follows by substituting $m_0 = qrgu$ and $t = q + rgv$ into $N = 2m_0 + t$ and $p_0 = m_0 + t$. For D_r use $D_r = \gcd(B_r, t - r) = \gcd(qgu, q + (gv-1)r)$. Since $\gcd(t, m_0) = 1$ we have $q \nmid gv$. [BIB, thm:v75-primitive-normal] □

Theorem 7.40 (High-gap amplification, `B-HIGH-GAP`). **PROVED** *under the Exact-D pair-failure hypothesis. With $d := v_2(r-q) \geq 1$, every Exact-D pair failure in primitive normal form satisfies*

$$\boxed{r > 2^{2d+1}(2qu+v) > 2^{2d+2}qu} \tag{HG-r}$$

and

$$\boxed{\frac{h^2}{m_0} > 4^d gv^2 + 2^{2d-1}\frac{gv^3}{qu} + \frac{v}{2u} > 4^d gv^2 \geq 4,} \tag{HG}$$

so in particular $\boxed{h^2 > 4m_0}$.

Proof. Put $A = 2qu + v$. From $D_r \leq B_r = qgu$ we get $M_r \geq (2 + \frac{gv-1}{qgu})r + \frac{1}{gu} > 2r$. Failure $M_q M_r \leq (r-q)^2/4^d$ then gives $2 \cdot 4^d Ar < r^2$, which is (HG-r). Moreover $\frac{h^2}{m_0} = \frac{rgv^2}{4qu} + \frac{v}{2u} + \frac{q}{4rgu}$; substituting $r > 2^{2d+1}(2qu+v)$ into the first term gives (HG). [BIB, thm:v75-high-gap] ☐

Warning 7.41 (Hypothesis, restated). Warning 7.5 applies verbatim: the hypothesis is *Exact-D pair failure*, not CRD failure. Using $h^2 > 4m_0$ under canonical failure alone is an error.

Corollary 7.42 (Quadratic canonical gap filter). **PROVED**. *With $\Lambda := 4^d gv^2 + 2^{2d-1}\frac{gv^3}{qu} + \frac{v}{2u}$ and $h < G$,*

$$\boxed{p_0 < 2G + \frac{G^2}{\Lambda} < 2G + \frac{G^2}{4^d gv^2} \leq 2G + \frac{G^2}{4}.}$$

Finite check of the quadratic filter. **VERIFIED COMPUTATION** *as reported by the source.* An exact check to $p_0 \leq 10^9$ found no canonical failure satisfying the filter $p_0 < 2G + G^2/4$; of the candidates only $p_0 = 11$ passed, and it admits no permissible pair of distinct roots. This is a finite verification, not a global proof. — *reported computational result, not reproduced in this document.*

7.5.1 The branch $k = 1$

For $k = 1$ one has $g = v = 1$, so $t = q + r$, $m_0 = qru$ with $u \geq 3$ odd, and exactly $D_q = r$, $M_q = 2qu + 1$, $D_r = q$, $M_r = 2ru + 1$.

Theorem 7.43 ($k = 1$ high-gap strengthening). **PROVED**. *For $k = 1$ every pair failure satisfies*

$$\boxed{r > 4^{d+1}qu^2, \qquad \frac{h^2}{m_0} > 4^d u,} \qquad \textit{hence} \qquad \boxed{p_0 < 2G + \frac{G^2}{4^d u} \leq 2G + \frac{G^2}{12}.}$$

The only non-automatic SMOOTH condition on the larger row is $\boxed{P^+(2ru+1) < qu}$.

Proof. Failure gives $(2qu+1)(2ru+1) \leq (r-q)^2/4^d < r^2/4^d$; the lower bounds on both factors give $r > 4^{d+1}qu^2$. Then $\frac{h^2}{m_0} = \frac{(q+r)^2}{4qru} > \frac{r}{4qu} > 4^d u$, and the filter for p_0 follows from $h < G$. The condition $u \geq 3$ is necessary: at $u = 1$ the factor $r = D_q = B_q$ breaks the strict SMOOTH threshold. [BIB, thm:v75-k1-gap] ☐

Theorem 7.44 ($k = 1$ quotient normal form). **PROVED**. *With $A := 2qu + 1$, $B := 2ru + 1$ one has exactly*

$$q = \frac{A-1}{2u}, \qquad r = \frac{B-1}{2u}, \qquad A \equiv B \equiv 3 \pmod 4,$$

$$\boxed{4um_0 = (A-1)(B-1)}, \quad \boxed{4uh = A + B - 2}, \quad \boxed{4ux = AB - 1}, \quad \boxed{4up_0 = AB + A + B - 3},$$

$$\boxed{up_0 + 1 = \frac{(A+1)(B+1)}{4} = (qu+1)(ru+1).}$$

Moreover $d = v_2(B-A) - 1$ *and* $\delta = \frac{B-A}{2^{d+1}u}$*; pair failure is equivalent to*

$$\boxed{4^{d+1}u^2 AB \leq (B-A)^2,}$$

SMOOTH of the larger row reads $\boxed{P^+(B) < \frac{A-1}{2}}$*, and canonicity reads*

$$\boxed{\left[\frac{AB-1}{4u}, \frac{AB+A+B-3}{4u}\right) \cap \mathbb{P} = \varnothing}$$

with prime endpoint $p_0 = \frac{AB+A+B-3}{4u}$. [BIB, thm:v75-k1-quotient]

Remark 7.45 (The continued-fraction relation, and why it is not a descent). With $a := qu+1 = \frac{A+1}{2}$ and $b := ru+1 = \frac{B+1}{2}$ one has $\boxed{ab - up_0 = 1}$, and the Euclidean algorithm gives explicitly

$$\boxed{\frac{a}{p_0} = [0; r, u, q], \qquad \frac{u}{b} = [0; r, u].}$$

The modular-inverse relation is therefore an exact relation between consecutive convergents — but by itself it supplies no descent. Compare the *Matched Euclidean Gluing Lemma* recorded as missing in §15.5.

Proposition 7.46 (Structured composite coverage in the prime-free interval). ***PROVED***. *For odd* s *with* $\frac{r-q}{2q} < s < \frac{r}{q}$*, the number* $z = r - sq$ *satisfies* $0 < z < h$ *and*

$$\boxed{p_0 - z = q(ru + s + 1).}$$

Moreover for every prime $\ell \mid B$, $\boxed{2up_0 \equiv qu - 1 \pmod{\ell}}$*, hence* $\boxed{\gcd(B, qu-1) = 1}$*. Each such* ℓ *covers exactly one residue class among the shifts* $p_0 - 2j$*, and under pair failure the number of its hits in* $0 < 2j < h$ *is at least* $4^d u$. [BIB, prop:v75-k1-coverage]

Open target 7.47 (Canonical $k = 1$ gap/covering barrier). **OPEN**. The canonical branch $k = 1$ has *not* been eliminated. The present apparatus produces a gap filter, a smooth-shift condition and a covering requirement, but no contradiction. This is the sharpest surviving remnant of the Exact-D programme and is a self-contained problem: exhibit a canonical $k = 1$ Exact-D pair failure, or prove none exists. [BIB, Canonical $k = 1$ Gap/Covering Barrier]

7.6 Sparse geometric core locks (FFS layer)

The factor-flow-system (FFS) layer of the Biblia proves that *collision compression must be paid for with geometry.* These results are stated for a finite factor-closed core C of a prime affine FFS; the Goldbach specialisations take $\delta = 2$ because all active primes are odd. Statuses are the source's own: `tagN` = source claim "new, proved" and `tagG` = Goldbach specialisation, both imported as **PROVED – PENDING RED TEAM**.

Definition 7.48 (Collision multiplexing ratio). For a finite factor-closed core C let $\eta(C) = \Omega(J_C) > 0$, let $\chi(C)$ be the minimal number of distinct parent-pairs in a collision witness forest, and put

$$\boxed{\kappa_{\text{col}}(C) := \frac{\eta(C)}{\chi(C)} \geq 1.}$$

$\kappa_{\text{col}} = 1$ means no multiplexing: each unit of collision excess needs its own pair. A value above 1 means some witness pairs carry several distinct prime labels at once.

Theorem 7.49 (Multiplex collision rigidity, `B-MULTIPLEX-RIGID`). **PROVED – PENDING RED TEAM**. *With $m = \min C$, $D = \max C - \min C$, $\eta = \Omega(J_C) > 0$ and $\chi = \chi(C)$,*

$$\boxed{D \;\geq\; \delta\, m^{\eta/\chi} = \delta\, m^{\kappa_{\mathrm{col}}(C)}.}$$

Proof. Closure puts every child in C, so every prime label occurring in J_C is at least m and $J_C \geq m^\eta$. Choose a collision witness forest realising χ; lattice collision divisibility bounds each of its χ aggregated labels by D/δ, and the product of the aggregated labels is exactly J_C, so $J_C \leq (D/\delta)^\chi$. Combine and take χ-th roots. [BIB, Multiplex Collision Rigidity] □

Reading. If many independent collision-excess labels are packed into few parent-pairs, those pairs must be extremely far apart: at $\kappa_{\mathrm{col}} = 2$ the diameter grows at least quadratically in the minimal atom, at 3 at least cubically. Topological excess is converted into a power lower bound for the diameter.

Corollary 7.50 (Goldbach collision-compression barrier). **PROVED – PENDING RED TEAM**. *Let C be a terminal bad SCC of the graph for even N that is not a pure support cycle, $m = \min C$, $D = \max C - \min C$. Then $\boxed{D \geq 2m^{\eta/\chi}}$, and since $\max C < N/3$,*

$$\boxed{m < \left(\frac{N}{6}\right)^{\chi/\eta}.}$$

In particular for every integer $t \geq 2$: $\boxed{\eta \geq t\chi \Rightarrow m < (N/6)^{1/t}}$; equivalently a hypothetical core with $m > N^{1/t}$ must satisfy $\boxed{\eta < t\chi}$.

Proof. All active primes are odd, so $\delta = 2$; apply Theorem 7.49 and $\max C < N/3$ (Theorem 5.2). □

Reading. This converts a structural question into a very concrete one: instead of killing all large cores directly, it suffices to exhibit *triple* collision multiplexing in the relevant class. It does not solve Goldbach — χ may be close to η, recovering only a linear diameter barrier.

Corollary 7.51 (Sparse Geometric Core Lock, `B-SPARSE-CORE-LOCK`). **PROVED – PENDING RED TEAM**. *Under the same hypotheses, for every collision witness forest with χ_τ distinct parent-pairs,*

$$\boxed{N-1 \;\leq\; U_C J_C \;\leq\; U_C\left(\frac{D}{2}\right)^{\chi_\tau},} \qquad \textit{hence} \qquad \boxed{D \;\geq\; 2\left(\frac{N-1}{U_C}\right)^{1/\chi_\tau}.}$$

Proof. For a terminal bad SCC the unit-slope product lock gives $T_C = U_C J_C \geq N - 1$, and lattice collision divisibility with $\delta = 2$ gives $J_C \leq (D/2)^{\chi_\tau}$. [BIB, Sparse Geometric Core Lock] □

Remark 7.52. Unlike the older bound $U_C G_C \geq N - 1$, the right-hand side here uses only the *sparse* family of parent-pairs needed to realise the collision fibres. It is a geometric version of exact-J, not a product over all pairwise gcds of rows.

Theorem 7.53 (Quantitative support explosion). **PROVED – PENDING RED TEAM**. *Let S be a finite subsystem of an exact prime affine FFS and $K_\tau(S)$ any collision witness product. Put*

$$\mathcal{M}(S) = P_{\mathrm{src}}(S)\frac{\prod_{p\in S} R(p)}{\prod_{p\in S} p}, \qquad B_S = \max_{p\in S} R(p).$$

If $A_S := \mathcal{M}(S)/(U_S K_\tau(S)) > 1$ then the number of distinct boundary atoms satisfies

$$\boxed{|\partial S| \;\geq\; \left\lceil \frac{\log A_S}{\log B_S} \right\rceil,}$$

and more generally $\mathcal{M}(S) > U_S K_\tau(S) B_S^r \Rightarrow |\partial S| \geq r + 1$.

Proof. The geometric escape criterion gives $R_\partial(S) \geq A_S$. Every boundary atom divides some $R(p)$, hence is at most B_S, and R_∂ is a product of $|\partial S|$ distinct primes, so $R_\partial(S) \leq B_S^{|\partial S|}$. Take logarithms and use integrality. [BIB, Quantitative Support Explosion] □

Reading. The escape criterion is not merely "$R_\partial > 1$"; the same computation can force *many* new boundary atoms. Together with Corollary 7.51 the FFS layer therefore has two independent elimination mechanisms, one for existence of escape and one for its multiplicity.

Chapter 8

Canonical Root Dynamics (CRD)

8.1 Epistemic convention

CRD is a *reformulation*, not a mechanism. Its content is:

1. the failed world is a closed bad set with a unique source (Lemma 3.26);
2. the union of all closed bad sets below $N/3$ is computable by a decreasing filtration (Definition 3.28);
3. canonical failure is exactly membership of $\mathrm{PF}(N - p_0)$ in that fixed point (Theorem 3.29).

Anything beyond that — capacity arguments, container arguments, spectral arguments — belongs to the successor theories of §12.2 and is **KILLED**. When reading older CRD material, keep this boundary in mind: the three items above survived every audit; the rest did not.

8.2 The architecture wall

Theorem 8.1 (Finite Affine Canonical Programmability)**.** ***PROVED*** *in [PCRD] (proof not reproduced here), class A as a statement about canonicity. For any finite family of affine forms $L_j(P) = a_jP + b_j$ and any prescribed finite valuation and compositeness conditions, under the standard coprimality conditions there exist — by the Chinese remainder theorem and Dirichlet's theorem — infinitely many primes P realising the whole pattern. Setting $N = 2(P - 1)$ gives canonical $p_0 = P$, $h = 1$.*

Theorem 8.2 (Partial BAD-transcript programmability; scope-corrected)**.** ***SCOPE-CORRECTED***. *Compatible prescribed finite BAD edges and edge valuations can be programmed in canonical $h = 1$ worlds by CRT/Dirichlet methods, as formalised by `CRT-TRANSCRIPT-EXACT-SCOPE-20260905`. The theorem controls the selected requirements only. It does* not *assert that every unprescribed prime factor of an expanded row is BAD, nor that a complete BFS neighbourhood of a prescribed depth is BAD. The earlier blanket reading of FBTU is withdrawn.*

Corollary 8.3 (Partial-transcript no-go; not a complete-BFS wall)**.** ***SCOPE-CORRECTED***. *A contradiction cannot follow merely from the existence of a finite compatible* selected *divisibility/valuation transcript, because such a transcript may be realised in canonical worlds with extra cofactors. This corollary does* ***not*** *rule out a theorem controlling a complete bounded-depth BFS frontier. The exact $N = 92$ certificate in §10.2 shows why the quantifiers differ: a selected BAD cycle coexists with a depth-1 GOOD child.*

Remark 8.4 (Link to the phase diagram). All $h = 1$ worlds with $G \geq 4$ lie in phase Type I (Theorem 6.12), hence have m_0 composite and $\rho_0 = 0$. The whole $h = 1$ programme of [PCRD] therefore lives *inside a single phase* — the one in which the reflection gate gives no guarantee. Two reported witnesses, verified here:

N	p_0	G	m_0	type, depth
103 344	51 673	14	$51\,671 = 163 \cdot 317$	Type I, $d = 4$
364 844	182 423	6	$182\,421 = 3^2 \cdot 20\,269$	Type I, $d = 6$

VERIFIED COMPUTATION.

8.3 C0a — the small-gap canonical obligation

Theorem 8.5 (Small-gap residual closure, `C0a`). ***PROVED***, *class C.* Hypotheses: *canonical failure, $h \le 2$. Then $h \in \{1, 2\}$, m_0 is odd composite, $Q \ne \varnothing$, and*

$$\boxed{\max\{U_R, J_R, B_{\mathrm{abs}}\} > \left[\left(\tfrac{5}{3}\right)^{|Q|} \operatorname{rad}(m_0)\right]^{1/3} > 1.}$$

[CORE, C0a-RESIDUAL-CLOSURE]

Reading. For $h \le 2$ the reflected block is empty, so boundary geometry gives nothing; the theorem replaces it by a nonzero obligation on the row side. *Limitation:* it yields neither a path to GOOD nor a capacity contradiction.

8.4 C0b — the exact canonical reflection-defect ledger

Theorem 8.6 (`C0b-RD`). ***PROVED***, *class $A + C$.* Hypotheses: *canonical failure, $N \ge 50$, $h \ge 3$. With $t = \lfloor (h-1)/2 \rfloor$ and*

$$\mathcal{I} = \{j : \mathrm{PF}(c_j) \cap R \ne \varnothing,\ a_j \in \mathcal{A}_N\}, \qquad \mathcal{D} = \{j : \mathrm{PF}(c_j) \cap R \ne \varnothing,\ a_j \notin \mathcal{A}_N\},$$
$$\mathcal{S} = \{j : \mathrm{PF}(c_j) \cap R = \varnothing,\ P^+(c_j) < h\}, \quad \mathcal{F} = \{j : \mathrm{PF}(c_j) \cap R = \varnothing,\ P^+(c_j) \ge h\},$$

one has

$$\boxed{\mathcal{J}_h = \mathcal{I} \uplus \mathcal{D} \uplus \mathcal{S} \uplus \mathcal{F}}$$

and for $j \in \mathcal{F}$ the number $\ell_j = P^+(c_j)$ is active, lies outside R, and occurs in exactly one element of $\mathfrak{B}(p_0)$. Moreover, if $\mathcal{I} = \varnothing$ and $k = \lceil t/3 \rceil$, then at least one of

$$\sum_{j \in \mathcal{D}} \Omega(a_j) \ge 2k, \qquad \sum_{j \in \mathcal{S}} \Omega(c_j) > k \frac{\log x}{\log h}, \qquad \prod_{\ell \in \Lambda_{\mathcal{F}}} \ell \ge h^k$$

holds. With $Q_{\le} = \{q \in Q : q \le t+1\} \ne \varnothing$ and $j(q) \equiv h \pmod q$, $1 \le j(q) \le q-1$, one has $q \mid c_{j(q)}$, $j(q) \in \mathcal{I} \cup \mathcal{D}$ and $B_{\le} < h^{|\mathscr{I}_{\le}|}$; and if $Q_{\le} = \varnothing$ then $\mathfrak{F}_Q > (5(t+1)/3)^{|Q|}$. [CORE, C0b-RD]

Reading. This is the strongest *quantitative* obligation that canonical boundary geometry issues at all: every coordinate of the reflected block must land in one of the four classes. *Known limitation:* `C0b-RD` supplies neither a path to GOOD nor a capacity bound.

Scope note 8.7 (Why the hypothesis $Q_{\le} \ne \varnothing$ is needed). SCOPE-CORRECTED. In version v1 this condition was omitted; with $Q_{\le} = \varnothing$ the inequality would read $1 < 1$. The configuration is not rare: among canonical instances with $h \ge 3$ and $N < 2 \cdot 10^5$, 47 551 of 72 446 have $Q_{\le} = \varnothing$; the smallest case is $N = 50$, $p_0 = 29$, $h = 4$, $t = 1$, $m_0 = 21$, $Q = \{3, 7\}$. [CORE, C0b-RD audit repair] — *reported computational result, not reproduced in this document.*

8.5 The C0b branch targets: TI, TD, TS, TF

`COb-RD` says one of four classes must carry the obligation. The four branch targets are the statements "if the obligation lands in class X, then the dynamics must realise it". All four are **OPEN**. They are the entry point to the frontier; full frontier entries with obstructions and success criteria are in Chapter 9.

ID	Branch	Obligation to be discharged
`TI`	$\mathcal{I}$ (incident, active mirror)	Turn an active reflected row into realised production: the mirror a_j is an active parent of c_j, so it *could* feed the world; TI asks that it must feed it *measurably*.
`TD`	$\mathcal{D}$ (incident, inactive mirror)	The root-anchored dominance obligation. The mirror is not active, so the mass must be recovered from $\Omega(a_j)$; TD asks for a lower bound on realised production anchored at the root. **This is the smallest missing bridge** (§9.11).
`TS`	$\mathcal{S}$ (h-smooth mirror)	The coordinate is smooth, so its factors are small and lie in the low range; TS asks that smooth coordinates cannot all be absorbed.
`TF`	$\mathcal{F}$ (fresh large factor)	$\ell_j = P^+(c_j) \geq h$ is active and outside R; TF asks that a fresh external atom forces an escape or a coupling.
`TD-DENSITY`	density form of TD	**SUPERSEDED**: replaced by the root-anchored form after the density version was found to require a bound of the same strength as its conclusion.

Scope note 8.8 (TF is local, and locality is not enough). **SCOPE-CORRECTED**. "A local $\mathcal{F}$ certificate together with $h < G$ forces routing" is false: at $N = 128$, $\ell = 13$ the certificate holds, but p_0 wins and $R = \varnothing$. `FAIL-TF-LOCAL-SCOPE-001`

Chapter 9

Current Proof Frontier

This is the only chapter where new work should start. Everything before it is either settled or dead. Each target below carries seven fields: exact statement, why it matters, what is already proved, the exact remaining obligation, known obstructions, what would count as success, and the failure-log entries that constrain it.
Before reading further, internalise the shape of the problem. The chain is

$$\underbrace{\texttt{C0a}}_{\text{PROVED}} \Rightarrow \underbrace{\texttt{C0b-RD}}_{\text{PROVED}} \Rightarrow \underbrace{\{\texttt{TI},\texttt{TD},\texttt{TS},\texttt{TF}\}}_{\text{OPEN}} \Rightarrow \underbrace{\texttt{P1}^{\mathrm{br}}}_{\text{OPEN}} \Rightarrow \underbrace{\{\texttt{L1},\texttt{E1},\texttt{W1},\texttt{K1}\}}_{\text{OPEN}} \Rightarrow \underbrace{\texttt{X1}}_{\text{OPEN}}.$$

The chain does *not* end at Goldbach even if every link is proved: it ends at "the canonical traversal succeeds", which is strictly stronger than Goldbach for each N but is not implied by it. The architecture wall (§8.2) says the chain cannot be closed by finite local data alone, so at least one link must import global input.

Read §9.2 first. That chain is the historical frontier. The project's *current* frontier is the canonical threshold programme of v4.79–v4.81 (§9.2), whose target is deliberately weaker than X1 — an asymptotic $N_0^{p_0}$ rather than all N — and which has already converted two of its own demanded outputs into proved impossibilities. The two live targets are named there. The older chain is retained because its statements are correct and its obstructions are real, and because `C0b-RD` still feeds it.

9.1 The exact Goldbach boundary

What is proved. For every even N in the non-diagonal case: the canonical root is well defined and unique; if it fails, the failed world is closed, all-BAD, has p_0 as its unique source, and lies below $N/3$; the reflected boundary is defect-free and its ledger has exactly four classes; and the row side satisfies `C0a` when $h \le 2$ and `C0b-RD` when $h \ge 3$.
What would imply Goldbach. If for every N the canonical traversal succeeded, then in particular $\mathrm{Good}_N \neq \varnothing$ for every N, which is binary Goldbach. So a proof of universal canonical success is *sufficient*.
What does not follow. Goldbach does *not* imply universal canonical success; the six N of §6.6 have $B_N^{\mathrm{low}} \neq \varnothing$ while satisfying Goldbach comfortably, and 301 non-canonical upper roots fail while their N satisfy Goldbach. Every statement conditioned on $\mathrm{Good}_N = \varnothing$ is class D and is unusable under canonical failure alone. `FAIL-UPPERHALF-SCOPE-001`
The current smallest missing bridge. The root-anchored `TD` obligation feeding $\texttt{P1}^{\mathrm{br}}$. Everything upstream is proved; everything downstream is blocked on it or on the same defect in

another guise.

9.2 The canonical threshold programme (post-v4.79/4.80/4.81)

This is the project's actual current frontier, and it is newer than the TI/TD/TS/T-F/P1BR/K1/X1 chain of §9.3–§9.9. The older chain is not wrong; it is simply not where the work is. A model starting fresh should read this section first.
The target here is deliberately weaker than X1 and therefore more attackable:

$$\boxed{\exists\, N_0^{p_0} < \infty \quad \forall \text{ even } N \geq N_0^{p_0} : \quad p_0(N) \text{ succeeds.}}$$

The verdict after v4.81 is: *yes in architecture, not in closure*. Two of the four demanded outputs of the previous frontier were *proved impossible*; exactly three routes remain, and the preferred one is named.

9.2.1 The v4.79/v4.80 engine

Throughout: $N = 2x$, $p_0 = x + h$, $m = N - p_0$, $Q = \mathrm{PF}(m)$, $R = R_N(p_0)$, canonical failure, $N \geq 50$.

Proposition 9.1 (Nagura window). **PROVED**. *For $N \geq 50$, Nagura's explicit interval theorem together with canonical minimality gives*

$$\boxed{p_0 < \frac{3N}{5}, \qquad m > \frac{2N}{5}.}$$

[CORE, v4.79 (V79.2); Nagura [1]]

Reading. This is why the two-sheet argument (Theorem 5.34) may use $m_0 > 2N/5$; it is the only place an external prime-counting input enters the canonical layer, and it is completely explicit.

Theorem 9.2 (Root-row vertical compensation and weighted residual floor). **PROVED**. *The root row itself contributes $V_0 := m/\operatorname{rad}(m)$ to the canonical vertical defect U_R. Combined with full-Q residual production (Theorem 5.13) and the rooted residual–absorption split,*

$$\boxed{U_R^2 J_R B_{\mathrm{abs}} > m\left(\frac{5}{3}\right)^{|Q|} > \frac{2N}{3}.} \tag{V79.4}$$

[CORE, vNEXT4.79]

Theorem 9.3 (First-front coefficient dichotomy). **PROVED**. *With the exact first-front coefficient defect*

$$F_Q^{\mathrm{coeff}} = \frac{m}{\operatorname{rad}(m)} \prod_{\substack{q \mid m \\ \kappa_R(q) \geq 2}} q,$$

one has the sharp dichotomy

$$\boxed{F_Q^{\mathrm{coeff}} = 1 \iff m \text{ squarefree and } \kappa_R(q) = 1 \ (q \mid m),}$$

while

$$\boxed{F_Q^{\mathrm{coeff}} > 1 \implies T_R > \tfrac{5}{2}N,}$$

which forces a half-power U/J channel. Nonprimitive first-front failure is therefore already coefficient-heavy. [CORE, vNEXT4.79–4.80]

Definition 9.4 (Primitive branch)**.** The canonical failed world is *primitive* when $F_Q^{\mathrm{coeff}} = 1$, i.e. m is squarefree and every first-front label has load one. Otherwise it is *nonprimitive*.

Theorem 9.5 (Primitive load-one and selected first-front production)**.** **PROVED**. *On the primitive branch the load-one condition is exactly*

$$\boxed{D_q = \gcd(m, N-q) = 1 \qquad (q \in Q),}$$

so the selected first-front product is exactly

$$\boxed{U_F(J_F B_{\mathrm{re}})B_{\mathrm{abs}} = \prod_{q\in Q}(N-q).}$$

With the adaptive least non-dividing prime $\lambda_N = \min\{\ell \in \mathbb{P} : \ell \geq 3,\ \ell \nmid N\}$ *one has* $q < N/(2\lambda_N)$ *for every* $q \in Q$*, hence*

$$\boxed{\prod_{q\in Q}(N-q) > \Big(1 - \frac{1}{2\lambda_N}\Big)^{|Q|} N^{|Q|} \;\geq\; \frac{25}{36}N^2.}$$

The primitive selected production therefore has leading logarithmic coefficient $|Q| \geq 2$*, not merely one.* [CORE, vNEXT4.80]

Theorem 9.6 (Owner-preserving quadratic *UJD* charge, `C-P0-PRIMITIVE-OWNERPRESERVING-QUADRATIC-UJD-CHARGE`)**.** **PROVED**. *The adaptive high range forms a unique-parent acyclic forest. Each absorbed high atom is replaced by a selected terminal descendant whose support lies in* $C_{\lambda_N} \setminus Q$*; low absorbed support and selected terminal support are disjoint, so vertical, horizontal and denominator support can each be single-charged. The resulting owner-preserving selected divisor* $\mathscr{C}_Q$ *satisfies*

$$\boxed{\mathscr{C}_Q \;\Big|\; U_R J_R D_{\lambda_N}.}$$

[CORE, vNEXT4.80]

> **Consequences recorded by v4.80.** `C-P0-PRIMITIVE-BASEABS-ESCAPE-OR-REUSE-CHARGE` is **SUPERSEDED**: no external-parent escape/reuse theorem is needed to charge primitive base absorption. `C-P0-PRIMITIVE-FIRSTFRONT-UNITMARGIN-PRODUCTION` ($U_F(J_F B_{\mathrm{re}})B_{\mathrm{abs}} > \frac{10N}{9}$, **PROVED** in v4.79) is *retained* but superseded *as the binding primitive floor*. `C-P0-PRIMITIVE-SELECTEDFAMILY-CAPACITY` is refined to `C-P0-PRIMITIVE-SELECTED-UJD-CAPACITY`, which remains **OPEN**. `G-CONDITIONAL-EFFECTIVE-P0-N0` is **PROVED** *only as a conditional theorem*: no numerical $N_0^{p_0}$, no universal p_0-success theorem and no Goldbach proof is claimed.

9.2.2 The v4.81 container theorem and what it kills

The v4.80 preferred frontier `C-P0-PRIMITIVE-FIRSTFRONT-REFLECTION-DISJOINT-CAPACITY` asked for one of four outputs: (1) reflected production disjoint from $\mathscr{C}_Q$ at fixed positive logarithmic scale; (2) same-label overlap forcing an additional exact J/U layer; (3) a GOOD event; (4) strict scale descent. v4.81 proves that **(1) and (2) cannot exist**, by exact identity rather than heuristic.

Theorem 9.7 (Exact rooted container identity)**.** **PROVED**. *On the primitive canonical failed branch,*

$$\boxed{U_R J_R D_{\lambda_N} = \frac{\prod_{u\in R}(N-u)}{m\prod_{r\in H_\lambda} r}.} \tag{V81.1}$$

So the "container" of the v4.80 selected charge is, up to an explicit denominator, the row product of the world itself. [CRD v4.81, Thm. container]

Theorem 9.8 (Container slack for $\mathscr{C}_Q$). **PROVED**. *For every $N \geq 500$,*

$$\boxed{\frac{U_R J_R D_{\lambda_N}}{\mathscr{C}_Q} = \frac{\prod_{r\in C_\lambda\setminus Q}(N-r)\ \prod_{r\in H_\lambda\setminus Z}(N-r)}{\prod_{r\in H_\lambda\setminus A_H} r} > \frac{5N}{6}.} \tag{V81.2}$$

The selected charge therefore never *approaches its container: the gap is at least one full logarithmic unit, and grows by one further unit for every low non-front state.* [CRD v4.81, Thm. slackCQ]

Corollary 9.9 (Death of the size route). **PROVED**. *Two consequences close the archimedean size route completely.*

1. Trivially*: a divisor never exceeds what it divides. Once $\mathscr{C}_Q \mid U_R J_R D_{\lambda_N}$ is a theorem (Theorem 9.6), no lower bound for $\mathscr{C}_Q$ can contradict any true upper capacity for the container.*
2. Quantitatively*: any adjoined production P with $\mathscr{C}_Q P \mid U_R J_R D_{\lambda_N}$ is priced exactly by (V81.2) — it must supply one factor $\geq \frac{5}{6}N$ for every low non-front state before it can begin to bite.*

Hence "new selected storage disjoint from $\mathscr{C}_Q$ at fixed positive logarithmic scale" — the frontier's own demanded output (1) — is insufficient by construction, *not merely unproved.*

Theorem 9.10 (Exact overlap normal form: overlaps are charge-free). **PROVED**. *If a prime ℓ divides a selected row $N-u$ ($u \in Q\cup Z$) and a strict reflected carrier $c_j = p_0 - 2j$, then the entire content of that coincidence is*

$$\boxed{\ell \mid a_j - u, \qquad 0 < a_j - u < x,} \tag{V81.3}$$

which is equivalent *to the already-counted divisibility $\ell \mid N-u$ together with the residue condition $2j \equiv p_0 \pmod{\ell}$. Because $a_j \notin R$, the coincidence creates no rooted parent, no κ_R increment, no J-layer and no U-layer.*

Corollary 9.11. *Alternative (2) of the v4.80 frontier is* empty*.*

Reading. This is the sharpened, final form of the reflection-gate warning: not only does the prime-free boundary fail to *activate* a reflected row (Warning 3.16, $N = 242$), an overlap between the reflected block and the selected family carries *no charge at all.*

9.2.3 New positive structure surviving the attack

All of the following is owner-preserving and reconstructible from the rooted CRD definitions.

Theorem 9.12 (C1: gap-scale shadow normal form). **PROVED**. *$D_q = \gcd(m, 2h-q)$, and primitive load-one is* equivalent *to*

$$\boxed{\gcd(m,\ 2h-u) = 1 \qquad (u \in R \setminus \{p_0\}).} \tag{V81.11}$$

This puts the first front and the reflected block on a single scale $\delta = 2h$.

Remark 9.13. This is the Prime-Gap Shadow Identity (Theorem 5.22) re-proved from the v4.80 definitions and used for the first time in the v4.79/v4.80 chain. The v4.81 salvage audit records it as *the one genuinely missing tool* the older corpus contributed to the current frontier.

Theorem 9.14 (C2: primitive residual world). **PROVED**. *$R' := R \setminus (\{p_0\} \cup Q)$ is child-closed, nonempty, satisfies $\mathrm{Src}(R') \subseteq \Gamma_N(Q)$ and $\min R' < \sqrt{N}$.*

Theorem 9.15 (C3: owner-preserving capacity factorisation). **PROVED**. *With $\Sigma' = \prod_{d\in\mathrm{Src}(R')} d$,*

$$\boxed{T_R\,\Sigma' = \Big(\prod_{q\in Q}(N-q)\Big) T_{R'}.}$$

Theorem 9.16 (C4: selected residue lock)**.** ***PROVED***.

$$\boxed{\mathscr{C}_Q B_H \equiv (-1)^{|Q|+|Z|}\, m \prod_{z\in Z} z \pmod N.} \tag{V81.13}$$

Theorem 9.17 (C5: adaptive fringe cofactor alphabet)**.** ***PROVED***. *Every edge into H_λ has cofactor a* prime *in* $[\lambda, 2\lambda)$ *coprime to N, and that prime is itself a low state. The alphabet is the full prime range $\mathcal{L}_N \subseteq [\lambda, 2\lambda)$.*

Warning 9.18. This is *not* the rejected two-letter claim $\{\lambda, 2\lambda-1\}$, which is **KILLED** as `FAIL-FRINGE-ALPHABET-TRANSF` Theorem 9.17 is precisely the "what survives" clause of that failure entry.

9.2.4 The minimal hostile survivor and the surviving routes

Theorem 9.19 (Product-comparison immunity)**.** ***PROVED***. *A primitive canonical failed world survives every* archimedean product comparison *between a single-charged selected family and the rooted $U/J/D$ container. The reason is now exactly quantified: a selected family can beat its container only if it leaves fewer than $|H_\lambda|$ rows unselected, whereas $\mathscr{C}_Q$ leaves at least $|C_\lambda \setminus Q| + |H_\lambda| - k$.*

Theorem 9.20 (Trichotomy of surviving routes)**.** ***PROVED***. *Any proof that the minimal hostile survivor class is empty must proceed through at least one of*

$$\boxed{\textbf{\textit{(A) population capacity}} \ \vee\ \textbf{\textit{(B) residue/congruence obstruction}} \ \vee\ \textbf{\textit{(C) a GOOD witness}}.}$$

(A) *an upper bound for $|C_\lambda|$ (or $|R|$) contradicting a lower bound for the same quantity.* This is the only archimedean route left, *because by (V81.1) every product container scales with $|R|$.*

(B) *an obstruction inside a container of* bounded *size. The only such containers in the corpus are the residue class modulo N (Theorem 9.16, Theorem 5.8) and the gap-scale divisors $h - j < h$ of* `COb-RD(3)`.

(C) *an actual reachable $g \in R$ with $N - g$ prime.*

Proof. Immediate from Theorem 9.19 together with the observation that every capacity object in the corpus (T_R, U_R, J_R, D_λ, the FC and LCAP envelopes) is a product over rows or labels of R, hence has logarithm $\Theta(|R| \log N)$, whereas any selected subfamily has logarithm at most $|\mathcal{S}| \log N$ with $\mathcal{S} \subseteq R$. □

`C-PO-PRIMITIVE-LOWCORE-SMOOTHROW-POPULATION-CAPACITY` — the current preferred target

Status. **OPEN**.
Statement. Primitive canonical failed world, $N \geq 500$, with the partition $R = \{p_0\} \uplus C_\lambda \uplus H_\lambda$. By Theorem 9.17 exactly $|H_\lambda|$ rows carry a high factor, each of the form $L\ell$ with $L \in \mathcal{L}_N$. Hence

$$\boxed{\text{exactly } |R| - |H_\lambda| \text{ distinct integers } N - u \ (u \in R) \text{ are } C_\lambda\text{-smooth and lie in } [m, N),}$$

all of them $> (1 - \frac{1}{2\lambda})N$ except the root row $m > 2N/5$. Wanted: an effective upper bound

$$|C_\lambda| \ \leq\ \Psi^*(N, \lambda_N)$$

for the number of integers in the window $\big((1 - \frac{1}{2\lambda})N,\ N\big)$ all of whose prime factors lie in a prescribed set of $|C_\lambda|$ primes below $N/(2\lambda)$, contradicting the matching lower bound $|C_\lambda| \geq |R| - |H_\lambda| - 1$ forced by child-closure.
Why it matters. By Theorem 9.20 route (A) is the only archimedean route left; by Theorem 9.17 the high band is completely described, so the whole unknown is the low core; and by (V81.1) a bound on $|C_\lambda|$ is exactly what converts every existing product capacity into a finite quantity.

Already proved. The container identity (V81.1), the slack bound (V81.2), the fringe alphabet, the shadow normal form (V81.11), the residual world R', and the capacity factorisation.
Honest difficulty statement. This is an analytic question about S-smooth integers in a short interval with prescribed *sparse* support. $\Psi(x, y)$ technology does not apply directly, because the support set is not an initial segment of the primes.
Known obstruction. Evertse's S-unit bound is *vacuous* here: the equation $u + w = N$ with $u \in S$, w S-smooth, $S = R \setminus \{p_0\}$ has $|R| - 1$ solutions, while the bound is $3 \cdot 7^{2|S'|+3}$ with $|S'| \leq |S| + \omega(N)$ — exponential in the very quantity to be bounded. Recorded as `FAIL-SUNIT-COUNT-BOUNDS-THE-LOWCORE-001`.
Success criterion. The effective bound Ψ^* above, or a proof that no such bound can exist at this scale.

`C-P0-PRIMITIVE-SELECTED-RESIDUE-LOCK-CAPACITY` — **the secondary target**

Status. OPEN.
Statement. Produce a second, *independent* determination of $\mathscr{C}_Q \bmod N$ (or of $T_{R'} \bmod N$) from the reflection-complete boundary, incompatible with (V81.13) above a computable threshold.
Input. (V81.13), `G-ROOT-LOCK` (Theorem 5.8), the capacity factorisation (Theorem 9.15), and the gap-scale normal form (V81.11). Reflected carriers entering the exact lock are the natural source of a second determination.
Guard — read before attempting. A congruence that is a formal consequence of the *same* factorisation is vacuous. The check performed in Theorem 9.16 shows that combining (V81.12), (V81.13) and `G-ROOT-LOCK` reproduces only $m \equiv -p_0$. Any candidate must import an *independent* evaluation.
Why it is route (B). It lives inside a container of bounded size (the residue class modulo N), which is the only kind of container Theorem 9.20 leaves for a non-archimedean argument.

The nonprimitive branch. Separately alive but without a consumer. The coefficient lock gives $T_R > \frac{5}{2}N$; a large F_Q^{coeff} yields a superlinear amplifier, while cheap nonprimitive defect bases localise below a small power and into a Baker-accessible reduction. *These are reductions, not a capacity theorem*, and the same container degeneracy (Corollary 9.9) forbids the scalar route there too. The exact Baker requirement recorded in v4.81 is *not* met by the available effective machinery.

Status line, verbatim from the source. Binary Goldbach remains OPEN; universal canonical p_0-success remains OPEN; $N_0^{p_0}$ is **not proved**, neither effectively nor ineffectively; no status of `G-CONDITIONAL-EFFECTIVE-P0-N0` is changed.

9.2.5 Updated proof DAG

$p_0(N)$ fails, $\quad N \geq 500$

$\Downarrow$

$F_Q^{\mathrm{coeff}} > 1 \ \vee \ F_Q^{\mathrm{coeff}} = 1$

$\Downarrow$

$$\begin{cases} \textbf{nonprimitive: } T_R > \frac{5}{2}N, \text{ split at } N^{3/40} & \text{(no consumer; the scalar route is forbidden)} \\ \textbf{primitive: } m \text{ squarefree}, \ \kappa_R(q) = 1 & \end{cases}$$

$\Downarrow$ (primitive)

$D_q = 1 \iff \gcd(m, 2h - q) = 1 \iff \gcd(m, 2h - u) = 1 \ \ \forall u \in R \setminus \{p_0\}$ (V81.11)

$\Downarrow$

$$U_F(J_F B_{\mathrm{re}}) B_{\mathrm{abs}} = \prod_{q \in Q} (N - q) > \left(\tfrac{5N}{6}\right)^{|Q|}$$

$\Downarrow$

$\mathscr{C}_Q \mid U_R J_R D_{\lambda_N}$ and $\dfrac{U_R J_R D_{\lambda_N}}{\mathscr{C}_Q} > \dfrac{5N}{6}$ (V81.1, V81.2)

$\Downarrow$

archimedean product comparison is immune $\Longrightarrow$ routes (A), (B), (C) only

9.3 The C0b branch targets

TI — injective branch obligation

Statement. If the `C0b-RD` obligation is carried by the class $\mathcal{I}$ (incident coordinates with an *active* lower mirror), then the map $j \mapsto (a_j, \mathrm{PF}(c_j))$ realises production in the world: the active mirrors inject into distinct parent–child incidences, and the number of such incidences is bounded below by $|\mathcal{I}|$ up to an explicit constant.

Why it matters. $\mathcal{I}$ is the only class whose coordinates are already *active graph rows*. If the obligation lands there and TI holds, the boundary obligation converts directly into world structure with no further input. It is the cheapest of the four branches.

Already proved. The reflection gate (Lemma 3.15): for $j \in \mathcal{I}$ the edge $a_j \to \mathrm{PF}(c_j)$ exists and $\Gamma_N(\{a_j\}) = \mathrm{PF}(c_j)$. Strict non-reuse (Lemma 3.13): every prime $\ell \geq h$ occurs in at most one block element, so distinct $j \in \mathcal{I}$ with a large factor give distinct incidences.

Remaining obligation. Injectivity for the *small* factors, i.e. for primes $\ell < h$, where strict non-reuse gives nothing. Equivalently: bound $\#\{j \in \mathcal{I} : \mathrm{PF}(c_j) \subseteq [2, h)\}$ from above by a fraction of $|\mathcal{I}|$.

Obstructions. (i) The class $\mathcal{I}$ can be empty — $N = 242$ has no active reflected row at all (Warning 3.16); TI is then vacuous and the obligation moves to $\mathcal{D}$, $\mathcal{S}$, $\mathcal{F}$. (ii) The guaranteed active mirror exists only in phase Type III (Corollary 6.14), so TI can only be the operative branch on $\theta > 1/2$. (iii) Small-factor collisions are exactly what the widened block would have controlled, and the widened block is refuted (Warning 3.14).

Success criterion. An unconditional inequality of the form $\#\{\text{distinct incidences from } \mathcal{I}\} \geq c\,|\mathcal{I}|$ with $c > 0$ absolute, valid in all phases where $\mathcal{I} \neq \varnothing$, and *stated in the same functional* that $\mathrm{P1}^{\mathrm{br}}$ consumes.

Constraining log entries. `FAIL-REFLECTION-ACTIVATION-001`, `FAIL-BROADBLOCK-ENDPOINT`, `FAIL-TF-LOCAL-SCOPE-001`.

TD — root-anchored dominance obligation

Statement. If the obligation is carried by $\mathcal{D}$ (incident coordinates whose lower mirror is *inactive*), then the recovered mass $\sum_{j\in\mathcal{D}} \Omega(a_j) \geq 2\lceil t/3 \rceil$ of `COb-RD` translates into a lower bound on *realised production in the world*, anchored at the root, measured by the same functional that bounds capacity.

Why it matters. This is **the smallest missing bridge in the whole project**. `COb-RD` is proved and hands over a genuine quantitative obligation; TD is the first step that turns a boundary count into a dynamical count. Every downstream target ($\mathrm{P1}^{\mathrm{br}}$, L1, W1, K1, X1) either depends on TD or fails for the same reason TD fails.

Already proved. `COb-RD` (Theorem 8.6) including the $\mathcal{D}$-branch inequality $\sum_{j\in\mathcal{D}} \Omega(a_j) \geq 2k$ under $\mathcal{I} = \varnothing$; the closed-world and unique-source structure (Lemma 3.26); no-bad-return (Theorem 5.33); the exact factorization $T_S = U_S J_S$ (Theorem 3.19).

Remaining obligation. Produce a lower bound

$$\text{(realised production of the world)} \;\geq\; \gamma(\text{obligation mass of } \mathcal{D})$$

in which γ is *the same* functional as the one appearing on the capacity side of $\mathrm{P1}^{\mathrm{br}}$ and K1. Not a comparable functional, not a functional that dominates it on average: the same one.

Obstructions. (i) a_j inactive means a_j is composite or divides N; the mass $\Omega(a_j)$ is therefore not attached to any graph vertex, and there is no proved route from it to an incidence. (ii) The density formulation `TD-DENSITY` is SUPERSEDED precisely because it needed a bound of the same strength as its own conclusion. (iii) The scope-corrected partial-transcript wall (Corollary 8.3) forbids inferring the bound from selected programmed edges alone; a theorem controlling the complete frontier is not excluded. (iv) Every recorded attempt to substitute a different but "morally equivalent" functional is in the log as a scope error.

Success criterion. A proof of the displayed inequality with an explicit γ, plus a verification that the γ used is literally the one in §9.4. A proof with a different functional is *not* partial progress; it is a new open problem.

Constraining log entries. `FAIL-ETA-GAMMA-IDENTIFY-001`, `FAIL-POTENTIAL-CAPACITY-001`, `FAIL-CAB-001`, `FAIL-CAB-INTERNAL-001`, `FAIL-BABS-PRODUCTION-001`.

TS — smooth branch obligation

Statement. If the obligation is carried by $\mathcal{S}$ (coordinates with $P^+(c_j) < h$ and $\mathrm{PF}(c_j) \cap R = \varnothing$), then the smooth mass $\sum_{j\in\mathcal{S}} \Omega(c_j) > k \log x / \log h$ cannot be entirely absorbed inside the world: some smooth coordinate must contribute an escape or a coupling.

Why it matters. $\mathcal{S}$ is the branch where the mass bound is largest (it grows like $\log x/\log h$ per unit), so if the obligation lands there the quantitative margin is greatest. It is the branch most likely to be provable by elementary means.

Already proved. The $\mathcal{S}$-branch inequality of `COb-RD`; minimum-factor escape (Corollary 5.25); shadow cycle localisation (Theorem 5.27), which confines a terminal core inside Q to $(0, 2h/3)$ — exactly the range in which smooth coordinates live.

Remaining obligation. Convert "c_j is h-smooth and its factors avoid R" into a statement about the world. The factors of c_j are active primes outside R; they are candidate escape targets, but their rows have not been shown to reach GOOD.

Obstructions. (i) A factor outside Q or outside R does not mean success — only that the current alphabet was left (warning after Corollary 5.25). (ii) Smoothness bounds *numbers*, not *vertices*; the passage from $\Omega(c_j)$ to distinct active vertices needs the same injectivity that TI lacks.

Success criterion. An unconditional statement: if $\mathcal{I} = \varnothing$ and the $\mathcal{S}$ inequality is the operative one, then $G_{\mathrm{esc}}(N) > 0$. That alone would be a real theorem, since it would give Goldbach for every N in that regime.

Constraining log entries. `FAIL-HALL-SUPPORT-001`, `FAIL-DIRECT-ADJACENCY-001`.

TF — fresh branch obligation

Statement. If the obligation is carried by $\mathcal{F}$ (coordinates with $\ell_j = P^+(c_j) \geq h$, active and outside R), then the product bound $\prod_{\ell \in \Lambda_{\mathcal{F}}} \ell \geq h^k$ forces an escape edge or an external coupling.
Why it matters. $\mathcal{F}$ is the only branch that hands over *named active primes outside the world.* Those are exactly the objects an escape argument needs.
Already proved. The $\mathcal{F}$-branch product inequality; that each ℓ_j is active, outside R, and occurs in exactly one block element (Theorem 8.6 combined with Lemma 3.13).
Remaining obligation. Show that an active prime outside R occurring as a large factor of a reflected coordinate must be, or must lead to, a GOOD vertex.
Obstructions. (i) **Locality is not enough**: at $N = 128$, $\ell = 13$, the local $\mathcal{F}$ certificate holds but the canonical root wins outright and $R = \varnothing$, so the certificate constrains nothing. `FAIL-TF-LOCAL-SCOPE-001` (ii) An external parent need not be a success root: $(6142, 877, 3511)$ has 3511 active and above p_0, yet its own traversal fails. `FAIL-EXTERNAL-SUCCESS-001` (iii) External reuse is not local: $(N, \ell) = (1\,825\,372, 52\,153)$ at distance $53\,750$. `FAIL-LOCAL-REUSE-001`
Success criterion. A theorem of the form: under canonical failure with $\mathcal{I} = \mathcal{D} = \mathcal{S} = \varnothing$, some $\ell \in \Lambda_{\mathcal{F}}$ has $N - \ell$ prime, or has an active parent outside R whose row meets Good_N. The second alternative must be included — it is provably necessary.
Constraining log entries. `FAIL-TF-LOCAL-SCOPE-001`, `FAIL-EXTERNAL-SUCCESS-001`, `FAIL-LOCAL-REUSE-001`.

9.4 P1$^{\mathrm{br}}$ — branchwise production upgrade

P1BR

Statement. Let γ be the capacity functional of the canonical world. P1$^{\mathrm{br}}$ asserts that the production realised branchwise — summed over the branches of the first front, each measured on its own subtree — admits a lower bound in γ that exceeds the capacity bound in the same γ:

$$\sum_{\text{branches } \beta} \gamma(\text{production}(\beta)) \;>\; \gamma(\mathcal{C}_N(R)).$$

Why it matters. P1$^{\mathrm{br}}$ is the node where the boundary obligations become a contradiction. Everything downstream (L1, E1, W1, K1, X1) consumes its conclusion.
Already proved. The branchwise decomposition itself is legitimate: the world is closed with a unique source, so the first front partitions it into branches. The exact ledger $T_S = U_S J_S$ holds on every branch separately. The capacity side $\mathcal{C}_N(R)$ is well defined.
Remaining obligation. A *positive external packet* feeding the *same* P1 capacity — the corpus target `C-POSITIVE-EXTERNAL-PACKET-TO-SAME-P1-CAPACITY`. Concretely: exhibit a quantity that (a) is provably positive under canonical failure, (b) is external to the world (so it is not already counted in $\mathcal{C}_N(R)$), and (c) is measured in γ rather than in a functional that merely correlates with γ.
Obstructions. (i) The status of P1$^{\mathrm{br}}$ was overridden several times in the corpus ([CORE, vNEXT4.25–4.33]) as successive candidate packets were found to be either non-positive or already internal. (ii) $\mathcal{C}_S < \mathcal{D}_S$ from the first front is **KILLED** by $N = 4736$ ($\mathcal{C}_S \approx 22.60 > \mathcal{D}_S \approx 20.16$). `FAIL-POTENTIAL-CAPACITY-001` (iii) Unconditional CAB is **KILLED** by $N = 322$ and $N = 858$. (iv) The naive support expansion of all sources is **KILLED** by Hall compression $456 \to 17$ at $N = 51\,032$. `FAIL-HALL-SUPPORT-001`
Success criterion. A packet satisfying (a), (b), (c) above, with the positivity proved and the functional identity checked symbol by symbol. Any proof that silently changes γ between the

production and capacity sides is invalid; that substitution is the single most repeated error in the log.
Constraining log entries. `FAIL-POTENTIAL-CAPACITY-001`, `FAIL-CAB-001`, `FAIL-CAB-INTERNAL-001`, `FAIL-HALL-SUPPORT-001`, `FAIL-ETA-GAMMA-IDENTIFY-001`, `FAIL-CHILD-C1-001`.

9.5 L1 — actual layer realisation

L1

Statement. The layer part J_R of the canonical world is realised by *actual* distinct layers of the internal filtration, i.e. the support reuse counted by J_R corresponds to genuinely distinct BFS levels rather than to repeated incidences inside one level.
Why it matters. Every capacity argument that uses J_R implicitly assumes this. Without L1, J_R is a bookkeeping quantity with no dynamical meaning, and the Euler-defect route collapses — as the computation in §5.7 shows it does under the naive normalisations.
Already proved. $T_S = U_S J_S$ exactly (Theorem 3.19); the world is closed with a unique source, so the BFS filtration is well defined and canonical.
Remaining obligation. A lower bound on the number of layers in terms of $\log J_R$, or a counterexample. Note that the two false normalisations (N1) and (N2) of §5.7 are exactly failed attempts at L1 in disguise.
Obstructions. The measured data go the wrong way: 3 804 of 21 093 canonical worlds violate the (N2) inequality, so support reuse is systematically *larger* than the layer count can absorb.
Success criterion. Either a proof of $\nu(R) \geq c \log J_R$ for some $c > 0$, or an explicit family showing $\nu(R) = O(1)$ with J_R unbounded (which would kill L1 and, with it, every capacity route that passes through J). The second is probably easier and is listed as an experiment in Chapter 17.
Constraining log entries. `FAIL-ETA-GAMMA-IDENTIFY-001` and the computational refutations of (N1), (N2) recorded in §5.7.

9.6 E1 — escape or coupling

E1

Statement. Under canonical failure, every internal base atom $\ell \in B_{\mathrm{abs,int}}$ either has a GOOD edge reachable inside one of its first-front branches, or has an active parent outside R. (This is the *escape-or-reuse* alternative promoted to a theorem.)
Why it matters. E1 is the only frontier target with overwhelming computational support and a completely explicit statement: 61 452 instances with 0 orphans, plus an independent red team over 60 413 atoms with 0 orphans. If any target is true, this one is.
Already proved. Retention–escape (Theorem 5.24); minimum-factor escape (Corollary 5.25); the prime-power witness (Proposition 5.29); $B_{\mathrm{abs,int}} = B_{\mathrm{abs}}$ without vacuity [CORE, ARS-BABS-INTERNAL].
Remaining obligation. The disjunction itself, unconditionally. Both alternatives are provably necessary: $(6142, 877, 3511)$ shows the second cannot be dropped.
Obstructions. (i) External reuse is not local (`FAIL-LOCAL-REUSE-001`), so the "active parent outside R" cannot be found by bounded search. (ii) The external parent need not itself be a success root (`FAIL-EXTERNAL-SUCCESS-001`). (iii) Escape need not be directly adjacent (`FAIL-DIRECT-ADJACENCY-001`: at $N = 14198$ the atoms 19 and 167 escape at distances 8 and 7).

(iv) Charging multiplicity is observed to be 1 but is *not* proved bounded, and the constant-$C = 1$ charging scheme is **KILLED** (53 violations, maximum 9.42243, `FAIL-CHILD-C1-001`).
Success criterion. A proof of the disjunction, or a bounded-charging lemma (multiplicity $\leq C$ for an absolute C) which the corpus identifies as the missing ingredient.
Constraining log entries. `FAIL-LOCAL-REUSE-001`, `FAIL-EXTERNAL-SUCCESS-001`, `FAIL-DIRECT-ADJACENCY-001`, `FAIL-CHILD-C1-001`.

9.7 W1 — simultaneous clean witnesses

`W1` — corrected

Statement. Several clean witnesses can be selected simultaneously at the *actual* layer scale of the world: a selection of parents, one per branch, simultaneously clean (no shared support, no double charging), covering the layers that really occur in R rather than only the polylogarithmic ones.
Correction to an earlier reading. **SCOPE-CORRECTED**. Earlier compressions listed `G-CLEAN-POLYLOG-PARENT-SELECTION` as W1's open target. It is *not* open: it is a **PROVED** theorem of the CRD Core (Theorem 5.15), giving exactly such a clean selection for every finite layer family with $\ell^k \leq (\log N)^A$, with global support load at most $\rho(\mathcal{L})$ — and load one when the base primes are distinct. See Warning 5.16.
Why it matters. Capacity arguments need many witnesses at once, and the selection problem is *solved* in the polylogarithmic range. What is missing is the range, not the selection.
Already proved. Theorem 5.15 (the selection itself); strict non-reuse (Lemma 3.13) for primes $\ell \geq h$; the global two-sheet collision budget (Theorem 5.34).
Remaining obligation. Extend the selection from $\ell^k \leq (\log N)^A$ to the actual layers of R. That is precisely the content of L1 (§9.5): **W1 is blocked on L1, not on a selection lemma.** If L1 is refuted — and §5.7 suggests it may be — then W1 is not merely open but misconceived, and the correct object is the polylogarithmic family alone.
Obstructions. Hall-type compression destroys naive selections outside the proved range: at $N = 51\,032$ the naive support expansion of all sources compresses $456 \to 17$ (`FAIL-HALL-SUPPORT-001`). This is not a technical nuisance — it says the bipartite incidence structure has small vertex covers, which is exactly what a simultaneous selection must avoid, and it is why Theorem 5.15 needs Siegel–Walfisz and a lower-bound sieve rather than a combinatorial argument.
Success criterion. Either a clean selection at the actual layer scale, or a proof that the polylogarithmic range is the maximum possible — which would close W1 negatively and remove it from the frontier.
Constraining log entries. `FAIL-HALL-SUPPORT-001`, `FAIL-CHILD-C1-001`.

9.8 K1 — source-sensitive capacity

`K1`

Statement.

$$\boxed{K_\times(R,\mathcal{P}) + K(\mathcal{P}) \;\leq\; \mathcal{C}_N(R)}$$

where $K_\times$ is the cross term between the world R and the selected parent set $\mathcal{P}$, $K(\mathcal{P})$ the internal term of $\mathcal{P}$, and $\mathcal{C}_N(R)$ the capacity of the world — all three measured in the *same* functional as $\mathrm{P1}^{\mathrm{br}}$.
Why it matters. K1 is the quantitative closure: with $\mathrm{P1}^{\mathrm{br}}$ above and K1 below, the two bounds cross and canonical failure is excluded.

Already proved. The capacity side $\mathcal{C}_N(R)$ is well defined and computable; the world's ledger identities (IL1–IL6) hold on Q_{smooth}.
Remaining obligation. *Exactly the same defect as TD.* K1 needs the cross term and the capacity to be expressed in one functional. Every recorded attempt produced two functionals with a comparison that holds on average and fails pointwise. The corpus's vertical-origin overrides ([CORE, vNEXT4.8–4.13]) end at the target `C-NATURAL-SCALE-WEIGHTED-CONSUMER`: a consumer of the packet weighted by a natural scale, which has never been produced.
Obstructions. (i) The same-functional obstruction, shared with TD and P1^{br} — these three are not independent problems. (ii) $N = 4736$ kills the first-front form of the capacity inequality. (iii) $N = 322$ and $N = 858$ kill the unconditional CAB form.
Success criterion. The displayed inequality with all three terms written in one explicit functional and proved unconditionally under canonical failure. Equivalently: solve TD and K1 together, since a solution to one in the correct functional very likely gives the other.
Constraining log entries. `FAIL-POTENTIAL-CAPACITY-001`, `FAIL-CAB-001`, `FAIL-CAB-INTERNAL-001`, `FAIL-ETA-GAMMA-IDENTIFY-001`.

9.9 X1 — canonical failure exclusion

X1

Statement. For every even N in the non-diagonal case, the canonical traversal from p_0 terminates in SUCCESS. Equivalently: $\mathrm{PF}(N - p_0) \not\subseteq B_N^{\text{low}}$ whenever $N - p_0$ is composite (Theorem 3.29).
Why it matters. X1 is the terminal node. It implies binary Goldbach. It is the central hypothesis of the whole project, written here as **(H-p_0)**.
Already proved. Nothing towards X1 itself. The fixed-point reformulation (Theorem 3.29) is exact but not progress.
Remaining obligation. Everything above, plus the passage from "the capacity inequality fails" to "the traversal succeeds".
Obstructions. The architecture wall (§8.2) — no finite local certificate can do it; the parity barrier (§12.4) — no purely sieve-theoretic route reaches a prime complement; and the fact that X1 is *strictly stronger* than Goldbach for each N, so any proof strategy that would also prove Goldbach must be at least as hard as Goldbach.
Success criterion. A proof. Nothing weaker counts; in particular no finite scan, however large, changes the status of X1.
Computational status. VERIFIED COMPUTATION: no canonical failure for even $N \le 10^7$ (this project, exhaustive). Independent Goldbach verification reaches $4 \cdot 10^{18}$ [12] but says nothing about X1.

9.10 RPD — repeated prime-power redundancy

RPD

Statement. The repeated external prime-power layer redundancy is bounded:

$$D_{\text{high}}(H_Q) \;\le\; C \log\log N \cdot \sum_{\text{layers}} (\kappa - 1)_+ \;+\; o(M),$$

equivalently $D_{\text{high}}(H_Q) \le O(\log\log N) \cdot \eta_{\text{high}} + o(M)$, where M is the total mass and η_{high} the high-range excess valuation.

Why it matters. If RPD is true then $\Delta_J \gg M$: the layer defect dominates the mass, which is exactly the quantitative separation that the capacity route needs. RPD is the one open target that would supply the *margin* rather than merely the structure.
Already proved. $T_S = U_S J_S$; the high/low decomposition of the valuation mass; the prime-power witness (Proposition 5.29).
Remaining obligation. The $\log\log N$ bound. The natural approach is a divisor-type bound on how often a fixed prime power can recur across layers, which is where $\log\log N$ would come from.
Obstructions. (i) The bound is asserted for the *external* redundancy; the internal part is already controlled and is not the difficulty. (ii) The $o(M)$ term hides the interaction between layers, and no proof technique in the corpus separates layers without losing a factor of the size that RPD is supposed to save.
Success criterion. A proof of the displayed inequality with an explicit C, or a family of N showing D_{high} can exceed $\eta_{\text{high}} \log\log N$ by an unbounded factor.
Provenance note. RPD is recorded in the corpus as a v4.45 open target inside a commented archive block of the CRD failure log; it is reproduced here because it is the only surviving open target that carries an explicit quantitative margin. Its exact original phrasing should be checked against the source before it is used as a hypothesis.

9.11 Open targets specific to p_0 alone

These are stated *without* reference to a Goldbach proof and each is a self-contained problem.

`C-SAMEFUNC-BRIDGE` — global same-functional margin bridge

Statement. There is a single functional γ in which both the production lower bound (TD, $\mathrm{P1}^{\mathrm{br}}$) and the capacity upper bound (K1) can be stated, with a positive margin between them.
Why it matters. This is the abstract form of the obstruction shared by TD, $\mathrm{P1}^{\mathrm{br}}$ and K1. Solving it solves all three; failing to solve it blocks all three. Anyone attacking one of them individually should first decide whether they are really attacking this.
Already proved. Nothing. Several *pairs* of functionals are known in which one side is provable; no pair is known in which both are.
Remaining obligation / obstruction / success criterion. As for TD; see §9.3.
Constraining log entries. `FAIL-ETA-GAMMA-IDENTIFY-001` and every entry filed under capacity.

`C-GENEALFREE-DYN` — boundary to dynamic escape

Statement. Does the absence of any genealogical boundary defect ($\kappa_{\text{gen}} = 0$) force the whole `C0b-RD` obligation into *actual* dynamical channels?
Why it matters. This is the single missing implication between Chapter 6 (exact boundary structure) and any dynamical statement. The entire "why p_0 works" question reduces to it.
Already proved. The boundary side completely (Theorems 6.2–6.6); the obligation side completely (Theorem 8.6). The implication between them: nothing.
Minimum acceptable form. A theorem whose conclusion *actually fails*, or acquires an explicit leakage term, when the defect rank is one. Anything that also holds at rank one is not progress — it is a class-B result in disguise.
Obstruction. Warning 6.7(2): the statement $\kappa_{\text{gen}}(r) = 0$ may be informationally equivalent to "r is the first upper prime". If so, no theorem can extract more from it than from the ordering of the primes, and the target should be restated.
Constraining log entries. `FAIL-P0-ZEROHOLE-ESSENTIAL-001`, `FAIL-P0-INFORMATION-SELECTOR-001`.

C-POP1-SEP — one-defect dynamic separation

Statement. Prove a dynamical difference between p_0 and p_1 that is caused by the single embedded channel of the earlier root, not by aggregate size.
Why it matters. This is the sharpest available test of C-GENEALFREE-DYN and the one with the best data.
Already proved. The categorical separation on the boundary (κ_{gen}: 0 versus 0.9041 mean, maximum 1) and the aggregate near-identity of every other observable (§6.5.2).
Remaining obligation. Isolate causality. The aggregate observables differ by 6–24%, which is consistent with pure size effects; the defect rank differs categorically but has negligible logarithmic mass.
Obstruction. FAIL-P0-ZEROHOLE-ESSENTIAL-001: the mass of the hole is $O(\log N)$, so any argument that routes the separation through mass is dead before it starts.
Success criterion. A statement true for p_0 and false for p_1 whose proof uses the hole *structurally* (as absence of a multiplicative carrier, [CORE, V71.44]) and not quantitatively.

9.12 Smaller open questions about p_0

Each of the following is a self-contained problem that does not mention Goldbach.

Q1. Does ρ_0 have dynamical consequences? **OPEN**. The signature $(\kappa^+, \kappa^-) = (0, \rho_0)$ is the first boundary invariant that varies nontrivially between canonical worlds. Does $\rho_0 \geq 1$ (Type III, where a guaranteed active reflected row exists) correlate with, or imply, anything about the traversal? Preliminary data: immediate success occurs in 18.4% of Type III cases and is *impossible* in Type I — but that is a consequence of the definition, not of dynamics.
Q2. Does the escape margin grow? **OPEN**. Is $G_{\mathrm{esc}}(N) \to \infty$? Data (§6.7): the per-dyadic-block minimum runs $1, 1, 1, 1, 1, 1, 2, 2, 3, 3, 3$ up to $2 \cdot 10^5$. A theorem "$G_{\mathrm{esc}}(N) \geq f(N)$ for increasing f" would be strictly stronger than the existence of a Goldbach pair and is, as far as is known, untouched.
Q3. Is there a missing state variable? **OPEN**. Four pairs of structural twins survived enrichment by all global-source information; they are separated only by the *internal witness signature* (localisation, nearest escape distance in the same branch, external-reuse flag) [F3]. Is there a scalar invariant, or is the system state intrinsically a profile?
Q4. A second canonical anchor? **OPEN**. Let $p^\star = \max\{p \in \mathbb{P} : p < N\}$ and $g = N - p^\star$. Then $g \leq N - p \leq m_0$ for upper sources, $p^\star - p_0 = m_0 - g$, and $\gcd(m_0, g) = \gcd(m_0, p^\star - p_0) = \gcd(g, p^\star - p_0)$ **PROVED** [BIB]. Does $p^\star$ have its own rigidity? Evidence against: $p^\star$ finances on average only 25.07% of the B_{abs} atoms against 81.44% for the whole ensemble of upper sources; priority assessed as low [D4, q. 11, 21].
Q5. Is p_0 exceptional outside this one graph? **OPEN**. An analogous selector appears in the Erdős–Straus model and in the Schinzel model [BIB]. Does "the least element above the midpoint, chosen unconditionally" have the same reflection-completeness property there? The definition of the reflected block transfers verbatim, so the question is well posed.
Q6. Diagonal closure. **OPEN** (Scope note 3.17). What is the right formulation of the genealogical layer covering $x \in \mathbb{P}$? Replacing $[x, r)$ by (x, r) makes the theorem unconditionally true, but then p_0 is no longer uniquely characterised. Is there a simultaneous formulation?
Q7. Invariants of the boundary filtration. **OPEN**. The filtration $\mathfrak{B}_0 \subset \mathfrak{B}_1 \subset \cdots$ (Theorem 6.8) with the coupling-matrix embedding (Corollary 6.9) is a purely combinatorial structure independent of Goldbach. Does the sequence $(M^{(k)})_k$ have invariants — rank, spectrum, zero pattern — that do not reduce to the gap sequence $g_1, g_2, \ldots$?
Q8. Prove the active-mass Euler bound. **OPEN**. Normalisation (N3) of §5.7, $\sum_q d_R(q) \geq |R| + 1$, held in all 21 093 canonical worlds tested. It is the only nontrivial surviving form; a proof would be small but real, and would repair the part of the corpus that used the false normalisations.

Chapter 10

Post-v4.81 Research Continuation: 5–7 September 2026

Binding override. This chapter integrates the lossless 2026-09-05–06B archive and the final 2026-09-07C checkpoint into the project bible. It does not promote novelty claims. Binary Goldbach and universal canonical p_0-reachability remain **OPEN**. The exact final raw checkpoint is preserved in Appendix B; later entries in that transcript override earlier frontier summaries when they conflict.

10.1 Merge provenance and source dominance

The merge used three supplied sources: the v1.0 master, the standalone lossless checkpoint archive through 2026-09-06B, and the final 2026-09-07C checkpoint. The archive's 5108-line raw payload is an *exact prefix* of the final 6420-line checkpoint, so duplicating both raw transcripts would add no information. The final checkpoint is therefore the canonical raw payload; the earlier archive contributes its provenance discipline and strict prefix genealogy.

Master source — 13,744 lines

`goldbach_master_for_future_ai(5)(1).tex`

`SHA-256: 685dd184 103a54d7 0d32d9f5 0b440520 9c287c39 c1a41288 8e4f717a 47fcb197`

Lossless archive — 5,346 lines

`goldbach_checkpoint_archive_2026-09-05_to_2026-09-06B(1).tex`

`SHA-256: d8a4cfe0 d9db56eb e7c4bf45 eed1becc 01f5cad6 1d8784e9 51a0661b 96ad2d54`

Final research checkpoint — 6,420 lines

`goldbach_research_checkpoint_2026-09-07(3).txt`

`SHA-256: 49d1ce74 1f76bec6 297e46c2 546c562c 7089e5b9 e0be31c2 a3d18e67 1a814392`

10.2 Binding FBTU scope correction

The audit `AUDIT-FBTU-PARTIAL-VERSUS-COMPLETE-20260905` is binding. Prescribed CRT constraints force selected edges; they do not force the entire factorisation of an expanded row. At

$$N = 92, \qquad p_0 = 47, \qquad 92 - 47 = 3^2 \cdot 5,$$

the selected walk $47 \to 5 \to 29 \to 7$ and the BAD cycle $5 \to 29 \to 7 \to 5$ are real, while $92 - 3 = 89$ is prime, so the complete BFS succeeds at depth 1. This kills the blanket implication "programmable selected BAD transcript $\Rightarrow$ programmable complete BAD neighbourhood". The repaired theorem `CRT-TRANSCRIPT-EXACT-SCOPE- 20260905` survives.

10.3 Closed-support counting, height and forced envelope escape

The September 5 continuation produced a coherent fixed-/bounded-support layer. For a fixed prime alphabet S of size s, coprime S-smooth solutions of a fixed affine equation are bounded by 2^{9s+16} via the verified Beukers–Schlickewei input. This yields both `COMPLETE-H1-TWO-ROW-FINITENESS-20260905` and `CLOSED-ALPHABET-COUNT-20260905`: one fixed finite alphabet cannot support infinitely many complete closed active BAD worlds.

The Matveev route was then repaired to remove the old branching/pure-row hypothesis. If C is any nonempty closed active BAD set, $s = |C|$ and $M = \max C$, then

$$\log \frac{N-M}{M} < 1.4\, 30^{s+3} s^{9/2} \prod_{q \in C} \log q \left(1 + \log \frac{\log N}{\log 3}\right).$$

Consequences retained as proved are the explicit bounded-core threshold and the pointwise lower growth

$$M \geq (1 - o(1)) \frac{(\log\log N)(\log\log\log N)}{\log\log\log\log N}$$

along any sequence carrying a closed active BAD set. These are necessary conditions, not contradictions: moving support remains possible.

In the canonical $h = 1$ family, bounded first support also forces complete-row escape from a fixed envelope: beyond the explicit threshold $F_2(y)$ every first child row has a prime factor $> y$, and the large factors arising from distinct first children are pairwise disjoint. The inference that this automatically iterates was explicitly killed by `ENVELOPE-ESCAPE-AUTOMATIC-ITERATION-20260905`.

10.4 Shallow asymptotics and same-N no-gos

The September 6/06B continuation establishes several asymptotic traversal constraints, all with novelty still unaudited externally.

1. `COMPLETE-H1-RADIUS-TWO-RARE-SUCCESS-20260906`: for almost every prime $P \in [X, 2X]$, the canonical $h = 1$ start has complete BADness through radius two; the exceptional relative proportion has an explicit decaying bound.
2. `SAME-N-GROWING-ROOTS-COMPLETE-RADIUS-TWO-20260906B`: the same complete radius-two BADness holds simultaneously for a growing prefix of nearby same-N roots. Thus shallow BADness is not a p_0-only signature.
3. `SAME-N-GROWING-DEPTH-DUAL-PRODUCT-BARRIER-20260906B`: for fixed $c + \kappa < 1$, almost every prime start and every same-N root $k \leq (\log X)^\kappa$ has no short successful simple path whose label product or cofactor product is $\leq X^{1-\varepsilon}$, up to depth $c \log\log X / \log\log\log X$.
4. Exact non-dominance survives hostile checking: at $N = 2180$, p_1 is directly GOOD while p_0 is BAD through radius two; at $N = 2024$, p_0 succeeds earlier while p_1 is BAD through radius two.

The product barriers quantify over *successful paths*. They are vacuous inside a completely closed failed core, so the attempted inference from those barriers to closed-core height was correctly killed by `SUCCESS-PRODUCT-BARRIER-TO-CLOSED-CORE-HEIGHT-20260906B`.

10.5 Exact $h = 1$ conjugacy, cycle realisation and canonical embedding

Prime-lift full-BFS conjugacy. Let $N = dx$, let $m = x - 1 \geq 2$, and suppose $R = (d-1)x + 1$ is prime. In the composite-m case the complete BFS from the inactive prime divisor d and from

R have identical ordered transcripts after replacing the initial root. For $d = 2$ this gives the exact all-generation normal form

$$N = 2(P-1), \quad P = p_0(N) \qquad \Longrightarrow \qquad \text{BFS from } P \ \equiv\ \text{BFS from } 2$$

for every non-direct $h = 1$ case. The tempting general implication "prime lift ⇒ success" is false: $(N, d, R) = (1862, 7, 1597)$ gives a complete failed 22-test conjugate pair.

Canonical first-support cycles. For every prescribed ordered list of distinct odd primes there is a fixed gap h and infinitely many canonical worlds whose complete first support contains a directed BAD cycle through those primes. Extra factors and GOOD branches are uncontrolled. Hence canonical first-support acyclicity and any uniform bound on cycle length are false outside the special $h = 1$ situation.

Fixed-gap canonical valuation embedding. Take any actual upper BAD root (N_0, R_0) with composite first complement and gap h_0. Its complete induced first-support graph on $S = \mathrm{PF}(N_0 - R_0)$, including root valuations and all internal row valuations, embeds exactly into infinitely many canonical BAD-root first-support graphs with the same numerical gap h_0. Additional support outside S is forced and remains uncontrolled. Therefore every hereditary property of this labelled induced graph that holds for all canonical BAD roots must also hold for every upper BAD root. This is the rigorous replacement for the old overbroad finite-local wall: finite hereditary first-support graph/valuation invariants cannot universally separate p_0, while complete support and later reachability remain outside the theorem.

10.6 Low-gap complete-support restrictions

Let $r = N/2 + h$ be an upper prime root, $m = N - r$ composite and $q = \min \mathrm{PF}(m)$ with $0 < h < q$.

1. If no GOOD vertex occurs through depth 2, then at least three distinct nonroot primes have been discovered by depth 3 (`LOW-GAP-COMPLETE-THREE-LABEL-ALTERNATIVE-20260907`).
2. If q participates in a reciprocal BAD two-cycle $q \to b \to q$, then

$$b = 2h + (2j+1)q \quad (j \geq 1), \qquad b \geq 3q + 2h,$$

   and this numerical bound is sharp for both canonical and noncanonical roots.
3. If the complete row is a prime square, $N - q = b^2$, then $q \nmid N - b$ and $(b-1)^2 < N - b < b^2$. In the $h = 1$ singleton-square subcase, necessarily u is odd, $u \geq 3$, and $q \equiv 23 \pmod{24}$ when $m = q^u$.
4. If $m = q^u$, $N - q = b^v$ is a complete prime-power row and the reciprocal return occurs, then

$$3 \leq v \leq u - 1.$$

   Thus $u = 2, 3$ exclude this immediate monomial return.

The attempted universal FIVE-label strengthening was refuted exactly at $N = 44670$, root rank 12, where the depth-three nonroot labels are precisely $\{149, 211, 23, 1933\}$. The proved lower bound THREE survives. The final exact reduction `LOW-GAP-THREE-LABEL-EXACT-OBSTRUCTION-SPLIT-20260907C` says that a genuine three-label case must be either (A) a complete closed active BAD world on exactly three primes, or (B) an explicit prime-power system

$$m = q^u, \qquad N - q = b^v, \qquad N - b = q^a c^w,$$

with the restrictions recorded in the raw theorem. In the singleton square branch the remaining equations can be written

$$2q^u + 2h = q + b^2 = b + c^w, \qquad w \geq 3 \text{ odd};$$

for $h = 1$, also u is odd and $q \equiv 23 \pmod{24}$.

10.7 Negative structural diagnostics

The multiplicative subgroup generated modulo N by a closed BAD support necessarily contains -1, and contains any upper root whose complete first support lies in that closed set. As a general explanation of p_0 this route fails: at five of the six historical hosts the maximal closed low BAD support generates the entire unit group. At $N = 2200$ the generated subgroup is proper, but membership is still not sufficient for failure. This kills simple multiplicative-character separation as a general p_0 mechanism while leaving more structured valuation/additive residue arguments open.

10.8 Binding current frontier

Global exact question: the $h = 1$ closed-support obstruction

Can complete factor search from 2 fail when

$$N = 2(P-1), \qquad P \geq 5 \text{ prime}, \qquad P-2 \text{ composite?}$$

Equivalently, can there exist a finite set C of odd active primes below $2(P-1)/3$ such that $\mathrm{PF}(P-2) \subseteq C$ and, for every $q \in C$, $2(P-1) - q$ is composite with its *entire* prime support contained in C? By the full-BFS conjugacy this is exactly the unresolved non-direct canonical $h = 1$ failure question. Selected cycles, finite internal valuations and support-height lower bounds do not answer it.

Local exact question: three-label obstruction

Can either branch of `LOW-GAP-THREE-LABEL-EXACT-OBSTRUCTION-SPLIT-20260907C` actually occur? The singleton-square branch reduces to the displayed exponential/prime-power system above. A proof of impossibility would strengthen the current THREE-label theorem to FOUR in the stated low-gap domain; an exact example would kill that strengthening cleanly.

Asymptotic radius-three question

The previous 2026-09-06B frontier remains open: control the canonical depth-3 successful-path system in the region where *both* the label product and the cofactor product exceed $X^{9/10}$. A suitable $o(X/\log X)$ count would give almost-everywhere complete radius-three BADness in the $h = 1$ family, not a Goldbach proof.

10.9 p_0 status after the continuation

The empirical eventual reliability of p_0 remains unexplained. Several simple candidate explanations are now rigorously unavailable: shallow BADness and product barriers are shared by nearby roots; finite first-support acyclicity can reverse between p_0 and p_1; arbitrary first-support BAD cycles occur in canonical worlds; and hereditary first-support label/valuation invariants cannot separate all canonical BAD roots from all upper BAD roots. What may still distinguish p_0 must use information outside that hereditary local class — complete support, leakage/external cofactors, absolute root location/size, or truly multigeneration reachability.

Chapter 11

Computational Evidence

Everything in this chapter is **VERIFIED COMPUTATION** or **EMPIRICAL PATTERN**. **Nothing here is evidence for a universal statement.** A range is part of a claim: quoting a number without its range is an error, and quoting a number without its *domain* (which roots, which complements, whether unit rows are included) is the error that produced the “301” confusion (`FAIL-CENSUS-DOMAIN-001`).
Every number below is regenerated by the single script in Appendix A, which has no external dependencies. The block IDs EXP1–EXP8 are the script’s own.

11.1 Provenance and reproduction

ID	What it computes	Range	Key result
EXP1	exhaustive canonical scan, `census` mode	even $N \le 10^7$	0 failures; mean inclusive work $W = 7.0527$ tested vertices
EXP2	verdict of roots $p_0, \dots, p_6$	even $N \le 2 \cdot 10^5$	0 for $k = 0$; $2, 4, 5, 3, 2, 1$ for $k \ge 1$
EXP3	census of all failing upper roots	even $N \le 2 \cdot 10^4$	301 roots, six N, 0 canonical
EXP4	escape margin G_{esc}	even $N \le 2 \cdot 10^5$	52 numbers with margin 1; 10 with $\lvert W\rvert \ge 3$
EXP5	phase trichotomy and ρ_0	even $N \le 2 \cdot 10^5$	0 violations; $\rho_0 \ge 1 \iff$ Type III
EXP6	embedding of reflected blocks	$N \le 2 \cdot 10^4$, $k \le 6$	731 446 indices, 0 violations
EXP7	fixed point B_N^{low}	even $N \le 6 \cdot 10^4$	nonempty for exactly six N
EXP8	observables p_0 vs p_1	even $N \le 2 \cdot 10^5$	aggregates $\pm$6–24%; κ_{gen}: $0 \to 0.9041$
EULER	Euler-defect normalisations (N1)–(N4)	even $50 \le N \le 6 \cdot 10^4$	(N1) 6 044 and (N2) 3 804 violations; (N3), (N4) none
CERT	the 34 hostile certificates of the corpus	fixed witnesses	34/34 pass

Exact invocation reproducing the numbers in this file:

```
python3 p0_experiments.py -scan 10000000 -ablation 200000 -census 20000
                          -margin 200000 -phase 200000
                          -filtration 20000 -fixedpoint 60000
                          -observables 200000 -json data_p0.json
```

EXP1 at `-scan 10000000` takes about 75 s and about 50 MB; EXP7 at `-fixedpoint 60000` about 70 s. The defaults (`-scan 1000000 -fixedpoint 20000`) finish in a few minutes and reproduce everything except the 10^7 scan records.

11.2 The canonical scan

Scope note 11.1 (Work convention audit, 2026-09-07). The binding convention is inclusive work W: every dequeued/tested vertex, including the successful endpoint, is counted. The regenerated scan gives mean $W = 7.052697210539442$, maximum $W = 151$ first at $N = 3\,542\,948$, and maximum success depth 10 first at $N = 11\,822$. The older pseudocode variable "processed" counted only expanded composite rows E; on success $W = E + 1$. This is a source/counting repair only and changes no success verdict.

EXP1. VERIFIED COMPUTATION. All even $4 \le N \le 10^7$, canonical root, `census` mode:

cases	4 999 999
failures of p_0	**0**
mean number of tested vertices W (successful endpoint included)	7.0527
record vertex count	151 $(N = 3\,542\,948)$
record success depth	10 $(N = 11\,822)$

The same scan restricted to $N \le 10^6$ gives 499 999 cases, 0 failures, mean 5.8044, record 133 $(N = 976\,904)$, record depth 10 $(N = 11\,822)$ — reproducing the table of [BIB] to the last digit. The depth record 10 is attained only by $N = 11\,822$ and remains stable through the whole 10^7 range.

Distribution of the success depth $d(N)$ over all 4 999 999 cases:

d	0	1	2	3	4	5	6	7	8	9	10
$\#N$	983 722	1 426 153	1 500 435	802 099	240 796	42 059	4 436	276	18	4	1

$d = 0$ means immediate success ($N - p_0$ prime, or the diagonal case).

Warning 11.2 (What the scan does not say). 0 failures up to 10^7 is consistent with a failure at $10^7 + 2$. The scan is VERIFIED COMPUTATION and stays VERIFIED COMPUTATION. In particular it is *not* evidence for X1 (§9.9); it is the statement that X1 has no small counterexample.

11.3 Root ablation and the census

EXP2 — verdict of the root p_k. VERIFIED COMPUTATION. All even $N \le 2 \cdot 10^5$, roots $p_0, \ldots, p_6$; failures on a *composite* row and degenerate failures on the unit row $N - p_k = 1$

counted separately:

k	0	1	2	3	4	5	6
failures, composite row	**0**	2	4	5	3	2	1
failures, unit row	0	4	2	3	3	2	2

Smallest failures on a composite row: $k = 1$: $(N, p_1) = (1862, 941)$; $k = 2, \dots, 5$: $(128, 73), (128, 79), (128, 83), (128, 89)$; $k = 6$: $(1862, 977)$.

The full census EXP3 and the six special N are in §6.6; the escape margin EXP4 is in §6.7; the phase census EXP5 in §6.3; the filtration check EXP6 in §6.2; the fixed point EXP7 in §6.6; the ablation EXP8 in §6.5.2. They are not repeated here.

11.4 Hostile certificates

CERT — 34/34 pass. VERIFIED COMPUTATION. Every finite certificate cited anywhere in the corpus was re-run against the reference implementation of §3.4. Groups:
A1 regression on the record cases $N = 11\,822$ (depth 10) and $N = 3\,542\,948$ (151 vertices).
B1 gap equivalence (Theorem 3.10) on all even $N \le 2 \cdot 10^5$.
C1–C5 genealogy, reflection completeness, filtration embedding, and the diagonal calibration ($N = 14$, $r = 11$).
D1–D5 canonical geometry, shadow identity, Bézout invariant, no-bad-return, unique source.
E1–E14 the hostile counterexamples: $N = 242$ (reflection activation), $N = 128$ (TF local scope), $N = 322$ and $N = 858$ (CAB), $N = 4736$ (potential capacity), $N = 6142$ (external success), $N = 14198$ (direct adjacency), $N = 20$ (four-channel ledger), $N = 1718$ (generic absorption), $N = 51\,032$ (Hall compression), Exact-D countermodels, the $K_{3,3}$ grid at $N = 10^6$, the $N = 2200$ two-core, and the $h = 1$ witnesses $103\,344$ and $364\,844$.
F1–F5 ablation and the 301 census with its exact domain.
All 34 pass. Where a corpus claim and a certificate disagreed, the disagreement is recorded as a scope correction in §12.1, not silently resolved.

11.5 Other observables

Depth. EMPIRICAL PATTERN. The success depth of the canonical traversal is ≤ 2 for $3\,910\,310$ of $4\,999\,999$ cases (78.21%) and ≤ 4 for $4\,953\,205$ cases (99.06%). The tail is extremely thin: a single case reaches $d = 10$. No mechanism is known that would force a bounded depth, and the depth is *not* the right measure of difficulty — see §6.7.

p_0 is not measurably better at escaping. EMPIRICAL PATTERN. Beyond the aggregate table of §6.5.2: within the 301 failing upper roots of EXP3, the canonical root wins at all six N, but the set of failing roots is neither an initial nor a final segment of $p_1, p_2, \dots$ (Example 6.24: p_1 wins, p_2 fails). Ordering by height does not order the outcome.

Two censuses agree. EMPIRICAL PATTERN. The six N with $B_N^{\mathrm{low}} \neq \varnothing$ (EXP7, $N \le 6 \cdot 10^4$) coincide with the six N carrying failing upper roots (EXP3, $N \le 2 \cdot 10^4$) and with the six numbers with an exceptional autonomous component reported to 10^8 by Bożek [8]. The three lists measure different things; their coincidence is data, not a theorem, and it has not been checked beyond $6 \cdot 10^4$ in this project.

Chapter 12

Failed and Killed Approaches

This chapter is the curated narrative; Chapter 13 is the complete raw record. Read this one first: it groups 239 individual entries into the dozen mistakes that actually recur. A model that internalises this chapter will not need the raw log except to check a specific ID.

The five recurring shapes of error.

1. **Domain mixing** — quoting a result outside the class (A/B/C/D) on which it was proved. Most common single failure.
2. **Functional substitution** — proving a bound in one functional and consuming it in another. Kills TD, $P1^{br}$, K1.
3. **Vacuity mistaken for canonicity** — a construction that "only works for p_0" because for p_k it is undefined on k coordinates.
4. **Finite certificate** — deriving a contradiction from a bounded canonical transcript. Provably impossible (§8.2).
5. **Promotion** — turning a scan with zero counterexamples into a theorem, or an exact identity into a mechanism.

12.1 Curated ledger of negative results

The table is complete with respect to all sources. Rows are grouped by theme; each entry carries an exact counterexample or an exact reason.

Claim / route	Status	Exact reason
Exceptionality of p_0		
Zero hole count as a mass mechanism	KILLED	loss $O_k(\log N)$ against block mass $t_k \log(x/h_k)$ `FAIL-P0-ZEROHOLE-ESSENTIAL-001`
p_0 as information selector	SUPERSEDED	every clause survives ablation; not refuted, but **not established** `FAIL-P0-INFORMATION-SELECTOR-001`
MR2 as a canonical identity	KILLED	$q \mid (r-2j) \iff q \mid (h_r - j)$ for any upper root `FAIL-MR2-CANONICAL-ONLY-001`
Least-upper-parent as selector	KILLED	eleven non-canonical roots share the property [CORE, V71.41]
Prime-free boundary activates a reflected row	KILLED	$N = 242$: $a_1 = 117$, $a_2 = 119$ both composite `FAIL-REFLECTION-ACTIVATION-001`
Statistical dominance of p_0	KILLED	median rank percentile 0.20982 [D4, q. 12]

Claim / route	Status	Exact reason
Minimality determines local geometry	KILLED	23+15 transitions $B_{\text{abs}} \leftrightarrow B_{\text{re}}$ at fixed p_0 [D4, q. 17]
Largest-factor heuristic	KILLED	$N = 124$ [PCRD, FAIL-P0-LARGEST-FACTOR-TWOSTEP]
Finite canonical data give a contradiction	KILLED	affine programmability and finite bad-tree universality [PCRD]
Mechanisms around p_0		
Literal label conservation along a path	KILLED	$v_\ell(N-\ell) = v_\ell(N) = 0$ [LOG]
Absorption trichotomy from factor-flow alone	KILLED	$N = 1718$, $\ell = 59$: both payments have zero valuation FAIL-GENERIC-ABSORPTION-001
Four-channel ledger on all of Q_0	KILLED	$N = 20$: a GOOD row outside R gives $\kappa_R < m_\ell$ [ILA]
Unconditional CAB	KILLED	$N = 322$: $U_R = J_R = 1$, $B_{\text{abs}} = 11 \cdot 29$; internal version $N = 858$ FAIL-CAB-001, FAIL-CAB-INTERNAL-001
$\mathcal{C}_S < \mathcal{D}_S$ from the first front	KILLED	$N = 4736$: $\mathcal{C}_S \approx 22.60 > \mathcal{D}_S \approx 20.16$ FAIL-POTENTIAL-CAPACITY-001
Escape must be directly adjacent	KILLED	$N = 14198$: atoms $19, 167$ at distances 8 and 7 FAIL-DIRECT-ADJACENCY-001
External reuse is local	KILLED	$(N, \ell) = (1\,825\,372, 52\,153)$, distance $53\,750$ FAIL-LOCAL-REUSE-001
External parent is a success root	KILLED	$(6142, 877, 3511)$: 3511 active, above p_0, own traversal fails FAIL-EXTERNAL-SUCCESS-001
Constant $C = 1$ in charging	KILLED	53 violations, maximum 9.42243 FAIL-CHILD-C1-001
Naive support expansion of all sources	KILLED	Hall compression $456 \to 17$ at $N = 51\,032$ FAIL-HALL-SUPPORT-001
Exact-D local algebra replaces minimality	KILLED	non-canonical models die only on $h < G$ FAIL-EXACTD-NONCANONICAL-001
Failure always produces nontrivial B_{abs}	KILLED	FAIL-BABS-PRODUCTION-001
Widened block in the non-reuse lemma	KILLED	the endpoint pair has difference $2h$ FAIL-BROADBLOCK-ENDPOINT
Euler defect in normalisation (N1) or (N2)	KILLED	$6\,044$ resp. $3\,804$ violations for $N \le 6 \cdot 10^4$ (§5.7)
Scope corrections		
Upper-half compression under p_0 failure	SCOPE-CORRECTED	the theorem assumes $\text{Good}_N = \varnothing$ FAIL-UPPERHALF-SCOPE-001
Local $\mathcal{F}$ certificate + $h < G$ forces routing	SCOPE-CORRECTED	$N = 128$, $\ell = 13$: certificate holds, p_0 wins, $R = \varnothing$ FAIL-TF-LOCAL-SCOPE-001
Source-free kernel = EAC = terminal component	SCOPE-CORRECTED	three different constructions FAIL-KERNEL-EAC-001
Parent lattice over all primes	SCOPE-CORRECTED	the correct object is the intersection with $\mathcal{A}_N$ FAIL-PARENT-LATTICE-DOMAIN-001

Claim / route	Status	Exact reason
"301" counts all failing roots	SCOPE-CORRECTED	the census covers composite complements only `FAIL-CENSUS-DOMAIN-001`
Genealogy and reflection-completeness without hypothesis	SCOPE-CORRECTED	diagonal case, $N = 14$ (Scope note 3.17)
Formula $\rho_0 = \#\{D_s < 2h\}$	SCOPE-CORRECTED	missing lower condition $D_s > h$ (Warning 6.18)
High-gap amplification as unconditional	SCOPE-CORRECTED	hypothesis is *pair failure*, not CRD failure (Warning 7.5)
$q < m_0/3$	SCOPE-CORRECTED	correct is $q \le m_0/3$; $N = 32$ attains equality (Scope note 5.4)
IL1–IL6 on all of Q	SCOPE-CORRECTED	domain is Q_{smooth} (Warning 7.7)

12.1.1 What survived hostile testing

- Incidence ledger IL1–IL6 on the domain $S = Q_{\text{smooth}}$: two independent reconstruction routes, 1 000 worlds, 0 errors [ILA] — **AUDITED IDENTITY**.
- The escape-or-reuse alternative: 61 452 instances, 0 orphans; independent red team 60 413 atoms, 0 orphans [ORPH, RT] — **VERIFIED COMPUTATION**, as a theorem **OPEN**.
- The whole genealogical layer (§6.1) and the filtration (§6.2) — **PROVED**.
- The phase diagram and the bidirectional rank (§6.3, §6.4) — **PROVED**.
- Global two-sheet collision budget, no-bad-return-to-p_0, the B_N^{low} characterisation — **PROVED**.
- All 34 hostile certificates of the corpus were reproduced here and pass (§11.4).

12.2 Post-CRD successor theories

[PCRD] documents a sequence of theories built after CRD1, each intended to preserve information lost by its predecessor. All reached a wall. The table summarises them; their proofs are *not* reproduced here.

Theory	Fundamental object	What it was to recover	What stopped it
CRD1	support/valuation dynamics from p_0	reachability, shadow, capacity	the "same container" argument, generic smooth counting, residue saturation, exact self-consistency of the fixed point
CMFT	lossless mirror-factorization tensor (δ, q, a, b)	parent–valuation–cofactor correlation	the least-factor tree *is* the Buchstab/sieve decomposition; the cofactor star at fixed q is an affine bijection
CGST	canonical cell N_h and the convolution $a_P * a_P$	exact additive pairing + phase	the factorization layer collapses diagonally in (h, t); exception propagation reduces to prime-pair geometry
CSD	exceptional spectral configurations + the gap	a pointwise inverse anomaly	finite conductor patterns are consistent with an arbitrarily long *fixed* canonical gap

Theory	Fundamental object	What it was to recover	What stopped it
CAPT	Li–Liu almost-prime witnesses	a massive real family of sources	at fixed (r, s) the system reduces to two linear prime forms; $(3, 3)$ is the natural degeneration
RSD	rough reservoir + Liouville sign + P_3 correction	explicit exposure of parity	either level of distribution $> 2/3$, or a pointwise binary prime–Liouville correlation
rough-P_2 matrix	$A_N(q, r) = \mathbf{1}_{\mathbb{P}}(N - qr)$	global higher-order correlation	boundary blindness, local spectrum = singular series, codegrees = prime-tuple problem, $K_{3,3}$ grids exist arithmetically
H1-FP	root-factor descent at $h = 1$ with shift-2 orthogonality	elementary closure of the 2-core	reduces to a Pillai/Bennett equation $a^\alpha - a = b^\beta - b$

Example 12.1 (A $K_{3,3}$ grid at $N = 10^6$). VERIFIED COMPUTATION. For $q \in \{61, 67, 73\}$ and $r \in \{113, 179, 599\}$ all nine numbers $N - qr$ are prime, and all six factor vertices are BAD. A full rank-two bipartite rough-P_2 grid is therefore arithmetically real and is *not* a contradiction. [PCRD, hostile exact survivor]

Updated verdict on the p_0 programme. The *old local p_0-exceptionality proof architecture* is SUPERSEDED, but the stronger historical claim that finite complete-BFS neighbourhoods are universally programmable was scope-corrected in September 2026. Consequently universal p_0 reachability and the $h = 1$ complete-support problem remain live open questions. What is ruled out rigorously is narrower: selected partial transcripts by themselves, and hereditary first-support graph/valuation separators, cannot explain universal p_0 reliability.
What remains alive: p_0 as a selector, the exact full-BFS/root-2 conjugacy at $h = 1$, complete-support/moving-support analysis, non-hereditary selector invariants, and the full reachability problem.

12.3 Surviving lemmas of the killed programmes

A killed theory is not a theory whose theorems are false. Each successor programme of §12.2 died at an *architecture* wall, and each left behind correct, reusable lemmas. Compression is exactly where those get thrown out with the dead programme, so they are collected here with their statuses intact. *Every item below is* ***PROVED*** *in [PCRD] unless marked otherwise; the proofs are not reproduced here.* Using one of them is legitimate; restarting its parent programme is not.

12.3.1 From CMFT (Canonical Mirror Factorization Theory)

CMFT's objects: for $N = 2x$, $\mathcal{U}_N = \{p \in \mathbb{P} : x \le p < N\}$, $\mathcal{M}_N = \{m \le x : N - m \in \mathbb{P}\}$, so that Goldbach is exactly $\mathcal{M}_N \cap \mathbb{P} \neq \varnothing$; the canonical root has mirror anchor $m_0 = N - p_0 = \max \mathcal{M}_N$; and the CMFT atom is $\mathfrak{e} = (\delta, q, a, b)$ with $m_0 - \delta = q^a b$ and $p_0 + \delta + q^a b = N$.

Theorem 12.2 (Lossless anchor filtration, `PCRD-CMFT-1`). ***PROVED***. *For upper primes $p_0 < p_1 < \cdots$ with $m_j = N - p_j$, the full left history of the anchor p_j grows by adjoining the* entire *hyperedge row*

m_j:

$$\boxed{\mathscr{H}_{j+1} = \mathscr{H}_j \sqcup \mathscr{A}_j, \qquad L_{j+1}(\ell,k) - L_j(\ell,k) = \mathbf{1}_{\ell^k | m_j}.}$$

The ablation $p_0 \to p_1$ loses exactly the full factorization of m_0.

Why it is worth keeping. This is the precise *information-theoretic* form of the ablation result of §6.5.2: it says what is lost, not merely that something is.

Theorem 12.3 (Tail hard-edge universality, `PCRD-CMFT-2`). ***PROVED*** */ no-go. For any later anchor p_j, after truncating the earlier prefix its complement m_j becomes the maximal element of the tail mirror field and satisfies* identical *local gcd, offset and bilinear identities. Hence*

$$\boxed{\textit{the local geometry of the tail does not distinguish } p_0.}$$

The genuinely unique information carried by p_0 is only the zero lost prefix*.*

Reading. This is the sharpest available formulation of what §6.5 establishes empirically, and it is a theorem. Combined with Warning 6.7(2) it is the strongest reason to doubt that any purely local property of p_0 can do dynamical work.

Theorem 12.4 (Canonical least-factor split, `PCRD-CMFT-3`). ***PROVED****. For a composite mirror row m choose uniquely $q = P^-(m)$, $b = m/q$. If $q > x^{1/3}$ then b must be prime, hence*

$$\boxed{m \textit{ composite} \implies P^-(m) \le x^{1/3} \quad \vee \quad m = qr \textit{ a balanced semiprime.}}$$

Why it is worth keeping. It repairs the divisor-multiplicity bias of any raw bilinear form over rows: without it, divisor-rich rows are counted $\tau(m)$ times.

Theorem 12.5 (All-root collision identity, `PCRD-CMFT-4`). ***PROVED****. For a composite row family S,*

$$\boxed{\sum_{\ell} \sum_{k \ge 1} \binom{\nu_{\ell,k}(S)}{2} \log \ell = \sum_{i<j} \log \gcd(m_i, m_j).}$$

A pigeonhole over least factors yields a positive collision floor.

Limitation, recorded by the source. The floor has *no consumer*: nothing in the corpus converts it into a contradiction. Compare Theorem 7.30, which is the residual-row analogue and does have one.

Theorem 12.6 (Prime-pressure failure ceiling, `PCRD-CMFT-5`). ***PROVED****. With $\mathcal{P}_N = \sum_{m \in \mathcal{M}_N} \Lambda(m)$, a hypothetical Goldbach failure eliminates every $k = 1$ contribution, so*

$$\mathcal{P}_N \le E(x) := \sum_{r^k \le x,\ k \ge 2} \log r \le \sqrt{x} \log x, \qquad \textit{whence} \qquad \boxed{\mathcal{P}_N > \sqrt{x} \log x \implies \textit{Goldbach.}}$$

Reading. A clean quantitative reformulation of Goldbach for a single N, entirely elementary. It did not supply the matching lower bound, which is why CMFT died — but the reformulation itself is correct and reusable, and it is a class-D-free statement (it does not assume $\mathrm{Good}_N = \varnothing$; it concludes Goldbach).

> **The CMFT wall.** Raw bilinear mass counted divisor-rich rows repeatedly. After the least-factor repair the machinery turned out to be *exactly* the Buchstab/sieve decomposition, and the fixed-q cofactor star $p \leftrightarrow b = (N-p)/q$ is an affine bijection, not an independent Type-II correlation. Giant stars have the natural scale of primes in arithmetic progressions and can be far larger than the "forced" $x^{2/3}$ with no anomaly at all.

12.3.2 From CGST (Canonical Gap-Spectral Theory)

CGST's reformulation: with $A_P = \{u \in \mathbb{Z} : P + 2u \in \mathbb{P}\}$ and $N_h = 2(P - h)$,

$$\boxed{N_h \text{ is Goldbach} \iff -h \in A_P + A_P.}$$

Theorem 12.7 (Canonical straddle theorem, `PCRD-CGST-1`)**.** **PROVED**. *If no direct pair through P exists, every Goldbach representation has the form*

$$P + 2t + \bigl(P - 2(h + t)\bigr) = 2(P - h), \qquad t \geq 1,$$

and the lower prime must lie below *the previous prime P_-.*

Reading. This is the additive counterpart of the phase trichotomy (Theorem 6.12) and explains why Type I forces the search strictly past p_-.

Theorem 12.8 (Arbitrarily deep finite reflection obstruction, `PCRD-CGST-2`)**.** **PROVED**. *For any fixed T, CRT and Dirichlet construct infinitely many canonical $h = 1$ worlds in which $P - 2, P - 4, \ldots, P - 2(T + 1)$ are all composite. Hence* no bounded-depth reflection search can be a universal closer.

Reading. The reflection-specific instance of the architecture wall (§8.2). If a proposed argument inspects the reflected block to any fixed depth, this theorem kills it immediately.

Theorem 12.9 (Diagonal collapse, `PCRD-CGST-3`)**.** **PROVED**. *In cell coordinates the reflection row is $C_P(h, t) = P - 2(h + t)$, so the entire valuation/factor tensor on the (h, t) plane depends only on $h + t$. The extra "cell dimension" is a* false degree of freedom *for the factorization layer.*

Warning 12.10. Any construction that treats (h, t) as two independent parameters at the factorization layer is already dead. The exception-amplification wall follows: two cells h and $h+s$ share exact rows only when the upper-prime offsets form a prime pair of difference $2s$, and same-row factor certificates satisfy $\gcd(C_P(h, t), C_P(h + s, t)) = \gcd(C_P(h, t), s)$, which is 1 for $s = 1$. Factorization-driven propagation of one Goldbach exception to neighbouring N is therefore far too weak.

12.3.3 From CSD (Canonical Spectral Defect)

Theorem 12.11 (Canonical conductor–gap compatibility, `PCRD-CSD-1`)**.** **PROVED**. *For any fixed modulus pattern M and any fixed length L, CRT and Dirichlet construct infinitely many canonical $h = 1$ worlds with $M \mid N$ and a prime-free interval of length at least L immediately before p_0. Hence* any finite conductor pattern is compatible with an arbitrarily long fixed canonical gap.

Reading. The bridge "exceptional characters ⇒ canonical gap contradiction" does not exist, and this theorem says why in one line. Keep it: it is the cheapest available refutation of an entire family of circle-method hybrids.

12.3.4 From CAPT (Canonical Almost-Prime Transfer)

Theorem 12.12 (Fixed-(r, s) compression, `PCRD-CAPT-1`)**.** **PROVED**. *After freezing (r, s) in the Li–Liu representation $N = p + rq$ with $N - q = sb$, one has $q = N - sb$ and*

$$\boxed{p = rsb - (r - 1)N,}$$

so the system becomes two linear prime forms in the single variable b.

Remark 12.13 (The $(3, 3)$ degeneration). For $(r, s) = (3, 3)$: $q = N - 3b$, $p = 9b - 2N$ — an ordinary two-linear-prime pattern that occurs abundantly in real canonical worlds. Second-layer compression is therefore not a contradiction. Large-factor consequences of Li–Liu that remain correct: under Goldbach failure $r \neq 1$, and from $rq < N$ with $r \leq q^{9/10}$ one gets $r < N^{9/19}$ and $q > N^{10/19 - o(1)}$ for almost all of the relevant mass.

12.3.5 From RSD (Rough Signed Decomposition)

Theorem 12.14 (Exact rough signed identity, `PCRD-RSD-1`). ***PROVED***. *On the rough support ($\rho_z(n) = \mathbf{1}_{(n,P(z))=1}$, $z > X^{1/4}$, so $\Omega(n) \le 3$),*

$$\boxed{\mathbf{1}_{\mathbb{P}}(n)\rho_z(n) = \rho_z(n)\Big[\frac{1-\lambda(n)}{2} - \mathbf{1}_{\Omega(n)=3}\Big],}$$

and for shifted-prime weights

$$\boxed{G_z(N) = \tfrac{1}{2}R_z(N) - \tfrac{1}{2}L_z(N) - T_{3,z}(N),}$$

so that under Goldbach failure $\boxed{L_z(N) = R_z(N) - 2T_{3,z}(N)}$.

Reading. This is the cleanest *exact* statement in the corpus of where the parity obstruction sits for this problem. It is worth keeping precisely because it is an identity, not an estimate.

Theorem 12.15 (Distribution–roughness phase law, `PCRD-RSD-2`). ***PROVED***. *If the usable distribution level is $D = X^\theta$ and one sieves to $z = X^\beta$, the lower linear sieve needs $\beta < \theta/2$, while forcing $\Omega(n) \le k$ by roughness needs $\beta > 1/(k+1)$. The two are compatible* iff

$$\boxed{\theta > \frac{2}{k+1}.}$$

For $k = 2$ the natural threshold is $\theta > 2/3$.

Reading. A quantitative triage rule: any proposal in this family can be checked against it in one line. Pascadi's $5/8 - o(1)$ [93] for special factorable weights is below $2/3$, so branch A is closed at current technology.

Remark 12.16 (The parity endpoint, exactly). At $z = X^{5/16-\varepsilon}$ the surviving P_3 sits in a narrow balanced box that switching/modern Chen sieves control well, but after reduction to P_1/P_2 the exact identity

$$P_1(N) = \frac{D_{1,2}(N) - \Delta_2(N)}{2}, \qquad \Delta_2 = P_2 - P_1$$

remains, and Goldbach failure means full saturation $\Delta_2 = D_{1,2}$. Any theorem giving "even a little desaturation" is *already equivalent* to finding a prime complement on that family. This is the precise sense in which the parity barrier is the endpoint (§12.4).

12.3.6 From the global rough-P_2 matrix route

With $A_N(q,r) = \mathbf{1}_{\mathbb{P}}(N - qr)$:

Theorem 12.17 ($K_{3,3}$ explosion, `PCRD-M-1`). ***CONJECTURE*** *(conditional* ***PROVED****). At rough-P_2 mass of order $X/\log^2 X$, Kővári–Sós–Turán forces a $K_{3,3}$ at the natural scales $|Q| \sim X^{1/2}$, $|R| \sim X^{11/16}$. $K_{4,4}$* is not forced *by that density scale.*

Theorem 12.18 (Boundary blindness, `PCRD-M-2`). ***PROVED***. *Extending the bulk matrix by the boundary row/column $r = 1$ encodes the Goldbach entries. Zeroing the whole boundary axis is a perturbation of rank at most 2, whose operator norm is asymptotically small against the bulk scale. A generic operator norm or energy can therefore* almost fail to notice *the loss of every Goldbach entry.*

Theorem 12.19 ($h = 1$ matrix invisibility, `PCRD-M-3`). ***PROVED***. *For $N = 2(P-1)$ we have $m_0 = P - 2 = X - 1$, so every odd product $qr \le X$ automatically satisfies $qr \le X - 1 = m_0$. The canonical hard edge removes* no *rough-P_2 matrix cell: p_0 disappears from the domain geometry entirely.*

Theorem 12.20 (Fixed-modulus spectrum, `PCRD-M-4`). **PROVED**. *The local admissibility matrix modulo a prime $\ell \nmid N$ has the form $J - P_N$; its singular values are explicit and recover* only *local density / singular-series data. The fixed-modulus spectrum does not see Goldbach failure.*

Remark 12.21 (The codegree wall). The pointwise codegree $C_N(q_1, q_2) = \sum_r A_N(q_1, r) A_N(q_2, r)$ counts simultaneous primality of the three linear forms r, $N - q_1 r$, $N - q_2 r$, i.e. it enters the prime-tuple problem; average codegrees are boundary-blind by Theorem 12.18. Only a specially constructed bulk-to-boundary inverse theorem would remain, and no candidate is known that does not reduce to an earlier binary-correlation wall.

12.3.7 From the p_0 kill programme and the H1-FP revival

Theorem 12.22 (Gap scalar freedom, `PCRD-P0-3`). **PROVED** *(from known facts). For a fixed prime P with gap $G = P - P_-$, all $N_h = 2(P - h)$, $0 \le h < G$, have canonical root P. With unbounded prime gaps the ratio h/G can approximate all of $[0, 1]$; with bounded-gap results there are infinitely many small G. Hence* the scalars h, G, h/G have no universal closing regime*.*

Warning 12.23. This is the exact statement that kills every attempt to close the problem by a size condition on $\theta = h/G$ (§6.3). The phase diagram is a correct classification; it is not a filter.

Theorem 12.24 (H1 root-factor descent, `PCRD-H1-1`). **PROVED**. *If $q, r \mid m_0 = P - 2$ then*

$$\boxed{r \mid N - q \iff r \mid q - 2.}$$

The root-factor subsystem is therefore a strictly decreasing DAG, and the least root factor q_* satisfies

$$\boxed{\mathrm{PF}(N - q_*) \cap \mathrm{PF}(m_0) = \varnothing,}$$

so failure must immediately *produce external factors.*

Theorem 12.25 (H1 shift-2 orthogonality, `PCRD-H1-2`). **PROVED**. *For $N = 2(P - 1)$ with P prime and $u < N/3$ prime,*

$$\boxed{\gcd(N - u,\ u + 2) = 1.}$$

Every terminal fixed core must therefore satisfy full shift-2 coprimality on all of its hyperedges.

Remark 12.26 (The 3-core normal form). The two smallest rows of a terminal core have disjoint factor supports. For $K = \{a < b < c\}$ this leads to one of two coupled forms, e.g.

$$N - b = a^\alpha, \qquad N - a = b^\beta c^\gamma, \qquad N - c = a^u b^v.$$

A hostile reverse search over large finite boxes found no 3-core; the status is **VERIFIED COMPUTATION**, *not* a theorem.

> **Why the H1-FP revival was not enough.** Closing $N = 2(P - 1) \Rightarrow B_N^{\mathrm{low}} = \varnothing$ requires classifying complete self-consistent shifted S-unit systems *and* the primality of the linear form $N/2+1$. Already the minimal case $|K| = 2$ touches an unresolved Pillai/Bennett-type uniformity ($a^\alpha - a = b^\beta - b$; Remark 6.25). p_0 supplies no mechanism of its own for that problem.

12.3.8 Live CRD results retained through every pivot

[PCRD] records the following as explicitly retained when CRD1 was frozen. They are the intersection of everything that survived; all appear elsewhere in this file and are listed together here so a future model can see the retained core at a glance.

- the fixed-point characterisation B_N^{low} (Theorem 3.29);
- the C0b decomposition and its correct gap-scale congruence consequences (Theorem 8.6);
- T2 first-front residue closure;
- T3 mod-3 exclusion/rigidity (Theorems 5.17, 5.18);
- the source-free core K with its shifted S-unit equations (Remark 6.25, Remark 12.26);
- the exact container/product identities (§5.3);
- the correct prime-power collision geometry (§7.4);
- the empirical six-EAC corpus (§6.6).

What killed CRD1 — the convergence, not a single false lemma.

1. Selected-divisor capacity routes were *same-container* arguments: one cannot beat a container with one of its own divisors. (This is exactly what Corollary 9.9 later made quantitative.)
2. Generic smooth-population bounds lost self-consistency: the factor alphabet is simultaneously the set of state labels.
3. The pure local-residue route saturates for $q \geq 5$; modulus 3 is an exception, not a mechanism (Theorem 5.17).
4. Reflection is not a universal closer: $h = 1$ gives an *empty* strict reflected block.
5. Nonprimitive large product/reuse is compatible with a failed world and gives no capacity contradiction.
6. A source-free core is not itself contradictory: $N = 2200$, $K = \{3, 13\}$ is an exact witness.
7. After removing the intermediate ledgers the frontier returned to the exact self-consistency $\mathrm{PF}(m_0) \subseteq B_N^{\text{low}}$ with no independent intermediate parameter.

12.4 The parity barrier and why it is the real wall

Two distinct obstructions are present and must not be conflated.
The partial-transcript wall (§8.2) is internal and scope-corrected: compatible selected finite divisibility/valuation patterns can be realised canonically, so those selected patterns alone cannot supply a contradiction. It does *not* rule out a theorem about a complete bounded-depth frontier.
The parity barrier is external and classical: sieve methods cannot distinguish numbers with an even number of prime factors from those with an odd number, so no purely sieve-theoretic argument produces a *prime* complement.
Thus selected-edge programming cannot replace complete-support control, while pure sieve input cannot by itself force primality. The current moving-support frontier may require a combination of exact complete-row arithmetic and analytic input; the old slogan "not locally" was too broad.

RSD and parity. RSD attempted to expose parity explicitly through a rough reservoir with a Liouville sign and a P_3 correction. It requires either level of distribution $> 2/3$ — beyond Pascadi's $5/8$ [93] — or a pointwise binary correlation between primes and the Liouville function, which is at least as hard as the problems calibrated by Mangerel [95]. KILLED.
The weaker P_2 desaturation target. Also KILLED: at $\Omega \leq 2$ a sign-(-1) witness *is* a prime complement, so the "weaker" target is the original problem.
Chen-type input. R. Li's bound $D_{1,2}(N) \geq 1.9728\, C(N)N/\log^2 N$ [94] and Li–Liu's $(1+1.9)$ [15] supply almost-prime representations in quantity, but not the last bit distinguishing P_1 from P_2. That last bit is the parity barrier.
Friedlander–Iwaniec. The asymptotic sieve for primes [38] is the template for what the missing input looks like: a bilinear form estimate. No bilinear input of the required shape has been produced in this project, and the rough-P_2 matrix was the attempt to manufacture one.

Warning 12.27 (Do not try to route around parity by enlarging the graph). Several killed routes are variations on "add more structure to the graph until the parity obstruction disappears". The obstruction is not a property of the graph; it is a property of the sieve information being used.

Adding vertices, weights or higher-order correlations to a construction whose only input is sieve-type data cannot remove it.

Chapter 13

Full Failure Log

This chapter is reproduced in full and unabridged. It contains the 239 legacy entries of the project failure log plus seven post-v4.81 binding additions. The exact source text for the new additions is also preserved in Appendix B. No entry was removed for appearing unimportant; the record's value lies precisely in its completeness.

How to use it.

- The 239 legacy entries remain alphabetically ordered by `FAIL-ID`; the seven post-v4.81 additions are appended in their research chronology and are also listed in the index.
- The index in §13.1 gives ID, status and page.
- Every entry is labelled `fail:FAIL-ID`, so `\ref{fail:FAIL-XXX-001}` resolves.
- Cross-references that pointed into the original document and cannot be resolved here appear as [target] rather than being silently dropped.
- Legacy status macros of the source were mapped onto the status vocabulary of §1; the mapping is stated below.

Legacy status mapping. `R`, `RI`, `RM` → KILLED; `SC` → SCOPE-CORRECTED; `SU`, `RET` → SUPERSEDED; `SR`, `C` → VERIFIED COMPUTATION; `P` → PROVED; `PA` → PROVED WITH REPAIR; `H` → EMPIRICAL PATTERN; `O` → OPEN. The mapping is status-preserving in the sense that no entry was moved to a stronger status.

13.1 Index of failure-log entries

ID	Status	Str.
FAIL-BAKER-CORE-FAN-DIRECTLY-GIVES-LEVELC-SPD-WITNESS-001	SCOPE-CORR.	112
FAIL-BOUNDED-ARITY-WEIGHT-001	KILLED (infer.).	112
FAIL-BUNDLE-ENLARGES-CAPACITY-001	KILLED.	113
FAIL-BV-BOUNDARY-DESTROYS-MASS-001	KILLED.	114
FAIL-BV-INACCESSIBILITY-HIDES-MIRROR-MASS-001	SUPERSEDED.	114
FAIL-C-SYMBOL-COLLISION-001	SCOPE-CORR.	114
FAIL-CAB-001	KILLED without a failure condition.	115
FAIL-CAB-INTERNAL-001	KILLED without a failure condition.	116
FAIL-CANONICAL-BT-CAPACITY-TRANSPORT-TO-EXTERNAL-HQ-001	SCOPE-CORR (domain failure).	116
FAIL-CANONICAL-EXACT-LAYER-TRANSFER-MISSING-001	SCOPE-CORR.	117
FAIL-CASCADE-DIRECTION-001	KILLED as stated and as proved.	117
FAIL-CENSUS-DOMAIN-001	SCOPE-CORR.	118
FAIL-CHILD-C1-001	KILLED by source-recorded computation.	118
FAIL-CLEAN-ROOTED-CAP-P1-001	KILLED (infer.).	119
FAIL-COFACTOR-ENTROPY-ALONE-CLOSES-LOADONE-001	KILLED (infer.).	120
FAIL-COFACTOR-SIBLING-PRIME-UNSCOPED-001	KILLED (infer.).	120
FAIL-CRD2-NEEDED-FOR-FIRST-OCCURRENCE-001	KILLED as a present engineering requir	121
FAIL-CRD2-PREMATURE-001	DESIGN PROHIBITION / FRONTIER CERTIFIC	121
FAIL-CRITICAL-VERTICAL-CARRIES-TLOGN-001	KILLED.	121
FAIL-CRITICAL-VERTICAL-GENERIC-001	KILLED as a description.	122
FAIL-CUBEROOT-DEFECT-PIGEONHOLE-GIVES-POLYNOMIAL-ISTAR-001	KILLED.	122
FAIL-CUBEROOT-PIGEONHOLE-IS-BOUNDARY-CASE-001	SCOPE-CORR.	123
FAIL-CYCLE-COFACTOR-LOCK-ALONE-CLOSES-DEEP-CYCLES-001	KILLED as a sufficient uniform closure	123
FAIL-DC-CAPACITY-COMPRESSION-001	SCOPE-CORR.	123
FAIL-DEADSMOOTH-CAN-CARRY-POLYNOMIAL-HIGHGAP-PACKET-001	KILLED.	123
FAIL-DEADSMOOTH-PILLAI-001	SCOPE-CORR. The search result stands a	124
FAIL-DEADSMOOTH-UNBOUNDED-001	KILLED.	124
FAIL-DEEP-AFFINE-PATH-RESIDUE-IS-CAPACITY-001	KILLED for the capacity inference.	125
FAIL-DEEP-FIBRE-NEEDS-FIRSTENTRY-OWNER-ESCAPE-001	KILLED as a proof-design necessity.	126
FAIL-DEEP-FIBRE-TO-SUBPOLY-BY-ANCESTRY-ALONE-001	KILLED as a proof mechanism from one r	126
FAIL-DEEP-ONE-AP-PACKING-CLOSES-001	KILLED as a sufficient closing mechani	126
FAIL-DEEP-PURE-QMONOMIAL-CAN-CYCLE-001	KILLED.	126
FAIL-DEEP-SELFRENEWAL-IS-INDEPENDENT-TERMINAL-001	KILLED as a proof-design description.	127

ID	Status	Str.
FAIL-DEEP-U-TO-U-RETURN-AS-PRIMARY-FRONTIER-001	SUPERSEDED as a primary frontier.	127
FAIL-DESCENT-ONE-STATE-001	SUPERSEDED.	127
FAIL-DIRECT-ADJACENCY-001	KILLED for direct adjacency.	128
FAIL-ETA-EQUIVALENT-TO-WEIGHTED-J-001	KILLED (infer.).	128
FAIL-EXACTD-NONCANONICAL-001	KILLED as a substitute for minimality;	129
FAIL-EXACTD-RESIDUAL-SURVIVES-SUBROOT-001	KILLED.	129
FAIL-EXTERNAL-ETA-IDENTIFICATION-001	KILLED.	130
FAIL-EXTERNAL-FIXED-POLYLOG-SQUAREFREE-PENDING-001	CLOSED PENDING INDEPENDENT AUDIT .	130
FAIL-EXTERNAL-MERGER-FIRST-SPLIT-001	SUPERSEDED as proof organisation.	131
FAIL-EXTERNAL-MULTISOURCE-CRT-IS-THE-NEXT-CAPACITY-BRIDGE-001	KILLED (mech.) relative to the current	131
FAIL-EXTERNAL-SOURCE-MASS-IS-INDEPENDENT-RADICAL-RESERVOIR-001	SUPERSEDED as a primary factorisation	131
FAIL-EXTERNAL-SUCCESS-001	KILLED.	132
FAIL-EXTERNAL-UT-IDENTITY-IS-UPPER-CAPACITY-001	KILLED (infer.).	132
FAIL-F1-DOMAIN-MIX-001	SCOPE-CORR.	133
FAIL-FC8-FLOOR-FOR-RHS-001	KILLED (infer.).	133
FAIL-FC8-SELF-CONTRADICTION-001	KILLED (infer.). (FC8) itself is PROVE	134
FAIL-FC8-SLACK-IS-VERTICAL-MASS-001	SCOPE-CORR.	134
FAIL-FIRSTENTRY-SQUARE-ONLY-HAS-Q-SPACING-001	KILLED.	135
FAIL-FIRSTFRONT-CONTAINMENT-CLAIMED-NEW-001	KILLED (infer.).	135
FAIL-FIRSTFRONT-RESIDUAL-SCALE-IS-CONTROLLED-ONLY-BY-Q-001	KILLED on [illegible]	135
FAIL-FIRSTFRONT-SCALAR-ROUTE-001	KILLED (mech.) with the scope below.	136
FAIL-FIXED-FINITE-MIRROR-HUB-001	KILLED.	136
FAIL-FIXED-K-UNIFORM-SELBERG-FREE-001	KILLED (infer.).	136
FAIL-FRESH-BRANCH-PRIMACY-001	SCOPE-CORR.	137
FAIL-FRINGE-ALPHABET-TRANSFER-001	SCOPE-CORR.	137
FAIL-FRINGE-BOUNDED-MULTIPLICITY-001	KILLED (mech.).	138
FAIL-FRINGE-BRANCH-DOUBLE-CHARGE-001	SCOPE-CORR.	138
FAIL-FRINGE-CONGRUENCE-CONTRADICTION-001	SCOPE-CORR.	139
FAIL-GAMMA-BOUND-ROUTES-001	KILLED for all four, as routes; the un	139
FAIL-GAMMA-POLYLOG-ASSUMED-UNIVERSAL-001	KILLED (infer.).	139
FAIL-GENERIC-ABSORPTION-001	KILLED on the generic/noncanonical dom	140
FAIL-GENERIC-AVERAGE-DEPTH-O-LOGLN-001	SUPERSEDED.	140
FAIL-GENERIC-PRIME-ROW-RPD-001	KILLED as a generic theorem.	141
FAIL-GREEDY-AVOIDANCE-NAIVE-001	KILLED as an argument. The conclusion	141

ID	Status	Str.
FAIL-HALL-SUPPORT-001	KILLED (mech.).	142
FAIL-HIGH-LOADONE-COFACTOR-CAN-HAVE-ARBITRARY-ARITY-001	KILLED.	143
FAIL-HIGH-SEMIPRIME-SIBLING-REUSE-IS-CHEAP-001	KILLED.	143
FAIL-HIGH-VERTICAL-IS-INDEPENDENT-EXTERNAL-CHANNEL-001	KILLED.	143
FAIL-HIGHFLOOR-DOMAIN-VACUITY-001	SCOPE-CORR. The theorem is PROVED; its	144
FAIL-IMPORT-V32-POSITIVE-K-TO-DEEP-LOW-OWNER-001	KILLED.	144
FAIL-INCREMENTAL-J-IS-BOUNDED-BY-OLD-J-001	KILLED.	145
FAIL-INDIVIDUAL-MESOSCALE-BV-001	KILLED (infer.).	145
FAIL-INVERSE-SIBLING-INJECTIVE-001	KILLED (infer.).	146
FAIL-KERNEL-EAC-001	SCOPE-CORR.	146
FAIL-KPOWER-SPECTRUM-IS-OPTIMAL-DJ-BOUND-001	SUPERSEDED.	147
FAIL-LARGE-FRESH-SUPPORT-MAY-BE-MOSTLY-REPEATED-001	KILLED.	147
FAIL-LARGE-MONOMIAL-PACKET-CAN-OCCUR-IN-BOTH-MOD3-CLASSES-001	KILLED.	148
FAIL-LARGEBASE-DEEP-VERTICAL-ESCAPE-001	KILLED as a live frontier on q>N^ 1/3	148
FAIL-LAYERED-GAMMALOGN-IS-CONTRADICTION-001	KILLED (infer.).	148
FAIL-LEDGER-INDEPENDENCE-LI5-SCOPE-001	SCOPE-CORR. G-POSITIVE-ROUND-LEDGER-IN	149
FAIL-LEGACY-CONTINUUM-001	KILLED.	149
FAIL-LEGACY-ENCODING-001	KILLED.	150
FAIL-LEGACY-PIGEONHOLE-001	KILLED as a mechanism.	150
FAIL-LEGACY-WAVES-001	KILLED.	151
FAIL-LF2-AS-UPPER-BOUND-001	KILLED.	151
FAIL-LINEAR-SMOOTH-MIRROR-PACKET-CAN-STAY-FULLY-SUPPORTED-001	KILLED asymptotically on the recorded	152
FAIL-LITERAL-LINEAGE-001	KILLED.	152
FAIL-LOCAL-REUSE-001	KILLED by source-recorded computation.	153
FAIL-LOSSLESS-INTERFACE-001	KILLED for lossless transfer, under ev	153
FAIL-LOWBASE-POWERFUL-ROWS-AS-INDEPENDENT-VERTICAL-WALL-001	KILLED (mech.) as an independent prima	154
FAIL-LOWCORE-CARDINALITY-FC8-WEIGHT-001	KILLED (infer.).	154
FAIL-LOWCORE-CHILDCLOSED-001	KILLED (infer.).	155
FAIL-LOWPOLYLOG-MULTIPLICITY-IS-PARENT-EXISTENCE-001	KILLED as a description of the current	156
FAIL-MANY-FIRSTFRONT-ROOTS-CAN-ALL-LIE-ABOVE-4H-001	KILLED for h>N^ 1/3 .	156
FAIL-MATCHING-CAPACITY-001	KILLED (infer.) — the matching theor	156
FAIL-MERGER-COUNT-IMPLIES-AMBIENT-J-001	KILLED (infer.).	157
FAIL-MERGER-WEIGHT-TO-J-TOPOLOGY-001	KILLED (infer.).	157
FAIL-MIRROR-FIXED-POLYLOG-RESERVOIR-001	SCOPE-CORR.	158
FAIL-MIRROR-MASS-ONE-SCALE-001	KILLED.	158
FAIL-MIRROR-MULTIPLICITY-BY-REPEATED-BASE-001	KILLED (infer.).	158

ID	Status	Str.
FAIL-MONOMIAL-MOD3-NOVELTY-vNEXT4.11	SCOPE-CORR (provenance correction).	159
FAIL-MONOMIAL-NEEDS-SPARSITY-001	KILLED (mech.).	159
FAIL-MONOMIAL-NODESCENT-CHEAP-001	KILLED.	159
FAIL-MONOMIAL-PATH-LENGTH-TWO-IS-ONLY-EMPIRICAL-001	KILLED.	160
FAIL-MONOMIAL-VERTICAL-SELF-SUFFICIENT-001	KILLED.	160
FAIL-MR2-CANONICAL-ONLY-001	KILLED as a scope claim.	160
FAIL-N1/4-IS-MULTIPLICITY-REALISATION-CEILING-001	KILLED.	160
FAIL-N1/6-DESIGNATED-BASE-BARRIER-001	KILLED (infer.).	161
FAIL-NAIVE-MU-LE-3ETA-001	KILLED.	161
FAIL-NATURAL-DENOMINATOR-IS-ONLY-BASELINE-001	KILLED.	161
FAIL-NESTED-PRIMEPOWER-PACKING-GIVES-CUMULATIVE-SAVING-001	KILLED (infer.).	162
FAIL-NONMONOMIAL-VERTICAL-IS-INDEPENDENT-001	KILLED as a description.	162
FAIL-ONE-REPRESENTATIVE-CONSUMES-MIRROR-MULTIPLICITY-001	KILLED (infer.).	162
FAIL-ORTHOGONAL-BRIDGE-001	KILLED.	163
FAIL-ORTHOGONALITY-AS-HYPOTHESIS-001	SCOPE-CORR.	164
FAIL-OWNED-RETURN-FIRST-EDGE-IS-FREE-SOURCE-LOSS-001	KILLED on the v18 owner-completed doma	164
FAIL-OWNER-SEALED-RETURN-CAN-STAY-SQUAREFREE-001	KILLED.	164
FAIL-OWNERLESS-FIRST-OCCURRENCE-HAS-NO-ADDITIVE-STRUCTURE-001	KILLED.	165
FAIL-P0-INFORMATION-SELECTOR-001	KILLED (mech.). Not refuted; not estab	165
FAIL-P0-ZEROHOLE-ESSENTIAL-001	KILLED as a mechanism for the active/o	166
FAIL-P4-IS-THE-BEST-BV-ARITY-001	KILLED.	166
FAIL-PAIR-LOADED-DESCENT-OPEN-001	CLOSED AS A POLYLOG-( ) SCALE-REENTRY	167
FAIL-PAIRWISE-K-IS-THE-ONLY-POSSIBLE-X1-CAPACITY-001	KILLED.	167
FAIL-PARENT-LATTICE-DOMAIN-001	SCOPE-CORR.	167
FAIL-PATH-PRODUCT-AGGREGATION-ERASES-DEEP-VALUATION-001	KILLED for the inference [remote refl	168
FAIL-PER-ROW-RADC-COMPRESSION-001	KILLED as a failure certificate.	168
FAIL-PLAIN-AP-COUNT-KILLS-VERTICAL-001	SCOPE-CORR.	168
FAIL-POLYLOG-IS-NEEDED-FOR-DETERMINISTIC-EXACT-STAR-EXPORT-001	SCOPE-CORR.	169
FAIL-POSITIVE-ROUND-BLOCKS-REPEATED-FAN-PROGRESS-001	KILLED.	169
FAIL-POSITIVE-ROUND-CONSEQUENCE-001	KILLED (infer.). The depletion apparat	169
FAIL-POSITIVE-ROUND-EXHAUSTION-001	KILLED, quantitatively.	170
FAIL-POSITIVE-ROUND-LEDGER-CONSEQUENCE-001	KILLED for ledger-only derivations. No	171
FAIL-POSITIVE-SEMIPRIME-ROUND-IS-NEXT-PRIMARY-001	SUPERSEDED.	171
FAIL-POTENTIAL-CAPACITY-001	KILLED.	172
FAIL-POWER-THRESHOLD-TO-POLYLOG-001	SCOPE-CORR.	173
FAIL-PRODUCT-BUDGET-ALONE-CANNOT-PRESERVE-HIERARCHICAL-STAR	KILLED (mech.)	173
FAIL-PROFILE-CARRIER-001	SUPERSEDED.	173

ID	Status	Str.
FAIL-PURE-PATH-ONLY-RETAINS-MOD-Q-001	KILLED.	174
FAIL-PURE-Q-GRAPH-CAN-BRANCH-001	KILLED.	174
FAIL-Q0-LEDGER-001	KILLED.	175
FAIL-Q0-ONLY-OWNER-SEALED-RETURN-IS-CHEAP-001	KILLED.	175
FAIL-QFREE-FIRST-COPY-CAN-HIDE-BELOW-CARDINALITY-001	KILLED.	176
FAIL-QUOTIENT-SIEVE-POLYNOMIAL-RANGE-001	KILLED (mech.).	176
FAIL-RDS-GEOMETRY-DENSITY-001	KILLED (infer.).	176
FAIL-REFLECTION-ACTIVATION-001	KILLED.	177
FAIL-REPEATED-BASE-MUST-BE-FIXED-POLYLOG-001	KILLED.	178
FAIL-REPEATED-BASE-QUADRATIC-COLLISION-UNAVOIDABLE-001	SCOPE-CORR.	178
FAIL-REPEATED-HUB-QUADRATIC-COST-EQUALS-ACTUAL-STORAGE-001	KILLED.	178
FAIL-RETURN-COEFFICIENT-K-IS-THE-INDEPENDENT-BINDING-WALL-001	SUPERSEDED.	179
FAIL-RETURN-COFACTOR-IS-FREE-001	KILLED as a description.	179
FAIL-RETURNED-SEED-FIRST-OCCURRENCE-IS-J-001	KILLED.	179
FAIL-RETURNING-BASIN-IS-ABSOLUTELY-COSTFREE-001	SCOPE-CORR.	179
FAIL-REUSE-DOUBLE-CHARGE-001	KILLED (infer.).	180
FAIL-ROOT-GENERATION-AC-ALONE-CLOSES-HQ-RPS-001	SCOPE-CORR.	180
FAIL-ROOTED-AMBIENT-001	KILLED.	180
FAIL-ROUTING-INDIFFERENCE-SCOPE-001	SCOPE-CORR. The theorem survives; its	181
FAIL-ROUTING-LOSES-SEED-WEIGHT-001	KILLED.	182
FAIL-ROUTING-RESCUES-P1BR-001	KILLED (mech.). The routing statement	182
FAIL-SATURATION-DOUBLE-CHARGE-001	KILLED (infer.).	183
FAIL-SEALED-MULTISOURCE-LOCK-001	KILLED (mech.).	183
FAIL-SELECTOR-CLOSES-CASCADE-001	KILLED (infer.). The selector is PROVE	184
FAIL-SHORTEN-PARENT-INTERVAL-TO-GET-P3-001	KILLED (mech.) relative to that stated	185
FAIL-SIMPLE-STATE-001	SUPERSEDED.	185
FAIL-SINGLE-BASE-CONCENTRATION-NEEDED-FOR-VERTICAL-MIGRATION-001	KILLED.	186
FAIL-SINGLETON-STAR-LOCAL-WINDOW-001	KILLED (infer.).	186
FAIL-SIZE-COMPARISON-AGAINST-RECORDED-CAPACITY-001	SCOPE-CORR.	187
FAIL-SMALL-GAMMA-MAKES-CANONICAL-BLOCK-INVISIBLE-001	KILLED.	187
FAIL-STAR-SELF-REENTRY-001	KILLED.	187
FAIL-STRIP-COMMON-BASE-THEN-IMPORT-CLEAN-ROUND-001	KILLED.	188
FAIL-TAGGED-P3-RESIDUE-FORCES-RETURN-001	KILLED (infer.) for every prime q5 .	188
FAIL-TD-NAIVE-001	KILLED.	188
FAIL-TD-SMOOTH-INACTIVE-001	KILLED as a proof route resting on the	189
FAIL-TF-LOCAL-SCOPE-001	SCOPE-CORR.	191
FAIL-TWOVARIABLE-FULL-LEVEL-001	KILLED (infer.).	191
FAIL-UNIQUE-FRESH-TARGETS-MAY-FUNNEL-INTO-O(1)-PARENTS-001	KILLED against the high layer extracted in	192
FAIL-UNIQUE-MISSING-DATUM-001	SCOPE-CORR.	192

ID	Status	Str.
FAIL-UNIT-COMPLEMENT-001	KILLED.	193
FAIL-UNIVERSAL-BRIDGE-IMPOSSIBILITY-001	SCOPE-CORR. The mathematics is correct	193
FAIL-UPPERHALF-SCOPE-001	SCOPE-CORR.	194
FAIL-V13-EXTERNAL-VALUATION-AS-SEED-WEIGHT-001	SCOPE-CORR.	194
FAIL-V18-SOURCE-STAR-COUNTEREXAMPLE-SPECIFIC-001	KILLED (infer.).	195
FAIL-V410-ANONYMOUS-STAR-COFACTOR-001	SUPERSEDED on the monomial branch.	195
FAIL-V47-DEEP-RPD-GLOBAL-GENERATION-LOCALISATION-001	SCOPE-CORR.	196
FAIL-V48-DIRECT-REFLECTED-TAG-INHERITANCE-001	SCOPE-CORR.	196
FAIL-V48-HIGHTARGET-LOADONE-NECESSARY-001	SCOPE-CORR.	196
FAIL-V48-LOCAL-J-AS-AMBIENT-M-SCALE-001	SCOPE-CORR.	197
FAIL-V48-LOWCHILD-VERBATIM-DOMAIN-TRANSFER-001	KILLED as a direct domain application.	197
FAIL-V48-MONOMIAL-NOVELTY-001	SCOPE-CORR (provenance correction).	197
FAIL-V48-MULTIPARENT-SEPARATE-BRANCH-NECESSARY-001	SUPERSEDED as a necessary proof split.	197
FAIL-V48-POSITIVE-DEEP-SEMIPRIME-CLOSES-P1BR-001	KILLED for immediate closure.	198
FAIL-V48-VARIABLE-M-SELFREENTRY-NEW-001	SUPERSEDED.	198
FAIL-V49-CROSSGEN-GCD-IS-P1BR-BRIDGE-001	KILLED as a sufficient general bridge.	198
FAIL-V49-DEEP-SEMIPRIME-AUTOMATIC-POSITIVE-ROUND-CLOSURE-001	SCOPE-CORR.	199
FAIL-V49-DIRECT-REFLECTED-FACTOR-INHERITANCE-001	KILLED/SCOPE-CORR.	199
FAIL-V49-LATER-DESCENDANTS-DESTROY-ALL-REFLECTED-ARITHMETIC-001	KILLED.	199
FAIL-V49-RELATIVE-T-OR-FRESH-MASS-ALONE-CLOSES-001	KILLED as an immediate contradiction r	199
FAIL-V50-ANCESTRY-DIAGONAL-TRANSPORTS-DIRECT-CAPACITY-001	SCOPE-CORR.	200
FAIL-V50-BOUNDED-BRANCHING-CLOSES-SINGLESEED-001	KILLED as a weighted-capacity inferenc	200
FAIL-V50-REFLECTED-PROVENANCE-USELESS-AFTER-SELECTION-001	KILLED on the return-heavy cheap- (J)	200
FAIL-V50-RETURNED-SEED-IMPLIES-CYCLE-001	KILLED.	200
FAIL-V50-SOURCEHEAVY-REFLECTED-SEEDS-CLOSE-001	KILLED.	201
FAIL-V51-ALTERNATE-PATH-CONGRUENCE-AMPLIFIES-DEEP-POWER-001	KILLED.	201
FAIL-V51-ANCESTRY-LOCALISATION-FAILED-001	KILLED.	201
FAIL-V51-CYCLE-LOCK-FALSE-001	KILLED as a description of the theorem	202
FAIL-V51-UNWEIGHTED-REFLECTED-MATCHING-CARRIES-DEEP-MASS-001	SCOPE-CORR.	202
FAIL-VERTICAL-DESCENT-AUTOMATIC-BV-001	KILLED (infer.).	202
FAIL-VERTICAL-FC8-SEPARATION-001	SCOPE-CORR / KILLED (mech.).	203
FAIL-VERTICAL-HIGHGAP-UNBOUNDED-CAPACITY-001	KILLED (mech.).	203
FAIL-VERTICAL-SQRTLOG-WALL-001	SUPERSEDED within that conditional bra	203
FAIL-VERTICAL-TO-ETA-LOSES-LOG-001	KILLED (mech.).	204

ID	Status	Str.
FAIL-VERTICAL-UNLOCALISED-001	KILLED as a description.	204
FAIL-VNEXT433-UNIFORM-FLOOR-GAP-001	KILLED as a failure certificate; it mu	205
FAIL-WEIGHTED-REUSE-AMPLIFICATION-001	SCOPE-CORR.	205
FAIL-WHOLE-BLOCK-EXTERNAL-SET-LOSES-REFLECTED-PROVENANCE-001	KILLED.	205
FAIL-WORLD-SIZE-NAIVE-001	KILLED as stated.	206
FAIL-v40-HIGH-FLOOR-N-OVER-LOG2-IS-SHARP-001	SUPERSEDED.	206
FAIL-v41-MONOMIAL-LENGTH-TWO-NOVELTY-001	SCOPE-CORR.	207
FAIL-BOUNDED-COFACTOR-ROWSLOTS-ARE-UNCONSTRAINED-001	KILLED.	207
FAIL-BOUNDED-COFACTOR-SUM-MUST-PAY-ONE-PER-J-001	KILLED.	207
FAIL-CHEAP-RADIC-OSCILLATION-HAS-A-HIDDEN-ROUTING-SINK-001	KILLED as a routing diagnosis.	207
FAIL-COFACTOR-RENEWAL-IMPLIES-DIVERGENCE-001	KILLED.	208
FAIL-FIRST-COPY-HIGH-ROW-POPULATION-IS-BY-ITSELF-A-CONTRADICTION-001	KILLED.	208
FAIL-FIXED-COFACTOR-PRIMALITY-GIVES-NO-MORE-THAN-SPACING-001	KILLED.	208
FAIL-HIGH-BASE-R-GREATER-Y-IMPLIES-RADIC-MONOTONICITY-001	KILLED.	208
FAIL-LARGE-COFACTOR-IS-AN-ANONYMOUS-COEFFICIENT-SINK-001	KILLED.	209
FAIL-LARGE-COFACTOR-RENEWAL-IS-DIVERGENCE-001	KILLED.	209
FAIL-PRODUCT-MODULUS-COMPRESSION-ALONE-CLOSES-DEEP-001	KILLED.	209
FAIL-RADIC-HALVING-IS-AUTOMATIC-DYNAMICAL-DESCENT-001	SCOPE-CORR.	209
FAIL-RADIC-LEVEL-IS-MONOTONE-POTENTIAL-001	KILLED.	209
FAIL-RADIC-LEVEL-RETURN-ITSELF-PAYS-J-001	KILLED.	210
FAIL-SAME-EXACT-LEVEL-IS-A-NEW-EXTERNAL-RESTART-001	KILLED as a packet-design inference.	210
FAIL-SELBERG-LOGSAVING-ALONE-CLOSES-DEEP-001	KILLED.	210

13.2 Entries

`FAIL-ABOVE-BV-MASS-INVISIBLE-001`

FAIL-ID:
FAIL-ABOVE-BV-MASS-INVISIBLE-001.
Date / iteration:
vNEXT4.19.
Old description:
Mirror mass above one scalar BV boundary is unavailable to clean selection.
Status:
SCOPE-CORRECTED.
Replacement:
Track both designated-base scale Q and roughness scale z. With polylogarithmic z, the clean designated-base range reaches $Q_* = N^{1/2}/\operatorname{polylog} N$. Significant mass beyond Q_* forces a polynomial actual-state packet rather than disappearing.

`FAIL-ABSORBED-FRINGE-RESIDUAL-HAS-ARBITRARY-COFACTOR-001`

FAIL-ID:
FAIL-ABSORBED-FRINGE-RESIDUAL-HAS-ARBITRARY-COFACTOR-001.

Status:
KILLED.
Rejected statement:
A first-front residual prime $\ell \in (N/6, N/3)$ may occur with an arbitrary cofactor in $N - q$.
Certificate:
since $q < N/6$, $(N - q)/\ell$ is an odd integer strictly between 1 and 6, hence $N - q = 3\ell$ or $N - q = 5\ell$. The root-supported gcd then divides 3 or 5.

FAIL-AP2-HIDDEN-OMEGA-DEPENDENCY-001

FAIL-ID:
FAIL-AP2-HIDDEN-OMEGA-DEPENDENCY-001.
Date / iteration:
vNEXT4.5 audit of vNEXT4.2.
Claim:
The recorded AP1/AP2 consequence follows merely from "standard Mertens/PNT estimates".
Status:
SCOPE-CORRECTED.
Exact repair:
The support-capacity inequality contains $\omega_h(x) \log c_{\max}/t$, so one also needs

$$\omega_h(x) \le \omega(x) = O\left(\frac{\log x}{\log \log x}\right).$$

With that explicit dependency, the quantified population conclusion survives.
Do not repeat:
Do not suppress the $\omega(x)$ dependency when invoking this population lemma.

FAIL-BABS-PRODUCTION-001

FAIL-ID:
FAIL-BABS-PRODUCTION-001.
Date / iteration:
Canonical bridge audit §4, 31 August 2026.
Claim:
Every failed world produces a nontrivial absorbed base product $B_{\text{abs}} > 1$, which is therefore the universal carrier of failure mass.
Original motivation:
B_{abs} is the natural object of ambient promotion and was observed nontrivial in most instances.
Status:
SUPERSEDED.
Minimal / strong counterexample:
$(N, r) = (128, 79)$.
Exact certificate:
Full-Q production is nontrivial for this root while the recorded B_{abs} equals 1. Production and absorption are different channels, and only the former is always nonvacuous on the composite-complement domain.
Failed hypothesis or mechanism:
One channel of a four-channel ledger was promoted to the whole ledger.
Why the argument failed:
B_{abs} collects exactly the residual primes with $\kappa_R(\ell) = m_\ell$; nothing forces that set to be nonempty.
What survives:
`ARS-BABS-INTERNAL`, that $\mathcal{B}_{\text{abs,int}} = \mathcal{B}_{\text{abs}}$ under failure — which explicitly yields *no* nonvacuity

conclusion — and B_{abs} as one channel and the proper object of ambient promotion.

Safe weaker version:
Carry failure mass on $\mathfrak{F}_Q = \prod_{q\in Q_r} M_q$, not on B_{abs}.

Current replacement:
Theory Core, `G-FULLQ-PRODUCTION`, which gives $\mathfrak{F}_{Q_r} > (5/3)^{|Q_r|}\,\mathrm{rad}(m_r) > 1$ unconditionally on the composite-complement domain.

Do not repeat:
Do not make one ledger channel universal.

Related current target:
$P1^{\text{br}}$, K1.

`FAIL-BAKER-CORE-FAN-DIRECTLY-GIVES-LEVELC-RPD-WITNESS-001`

FAIL-ID:
FAIL-BAKER-CORE-FAN-DIRECTLY-GIVES-LEVELC-RPD-WITNESS-001.

Status:
SCOPE-CORRECTED.

Correction:
`G-BAKER-SPARSE-EXACT-LAYER-FAN` packages Baker's analytic theorem together with extra CRD domain assumptions and an upper-half conclusion. The v4.46 witness uses only the analytic sparse-moduli input and repeats the endpoint subtraction on a different fixed proportional interval.

What survives:
the Level-C prime-row no-go is a legitimate new arithmetic consequence, but its provenance must be stated correctly.

`FAIL-BOUNDED-ARITY-WEIGHT-001`

FAIL-ID:
FAIL-BOUNDED-ARITY-WEIGHT-001.

Date / iteration:
Bombieri–Vinogradov four-ary continuation, 1 September 2026.

Claim:
Once every selected parent has at most four unintended children, the large-state packet E is automatically a lower-order cost, or the clean parent family closes P1BR by cardinality alone.

Status:
KILLED.

Minimal / strong counterexample:
Structural; one unintended prime factor is already enough.

Exact certificate:
A squarefree selected row may have

$$N - p = \ell r, \qquad r > N^{1/4-\varepsilon},$$

with $r \in R$. The four-ary conclusion permits this configuration, and the only universal weight estimate it supplies is

$$\log r = \log(N-p) - \log \ell.$$

Thus even $\omega(E_\ell) = 1$ permits the unintended label to carry the entire non-base cofactor mass. More generally

$$\log E \le L(\mathscr{P}) - \log \prod \ell,$$

with equality when every unintended child lies in R.

Failed hypothesis or mechanism:
An unweighted degree bound was used as a weighted logarithmic capacity bound.

Why the argument failed:
The number of factors and the sum of their logarithms are different functionals. Roughness makes each surviving factor larger, not cheaper.

What survives:
The four-ary theorem turns E into a bounded-degree dynamic packet. The exact clean source-surplus identity then cancels the base copy of every internal unintended factor and exposes its true cost as $(\kappa_R(r) - 1)\log r$.

Safe weaker version:
Use bounded arity only for dynamic branching and population statements. For P1BR prove an external-mass-versus-rooted-reuse inequality with logarithmic weights.

Current replacement:
Theory Core v1.6, `G-CLEAN-MULTIPARENT-SURPLUS-IDENTITY` and `G-BV-FOUR-ARY-DISPOSITION`.

Do not repeat:
$\omega(E) = O(|A_0|)$ does not imply $\log E = o(L(\mathscr{P}))$.

Related current target:
P1BR, K1, large-layer disposition.

FAIL-BUNDLE-ENLARGES-CAPACITY-001

FAIL-ID:
FAIL-BUNDLE-ENLARGES-CAPACITY-001.

Date / iteration:
vNEXT4.1, Phase 7.

Claim:
That preserving a canonical bundle D in the lifted row forces a canonical contribution — to the denominator of (RC4′), or to a state-difference packing budget — and so extracts more from R than an arbitrary external row does.

Original motivation:
The lifted row shares exact prime powers with a canonical reflected row, and empirically bundle rows are much more often nonpositive than single-base rows on the same reflected row (87% against 72% in `COMP-VNEXT41-DISPOSITION`). Nonpositive rounds are the ones with a recorded consumer, so this looked exploitable.

Status:
KILLED.

Exact certificate:
Theory Core `G-EXTERNAL-INTERFACE-CAPACITY`. For any globally coprime family $\mathscr{P}$ of external rows and any failed world R, the integers $\gcd(N-p, N-u)$, $p \in \mathscr{P}$, are pairwise coprime divisors of $N-u$, so $\sum_p \sum_u \log\gcd(N-p, N-u) \le \sum_u \log(N-u) < |R| \log N$. The bound has *no* hypothesis on how $\mathscr{P}$ was chosen. Consequently at most $2|R| < 6.6\gamma \log N$ members with $N - p > \sqrt{N}$ can be nonpositive, whatever the provenance; the empirical bias towards nonpositivity is therefore not exploitable beyond that ceiling.

Failed hypothesis or mechanism:
Confusing a richer *description* of a row with a larger *charge* on R. The charge is a gcd against the fixed integers $N-u$, and those integers have finitely many divisors.

What survives:
The bundle lift, the exact valuations and the mirror relation, all PROVED; and the ceiling (IC3), which is new and which also strengthens the vNEXT4 statement (MD1) by dropping the designated-base hypothesis.

Do not repeat:
Before claiming a selection strategy extracts more from R, check it against (IC1), which is provenance-blind.

Related current target:
P1BR-ADDITIVE-COUPLING.

FAIL-BV-BOUNDARY-DESTROYS-MASS-001

FAIL-ID:
FAIL-BV-BOUNDARY-DESTROYS-MASS-001.
Date / iteration:
vNEXT4.15.
Claim:
Low-child states above the current BV-accessible range lose the natural reflected mass because they cannot be cleanly selected.
Status:
KILLED.
Exact correction:
Above $Y_{\mathrm{BV}} = N^{1/(6+2\vartheta)}(\log N)^{-O(1)}$, each distinct state already carries $(1/6 - o(1)) \log N$ weight. A proportional family therefore retains the original $t \log \Lambda$ scale as weighted state mass.
Do not repeat:
The analytic boundary changes the type of object, not its total natural logarithmic weight.

FAIL-BV-INACCESSIBILITY-HIDES-MIRROR-MASS-001

FAIL-ID:
FAIL-BV-INACCESSIBILITY-HIDES-MIRROR-MASS-001.
Date / iteration:
vNEXT4.17; interpretation superseded by vNEXT4.19.
Old description:
Above the then-used same-scale BV threshold the mirror packet becomes analytically invisible.
Status:
SUPERSEDED.
Replacement:
Already vNEXT4.17 converted significant above-threshold mass to distinct active states. vNEXT4.19 further shows that the relevant selector geometry has two independent scales Q, z, so the designated-base range extends almost to $\sqrt{N}$ when z is polylogarithmic.

FAIL-C-SYMBOL-COLLISION-001

FAIL-ID:
FAIL-C-SYMBOL-COLLISION-001.
Date / iteration:
vNEXT2 independent audit.
Claim:
The symbol c may be read uniformly across the Theory Core.
Original motivation:
Both quantities are small integers attached to a packet and were introduced in different rounds.
Status:
SCOPE-CORRECTED.
Minimal / strong counterexample:
Internal. In (LE1)–(LE6), $c = |D_S^{\mathrm{src}}|$ is the *source count*; in (FC7)–(FC10), $c = |C|$ is the *low-core cardinality*. For a canonical R the first is 1 and the second is typically large, so any statement that combines an LE identity with an FC bound while reading c uniformly is wrong by a factor that grows with the world.

Exact certificate:
(LE4) reads $\eta(S)+\nu(S) \geq |S|+c$ with $c = 1$ for a canonical rooted world, whereas (FC8) reads $J_R < N^{c+\tau}(25/18)^\tau / \prod_{q\in C\setminus Q} q$ with $c = |C|$. Substituting one into the other is a category error.

Failed hypothesis or mechanism:
Symbol reuse across rounds without a concordance entry.

What survives:
Both families of statements, unchanged, on their own readings.

Safe weaker version:
Use $\gamma = |C|$ for the low core and reserve c for the source count, as fixed by Convention [conv:c-collision] in the Theory Core. (FC7)–(FC10) are not restated; their c is γ.

Current replacement:
Theory Core, Convention [conv:c-collision].

Do not repeat:
Do not introduce a symbol already bound in an earlier round without a concordance line.

Related current target:
(FC8), K1.

`FAIL-CAB-001`

FAIL-ID:
FAIL-CAB-001.

Date / iteration:
Post-v7.5 CAB audit, 30 August 2026.

Claim:
For every canonical stopped world, $\log B_{\text{abs}} \leq A(\log U_R + \log J_R)$ for a fixed finite A.

Original motivation:
Every absorbed base atom was expected to be financed by power excess or support reuse.

Status:
KILLED without a failure condition.

Minimal / strong counterexample:
$N = 322, p_0 = 163$.

Exact certificate:
$N - p_0 = 159 = 3\cdot 53$, and $322 - 3 = 319 = 11\cdot 29$. The audited stopped world has $U_R = J_R = 1$ but $B_{\text{abs}} = 11 \cdot 29 > 1$. The other first-front state 53 is GOOD because $322 - 53 = 269 \in \mathbb{P}$.

Failed hypothesis or mechanism:
A successful-world local ledger was treated as a failure-world capacity law.

Why the argument failed:
Absorption can occur on the bad branch while another first-front branch already succeeds, leaving zero proposed denominator.

What survives:
This witness does not refute a theorem explicitly conditioned on a closed canonical failure.

Safe weaker version:
Any CAB repair must state the failure domain and a nonzero source term; no such repair is currently proved.

Current replacement:
C0b-RD and the branchwise production program.

Do not repeat:
Never cite $N = 322$ as refuting a failure-conditioned theorem, and never cite CAB unqualified.

Related current target:
$P1^{\text{br}}, K1$.

`FAIL-CAB-INTERNAL-001`

FAIL-ID:
FAIL-CAB-INTERNAL-001.
Date / iteration:
Post-v7.5 CAB audit, 30 August 2026.
Claim:
CAB becomes valid in every canonical stopped world after discarding boundary atoms and retaining only internal absorbed atoms.
Original motivation:
The $N = 322$ defect looked purely external.
Status:
KILLED without a failure condition.
Minimal / strong counterexample:
$N = 858, p_0 = 431, \ell = 23$.
Exact certificate:
$N - p_0 = 427 = 7 \cdot 61$, $858 - 7 = 851 = 23 \cdot 37$, and the audited stopped world has $U_R = J_R = 1$. The atom 23 is internal and absorbed; 37 is a boundary atom. The state 61 is GOOD because $858 - 61 = 797 \in \mathbb{P}$.
Failed hypothesis or mechanism:
Boundary deletion was assumed to remove the zero-payment phenomenon.
Why the argument failed:
The same phenomenon occurs at an internal atom on a bad branch of a globally successful rooted traversal.
What survives:
Again, this successful-world witness does not refute a statement explicitly conditioned on canonical failure.
Safe weaker version:
None is currently proved; require a genuine failure-specific source mechanism.
Current replacement:
C0b-RD and targets $L1, E1, K1$.
Do not repeat:
"Internal" does not mean "already paid".
Related current target:
$K1$.

`FAIL-CANONICAL-BT-CAPACITY-TRANSPORT-TO-EXTERNAL-HQ-001`

FAIL-ID:
FAIL-CANONICAL-BT-CAPACITY-TRANSPORT-TO-EXTERNAL-HQ-001.
Status:
SCOPE-CORRECTED (domain failure).
Killed transfer:
apply the canonical fringe Brun–Titchmarsh prime-power packing theorem unchanged to all of $H_Q \setminus R$.
Certificate:
its decisive population cap is tied to canonical fringe paths / the γ-controlled canonical domain. External descendants do not inherit that cap.
What survives:
the theorem applies on its canonical domain and the overlap $H_Q \cap R$ is negligible in the v4.45 high-band accounting.
Do not repeat:
replacing γ by $|H_Q|$ is a new crude estimate, not a legal transport of the canonical theorem.

`FAIL-CANONICAL-EXACT-LAYER-TRANSFER-MISSING-001`

FAIL-ID:
FAIL-CANONICAL-EXACT-LAYER-TRANSFER-MISSING-001.

Status:
SCOPE-CORRECTED .

Killed formulation:
the v18–v32 exact-layer/star machinery is external and cannot be applied to an exact layer already contained in the canonical world R.

Certificate:
v4.26 already contains `C-CANONICAL-POLYLOG-STAR-ABSORPTION`; exact-layer parent slots, quotient inflation and cofactor conservation are generic exact-layer results. Moreover v4.35 shows that an already extracted polynomial canonical vertical layer has only the full-scale receiving channels J_R, U_R^{nr} and D_C.

What was actually missing at v4.35:
a termination/payment theorem for repeated q-free $U \to U$ migration, or a legal generation/owner transfer to the v28–v32 package. v4.36 then supersedes this dynamic issue as the primary vertical frontier by giving a static upper capacity for the whole U-mass.

`FAIL-CASCADE-DIRECTION-001`

FAIL-ID:
FAIL-CASCADE-DIRECTION-001.

Date / iteration:
vNEXT3.1 correction round.

Claim:
`G-CASCADE-GAMMA-TERMINUS`: the (FC10) cascade closes precisely on $\gamma \le (\log N)^A$, proved by bounding $\sum_i M_i \log z_i \ll (\log N)^A \log N$.

Original motivation:
The selector supplies at most one clean row per designated base and the bases are polylogarithmically bounded, so the total production looked easy to estimate.

Status:
KILLED as stated and as proved.

Minimal / strong counterexample:
The proof itself. (DP5) states that *simultaneous nonpositivity implies* $\sum_i M_i \log z_i < \log J_R$; to conclude that some round is positive one must therefore exhibit a **lower** bound $\sum_i M_i \log z_i \ge \log J_R$. The vNEXT3 proof supplied an **upper** bound $\sum_i M_i \log z_i \ll (\log N)^A \log N$ and used it in that place. An upper bound on a quantity can never certify that the quantity is large; the argument establishes nothing, in either direction.

Failed hypothesis or mechanism:
The two roles of $\sum_i M_i \log z_i$ — what (DP5) concludes about it, and what must be supplied to invoke (DP5) contrapositively — were interchanged.

Why the argument failed:
Counting the *number* of available bases bounds production from above. Forcing a positive round needs the *logarithmic mass* of the bases actually used, bounded from below.

What survives:
The intended conclusion, correctly proved, with the quantifiers made explicit and the domain moved to the global-failure DAG: see `G-SELECTABLE-CLEAN-LOG-MASS` and `G-GAMMA-POLYLOG-FORCES-POSITIVE-ROUND` in the Theory Core. The correct relation between the exponents is $A > B + 1$, with B fixed first.

Safe weaker version:
None; the statement must be reproved from a lower bound, which is what the replacement does.

Current replacement:
Theory Core, Theorem [thm:clean-log-mass] and Theorem [thm:gamma-forces-positive].
Do not repeat:
Before invoking a contrapositive, check which side of the inequality must be supplied and which side is concluded.
Related current target:
γ, `P1BR-POSITIVE-ROUND-CONSEQUENCE`.

`FAIL-CENSUS-DOMAIN-001`

FAIL-ID:
FAIL-CENSUS-DOMAIN-001.
Date / iteration:
Canonical bridge audit §12, item 2.
Claim:
The recorded figure "301" counts all foundational failed upper roots in the scanned range.
Original motivation:
The label was abbreviated in later reports.
Status:
SCOPE-CORRECTED.
Minimal / strong counterexample:
The unit-complement row, already registered as `FAIL-UNIT-COMPLEMENT-001`.
Exact certificate:
The scan enumerated *composite-complement* failed upper roots. A row with $N - r = 1$ also fails, has no children, and is outside that census by construction; $(N, r) = (20, 19)$ is the smallest instance.
Failed hypothesis or mechanism:
A domain-qualified count was quoted without its qualifier.
Why the argument failed:
Every theorem keyed to that census inherits the composite-complement hypothesis.
What survives:
The number itself, with its domain label attached, as a regression fixture. Status VERIFIED COMPUTATION.
Safe weaker version:
Always write "301 composite-complement failed upper roots in the recorded scan".
Current replacement:
Benchmark suite below, where the domain label is preserved.
Do not repeat:
Never quote a census figure without its domain.
Related current target:
C0a and every generic failed-root lemma.

`FAIL-CHILD-C1-001`

FAIL-ID:
FAIL-CHILD-C1-001.
Date / iteration:
Escape-or-Reuse campaign, 30 August 2026.
Claim:
The selected child-prime normalization has universal charging constant $C = 1$.
Original motivation:
Every selected parent financed at most one audited base atom in the focused incidence table.
Status:
KILLED by source-recorded computation.

Minimal / strong counterexample:
The campaign records 53 violations; the empirical maximum is 9.42243 at $N = 406996$.
Exact certificate:
The numerical ratio and the location of its maximum are preserved from the final report. The individual raw child-mass table was unavailable in this extraction, so this certificate was not independently recomputed.
Failed hypothesis or mechanism:
A one-to-one incidence observation was used as a weighted logarithmic inequality.
Why the argument failed:
Multiplicity of financed atoms and size of their weights are different quantities.
What survives:
A more generous partner-prime normalization had empirical maximum 1, but remains computational only.
Safe weaker version:
Treat every finite observed constant as a conjectural target until an exact source-sensitive bound is proved.
Current replacement:
Theory Core target $K1$.
Do not repeat:
Never promote the best observed normalisation to a universal constant.
Related current target:
$K1$.

FAIL-CLEAN-ROOTED-CAP-P1-001

FAIL-ID:
FAIL-CLEAN-ROOTED-CAP-P1-001.
Date / iteration:
Clean-support-load continuation, 1 September 2026.
Claim:
Once one selects pairwise-coprime upper-half external parents, the rooted capacity $\mathcal{C}_N(R) = L(R)$ supplied by the clean base-layer theorem can be beaten by the cofactor production of those same parents, closing P1BR.
Status:
KILLED.
Minimal / strong counterexample:
Structural; no finite witness is needed.
Exact certificate:
In a canonical failed world, every non-root state $r \in R \setminus \{p_0\}$ lies below $N/3$, hence $N - r > 2N/3$. Every selected upper-half parent satisfies $p > p_0$, hence $N - p < m = N - p_0 < N/2$. With at most one selected parent per distinct internal base label, $|\mathscr{P}| \leq |R| - 1$, so

$$L(\mathscr{P}) < (|R| - 1) \log m < \log m + (|R| - 1) \log(2N/3) < L(R).$$

Thus the exact rooted upper bound $K_\times + K(\mathscr{P}) \leq L(R)$ is too large for this naive production choice to exceed.
Failed hypothesis or mechanism:
Solving the collision-selection problem was mistaken for solving the production-versus-capacity comparison.
Why the argument failed:
The clean theorem makes external complements small and noncolliding, but the failed-world rows are individually larger: non-root rows exceed $2N/3$, whereas selected upper-half-parent complements are below $m < N/2$. The order geometry points in the wrong direction for $L(\mathscr{P}) > L(R)$.

What survives:
G-CLEAN-POLYLOG-PARENT-SELECTION and G-CLEAN-BASE-ROOTED-CAPACITY remain exact PROVED statements. They close the W1/K1 collision side on the fixed-polylog base-layer subdomain.

Safe weaker version:
Use the clean selection to isolate the remaining cost: the base-copy surcharge from unintended labels of R. A future P1BR route must control that smaller support set, or use a different common production/capacity functional; it may not simply take $L(R)$.

Do not repeat:
"Pairwise coprime parents" is a capacity statement, not a production surplus.

Related current target:
P1BR, W1, K1.

FAIL-COFACTOR-ENTROPY-ALONE-CLOSES-LOADONE-001

FAIL-ID:
FAIL-COFACTOR-ENTROPY-ALONE-CLOSES-LOADONE-001.

Status:
KILLED.

Killed inference:
once cross-row cofactor radical reuse is identified with incremental J, a large load-one packet automatically has ambient $\Omega(M \log N)$ cofactor entropy.

Certificate:
a common tiny factor such as 3 across M rows carries only $(M-1)\log 3 = O(M)$ entropy while designated high labels have $\Theta(M \log N)$ mass.

What survives:
cofactor entropy is exact horizontal storage, but the correct next reduction is the cube-root row trichotomy / boundary argument.

FAIL-COFACTOR-SIBLING-PRIME-UNSCOPED-001

FAIL-ID:
FAIL-COFACTOR-SIBLING-PRIME-UNSCOPED-001.

Date / iteration:
v1.22 cofactor-window continuation, 1 September 2026.

Claim:
For every internal returned fresh seed, the cofactor $r_j = (N - w_j)/\ell_j$ is prime.

Status:
KILLED.

Why the inference fails:
v1.21 proves only that all prime factors of r_j lie in U and are at least $s = \min U$. Primality follows in v1.22 only when $r_j < s^2$, supplied by the bounded window $r_j < 2\Lambda h < h^2 \le s^2$ under no descent.

What survives:
On $b_j \le \Lambda h$ with $s \ge h > 2\Lambda$, the return cofactor is prime and yields the sibling packet of TF-COFACTOR-WINDOW-SIBLING-RENEWAL.

Do not repeat:
Do not replace cofactor roughness by primality outside the explicit short-window inequality.

Related current target:
TF, above-BV return transport.

`FAIL-CRD2-NEEDED-FOR-FIRST-OCCURRENCE-001`

FAIL-ID:
FAIL-CRD2-NEEDED-FOR-FIRST-OCCURRENCE-001.
Status:
KILLED as a present engineering requirement.
Killed inference:
first/second occurrence, reflected provenance, parenthood, cofactor data or no-double-charge ownership currently require a new fundamental dynamic state variable.
Certificate:
the v4.39 audit finds all theorem-relevant objects reconstructible from full CRD1 together with derived subsets/matchings. No pair with equal full CRD1 state and different theorem-relevant deterministic outcome has been exhibited.
What survives:
introduce CRD2 only if a future proof exhibits non-reconstructible data, genuine history dependence, same-state/different- outcome behaviour, or irreducible persistent ownership. Until then use CRD1-R / CRD1.5 as a derived relation-preserving interface.

`FAIL-CRD2-PREMATURE-001`

FAIL-ID:
FAIL-CRD2-PREMATURE-001.
Date / iteration:
vNEXT4.3–vNEXT4.4.
Claim:
Because the $J, U, \eta, \kappa, \Delta$ ledgers forget parent or bundle provenance, CRD1 must be replaced immediately by a richer fundamental CRD2 state space.
Status:
DESIGN PROHIBITION / FRONTIER CERTIFICATE; not a claim that CRD2 will never be needed.
Exact certificate:
The full ambient object

$$\mathfrak{C}_N = (N, \mathcal{A}_N, p_0, \nu_N), \qquad \nu_N(p,q) = v_q(N-p),$$

already reconstructs active supports/valuations, parent sets, sibling and factorisation hyperedges, exact cofactors $(N-p)/q^e$, inactive residuals, and reflected provenance (j, a_j, c_j). The proven vNEXT4 independence certificates concern compressed ledgers, not equality of the full CRD1 state.
What survives:
Use a relation-preserving derived view (CRD1-R/CRD1.5) when a theorem must keep provenance alive.
Migration trigger:
CRD2 becomes justified only after a theorem-relevant non-reconstructible datum, genuine history dependence, a full-state information-loss pair, or an irreducible ownership state is exhibited.
Related current target:
`C-VERTICAL-LOWCORE-CAPACITY`; CRD2 is dormant.

`FAIL-CRITICAL-VERTICAL-CARRIES-TLOGN-001`

FAIL-ID:
FAIL-CRITICAL-VERTICAL-CARRIES-TLOGN-001.
Date / iteration:
vNEXT4.7.

Claim:
On $h \geq \sqrt{N/3}$, the block-derived vertical family can still absorb a positive proportion of a $t \log N$-scale obligation.
Status:
KILLED.
Exact certificate:
The monomial collapse puts every vertical base in $(\sqrt{2N/3}, \sqrt{N})$, so standard prime counting gives
$$|\mathcal{V}| \ll \sqrt{N}/\log N = O(t/\log N),$$
and exactly one vertical layer per base gives
$$\log U_{\mathcal{V}} \ll \sqrt{N} = O(t) = o(t \log N).$$
What survives:
The critical square-root branch itself is not closed, because production of enough large block factors degenerates there. What is closed is the possibility that an already-produced critical block family can hide its logarithmic cost in vertical layers.
Related current target:
`C-MONOMIAL-REFLECTED-PAIR-SPARSITY`.

`FAIL-CRITICAL-VERTICAL-GENERIC-001`

FAIL-ID:
FAIL-CRITICAL-VERTICAL-GENERIC-001.
Date / iteration:
vNEXT4.7.
Claim:
Near the critical square-root gap, the surviving vertical branch still has arbitrary prime-power/cofactor structure.
Status:
KILLED as a description.
Exact certificate:
On
$$h \geq \sqrt{N/3},$$
every block-derived vertical event is exactly
$$N - w = q^2.$$
What survives:
A specific monomial reflected/quadratic prime-pair correlation, not generic U_R-capacity.

`FAIL-CUBEROOT-DEFECT-PIGEONHOLE-GIVES-POLYNOMIAL-STAR-001`

FAIL-ID:
FAIL-CUBEROOT-DEFECT-PIGEONHOLE-GIVES-POLYNOMIAL-STAR-001.
Status:
KILLED.
Killed inference:
a linear aggregate defect on bases $q \leq N^{1/3}$ automatically yields one base of polynomial multiplicity.
Certificate:
there are $\asymp N^{1/3}/\log N$ available prime bases; for a packet of size $M \asymp N^{1/3}$, plain pigeonhole forces only $\gg \log N$ average multiplicity.

Do not repeat:
do not replace the current low-layer consumer by an unweighted single-base pigeonhole.

`FAIL-CUBEROOT-PIGEONHOLE-IS-BOUNDARY-CASE-001`

FAIL-ID:
FAIL-CUBEROOT-PIGEONHOLE-IS-BOUNDARY-CASE-001.

Status:
SCOPE-CORRECTED.

Earlier failure:
`FAIL-CUBEROOT-DEFECT-PIGEONHOLE-GIVES-POLYNOMIAL-STAR-001` correctly kills raw cube-root pigeonhole as a concentration theorem.

New explanation:
this is exactly the boundary value $m = 2$ of `G-POWER-THRESHOLD-DEFECT-COMPRESSION`: the threshold $N^{1/(m+1)}$ is then $N^{1/3}$ and the guaranteed star exponent $1/3 - 1/(m+1)$ is zero. Every fixed $m \geq 3$ gives a nontrivial polynomial star.

What survives:
do not infer polynomial concentration from cube-root pigeonhole alone; first split the alphabet at a lower power threshold.

`FAIL-CYCLE-COFACTOR-LOCK-ALONE-CLOSES-DEEP-CYCLES-001`

FAIL-ID:
FAIL-CYCLE-COFACTOR-LOCK-ALONE-CLOSES-DEEP-CYCLES-001.

Source:
explicit v4.53 Failure Log addition.

Status:
KILLED as a sufficient uniform closure mechanism.

Reason:
the exact relation

$$Q^L D = mN + (-1)^L$$

is useful, but in the small-Q regime it does not by itself bound the population of non-pure cycles at the required polynomial scale.

`FAIL-DC-CAPACITY-COMPRESSION-001`

FAIL-ID:
FAIL-DC-CAPACITY-COMPRESSION-001.

Status:
SCOPE-CORRECTED.

Killed form:
within the Q-indexed scalar route of `C-Q-INDEXED-SCALAR-PRODUCTION-SUBCRITICALITY`, increasing D_C cannot lower FC8 below the hard $|Q|$-scale floor needed by that same production.

What survives:
once a proposed production violates the one-logarithmic-unit per-Q-index hypothesis, the exact term $-\log D_C$ in A_R matters again. Do not promote this entry to a universal ban on denominator compression.

`FAIL-DEADSMOOTH-CAN-CARRY-POLYNOMIAL-HIGHGAP-PACKET-001`

FAIL-ID:
FAIL-DEADSMOOTH-CAN-CARRY-POLYNOMIAL-HIGHGAP-PACKET-001.

Status:
KILLED.

Killed inference:
the dead-smooth carrier or reflected classes may carry a fixed polynomial fraction of a high-gap reflected packet.

Certificate:
v4.40 counts integers supported on the fixed alphabet $\mathrm{PF}(x)$ by exponent vectors and obtains only $x^{o(1)}$ possibilities below $2x$. Hence $|\mathrm{DS}|, |\mathrm{RDS}| = N^{o(1)} = o(t)$ on $h > N^{1/3}$.

Compatibility:
`FAIL-DEADSMOOTH-UNBOUNDED-001` remains valid. It rejected a finite/uniform inference from the old local ceiling. The new statement is asymptotic fixed-support counting and does not assert that two dead-smooth rows are impossible or that the class is absolutely bounded.

`FAIL-DEADSMOOTH-PILLAI-001`

FAIL-ID:
FAIL-DEADSMOOTH-PILLAI-001.

Date / iteration:
Phase-3 continuation, 31 August 2026.

Claim:
No canonical N carries two simultaneous dead-smooth carriers, since none was found below $3 \cdot 10^5$.

Original motivation:
The search returned nothing at all, and the configuration looks arithmetically absurd.

Status:
SCOPE-CORRECTED. The *search result* stands as `COMP-DEADSMOOTH-CENSUS`; the *exclusion* does not.

Minimal / strong counterexample:
None is needed: the claim has no proof.

Exact certificate:
By `TD-DEADSMOOTH-NORMALFORM`(5), two simultaneous dead-smooth carriers with $\omega(a_j) = 1$ are exactly a Pillai coincidence $|q^e - q'^{e'}| < h$ with $q^e, q'^{e'} > 4x/5$, $q, q' < h$ and $qq' \mid x$. Excluding that unconditionally requires effective linear forms in logarithms. The corpus contains no such argument.

Failed hypothesis or mechanism:
An empty finite search was read as an exclusion theorem.

Why the argument failed:
This is `PROHIB-STATUS-LADDER`: a finite computation, however clean, is not a universal statement.

What survives:
The exact reduction to a Pillai equation, which is a genuine normal form and the right object for a future attack.

Safe weaker version:
"No such pair occurs below $3 \cdot 10^5$", labelled VERIFIED COMPUTATION.

Current replacement:
Theory Core, `TD-DEADSMOOTH-NORMALFORM`(5) and `COMP-DEADSMOOTH-CENSUS`.

Do not repeat:
Never convert an empty search into an exclusion.

Related current target:
TD-SMALLSMOOTH.

`FAIL-DEADSMOOTH-UNBOUNDED-001`

FAIL-ID:
FAIL-DEADSMOOTH-UNBOUNDED-001.

Date / iteration:
Phase-3 continuation, 31 August 2026.

Claim:
The exceptional dead-smooth class is negligible for every N, so the side condition $\lceil t/3 \rceil > \sigma_N^+(h)$ of `TD-COB-CLOSURE` can be discharged and removed.

Original motivation:
Dead-smooth carriers are extraordinarily rare — 19 among 514 447 tested, none doubled — which invites the conclusion that the class is uniformly negligible.

Status:
KILLED.

Minimal / strong counterexample:
$N = 60\,060$, $p_0 = 30\,047$, $h = 17$, $t = 8$.

Exact certificate:
$x = 30\,030 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$ is a primorial, so $\mathcal{P}_N(h) = \{3, 5, 7, 11, 13\}$ — every odd prime below h divides x. The ceiling then degenerates: $\sigma_N^+(h) = 6.32$ while $\lceil t/3 \rceil = 3$. The side condition fails.

Failed hypothesis or mechanism:
Rarity of the class in a sample was confused with a bound valid for every N. The ceiling depends on how many small primes divide x, and x may legitimately be a primorial.

Why the argument failed:
σ^{dead} and σ^+ are small *because* x usually has few small prime factors, not because the dead-smooth normal form is intrinsically bounded. No theorem forbids $x = \prod_{\ell \le h} \ell$.

What survives:
`TD-DEADSMOOTH-CEILING` and `TD-COB-CLOSURE` exactly as stated, with the side condition retained as a hypothesis. It is an arithmetic condition on N alone, checkable in $O(\pi(h) \log x)$ operations, and it held in every tested canonical instance with $t \ge 16$.

Safe weaker version:
State the closure with the side condition, and treat the small-smooth-x regime as a separate open case.

Current replacement:
Theory Core, `TD-COB-CLOSURE` and the retained target TD-DENSITY.

Do not repeat:
Do not discharge a hypothesis because a sample makes it look automatic; check the extremal arithmetic first.

Related current target:
TD-DENSITY, $P1^{\text{br}}$.

`FAIL-DEEP-AFFINE-PATH-RESIDUE-IS-CAPACITY-001`

FAIL-ID:
FAIL-DEEP-AFFINE-PATH-RESIDUE-IS-CAPACITY-001.

Alias:
v4.50 R50-1.

Status:
KILLED for the capacity inference.

Exact theorem retained:
for any actual path $v_0 \rightsquigarrow p$,

$$Q \mid N - p \Longrightarrow \Xi(\mathcal{P}) \equiv v_0 N^{-1} \pmod{Q}.$$

Why rejected as capacity:
this residue is forced by the endpoint through the exact affine path identity and imposes no independent restriction on intermediate vertices or cofactors.

Replacement theorem:
`G-DEEP-PATH-RESIDUE-ENDPOINT-TAUTOLOGY`.

`FAIL-DEEP-FIBRE-NEEDS-FIRSTENTRY-OWNER-ESCAPE-001`

FAIL-ID:
FAIL-DEEP-FIBRE-NEEDS-FIRSTENTRY-OWNER-ESCAPE-001.
Source:
explicit v4.54 Failure Log addition and correction to v4.53.
Status:
KILLED as a proof-design necessity.
Reason:
every pure path terminates at a non-pure row, its length is $N^{o(1)}$, and the refined fibre theorem already forces polynomial boundary defect/off-layer support without first passing through ancestry predecessors.
Supersession:
`C-DEEP-FIRST-ENTRY-OWNER-CAPACITY` is SUPERSEDED as the binding frontier. The v4.53 first-entry matching remains a valid secondary local diagnostic.

`FAIL-DEEP-FIBRE-TO-SUBPOLY-BY-ANCESTRY-ALONE-001`

FAIL-ID:
FAIL-DEEP-FIBRE-TO-SUBPOLY-BY-ANCESTRY-ALONE-001.
Source:
explicit v4.52 Failure Log addition.
Status:
KILLED as a proof mechanism from one reflected root, common-sink topology, total path cofactor products, and bounded branching alone.
Reason:
the current corpus yields renewal alternatives rather than a subpolynomial fibre bound.
Scope guard:
this does not refute a stronger arithmetic fibre theorem using edge-local valuation or additional selected-family structure.

`FAIL-DEEP-ONE-AP-PACKING-CLOSES-001`

FAIL-ID:
FAIL-DEEP-ONE-AP-PACKING-CLOSES-001.
Status:
KILLED as a sufficient closing mechanism.
Killed route:
use only the fact that a selected row/index lies in one residue class modulo the deep modulus Q.
Reason:
the resulting ceiling of shape $1 + N/(6Q)$ is too weak when $Q = N^{o(1)}$.

`FAIL-DEEP-PURE-QMONOMIAL-CAN-CYCLE-001`

FAIL-ID:
FAIL-DEEP-PURE-QMONOMIAL-CAN-CYCLE-001.
Source:
explicit v4.53 Failure Log addition.
Status:
KILLED.

Killed possibility:
the pure subgraph defined by $N - p = Qq$ can support a nontrivial active cycle.
Reason:
the affine fixed-point identity of v4.53 forbids every nontrivial active cycle in the pure $d = 1$ subgraph.
Scope guard:
non-pure cycles with variable multipliers d_i are not ruled out by this statement; they satisfy the separate multiplier law.

`FAIL-DEEP-SELFRENEWAL-IS-INDEPENDENT-TERMINAL-001`

FAIL-ID:
FAIL-DEEP-SELFRENEWAL-IS-INDEPENDENT-TERMINAL-001.
Source:
explicit v4.53 Failure Log addition and v4.53 frontier correction.
Status:
KILLED as a proof-design description.
Killed description:
a cheap polynomial deep fibre may require an independent global theorem controlling exact-layer self-renewal as a terminal phenomenon.
Reason:
the v4.53 stability theorem forces the cheap fibre into short pure affine paths whose starts expose boundary owners / off-layer support.
Supersession:
`C-DEEP-SELF-RENEWAL-QCOFACTOR-CAPACITY` is SUPERSEDED as the standalone binding frontier. This does not say all renewed external population is contradictory.

`FAIL-DEEP-U-TO-U-RETURN-AS-PRIMARY-FRONTIER-001`

FAIL-ID:
FAIL-DEEP-U-TO-U-RETURN-AS-PRIMARY-FRONTIER-001.
Status:
SUPERSEDED as a primary frontier.
Superseded description:
v4.35 temporarily treated repeated migration of deep vertical mass to new q-free bases as the smallest obstruction.
Certificate:
v4.36 proves, pending audit, the static global capacity

$$\log U_R^{\mathrm{nr}} \ll \sqrt{\frac{N\gamma}{\log(2+\gamma)}},$$

without proving dynamic termination.
What survives:
nested peeling remains useful for structural analysis of near-extremal packets, but it is not necessary merely to prevent unbounded vertical storage.

`FAIL-DESCENT-ONE-STATE-001`

FAIL-ID:
FAIL-DESCENT-ONE-STATE-001.
Date / iteration:
vNEXT4.15.
Old description:
Passing immediately to the full descendant closure reduces the descent branch to the existence

of one state $r \le h$.

Status:

SUPERSEDED.

Replacement:

G-FIRST-ROW-HIGHLOW-DISPOSITION retains the selected first parent rows and yields full logarithmic defect, weighted state expansion, explicit horizontal reuse, a fixed-polylog population, or a quantitative reciprocal band.

FAIL-DIRECT-ADJACENCY-001

FAIL-ID:

FAIL-DIRECT-ADJACENCY-001.

Date / iteration:

Escape-or-Reuse campaign, 30 August 2026.

Claim:

Every internally absorbed atom is incident, or uniformly adjacent, to a GOOD escape edge in its first-front branch.

Original motivation:

Most observed witnesses were shallow.

Status:

KILLED for direct adjacency.

Minimal / strong counterexample:

$N = 14198$, internal atoms 19 and 167.

Exact certificate:

The audited sets are $B_{\mathrm{abs,int}} = \{19, 167\}$ and $B_{\mathrm{re}} = \{17, 83, 149\}$. Both internal atoms use the same merged GOOD edge $71 \to 277$, reached only at branch distances 8 and 7, respectively. Atom 19 has 93 external parents and atom 167 has 12; neither is directly incident to that GOOD edge.

Failed hypothesis or mechanism:

Typical shallow depth was promoted to a local theorem.

Why the argument failed:

Branches can merge long before their eventual escape.

What survives:

A nonlocal branch-escape disjunct remains viable.

Safe weaker version:

Reachability of a GOOD edge, with no fixed adjacency bound.

Current replacement:

Theory Core target $E1$.

Do not repeat:

Do not charge only the nearest one or two edges.

Related current target:

$E1, W1$.

FAIL-ETA-EQUIVALENT-TO-WEIGHTED-J-001

FAIL-ID:

FAIL-ETA-EQUIVALENT-TO-WEIGHTED-J-001.

Status:

KILLED.

Rejected inference:

an unweighted lower bound on $\eta(R) = \sum_q (\kappa_R(q) - 1)$ is equivalent to the corresponding lower bound on $\log J_R$.

Reason:
exactly

$$\log J_R = \sum_q (\kappa_R(q) - 1) \log q,$$

while only $\log J_R \geq \eta(R) \log 3$ follows from the unweighted count.

`FAIL-EXACTD-NONCANONICAL-001`

FAIL-ID:
FAIL-EXACTD-NONCANONICAL-001.

Date / iteration:
Exact-D audit; reaffirmed in the canonical bridge audit §11.2.

Claim:
The Exact-D local row algebra — pair compensation, smoothness, primitive normal form, prime endpoint — is enough to isolate the canonical root, so canonicity need not be used as boundary transport.

Original motivation:
The local conditions are sharp and eliminate most models.

Status:
KILLED as a substitute for minimality; SCOPE-CORRECTED as machinery.

Minimal / strong counterexample:
The recorded noncanonical normal forms, including $(q, r, m_0) = (3,\ 5\,413\,103,\ 81\,196\,545)$ — note $81\,196\,545 = 3 \cdot 5 \cdot 5\,413\,103$ — together with the two recorded $k = 1$ examples.

Exact certificate:
These models satisfy the local D_q, D_r conditions, smoothness, pair failure and a prime endpoint, and are eliminated *only* by $h < G$. Removing $h < G$ restores them.

Failed hypothesis or mechanism:
Row algebra was asked to do the work of the empty prime interval.

Why the argument failed:
Local normal forms constrain a row; they do not constrain the position of the root relative to x.

What survives:
`ARS-EXACTD`: pair compensation outside $q < r < 2h$, the primitive normal form, and $h^2 > 4m$ in the residual regime — all under an Exact-D *pair failure* hypothesis, not under CRD failure alone.

Safe weaker version:
Use Exact-D only where the pair-failure hypothesis is explicit, and state $h < G$ as a separate input.

Current replacement:
Theory Core, `ARS-EXACTD` and the ablation discussion of $h < G$.

Do not repeat:
Do not attempt to repair Exact-D by improving a constant; the corpus already localises the missing step as interval-to-ledger transport.

Related current target:
C0a, T_D, $P1^{\text{br}}$.

`FAIL-EXACTD-RESIDUAL-SURVIVES-SUBROOT-001`

FAIL-ID:
FAIL-EXACTD-RESIDUAL-SURVIVES-SUBROOT-001.

Date / iteration:
vNEXT4.20.

Branch:
Retain Exact-D Pair Compensation failure as an obstruction when $h \le \sqrt{x}$.
Status:
KILLED .
Certificate:
Existing Exact-D high-gap amplification says pair failure forces $h^2 > 4m$. But $h \le \sqrt{x}$ gives $h^2 \le x < 4m$ for sufficiently large x. Thus all first-front SMOOTH pairs satisfy Exact-D compensation on this domain.
Scope guard:
This closes only the pair-failure branch; it does not by itself strengthen the global weighted-cut/K1 bound.

FAIL-EXTERNAL-ETA-IDENTIFICATION-001

FAIL-ID:
FAIL-EXTERNAL-ETA-IDENTIFICATION-001.
Date / iteration:
vNEXT2 independent audit, Phase 3.
Claim:
The high-floor reuse $\eta(U) \ge \lceil n/3 \rceil + c$ of an external orthogonal packet can be charged to the canonical capacity ledger.
Original motivation:
η is the same functional in both places, and a large value looks like a large obligation wherever it occurs.
Status:
KILLED .
Minimal / strong counterexample:
Structural, and it is exactly the orthogonality that was thought to help.
Exact certificate:
The packet is orthogonal, $K_\times(R, U) = 0$. That identity says precisely that U contributes nothing to the rooted collision ledger. Any functional built from cross-collision is therefore identically zero on this configuration, so no such functional can carry $\eta(U)$ into $\mathcal{C}_N(R)$.
Failed hypothesis or mechanism:
Orthogonality was read as a licence to add the two ledgers; it is the obstruction to doing so.
What survives:
`TF-HIGH-FLOOR-HORIZONTAL-RIGIDITY` itself, now PROVED with both of its skeleton gaps closed.
Safe weaker version:
Report $\eta(U)$ as a property of U only.
Current replacement:
Theory Core, Proposition [prop:missing-bridge], which states the exact minimal lemma required: a *source-side* statement that re-closing $R \cup U$ changes the multisource product by a factor whose logarithm is bounded below in terms of $\eta(U)$.
Do not repeat:
Never add an orthogonal packet's ledger to the rooted one; the zero cross-term is the reason it cannot be done.
Related current target:
TF, K1, W1.

FAIL-EXTERNAL-FIXED-POLYLOG-SQUAREFREE-PENDING-001

FAIL-ID:
FAIL-EXTERNAL-FIXED-POLYLOG-SQUAREFREE-PENDING-001.

Date / iteration:
vNEXT4.15.
Old obstruction:
The fixed-polylog external BAD clean bridge lacked an explicit squarefree-parent theorem.
Status:
CLOSED PENDING INDEPENDENT AUDIT.
Repair:
Large-square thinning applies to the external BAD parent fans as well, yielding squarefree, pairwise-coprime complements with exact designated valuation one and no unintended fixed-polylog factors.

`FAIL-EXTERNAL-MERGER-FIRST-SPLIT-001`

FAIL-ID:
FAIL-EXTERNAL-MERGER-FIRST-SPLIT-001.
Date / iteration:
vNEXT4.14.
Strategy:
Partition external seeds into merger/disjoint subfamilies before analysing source retention.
Status:
SUPERSEDED as proof organisation.
Replacement:
Union all failed seed basins first. The union is child-closed below $N/3$, source retention applies to the whole family, and one overlap ledger covers either orthogonality or nonempty intersection with R.

`FAIL-EXTERNAL-MULTISOURCE-CRT-IS-THE-NEXT-CAPACITY-BRIDGE-001`

FAIL-ID:
FAIL-EXTERNAL-MULTISOURCE-CRT-IS-THE-NEXT-CAPACITY-BRIDGE-001.
Status:
KILLED relative to the current data.
Rejected strategic description:
The next optimal step is to force a contradiction from source-front CRT compatibility of the external v18 star.
Reason:
the source-front lock constrains one affine coefficient but does not upper-bound it, and the non-q_0 moduli are largely pairwise coprime, so CRT compatibility is not itself a capacity obstruction.
Current replacement:
attack the canonical root-tied target `C-FIRSTFRONT-RESIDUAL-TO-LOWCORE-CAPACITY`; retain K1-INC and owner-anchored incremental capacity as secondary global routes.

`FAIL-EXTERNAL-SOURCE-MASS-IS-INDEPENDENT-RADICAL-RESERVOIR-001`

FAIL-ID:
FAIL-EXTERNAL-SOURCE-MASS-IS-INDEPENDENT-RADICAL-RESERVOIR-001.
Status:
SUPERSEDED as a primary factorisation branch.
Superseded description:
external seed/source weight must remain as a third independent radical-storage reservoir in addition to fresh support and incremental J.

Certificate:
v4.39 proves the exact extension identity

$$\prod_{a \in A} \operatorname{rad}(N-a) = F_R(A)\frac{J_S}{J_R}.$$

All new complement radical mass is therefore stored in one baseline copy of fresh support plus rooted/repeated copies in incremental J.

What survives:
Σ_{ext} remains useful for source-sensitive congruence/product arguments, but it is not needed as an independent primary factorisation leaf.

FAIL-EXTERNAL-SUCCESS-001

FAIL-ID:
FAIL-EXTERNAL-SUCCESS-001.

Date / iteration:
Escape-or-Reuse campaign, 30 August 2026.

Claim:
If an internal atom has an active parent outside the current bad world, then that parent supplies a successful rooted escape.

Original motivation:
Leaving the current closed set was conflated with entering the global success basin.

Status:
KILLED.

Minimal / strong counterexample:
$(N, \ell, p) = (6142, 877, 3511)$.

Exact certificate:
$3511 \in \mathcal{A}_{6142}$, it lies outside the audited p_0-world, and $6142 - 3511 = 2631 = 3 \cdot 877$; hence it is an external parent of 877. The independently checked stopped traversal rooted at 3511 fails.

Failed hypothesis or mechanism:
"External to R" was replaced by "outside the global bad basin."

Why the argument failed:
A different failed basin can feed the same atom.

What survives:
The parent is an exact ambient incidence and can be promoted into an enlarged valuation ledger.

Safe weaker version:
External active parent *or* internal branch escape, with no success claim about the parent.

Current replacement:
Theory Core, G-AMBIENT-PROMOTION and targets $E1, W1, K1$.

Do not repeat:
Existence of a parent is not success of that parent.

Related current target:
$E1, W1$.

FAIL-EXTERNAL-UT-IDENTITY-IS-UPPER-CAPACITY-001

FAIL-ID:
FAIL-EXTERNAL-UT-IDENTITY-IS-UPPER-CAPACITY-001.

Status:
KILLED.

Killed inference:
the exact relative identity $T_{S^*}/T_R = (U_{S^*}/U_R)(J_{S^*}/J_R)$ and source lock automatically bound

the external vertical factor from above.

Certificate:
increasing an exact valuation from 1 to e enlarges Δ_U without changing the support graph, source set, or source-lock residue class. The known canonical upper capacity is for canonical U_R^{nr}, not arbitrary external incremental U.

What survives:
the identity is a strong exact payment ledger, but a new receiving-capacity theorem would be required to close it.

`FAIL-F1-DOMAIN-MIX-001`

FAIL-ID:
FAIL-F1-DOMAIN-MIX-001.

Date / iteration:
Interrupted T_D round, dry run, 31 August 2026.

Claim:
A falsifier run over mixed controls can refute a candidate statement whose domain is "closed failed world".

Original motivation:
A single sweep over many roots was cheaper than three separate domains.

Status:
SCOPE-CORRECTED .

Minimal / strong counterexample:
The dry run itself.

Exact certificate:
The recorded log states that the candidate F1 had domain "closed failed world" while the code mixed in successful-envelope controls; the domain filter and the counter for fresh states belonging to R were corrected, and three separate domains were preserved thereafter. This is a VERIFIED COMPUTATION record of a tooling defect, not a mathematical failure of F1.

Failed hypothesis or mechanism:
Domain bookkeeping inside the falsifier, not the statement under test.

Why the argument failed:
A counterexample from the wrong domain is not a counterexample.

What survives:
The candidate statement, untouched, still awaiting a correctly scoped test.

Safe weaker version:
Every falsifier must declare and enforce its domain before it may report a kill.

Current replacement:
The benchmark suite below, in which every row carries its domain and evidence class.

Do not repeat:
Never let a falsifier report a kill it obtained outside the declared domain.

Related current target:
TF, T_D.

`FAIL-FC8-FLOOR-FOR-RHS-001`

FAIL-ID:
FAIL-FC8-FLOOR-FOR-RHS-001.

Status:
KILLED .

Rejected inference:
proving $\log J_R > (|Q| + \tau) \log N$ automatically contradicts FC8.

Reason:
$(|Q|+\tau)\log N$ is only a lower floor for

$$A_R = (\gamma+\tau)\log N + \tau \log \frac{25}{18} - \sum_{q \in C\setminus Q} \log q.$$

The exact target must use the exact right-hand side, not a lower bound for it.

FAIL-FC8-SELF-CONTRADICTION-001

FAIL-ID:
FAIL-FC8-SELF-CONTRADICTION-001.

Date / iteration:
vNEXT2 independent audit, Phase 2.

Claim:
The low-core capacity bound (FC8), being an upper bound on J_R under canonical failure, can be contradicted by the forced lower bounds already available, closing the $\mathcal{D}$/low-core front.

Original motivation:
(FC8) is the first genuine upper capacity for J_R, and upper capacities are what a contradiction architecture wants.

Status:
KILLED. (FC8) itself is PROVED; the inference from it is not.

Minimal / strong counterexample:
Structural; no finite witness needed.

Exact certificate:
The only unconditional lower bound on J_R available from canonical failure is $J_R \geq 3^{\eta(R)}$ by (LE3). A contradiction with (FC8) therefore requires $\eta(R) > (\gamma+\tau)\log_3 N + \tau \log_3(25/18) - \sum_{C\setminus Q} \log_3 q$, that is $\eta(R) \gtrsim \gamma \log N$. Nothing in the corpus forces a reuse count carrying a factor $\log N$; the strongest available floor is $\eta(R) \geq |Q| + |H|$ (Theorem [thm:lowcore-floor]), which is bounded by $|R|$.

Failed hypothesis or mechanism:
An upper capacity was treated as self-contradictory without a matching lower bound of the same order.

Why the argument failed:
The two sides differ by a factor $\log N$ per low-core element; that gap is not an accident of effort but of what the available identities can produce.

What survives:
(FC8) as an upper capacity usable through the external comparison (FC10), exactly as its own capacity paragraph states.

Safe weaker version:
Report (FC8) only in the (FC10) form.

Current replacement:
Theory Core, Proposition [prop:fc8-barrier].

Do not repeat:
Do not attempt another self-contained contradiction from (FC8); the barrier is at the level of available bounds.

Related current target:
K1, P1BR.

FAIL-FC8-SLACK-IS-VERTICAL-MASS-001

FAIL-ID:
FAIL-FC8-SLACK-IS-VERTICAL-MASS-001.

Status:
SCOPE-CORRECTED .
Killed inference:
a large $\Delta_{\mathrm{FC8}} = B_R - \log(D_C J_R)$ by itself forces a comparably large $\log U_R^{\mathrm{nr}}$.
Certificate:
v4.35 gives the exact identity

$$\Delta_{\mathrm{FC8}} = \log U_R^{\mathrm{nr}} + \Delta_{\mathrm{full}}, \qquad \Delta_{\mathrm{full}} > 0.$$

The positive full-layer row/path envelope slack is an independent part of the FC8 slack.
What survives:
LCAP1/LCAP4 may still require vertical mass after the size of $D_C J_R$ has been specified.

FAIL-FIRSTENTRY-SQUARE-ONLY-HAS-Q-SPACING-001

FAIL-ID:
FAIL-FIRSTENTRY-SQUARE-ONLY-HAS-Q-SPACING-001.
Source:
explicit v4.54 Failure Log addition.
Status:
KILLED .
Killed description:
the square first-entry packet can only be spaced using one residue class modulo Q.
Correction:
a pure-path start satisfies

$$p_i \equiv N(1-Q) \pmod{Q^2},$$

so distinct square first-entry targets have spacing at least $2Q^2$.

FAIL-FIRSTFRONT-CONTAINMENT-CLAIMED-NEW-001

FAIL-ID:
FAIL-FIRSTFRONT-CONTAINMENT-CLAIMED-NEW-001.
Status:
KILLED .
Rejected description:
$Q \subseteq C$ was missing from the previous Core.
Certificate:
it is explicitly used in the independently audited FC7/FC8 proof and audit. v4.34 only extracted it for a new scale comparison.

FAIL-FIRSTFRONT-RESIDUAL-SCALE-IS-CONTROLLED-ONLY-BY-Q-001

FAIL-ID:
FAIL-FIRSTFRONT-RESIDUAL-SCALE-IS-CONTROLLED-ONLY-BY-Q-001.
Status:
KILLED on $h \le \sqrt{x}$.
Rejected statement:
The only general lower scale for $M_q = (N-q)/\gcd(m, N-q)$ is the old $M_q > (5/3)q$ bound.
Certificate:
canonical shadow geometry gives $D_q = \gcd(m, |2h-q|)$ and uniformly $M_q > (5/6)h$; when $q \le 4h$, $M_q > (5/6)\Lambda h$.

`FAIL-FIRSTFRONT-SCALAR-ROUTE-001`

FAIL-ID:
FAIL-FIRSTFRONT-SCALAR-ROUTE-001.
Status:
KILLED with the scope below.
Killed inference:
a production indexed by $S \subseteq Q$, contributing less than one ambient logarithmic unit per index, can close the current FC8 capacity without multiplicity amplification or a new structural upper bound.
Certificate:
$Q \subseteq C$ is already proved and the exact FC8 right-hand side A_R satisfies $A_R > |Q| \log N$, whereas such a production is $< |Q| \log N$.
What survives:
first-front shadow geometry, H_3/H_5 structure, multiplicity-amplified arguments, independent capacity compression, analytic input, and non-scalar contradictions are not killed.

`FAIL-FIXED-FINITE-MIRROR-HUB-001`

FAIL-ID:
FAIL-FIXED-FINITE-MIRROR-HUB-001.
Date / iteration:
vNEXT4.16.
Escape:
Store all active smooth-mirror valuation mass on an N-independent finite family of small primes.
Status:
KILLED.
Exact certificate:
The mirror support capacity

$$M_{\text{act}}^{-} \leq t \sum_{r \in P_{\text{mir}}} \frac{\log r}{r-1} + |P_{\text{mir}}| \log N$$

is only $O(t) + O(\log N)$ for fixed finite support, while the natural smooth branch requires $M_{\text{act}}^{-} \gg t \log \Lambda$ with $\Lambda \to \infty$.
Scope guard:
This does not say the direct least-prime children of the original parent rows cannot all equal 3; it says the mirror side must then create additional growing active support.

`FAIL-FIXED-K-UNIFORM-SELBERG-FREE-001`

FAIL-ID:
FAIL-FIXED-K-UNIFORM-SELBERG-FREE-001.
Date / iteration:
vNEXT4.12.
Claim:
A standard fixed-k Selberg sieve immediately gives

$$\#\{q : q,\ N - kq^2 \text{ prime}\} \ll \sqrt{N/k}/\log^2 N$$

with one absolute constant uniformly over all polylogarithmic odd k.
Status:
KILLED.
Exact obstruction:
The fixed-k local root count contains the quadratic character (kN/p); uniform singular-series

control is an additional theorem obligation.

Replacement:

Sum over k first and use the hybrid dimension-one linear sieve in the coefficient variable.

`FAIL-FRESH-BRANCH-PRIMACY-001`

FAIL-ID:

FAIL-FRESH-BRANCH-PRIMACY-001.

Date / iteration:

Phase-3 continuation, 31 August 2026.

Claim:

The external fresh branch $\mathcal{D}_{\mathrm{fr}}^{\mathrm{ext}}$ is where the $\mathcal{D}$ obligation is strongest, so cut statements should be developed there first and restricted to it.

Original motivation:

External carriers are candidate *new sources* and therefore the ones that bear on $P1^{\mathrm{br}}$; that relevance was transferred to the cut itself.

Status:

SCOPE-CORRECTED .

Minimal / strong counterexample:

Structural, by `TD-LOCALIZED-CUT`.

Exact certificate:

For $\tau \geq 2$ distinct active primes bounded by y, $C_{\mathrm{cut}} > \tau(\tau-1)\log((N-y)/y)$. A fresh carrier gives only $y = N/6$ and hence the constant $\log 5$; a live smooth carrier gives $y < h$ and hence $\log((N-h)/h)$, larger by orders of magnitude once $h = o(N)$. The fresh branch is therefore the *weakest* of the three for cut purposes.

Failed hypothesis or mechanism:

Relevance to $P1^{\mathrm{br}}$ was conflated with strength of the cut.

Why the argument failed:

Two different functionals were being compared. Only external carriers can be new sources; but the cut constant is governed by the *localisation* of the primes, and small primes localise better.

What survives:

Both facts, kept apart. Only the external branch bears on $P1^{\mathrm{br}}$; the live-smooth branch gives the largest cut defect.

Safe weaker version:

State the cut with the explicit bound y and read the constant off it, rather than fixing $\log 5$.

Current replacement:

Theory Core, `TD-LOCALIZED-CUT`, with `TD-FRESH-CUT` and `TD-FIVEFOLD-CUT` as its $y = N/6$ cases.

Do not repeat:

Do not restrict a theorem to the sub-branch that happens to be relevant downstream; state it where its proof lives.

Related current target:

T_D, $P1^{\mathrm{br}}$.

`FAIL-FRINGE-ALPHABET-TRANSFER-001`

FAIL-ID:

FAIL-FRINGE-ALPHABET-TRANSFER-001.

Date / iteration:

vNEXT2 independent audit, Phase 4.

Claim:

For $\lambda_N > 3$ the upper fringe is charged by the alphabet $\{\lambda, 2\lambda - 1\}$, by analogy with $\{3, 5\}$.

Original motivation:
$\{3, 5\} = \{\lambda, 2\lambda - 1\}$ at $\lambda = 3$, and (AN7) features $2\lambda - 1$.
Status:
SCOPE-CORRECTED .
Exact certificate:
The admissible coefficients are determined by (AN5), $N(1 - 1/\lambda)/q < b < N/q$, together with b odd, $(b, N) = 1$, $b \geq \lambda$. For q ranging over $(N/(2\lambda), N/\lambda)$ this interval has length $N/(\lambda q) \in (1, 2)$ only near the top of the band; lower in the band it admits several odd values, so the alphabet is a range, not a pair. The two-letter alphabet at $\lambda = 3$ is an accident of $2 < b < 6$ admitting exactly $\{3, 5\}$.
Failed hypothesis or mechanism:
A coincidence at one value of λ was read as a pattern.
What survives:
(AN5) itself, which is the correct general statement and the right starting point: compute the alphabet from the interval before proposing any potential.
Safe weaker version:
For general λ, state the admissible coefficient set as the odd integers coprime to N in the (AN5) interval.
Current replacement:
Theory Core, Phase-4 note in Section [sec:vnext2].
Do not repeat:
Do not propose a monotone potential of $D(q) = 4q - N$ type for $\lambda > 3$ before the alphabet is computed; D was built from the coefficient 3.
Related current target:
$\lambda_N > 3$ fringe capacity.

FAIL-FRINGE-BOUNDED-MULTIPLICITY-001

FAIL-ID:
FAIL-FRINGE-BOUNDED-MULTIPLICITY-001.
Status:
KILLED .
Certificate:
the first-front-to-fringe map is already injective and the H_3, H_5 bands are disjoint. Thus there is no hidden large fibre to overflow.
What survives:
an independent theorem proving that the relevant incidences cannot exist at all, or cannot coexist with canonical high-gap geometry, remains possible.

FAIL-FRINGE-BRANCH-DOUBLE-CHARGE-001

FAIL-ID:
FAIL-FRINGE-BRANCH-DOUBLE-CHARGE-001.
Status:
SCOPE-CORRECTED .
Killed inference:
count the same fringe entrance/path population τ as a new independent scalar payment after FC8/LCAP4 already price those rows.
What survives:
exact 3/5 arithmetic may still yield a forbidden configuration, a new nonlocal relation, or GOOD. Structural use is not double charging.

FAIL-FRINGE-CONGRUENCE-CONTRADICTION-001

FAIL-ID:
FAIL-FRINGE-CONGRUENCE-CONTRADICTION-001.

Status:
SCOPE-CORRECTED.

Certificate:
the finite exact example $N = 4298$ in the revised report shows that the bare congruence pattern $q \mid m$, $N - q = b\ell$ with $b \in \{3, 5\}$ is arithmetically realisable with several incidences.

Scope:
that example is outside the canonical high-gap/sub-root failed domain. It kills a contradiction from the bare congruences alone, not a future theorem using the missing canonical/high-gap or primality input.

FAIL-GAMMA-BOUND-ROUTES-001

FAIL-ID:
FAIL-GAMMA-BOUND-ROUTES-001.

Date / iteration:
vNEXT2 independent audit, Phase 2.

Claim:
The adaptive indegree bound, the denominator of (FC8), a distinct resource per low-core element, or the layered Euler defect bounds $\gamma = |C|$.

Original motivation:
Each route looks like it converts local structure into a global count.

Status:
KILLED for all four, as routes; the underlying statements are untouched.

Exact certificate:
(i) Summing (AN2) against Theorem [thm:lowcore-floor] gives $\gamma + |H| \le \sum_{q \in C} \lceil N/(6q) \rceil$, which every summand already satisfies term by term. (ii) Nothing distinguishes $C \setminus Q$ from an arbitrary set of distinct odd primes, so only the primorial bound applies, and Corollary [cor:primorial-ceiling] then gives $\gamma \log \gamma \lesssim 2\gamma \log N$, i.e. $\gamma \lesssim N^2$. (iii) A single row can host $\log_3 N$ low prime factors, so the parent count bounds γ only by $|R| \log_3 N$. (iv) The layered Euler route produces Theorem [thm:lowcore-floor] itself, which bounds $\eta(R)$ below and the low-core primorial mass above, and γ in neither direction.

What survives:
Theorem [thm:lowcore-floor] and Corollary [cor:primorial-ceiling], both new and both genuine.

Do not repeat:
Do not retry these four; a bound on γ needs a mechanism that is not a summation of an existing local bound.

Related current target:
(FC8), K1.

FAIL-GAMMA-POLYLOG-ASSUMED-UNIVERSAL-001

FAIL-ID:
FAIL-GAMMA-POLYLOG-ASSUMED-UNIVERSAL-001.

Status:
KILLED.

Rejected inference:
$\gamma = |C| = O(\log N / \log \log N)$, and hence $t \gg \gamma$, holds throughout the relevant canonical domain.

Reason:
the standard divisor-count bound concerns $\omega(m)$ and hence $|Q|$, not γ. Existing polylogarithmic statements for γ are hypotheses on subdomains, not universal conclusions.

Do not repeat:
do not compare reflected population t with γ without a proved theorem giving that comparison.

`FAIL-GENERIC-ABSORPTION-001`

FAIL-ID:
FAIL-GENERIC-ABSORPTION-001.
Date / iteration:
Post-v7.5 generic absorption audit.
Claim:
Every residual label in a closed world is paid by within-row power, cross-row reuse, or the proposed generic absorption channel.
Original motivation:
$T = UJ$ suggested that no residual singleton could remain unpaid.
Status:
KILLED on the generic/noncanonical domain.
Minimal / strong counterexample:
$N = 1718, r = 1109$, not the canonical root (the canonical selector is the diagonal 859).
Exact certificate:
$N - r = 609 = 3 \cdot 7 \cdot 29$, and

$$1718 - 7 = 1711 = 29 \cdot 59, \quad D_7 = 29, \quad M_7 = 59.$$

The chosen closed recurrent core contains 59, only the row 7 contains the factor 59, and $1718 - 59 = 1659 = 3 \cdot 7 \cdot 79$. Thus the audited 59-valuations of both the within-row and cross-row payment terms are zero.
Failed hypothesis or mechanism:
A generic identity was asked to produce canonical source information.
Why the argument failed:
A singleton residual can enter the state set without being paid by the proposed local channels.
What survives:
$T = UJ$ itself and any future statement that uses genuine canonical minimality essentially.
Safe weaker version:
Keep the generic identity; add a separately proved canonical transport statement before asserting absorption.
Current replacement:
Theory Core, C0b-RD and branch targets.
Do not repeat:
Do not derive a canonical trichotomy from Factor–Flow alone.
Related current target:
$T_D, T_F, P1^{\mathrm{br}}$.

`FAIL-GENERIC-AVERAGE-DEPTH-O-LOGLN-001`

FAIL-ID:
FAIL-GENERIC-AVERAGE-DEPTH-O-LOGLN-001.
Status:
SUPERSEDED.
Old diagnostic:
size, exact valuation, parent-slot and scalar Brun–Titchmarsh admit Level-B integer packets with full-scale vertical depth but too little horizontal multiplicity for an $O(\log\log N)$ average-depth bound.
Superseded by:
`FAIL-GENERIC-PRIME-ROW-RPD-001`, which realises the same obstruction on actual prime

rows.

FAIL-GENERIC-PRIME-ROW-RPD-001

FAIL-ID:
FAIL-GENERIC-PRIME-ROW-RPD-001.

Status:
KILLED as a generic theorem.

Killed statement:
for every packet X of genuine prime rows below $N/3$,

$$\sum_{Y<r\leq N^{1/3}} \sum_{k\geq 2}(n_k^X(r)-1)_+ \ll (\log\log N) \sum_{Y<r\leq N^{1/3}} (\kappa_X(r)-1)_+ + o(|X|).$$

Certificate:
using the analytic sparse-moduli theorem underlying Baker's exact-layer fan, choose a mesoscale prime $r > Y$ and an exact depth $e \asymp (\log\log N)^3$. A fixed proportional interval below $N/3$ contains $N^{1-o(1)}$ primes in the required exact residue class. After Brun–Titchmarsh/Mertens control of unintended band supports one can select $h = N^{1/3+o(1)}$ prime rows with

$$R_X \gg h(\log\log N)^3, \qquad \eta(X) = O(h\log\log N).$$

Scope guard:
this is Level C only. It does not show that the rows are actual members of $H_Q \setminus R$ or descendants of the designated v4.41 matching.

Do not repeat:
generic primality, spacing, nested packing and standard BT cannot establish the Level-D RPD theorem.

FAIL-GREEDY-AVOIDANCE-NAIVE-001

FAIL-ID:
FAIL-GREEDY-AVOIDANCE-NAIVE-001.

Date / iteration:
vNEXT3 hostile proof of the external selector.

Claim:
In the external selector one may choose complements greedily, avoiding the primes already used, and the surviving proportion of the fan is $\prod_{r\in\mathcal{E}}(1-\frac{1}{r-1}) \gg 1/\log\log N$.

Original motivation:
$|\mathcal{E}| = O((\log N)^{A+1})$, so the avoided set is small compared with the fan, and the product looks Mertens-like.

Status:
KILLED as an argument. The *conclusion* of the selector is true — see `G-BAD-BASE-SQUAREFREE-POLYLOG-EXTENSION-PROVED` — but not by this route.

Minimal / strong counterexample:
Structural, two independent failures.

Exact certificate:
(i) Without a prior roughness condition, $\mathcal{E}$ may contain small primes. If $3\in\mathcal{E}$ then $\sum_{r\in\mathcal{E}} 1/(r-1) \geq \frac{1}{2}$, and the Brun–Titchmarsh union bound over $\mathcal{E}$ already exceeds the whole fan, so the union bound gives nothing. (ii) To use $\prod_{r\in\mathcal{E}}(1-\frac{1}{r-1})$ as a density one needs a *sieve* over $\mathcal{E}$, not a union bound. The combined modulus is $\prod_{r\in\mathcal{E}} r = \exp(\theta(y))$ with $y \approx (\log N)^{A+2}$, that is $\exp((\log N)^{A+2})$, which vastly exceeds N; Siegel–Walfisz does not apply there, and a sieve at that level would require a level-of-distribution input such as Bombieri–Vinogradov, which was never declared.

Failed hypothesis or mechanism:
A product of local densities was used where only a union bound was available, and the modulus range needed for the alternative was never checked.

What survives:
Everything, after reordering. Imposing w-roughness *first* — no prime below $w = (\log N)^{A+2}$ divides a complement except its own designated base — forces $\mathcal{E}$ to consist only of primes $\geq w$, whence $\sum_{r\in\mathcal{E}} 1/(r-1) \leq |\mathcal{E}|/(w-1) \ll 1/\log N$ and a plain union bound suffices. The sieve then used has polylogarithmic sifting limit and is truncated at a polylogarithmic level, so every modulus stays inside the Siegel–Walfisz range and no level-of-distribution theorem is imported.

Attribution correction:
the $1/\log\log N$ loss is real, but it belongs to the roughness sieve, where it is $\asymp 1/\log w$. The coprimality step costs only $1 + O(\log w/\log N)$. vNEXT2 attributed the loss to the wrong step.

Safe weaker version:
State the selection as: sieve for roughness, then delete squares, then union-bound the coprimality classes.

Current replacement:
Theory Core, Theorem [thm:selector].

Do not repeat:
Do not multiply local densities across conditions that are only accessible by a union bound, and always check that the combined modulus stays in the range of the distribution theorem being used.

Related current target:
(FC10), K1.

FAIL-HALL-SUPPORT-001

FAIL-ID:
FAIL-HALL-SUPPORT-001.

Date / iteration:
Escape-or-Reuse campaign, 30 August 2026.

Claim:
A Hall expansion over the support counts of all sources yields the required capacity bound.

Original motivation:
Every source contributes support, so a global counting argument looked available.

Status:
KILLED.

Minimal / strong counterexample:
The recorded compression figure: 456 sources compress to 17 factor atoms. Status VERIFIED COMPUTATION; the raw table was not reloaded in this extraction.

Exact certificate:
With that degree of compression the naive expansion has no room to run: the support side is far smaller than the source side, so no Hall-type surplus exists in the required direction.

Failed hypothesis or mechanism:
Sources and atoms were counted as if they were in near-bijection.

Why the argument failed:
Compression is exactly the regime in which a counting surplus disappears.

What survives:
The precise asymptotic polylog Hall matching theorem is *not* refuted by this: `G-POLYLOG-MATCHING` runs on a fixed finite family of polylog prime-power layers with an enormous per-layer fan, not on an all-source support count.

Safe weaker version:
Restrict any Hall argument to the polylog layer family where the fan is proved.
Current replacement:
Theory Core, `G-POLYLOG-PARENTS`, `G-POLYLOG-MATCHING`, target W1.
Do not repeat:
Do not run a counting argument across a compression.
Related current target:
W1, K1.

`FAIL-HIGH-LOADONE-COFACTOR-CAN-HAVE-ARBITRARY-ARITY-001`

FAIL-ID:
FAIL-HIGH-LOADONE-COFACTOR-CAN-HAVE-ARBITRARY-ARITY-001.
Status:
KILLED.
Killed inference:
after selecting a unique-parent target $q > N^{1/3}$, the remaining complement may still contain an uncontrolled high- arity cofactor supported entirely above the cube root.
Certificate:
v4.41 proves exact valuation depth $e \in \{1, 2\}$ and the row trichotomy: a low child $r \leq N^{1/3}$, the square q^2, or the squarefree semiprime qr with both factors above the cube root.
What survives:
low boundary layers, monomial squares and high semiprime dynamics.

`FAIL-HIGH-SEMIPRIME-SIBLING-REUSE-IS-CHEAP-001`

FAIL-ID:
FAIL-HIGH-SEMIPRIME-SIBLING-REUSE-IS-CHEAP-001.
Status:
KILLED.
Killed inference:
a linear high-semiprime packet may heavily reuse its second factors while keeping incremental J negligible.
Certificate:
every repeated sibling $r > N^{1/3}$ contributes more than $(\log N)/3$ per excess row occurrence. If that ambient payment is absent, a $> 3/4$ subfamily has distinct siblings and therefore pairwise-coprime squarefree two-prime complements.
What survives:
the clean positive-sign branch itself was not a contradiction; v4.42 later supersedes it as the primary node.

`FAIL-HIGH-VERTICAL-IS-INDEPENDENT-EXTERNAL-CHANNEL-001`

FAIL-ID:
FAIL-HIGH-VERTICAL-IS-INDEPENDENT-EXTERNAL-CHANNEL-001.
Status:
KILLED.
Killed inference:
above the exact cube-root floor, a large $\nu_{>N^{1/3}}$ term may remain a third independent storage channel alongside horizontal reuse and low boundary descent.
Certificate:
every high squared prime either forms a monomial row or has a cofactor $< N^{1/3}$ and therefore already crosses the low boundary. Together with monomial-component capacity and floor-

truncated conservation,

$$\eta_{>N^{1/3}}(H) + 2D_{\le N^{1/3}}(H) \ge (|H| - 1)/2 + c_{N^{1/3}}.$$

What survives:
a genuine aggregate low-layer defect on bases $q \le N^{1/3}$.

`FAIL-HIGHFLOOR-DOMAIN-VACUITY-001`

FAIL-ID:
FAIL-HIGHFLOOR-DOMAIN-VACUITY-001.
Date / iteration:
vNEXT2 independent audit, Phase 1E.
Claim:
`TF-HIGH-FLOOR-HORIZONTAL-RIGIDITY` applies to packets that actually occur.
Original motivation:
The theorem is stated unconditionally on its hypothesis and its proof is complete.
Status:
SCOPE-CORRECTED. The theorem is PROVED; its *antecedent* is unverified.
Exact certificate:
An exhaustive search over even $N < 3 \cdot 10^4$ and all admissible seeds found *no* bad child-closed packet with $\min U > N^{1/3}$ and $U \subset (0, N/3)$. Status of that search: VERIFIED COMPUTATION.
Failed hypothesis or mechanism:
None mathematically; the risk is rhetorical — quoting $\eta(U) \ge \lceil n/3 \rceil + c$ as though it were about an exhibited object.
What survives:
The theorem in full, with both skeleton gaps now proved: σ has no cycles (a 1-cycle forces $N = u(u+1)$, hence $u \mid N$ and u inactive; a 2-cycle forces $(v-u)(v+u+1) = 0$; longer cycles are excluded because $f(x) = \sqrt{N-x}$ is strictly decreasing), and there are no three consecutive square rows (the contraction constant of f puts u_2, u_3 within 0.111 of the fixed point for $N \ge 50$, forcing $u_2 = u_3$ and hence a 1-cycle).
Safe weaker version:
Always state it as a conditional.
Do not repeat:
An empty finite search is not evidence that the hypothesis is unsatisfiable, and a complete proof is not an exhibited instance.
Related current target:
TF, K1.

`FAIL-IMPORT-V32-POSITIVE-K-TO-DEEP-LOW-OWNER-001`

FAIL-ID:
FAIL-IMPORT-V32-POSITIVE-K-TO-DEEP-LOW-OWNER-001.
Source:
v4.53 scope correction relative to v4.32.
Status:
KILLED.
Killed import:
transfer the old lower bound

$$k_i > \frac{N}{6s_i}$$

from the upper-half owner geometry directly to the present low descendant basin.
Reason:
the algebraic owner-return normal form transfers, but the old sign and positive lower bound

used an upper-half owner which is absent here.

What survives:

the signed deep owner-return normal form and the same-layer second-parent inverse bound remain valid in their stated domains.

`FAIL-INCREMENTAL-J-IS-BOUNDED-BY-OLD-J-001`

FAIL-ID:

FAIL-INCREMENTAL-J-IS-BOUNDED-BY-OLD-J-001.

Status:

KILLED.

Rejected statement:

$J_{R\cup P}/J_R \leq J_R$ for arbitrary external extensions.

Certificate:

vNEXT4.31 gives the exact primewise increment and permits new support multiplicity to create genuinely new J storage.

Do not repeat:

K1-INC itself is an open upper-capacity theorem, not a consequence of the definition of J.

`FAIL-INDIVIDUAL-MESOSCALE-BV-001`

FAIL-ID:

FAIL-INDIVIDUAL-MESOSCALE-BV-001.

Date / iteration:

v1.10 scale-wall continuation, 1 September 2026.

Claim:

The v1.6 “sub-sum of ordinary Bombieri–Vinogradov” argument can be transferred unchanged from a polylogarithmic base modulus to one fixed base $q = N^\beta$, $\beta > 0$, and still prove a positive lower- sieve fan.

Original motivation:

Extend the clean-parent theorem directly from polylogarithmic labels to polynomially sized states.

Status:

KILLED.

Exact certificate:

For one polynomial base the expected prime/sieve main term is of size at most

$$\asymp \frac{N}{q\ \log N\ \log z} = N^{1-\beta}\,\mathrm{polylog}(N)^{-1},$$

whereas the ordinary Bombieri–Vinogradov all-moduli error available from simply taking a sub-sum is

$$O_B\left(\frac{N}{(\log N)^B}\right).$$

For every fixed B and every fixed $\beta > 0$, the latter is larger by a factor $N^\beta/\,\mathrm{polylog}(N)$. Therefore the v1.6 error comparison does not prove positivity for an individual mesoscopic base.

Failed hypothesis or mechanism:

A global mean error was treated as though it scaled with the individual base modulus.

What survives:

Aggregate use over many bases does work at the unsieved parent-incidence level. `G-BV-HARMONIC-PARENT-MASS` proves that fixed positive reciprocal mass below the BV level yields $\gg N/\log N$ parent incidences and $\gg N/\log^2 N$ distinct external parents. `TD-HARMONIC-SUPPORT-DISPOSITION` supplies the reciprocal-mass input on the D branch.

Safe weaker version:
Use the existing individual selector only while the prescribed base modulus is polylogarithmic, or prove a genuinely aggregate many-base theorem with its own remainder accounting.
Current replacement:
v1.10 adaptive lift remains the individual polylog tool. v1.11 adds genuinely aggregate clean extraction: `G-BV-HARMONIC-CLEAN-EXTRACTION` selects a clean load-one subfamily from mesoscopic reciprocal mass, and `G-BV-POLYNOMIAL-HARMONIC-CLEAN-EXTRACTION` gives a polynomially large clean family on fixed power bands. Neither theorem supplies a fan for every prescribed mesoscopic base.
Do not repeat:
Do not cite ordinary BV's total error as an individual polynomial-modulus lower bound.
Related current target:
TD scale bridge, W1, P1BR/K1.

`FAIL-INVERSE-SIBLING-INJECTIVE-001`

FAIL-ID:
FAIL-INVERSE-SIBLING-INJECTIVE-001.
Date / iteration:
v1.23 inverse-sibling continuation, 1 September 2026.
Claim:
Distinct returned fresh seeds ℓ necessarily produce distinct prime siblings s satisfying $\ell s < N$.
Status:
KILLED.
Why the inference fails:
A single packet state can divide the return cofactors of several distinct parent rows. The inverse product inequality contains no injectivity information.
What survives:
After first passing to distinct return parents, repeated choice of the same sibling s gives genuine rooted reuse inside U and is charged by

$$\eta(U) \geq |P| - |S|.$$

Thus collisions are an explicit alternative rather than an ignored loss.
Current replacement:
Theory Core, `TF-HIGH-RETURN-INVERSE-PACKET`.
Do not repeat:
Never convert the inverse-scale map directly into a cardinality lower bound without the $\eta(U)$ collision term.
Related current target:
TF inverse-scale analytic re-entry.

`FAIL-KERNEL-EAC-001`

FAIL-ID:
FAIL-KERNEL-EAC-001.
Date / iteration:
v0.1 audit, notation and object review.
Claim:
The iterated source-free core $C_{\mathrm{sf}} = \bigcap_{j\geq 0} \Gamma_N^j(R)$ is an EAC, equivalently a terminal strongly connected component.
Original motivation:
Both objects are "what is left at the bottom".

Status:
SCOPE-CORRECTED.
Minimal / strong counterexample:
Definitional; the two constructions are not the same operation.
Exact certificate:
C_{sf} is defined by iterated image and need not be strongly connected, nor a single component. A terminal SCC is defined by the condensation having no outgoing edge. `G-FAIL-CLOSURE` produces a terminal SCC; it does not produce C_{sf}, and neither statement implies the other.
Failed hypothesis or mechanism:
Two distinct objects shared an informal name.
Why the argument failed:
A theorem about one was used as a theorem about the other.
What survives:
Both objects, kept separate, as in the Theory Core notation registry.
Safe weaker version:
Name the construction whenever either object is used.
Current replacement:
Theory Core, `G-FAIL-CLOSURE` and the notation entry for C_{sf}.
Do not repeat:
Existence of a terminal bad SCC somewhere does not show that p_0 reaches it.
Related current target:
X1.

`FAIL-KPOWER-SPECTRUM-IS-OPTIMAL-DJ-BOUND-001`

FAIL-ID:
FAIL-KPOWER-SPECTRUM-IS-OPTIMAL-DJ-BOUND-001.
Status:
SUPERSEDED.
Superseded description:
the coefficient $(k\alpha-1)/(k-1)^2$ from v4.37 is the primary canonical D/J lower bound.
Certificate:
v4.38 constructs exactly γ fully-C-supported rows and uses the uniform radical-fibre count to obtain

$$\log(D_C J_R) \geq \gamma \log \gamma - o(\gamma \log N).$$

For $\gamma = N^{\alpha+o(1)}$ the coefficient is $\alpha - o(1)$, and at the cube-root splice it is $1/3 - o(1)$ instead of $1/24 - o(1)$.
What survives:
the k-power theorem remains valid and useful for explicit powerful-divisor structure.

`FAIL-LARGE-FRESH-SUPPORT-MAY-BE-MOSTLY-REPEATED-001`

FAIL-ID:
FAIL-LARGE-FRESH-SUPPORT-MAY-BE-MOSTLY-REPEATED-001.
Status:
KILLED.
Killed inference:
a large fresh-support factor may consist almost entirely of fresh labels repeated on many extension rows while incremental J stays small.
Certificate:
for fresh labels with $\kappa_A(q) \geq 2$, at least one full copy of $\log q$ already belongs to incremental J. Hence

$$W_1(A) \geq \log F_R(A) - \log(J_S/J_R),$$

and the relative radical floor gives

$$\log(J_S/J_R) \geq \mathcal{R}(A)/3 \quad \vee \quad W_1(A) \geq \mathcal{R}(A)/3.$$

What survives:
the cheap branch is not arbitrary fresh support but the exact unique-parent layer.

FAIL-LARGE-MONOMIAL-PACKET-CAN-OCCUR-IN-BOTH-MOD3-CLASSES-001

FAIL-ID:
FAIL-LARGE-MONOMIAL-PACKET-CAN-OCCUR-IN-BOTH-MOD3-CLASSES-001.
Status:
KILLED.
Killed inference:
for $3 \nmid N$, a polynomial family $N - w = q^2$ with $q > N^{1/3}$ may occur for either nonzero residue of N modulo 3.
Certificate:
if $N \equiv 1 \pmod 3$, then $w = N - q^2 \equiv 0 \pmod 3$; prime w must equal 3, so at most one such edge exists.
What survives:
a polynomial monomial packet can only remain when $N \equiv 2 \pmod 3$.

FAIL-LARGEBASE-DEEP-VERTICAL-ESCAPE-001

FAIL-ID:
FAIL-LARGEBASE-DEEP-VERTICAL-ESCAPE-001.
Status:
KILLED as a live frontier on $q > N^{1/3}$.
Killed description:
genuinely independent cubic-or-deeper vertical storage may persist on bases above $N^{1/3}$.
Certificate:
$q^3 < N$ is necessary for depth at least three. Above the cube-root scale vertical depth is at most two, and the one depth-two copy is termwise accompanied by the same-base radical support already counted by FC7.
What survives:
cubic-and-deeper layers on $q < N^{1/3}$.

FAIL-LAYERED-GAMMALOGN-IS-CONTRADICTION-001

FAIL-ID:
FAIL-LAYERED-GAMMALOGN-IS-CONTRADICTION-001.
Date / iteration:
vNEXT4.5.
Claim:
Once $\log D + \log J + \log U = \Omega(\gamma \log N)$, FC8 or Goldbach follows automatically.
Status:
KILLED.
Exact certificate:
The same functional has the natural sandwich

$$\log W_C^{\mathrm{nr}} = \gamma \log N + O(\gamma + |H|).$$

The lower scale and upper capacity match in the leading term.
Failed mechanism:
A conservation identity was mistaken for an independent overload inequality.

What survives:
An additive/canonical theorem may still force an incompatible $D/J/U$ partition.
Do not repeat:
Do not count the exact row mass twice as two independent obligations.

FAIL-LEDGER-INDEPENDENCE-LI5-SCOPE-001

FAIL-ID:
FAIL-LEDGER-INDEPENDENCE-LI5-SCOPE-001.
Date / iteration:
vNEXT4.1, Phase 0B.
Claim:
vNEXT4 introduced (LI5) under the heading "that trivial bound is the only one available", suggesting η_{supp} is the unique ledger-derivable lower bound on $\eta(S)$.
Original motivation:
The witness family drives $\eta(R_n)$ to infinity at constant round datum, which does refute every *growing* bound.
Status:
SCOPE-CORRECTED. `G-POSITIVE-ROUND-LEDGER-INDEPENDENCE` is preserved in full.
Exact certificate:
The $n = 1$ member has $\eta(R_1) = 2$. Hence any F with $F \leq \eta$ satisfies only $F(\mathfrak{d}) \leq 2$ at that datum; constants up to 2 are consistent with the family. So the family excludes unbounded growth in Δ, not the existence of some bounded lower bound.
What survives:
(LI5′): no universally valid lower bound on $\eta(S)$ expressible solely as a function of the round datum can grow unboundedly with Δ; in particular none supplies the $\gamma \log N$-scale bound the (LE3)+(FC8) route consumes. This is the only form used downstream, and it is what makes the route unavailable.
Do not repeat:
Distinguish "no bound of the required size" from "no bound". Only the former was proved.
Related current target:
`P1BR-ADDITIVE-COUPLING`.

FAIL-LEGACY-CONTINUUM-001

FAIL-ID:
FAIL-LEGACY-CONTINUUM-001.
Date / iteration:
v0.1 audit of legacy candidates.
Claim:
As N grows the discrete CRD structure becomes effectively continuous, and that transition is itself usable.
Original motivation:
Asymptotic densities do appear.
Status:
KILLED.
Exact certificate:
No density or covering statement is attached to the phrase. It must be replaced by a precise density or covering theorem before anything can be deduced.
Failed hypothesis or mechanism:
An informal limit replaced a quantified statement.
What survives:
Genuine asymptotics, such as `G-POLYLOG-PARENTS`, stated with explicit uniformity and an explicitly INEFFECTIVE threshold.

Current replacement:
Theory Core, the polylog theorems.
Do not repeat:
Do not use a limit as a lemma.
Related current target:
W1.

FAIL-LEGACY-ENCODING-001

FAIL-ID:
FAIL-LEGACY-ENCODING-001.
Date / iteration:
v0.1 audit of legacy candidates.
Claim:
A binary or positional encoding of the states supplies an established CRD invariant.
Original motivation:
Encodings are cheap to define.
Status:
KILLED.
Exact certificate:
No retained theorem uses such an encoding, and none of the exact identities — $T = UJ$, the source lock, the collision and cut identities — is expressible through one.
Failed hypothesis or mechanism:
A representation was mistaken for a structure.
What survives:
Nothing in this direction.
Current replacement:
Theory Core, `GEN-SBR` and the specialisation dictionaries, which is where representation-level statements belong.
Do not repeat:
A change of notation is not a theorem.
Related current target:
none.

FAIL-LEGACY-PIGEONHOLE-001

FAIL-ID:
FAIL-LEGACY-PIGEONHOLE-001.
Date / iteration:
v0.1 audit of legacy candidates.
Claim:
A parametric symmetric pigeonhole argument supplies the weighted CRD capacity bound.
Original motivation:
The finite lemma is genuinely true.
Status:
KILLED as a mechanism.
Minimal / strong counterexample:
Structural.
Exact certificate:
The bare finite lemma is true and supplies no weighted logarithmic inequality; capacity in CRD is a statement about $\sum \mu_{\ell,k} \log \ell$, which the lemma never reaches.
Failed hypothesis or mechanism:
A cardinality lemma was used where a weighted one is required.

Why the argument failed:
Counting objects is not bounding their logarithmic mass.
What survives:
The finite lemma itself, as an unrelated true statement.
Safe weaker version:
None in this direction.
Current replacement:
Theory Core, targets $P1^{\mathrm{br}}$ and K1.
Do not repeat:
Do not offer an unweighted lemma as a weighted bound.
Related current target:
$P1^{\mathrm{br}}$, K1.

`FAIL-LEGACY-WAVES-001`

FAIL-ID:
FAIL-LEGACY-WAVES-001.
Date / iteration:
v0.1 audit of legacy candidates.
Claim:
Describing residue behaviour as interfering sinusoidal waves supplies a rigorous CRD invariant.
Original motivation:
The picture is suggestive.
Status:
KILLED.
Exact certificate:
Only the underlying residue lattice has proved content; no retained theorem uses the wave description, and no invariant is derived from it anywhere in the corpus.
Failed hypothesis or mechanism:
Imagery in place of a statement.
What survives:
The residue lattice, used directly.
Current replacement:
Theory Core, prime-power layer formalism.
Do not repeat:
Do not carry a metaphor into a proof.
Related current target:
none.

`FAIL-LF2-AS-UPPER-BOUND-001`

FAIL-ID:
FAIL-LF2-AS-UPPER-BOUND-001.
Date / iteration:
vNEXT3.1 correction round.
Claim:
In the (FC8) frontier certificate, "(LF2) bounds $\eta(R)$ above by $|R|$".
Original motivation:
$|Q| + |H| \leq |R|$ is true, and the certificate needed $\eta(R)$ to be small.
Status:
KILLED.
Exact certificate:
(LF2) reads $\eta(R) \geq |Q| + |H|$. It is a *lower* bound. The inequality $|Q| + |H| \leq |R|$ says only that this recorded lower bound is itself at most $|R|$; it places no ceiling on $\eta(R)$, which by

(LE2) can exceed $|R|$ whenever rows carry repeated support.

Failed hypothesis or mechanism:
A bound on the right-hand side of a lower-bound inequality was read as a bound on the left-hand side.

What survives:
The certificate, correctly stated: to contradict (FC8) through (LE3) one needs a *lower* bound on $\eta(R)$ of order $\gamma \log N$, and the recorded (LF2) does not carry a factor $\log N$.

Current replacement:
Theory Core, Proposition [prop:fc8-corrected], status FRONTIER CERTIFICATE / RELATIVE TO RECORDED BOUNDS.

Do not repeat:
Never infer a ceiling from a floor.

Related current target:
K1, γ.

FAIL-LINEAR-SMOOTH-MIRROR-PACKET-CAN-STAY-POLYLOG-SUPPORTED-001

FAIL-ID:
FAIL-LINEAR-SMOOTH-MIRROR-PACKET-CAN-STAY-POLYLOG-SUPPORTED-001.

Status:
KILLED asymptotically on the recorded high-gap/BHP domain.

Killed inference:
a linear smooth-mirror family may carry the required active mass on only polylogarithmically many active support primes.

Certificate:
existing mirror active-mass plus support capacity gives, for $|J_{\mathrm{sm}}| \geq \rho t$,

$$|P_{\mathrm{mir}}| \geq N^{0.2375\rho - o(1)}$$

unless there is already a linear active-state population.

What survives:
this is asymptotic and does not replace finite small-gap case analysis.

FAIL-LITERAL-LINEAGE-001

FAIL-ID:
FAIL-LITERAL-LINEAGE-001.

Date / iteration:
Post-v7.5 layer-lineage audit.

Claim:
Once a prime label ℓ is produced, its valuation can be followed literally through the row indexed by ℓ.

Original motivation:
Residual factors were pictured as conserved tokens moving down a path.

Status:
KILLED.

Minimal / strong counterexample:
Universal for active ℓ.

Exact certificate:
Since $\ell \nmid N$, $N - \ell \equiv N \not\equiv 0 \pmod{\ell}$; hence

$$v_\ell(N - \ell) = 0.$$

Failed hypothesis or mechanism:
Row labels and factor labels were identified.

Why the argument failed:
The dynamics changes the row index; it does not conserve a prime token along a path.
What survives:
Global valuation accounting $T = UJ$, collision layers, and residual products.
Safe weaker version:
Track incidence across a finite set of rows, not literal pathwise conservation.
Current replacement:
Theory Core, G-TUJ and the prime-power layer formalism.
Do not repeat:
Never write a lineage law without an exact valuation identity at every transition.
Related current target:
$E1, K1$.

`FAIL-LOCAL-REUSE-001`

FAIL-ID:
FAIL-LOCAL-REUSE-001.
Date / iteration:
Escape-or-Reuse campaign, 30 August 2026.
Claim:
An upper-source reuse parent occurs within five source indices.
Original motivation:
A local window would make simultaneous witness selection inexpensive.
Status:
KILLED by source-recorded computation.
Minimal / strong counterexample:
The strongest recorded witness is $(N, \ell) = (1825372, 52153)$ with source-index distance 53750.
Exact certificate:
Only 0.0265727 of defined recorded witnesses lay within five indices; the cited distance is the recorded maximum. The raw row list was not reloaded in this extraction.
Failed hypothesis or mechanism:
Source order was treated as a metric controlling arithmetic incidence.
Why the argument failed:
Parent lattices are residue classes, not short intervals in the adaptive source enumeration.
What survives:
Polylogarithmic parent abundance and finite matching for prescribed prime powers.
Safe weaker version:
Use the full active parent lattice or prove a new quantitative selector theorem.
Current replacement:
Theory Core, G-POLYLOG-PARENTS, G-POLYLOG-MATCHING, and $W1$.
Do not repeat:
Do not impose an unproved absolute locality window.
Related current target:
$W1, K1$.

`FAIL-LOSSLESS-INTERFACE-001`

FAIL-ID:
FAIL-LOSSLESS-INTERFACE-001.
Date / iteration:
vNEXT4.1, Phase 4.
Claim:
That a canonical row c_j can be transferred to a global prime selection with its information preserved — the round's primary success criterion, third arrow.

Original motivation:
The bundle query does preserve strictly more than the old single-base query: exact valuations, the cofactor class and the mirror relation $D \mid p - a_j$. It was natural to hope the preserved part could be made all of A_j.

Status:
KILLED for lossless transfer, under every distribution theorem presently available. The bundle lift itself is PROVED.

Exact certificate:
Theory Core `P0-QUERY-CAPACITY`: if every member of a residue class $n \equiv a\ (M)$ satisfies $v_q(N-n) = e$, then $q^{e+1} \mid M$. Taking $n = a$ and $n = a - M$ forces $q^e \mid M$; and if $v_q(M) = e$ exactly then one k in each complete residue system mod q gives $q^{e+1} \mid N - a + kM$. Hence $D \operatorname{rad} D \mid M$ and $\log D \le \frac{E}{E+1} \log M$. With Siegel–Walfisz, $\log M \le K \log\log N$, so the preserved fraction is $< K \log\log N / \log(N/2) \to 0$. Under GRH the fraction is at most $\frac{1}{2}$ (at most $\frac{1}{4}$ for squarefree bundles), so GRH does not make it lossless either. Losslessness needs a level of distribution $\theta \ge 1$ for an *individually specified* modulus of size $N^{1-o(1)}$ — stronger than GRH, and not the average-level statement of Elliott–Halberstam.

Failed hypothesis or mechanism:
Treating "retain more information" as free. Retention is paid for in the modulus, at a rate fixed by (QC1), and the modulus is capped by the available distribution theorem.

What survives:
The exact capacity: $\log D \le \frac{E}{E+1} K \log\log N$, and everything the bundle does preserve, which is genuinely more than the old query. See `P0-EXACT-BUNDLE-LIFT` and `P0-ADAPTIVE-BUNDLE-DICHOTOMY`.

Do not repeat:
Do not propose an interface that preserves a positive proportion of a canonical row without first naming the level-of-distribution theorem that pays for it, and do not import Bombieri–Vinogradov silently — this corpus does not use it anywhere.

Related current target:
`P1BR-ADDITIVE-COUPLING`, `P0-BUNDLE-ROUGH-HIGHPOWER-BRANCH`.

`FAIL-LOWBASE-POWERFUL-ROWS-AS-INDEPENDENT-VERTICAL-WALL-001`

FAIL-ID:
FAIL-LOWBASE-POWERFUL-ROWS-AS-INDEPENDENT-VERTICAL-WALL-001.

Status:
KILLED as an independent primary frontier.

Killed description:
a polynomial-size canonical low core may keep an asymptotically negligible $D_C J_R$ while financing essentially all compulsory mass through diffuse low-base powerful rows.

Certificate:
v4.37 proves, pending audit, that for $\gamma = N^{\alpha+o(1)}$ and every fixed $\alpha > 0$,

$$\log(D_C J_R) = \Omega_\alpha(\gamma \log N).$$

Equivalently, near-pure vertical escape forces $\gamma = N^{o(1)}$.

What survives:
a subpolynomial low core or a non-negligible D/J payment; rowwise powerful-divisor structure remains useful downstream.

`FAIL-LOWCORE-CARDINALITY-FC8-WEIGHT-001`

FAIL-ID:
FAIL-LOWCORE-CARDINALITY-FC8-WEIGHT-001.

Date / iteration:
vNEXT4.4.
Claim:
Once the reflected block forces large $\gamma = |C|$, the layered Euler theorem automatically converts this into enough rooted reuse to contradict FC8.
Status:
KILLED .
Exact certificate:
Canonical layered Euler gives only

$$\eta(R) + \nu(R) \geq |R| + 1 \geq \gamma + 2,$$

hence only

$$\max\{\log J_R, \log U_R\} \geq \frac{\gamma + 2}{2} \log 3.$$

This is $O(\gamma)$ guaranteed log mass. The recorded FC8 route through $J_R \geq 3^{\eta(R)}$ needs a lower bound at scale $\gamma \log N$ unless a separate large denominator is supplied. Moreover the population may be paid vertically through prime-power layers U_R; the generic failed-world prime-power examples show that such vertical storage is a real mechanism.
Failed hypothesis or mechanism:
Cardinality was confused with weighted prime-base mass, and horizontal reuse was treated as if it were the only defect channel.
What survives:
vNEXT4.4 produces a genuine FC8 denominator on the sub-square-root large-factor branch, and the exact horizontal/vertical split remains available. A theorem forcing large bases or upper-bounding cheap vertical storage would reopen the route.
Do not repeat:
A new lower bound on γ, $\eta + \nu$, or an unweighted layer count is not a consumer unless its logarithmic bases are controlled.
Related current target:
`C-VERTICAL-LOWCORE-CAPACITY`.

FAIL-LOWCORE-CHILDCLOSED-001

FAIL-ID:
FAIL-LOWCORE-CHILDCLOSED-001.
Date / iteration:
v1.23 low-core continuation, 1 September 2026.
Claim:
Since every reused rooted label is below $N/6$, the set $C = R \cap (0, N/6)$ may be treated as a child-closed failed subsystem.
Status:
KILLED .
Why the inference fails:
The theorem controls the location of *repeated targets*, not all children of low rows. A low row $u < N/6$ has $N - u > 5N/6$, and a prime factor of this complement can lie in $(N/6, N/3)$. Thus an edge may leave C for the upper fringe.
What survives:
Every repeated target and the complete factor J_R are supported in C; every high state has a unique-parent forest structure and returns to C when parent ancestry is followed backwards.
Current replacement:
Theory Core, `G-CANONICAL-REUSE-LOWCORE` and `G-CANONICAL-UPPER-FRINGE-FOREST`.

Do not repeat:
Do not apply G-MULTISOURCE-LOCK, low-packet product locks, or child-closed component arguments directly to C without separately proving closure for the subset in question.
Related current target:
K1 low-core upper capacity.

FAIL-LOWPOLYLOG-MULTIPLICITY-IS-PARENT-EXISTENCE-001

FAIL-ID:
FAIL-LOWPOLYLOG-MULTIPLICITY-IS-PARENT-EXISTENCE-001.
Date / iteration:
vNEXT4.20.
Old description:
The low-polylog $t \log \Lambda$ mirror multiplicity cannot be consumed because there are not enough actual prime parent rows preserving the required layers.
Status:
KILLED as a description of the current obstruction.
Certificate:
After an $o(t \log \Lambda)$ high-power tail is removed, all retained polylog prime-power blocks can be simultaneously matched to distinct external BAD primes satisfying $p_{j,q} \equiv c_j \pmod{q^{e+1}}$.
Replacement:
The live obstruction is joint unintended support load / single charging, not parent abundance.

FAIL-MANY-FIRSTFRONT-ROOTS-CAN-ALL-LIE-ABOVE-4H-001

FAIL-ID:
FAIL-MANY-FIRSTFRONT-ROOTS-CAN-ALL-LIE-ABOVE-4H-001.
Status:
KILLED for $h > N^{1/3}$.
Rejected statement:
A large fraction of Q may lie above $4h$ without additional restriction.
Certificate:
three such distinct primes would divide $m < N$ but have product $> (4h)^3 > 64N$.
Current safe statement:
$\#\{q \in Q : q > 4h\} \le 2$.

FAIL-MATCHING-CAPACITY-001

FAIL-ID:
FAIL-MATCHING-CAPACITY-001.
Date / iteration:
Canonical bridge audit §5.3, 31 August 2026.
Claim:
An injective, load-one assignment of external parents to layers bounds the ambient collision cost, closing the capacity target.
Original motivation:
Load one removes repeated use of a single parent, which was mistaken for removing the cost.
Status:
KILLED — the matching theorem itself is PROVED; the inference drawn from it is not.
Minimal / strong counterexample:
Structural; no finite witness is needed.
Exact certificate:
Injectivity leaves two costs untouched. One selected row $N - p$ may meet many atoms of R, inflating $K_\times(R, \mathscr{P})$; and two distinct selected rows may share unintended new support,

inflating $K(\mathscr{P})$. Greedy selection of pairwise-coprime complements would have to avoid, for each previously chosen p', every class $p \equiv N \pmod t$ with $t \mid N - p'$; Siegel–Walfisz for a single polylog modulus gives no uniform positive count under a growing adaptive system of such prohibitions, and some t may be large.

Failed hypothesis or mechanism:
Existence and load were confused with cost.

Why the argument failed:
Abundance is a statement about one modulus at a time; capacity is a statement about a whole selected family at once.

What survives:
`G-POLYLOG-PARENTS` and `G-POLYLOG-MATCHING`, now for all polylog prime-power layers, both with an explicitly INEFFECTIVE threshold.

Safe weaker version:
Report load one and the two uncontrolled terms separately; never report their conjunction as capacity.

Current replacement:
Theory Core, target K1, which is exactly the missing bound $K_\times(R, \mathscr{P}) + K(\mathscr{P}) \le \mathcal{C}_N(R)$.

Do not repeat:
This is the precise point at which abundance stops becoming clean capacity; do not cross it silently.

Related current target:
W1, K1.

`FAIL-MERGER-COUNT-IMPLIES-AMBIENT-J-001`

FAIL-ID:
FAIL-MERGER-COUNT-IMPLIES-AMBIENT-J-001.

Status:
KILLED.

Killed inference:
e merger/attachment events automatically force $\Delta \log J = \Omega(e \log N)$.

Certificate:
v4.38 proves a real attachment charge, but if all events reuse a fixed small label q, its weight is only $O(e \log q)$. The ELCFS finite-realisation theorem shows that source counts/topology/reverse-generation bookkeeping alone cannot manufacture the missing ambient $\log N$ factor.

What survives:
attachment forces an unweighted horizontal defect and CRD-specific arithmetic may still retain scale.

`FAIL-MERGER-WEIGHT-TO-J-TOPOLOGY-001`

FAIL-ID:
FAIL-MERGER-WEIGHT-TO-J-TOPOLOGY-001.

Status:
KILLED.

Killed sufficient strategy:
prove $\Delta \log J \gg W(E)$ using only child closure, source counts, graph topology, support multiplicities and the v28 reverse-extension ledger.

Certificate:
ELCFS finite realisation permits arbitrarily large source weights attached through a fixed low target. The topology grows while the horizontal receiving weight remains only $e \log q$.

What survives:
CRD arithmetic absent from ELCFS—the map $N - p$, difference divisibility, reflected prove-

nance, exact cofactors and canonical root extremality—may still supply the missing scale.

FAIL-MIRROR-FIXED-POLYLOG-RESERVOIR-001

FAIL-ID:
FAIL-MIRROR-FIXED-POLYLOG-RESERVOIR-001.
Date / iteration:
vNEXT4.17, sharpened by vNEXT4.19.
Escape:
Store all natural mirror mass below $(\log N)^D$ for super-polylogarithmic Λ.
Status:
SCOPE-CORRECTED.
Certificate:
For fixed D,

$$M^-_{\leq(\log N)^D} = O_D(t \log\log N).$$

Thus this reservoir can remain competitive only when $\log\Lambda = O(\log\log N)$. That low-polylog balance is the current open frontier rather than a killed escape.

FAIL-MIRROR-MASS-ONE-SCALE-001

FAIL-ID:
FAIL-MIRROR-MASS-ONE-SCALE-001.
Date / iteration:
vNEXT4.17.
Escape:
Store $\gg t\log\Lambda$ active mirror mass in one dyadic base band.
Status:
KILLED.
Certificate:
Uniformly $M_Y^- \ll t$. Natural mass therefore occupies $\Omega(\log\Lambda)$ scales.

FAIL-MIRROR-MULTIPLICITY-BY-REPEATED-BASE-001

FAIL-ID:
FAIL-MIRROR-MULTIPLICITY-BY-REPEATED-BASE-001.
Date / iteration:
vNEXT4.17.
Claim:
Represent reflected multiplicity m_r^- by the same number of clean selected parent copies carrying designated base r in one family.
Status:
KILLED.
Exact obstruction:
The clean multiparent identity contains

$$-\binom{\rho_r}{2}\log r,$$

a quadratic selected-family self-collision cost.
Safe replacement:
Convert aggregate reflected mass first to distinct-base reciprocal mass/population, or use separate rounds with a proved global single-charge mechanism.

FAIL-MONOMIAL-MOD3-NOVELTY-vNEXT4.11

FAIL-ID:
FAIL-MONOMIAL-MOD3-NOVELTY-vNEXT4.11.
Date / iteration:
vNEXT4.11.
Claim:
The mod-3 collapse used in the repeated-b classification is new in vNEXT4.11.
Status:
SCOPE-CORRECTED (provenance correction).
Exact certificate:
Its source is the earlier `C-MONOMIAL-MOD3-COLLAPSE` theorem from vNEXT4.7.
What survives:
The repeated-entry-cofactor classification remains valid subject to its own audit; only the novelty/provenance claim is corrected.

FAIL-MONOMIAL-NEEDS-SPARSITY-001

FAIL-ID:
FAIL-MONOMIAL-NEEDS-SPARSITY-001.
Date / iteration:
vNEXT4.8.
Claim:
The prime-square branch can only be handled by proving that q and $N - q^2$ are sparse.
Status:
KILLED.
Exact certificate:
Monomial square edges form components with at most two edges. A large monomial family therefore forces many component starts; every nonroot start has a nonmonomial entry edge, and on $h > N^{1/3}$ those starts produce a linear packet of actual entry-parent rows. Existing rooted transport then routes the packet into horizontal reuse, low-scale states, or reciprocal mass.
What survives:
Prime-square sparsity remains an optional analytic strengthening, not the necessary binding step.

FAIL-MONOMIAL-NODESCENT-CHEAP-001

FAIL-ID:
FAIL-MONOMIAL-NODESCENT-CHEAP-001.
Date / iteration:
vNEXT4.8.
Claim:
Above $N^{1/3}$, a large monomial entry packet whose descendant closure stays above h can remain supported only by cheap vertical layers.
Status:
KILLED.
Exact certificate:
If P is the selected entry-parent packet and its descendant closure U satisfies $\min U > h > N^{1/3}$, then

$$\eta(U) \geq |P|, \qquad \log J_R > |P| \log h.$$

What survives:
The downstream question is how to consume this horizontal payment inside K1.

FAIL-MONOMIAL-PATH-LENGTH-TWO-IS-ONLY-EMPIRICAL-001

FAIL-ID:
FAIL-MONOMIAL-PATH-LENGTH-TWO-IS-ONLY-EMPIRICAL-001.
Status:
KILLED.
Killed inference:
the observed maximum monomial path length two is only a finite-census phenomenon.
Certificate:
the pre-v4.41 Core theorem already proves path length at most two. The interval $(\sqrt{N-\sqrt{N}}, \sqrt{N})$ additionally yields uniqueness of a two-edge component.

FAIL-MONOMIAL-VERTICAL-SELF-SUFFICIENT-001

FAIL-ID:
FAIL-MONOMIAL-VERTICAL-SELF-SUFFICIENT-001.
Date / iteration:
vNEXT4.8.
Claim:
A large monomial vertical family may account for the whole canonical layered defect by itself.
Status:
KILLED.
Exact certificate:

$$\eta(R) + \nu_{\neq\square}(R) \geq \left\lceil \frac{M_\square}{2} \right\rceil + 1.$$

Every monomial family forces a companion horizontal or nonmonomial vertical defect.

FAIL-MR2-CANONICAL-ONLY-001

FAIL-ID:
FAIL-MR2-CANONICAL-ONLY-001.
Date / iteration:
vNEXT4.2.
Claim:
For $q \mid m$, the equivalence $q \mid c_j \iff q \mid h - j$ is a genuinely canonical p_0-only relation.
Status:
KILLED as a scope claim.
Exact certificate:
For an arbitrary upper prime root $r = x + h_r$, $m_r = x - h_r$, one has $r = m_r + 2h_r$. Hence for every odd $q \mid m_r$,

$$q \mid (r - 2j) \iff q \mid h_r - j.$$

No minimality of p_0 is used.
What survives:
The relation itself remains valid and useful; only its classification changes from "canonical selector information" to generic upper-root geometry.
Do not repeat:
Reserve "genuinely canonical" for conclusions using minimality/prime-freeness, e.g. $h < G$, not identities valid for every upper root.

FAIL-N1/4-IS-MULTIPLICITY-REALISATION-CEILING-001

FAIL-ID:
FAIL-N1/4-IS-MULTIPLICITY-REALISATION-CEILING-001.

Date / iteration:
vNEXT4.20.
Killed description:
$N^{1/4-\varepsilon}$ is an intrinsic ceiling on simultaneous realisation of the low-polylog mirror multiplicity.
Status:
KILLED.
Correction:
That scale belongs to the clean rough-support/lower-sieve architecture. Without demanding global complement coprimality, the full low-polylog block reservoir, of size up to $N^{1/2-o(1)}$, has distinct prime parent representatives.

FAIL-N1/6-DESIGNATED-BASE-BARRIER-001

FAIL-ID:
FAIL-N1/6-DESIGNATED-BASE-BARRIER-001.
Date / iteration:
vNEXT4.19 correction of vNEXT4.15–vNEXT4.17.
Killed claim:
Variable-mass clean selection cannot act on designated bases larger than $N^{1/6-o(1)}$.
Status:
KILLED.
Exact correction:
The selector requires only

$$Qz^{2+\vartheta} \ll N^{1/2}/\operatorname{polylog} N.$$

With $z = (\log N)^D$, designated bases may reach

$$Q = N^{1/2}/\operatorname{polylog} N.$$

Call $N^{1/6}$ only the same-scale specialization $z \asymp Q$.

FAIL-NAIVE-MU-LE-3ETA-001

FAIL-ID:
FAIL-NAIVE-MU-LE-3ETA-001.
Status:
KILLED.
Killed inference:
for $q > N^{1/4}$, valuation depth at most three gives $\mu_H(q) \le 3(\kappa_H(q) - 1)$.
Certificate:
if $\kappa_H(q) = 1$ and $d_H(q) = 3$, then $\mu_H(q) = 2$ while $\eta_q = 0$.
Safe replacement:
$\mu_H(q) \le 3\eta_q + 2$; for general depth m, $\mu_H(q) \le m\eta_q + (m-1)$.
Do not repeat:
never discard the occupied-base additive term in a power-threshold argument.

FAIL-NATURAL-DENOMINATOR-IS-ONLY-BASELINE-001

FAIL-ID:
FAIL-NATURAL-DENOMINATOR-IS-ONLY-BASELINE-001.
Date / iteration:
vNEXT4.14.
Claim:
A high reflected denominator family may carry only its first copies in D_C, with no further canonical consequence.

Status:
KILLED .
Exact certificate:
If its selected parent descendants do not reach $\le h$, then

$$\log J_R > \frac{\log h}{3\log N} L(E),$$

so a fixed positive fraction of the same logarithmic family weight is paid horizontally.

FAIL-NESTED-PRIMEPOWER-PACKING-GIVES-CUMULATIVE-SAVING-001

FAIL-ID:
FAIL-NESTED-PRIMEPOWER-PACKING-GIVES-CUMULATIVE-SAVING-001.
Status:
KILLED .
Killed inference:
simultaneous use of the nested spacing bounds for $r^2, r^3, \ldots, r^e$ must create an extra cumulative saving beyond the single-layer parent-slot envelope.
Certificate:
if κ rows lie in one exact layer $v_r(N-p) = e$, then $n_k(r) = \kappa$ for every $1 \le k \le e$. Consequently all nested layers are realised by the same packet and

$$\sum_{k\ge 2} (n_k(r) - 1)_+ = (e-1)(\kappa - 1).$$

The support graph sees only $\kappa - 1$ repeated r-edges.
Do not repeat:
nested packing alone cannot turn valuation depth into an $O(1)$ or $O(\log\log N)$ bounded-multiplicity graph charge.

FAIL-NONMONOMIAL-VERTICAL-IS-INDEPENDENT-001

FAIL-ID:
FAIL-NONMONOMIAL-VERTICAL-IS-INDEPENDENT-001.
Date / iteration:
vNEXT4.7.
Claim:
For $h > N^{1/3}$, a nonmonomial event $N - w = kq^2$, $k > 1$, may remain an independent vertical obstruction.
Status:
KILLED as a description.
Exact certificate:
Every prime $r \mid k$ is rooted and satisfies

$$r < N/h^2 < h.$$

For a family of such events, repeated siblings force horizontal/collision mass, while distinct siblings create an explicit lower-scale rooted packet.
What survives:
A downstream consumer/upper-capacity problem for the exported sibling/collision alternatives.

FAIL-ONE-REPRESENTATIVE-CONSUMES-MIRROR-MULTIPLICITY-001

FAIL-ID:
FAIL-ONE-REPRESENTATIVE-CONSUMES-MIRROR-MULTIPLICITY-001.

Date / iteration:
vNEXT4.16.
Claim:
The existing clean selector closes the smooth-mirror harmonic branch merely by selecting one representative incidence for each active base.
Status:
KILLED.
Exact obstruction:
Reflected packing permits about t/r incidences at a small base r. Present clean selectors retain one incidence per base, or repeated multiplicity only for a fixed-polylog total multiset.
Replacement target:
Preserve a positive fraction of the weighted multiplicity before applying the signed K1 ledger.

`FAIL-ORTHOGONAL-BRIDGE-001`

FAIL-ID:
FAIL-ORTHOGONAL-BRIDGE-001.
Date / iteration:
vNEXT3, Phases 2–3.
Claim:
Although cross-collision is dead under $K_\times(R, U) = 0$, a source- or product-side functional can still carry $\eta(U)$ from an external orthogonal packet into a capacity determined by R.
Original motivation:
The source product of `G-MULTISOURCE-LOCK` does change when a packet is adjoined, so it looked like a live channel.
Status:
KILLED.
Minimal / strong counterexample:
Structural, by exact decoupling.
Exact certificate:
For disjoint bad child-closed R, U, `G-ORTHOGONAL-DECOUPLING` gives $\Gamma_N(S) = \Gamma_N(R)\dot\cup\Gamma_N(U)$, $\mathrm{Src}(S) = \mathrm{Src}(R)\dot\cup\,\mathrm{Src}(U)$, $T_S = T_R T_U$, $J_S = J_R J_U$, $U_S = U_R U_U$, $\eta(S) = \eta(R) + \eta(U)$, $\nu(S) = \nu(R) + \nu(U)$, and the multisource lock for S factors as the product of the two locks. Every recorded functional is therefore a disjoint-union homomorphism, and the recorded state of R is *identical* for every admissible U. The target inequality $\log(T_{R\cup U}/T_R) \geq c_0\eta(U)$ does hold, with $c_0 = \log 3$, and is worthless: its left side equals $\log T_U$.
Failed hypothesis or mechanism:
The source product does change, but it changes by exactly U's own source product; nothing is shared.
Why the argument failed:
$K_\times(R, U) = 0$ is not an extra hypothesis to be worked around — it is *automatic* for disjoint child-closed packets, and it is a symptom of total decoupling rather than of one blocked channel.
What survives:
`G-BRIDGE-ROUTING-EQUIVALENCE`: a bridge can exist only if $U \cap R \neq \varnothing$, which is precisely the TD/TF routing target. The two fronts are one.
Safe weaker version:
Report $\eta(U)$ as a property of U alone.
Current replacement:
Theory Core, Theorem [thm:decoupling], Corollary [cor:worthless-coupling], Theorem [thm:bridge-routing].
Do not repeat:
Do not seek any functional coupling two disjoint child-closed packets; all recorded functionals

provably factor.
Related current target:
TD, TF, K1.

FAIL-ORTHOGONALITY-AS-HYPOTHESIS-001

FAIL-ID:
FAIL-ORTHOGONALITY-AS-HYPOTHESIS-001.
Date / iteration:
vNEXT3, Phase 2.
Claim:
$K_\times(R, U) = 0$ is a special favourable hypothesis on the external packet, to be secured by construction.
Original motivation:
It is written as a hypothesis throughout the external-packet material.
Status:
SCOPE-CORRECTED.
Exact certificate:
(OD4). If R and U are disjoint and both child-closed, then $\Gamma_N(R) \subseteq R$ and $\Gamma_N(U) \subseteq U$ are disjoint, so no prime divides a complement on both sides and every cross gcd is 1. Orthogonality is a consequence of disjointness plus child-closure, not an independent assumption.
What survives:
Every statement conditioned on orthogonality, now with the hypothesis recognised as redundant given the other two.
Do not repeat:
Do not spend effort "arranging" orthogonality, and do not read it as evidence of a favourable configuration; it signals decoupling.
Related current target:
TF, K1.

FAIL-OWNED-RETURN-FIRST-EDGE-IS-FREE-SOURCE-LOSS-001

FAIL-ID:
FAIL-OWNED-RETURN-FIRST-EDGE-IS-FREE-SOURCE-LOSS-001.
Status:
KILLED on the v18 owner-completed domain.
Rejected statement:
The first later return edge into an external rough seed can always be absorbed solely by destroying that seed as a source.
Certificate:
the owner row p_i already supplies $p_i \to r$; a later $u \to r$ is a second occurrence and creates a full $+\log r$ incremental J charge.
Scope guard:
This does not invalidate the generic vNEXT4.28 ledger; it strengthens it on the retained-owner domain.

FAIL-OWNER-SEALED-RETURN-CAN-STAY-SQUAREFREE-001

FAIL-ID:
FAIL-OWNER-SEALED-RETURN-CAN-STAY-SQUAREFREE-001.
Status:
KILLED.
Rejected statement:
A return row using only the owner support $\{q_0\} \cup \mathrm{PF}(b_i)$ may be a cheap squarefree recycle.

Certificate:
its complement is larger than the squarefree owner complement while its radical divides that owner complement; verticality is forced.
Do not repeat:
Owner support preservation is a verticality certificate, not a squarefree escape.

FAIL-OWNERLESS-FIRST-OCCURRENCE-HAS-NO-ADDITIVE-STRUCTURE-001

FAIL-ID:
FAIL-OWNERLESS-FIRST-OCCURRENCE-HAS-NO-ADDITIVE-STRUCTURE-001.
Status:
KILLED.
Killed description:
a fresh returned reflected factor with no prior owner is arithmetically anonymous apart from its support incidence.
Certificate:
for $c_j = sq$ and a low active parent $N - w = bq$, v4.39 gives

$$\frac{10}{9}s < b < 2s, \qquad a_j - w = (b - s)q,$$

with $b - s$ positive even and

$$a_j - w > N/18.$$

What survives:
this additive relation still needs a capacity / second- occurrence consumer.

FAIL-P0-INFORMATION-SELECTOR-001

FAIL-ID:
FAIL-P0-INFORMATION-SELECTOR-001.
Date / iteration:
vNEXT4.1, Phase 9.
Claim:
The central hypothesis of the round: that p_0 identifies the particular additive–multiplicative correlations a global prime-distribution theorem should preserve.
Original motivation:
The canonical reflected block is the only place in CRD where a *composite* row of size $> N/2$ with a canonical mirror is produced deterministically.
Status:
KILLED. Not refuted; **not established**.
Exact certificate:
Theory Core `P0-CANONICITY-ABLATION`. Every clause of `P0-EXACT-BUNDLE-LIFT` classifies as (AB-A) "fixed N only" or (AB-B) "an arbitrary reflected pair $a + c = N$". Replacing p_0 by any upper-half prime p_1 leaves (BL3)–(BL6) and (BL5) word for word intact on every row whose c'_j is composite. Canonicity is used in exactly one place: the prime-free interval $[x, p_0)$ makes *every* c_j composite, so branch (MC1) is available on the whole block. Measured on the 301 recorded noncanonical failed upper roots, 32,766 of 130,090 reflected rows — 25.2% — have c'_j prime and are unbundleable (their single block costs $c_j'^2 > N$).
Failed hypothesis or mechanism:
A non-vacuity guarantee is not an information-selection principle. Canonicity guarantees the construction is always available; it does not make the construction, or its conclusion, different.
What survives:
One genuinely canonical statement, (MR2): for $q^e \| m$, $q \mid c_j \iff q \mid h - j$, from `ARS-CANON-GEOMETRY`. No consequence was built on it in this round.

Do not repeat:
Do not record a `P0-INFORMATION-SELECTOR-CERTIFICATE` without an ablation in which the conclusion actually fails after the canonical datum is deleted. None was found.
Related current target:
`P1BR-ADDITIVE-COUPLING`, `P0-BUNDLE-ROUGH-HIGHPOWER-BRANCH`.

`FAIL-P0-ZEROHOLE-ESSENTIAL-001`

FAIL-ID:
FAIL-P0-ZEROHOLE-ESSENTIAL-001.
Date / iteration:
vNEXT4.2.
Claim:
The fact that the canonical reflected block has no prime holes is, by itself, the essential p_0-specific additive–multiplicative glue.
Status:
KILLED as a mechanism for the active/overflow mass argument.
Exact certificate:
If p_1 is the immediate next prime after canonical $p_0 > x$, the strict p_1-reflected block has exactly one prime hole, namely p_0, at index $(p_1 - p_0)/2$. More generally the fixed p_k block has exactly the k earlier primes as holes. Iterating the recorded pointwise short-interval bound a fixed number of times gives $p_k - x = O_k(x^{0.525})$, so deleting the holes costs only $O_k(\log x)$ in the block active-mass identity when the width tends to infinity.
Why the argument failed:
The earlier 25.2% statistic concerned a broad family of noncanonical upper roots; it was not the immediate $p_0 \to p_1$ ablation. The correct immediate ablation is a one-hole perturbation, not a positive-density destruction of the block.
What survives:
p_0 still gives the exact zero-hole formulation and may be essential through stronger canonical information such as $h < G$, Exact-D, or another conclusion which actually fails already at p_1.
Do not repeat:
Do not infer a `P0-INFORMATION-SELECTOR-CERTIFICATE` from the absence of prime holes alone.
Related current target:
`C-CRITICAL-SQRT-GAP-BRANCH`.

`FAIL-P4-IS-THE-BEST-BV-ARITY-001`

FAIL-ID:
FAIL-P4-IS-THE-BEST-BV-ARITY-001.
Date / iteration:
vNEXT4.21.
Killed claim:
Ordinary full-interval Bombieri–Vinogradov can support only the old P_4 rough cofactor output.
Status:
KILLED.
Replacement:
The explicit Richert-weighted dimension-one sieve in vNEXT4.21 gives a tagged squarefree P_3 fan with rough floor $N^{6/25-o(1)}$, pending independent analytic audit of the weighted-sieve interface.

FAIL-PAIR-LOADED-DESCENT-OPEN-001

FAIL-ID:
FAIL-PAIR-LOADED-DESCENT-OPEN-001.
Date / iteration:
vNEXT4.15.
Old obstruction:
Rows $N - w = Bq_1q_2$ with $q_i \geq h$ and $B > 1$ might create an uncontrolled low residual packet.
Status:
CLOSED AS A POLYLOG-Λ SCALE-REENTRY PROBLEM.
Exact certificate:
If $\Lambda \leq (\log N)^A$, then $B < 2\Lambda$; residual prime factors are fixed-polylog BAD states. A large family therefore produces reuse or enters the repaired rooted/external clean ledger.

FAIL-PAIRWISE-K-IS-THE-ONLY-POSSIBLE-X1-CAPACITY-001

FAIL-ID:
FAIL-PAIRWISE-K-IS-THE-ONLY-POSSIBLE-X1-CAPACITY-001.
Status:
KILLED.
Rejected statement:
The final X1 stitching must upper-bound the full pairwise functional $K_\times(R, P) + K(P)$.
Certificate:
vNEXT4.31 proves the exact promotion hierarchy

$$L(A) \leq \log \frac{J_{R\cup P}}{J_R} \leq K_\times(R, P) + K(P).$$

Thus the weaker target K1-INC, bounding only the actual incremental J by the same P1BR functional, is logically sufficient.
Do not repeat:
Do not identify "K1 is sufficient" with "K1 is logically necessary."

FAIL-PARENT-LATTICE-DOMAIN-001

FAIL-ID:
FAIL-PARENT-LATTICE-DOMAIN-001.
Date / iteration:
v0.1 audit, canonical–ambient split correction.
Claim:
The parent lattice of a layer may be treated as the full residue class $p \equiv N \pmod{\ell^k}$ over all primes.
Original motivation:
The analytic count is naturally stated over all primes in an interval.
Status:
SCOPE-CORRECTED.
Minimal / strong counterexample:
Definitional.
Exact certificate:
The correct object is $\mathcal{P}_N(\ell^k) = \{p \in \mathcal{A}_N : \ell^k \mid N - p\}$, the intersection of the congruence class with the *finite active prime universe* $\mathcal{A}_N$. Primes dividing N, and primes at least N, are not states.
Failed hypothesis or mechanism:
An ambient analytic set was used as a set of CRD states.

Why the argument failed:
Counting in the class is legitimate; treating every counted prime as a usable parent is not.
What survives:
`G-POLYLOG-PARENTS`, which restricts to $p \in (N/2, N)$ and proves activity and externality explicitly.
Safe weaker version:
Intersect with $\mathcal{A}_N$ at the moment the class is introduced.
Current replacement:
Theory Core, notation entry $\mathcal{P}_N(\ell^k)$ and `G-POLYLOG-PARENTS`.
Do not repeat:
A residue class is not a state set.
Related current target:
W1, K1.

`FAIL-PATH-PRODUCT-AGGREGATION-ERASES-DEEP-VALUATION-001`

FAIL-ID:
FAIL-PATH-PRODUCT-AGGREGATION-ERASES-DEEP-VALUATION-001.
Source:
explicit proposed Failure Log item in v4.51; also v4.51 R51-2.
Status:
KILLED for the inference

$$\text{remote reflected root+aggregate affine path product} \Longrightarrow \text{control of the terminal exponent } E.$$

Certificate:
after appending the last edge $p \to r$, the aggregate affine identity recombines the terminal row into the single integer $N - p$; modulo N, the factor r^{E-1} in the total path product is cancellable because $r \nmid N$.
Scope guard:
this is not a no-go for CRD1-R. The full valuation tensor still knows $v_r(N - p) = E$. The loss occurs only after compressing the terminal edge into the aggregate path product.
Replacement:
retain the final row/layer datum (p, r, E) edge-locally.

`FAIL-PER-ROW-RADC-COMPRESSION-001`

FAIL-ID:
FAIL-PER-ROW-RADC-COMPRESSION-001.
Status:
KILLED as a failure certificate.
Reason:
LCAP4 controls the full product $D_C J_R U_R^{\mathrm{nr}}$, not the radical factor $D_C J_R$ alone. Large mass may be stored in U_R^{nr}. The revised report had conflated the three-channel product with one channel.

`FAIL-PLAIN-AP-COUNT-KILLS-VERTICAL-001`

FAIL-ID:
FAIL-PLAIN-AP-COUNT-KILLS-VERTICAL-001.
Status:
SCOPE-CORRECTED.
Killed sufficient strategy:
use Brun–Titchmarsh or Bombieri–Vinogradov pointwise to rule out one dynamically selected canonical exact layer or to infer non-reachability in R.

Safe surviving statement:
ambient AP estimates do not see canonical reachability and need not kill singleton or exceptional layers.
v4.36 correction:
Brun–Titchmarsh becomes effective after summing over *all* repeated prime-power layers together with the canonical γ-row budget and the weighted layer count $A_2(Y) \ll \sqrt{Y}$. Thus the old no-go was correct pointwise and too broad as an aggregate prohibition.

FAIL-POLYLOG-IS-NEEDED-FOR-DETERMINISTIC-EXACT-STAR-EXPORT-001

FAIL-ID:
FAIL-POLYLOG-IS-NEEDED-FOR-DETERMINISTIC-EXACT-STAR-EXPORT-001.
Status:
SCOPE-CORRECTED .
Scope correction:
the polylogarithmic restriction belongs to some analytic star-selection / finite-alphabet stages of the old v4.26 apparatus. If a fixed active base already has polynomial row multiplicity, exact-exponent pigeonhole loses only $N^{o(1)}$, and parent-slot / quotient inflation gives a q-free residual export of $\Omega(n \log N)$ for arbitrary base size.
What survives:
the hard part is producing polynomial concentration from the dispersed cube-root defect, not consuming such concentration after it is known.

FAIL-POSITIVE-ROUND-BLOCKS-REPEATED-FAN-PROGRESS-001

FAIL-ID:
FAIL-POSITIVE-ROUND-BLOCKS-REPEATED-FAN-PROGRESS-001.
Date / iteration:
vNEXT4.18.
Claim:
Because a positive signed singleton round is a dead end, the repeated-base fan gives no progress unless all rounds are nonpositive.
Status:
KILLED .
Correction:
Independently of signs, every selected row supplies a distinct rough active child above $N^{1/4-\varepsilon}$. The polynomial state/source packet therefore survives the positive-round wall, although it still is not an upper-capacity contradiction.

FAIL-POSITIVE-ROUND-CONSEQUENCE-001

FAIL-ID:
FAIL-POSITIVE-ROUND-CONSEQUENCE-001.
Date / iteration:
vNEXT3.1, Phase-3 semantic audit.
Claim:
Forcing a positive signed round is progress toward a contradiction, so the (FC10) route closes once γ is controlled.
Original motivation:
The (DP1)–(DP5) depletion apparatus is elaborate and its conclusion is that nonpositive rounds cannot continue; the natural reading is that the alternative must be favourable.
Status:
KILLED . The depletion apparatus is untouched; the inference from its escape branch is not supported.

Exact certificate:

By (EB6), $\Delta > 0$ says exactly $\sum_{r\in B_0} \log r > W_R(B_-)$: the fresh support mass of the selected rows exceeds the rooted reuse mass they touch. A mechanical audit of every `DOMAIN` line and every proof in the Theory Core found three theorems assuming *nonpositive* rounds — `G-ROW-SATURATION-ROOTED-DEFECT`, `G-EXTERNAL-BAD-SUPPORT-SATURATION-DISPOSITION`, `G-GLOBALLY-CLEAN-EXTERNAL-SUPPORT-BUDGET` — and *none* assuming a positive round. Nothing derives a GOOD state, a capacity violation, or any canonical obligation from $\Delta > 0$.

Failed hypothesis or mechanism:

The escape branch of a dichotomy was treated as a conclusion. Structurally $\Delta > 0$ is *growth* of the enlarged world, not contradiction; and growth cannot be iterated to exhaustion, because designated bases live below $(\log N)^A$ while $|\mathcal{A}_N| \sim N/\log N$.

What survives:

Everything about nonpositive rounds, and `G-GAMMA-POLYLOG-FORCES-POSITIVE-ROUND`, which correctly delivers the positive round. What is corrected is only what that delivery is worth.

Safe weaker version:

"γ polylog forces a positive signed round." Never "… forces a contradiction".

Current replacement:

Theory Core, Proposition [prop:positive-dead-end] and the new open target `P1BR-POSITIVE-ROUND-CONSEQUENCE`.

Do not repeat:

Do not announce that a wall has moved because an object is now forced to exist; check first that something consumes that object.

Related current target:

`P1BR-POSITIVE-ROUND-CONSEQUENCE`, P1BR, K1.

`FAIL-POSITIVE-ROUND-EXHAUSTION-001`

FAIL-ID:

FAIL-POSITIVE-ROUND-EXHAUSTION-001.

Date / iteration:

vNEXT4, Phase 9.

Claim:

That repeated positive rounds must eventually run out of fresh primes, giving a contradiction by exhaustion.

Original motivation:

Nonpositive rounds deplete the integer potential $\mathfrak{J}_k$ of (DP4); by symmetry one hopes positive rounds deplete something too.

Status:

KILLED, quantitatively.

Exact certificate:

All fresh support of an external upper-half round lies below $N/6$ by (RC2). The total logarithmic demand of the whole selector output at exponent A is at most $(\log N)^{A+1}$; the supply in that band is $\vartheta(N/6) \geq N/12$. The ratio tends to 0 like $(\log N)^{A+1}/N$. Theory Core, `G-AMBIENT-SUPPLY-NONDEPLETION`.

Failed hypothesis or mechanism:

$\Delta > 0$ is a statement of growth. Growth has no terminal state when the reservoir is superpolynomially larger than the demand.

What survives:

The depletion statement for *nonpositive* rounds, (DP1)–(DP4), which is unaffected and remains the only monotone integer potential in the DAG.

Do not repeat:
Do not argue by exhaustion without comparing demand to ϑ of the relevant band.
Related current target:
`P1BR-ADDITIVE-COUPLING`.

`FAIL-POSITIVE-ROUND-LEDGER-CONSEQUENCE-001`

FAIL-ID:
FAIL-POSITIVE-ROUND-LEDGER-CONSEQUENCE-001.
Date / iteration:
vNEXT4, Phase 7.
Claim:
That the failure recorded at `FAIL-POSITIVE-ROUND-CONSEQUENCE-001` was a gap in the present list of theorems, so that a consequence of $\Delta > 0$ could be found by writing more theorems in the same language.
Original motivation:
vNEXT3.1 established option F by a mechanical audit of the file's headers. An audit of a text is silent about what could be added to the text.
Status:
KILLED for ledger-only derivations. **Not** a refutation of `P1BR-POSITIVE-ROUND-CONSEQUENCE` itself.
Exact certificate:
Define the round datum $\mathfrak{d}(S, \mathscr{P})$ as the rows, their supports, the heights on those supports and κ_S restricted to those supports. Every quantity in (EB6)–(EB10), (DP1)–(DP5) and (PD1) is a function of $\mathfrak{d}$. The family $\mathfrak{S}_n$ of Theory Core `G-POSITIVE-ROUND-LEDGER-INDEPENDENCE` has constant $\mathfrak{d}$, constant $\Delta > 0$, and $\eta(R_n) = n + 1 \to \infty$; it is child-closed with a unique source, has no unit rows, and is routing-compatible. Its member $n = 1$ has $\eta(R_1) = 2$ while Δ can be made arbitrarily large by raising the two fresh heights. Hence the only lower bound on η expressible in $\mathfrak{d}$ is the trivial $\eta_{\text{supp}} = \sum_{r \in B_-}(\kappa(r) - 1)$, which is 0 whenever $B_- = \varnothing$; in particular no such bound grows with Δ. The witnesses exist by `GEN-FINITE-REALISATION`.
Failed hypothesis or mechanism:
The (LE3)+(FC8) route consumes a *lower* bound on $\eta(R)$ of order $\gamma \log N$. A nonpositive round supplies one by (EB8). A positive round supplies an *upper* bound on the rooted mass of its own support and constrains $\Gamma_N(R)$ nowhere else — which is exactly where the witnesses place all their incidence.
What survives:
The asymmetry is proved, not assumed: for a nonpositive round the same datum does bound $\eta(S)$ below. The obstruction is the sign, not the bookkeeping.
Safe weaker version:
A consequence may still be derivable using an axiom that the witness family violates. Three are identified and recorded as live: (L1) $\mathcal{D}(p) = N - p$; (L2) $\gcd(N - p, N - p') \mid p' - p$; (L3) band localisation plus prime counting. All three are additive.
Do not repeat:
Loss-control rule (LR1): a claimed consequence of a positive round must name a quantity that is not a function of the round datum.
Related current target:
`P1BR-ADDITIVE-COUPLING`.

`FAIL-POSITIVE-SEMIPRIME-ROUND-IS-NEXT-PRIMARY-001`

FAIL-ID:
FAIL-POSITIVE-SEMIPRIME-ROUND-IS-NEXT-PRIMARY-001.
Status:
SUPERSEDED.

Superseded description:
after v4.41, the binding branch is to convert a positive clean semiprime round directly through P1BR additive coupling.
Certificate:
v4.42 continues through the actual designated BAD states q_i and their stopped descendant closure. It forces ambient incremental J or a linear cube-root boundary load before any positive-sign consumer is needed.
What survives:
positive sign remains a valid certificate where proved, and P1BR-ADDITIVE-COUPLING remains open in its own right.

FAIL-POTENTIAL-CAPACITY-001

FAIL-ID:
FAIL-POTENTIAL-CAPACITY-001.
Date / iteration:
v7.5 first-front audit; retained in v0.1.
Claim:
A first-front-only potential capacity $\mathcal{C}_S < \mathcal{D}_S$ holds canonically and supplies the needed contradiction.
Original motivation:
Pairwise row isolation seemed to make the available factor capacity smaller than root demand.
Status:
KILLED.
Minimal / strong counterexample:
$N = 4736, p_0 = 2371, h = 3$, $m = 2365 = 5 \cdot 11 \cdot 43$.
Exact certificate:
The three first-front rows are

$$4736 - 5 = 4731 = 3 \cdot 19 \cdot 83, \quad 4736 - 11 = 4725 = 3^3 \cdot 5^2 \cdot 7,$$

$$4736 - 43 = 4693 = 13 \cdot 19^2.$$

With the source definitions, the audited values are $\mathcal{C}_S \approx 22.60 > \mathcal{D}_S \approx 20.16$.
Failed hypothesis or mechanism:
Potential parents were counted as if their availability already encoded closure and actual incidence.
Why the argument failed:
Pair isolation controls collisions, not which rows are dynamically realised or simultaneously usable.
What survives:
Exact collision identities for actual rows.
Safe weaker version:
Introduce closure/activation first, then charge only realised rows.
Current replacement:
Theory Core, G-COLLISION-CUT, G-AMBIENT-PROMOTION, and target $P1^{\mathrm{br}}$.
Do not repeat:
Potential incidence is not actual incidence.
Related current target:
$L1, W1, K1$.

FAIL-POWER-THRESHOLD-TO-POLYLOG-001

FAIL-ID:
FAIL-POWER-THRESHOLD-TO-POLYLOG-001.

Status:
SCOPE-CORRECTED.

Original killed formulation:
power-threshold compression can never force a polylogarithmic / near-logarithmic star, because reaching that scale requires $m \to \infty$.

Correction:
variable m is legal with its explicit occupied-base error. Taking $m \asymp \log N/\log\log N$ can force $q_0 \le (\log N)^{1+o(1)}$ with multiplicity $N^{1/3-o(1)}$ on the low branch.

What the no-go really says:
one threshold cannot simultaneously reach that star scale and retain the old $M \log N$-scale generic high-J lower bound. The high branch falls to $M(\log\log N)^2/\log N$.

Do not repeat:
do not interpret the v4.43 fixed-m calculation as a ban on growing m.

FAIL-PRODUCT-BUDGET-ALONE-CANNOT-PRESERVE-NEARCRITICAL-STAR

FAIL-ID:
FAIL-PRODUCT-BUDGET-ALONE-CANNOT-PRESERVE-NEARCRITICAL-STAR.

Status:
KILLED.

Killed mechanism:
preserve $N^{1/3-o(1)}$ rows under repeated exact-layer peeling using only distinct peeled bases, growth of the common modulus, parent-slot, and separate valuation-layer counts.

Sharp cumulative-loss bound:
writing $X = \log N$, $B_i = \log Q_i$ and $\ell_i = \log q_i$,

$$\mathcal{L}_t := \sum_{i<t} \log E_i \le \sum_{i<t} \log \frac{X - B_i}{\ell_i} \le \log \frac{Q_t}{Q_{\mathrm{in}}} + (1 + o(1)) \frac{\log N}{(\log\log N)^2}.$$

The coefficient 1 before $\log(Q_t/Q_{\mathrm{in}})$ is asymptotically sharp from the inspected scalar data alone.

Certificate:
a primorial hostile sequence with consecutive small primes up to $a \log N$, $1/3 < a < 2/3$, satisfies simultaneously

$$Q_t = N^{a+o(1)}, \qquad \prod_i E_i = N^{a+o(1)}.$$

Thus the formal cardinality can collapse before the modulus reaches the $N^{2/3}$ parent-slot scale.

Sharp scope guard:
the no-go ignores global compatibility of all valuations in the same integer $N - p$ and therefore does not apply to simultaneous joint-valuation compression.

Current replacement:
`G-SUBCRITICAL-ALPHABET-SIMULTANEOUS-VALUATION-FREEZE`.

FAIL-PROFILE-CARRIER-001

FAIL-ID:
FAIL-PROFILE-CARRIER-001.

Date / iteration:
Interrupted T_D round, 31 August 2026.

Claim:
The cardinality profile of the four C0b classes separates canonical from noncanonical roots and can therefore carry a theorem.
Original motivation:
Hard canonical controls were reported as D-heavy and $\mathcal{F}$ as associated with easy or direct-GOOD controls.
Status:
SUPERSEDED .
Minimal / strong counterexample:
The recorded profiles themselves.
Exact certificate:
The recovered log states that raw profiles did not separate the relevant outcomes and that the reported associations were explicitly not promoted to theorems. The associated 97.81% D-fresh rate is VERIFIED COMPUTATION , with no surviving artifact.
Failed hypothesis or mechanism:
A correlation was treated as a mechanism.
Why the argument failed:
Cardinalities discard the arithmetic that produces the classes.
What survives:
The *negative* conclusion only: cardinality-only theorem design is rejected as a current route.
Safe weaker version:
Select a branch by the theorem it produces, not by the correlation it exhibits.
Current replacement:
Theory Core, `TD-CARRIER-SPLIT`. T_D is selected because it factorises, not because it correlates.
Do not repeat:
Never promote a computational rate to a mechanism.
Related current target:
T_D.

`FAIL-PURE-PATH-ONLY-RETAINS-MOD-Q-001`

FAIL-ID:
FAIL-PURE-PATH-ONLY-RETAINS-MOD-Q-001.
Source:
explicit v4.54 Failure Log addition.
Status:
KILLED .
Killed description:
repeated pure edges preserve only the original endpoint congruence modulo Q.
Correction:
the fixed cofactor Q on every pure edge lifts the path-start congruence to modulus Q^{L+1}.
Scope guard:
this does not contradict the earlier aggregate affine-path no-go, which allowed arbitrary path cofactors and therefore could lose the deep exponent.

`FAIL-PURE-Q-GRAPH-CAN-BRANCH-001`

FAIL-ID:
FAIL-PURE-Q-GRAPH-CAN-BRANCH-001.
Source:
explicit v4.54 Failure Log addition.
Status:
KILLED .

Killed possibility:
a pure target q can have two distinct pure parents.
Reason:
a pure parent is uniquely determined by

$$p = N - Qq.$$

Thus the pure graph is injective.

FAIL-Q0-LEDGER-001

FAIL-ID:
FAIL-Q0-LEDGER-001.
Date / iteration:
Post-v7.5 incidence-ledger audit, 30 August 2026.
Claim:
The four channels proved for the smooth first-front domain cover every $q \in Q_0 = \mathrm{PF}(N - p_0)$ without an under-parented channel.
Original motivation:
The same residual split appeared syntactically available for all root factors.
Status:
KILLED.
Minimal / strong counterexample:
$N = 20, p_0 = 11, Q_0 = \{3\}$.
Exact certificate:
$20 - p_0 = 9$, $20 - 3 = 17 \in \mathbb{P}$, so 3 is GOOD. Under stopped traversal $R = \{11\}$. Moreover $D_3 = \gcd(9, 17) = 1$, $M_3 = 17$, while the required 17-incidence in R is zero although the root demand is one.
Failed hypothesis or mechanism:
A successful first-front state was treated as though it were an expanded bad-world row.
Why the argument failed:
Stopping at GOOD deletes precisely the row needed by the proposed channel accounting.
What survives:
The exact ledger on its verified smooth/failed domain; the arithmetic identity $D_q M_q = N - q$.
Safe weaker version:
Retain the original domain or add an explicit under-parented/success channel.
Current replacement:
Theory Core, Full-Q Failure Ledger and C0b-RD, both with explicit stopped-world hypotheses.
Do not repeat:
Never enlarge a ledger domain merely because its formulas remain syntactically defined.
Related current target:
T_I, T_D, T_S, T_F.

FAIL-Q0-ONLY-OWNER-SEALED-RETURN-IS-CHEAP-001

FAIL-ID:
FAIL-Q0-ONLY-OWNER-SEALED-RETURN-IS-CHEAP-001.
Status:
KILLED.
Rejected statement:
If owner-sealed verticality occurs only at the small common hub q_0, the branch is cheap.
Certificate:
polynomial q_0^2 occupancy enters the already retained vNEXT4.26 exact-layer star peeling and exports full q_0-free logarithmic mass.

FAIL-QFREE-FIRST-COPY-CAN-HIDE-BELOW-CARDINALITY-001

FAIL-ID:
FAIL-QFREE-FIRST-COPY-CAN-HIDE-BELOW-CARDINALITY-001.
Source:
explicit v4.52 Failure Log addition.
Status:
KILLED.
Killed possibility:
on a polynomial exact fibre with cheap J/U_{res}, all unique external q-free first-copy support can remain at primes $\leq n$.
Reason:
the unique external q-free support has $\Omega(n \log N)$ weight, while all primes at most n carry only $O(n)$ total Chebyshev weight.
Replacement:
a constant-scale portion escapes above n, yielding `C-DEEP-CARDINALITY-SCALE-SIBLING-RENEWAL`.

FAIL-QUOTIENT-SIEVE-POLYNOMIAL-RANGE-001

FAIL-ID:
FAIL-QUOTIENT-SIEVE-POLYNOMIAL-RANGE-001.
Date / iteration:
vNEXT4.6.
Claim:
The fixed-(s, b) two-linear-form upper sieve can be union-bounded without loss over all $s \leq N^{\delta}$.
Status:
KILLED.
Exact certificate:
For a given s, there are $O(s)$ possible parent coefficients b, while the reflected progression has only $O(t/s)$ indices. These factors cancel, leaving roughly one $t/\log^2 t$ cost per quotient band. The present union bound therefore gives no polynomial range.
What survives:
Fixed and sufficiently slowly growing S satisfying (QS4).
Do not repeat:
Do not spend another round merely optimising constants in QS3 in hopes of reaching N^{δ}.

FAIL-RDS-GEOMETRY-DENSITY-001

FAIL-ID:
FAIL-RDS-GEOMETRY-DENSITY-001.
Date / iteration:
v1.13 support-free continuation, 1 September 2026.
Claim killed:
From canonical reflected geometry and $\mathrm{rad}(c_j) \mid d_j$ alone, one may infer a universal bound $|\mathrm{RDS}| \leq t/8$.
Status:
KILLED.
Exact certificate:
Take

$$N = 4368, \quad x = 2184, \quad p_0 = 2203, \quad h = 19, \quad \mathrm{prevprime}(p_0) = 2179, \quad G = 24, \quad t = 9.$$

Thus $h < G$, so the root is canonical. The strict reflected block contains

$$j = 3: \; d_j = 13, \quad c_j = 2197 = 13^3,$$

and

$$j = 8: \ d_j = 3, \quad c_j = 2187 = 3^7.$$

Hence both indices satisfy $\mathrm{rad}(c_j) \mid d_j$, so $|\mathrm{RDS}| \geq 2 > 9/8 = t/8$.

Scope warning:

This is a successful-world certificate: $m = N - p_0 = 2165 = 5 \cdot 433$ and the first-front state 5 is GOOD because $N - 5 = 4363$ is prime. Therefore the witness kills only a *geometry-only* density inference; it does not refute a stronger theorem conditioned on a genuinely stopped canonical failed world.

What survives:

`TS-REFLECTED-DEADSMOOTH-CEILING` is exact and remains PROVED. Any stronger failure-conditioned density bound requires additional dynamic input, not the reflected geometry alone.

Do not repeat:

Do not infer a small universal RDS density from powerful-number rarity or from finite empirical scarcity without using a proved failure-conditioned mechanism.

`FAIL-REFLECTION-ACTIVATION-001`

FAIL-ID:

FAIL-REFLECTION-ACTIVATION-001.

Date / iteration:

Canonical bridge audit, 31 August 2026.

Claim:

Canonical $h < G$ alone forces at least one reflected index j with $a_j = N - c_j$ an active prime, hence an actual reflected CRD row.

Original motivation:

The Reflection Gate Lemma makes every c_j composite, and compositeness was read as if it also produced a parent row.

Status:

KILLED.

Minimal / strong counterexample:

$N = 242$, $x = 121$, $p_0 = 127$, previous prime 113, so $h = 6 < G = 14$.

Exact certificate:

The two internal offsets give $(c_1, a_1) = (125, 117)$ and $(c_2, a_2) = (123, 119)$, with $125 = 5^3$, $117 = 3^2 \cdot 13$, $123 = 3 \cdot 41$, $119 = 7 \cdot 17$. Both reflections are composite, so no reflected row exists at all. Verified exactly.

Failed hypothesis or mechanism:

The gate was treated as a state activation statement.

Why the argument failed:

`ARS-REFLECTION-GATE` is an *if and only if*: the factorisation of c_j is a CRD row precisely when a_j is an active prime, and nothing in $h < G$ supplies that.

What survives:

Compositeness of every c_j, and the classification `COb-RD` which does not require any a_j to be prime.

Safe weaker version:

State separately *boundary* $\to$ *active reflected state* as an open implication.

Current replacement:

Theory Core, `ARS-REFLECTION-GATE`; target BIA remains **OPEN**.

Do not repeat:

Do not hide a missing activation step under the word "reflection".

Related current target:

T_I, $P1^{\mathrm{br}}$.

Domain note:
$N = 242$ is not a failed canonical instance, so it does not refute BIA itself; it refutes any proof of BIA that uses only selector geometry before invoking failure.

FAIL-REPEATED-BASE-MUST-BE-FIXED-POLYLOG-001

FAIL-ID:
FAIL-REPEATED-BASE-MUST-BE-FIXED-POLYLOG-001.
Date / iteration:
vNEXT4.18.
Restriction:
The fixed-polylog multiset limit of the four-ary multiparent selector is a fundamental limit on repeated rows from one base.
Status:
KILLED.
Correction:
A greedy rough-support matching argument selects

$$\gg N^{1/4-\varepsilon}/(\log N)^2$$

rows from one already-constructed base fan while keeping unintended rough supports pairwise disjoint.

FAIL-REPEATED-BASE-QUADRATIC-COLLISION-UNAVOIDABLE-001

FAIL-ID:
FAIL-REPEATED-BASE-QUADRATIC-COLLISION-UNAVOIDABLE-001.
Date / iteration:
vNEXT4.18.
Claim:
Repeating a designated base necessarily incurs $\binom{\rho_q}{2}\log q$ collision even when rows are separated.
Status:
SCOPE-CORRECTED.
Correction:
The quadratic term is unavoidable inside one multiparent round. Separate singleton rounds with planned designated overlap and disjoint unintended rooted payment supports admit exact single charging.

FAIL-REPEATED-HUB-QUADRATIC-COST-EQUALS-ACTUAL-STORAGE-001

FAIL-ID:
FAIL-REPEATED-HUB-QUADRATIC-COST-EQUALS-ACTUAL-STORAGE-001.
Status:
KILLED.
Rejected statement:
A fresh repeated hub of multiplicity M creates $\binom{M}{2}\log q$ units of horizontal factor-flow storage.
Certificate:
actual incremental J at that layer is $(M-1)\log q$; the exact difference $((M-1)(M-2)/2)\log q$ is pairwise curvature.
Do not repeat:
Pairwise-gcd statistics and incremental storage answer different questions.

FAIL-RETURN-COEFFICIENT-K-IS-THE-INDEPENDENT-BINDING-WALL-001

FAIL-ID:
FAIL-RETURN-COEFFICIENT-K-IS-THE-INDEPENDENT-BINDING-WALL-001.
Status:
SUPERSEDED.
Superseded description:
Analyse $k_i = (p_i - u_i)/r_i$ in isolation.
Replacement:
Use the full owner-return relation $a_i = q_0 c_i + k_i$ and compare the support of a_i with the owner support. Preserved support forces verticality; changed support gives a foreign sibling and hence reuse or renewed state population.

FAIL-RETURN-COFACTOR-IS-FREE-001

FAIL-ID:
FAIL-RETURN-COFACTOR-IS-FREE-001.
Date / iteration:
vNEXT4.6.
Claim:
In $N - w = bq$, the cofactor b can be discarded as an unstructured residual.
Status:
KILLED as a description.
Exact certificate:
For a block-derived low-core target,

$$\frac{10}{9}s < b < 2s, \qquad (b, N) = 1, \qquad \mathrm{PF}(b) \subset R.$$

Hence b is a rooted cofactor packet of comparable scale and its support must be retained.
Do not repeat:
Do not project a quotient-return event to the base q alone.

FAIL-RETURNED-SEED-FIRST-OCCURRENCE-IS-J-001

FAIL-ID:
FAIL-RETURNED-SEED-FIRST-OCCURRENCE-IS-J-001.
Status:
KILLED.
Killed inference:
if $\ell \notin R$ is returned by an extension row, that occurrence immediately contributes $\log \ell$ to $\log(J_{R\cup A}/J_R)$.
Certificate:
when $\kappa_R(\ell) = 0$, the first occurrence is the baseline copy in the exact fresh factor $F_R(A)$. Only a second occurrence contributes full-weight incremental J.
What survives:
owner-anchored labels or later returns do enter incremental J exactly as recorded by the v32 second-occurrence theorem.

FAIL-RETURNING-BASIN-IS-ABSOLUTELY-COSTFREE-001

FAIL-ID:
FAIL-RETURNING-BASIN-IS-ABSOLUTELY-COSTFREE-001.
Status:
SCOPE-CORRECTED.
Correction:
A returning basin may leave a functional evaluated on the fixed set R unchanged, but relative

to the enlarged child-closed world it has the exact incremental TUJ/source-lock cost. A nonempty source-free BAD extension satisfies $T_S/T_R \geq N-1$.

Do not repeat:
Distinguish fixed-R invisibility from zero incremental cost.

`FAIL-REUSE-DOUBLE-CHARGE-001`

FAIL-ID:
FAIL-REUSE-DOUBLE-CHARGE-001.

Date / iteration:
v1.21 depletion audit, 1 September 2026.

Claim:
If one nonpositive clean round forces a rooted reuse payment $J_* \mid J_R$, then running clean selection again automatically forces a new, disjoint payment and therefore decreases the remaining rooted budget.

Status:
KILLED.

Exact logical obstruction:
The second "round" may select the identical family or may use complements sharing exactly the same rooted negative labels. Then the same set B_- and the same factor J_* are being observed again. Nothing implies $J_*^2 \mid J_R$ or any decrease of a residual quotient.

What survives:
If complete complement supports are pairwise disjoint across rounds, then the negative rooted label sets are disjoint and the product of the corresponding J_i divides J_R.

Current replacement:
Theory Core, `G-MULTIROUND-EXTERNAL-SINGLE-CHARGE` and `G-BAD-BASE-FORBIDDEN-SUPPORT-CLEAN-CONTINUATION`.

Do not repeat:
Never infer depletion from repeated invocation of a selector alone. First prove cross-round support freshness or enter the explicit support-saturation alternative.

Related current target:
K1, P1BR, TF renewal termination.

`FAIL-ROOT-GENERATION-AC-ALONE-CLOSES-HQ-RPD-001`

FAIL-ID:
FAIL-ROOT-GENERATION-AC-ALONE-CLOSES-HQ-RPD-001.

Status:
SCOPE-CORRECTED.

Killed overreach:
a distribution estimate for $m_k(r) = \#\{i : r^k \mid N - q_i\}$ on the initial v4.41 targets automatically proves RPD for H_Q.

Certificate:
$H_Q = S_Q \cap (N^{1/3}, N/3)$ contains later high descendants of the stopped basins as well as the initial q_i.

What survives:
an initial-target estimate is a valid closure input only after an ancestry-localisation theorem shows that every uncharged full-scale deep obstruction can be moved to an equivalent designated generation.

`FAIL-ROOTED-AMBIENT-001`

FAIL-ID:
FAIL-ROOTED-AMBIENT-001.

Date / iteration:
Canonical bridge audit, 31 August 2026.
Claim:
After adjoining a selected set $\mathscr{P}$ of external active parents, the old rooted lock still controls the enlarged system, so T_R, U_R, J_R bound the promoted valuation mass.
Original motivation:
The rooted congruence was the strongest available global identity and was reused outside its domain.
Status:
KILLED.
Minimal / strong counterexample:
Structural, by `G-MULTISOURCE-LOCK`.
Exact certificate:
For child-closed S, $T_S \equiv (-1)^{|S|} \prod_{d \in D_S^{\mathrm{src}}} d \pmod{N}$. Adjoining $\mathscr{P}$ and re-closing changes the source set from $\{p_0\}$ to a set containing $\mathscr{P}$, so the right-hand side changes. A large ambient J is consistent with a large source product.
Failed hypothesis or mechanism:
A one-source identity was applied to a multi-source system.
Why the argument failed:
The lock is an identity *about the sources*, not a bound on the collision channel.
What survives:
The exact multi-source congruence itself, and the promotion inequality $\log(J_{R\cup\mathscr{P}}/J_R) \le K_\times(R, \mathscr{P}) + K(\mathscr{P})$.
Safe weaker version:
Any future capacity theorem must book the new sources explicitly.
Current replacement:
Theory Core, `G-MULTISOURCE-LOCK`, `G-AMBIENT-PROMOTION`, target K1.
Do not repeat:
Never carry a rooted bound across a change of source set.
Related current target:
K1, W1.

`FAIL-ROUTING-INDIFFERENCE-SCOPE-001`

FAIL-ID:
FAIL-ROUTING-INDIFFERENCE-SCOPE-001.
Date / iteration:
vNEXT4.1, Phase 0A.
Claim:
vNEXT4 (RI1), `TD-UNIVERSAL-ROUTING` $\not\Rightarrow$ `P1BR`.
Original motivation:
Under routing every fresh seed returns, and the return branch was shown to move no recorded quantity of R.
Status:
SCOPE-CORRECTED. The theorem survives; its reading does not.
Exact certificate:
The proof enumerates the *present* functionals $\kappa_R, \Gamma_N(R), \eta(R), J_R, U_R, T_R, \gamma, C, H, \tau, Q$ and observes that each is a function of the set R, which does not change when a closure returns to it. That establishes a non-implication *through those functionals*. It says nothing about a theorem that uses the arithmetic of the return path — the actual integers $N-p$ along it, the congruences placing a returning seed inside R, or $\gcd(N-p, N-u) \mid p-u$. Quantifying over all future theorems was never proved and cannot be proved this way.

What survives:
Theory Core, `G-ROUTING-INDIFFERENCE-SCOPED`: routing alone together with the currently recorded CRD ledgers does not close P1BR.
Safe weaker version:
the scoped form above, which is what is used.
Do not repeat:
Loss-control rule (LR5) is amended: routing may not be cited as a *sufficient* route to P1BR through the recorded ledgers; it remains admissible as an *ingredient* of an additive argument.
Related current target:
`P1BR-ADDITIVE-COUPLING`, `TD-UNIVERSAL-ROUTING`.

`FAIL-ROUTING-LOSES-SEED-WEIGHT-001`

FAIL-ID:
FAIL-ROUTING-LOSES-SEED-WEIGHT-001.
Status:
KILLED.
Killed description:
after basin merging/returns, the original logarithmic weight of the external seed packet may disappear from all exact ledgers.
Certificate:
v4.38 proves

$$\prod_{\ell \in E} \ell \mid \Sigma_{\mathrm{ext}}(S) F_R(A),$$

so

$$W(E) \leq \log \Sigma_{\mathrm{ext}}(S) + \log F_R(A).$$

For $|E| \gg N^{1/3}$ one of the two reservoirs has ambient polynomial weight.
What survives:
neither reservoir yet has the required rooted upper consumer.

`FAIL-ROUTING-RESCUES-P1BR-001`

FAIL-ID:
FAIL-ROUTING-RESCUES-P1BR-001.
Date / iteration:
vNEXT4, Phase 6.
Claim:
That proving `TD-UNIVERSAL-ROUTING` — every BAD child-closed closure meets R — would remove the obstruction at `P1BR-POSITIVE-ROUND-CONSEQUENCE`, since it removes the disjoint-packet configuration "without requiring any new theory".
Original motivation:
vNEXT3.1 recorded, correctly, that routing removes the disjoint-packet discussion, the recorded-ledger bridge theorem and the CRD-2 question. It was natural to expect that it also unblocked the positive round.
Status:
KILLED. The routing statement itself is untouched and remains OPEN; only its claimed relevance to P1BR is rejected.
Exact certificate:
Under routing, $B_{\mathrm{ext}} = \varnothing$ and every fresh seed of a positive round returns. A returning seed r produces the enlarged world $R' = R \cup R_N(r)$, but κ_R, $\Gamma_N(R)$, $\eta(R)$, J_R, U_R, T_R, γ, $|R|$, C, H, τ, Q are functions of the set R alone and do not move; and the capacity bounds (FC7)–(FC10) do not transfer to R', whose source set is not a single upper-half root with composite complement. See Theory Core, `G-RETURN-BRANCH-NO-CANONICAL-CHARGE`

and `G-ROUTING-INDIFFERENCE`, (RI1).

Second, independent certificate:

The witness family of `G-POSITIVE-ROUND-LEDGER-INDEPENDENCE` is routing-compatible — every fresh state's closure meets R_n — and still has $\eta(R_n)$ unbounded at fixed round datum.

Failed hypothesis or mechanism:

Confusing "the seed reaches R" with "R is charged". Reaching a fixed set does not alter any functional of that set.

What survives:

Routing remains the right question for the bridge, the disjoint-packet configuration and the CRD-2 question, and the restricted form `TD-POSITIVE-ROUND-ROUTING` is recorded in the Core.

Do not repeat:

Loss-control rule (LR5): `TD-UNIVERSAL-ROUTING` may not be cited as a *sufficient* route to P1BR through the recorded ledgers. Scope-corrected at vNEXT4.1: see `FAIL-ROUTING-INDIFFERENCE-SCOPE-001`; routing remains admissible as an ingredient of an additive argument.

Related current target:

`P1BR-ADDITIVE-COUPLING`, `TD-UNIVERSAL-ROUTING`.

`FAIL-SATURATION-DOUBLE-CHARGE-001`

FAIL-ID:

FAIL-SATURATION-DOUBLE-CHARGE-001.

Date / iteration:

v1.22 support-saturation continuation, 1 September 2026.

Claim:

If a current external BAD packet substantially overlaps the support of an earlier nonpositive globally clean family, then the pivot payment (SP3) is an additional disjoint factor which can be multiplied into the old v1.21 depletion product.

Status:

KILLED.

Why the inference fails:

The pivot uses a subfamily of the *same old rows*. Its negative rooted support can therefore be a subset of the already charged payment set of those rows. Relabelling changes the certificate used to lower-bound the payment, not the underlying layer-profile balance.

What survives:

The overlap gives the quantitative localisation (SP1)–(SP5), and may reveal a positive signed subfamily. It becomes a new depletion charge only after a separate proof of support disjointness from all previous payment sets.

Current replacement:

Theory Core, `G-SATURATED-BASE-RELABEL-PIVOT` together with `G-MULTIROUND-EXTERNAL-SINGLE-CHARGE`.

Do not repeat:

Never add a pivot-derived rooted factor to $\prod_i J_i$ merely because the designated bases are new.

Related current target:

K1, globally clean renewal termination.

`FAIL-SEALED-MULTISOURCE-LOCK-001`

FAIL-ID:

FAIL-SEALED-MULTISOURCE-LOCK-001.

Date / iteration:

v1.6 sealed-source continuation, 1 September 2026.

Claim:
If many external upper-half bad parents have all children inside the canonical failed world R, adjoining them and applying `G-MULTISOURCE-LOCK` yields a new modular obstruction stronger than the rooted lock.

Original motivation:
The sealed branch creates genuine new sources, and the multi-source congruence explicitly records the source product.

Status:
KILLED.

Minimal / strong counterexample:
Structural; the proposed new congruence reduces identically to the old one.

Exact certificate:
Let $\mathscr{P}$ be finite, upper-half, external and bad with $\Gamma_N(\mathscr{P}) \subseteq R$, and put $S = R \cup \mathscr{P}$. Then

$$\mathrm{Src}(S) = \{p_0\}\dot{\cup}\mathscr{P}, \qquad T_S = T_R \prod_{p\in\mathscr{P}} (N-p).$$

The multi-source lock reads

$$T_R \prod_{p\in\mathscr{P}} (N-p) \equiv (-1)^{|R|+|\mathscr{P}|} p_0 \prod_{p\in\mathscr{P}} p \pmod{N}.$$

Using $N - p \equiv -p \pmod{N}$ and cancelling the active labels p, which are units modulo N, leaves exactly

$$T_R \equiv (-1)^{|R|} p_0 \pmod{N},$$

the original `G-ROOT-LOCK` congruence.

Failed hypothesis or mechanism:
A larger source set was assumed to create new modular information merely because the source product changed.

Why the argument failed:
The row factors contributed by the new sealed sources are congruent to the negatives of those same source labels, so their entire modular contribution cancels.

What survives:
`G-SEALED-SOURCE-LOCK-REDUCTION` is an exact PROVED structural statement. The sealed branch remains potentially useful through population size, valuation/reuse overload, or deeper topology, but not through the source-lock congruence alone.

Safe weaker version:
Use the sealed-source lift as bookkeeping only; seek a noncongruential invariant for the new sources.

Current replacement:
Theory Core v1.6, `G-SEALED-SOURCE-LOCK-REDUCTION`; see also the four-ary sign-overload disposition theorem in the Core.

Do not repeat:
Do not count new sealed sources as new modular constraints without simplifying the multi-source congruence first.

Related current target:
sealed-source branch, P1BR, K1.

`FAIL-SELECTOR-CLOSES-CASCADE-001`

FAIL-ID:
FAIL-SELECTOR-CLOSES-CASCADE-001.

Date / iteration:
vNEXT3, Phase 4.
Claim:
Once `G-BAD-BASE-SQUAREFREE-POLYLOG-EXTENSION` is proved, the (FC10) clean-round route yields a contradiction.
Original motivation:
The selector was the last analytic blocker on that route.
Status:
KILLED. The selector is PROVED; the inference is not.
Exact certificate:
The selector supplies at most one clean row per designated base, and the bases lie in $[3,(\log N)^A]$, so $\sum_i M_i \log z_i \ll (\log N)^A \log N$. (FC10) forces a positive signed round only once $\sum_i M_i \log z_i \geq \gamma \log(25N^2/18)$, i.e. once $\gamma \lesssim (\log N)^A$. The cascade therefore closes exactly on the subdomain $\gamma \leq (\log N)^A$ and on no larger one by this route.
Failed hypothesis or mechanism:
Removing an analytic obstruction was read as closing the chain; the counting obstruction downstream is untouched.
What survives:
`G-CASCADE-GAMMA-TERMINUS`, which states the subdomain exactly, and the selector itself.
Safe weaker version:
"Contradiction on $\gamma \leq (\log N)^A$."
Current replacement:
Theory Core, the retired `G-CASCADE-GAMMA-TERMINUS`, whose proof body is kept at Subsection [fail:cascade-body] of this document and no longer in the Core.
Do not repeat:
Do not announce a cascade closure without counting the number of rounds the selector actually supplies against the capacity it must exhaust.
Related current target:
γ, K1, X1.

`FAIL-SHORTEN-PARENT-INTERVAL-TO-GET-P3-001`

FAIL-ID:
FAIL-SHORTEN-PARENT-INTERVAL-TO-GET-P3-001.
Date / iteration:
vNEXT4.21.
Strategy:
Shorten the parent interval so the cofactor is smaller and use classical short-interval BV plus an unweighted lower linear sieve to force P_3.
Status:
KILLED relative to that stated analytic architecture.
Reason:
The loss in available distribution level dominates the gain in cofactor length. The successful vNEXT4.21 P_3 theorem instead uses weighted-sieve information at the full-interval BV level.
Scope guard:
This is not a universal impossibility result for stronger short-interval/switching technology.

`FAIL-SIMPLE-STATE-001`

FAIL-ID:
FAIL-SIMPLE-STATE-001.
Date / iteration:
v0.1 foundations review.

Claim:
The simple directed support graph on $\mathcal{A}_N$ is a faithful presentation of the CRD dynamics.
Original motivation:
Reachability questions are graph questions.
Status:
SUPERSEDED.
Minimal / strong counterexample:
Structural.
Exact certificate:
The support graph loses the valuations $\nu_N(p,q) = v_q(N-p)$, hence the prime-power layers $\mu_{\ell,k}$, hence the collision and cut functionals; and it carries no record of ambient external parents. `G-TUJ`, `G-FULLQ-LEDGER`, `G-COLLISION-CUT` and `G-AMBIENT-PROMOTION` are all invisible to it.
Failed hypothesis or mechanism:
Edge existence was taken for the whole structure.
Why the argument failed:
$T = UJ$, residual production, prime-power collision and ambient promotion all depend on valuations, not on the mere existence of an edge.
What survives:
Graph-level facts: closure, terminal SCC, source deletion.
Safe weaker version:
Use the support graph only for reachability statements.
Current replacement:
Theory Core, the rooted ambient valuation-incidence object $\mathfrak{C}_N = (N, \mathcal{A}_N, p_0, \nu_N)$.
Do not repeat:
Do not state a valuation theorem in graph language.
Related current target:
L1, K1.

`FAIL-SINGLE-BASE-CONCENTRATION-NEEDED-FOR-VERTICAL-MIGRATION-001`

FAIL-ID:
FAIL-SINGLE-BASE-CONCENTRATION-NEEDED-FOR-VERTICAL-MIGRATION-001.
Status:
KILLED.
Killed inference:
a polynomial exact star on one base must first be extracted before diffuse vertical storage can be forced into another CRD ledger.
Certificate:
v4.37 k-power divisor packing sums over all powerful parts simultaneously and forces a D/J lower bound without a common base.
What survives:
exact-star peeling is still useful when concentration is actually present.

`FAIL-SINGLETON-STAR-LOCAL-WINDOW-001`

FAIL-ID:
FAIL-SINGLETON-STAR-LOCAL-WINDOW-001.
Date / iteration:
vNEXT4.16.
Claim:
A fixed small prime r cannot divide parent cofactors for many high reflected targets because the legal coefficient interval is too short.
Status:
KILLED.

Exact obstruction:
For $c_j = sq$, the admissible cofactor interval has length $> 5s/9$. If $s > (18/5)r$, it already contains an odd multiple of r.
What remains:
Primality/rootedness, mirror feedback, or a signed multiplicity capacity must be used.

FAIL-SIZE-COMPARISON-AGAINST-RECORDED-CAPACITY-001

FAIL-ID:
FAIL-SIZE-COMPARISON-AGAINST-RECORDED-CAPACITY-001.
Status:
SCOPE-CORRECTED .
Killed inference:
derive $X \geq Y$ using only the same local size inequalities $x_i \leq y_i$ whose product already proves $X \leq Y$.
Reason:
this merely asks the local envelope to contradict itself.
What survives:
an independent global, canonical, high-gap, analytic or other nonlocal theorem may imply $X \geq Y$ on a special failed domain; together with the envelope this would prove that the domain is empty. Therefore the strong revised2 phrase "no global lower bound can ever contradict the envelope" is not an active prohibition.

FAIL-SMALL-GAMMA-MAKES-CANONICAL-BLOCK-INVISIBLE-001

FAIL-ID:
FAIL-SMALL-GAMMA-MAKES-CANONICAL-BLOCK-INVISIBLE-001.
Status:
KILLED .
Killed inference:
$\gamma < N^{1/3}/\log N$ makes the high-gap reflected block too small to yield a polynomial CRD object.
Certificate:
the whole-block disposition with any fixed $1/3 < \delta < 0.475$ forces

$$E \subset (\mathcal{A}_N \setminus R) \cap (0, N/5), \qquad |E| \gg N^{1/3}.$$

What survives:
the packet is external and requires a source/fresh consumer; it is not automatically absorbed by R.

FAIL-STAR-SELF-REENTRY-001

FAIL-ID:
FAIL-STAR-SELF-REENTRY-001.
Status:
KILLED .
Killed inference:
the star $N^{1/3-1/(m+1)-o(1)}$ produced at fixed m can be fed back into the same power-threshold theorem to obtain a terminating descent.
Certificate:
the theorem requires a packet carrying defect $\gg N^{1/3}$, while the fixed-m star has an exponent deficit $1/(m+1) > 0$.
Scope guard:
v4.44 does not use such self-reentry; it chooses growing m before concentration and then uses

simultaneous joint-profile freezing.

`FAIL-STRIP-COMMON-BASE-THEN-IMPORT-CLEAN-ROUND-001`

FAIL-ID:
FAIL-STRIP-COMMON-BASE-THEN-IMPORT-CLEAN-ROUND-001.
Source:
v4.52 scope correction and explicit Failure Log addition.
Status:
KILLED.
Killed inference:
remove the common r^e and then automatically apply `G-EXTERNAL-BASE-SIGNED-REUSE` as though the residuals were pairwise coprime and squarefree.
Reason:
the full complements are not pairwise coprime, and the stripped residuals b_p need not be pairwise coprime or squarefree.
Replacement:
use the exact r-stripped surplus ledger and split into paid residual collision/verticality or owned unique external sibling renewal. The stronger pairwise-coprime interface estimate remains conditional.

`FAIL-TAGGED-P3-RESIDUE-FORCES-RETURN-001`

FAIL-ID:
FAIL-TAGGED-P3-RESIDUE-FORCES-RETURN-001.
Date / iteration:
vNEXT4.21.
Killed inference:
Knowing $b \bmod q$ together with $\Omega(b) \leq 3$ forces some sibling $r \mid b$ to satisfy $r \equiv N \pmod q$.
Status:
KILLED for every prime $q \geq 5$.
Exact certificate:
With $G = (\mathbb{Z}/q\mathbb{Z})^\times$ and $a = N \bmod q$,

$$\boxed{(G \setminus \{a\})^2 = G.}$$

Thus two non-return sibling residues already realise every possible product residue.
Exceptional case:
$q = 3$ is different. If $N \equiv 2 \pmod 3$ and $b \equiv 2 \pmod 3$, at least one sibling returns to 3.

`FAIL-TD-NAIVE-001`

FAIL-ID:
FAIL-TD-NAIVE-001.
Date / iteration:
Interrupted T_D branch round, 31 August 2026.
Claim:
Every composite D-carrier a_j has an active prime factor, so every $\mathcal{D}$ index produces an actual CRD state.
Original motivation:
Fresh carriers do produce active states, and the smooth case was assumed to differ only quantitatively.
Status:
KILLED.

Minimal / strong counterexample:
$N = 1862,\ r = 1733,\ j = 2.$

Exact certificate:
$m = N - r = 129$ and $a_j = m + 2j = 133 = 7 \cdot 19$, while $7 \mid 1862$ and $19 \mid 1862$. Both prime factors of the carrier divide N, so neither is active and a_j contributes no CRD row. The reflected partner is $c_j = 1729 = 7 \cdot 13 \cdot 19$. Verified exactly.

Failed hypothesis or mechanism:
Compositeness of the carrier was confused with activity of its factors.

Why the argument failed:
Activity is a condition on N, not on a_j; a carrier can be built entirely from primes dividing N.

What survives:
The smooth/fresh split of `TD-CARRIER-SPLIT` covers this instance instead of hiding it: a *fresh* carrier, that is one with $P^+(a_j) \geq h$, does give an active prime state $s_j < N/6$ by strict reflected-block nonreuse. The exception lives in $\mathcal{D}_{\text{sm}}$.

Safe weaker version:
Assert activity only for $j \in \mathcal{D}_{\text{fr}}$.

Current replacement:
Theory Core, `TD-CARRIER-SPLIT` and `C-STRICT-NONREUSE`.

Do not repeat:
Never promote “composite” to “has an active factor”.

Related current target:
T_D, $P1^{\text{br}}$.

Domain note:
$r = 1733$ is a *noncanonical* upper root; the canonical selector for $N = 1862$ is $p_0 = 937$. The certificate is an exact arithmetic fact and is domain-independent, but it must not be quoted as a canonical-world witness.

`FAIL-TD-SMOOTH-INACTIVE-001`

FAIL-ID:
FAIL-TD-SMOOTH-INACTIVE-001.

Date / iteration:
Phase-3 T_D second-layer attack, 31 August 2026.

Claim:
The canonical geometry alone — p_0 canonical, $h < G$, $a_j = m+2j$ odd composite, c_j composite — forces a smooth reflected carrier to transport some cost: an active state, a prime-power layer, or a source term.

Original motivation:
`TD-CARRIER-SPLIT` gives $\Omega(a_j) > \log m / \log h$, a large factorisation mass, and mass was assumed to be transportable.

Status:
KILLED as a proof route resting on the canonical geometry. It is *not* a refutation of any statement conditioned on canonical failure; see the domain note below.

Minimal / strong counterexample:
$N = 168$, $x = 84$, $p_0 = 89$, $h = 5 < G = 6$, $m = 79$, $t = 2$, $j = 1$. Smallest instance with *composite* complement, hence with a genuine first front: $N = 2684$, $p_0 = 1361$, $h = 19 < G = 34$, $m = 1323 = 3^3 \cdot 7^2$, $j = 4$, $a_4 = 1331 = 11^3$, $c_4 = 1353 = 3 \cdot 11 \cdot 41$, $11 \mid 2684$.

Exact certificate:
$a_1 = m + 2 = 81 = 3^4$ and $c_1 = N - a_1 = 87 = 3 \cdot 29$. Since $3 \mid 168$, the only prime of a_1 divides N, so $3 \notin \mathcal{A}_N$. The carrier therefore contributes no active state, no prime-power layer of the CRD system, and no source. Its whole factorisation mass $\Omega(a_1) = 4$ is inert. Verified exactly.

Domain note — read before transferring this witness:
all six known instances are *successful* worlds. At $N = 168$ and $N = 492$ the complement m is prime, so p_0 is GOOD at depth zero and $R = \varnothing$; at $N = 2684$, 4380, 4386, 6260 the complement is composite but the traversal succeeds deeper. In none of them is R a failed world, so in none of them is the index j shown to lie in $\mathcal{D}$. By `PROHIB-DOMAIN-TRANSFER` this witness therefore kills only proofs that use the canonical geometry alone; the failure-conditioned form of the claim is untouched and remains open. This is the same shape of kill as `FAIL-TF-LOCAL-SCOPE-001`.

Further instances:
$(N, p_0, j, a_j) = (492, 251, 1, 3^5)$, $(4380, 2203, 5, 3^7)$, $(4386, 2203, 2, 3^7)$, $(6260, 3137, 1, 5^5)$. Over canonical $N < 1.2 \cdot 10^5$, 22.24% of composite reflected carriers have at least one prime factor dividing N.

Failed hypothesis or mechanism:
Ω-mass of a_j was identified with CRD mass. The two live in different objects: a_j is not a row, and its prime factors need not be states.

Why the argument failed:
Activity is a condition relative to N. A carrier assembled from primes dividing N is invisible to the CRD system.

Structural note:
such a carrier is necessarily *smooth*. If $\ell \mid a_j$ and $\ell \mid N$ then $\ell \mid N - a_j = c_j$, so ℓ divides two distinct members of the strict reflected block and `C-STRICT-NONREUSE` forces $\ell < h$. The fresh branch is therefore immune, which is exactly the content of `TD-CARRIER-SPLIT`: only $\mathcal{D}_{\mathrm{sm}}$ can be inert.

What survives:
The smooth branch is bounded in *count* rather than transported in *mass*. `TD-SMOOTH-CEILING` gives $|\mathcal{D}_{\mathrm{sm}}| \le \sigma_N(h)$ unconditionally, and that ceiling is what the second-layer trichotomy consumes.

Safe weaker version:
Never assert that a smooth carrier produces a state. Use only its cardinality, via the ceiling.

Current replacement:
Theory Core, `TD-SMOOTH-CEILING` and `TD-SECOND-LAYER`.

Do not repeat:
Do not attempt to transport smooth carrier mass to actual layers. That direction is closed by this certificate; the productive direction is the ceiling.

Relation to `FAIL-TD-NAIVE-001`:
complementary, not stronger. `FAIL-TD-NAIVE-001` uses a noncanonical root, $(N, r) = (1862, 1733)$ with canonical selector 937, and kills “every composite carrier has an active factor”. This entry uses canonical roots with $h < G$ and kills the same claim for any proof resting on that geometry, while strengthening the configuration from “some prime factor is inactive” to “the carrier is wholly inert”. Neither entry reaches the canonical *failed* domain, because no canonical failed world is known.

Related current target:
T_D, $P1^{\mathrm{br}}$.

Exact classification (Phase-3 continuation):
the failure mode is now completely understood. By `TD-CARRIER-DICHOTOMY`, a carrier produces no active state *precisely* when $\mathrm{rad}(a_j) \mid h - 2j$. All six instances above are of that kind, and each realises the normal form $x = q^e + d$: $84 = 3^4 + 3$, $246 = 3^5 + 3$, $1342 = 11^3 + 11$, $2190 = 3^7 + 3$, $2193 = 3^7 + 6$, $3130 = 5^5 + 5$. The class is bounded by `TD-DEADSMOOTH-CEILING` and is empty whenever no prime below h divides x — 48.0% of canonical instances with $N < 3 \cdot 10^5$. Across that whole range only 19 dead-smooth carriers exist among 514 447, and no N carries two.

`FAIL-TF-LOCAL-SCOPE-001`

FAIL-ID:
FAIL-TF-LOCAL-SCOPE-001.

Date / iteration:
Interrupted T_D branch round, hostile scope witness, 31 August 2026.

Claim:
A fresh reflected state ℓ_j, certified locally as $\mathcal{F}$-class together with $h < G$, routes to Good_N or into R.

Original motivation:
The recorded census showed $\mathcal{F}$ to be empirically rigid, and the local certificate looked self-sufficient.

Status:
SCOPE-CORRECTED.

Minimal / strong counterexample:
$N = 128,\ p_0 = 67,\ j = 1,\ c_1 = 65,\ \ell = 13.$

Exact certificate:
$x = 64$, $p_0 = 67$, previous prime 61, so $h = 3 < G = 6$; $t = 1$ and $c_1 = 65 = 5 \cdot 13$ with $P^+(c_1) = 13 \geq h$, so the local $\mathcal{F}$ certificate holds. The state 13 is active and not GOOD, since $128 - 13 = 115 = 5 \cdot 23$, and its stopped traversal fails with the separate closed world $\{3, 5, 7, 11, 13, 23, 29, 41\}$. Verified exactly.

Failed hypothesis or mechanism:
The certificate was evaluated without the hypothesis that defines R.

Why the argument failed:
Here $128 - 67 = 61$ is prime, so p_0 *succeeds* at depth zero and $R = R_N(p_0) = \varnothing$. The condition $\mathrm{PF}(c_j) \cap R = \varnothing$ is satisfied vacuously. The instance is a *successful world*, and R is not even defined as a failed world.

What survives:
The canonical-failure-conditioned form of TF is untouched. What is killed is every proof of it that uses only the local $\mathcal{F}$ certificate together with $h < G$.

Safe weaker version:
Always carry the failure hypothesis into the $\mathcal{F}$ certificate; never evaluate $\mathcal{I}, \mathcal{D}, \mathcal{S}, \mathcal{F}$ at a root that succeeds.

Current replacement:
Theory Core, target TF.

Do not repeat:
A vacuous R is not an empty support intersection in the intended sense.

Related current target:
TF, T_D.

`FAIL-TWOVARIABLE-FULL-LEVEL-001`

FAIL-ID:
FAIL-TWOVARIABLE-FULL-LEVEL-001.

Date / iteration:
vNEXT4.12.

Claim:
The exact local density $(2p - 1)/p^2 = 2/p - 1/p^2$ for the polynomial pair $q(N - kq^2)$ permits an elementary sieve level equal to a fixed power of N.

Status:
KILLED.

Exact certificate:
The coefficient coordinate has length only

$$K = N/h^2.$$

The elementary remainder term therefore restricts the second logarithmic saving to $\log K$, not $\log N$.
Do not repeat:
A local dimension-two identity is not a distribution level theorem.

FAIL-UNIQUE-FRESH-TARGETS-MAY-FUNNEL-INTO-O(n)-PARENTS-001

FAIL-ID:
FAIL-UNIQUE-FRESH-TARGETS-MAY-FUNNEL-INTO-O(n)-PARENTS-001.
Status:
KILLED on the high layer extracted in v4.40.
Killed inference:
even after extracting linearly many fresh load-one targets, they may all be fed by $o(n)$ parent rows.
Certificate:
after removing targets below $n/(\log N)^2$, one parent can feed at most three surviving targets because their product divides one complement $< N$. Therefore the parent population is at least $(1/27 - o(1))n$, and a greedy matching gives at least $(1/63 - o(1))n$ vertex-disjoint parent–target edges.
What survives:
the cofactor supports of those matched rows remain uncontrolled.

FAIL-UNIQUE-MISSING-DATUM-001

FAIL-ID:
FAIL-UNIQUE-MISSING-DATUM-001.
Date / iteration:
vNEXT3.1, Phase-5 reclassification.
Claim:
The vNEXT3 information-loss certificate named "a share of the ambient valuation tensor" as *the exact datum missing* for a coupling.
Original motivation:
It is the most natural non-additive quantity in sight.
Status:
SCOPE-CORRECTED.
Exact certificate:
What is proved is that all recorded packet functionals factor on disjoint child-closed unions, hence cannot express competition for an R-only capacity. What is *not* proved is that ambient tensor occupancy is the unique repair, that no other non-additive functional works, or that a CRD-2 is necessary. No uniqueness argument was given, and none is available.
What survives:
The diagnosis, reclassified as FRONTIER CERTIFICATE / INFORMATION-LOSS DIAGNOSIS, together with four candidates recorded as CONJECTURAL THEORY-DESIGN DIRECTIONS: ambient tensor occupancy, cofactor provenance, parent-pair correlations, non-additive resource ownership.
Do not repeat:
Do not call a design hypothesis an "exact missing datum" without a uniqueness proof.
Related current target:
`TD-UNIVERSAL-ROUTING`, CRD-2 question.

FAIL-UNIT-COMPLEMENT-001

FAIL-ID:
FAIL-UNIT-COMPLEMENT-001.
Date / iteration:
Domain audit, frozen 31 August 2026.
Claim:
Every failed active root has composite complement.
Original motivation:
Failure has no prime-complement success, so the complement was informally treated as composite.
Status:
KILLED .
Minimal / strong counterexample:
$N = 20, r = 19$.
Exact certificate:
$19 \in \mathcal{A}_{20}$, $20 - 19 = 1$, and $1 \notin \mathbb{P}$. The row has no prime children, hence the stopped traversal fails, although the complement is not composite.
Failed hypothesis or mechanism:
The implication "nonprime $\Rightarrow$ composite" silently assumed the complement exceeds one.
Why the argument failed:
The unit row was omitted from the domain.
What survives:
If $N - r > 1$ and r is not GOOD, then $N - r$ is composite.
Safe weaker version:
State the composite-complement hypothesis explicitly whenever $Q = \mathrm{PF}(N - r)$ or a root ledger is used.
Current replacement:
Theory Core, rooted algorithm and Theorems G-FULLQ-LEDGER/G-FULLQ-PRODUCTION on their stated domains.
Do not repeat:
Do not infer compositeness from failure before checking the unit row.
Related current target:
C0a and every generic failed-root lemma.

FAIL-UNIVERSAL-BRIDGE-IMPOSSIBILITY-001

FAIL-ID:
FAIL-UNIVERSAL-BRIDGE-IMPOSSIBILITY-001.
Date / iteration:
vNEXT3.1, Phase-4 scope correction.
Claim:
`G-BRIDGE-ROUTING-EQUIVALENCE` of vNEXT3, read as: no coupling of disjoint packets is possible at all.
Original motivation:
(OD1)–(OD4) really do make every recorded functional additive, which felt exhaustive.
Status:
SCOPE-CORRECTED . The mathematics is correct; the quantifier was too large.
Exact certificate:
(OD1)–(OD4) quantify over the recorded ledger $T, J, U_\bullet, \eta, \nu, K, \mathrm{Src}$. They say nothing about an arbitrary functional $\Psi_N(R, U)$, which need not be a disjoint-union homomorphism. Candidates not excluded: ambient valuation-tensor occupancy, parent or cofactor correlations, residue geometry, shared ambient prime supply, non-additive resource ownership.

What survives:
G-RECORDED-LEDGER-BRIDGE-ROUTING: no bridge built *solely from the recorded disjoint-union-homomorphic ledgers* can charge $\eta(U)$ against an R-only capacity, and any such bridge requires $U \cap R \neq \varnothing$.

Do not repeat:
Do not promote "every functional we have recorded" to "every functional".

Related current target:
TD/TF routing, TD-UNIVERSAL-ROUTING.

FAIL-UPPERHALF-SCOPE-001

FAIL-ID:
FAIL-UPPERHALF-SCOPE-001.

Date / iteration:
v0.1 hostile mathematical audit §3.

Claim:
Upper-half source compression applies to a failed canonical world, so every active upper-half prime is a source of the bad graph.

Original motivation:
Both settings are called "failure".

Status:
SCOPE-CORRECTED.

Minimal / strong counterexample:
Hypothesis mismatch; no finite witness is required.

Exact certificate:
The source theorem assumes $\mathrm{Good}_N = \varnothing$, that is, that N is a Goldbach counterexample. Only then do stopped and raw reachability coincide and every active $N/2 < p < N$ become a source. Canonical p_0-failure asserts nothing about other roots: the recorded corpus contains many failed worlds for arbitrary roots at values of N that have Goldbach representations.

Failed hypothesis or mechanism:
The global hypothesis was imported into the local one.

Why the argument failed:
Using a theorem whose hypothesis is global Goldbach failure in order to contradict p_0-failure assumes what is to be proved for every other root.

What survives:
ARS-UPPERHALF-COMPRESSION on its own domain, and the direct-Goldbach-failure DAG in which it is legitimate.

Safe weaker version:
Keep two separate DAGs — universal p_0-success, and direct contradiction under global failure — and never move a statement between them without annotation.

Current replacement:
Theory Core, ARS-UPPERHALF-COMPRESSION with its prohibition, and the exact Goldbach boundary section.

Do not repeat:
PROVED $\neq$ COMPUTATION $\neq$ OPEN $\neq$ REJECTED, and failure of the canonical traversal is not global Goldbach failure.

Related current target:
X1.

FAIL-V13-EXTERNAL-VALUATION-AS-SEED-WEIGHT-001

FAIL-ID:
FAIL-V13-EXTERNAL-VALUATION-AS-SEED-WEIGHT-001.

Date / iteration:
vNEXT4.14 correction of vNEXT4.13.
Claim:
The full weighted mass $W_{\text{ext}} = \sum v_q(c_j) \log q$ can be passed unchanged into a family consumer that retains one seed per prime label.
Status:
SCOPE-CORRECTED.
Exact correction:
On $h > N^{1/3}$, high reflected valuations are at most two, so for the distinct external labels

$$W(E_{\text{ext}}) \geq W_{\text{ext}}/2.$$

What survives:
The natural-scale external branch survives with an absolute factor-two loss.

FAIL-V18-SOURCE-STAR-COUNTEREXAMPLE-SPECIFIC-001

FAIL-ID:
FAIL-V18-SOURCE-STAR-COUNTEREXAMPLE-SPECIFIC-001.
Date / iteration:
vNEXT4.19 status repair of vNEXT4.18.
Killed inference:
A polynomial BAD source packet by itself is evidence specific to a Goldbach counterexample and should replace the mirror route as the primary proof target.
Status:
KILLED.
Reason:
The source-star selector is essentially a generic reduced residue-class/sieve construction once an active BAD base exists. Large source products are compatible with multisource defect identities and are lower obligations, not upper collision/source capacities.
What survives:
`G-POLYNOMIAL-REPEATED-BASE-ROUGH-FAN` and `G-PLANNED-BASE-OVERLAP-SINGLE-CHARGE` remain PROVED-PENDING-INDEPENDENT-AUDIT; `C-POLYNOMIAL-SOURCE-STAR-CAPACITY` becomes a secondary structural frontier.

FAIL-V410-ANONYMOUS-STAR-COFACTOR-001

FAIL-ID:
FAIL-V410-ANONYMOUS-STAR-COFACTOR-001.
Date / iteration:
vNEXT4.11 correction of the vNEXT4.10 frontier.
Claim/strategy:
For a large inverse-sibling star, treat $(N - v)/\ell$ as an anonymous large quotient and split it into generic rough/smooth cases.
Status:
SUPERSEDED on the monomial branch.
Exact certificate:
Retained provenance gives the two-level factorisation

$$N - v = bw, \qquad N - w = q^2, \qquad w > \sqrt{N},$$

with the whole entry cofactor b already small. Repeated values of b produce the exact entry comb, including $\operatorname{rad}(b)^{L-1} \mid J_R$.
Replacement:
Use the whole-entry-cofactor atom/comb, not an anonymous quotient split.

FAIL-V47-DEEP-RPD-GLOBAL-GENERATION-LOCALISATION-001

FAIL-ID:
FAIL-V47-DEEP-RPD-GLOBAL-GENERATION-LOCALISATION-001.
Source target:
`C-DEEP-RPD-ANCESTRY-LOCALISATION`.
Status:
SCOPE-CORRECTED.
Overstrong formulation:
a full uncharged deep-RPD obstruction can be pigeonholed into one common global generation.
Why not proved:
generation-edge amortisation charges cross-generation edges through horizontal surplus or source destruction but does not bound the total number of generations strongly enough for that pigeonhole.
Safe v4.47 replacement:
polynomial exact deep layer, followed by original v4.41 submatching, large incremental J, or a local immediate-ancestry restart.
v4.48 refinement:
the primary replacement is further sharpened to `C-DEEP-RPD-DESIGNATED-OWNER-LOCALISATION`; total unique parenthood is not required.

FAIL-V48-DIRECT-REFLECTED-TAG-INHERITANCE-001

FAIL-ID:
FAIL-V48-DIRECT-REFLECTED-TAG-INHERITANCE-001.
Alias:
v4.48 R48-4.
Status:
SCOPE-CORRECTED.
Killed automatic inference:
if an ancestor was born as a factor of a reflected carrier, a later descendant automatically divides that same carrier.
Certificate:
divisibility by the original carrier is not preserved by a generic edge $v_{j-1} = N - c_j v_j$.
What survives:
the v4.47 reflected congruences remain valid on the explicit intersection domain where a target really has both tags.
Replacement:
path-labelled affine ancestry transport, `G-CRD-PATH-AFFINE-COFACTOR-NORMAL-FORM` and `G-DEEP-ANCESTRAL-AFFINE-CONGRUENCE`.

FAIL-V48-HIGHTARGET-LOADONE-NECESSARY-001

FAIL-ID:
FAIL-V48-HIGHTARGET-LOADONE-NECESSARY-001.
Alias:
v4.48 R48-3.
Status:
SCOPE-CORRECTED.
Old scope:
treat the cube-root high-target row trichotomy as requiring global load one of the designated target.
Correction:
inspection of the proof uses only an actual BAD edge $u \to p$, $p > N^{1/3}$, and factorisation of $N - u$. The accepted replacement is `G-CUBEROOT-HIGH-TARGET-OWNER-TRICHOTOMY`.

FAIL-V48-LOCAL-J-AS-AMBIENT-M-SCALE-001

FAIL-ID:
FAIL-V48-LOCAL-J-AS-AMBIENT-M-SCALE-001.
Alias:
v4.48 R48-1.
Status:
SCOPE-CORRECTED.
Killed promotion:
from $|X| = N^{1/3-o(1)}$ and $M \gg N^{1/3}$, promote $\Omega(|X| \log N)$ automatically to an ambient/full-M-scale payment.
Correction:
retain only the explicit subpolynomial comparison delivered by the pigeonhole losses, recorded in v4.48 as $|X| \geq MN^{-o(1)}$. The local J-payment is valid; the stronger scale label was not justified.

FAIL-V48-LOWCHILD-VERBATIM-DOMAIN-TRANSFER-001

FAIL-ID:
FAIL-V48-LOWCHILD-VERBATIM-DOMAIN-TRANSFER-001.
Alias:
v4.48 R48-5.
Status:
KILLED as a direct domain application.
Killed transfer:
apply `C-LOWCHILD-J-OR-SMALLHUB` unchanged to the deep designated-owner family.
Domain mismatch:
the old theorem requires a family of size $\gg N^{1/3}$, whereas the v4.48 owner family is guaranteed only at $MN^{-o(1)} = N^{1/3-o(1)}$.
Replacement:
`G-DESIGNATED-LOWCHILD-SCALE-DISPOSITION` and the open consumer `C-DEEP-LOWCHILD-RENEWAL-CAPACITY`.

FAIL-V48-MONOMIAL-NOVELTY-001

FAIL-ID:
FAIL-V48-MONOMIAL-NOVELTY-001.
Date / iteration:
vNEXT4.9 audit of vNEXT4.8.
Claim:
`C-MONOMIAL-TWO-EDGE-RIGIDITY` introduced a new square-map theorem.
Status:
SCOPE-CORRECTED (provenance correction).
Exact certificate:
The injectivity/no-cycle/no-three-consecutive-square argument is already contained in the PROVED theorem `TF-HIGH-FLOOR-HORIZONTAL-RIGIDITY`.
What survives:
The extracted standalone formulation and its use of component starts in vNEXT4.8 remain valid.

FAIL-V48-MULTIPARENT-SEPARATE-BRANCH-NECESSARY-001

FAIL-ID:
FAIL-V48-MULTIPARENT-SEPARATE-BRANCH-NECESSARY-001.
Alias:
v4.48 R48-2.

Status:
SUPERSEDED as a necessary proof split.
Old design:
separate deep targets by total indegree and cash out the multi-parent part before constructing a restart matching.
Replacement:
choose one actual parent per target. Because one parent row cannot feed three distinct targets above $N^{1/3}$, the chosen-edge graph has bounded degree and yields a vertex-disjoint designated-owner matching directly.

`FAIL-V48-POSITIVE-DEEP-SEMIPRIME-CLOSES-P1BR-001`

FAIL-ID:
FAIL-V48-POSITIVE-DEEP-SEMIPRIME-CLOSES-P1BR-001.
Alias:
v4.48 R48-7.
Status:
KILLED for immediate closure.
Killed inference:
once the deep semiprime owner family enters a positive clean signed round, P1BR follows.
Reason:
the established positive-round independence theorem remains binding.
New datum retained:
every designated target additionally satisfies the common deep-layer relation $Q \mid N - p_i$, which was not part of the old round datum.
Live replacement:
`P1BR-DEEP-DESIGNATED-TARGET-COUPLING`.

`FAIL-V48-VARIABLE-M-SELFREENTRY-NEW-001`

FAIL-ID:
FAIL-V48-VARIABLE-M-SELFREENTRY-NEW-001.
Alias:
v4.48 R48-6.
Status:
SUPERSEDED.
Superseded idea:
restart the v4.43 fixed-m recurrence with growing $m \sim \log\log N$ as a new proof route.
Reason:
v4.44 already performed the variable-threshold correction and replaced the fixed-m recurrence by simultaneous valuation freezing, including the near-critical polylog star boundary.

`FAIL-V49-CROSSGEN-GCD-IS-P1BR-BRIDGE-001`

FAIL-ID:
FAIL-V49-CROSSGEN-GCD-IS-P1BR-BRIDGE-001.
Status:
KILLED as a sufficient general bridge.
Certificate:
the cheap semiprime orthogonal-broom branch admits a large selected subfamily with the relevant owner-support / later-target gcds equal to 1.
Consequence:
the common deep modulus does not automatically couple the old owner round to the descendant target rows through cross-generation gcd.

FAIL-V49-DEEP-SEMIPRIME-AUTOMATIC-POSITIVE-ROUND-CLOSURE-001

FAIL-ID:
FAIL-V49-DEEP-SEMIPRIME-AUTOMATIC-POSITIVE-ROUND-CLOSURE-001.
Status:
SCOPE-CORRECTED.
Killed promotion:
treat the localised deep semiprime family as automatically in the former ambient positive-round closure after the $N^{-o(1)}$ localisation loss.
Reason:
the scale/domain transfer is not automatic, and positive signed production remains insufficient for P1BR without an independent additive/capacity coupling.
Replacement:
v4.49 orthogonal-broom analysis and the later local reflected-anchor frontier.

FAIL-V49-DIRECT-REFLECTED-FACTOR-INHERITANCE-001

FAIL-ID:
FAIL-V49-DIRECT-REFLECTED-FACTOR-INHERITANCE-001.
Status:
KILLED / SCOPE-CORRECTED.
Killed inference:
from $\ell \rightsquigarrow p$ and $\ell \mid c_j^{\mathrm{ref}}$, infer $p \mid c_j^{\mathrm{ref}}$.
Reason:
direct divisor incidence is not hereditary along a generic CRD path.
Replacement:
the exact path-labelled identity `G-REFLECTED-ANCESTRY-EXACT-IDENTITY` and branch application `C-DEEP-ANCESTRAL-REFLECTED-TAG-TRANSPORT`.

FAIL-V49-LATER-DESCENDANTS-DESTROY-ALL-REFLECTED-ARITHMETIC-001

FAIL-ID:
FAIL-V49-LATER-DESCENDANTS-DESTROY-ALL-REFLECTED-ARITHMETIC-001.
Status:
KILLED.
Killed description:
once direct factor inheritance is lost, reflected arithmetic is lost entirely.
Counterdatum:
exact ancestry retains

$$N(2s\Xi - 1) - 2d_j = 2(-1)^t sC_t(N-p).$$

Scope guard from v4.50:
the exact identity is useful data, but its bare reduction modulo the terminal deep modulus is automatic and is not by itself capacity.

FAIL-V49-RELATIVE-T-OR-FRESH-MASS-ALONE-CLOSES-001

FAIL-ID:
FAIL-V49-RELATIVE-T-OR-FRESH-MASS-ALONE-CLOSES-001.
Status:
KILLED as an immediate contradiction route.
Reason:
the exact relative $T/U/J/F$ identities give lower reservoirs but no independent upper capacity for the external extension.
What survives:
the identities remain valid accounting tools.

FAIL-V50-ANCESTRY-DIAGONAL-TRANSPORTS-DIRECT-CAPACITY-001

FAIL-ID:
FAIL-V50-ANCESTRY-DIAGONAL-TRANSPORTS-DIRECT-CAPACITY-001.
Alias:
v4.50 R50-2.
Status:
SCOPE-CORRECTED.
What remains proved:
the exact identity

$$N(2s\Xi - 1) - 2d_j = 2(-1)^t sC_t(N - p).$$

What is removed:
the claim that reducing this identity modulo $Q \mid N - p$ creates a new reflected path-profile capacity condition.
Current distinction:
direct reflected incidence is a capacity datum; bare remote reflected ancestry is not, at residue level.

FAIL-V50-BOUNDED-BRANCHING-CLOSES-SINGLESEED-001

FAIL-ID:
FAIL-V50-BOUNDED-BRANCHING-CLOSES-SINGLESEED-001.
Alias:
v4.50 R50-5; consistent with `FAIL-BOUNDED-ARITY-WEIGHT-001`.
Status:
KILLED as a weighted-capacity inference.
Safe use:
branching may be combined with exact cofactor/path arithmetic or another single-charged quantity, but it is not a capacity theorem by itself.

FAIL-V50-REFLECTED-PROVENANCE-USELESS-AFTER-SELECTION-001

FAIL-ID:
FAIL-V50-REFLECTED-PROVENANCE-USELESS-AFTER-SELECTION-001.
Alias:
v4.50 R50-3.
Status:
KILLED on the return-heavy cheap-J branch.
Certificate:
`C-RETURNHEAVY-REFLECTED-CUBEROOT-LOADONE-MATCHING` recovers a positive cube-root-scale vertex-disjoint matching whose targets retain direct reflected tags with distinct reflected indices and quotient-return normal forms.

FAIL-V50-RETURNED-SEED-IMPLIES-CYCLE-001

FAIL-ID:
FAIL-V50-RETURNED-SEED-IMPLIES-CYCLE-001.
Alias:
v4.50 R50-4.
Status:
KILLED.
Killed inference:
if a reflected seed has some parent in the failed extension, then the parent lies in the same seed basin and closes a directed cycle.

Reason:
source retention supplies a parent somewhere in the basin union; it may belong to a different seed basin.
What survives:
when a genuine same-basin return or actual cycle is proved, `G-CRD-CYCLE-COFACTOR-LOCK` applies.

`FAIL-V50-SOURCEHEAVY-REFLECTED-SEEDS-CLOSE-001`

FAIL-ID:
FAIL-V50-SOURCEHEAVY-REFLECTED-SEEDS-CLOSE-001.
Alias:
v4.50 R50-6.
Status:
KILLED.
What is proved:
a source-heavy packet gives a large external source product / T_U-type lower defect.
What is missing:
an independent source-sensitive upper capacity.
Consequence:
full `C-REFLECTED-FIRST-OCCURRENCE-CONVERSION` remains open; only the return-heavy half is proved pending audit.

`FAIL-V51-ALTERNATE-PATH-CONGRUENCE-AMPLIFIES-DEEP-POWER-001`

FAIL-ID:
FAIL-V51-ALTERNATE-PATH-CONGRUENCE-AMPLIFIES-DEEP-POWER-001.
Source:
v4.51 R51-3.
Status:
KILLED.
Killed inference:
use the alternate-path cofactor congruence to obtain extra divisibility from the common factor r^{E-1}.
Reason:
$r \nmid N$, so the relevant factor is invertible modulo N and cancels.
What survives:
`G-ALTERNATE-PATH-COFACTOR-LOCK` remains PROVED.

`FAIL-V51-ANCESTRY-LOCALISATION-FAILED-001`

FAIL-ID:
FAIL-V51-ANCESTRY-LOCALISATION-FAILED-001.
Source:
v4.51 R51-5.
Status:
KILLED.
Killed description:
the inability of aggregate path products to see E means ancestry localisation itself failed.
Correction:
ancestry localisation remains useful and v4.51 strengthens it through `G-DEEP-ANCHOR-FIBRE-COUNTING-TRADEOFF` and `C-DEEP-LINKED-REFLECTED-ANCHOR-RECOVERY`. What fails is only an overly coarse path-product consumer.

FAIL-V51-CYCLE-LOCK-FALSE-001

FAIL-ID:
FAIL-V51-CYCLE-LOCK-FALSE-001.
Source:
v4.51 R51-4.
Status:
KILLED as a description of the theorem.
Correction:
`G-CRD-CYCLE-COFACTOR-LOCK` remains PROVED and is subsumed as a corollary of the stronger `G-ALTERNATE-PATH-COFACTOR-LOCK`.

FAIL-V51-UNWEIGHTED-REFLECTED-MATCHING-CARRIES-DEEP-MASS-001

FAIL-ID:
FAIL-V51-UNWEIGHTED-REFLECTED-MATCHING-CARRIES-DEEP-MASS-001.
Source:
v4.51 R51-1 / scope correction to v4.50.
Status:
SCOPE-CORRECTED.
Killed proof-DAG edge:

full cube-root reflected matching from v4.50 $\Longrightarrow$ its descendant closure carries the current common-deep ob

Reason:
the deep obstruction was extracted from the descendant closure of the v4.41/v4.42 high matching, while the v4.50 reflected matching was re-extracted from the original external seed packet. A large deep obstruction may be concentrated in basins omitted by that unweighted matching.
What survives:
the v4.50 reflected matching theorem itself remains PROVED-PENDING-INDEPENDENT-AUDIT; v4.51 repairs the transfer by selecting reflected anchors after the actual deep rows are known.

FAIL-VERTICAL-DESCENT-AUTOMATIC-BV-001

FAIL-ID:
FAIL-VERTICAL-DESCENT-AUTOMATIC-BV-001.
Date / iteration:
vNEXT4.9.
Claim:
A dyadic band produced at $Y \geq h > N^{1/3}$ automatically lies in the range of `G-DYADIC-RECIPROCAL-REENTRY`.
Status:
KILLED.
Exact certificate:
The current analytic theorem requires

$$2Y^{3+\vartheta} \leq \frac{N^{1/2}}{(\log N)^{C_0}},$$

so its upper scale is only roughly $N^{1/(6+2\vartheta)}$, modulo logarithms. The window between that scale and the cubic range is not covered.
Do not repeat:
Keep deterministic inverse-scale transport and analytic BV re-entry as separate arrows in the proof DAG.

FAIL-VERTICAL-FC8-SEPARATION-001

FAIL-ID:
FAIL-VERTICAL-FC8-SEPARATION-001.
Date / iteration:
vNEXT4.5.
Claim:
The nonroot vertical factor U_R is an essentially independent store which the exact FC8 low-core ledger does not see.
Status:
SCOPE-CORRECTED / KILLED.
Exact certificate:
In the proved $3 \nmid N$ fringe domain,

$$D_C J_R U_R^{\mathrm{nr}} = W_C^{\mathrm{nr}}.$$

Thus denominator, horizontal reuse, and nonroot vertical layers are exact factors of one conserved low-core mass. The root-only term $U_0 = m/\operatorname{rad}(m)$ is separate but has $\log U_0 < \log N$.
What survives:
The partition among D, J, U can still be the obstruction; what is killed is treating U^{nr} as outside the same functional.
Do not repeat:
Preserve the full C-part of each row before projecting to J and U.
Related current target:
C-MONOMIAL-REFLECTED-PAIR-SPARSITY.

FAIL-VERTICAL-HIGHGAP-UNBOUNDED-CAPACITY-001

FAIL-ID:
FAIL-VERTICAL-HIGHGAP-UNBOUNDED-CAPACITY-001.
Date / iteration:
vNEXT4.11.
Claim:
High-base block-derived vertical mass has no independent upper capacity and can only be controlled through J_R, $K(R)$, or a γ-dependent ledger.
Status:
KILLED.
Exact certificate:
For every $Y > N^{1/3}$, exact depth two and the square-parent slot interval give

$$m_q \le 1 + \frac{N}{6q^2}, \qquad V_{\ge Y}^{\mathrm{nr}} \ll \frac{N}{Y}.$$

In particular $V_{\mathrm{blk}} \ll N/h$ on the block-derived high-gap branch.
What survives:
This is an upper capacity on the vertical channel. It is not by itself a Goldbach contradiction and does not prove that the whole reflected obligation must enter that channel.

FAIL-VERTICAL-SQRTLOG-WALL-001

FAIL-ID:
FAIL-VERTICAL-SQRTLOG-WALL-001.
Date / iteration:
vNEXT4.12.

Claim:
Under the fixed-proportion vertical-heavy hypothesis, the best available confinement is $h \ll \sqrt{N/\log N}$.
Status:
SUPERSEDED within that conditional branch.
Replacement:
The hybrid linear sieve gives

$$\boxed{h \ll_\delta \sqrt{\frac{N}{\log N\ \log\log N}}}$$

when $V_{\mathrm{blk}} \geq \delta t \log h$.
Scope guard:
vNEXT4.13 shows that this remains a conditional vertical-heavy subroutine; it is not an unconditional ceiling on the whole canonical branch.

FAIL-VERTICAL-TO-ETA-LOSES-LOG-001

FAIL-ID:
FAIL-VERTICAL-TO-ETA-LOSES-LOG-001.
Date / iteration:
vNEXT4.9.
Claim:
Routing a linear vertical family only to $\eta(R) \gg h$ irretrievably loses the $\log h$ factor and therefore cannot preserve a $t \log h$-scale obligation.
Status:
KILLED.
Exact certificate:
The distinct parent rows are forced through a much smaller low-base sibling support. Cauchy/pigeonhole concentration yields

$$K^{(1)}_{<h}(R) \gg h \log h$$

for a linear family of distinct vertical bases, and the weighted theorem gives

$$V_{\mathrm{blk}} \gg t \log h \Longrightarrow \log J_R \gg V_{\mathrm{blk}} \ \vee\ K^{(1)}_{<h}(R) \gg t \log h.$$

What survives:
Large $K(R)$ is still not a contradiction; the missing ingredient is a source-sensitive consumer/upper capacity.

FAIL-VERTICAL-UNLOCALISED-001

FAIL-ID:
FAIL-VERTICAL-UNLOCALISED-001.
Date / iteration:
vNEXT4.6.
Claim:
Vertical storage may occur anywhere among block-derived rooted low-core factors.
Status:
KILLED as a description.
Exact certificate:
If $c_j = sq$ and $N - w = bq$, then

$$v_q(N-w) \geq 2 \iff q \mid b.$$

Since $b < 2s$ and $q > N/(2s)$, this forces

$$s > \sqrt{N}/2.$$

What survives:
The true vertical zone is the large-quotient region; below it every block-derived rooted factor is D/J-only.

`FAIL-VNEXT433-UNIFORM-FLOOR-GAP-001`

FAIL-ID:
FAIL-VNEXT433-UNIFORM-FLOOR-GAP-001.
Status:
KILLED as a failure certificate; it must not be used.
Incorrect claim:
the v4.33 uniform residual floor was allegedly unproved for $q > 4h$.
Repair:
the already proved `G-FULLQ-PRODUCTION` gives $M_q > (5/3)q$, hence $q > 4h$ gives $M_q > 20h/3$. For $q \leq 4h$, the shadow bound works after explicitly excluding $q = 2h$ by parity. Thus the full sub-root floor survives.
Do not repeat:
before recording a gap for a quantity, search the whole Core for an older bound on that same quantity, not only the newest report block.

`FAIL-WEIGHTED-REUSE-AMPLIFICATION-001`

FAIL-ID:
FAIL-WEIGHTED-REUSE-AMPLIFICATION-001.
Status:
SCOPE-CORRECTED.
Generic witness:
for $N = 2200$, root 1471 and $R = \{3, 13, 1471\}$, the complements are $3^6, 13^3, 3^7$ and $\log J_R = \log 3$; the vertical channel carries nearly all prime-power mass. This refutes a near-unit $\gamma \log N$ lower bound on $\log J_R$ over the full generic FC7/FC8 failed-root domain.
Critical scope:
the root is not canonical and lies outside the near-square-root subdomain. The witness does *not* refute a theorem in which independent canonical/high-gap information forces an incompatible lower bound; such a theorem would prove that its failed domain is empty.

`FAIL-WHOLE-BLOCK-EXTERNAL-SET-LOSES-REFLECTED-PROVENANCE-001`

FAIL-ID:
FAIL-WHOLE-BLOCK-EXTERNAL-SET-LOSES-REFLECTED-PROVENANCE-001.
Status:
KILLED.
Killed inference:
projecting the reflected block to a coarse set of external prime factors destroys the generating reflected indices so severely that no polynomial-weight tagged family can be recovered.
Certificate:
v4.39 assigns each prime to one reflected index, notes that one index carries $< \log N$ assigned weight, and recovers an injectively tagged subfamily preserving an $\alpha - o(1)$ fraction of the weight for a $N^{\alpha - o(1)}$ packet.
What survives:
the recovered packet may still lack an immediate second occurrence / upper consumer.

`FAIL-WORLD-SIZE-NAIVE-001`

FAIL-ID:
FAIL-WORLD-SIZE-NAIVE-001.
Date / iteration:
vNEXT4, Phase 1b.
Claim:
For a failed world R with composite complement, $|R| \le \eta(R) + 1$, on the ground that every row satisfies $\omega(N-u) \ge 2$ and that $\sum_{u \in R} \omega(N-u) = \eta(R) + |R| - 1$.
Original motivation:
Every state of a failed world is BAD, so its complement is composite; a composite number "has at least two prime factors".
Status:
KILLED as stated.
Minimal / strong counterexample:
$N = 2200$, root $r_* = 1471$,

$$R = \{3, 13, 1471\}, \qquad N - 1471 = 729 = 3^6, \quad N - 3 = 2197 = 13^3, \quad N - 13 = 2187 = 3^7.$$

Here $\Gamma_N(R) = \{3, 13\} = R \setminus \{r_*\}$, $\kappa_R(3) = 2$, $\kappa_R(13) = 1$, so $\eta(R) = 1$ while $|R| = 3 > \eta(R) + 1 = 2$.
Failed hypothesis or mechanism:
A composite number has at least two prime factors *counted with multiplicity*; it need not have two *distinct* ones. Every row of this world is a prime power, so $\omega = 1$ on all three.
Why the argument failed:
κ is layer-one incidence and therefore sees ω, not Ω. The step that requires $\omega \ge 2$ is the only place where badness was used, and badness gives $\Omega \ge 2$.
What survives:
The corrected count $|R| \le \eta(R) + 1 + 2a_0 + a_1$ with $a_0 \le 1$ and $a_1 \le \gamma(\lfloor \log_2 N \rfloor - 1)$, hence $|R| < 4\gamma \log N$ under the (FC9) hypotheses: Theory Core, `G-CANONICAL-WORLD-SIZE-CAP`, (WS1)–(WS3). Zero violations over the recorded 301 worlds.
Safe weaker version:
The corrected form above; nothing weaker is needed.
Do not repeat:
Whenever a count over the rows of a failed world uses $\omega(N-u) \ge 2$, the prime-power rows must be excluded explicitly. Recorded as loss-control rule (LR6).
Related current target:
`P1BR-ADDITIVE-COUPLING`, γ.

`FAIL-v40-HIGH-FLOOR-N-OVER-LOG2-IS-SHARP-001`

FAIL-ID:
FAIL-v40-HIGH-FLOOR-N-OVER-LOG2-IS-SHARP-001.
Status:
SUPERSEDED.
Superseded description:
the useful unique-parent target floor is only $q > n/(\log N)^2 = N^{1/3-o(1)}$.
Certificate:
the total log mass of all primes $q \le N^{1/3}$ is $O(N^{1/3}) = o(n \log N)$ on $n \gg N^{1/3}$. v4.41 therefore moves essentially all unique-parent mass to the strict floor $q > N^{1/3}$.
What survives:
the v4.40 matching is valid but its constants/floor are superseded by the v4.41 cube-root matching.

FAIL-v41-MONOMIAL-LENGTH-TWO-NOVELTY-001

FAIL-ID:
FAIL-v41-MONOMIAL-LENGTH-TWO-NOVELTY-001.
Status:
SCOPE-CORRECTED.
Scope correction:
v4.41 correctly proves/uses the interval mechanism, but the statement that every monomial component has at most two edges was already PROVED in the Theory Core as `C-MONOMIAL-TWO-EDGE-RIGIDITY`.
New part retained:
v4.42 records the stronger fact that for fixed N at most one monomial component can contain two edges.

FAIL-BOUNDED-COFACTOR-ROWSLOTS-ARE-UNCONSTRAINED-001

FAIL-ID:
`FAIL-BOUNDED-COFACTOR-ROWSLOTS-ARE-UNCONSTRAINED-001`.
Status:
KILLED.
Killed inference:
fixing the integer cofactor c leaves no useful deterministic capacity on the owned high row-slot packet.
Replacement:
fixed c forces

$$2QFc \mid p_i - p_j,$$

and summing over $c \leq C$ gives the v4.57 harmonic row-slot capacity.

FAIL-BOUNDED-COFACTOR-SUM-MUST-PAY-ONE-PER-C-001

FAIL-ID:
`FAIL-BOUNDED-COFACTOR-SUM-MUST-PAY-ONE-PER-C-001`.
Status:
KILLED.
Killed inference:
summing fixed-c Selberg bounds over a polynomial cofactor range necessarily creates an $O(C)$ loss comparable to the whole packet.
Replacement:
choose

$$C_\varepsilon = MN^{-1/(\log\log N)^\varepsilon} = o(M).$$

Then the total $O(1)$-per-c remainder is $o(M)$, while discarded cofactors still carry full logarithmic mass.

FAIL-CHEAP-RADIC-OSCILLATION-HAS-A-HIDDEN-ROUTING-SINK-001

FAIL-ID:
`FAIL-CHEAP-RADIC-OSCILLATION-HAS-A-HIDDEN-ROUTING-SINK-001`.
Status:
KILLED as a routing diagnosis.
Reason:
exact target-row entry accounting shows that cheap oscillation can survive only by consuming fresh target labels. Repeated targets are paid; the unresolved issue is population capacity, not missing routing bookkeeping.

FAIL-COFACTOR-RENEWAL-IMPLIES-DIVERGENCE-001

FAIL-ID:
`FAIL-COFACTOR-RENEWAL-IMPLIES-DIVERGENCE-001`.

Status:
KILLED.

Killed inference:
a renewed polynomial owned support population obtained from cofactor factorisation is already a contradiction.

Reason:
the standing renewal-is-not-divergence restriction remains binding; there is no proved global upper capacity for arbitrary fresh external population.

FAIL-FIRST-COPY-HIGH-ROW-POPULATION-IS-BY-ITSELF-A-CONTRADICTION-001

FAIL-ID:
`FAIL-FIRST-COPY-HIGH-ROW-POPULATION-IS-BY-ITSELF-A-CONTRADICTION-001`.

Status:
KILLED.

Killed inference:
a polynomial family of distinct external high target rows, each used once, is already an upper-capacity contradiction.

Reason:
this is the standing "renewal is not divergence" restriction in the present deep-owned domain. A fresh external population is an obligation until a CRD-specific capacity theorem consumes it.

FAIL-FIXED-COFACTOR-PRIMALITY-GIVES-NO-MORE-THAN-SPACING-001

FAIL-ID:
`FAIL-FIXED-COFACTOR-PRIMALITY-GIVES-NO-MORE-THAN-SPACING-001`.

Status:
KILLED.

Killed inference:
after v4.57 fixed-c spacing, primality of both endpoints cannot yield further capacity.

Replacement:
after parity parametrisation the endpoint primes are two explicit non-proportional affine forms with determinant $-2FN$, so the already audited classical dimension-two Selberg upper-bound sieve gives logarithmic-square compression.

FAIL-HIGH-BASE-R-GREATER-Y-IMPLIES-RADIC-MONOTONICITY-001

FAIL-ID:
`FAIL-HIGH-BASE-R-GREATER-Y-IMPLIES-RADIC-MONOTONICITY-001`.

Status:
KILLED.

Killed inference:
the large-base hypothesis $r > Y$ by itself forces monotone r-adic dynamics.

Certificate:
the explicit v4.55 high-base diagnostic witness with nonmonotone level pattern. It does not satisfy the full deep exponent hypothesis and therefore does not rule out stronger theorems using deep/reflected provenance.

FAIL-LARGE-COFACTOR-IS-AN-ANONYMOUS-COEFFICIENT-SINK-001

FAIL-ID:
`FAIL-LARGE-COFACTOR-IS-AN-ANONYMOUS-COEFFICIENT-SINK-001`.

Status:
KILLED.

Killed inference:
large c_i may absorb the row-slot obligation without producing a trackable CRD resource.

Replacement:
large cofactors carry explicit logarithmic mass with exact support/horizontal/vertical factorisation; this yields genuine residual J/U or owned fresh-support renewal.

FAIL-LARGE-COFACTOR-RENEWAL-IS-DIVERGENCE-001

FAIL-ID:
`FAIL-LARGE-COFACTOR-RENEWAL-IS-DIVERGENCE-001`.

Status:
KILLED.

Killed inference:
the v4.58 full-scale large-cofactor renewal branch alone gives a global contradiction.

Reason:
renewed support is a real relation-preserving CRD obligation, but no proved capacity theorem yet bounds its total fresh external population.

FAIL-PRODUCT-MODULUS-COMPRESSION-ALONE-CLOSES-DEEP-001

FAIL-ID:
`FAIL-PRODUCT-MODULUS-COMPRESSION-ALONE-CLOSES-DEEP-001`.

Status:
KILLED.

Killed inference:
the v4.57 product-modulus upper bound already contradicts the deep lower bound on $Q = r^e$.

Reason:
the deep hypothesis permits $Q = N^{o(1)}$, so even a bound of scale $QF \le N^{1-\alpha+o(1)}$ need not contradict the branch.

FAIL-RADIC-HALVING-IS-AUTOMATIC-DYNAMICAL-DESCENT-001

FAIL-ID:
`FAIL-RADIC-HALVING-IS-AUTOMATIC-DYNAMICAL-DESCENT-001`.

Status:
SCOPE-CORRECTED.

Overstrong inference:
organising threshold classes at $e, \lceil e/2\rceil, \lceil e/4\rceil, \ldots$ automatically produces strict forward descent at every owned transition.

Correction:
threshold layers may remain useful organisationally, but an edge may jump upward to a larger exact valuation. Any finite-return argument must use an independent payment/capacity statement.

FAIL-RADIC-LEVEL-IS-MONOTONE-POTENTIAL-001

FAIL-ID:
`FAIL-RADIC-LEVEL-IS-MONOTONE-POTENTIAL-001`.

Status:
KILLED.

Killed inference:
for an owned transition $p \to s$, the quantity $v_r(N-p)$ must decrease along the forward dynamics.
Certificate:
the exact v4.55 three-cycle template

$$N = r^{a+b+c} + 1, \quad N - p_0 = r^a p_1, \quad N - p_1 = r^b p_2, \quad N - p_2 = r^c p_0,$$

which realises arbitrary cyclic valuation pattern $a \to b \to c \to a$ whenever the states are active primes.
Safe replacement:
retain the exact adjacent-row transition identities and price actual repeated target rows rather than valuation-level recurrence.

`FAIL-RADIC-LEVEL-RETURN-ITSELF-PAYS-J-001`

FAIL-ID:
`FAIL-RADIC-LEVEL-RETURN-ITSELF-PAYS-J-001`.
Status:
KILLED.
Killed inference:
a return to a previously used value of $v_r(N-s)$ must create horizontal reuse.
Certificate:
horizontal reuse is indexed by target prime labels; the v4.55 cycle realises level recurrence through distinct target rows.
Safe replacement:
after the first copy at one target row, every later occurrence at that same target row is fully priced by incremental J.

`FAIL-SAME-EXACT-LEVEL-IS-A-NEW-EXTERNAL-RESTART-001`

FAIL-ID:
`FAIL-SAME-EXACT-LEVEL-IS-A-NEW-EXTERNAL-RESTART-001`.
Status:
KILLED as a packet-design inference.
Killed inference:
an owned edge landing on another high row with the same exact valuation e should be treated as a new external restart branch.
Replacement:
saturate the exact level in the stopped reflected-seed basin first; then every high exact-e target already belongs to the saturated layer.

`FAIL-SELBERG-LOGSAVING-ALONE-CLOSES-DEEP-001`

FAIL-ID:
`FAIL-SELBERG-LOGSAVING-ALONE-CLOSES-DEEP-001`.
Status:
KILLED.
Killed inference:
the v4.58 log-saved bound

$$QF \ll_\varepsilon \frac{N(\log\log N)^{1+\varepsilon}}{M \log N}$$

is already incompatible with the current deep modulus.
Reason:
Q may still be only $N^{o(1)}$; the logarithmic saving is real but not a contradiction.

13.3 Post-v4.81 failure-log additions (2026-09-05–2026-09-07C)

AUDIT-FBTU-PARTIAL-VERSUS-COMPLETE-20260905

Status. SCOPE-CORRECTED. The supplied CRT proof programs selected BAD edges/valuations but does not control every unprescribed factor of every expanded row. The blanket bounded-depth complete-BFS no-go is therefore not established. Exact witness: $N = 92$, $p_0 = 47$ has a selected BAD cycle $5 \to 29 \to 7 \to 5$ but succeeds at depth 1 via 3 because $92 - 3 = 89$ is prime. **Survives:** compatible partial-transcript programming.

ENVELOPE-ESCAPE-AUTOMATIC-ITERATION-20260905

Status. KILLED as an inference. The proved $h = 1$ envelope-escape theorem can force new primes $> y$ at depth 2, but after the envelope grows the threshold $F_2(y)$ changes and need not remain below the same N. Automatic repeated escape is not justified. **Survives:** the one-step and all-first-branch escape theorems under their explicit size hypotheses.

AUTOMATIC-THIRD-GENERATION-SIEVE-SWITCH-20260906

Status. KILLED as an inference. Radius-two upper-bound sieve estimates do not automatically iterate to a complete third generation: the number and arithmetic shape of residual forms change, and complete cofactors must be retained. No proved radius-two theorem is refuted.

SUCCESS-PRODUCT-BARRIER-TO-CLOSED-CORE-HEIGHT-20260906B

Status. KILLED as an inference. Label/cofactor product barriers quantify over successful paths. A completely closed BAD core contains no such path, so the successful-path theorem is vacuous there and cannot be combined with the closed-support height bound to obtain a contradiction.

PRIME-LIFT-PRIMALITY-FORCES-SUCCESS-20260907

Status. KILLED. Prime lift primality alone does not force BFS success. Exact counterexample: $(N, d, R) = (1862, 7, 1597)$, where the full BFS from inactive root 7 and from prime lift 1597 are conjugate and both fail with a complete closed active support certificate. **Survives:** PRIME-LIFT-FULL-BFS-CONJUGACY, and the special unresolved $d = 2/h = 1$ case.

LOW-GAP-FIVE-LABEL-OVERGENERALIZATION-20260907C

Status. KILLED by exact counterexample. At $N = 44670$, $r = 22469 = p_{12}$, $m = 149^2$, $h = 134 < 149$, the complete BAD frontiers through depth 2 discover exactly four nonroot labels by depth 3: $\{149, 211, 23, 1933\}$. Both depth-3 endpoints are GOOD. The proved universal lower bound THREE survives; FOUR remains open in the stated low-gap domain.

SUBGROUP-SEPARATION-OF-P0-AS-GENERAL-EXPLANATION-20260907C

Status. KILLED as a general explanation. For five of the six historical hosts, the subgroup of $(\mathbb{Z}/N\mathbb{Z})^\times$ generated by the maximal closed low BAD support is the entire unit group, so no character trivial on that support can separate p_0. At $N = 2200$ the subgroup is proper, but membership is still not sufficient for failure. The elementary necessary subgroup lemma survives as an exclusion test when a verified full BAD envelope is available.

Chapter 14

Research Agenda

14.1 How to choose what to work on

Three rules for triage, in order of importance.

1. **Check complete support first.** A finite *selected* transcript can be programmed, but the old blanket theorem about complete bounded-depth BFS was scope-corrected (Corollary 8.3). If an argument constrains only chosen divisibilities while ignoring residual cofactors, it cannot prove full failure or success. A complete-frontier theorem is still admissible, but it must control every factor of every expanded row.
2. **Check the functional.** If the argument proves a lower bound in one functional and consumes it in another, it is the $\mathrm{P1}^{\mathrm{br}}$/K1 error. Write down both functionals explicitly *before* starting.
3. **Check the scope class.** If the argument uses a class-B lemma to prove a class-A statement, it proves nothing about p_0. If it uses a class-D hypothesis under canonical failure, it is invalid.

2026-09-07C agenda override. The tier table below is retained as the historical v4.81 agenda. For new work, start instead with the three targets in §10.8: (1) the exact $h = 1$ full-BFS/root-2 closed-support question, (2) the low-gap three-label Diophantine obstruction split, and (3) the radius-three large-label/large-cofactor asymptotic region. The moving closed-support problem is the global wall.

14.1.1 Tiered targets

Tier	Target	Why it is in this tier
1	Prove or refute (N3): $\sum_q d_R(q) \geq \lvert R\rvert + 1$ (§5.7, Q8)	Small, self-contained, elementary in appearance, and it repairs the part of the corpus that used the false Euler normalisations. Zero counterexamples in 21 093 worlds. A model with a few hours should attempt this first.
1	Refute L1 by exhibiting $\nu(R) = O(1)$ with J_R unbounded (§9.5)	A *negative* result that would remove a whole family of capacity arguments. The data already point this way. Cheap to test computationally.
1	Extend the B_N^{low} census beyond $6 \cdot 10^4$	The coincidence of the three six-element lists (§11.5) is unexplained. A seventh N would be the single most informative new datum in the project.

Tier	Target	Why it is in this tier
2	`TS`: smooth branch to escape (§9.3)	The branch with the largest quantitative margin and the most elementary statement. Success would give Goldbach in a specific regime, which is a real theorem.
2	`E1`: escape-or-reuse, or the bounded-charging lemma (§9.6)	Overwhelming data, explicit statement, and the missing ingredient is named (bounded charging multiplicity). The most likely target to actually fall.
2	Invariants of the boundary filtration (Q7)	Purely combinatorial, independent of Goldbach, publishable on its own terms if anything nontrivial exists.
3	`TD` / `C-SAMEFUNC-BRIDGE` (§9.3)	The real bridge. Hard, and previous attempts all failed the same way. Attempt only with the same-functional constraint written down first.
3	`RPD` (§9.10)	The only open target carrying an explicit quantitative margin. Its original phrasing should be re-checked against the source before use.
4	`X1` / **(H-p_0)**	Strictly stronger than Goldbach. Do not attack directly.
4	Anything requiring level of distribution $> 2/3$ or a binary prime–Liouville correlation	These are the external walls (§12.4). Working on them is working on a different, famous problem.

14.1.2 Experiments that would change the picture

1. B_N^{low} **census to** 10^6**.** Currently $6 \cdot 10^4$. Cost: roughly quadratic in the limit as implemented; a smarter implementation should use the fact that B_0 can be built once per N from a shared sieve. A seventh nonempty B_N^{low}, or its absence at 10^6, is decisive information about whether the six are an artefact of smallness.
2. **Full upper-root census beyond** $2 \cdot 10^4$**.** The 301 number is the project's most-cited statistic and it is known only to $2 \cdot 10^4$. Extending it to 10^5 would test whether "failing roots cluster at $B_N^{\text{low}} \neq \varnothing$ numbers" persists.
3. J_R **versus** $\nu(R)$ **scatter.** Directly tests L1 and is a few lines on top of the existing world routine.
4. G_{esc} **to** 10^6**.** Tests Q2 (does the margin grow?). The current dyadic minima $1, 1, 1, 1, 1, 1, 2, 2, 3, 3, 3$ are suggestive but the range is small.
5. **Phase-conditioned traversal statistics.** Does anything about the traversal, beyond immediate success, depend on θ? Currently unknown, and ρ_0 is the only boundary invariant that varies between canonical worlds (Q1).

Chapter 15

Literature and Source Registry

15.1 Internal sources

Tag	Source	Role
[BIB]	the Goldbach Biblia (v7.5 and its post-v7.5 technical log)	Shadow, retention, path/winding invariants, prime-power witness, escape capacity, the 10^6 scan tables, the ten margin-one numbers.
[CORE]	CRD Core	Canonical geometry, C-GAP-EQUIVALENCE, C-STRICT-NONREUSE, the reflection gate, `C0a`, `C0b-RD`, the genealogical layer (V71.21–V71.44), the open-target registry, P0-CANONICITY-ABLATION.
[PCRD]	`GOLDBACH_POST_CRD_RESEARCH_FAILURE_LOG_20260904`	**The latest source; takes precedence on conflicts.** B_N^{low} fixed point and characterisation, the six N to $6 \cdot 10^4$, the successor-theory walls, finite affine canonical programmability, finite bad-tree universality, the canonical prime cell theorem, the H1 descent and the 2-core wall, the final verdict, its own theorem registry and 30-row log.
[FAIL]	the project failure log	All 239 live entries, reproduced in Chapter 13.
[LOG]	post-v7.5 technical log	D_q, M_q, the prime-power ledger, primitive residual normal form.
[ILA]	incidence-ledger audit	IL1–IL6, two reconstruction routes, 1 000 worlds.
[ORPH], [RT]	orphan census and red team	Escape-or-reuse data: 61 452 and 60 413 instances, 0 orphans.
[D4], [F3], [FRO], [FND], [V481], [NOV]	question-answer campaigns and the novelty note	Statistical ablations, structural twins, the rooted container identity, the prior-art boundary. Several of their numbers are *reported, not reproduced here* and are marked as such wherever they appear.

Warning 15.1 (Precedence)*.* Where sources conflict, [PCRD] takes precedence, then [CORE], then [BIB]. Where this file's own computation conflicts with any of them, this file states both and marks the discrepancy — it does not silently overwrite the source. The recorded discrepancies are: the Euler-defect normalisations (§5.7), the diagonal scope of the genealogical layer (Scope note 3.17), the lower bound in the ρ_0 formula (Warning 6.18), and the hypothesis of high-gap amplification (Warning 7.5).

15.2 Prior art and the novelty boundary

Warning 15.2. **No claim of novelty is made anywhere in this file.** Establishing novelty would require a separate literature review that has not been done. Any label of the form "derived here" means *derived from the corpus's own definitions in this document*, not "new in number theory".

The bibliography is a search record, not a citation list. It reproduces the Biblia's full list of 92 sources plus three later additions. Many entries are *not* cited anywhere in the running text: they are there so that a future model can see what has already been looked at and does not repeat the search. An entry's presence means "this was examined"; the disposition tables of §15.3–§15.4.1 say with what result. An entry's absence is the useful signal — it means the project has probably not looked there.

15.2.1 What must *not* be attributed to this project

The relation $p \to q$ for $q \mid N - p$; the notion of a Goldbach factorization graph; recursive search through complement factors; valuation weights $v_q(N-p)$; reachability, source and sink components, autonomous closed components; the study of Goldbach pairs near $N/2$; elementary parent residue classes and gcd identities. [NOV]

Topic	Source	Boundary
Factorization graph and exceptional autonomous components	Bożek [8]	The graph and the terminal pathology are not ours. Orientation matters: a source component in the paper's orientation corresponds to terminal behaviour after reversal (Warning 3.23).
Recursive factor search from a given root	OEIS A306746	Search from an arbitrary root is prior art; the root condition there is *not* the unconditional least prime $\geq N/2$.
First upper prime already forming a pair	OEIS A078496	$a(n) = \min\{p > n : p, 2n - p \in \mathbb{P}\}$ — a selector *conditioned on the outcome*. This is **not** p_0.
Nearest symmetric Goldbach pair	OEIS A082467	Goldbach near the midpoint is prior art.
Short intervals containing a prime	Baker–Harman–Pintz [13]	Calibrates $h \ll x^{0.525}$; it is not an escape lemma and is not used as one.
Verification of Goldbach	Oliveira e Silva et al. [12]	Says nothing about (H-p_0). Must never be cited as this project's own computation.
Linear forms in logarithms	Matveev [79]	Astronomical constants; asymptotic content only. Used only for the 2-core wall.
Almost-prime representations $(1 + 1.9)$	Li–Liu [15]	External input for CAPT; does not give the last bit P_1 versus P_2.
Exponent of distribution 5/8	Pascadi [93]	External input for RSD; below the 2/3 that RSD needs.
Chen-type lower bound	R. Li [94]	$D_{1,2}(N) \geq 1.9728\, C(N)N/\log^2 N$.
Asymptotic sieve for primes	Friedlander–Iwaniec [38]	The template for the missing bilinear information.

Topic	Source	Boundary
Goldbach problem for the Liouville function	Mangerel [95]	Calibrates the centred-correlation wall.

15.2.2 The safe formulation

Building on earlier work on Goldbach factorization graphs and on recursive search through complement factors, we study the *unconditional* selector $p_0(N) = \min\{p \in \mathbb{P} : p \geq N/2\}$ as a distinguished initial condition of the full dynamics. To the best of our knowledge we have not found earlier work using this selector — without requiring $N - p_0$ to be prime — as a distinguished root, and systematically comparing its reachable bad world with those of non-canonical roots. *This is a report on the extent of a literature search, not a claim of priority.*

15.3 Register of rejected literature applications

This section exists so that a future model does not re-import a tool the project has already tried and discarded. Each row records a specific *application* of a specific external result, and why it does not do the job here. A KILLED row means the application is dead, *not* that the cited theorem is wrong. An OPEN row means the tool may still be usable once a stated precondition is met.

15.3.1 First-front applications (Biblia v7.4)

Claim / tool	Status	Reason
$\lvert Q\rvert \sim N/\log N$	KILLED	$Q = \mathrm{PF}(m_0)$, so $\lvert Q\rvert = \omega(m_0)$; a global density comparison does not apply to the first front.
SMOOTH $\Rightarrow$ huge powerful part	KILLED	A SMOOTH row can be squarefree with many distinct small prime factors (Warning 7.13).
Positive density of squarefree values $\Rightarrow$ positive fraction of Q	KILLED	A result for all arguments, or all primes, does not transfer to the *adaptive* set of prime divisors of a single m_0.
Bombieri–Vinogradov on Q	KILLED	The theorem controls primes in arithmetic progressions *on average over moduli*, not the distribution of the prime divisors of one specific m_0.
Filaseta–Trifonov as a uniform lower gap between consecutive powerful numbers	KILLED	The short-interval squarefull-counting literature does not supply the universal lower gap barrier that was assumed.
Darmon–Granville $\Rightarrow$ uniform bound for a varying family	KILLED	Coefficients, the difference and the curve itself vary with N; finiteness for a fixed curve gives no uniformity.
Fixed-polynomial radical theorem with $F(X) = \prod_{q \in Q}(X - q)$	NEEDS VERIFICATION	Degree, height and the whole polynomial vary with N; an explicit uniform dependence is required.
Pairwise classical *abc* closes SMOOTH	KILLED	The shift $\lvert q - r\rvert$ can be of order N and consumes the radical; without extra global structure the required first-front lower bound does not arise.
Corvaja–Zannier greatest-prime-factor result	OPEN	Qualitatively supports escape, but gives no *moving* threshold $P^+(N - q) \geq m_0/q$.

Claim / tool	Status	Reason
Evertse–Schlickewei–Schmidt [18]	OPEN	Usable only after freezing a small alphabet/rank inside one counterexample. See also the vacuity result recorded in §9.2.4.
Gallagher's larger sieve [41]	OPEN	Requires first showing that a large part of Q occupies few classes modulo many primes.
Multiplicity-order literature	OPEN	Known generic / almost-all results give no deterministic lower side for specific ports modulo a specific N.

15.3.2 Iteration-12 audit of 52 primary sources

An honest protocol instead of a fictitious "1000". The scope and metadata of **52 distinct primary sources** were checked; for 18 of them the main theorems and the proof passages directly relevant to the project were read as well. This is *not* called "reading 1000 papers": automatically fetching a thousand bibliographic records is not reading and does not raise mathematical credibility. Below, mode `T` means "theorems and relevant sections", mode `Z` means "scope audit from abstract, introduction and statements of results".

Sources	Mode	Conclusion for the project
Bhowmik–Grimmelt [36]; Li–Liu [15]	T	Goldbach still open. The first localises the barrier in minor arcs and exceptional sets; the second gives $(1 + 1.9)$, not an adaptive traversal path. Contributed calibration of the main gap.
Lichtman [26]; Ford [30]	T	Strong distribution and the anatomy of shifted primes work *on average*. They do not give a pointwise choice from a reachable parent set.
Grimmelt–Teräväinen [45]; Halupczok [46]	Z	Transference and Goldbach in progressions / short intervals still leave an exceptional set or averaged conditions. Useful as a weighting template, insufficient pointwise.
Friedlander–Iwaniec [38]; Polymath8 [39]	T	**The most important analytic input:** breaking parity needs Type-II bilinear information, not a linear sieve alone.
Green–Tao [40]; Bienvenu–Shao–Teräväinen [44]	T/Z	Transference needs a pseudorandom majorant and positive relative density for a *fixed* configuration. Those hypotheses are not available for adaptive reachability.
Ford [31, 47]; Assing–Blomer–Li [28]	T	Pratt-type closures and uniform Titchmarsh problems are the closest structural analogues, but the relation $q \mid N - p$ moves with N.
Matomäki–Radziwiłł [48]	T/Z	"Almost all" short intervals give no certificate for every adaptive set.
Helfgott [16, 17]; Oliveira e Silva et al. [12]; Huang–Li [25]; Li [14]	T/Z	Circle method, finite verification, EH and short intervals are important calibration, but none synchronises a prime with the basin of p_0.
Evertse–Schlickewei–Schmidt [18]	T/Z	The number of S-unit solutions may depend only on dimension and rank, which is better than fixed-S constants. Application needs genuinely small rank; the known zero-wrap branch is empty.
Vaaler [49]	T/Z	Heights may give a short multiplicative dependence *after* an independent rank and height bound. They do not handle the present δ-forcing automatically.
Amoroso–Sombra–Zannier [50]	Z	Sparse-polynomial / unlikely-intersection results work for fixed families; Smith normal form is an excellent lattice algorithm but supplies no arithmetic contradiction by itself.

Sources	Mode	Conclusion for the project
Bugeaud [20]; Siksek–Stoll [21]; Bennett–Siksek [19, 5]	Z	Effective once a specific curve or support is frozen; no uniformity for a moving full SCC.
Bellman [51]; Karp [52]	T	The variational calculus of paths reduces to dynamic programming and a cycle criterion. Gave the cycle–potential duality, not prime selection.
Prajna–Jadbabaie [53]	Z	Barrier certificates and occupation measures are the right shape for reachability; on an SCC they need an arithmetic inequality that is missing.
Angeli–Philippe–Athanasopoulos–Jungers [54]	T/Z	A complete Lyapunov function is constant on recurrent classes, which justifies SCC compression. Multi-part versions do not evade the cycle barrier.
Friedman–Tillich [55]; Jiang–Lim–Yao–Ye [56]; De Leenheer [57]; Dodziuk [7]	T/Z	Calculus and Hodge theory on graphs formalise the gradient and the cyclic part. They contributed an exact no-go, not a primality theorem.
Bass [58]; Rangarajan [59]	Z	The graph zeta function and coker$(I - A)$ repackage determinant/Smith data. No new information about the rows $N - p$.
Z-transform / Koopman resolvent literature	Z	Appropriate for a *fixed* operator; the adaptive graph is not an LTI system.
Discrete variational calculus literature	Z	Gives stationarity equations, but for a sum of path costs it is not stronger than Bellman.

15.4 Post-v4.81 external-input audit (September 2026)

The continuation checked only the exact external statements it consumed; it did not perform a full literature-priority audit of the new project theorems.

Input	Source status	Exact use
`BS-1996`	primary statement verified	Beukers–Schlickewei's uniform bound for $x+y=1$ in finitely generated multiplicative groups; used only for solution counts, not heights.
`MATVEEV-IN-BMS-9.4`	exact product-minus-one form checked	Real/rational Matveev bound as stated in Bugeaud–Mignotte–Siksek, used for closed-support height and explicit envelope escape.
`FKL-2010-LEMMA-5.1`	exact statement and uniformity checked	Upper-bound sieve for a growing number of primitive linear forms; used for radius-two and multigeneration counting, never as a prime-tuples lower bound.
`PINTZ-GOLDBACH-EXCEPTIONAL-SET`	theorem statement/hypotheses checked	$E(Y) < Y^{0.72}$ above an ineffective constant, used only to show that ordinary Goldbach truth can coexist with complete shallow BADness on a density-one set of the special starts.

15.4.1 Frontier salvage audit (v4.81)

The v4.81 attack performed a targeted salvage audit of the older Biblia against the *current* frontier. Its verdict in one sentence: *the old corpus contributed exactly one genuinely missing tool — the gap-scale shadow identity — and one correct **shape** for the replacement frontier (the smoothness budget); everything else in the neighbourhood of the current obstruction is either already in the Core, or vacuous on the primitive branch, or domain-mismatched.*

Salvaged and used.

Old item	Restated form	Status and use
Prime-Gap Shadow Identity (Theorem 5.22)	$N = 2m+\delta$, $\delta = 2h$; for $r \mid m$: $r \mid N-y \iff r \mid \delta - y$	PROVED, re-proved from v4.80 definitions. Used to obtain the gap-scale form of primitive load-one (V81.11). *Not previously used in the v4.79/v4.80 chain.*
Retention–Escape (Theorem 5.24)	$\mathrm{PF}(N-q)\cap Q = \mathrm{PF}(d_q)$, $d_q = \gcd(m, \lvert\delta - q\rvert)$	SCOPE-CORRECTED. Identical to $D_q = \gcd(m, N-q)$, giving $D_q = \gcd(m, \lvert 2h-q\rvert)$. The old "escape" reading ("all children are new") must *not* be upgraded to success.
Gap/first-front disjointness	$\gcd(h, m) = 1$, $\mathrm{PF}(h)\cap Q = \varnothing$	PROVED, re-proved; used for the block/Q separation.
Multi-source product lock (Theorem 5.7)	$T_S \equiv (-1)^{\lvert S\rvert}\prod_{\mathrm{Src}} d \pmod N$	Already in the Core; the *new* use is the selected-family instance (V81.13) and the residual instance $T_{R'} \equiv (-1)^{\lvert R'\rvert}\Sigma'$.
Two-alphabet retention	$\gcd(N-q, mh) = \gcd(N-q, m)\gcd(N-q, h)$; $\gcd(E_q, E_r) \mid q-r$	PROVED (the split). On the primitive branch the two-alphabet retention of every first-front row collapses to the h-alphabet.
Smoothness budget (every row of a child-closed world is C-smooth)	The count of available C-smooth numbers in the row window is a necessary condition	SCOPE-CORRECTED to EMPIRICAL PATTERN: the old text carries it under an equidistribution assumption. Adopted here *only as the shape* of the new preferred frontier (§9.2.4). **The equidistribution assumption is not imported.**

Interesting but insufficient.

Old item	Content	Why insufficient at the frontier
Prime-power witness (Prop. 5.29)	$\mathrm{PF}(N-q) \subseteq Q \Rightarrow \exists r \in Q : v_r(m) = v_r(\delta - q)$	On the primitive branch the hypothesis is never satisfied ($\mathrm{PF}(N-q)\cap Q = \varnothing$): **vacuous domain** exactly where it is needed.
Shadow two-cycle (Prop. 5.28), $\delta = q+r+tqr$	Exact form of a Q-shadow two-cycle	The primitive branch makes the Q-shadow graph *empty*; no cycle exists to exploit. **Vacuous domain.**

Old item	Content	Why insufficient at the frontier
Quotient-rigidity master lemma	$\lvert u - v\rvert < 1.1u/m^2$, $\min(u, v) > 0.9m^2$ for row quotients	Proved inside a terminal-SCC band normalisation; its row-window and row-gap hypotheses are unavailable for primitive first-front rows. **Domain mismatch.**
One-escape residue lock	$g \equiv (-1)^{\lvert W\rvert} p_0(U_W J_W)^{-1} \pmod N$	Requires a unique good child; in a failed world there is none. The useful residue content is exactly G-MULTISOURCE-LOCK. Retained as motivation for the secondary frontier.
C-smooth budget calibration on the six EAC	Empirical ratios 0.62–0.79	VERIFIED COMPUTATION only; the six EAC are terminal-SCC objects, *not* canonical failed worlds. Not usable as evidence for the canonical branch.

Rejected or already superseded.

Old item	Disposition
"Terminal bad SCC ⇒ Goldbach contradiction" programme	Not revived. A terminal bad SCC is not by itself a Goldbach counterexample (Scope note 3.27). No lemma of that chain is used.
Exponent–determinant collapses	Domain is the two- or three-element terminal core with fixed Ω; the primitive first front has neither. SUPERSEDED by the adaptive low-core theory.
Fringe alphabet $\{\lambda, 2\lambda - 1\}$	FAIL-FRINGE-ALPHABET-TRANSFER-001. Theorem 9.17 is *not* that claim: it derives the alphabet as the full prime range $\mathcal{L}_N \subseteq [\lambda, 2\lambda)$, which is precisely the "what survives" clause of that failure entry.
"Prime-free boundary activates a reflected parent"	FAIL-REFLECTION-ACTIVATION-001 ($N = 242$, $p_0 = 127$) is respected throughout; Theorem 9.10 is its sharpened form.
Evertse S-unit counting as a closure device	Vacuous, as the Biblia itself noted; recorded as FAIL-SUNIT-COUNT-BOUNDS-THE-LOWCORE-001.

15.5 Literature verdicts and named missing lemmas

Iteration-157 verdict. Checked: the Euclidean algorithm for positive 2×2 matrices, the free monoid of positive SL_2, Conway–Coxeter friezes [86, 87], unit equations [88], graph toric ideals [81, 82], higher Lawrence configurations [89], cyclic congruence systems [90], and prime-escape theorems from subset products [91]. *None* supplies a dynamical transition preserving simultaneously N, the row identities, the full valuations and the self-label roles. The nearest missing theorem is named:

Matched Euclidean Gluing Lemma (OPEN). To the word or matrix of one packet one must assign an *actual* continuation to the next packet in the same system, preserving all roles, in such a way that a Euclidean measure strictly decreases or an escape is forced.

Without that lemma, word length, continued-fraction depth and matrix entry sums are only *static* coordinates. **They must not be used as Lyapunov functions for self-label dynamics.** Compare Remark 7.45, where the same continued-fraction structure appears exactly and still yields no descent.

The one external input whose shape is known. Across every audit the same conclusion recurs: what is missing is *Type-II bilinear information* in the sense of the asymptotic sieve for primes [38, 39]. Every internal construction that appeared to supply it turned out on inspection to be a Buchstab/sieve decomposition or an affine reparametrisation (§12.3.1). A future model should treat "does this really produce bilinear information, or is it an affine relabelling?" as the first question to ask of any new proposal.

Chapter 16

Historical and Superseded Results

Kept because the *history* matters: each item below was once load-bearing, and knowing why it stopped being load-bearing is what prevents its reconstruction under a new name. Nothing in this chapter may be used as a lemma. Where a superseded statement contains a correct sub-lemma, the sub-lemma appears in Chapter 4 under its own ID; the programme does not come back with it.

Item	Status	History and what replaced it
CRD1 as a proof engine	SUPERSEDED	Replaced by the fixed-point reformulation (Theorem 3.29), which is exact but is a computation, not a mechanism. What died: the "same container" argument, generic smooth counting, residue saturation, and the exact self-consistency of the fixed point.
`TD-DENSITY`	SUPERSEDED	The density formulation of TD required a bound of the same strength as its own conclusion. Replaced by the root-anchored form (§9.3).
p_0 as information selector	SUPERSEDED	Not refuted, but not established: every clause of the bundle-lift construction survives the ablation $p_0 \to p_1$ on rows with composite c'_j. The corpus verdict is "quantitatively useful, structurally inessential". `FAIL-P0-INFORMATION-SELECTOR-001`
Euler defect in normalisations (N1), (N2)	KILLED	Refuted here computationally (§5.7). Any earlier argument using $\eta + \nu \geq \lvert S\rvert + \lvert \mathrm{Src}(S)\rvert$ in those normalisations must be re-derived. The surviving forms are (N3) (VERIFIED COMPUTATION) and (N4) (trivial).
Prefix flux $B_k \equiv (-1)^k p_0$	AUDITED IDENTITY	Recorded in [BIB] (iterations 20, 22) with the flag "not auditable from packet". It has never been used downstream and is kept only so that it is not rediscovered and trusted.
Rooted container identity	AUDITED IDENTITY	[V481]. Closed the size-based route — i.e. its role was to *end* a line of attack, not to open one.
Four-channel ledger on all of Q_0	KILLED	Replaced by IL1–IL6 on Q_{smooth} after $N = 20$ refuted the wider domain. [ILA]
Unconditional CAB	KILLED	$N = 322$ and $N = 858$. No repaired version exists; the capacity route now goes through K1, which is open.

Item	Status	History and what replaced it
Upper-half source compression	PROVED (class D)	Correct, but its hypothesis is $\mathrm{Good}_N = \varnothing$. It was repeatedly invoked under canonical failure, which is invalid. `FAIL-UPPERHALF-SCOPE-001`
Symmetric fracture, forbidden offset band	PROVED (class D)	Same situation: correct under the minimal-counterexample hypothesis, unusable otherwise.
$h = 1$ local closure	SCOPE-CORRECTED	The blanket complete-BFS reading of finite bad-tree universality was withdrawn. The exact $h = 1$ full-BFS problem is live and is conjugate to root 2; selected transcript programming survives only with uncontrolled cofactors. [2026-09-05–07C]
CMFT, CGST, CSD, CAPT, RSD, rough-P_2	KILLED	§12.2. Their individual failure reasons remain binding (cofactor leakage, missing complete-support consumer, functional mismatch or parity/prime-correlation barriers); they must not now be justified by the old over-broad complete-BFS finite-local wall.
$q < m_0/3$	SCOPE-CORRECTED	Corrected to $q \leq m_0/3$; $N = 32$ attains equality. The consequence $q < N/6$ is unaffected.
$\rho_0 = \#\{D_s < 2h\}$	SCOPE-CORRECTED	Corrected here by adding the lower condition $D_s > h$ (Warning 6.18).
Genealogy stated for "every prime $r \geq x$"	SCOPE-CORRECTED	Corrected here to the non-diagonal domain; smallest witness $N = 14$ (Scope note 3.17).
High-gap amplification $h^2 > 4m_0$	SCOPE-CORRECTED	Hypothesis is Exact-D pair failure, not CRD failure (Warning 7.5).
`C0b-RD` v1 (without $Q_{\leq} \neq \varnothing$)	PROVED WITH REPAIR	The missing hypothesis made the inequality read $1 < 1$ in 47 551 of 72 446 canonical instances with $h \geq 3$ and $N < 2 \cdot 10^5$. Repaired version is Theorem 8.6 (Scope note 8.7).

Chapter 17

Handoff to the Next Model

If you read only one chapter, read this one — then Chapter 2, then Chapter 9.

17.1 What not to try again

1. **Do not confuse selected finite programming with complete BFS.** Compatible selected divisibility/valuation transcripts can be realised in canonical $h = 1$ worlds (Theorems 8.1, 8.2), but this does not control unprescribed cofactors. Therefore a selected BAD path/cycle is not a complete BAD neighbourhood. The old blanket finite-local wall is scope-corrected, not a prohibition on every bounded-depth complete-frontier theorem.
2. **Do not restart the programme in new coordinates.** Five successor theories have already done exactly that (§12.2). A new coordinate system does not change what information is available.
3. **Do not prove a bound in one functional and use it in another.** This is the single defect blocking TD, $\mathrm{P1}^{\mathrm{br}}$ and K1 simultaneously. "Morally the same" and "dominates on average" are not the same functional.
4. **Do not treat canonical failure as Goldbach failure.** Class-D theorems (upper-half compression, symmetric fracture, forbidden offset band) require $\mathrm{Good}_N = \varnothing$ and are unusable under canonical failure alone. `FAIL-UPPERHALF-SCOPE-001`
5. **Do not quote a count without its domain.** Which roots, which complements, unit rows included or not. `FAIL-CENSUS-DOMAIN-001`, `FAIL-UNIT-COMPLEMENT-001`
6. **Do not promote a scan.** 0 failures to 10^7 is not evidence for X1; it is the statement that X1 has no small counterexample.
7. **Do not use the mass of the boundary holes.** $O_k(\log N)$ against a block mass of order $t_k \log(x/h_k)$. Any argument routed through hole mass is dead before it starts. `FAIL-P0-ZEROHOLE-ESSENTIAL-001`
8. **Do not strengthen the reflection gate to activation.** $N = 242$ is final: both lower mirrors are composite and no active reflected row exists. The phase diagram explains why (Type I). `FAIL-REFLECTION-ACTIVATION-001`
9. **Do not widen the reflected block.** The endpoint pair has difference exactly $2h$ and is not covered by strict non-reuse. `FAIL-BROADBLOCK-ENDPOINT`
10. **Do not use the Euler defect in normalisation (N1) or (N2).** Refuted here: 6 044 and 3 804 violations respectively for $N \le 6 \cdot 10^4$ (§5.7).
11. **Do not attack X1 or Goldbach directly.** X1 is strictly stronger than Goldbach for each N.
12. **Do not route around parity by enlarging the graph.** The obstruction lives in the sieve information, not in the graph (§12.4).
13. **Do not re-prove what is already PROVED.** Use it by ID from Chapter 4.
14. **Do not raise a status.** If you prove something, add a row; do not edit an old one.

17.2 What currently looks best

1. `C-PO-PRIMITIVE-LOWCORE-SMOOTHROW-POPULATION-CAPACITY` — the actual current frontier (§9.2.4). It is the *only* archimedean route the trichotomy leaves open, the high band is completely described, and a bound on $|C_\lambda|$ converts every existing product capacity into a finite quantity. It is also honestly hard: an S-smooth counting problem in a short interval with sparse, non-initial support, where $\Psi(x, y)$ technology does not apply and Evertse's bound is vacuous.
2. **`E1` (escape-or-reuse).** The strongest data in the project (61 452 instances and an independent 60 413, zero orphans), a fully explicit statement, and a *named* missing ingredient: a bounded-charging lemma. If any frontier target falls, this is the one.
3. **The active-mass Euler bound (N3).** Small, elementary in appearance, zero counterexamples in 21 093 worlds, and it repairs the part of the corpus that used the false normalisations. High value per unit effort.
4. **`TS` (the smooth branch).** The largest quantitative margin of the four branch targets, and its success criterion — $G_{\mathrm{esc}}(N) > 0$ in a specified regime — would be a genuine theorem about Goldbach in that regime.
5. **Refuting `L1`.** A negative result that would cleanly remove a whole family of capacity arguments. The measured data already lean this way.
6. **The boundary filtration as pure combinatorics (Q7).** Independent of Goldbach, unexplored, and the only part of the project that is not blocked by any known wall.

Warning 17.1 (What does *not* look good, despite appearances). The genealogical layer is the most beautiful part of the project and the most likely to mislead. It is exact, it has genuine converses, and it does not imply anything dynamical — and Warning 6.7(2) raises the real possibility that it cannot, because "$\kappa_{\mathrm{gen}}(r) = 0$" may carry exactly the information "r is the first upper prime" and nothing more. Before building on it, decide whether your argument would survive that equivalence.

17.3 The most important dependency theorems

If any of the following were found to be wrong, large parts of the file would fall. They are listed in order of how much depends on them.

ID	Name	What depends on it
`C-GAP-EQ`	Canonical gap equivalence (Thm. 3.10)	Everything canonical. It is the definition of the canonical input and the only class-A hypothesis anywhere.
`ARS-CLOSED-WORLD`	Closure and unique source (Lem. 3.26)	The whole failure analysis, the fixed point, the Euler layer, every capacity statement.
`ARS-NORETURN`	No-bad-return-to-p_0 (Thm. 5.33)	Uniqueness of the source, hence `ARS-CLOSED-WORLD` in the canonical case, hence the two-sheet budget.
`CRD-FIXPOINT`	Fixed-point characterisation (Thm. 3.29)	The entire computational programme; the six special N; the tractability of the census.
`COb-RD`	Reflection-defect ledger (Thm. 8.6)	All four branch targets TI, TD, TS, TF, hence P1$^{\mathrm{br}}$, hence everything downstream. *Note its repaired hypothesis* (Scope note 8.7).
`V71.21`	Genealogical rank (Thm. 6.2)	Ledger completeness, uniqueness of the reflection-complete root, the lossless code, the ablation's interpretation. *Note its domain* (Scope note 3.17).
`C-STRICT-NONREUSE`	Strict non-reuse (Lem. 3.13)	TI's injectivity for large primes, W1's collision-freeness, the $\mathcal{F}$ branch of `COb-RD`.

ID	Name	What depends on it
`S-SHADOW-ID`	Shadow identity (Thm. 5.22)	Retention, D_q, M_q, cycle localisation, the prime-power witness — the whole multiplicative layer.
`V-TUJ`	$T_S = U_S J_S$ (Thm. 3.19)	Every ledger and capacity statement. Note that the *Euler defect* built on it does not inherit its reliability (§5.7).
`G-MULTISOURCE-LOCK`	Multi-source product lock (Thm. 5.7)	`G-ROOT-LOCK`, the selected residue lock (V81.13), and route (B) of the surviving-route trichotomy.
`V81-CONTAINER`	Rooted container identity (Thm. 9.7)	The whole current frontier: the slack bound, the death of the size route, and the trichotomy that names the only remaining routes.
`V81-C5-ALPHABET`	Adaptive fringe cofactor alphabet (Thm. 9.17)	The preferred target: it is what makes the high band completely known, so that the low core is the entire unknown.
`FACP` & scope-corrected `FBTU`	Partial-transcript constraint	FACP and the repaired transcript theorem constrain selected finite-pattern arguments only; they do not rule out complete-frontier theorems. See §10.2.

17.4 The five most valuable next moves

1. **Attack the exact $h = 1$ root-2 closed-support question** (§10.8). Either exhibit a complete failed support or prove that one cannot contain $\mathrm{PF}(P-2)$ when P is prime.
2. **Resolve LOW-GAP-THREE-LABEL-EXACT-OBSTRUCTION-SPLIT.** Kill or construct the three-prime closed world / prime-power system. This is compact, pointwise and hostile-audit friendly.
3. **Attack the radius-three large–large system.** The small label- product and cofactor-product regions are already excluded asymptotically; the remaining simultaneous large-product region is explicit.
4. **Find a complete-support moving-envelope consumer.** The fixed-S and Matveev results force support growth but do not iterate. A theorem coupling successive complete rows to the growing envelope would be a genuine global advance.
5. **Search for a non-hereditary p_0 invariant.** The canonical valuation embedding kills hereditary first-support graph/valuation separators. Any viable selector mechanism must use complete support, external leakage, absolute location or multigeneration reachability; test only invariants that really retain one of those data.

Warning 17.2 (Two moves demoted by the v4.81 result). Anything of the shape "find new selected storage disjoint from $\mathscr{C}_Q$ at fixed positive logarithmic scale" is now known to be *insufficient by construction*, not merely unproved (Corollary 9.9); and anything of the shape "exploit an overlap between the reflected block and the selected family" is known to be charge-free (Theorem 9.10). Do not spend a run on either.

Final orientation. Binary Goldbach is OPEN. Universal canonical success **(H-p_0)** is OPEN and is strictly stronger. The old finite-selected-transcript p_0 architecture is SUPERSEDED, but the complete-support/multigeneration reachability programme is live. The project owns an exact structural theory, a lossless record of what fails, and several sharply stated current questions. Any future proof attempt must respect the 2026-09-07C scope corrections and the moving-support frontier; the 246 legacy-plus-post-v4.81 failure/scope-correction entries exist precisely to prevent regression.

Appendix A

Reproduction Code

A.1 The data generator

The following script reproduces *every* number marked VERIFIED COMPUTATION in this file. It has no external dependencies (standard library, Python 3.9+). Blocks EXP1–EXP8 correspond to the identifiers in the provenance table of §11.1. Comments are the project's own, transliterated to ASCII; the executable code is unchanged.

Listing A.1: `p0_experiments.py` — complete self-contained generator for every VERIFIED COMPUTATION number in this file.

```python
#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""
p0_experiments.py -- JEDEN samowystarczalny skrypt odtwarzajacy KAZDA liczbe
oznaczona COMPUTATIONAL w dossier p_0 oraz dane wszystkich rysunkow.

Bez zaleznosci zewnetrznych (tylko biblioteka standardowa Pythona 3.9+).

    python3 p0_experiments.py --scan 10000000 --census 20000 --phase 200000 \
                           --fixedpoint 60000 --json dane.json

Domyslne, szybkie ustawienia (kilka minut):
    python3 p0_experiments.py

Kazdy blok EXP wypisuje dokladny zakres, ktory jest czescia twierdzenia
obliczeniowego -- zgodnie z regula "nigdy nie cytuj liczby bez jej domeny".
"""
from __future__ import annotations

import argparse
import json
from array import array
from collections import Counter, deque


# ===================================================================== sito
def build(limit):
    spf = array('i', range(limit + 1))
    i = 2
    while i * i <= limit:
        if spf[i] == i:
            for j in range(i * i, limit + 1, i):
                if spf[j] == j:
                    spf[j] = i
        i += 1
    isp = bytearray(limit + 1)
    for n in range(2, limit + 1):
        isp[n] = 1 if spf[n] == n else 0
    return spf, isp


def pf(n, spf):
```

```
    out = []
    while n > 1:
        q = spf[n]
        out.append(q)
        while n % q == 0:
            n //= q
    return out


def bigomega(n, spf):
    c = 0
    while n > 1:
        q = spf[n]
        while n % q == 0:
            n //= q
            c += 1
    return c


class Env:
    """Sito + selektory kanoniczne + dwa tryby przejscia."""

    def __init__(self, limit):
        self.L = limit
        self.spf, self.isp = build(limit)

    def nextp(self, n):
        m = n + 1
        while m <= self.L and not self.isp[m]:
            m += 1
        return m

    def prevp(self, n):
        m = n - 1
        while m >= 2 and not self.isp[m]:
            m -= 1
        return m if m >= 2 else -1

    def p0(self, N):
        x = N // 2
        return x if self.isp[x] else self.nextp(x)

    def census(self, N, r):
        """tryb census: (sukces, glebokosc, liczba wierzcholkow)"""
        if 2 * r == N:
            return True, 0, 1
        q = deque([(r, 0)])
        seen = {r}
        nodes = 0
        while q:
            p, d = q.popleft()
            nodes += 1
            n = N - p
            if n >= 2 and self.isp[n]:
                return True, d, nodes
            while n > 1:
                f = self.spf[n]
                while n % f == 0:
                    n //= f
                if N % f and f not in seen:
                    seen.add(f)
                    q.append((f, d + 1))
        return False, -1, nodes

    def world(self, N, r):
        """tryb world: (sukces, glebokosc, |R|, liczba krawedzi GOOD)"""
        if 2 * r == N or (N - r >= 2 and self.isp[N - r]):
            return True, 0, 0, 0
        q = deque([(r, 0)])
        queued = {r}
        R, esc, ok, dep = set(), 0, False, None
        while q:
            p, d = q.popleft()
            if p in R:
                continue
```

```
            n = N - p
            if n >= 2 and self.isp[n]:
                if not ok:
                    ok, dep = True, d
                continue
            R.add(p)
            while n > 1:
                f = self.spf[n]
                while n % f == 0:
                    n //= f
                if N % f:
                    if N - f >= 2 and self.isp[N - f]:
                        esc += 1
                        if not ok:
                            ok, dep = True, d + 1
                    elif f not in R and f not in queued:
                        queued.add(f)
                        q.append((f, d + 1))
        return ok, dep, len(R), esc


# ============================================= EXP1: wyczerpujacy skan p_0
def exp1_scan(limit):
    """Kazde parzyste 4<=N<=limit, korzen kanoniczny, tryb census."""
    e = Env(limit)
    fails, tot, s = [], 0, 0
    maxn, maxd, hist = (0, 0), (0, 0), Counter()
    for N in range(4, limit + 1, 2):
        r = e.p0(N)
        ok, d, nodes = e.census(N, r)
        tot += 1
        s += nodes
        if not ok:
            fails.append((N, r))
        else:
            hist[d] += 1
            if d > maxd[0]:
                maxd = (d, N)
        if nodes > maxn[0]:
            maxn = (nodes, N)
    return {"limit": limit, "cases": tot, "failures": len(fails),
            "failure_list": fails[:20], "mean_nodes": s / tot,
            "max_nodes": maxn, "max_depth": maxd,
            "depth_hist": {str(k): hist[k] for k in sorted(hist)}}


# ================================== EXP2: ablacja p_0 -> p_k (werdykt roota)
def exp2_ablation(limit, kmax=6):
    e = Env(limit)
    comp = [0] * (kmax + 1)
    unit = [0] * (kmax + 1)
    first = {}
    for N in range(4, limit + 1, 2):
        p = e.p0(N)
        for k in range(kmax + 1):
            if p >= N:
                break
            ok, _, _ = e.census(N, p)
            if not ok:
                if N - p == 1:
                    unit[k] += 1
                else:
                    comp[k] += 1
                    first.setdefault(k, (N, p))
            p = e.nextp(p)
    return {"limit": limit, "kmax": kmax, "composite_complement_failures": comp,
            "unit_row_failures": unit,
            "first": {str(k): v for k, v in sorted(first.items())}}


# ================= EXP3: census WSZYSTKICH zawodzacych gornych korzeni
def exp3_upper_census(limit):
    e = Env(limit)
    rows, unit = [], 0
    for N in range(4, limit + 1, 2):
```

```
        r0 = e.p0(N)
        p = r0
        while p < N:
            if e.isp[p] and N % p:
                if N - p == 1:
                    unit += 1
                elif not e.census(N, p)[0]:
                    rows.append((N, p, p == r0))
            p += 1
            while p < N and not e.isp[p]:
                p += 1
    Ns = sorted({r[0] for r in rows})
    return {"limit": limit, "failed_roots": len(rows), "unit_rows": unit,
            "N_values": Ns, "per_N": {str(n): sum(1 for r in rows if r[0] == n)
                                      for n in Ns},
            "canonical_among_them": sum(1 for r in rows if r[2])}


# ================================== EXP4: margines ucieczki G_esc
def exp4_margin(limit):
    e = Env(limit)
    ones, sizes = [], []
    hist = Counter()
    maxW = (0, 0)
    dyad = {}
    for N in range(4, limit + 1, 2):
        r = e.p0(N)
        ok, _, W, esc = e.world(N, r)
        if W == 0:
            continue
        hist[esc] += 1
        sizes.append(W)
        if W > maxW[0]:
            maxW = (W, N)
        if esc == 1:
            ones.append((N, r, W))
        if W >= 3:
            dyad.setdefault(N.bit_length(), []).append(esc)
    return {"limit": limit, "bad_p0_worlds": len(sizes),
            "G_esc_hist_small": {str(k): hist[k] for k in sorted(hist) if k <= 6},
            "margin_one_all": [o[0] for o in ones],
            "margin_one_W_ge_3": [(N, r, W) for N, r, W in ones if W >= 3],
            "max_world": maxW,
            "dyadic_min_G": {str(k): min(v) for k, v in sorted(dyad.items())}}


# ===================== EXP5: trichotomia fazowa i dwukierunkowa ranga rho_0
def exp5_phase(limit):
    e = Env(limit)
    primes = [n for n in range(2, limit + 1) if e.isp[n]]
    idx = {p: i for i, p in enumerate(primes)}
    types = Counter()
    imm = Counter()
    viol = Counter()
    rho_hist = Counter()
    rho_theta = {}
    theta_hist = Counter()
    for N in range(6, limit + 1, 2):
        x = N // 2
        P = e.p0(N)
        h, m0 = P - x, N - P
        Pm = e.prevp(P)
        G = P - Pm
        if m0 - Pm != G - 2 * h:
            viol["identity"] += 1
        t = "diag" if h == 0 else ("I" if 2 * h < G else
                                   ("II" if 2 * h == G else "III"))
        types[t] += 1
        theta_hist[int(20 * h / G)] += 1
        if e.isp[m0] or m0 == 1:
            imm[t] += 1
            if t == "I":
                viol["typeI_prime_m0"] += 1
        if t == "II" and m0 != Pm:
            viol["typeII"] += 1
```

```
        if t == "III":
            js, tt = h - G // 2, (h - 1) // 2
            if not (1 <= js <= tt and m0 + 2 * js == Pm):
                viol["typeIII_mirror"] += 1
            if N % Pm == 0:
                viol["typeIII_inactive"] += 1
        if h >= 3:
            tt = (h - 1) // 2
            direct = sum(1 for j in range(1, tt + 1) if e.isp[m0 + 2 * j])
            i0, f, r = idx[P], 0, 1
            while i0 - r >= 0:
                D = P - primes[i0 - r]
                if D >= 2 * h:
                    break
                if D > h:
                    f += 1
                r += 1
            if direct != f:
                viol["rho0"] += 1
            rho_hist[direct] += 1
            rho_theta.setdefault(round(h / G, 2), []).append(direct)
    return {"limit": limit, "type_counts": dict(types),
            "immediate_success_by_type": dict(imm),
            "violations": dict(viol),
            "theta_hist_20": {str(k): theta_hist[k] for k in sorted(theta_hist)},
            "rho0_hist": {str(k): rho_hist[k] for k in sorted(rho_hist)},
            "rho0_mean_by_theta": {str(k): round(sum(v) / len(v), 4)
                                   for k, v in sorted(rho_theta.items())
                                   if len(v) >= 50}}


# ===================================== EXP6: filtracja brzegu (embedding)
def exp6_filtration(limit, kmax=6):
    e = Env(limit)
    checked = viol = 0
    for N in range(50, limit + 1, 2):
        x = N // 2
        roots, p = [], e.p0(N)
        for _ in range(kmax + 1):
            if p >= N:
                break
            roots.append(p)
            p = e.nextp(p)
        for k in range(len(roots) - 1):
            pk, pk1 = roots[k], roots[k + 1]
            s = (pk1 - pk) // 2
            tk, tk1 = (pk - x - 1) // 2, (pk1 - x - 1) // 2
            if tk1 != tk + s:
                viol += 1
            for i in range(1, tk + 1):
                checked += 1
                if pk1 - 2 * (i + s) != pk - 2 * i:
                    viol += 1
    return {"limit": limit, "checked": checked, "violations": viol}


# ===================================== EXP7: fixed point B_N^low
def exp7_fixedpoint(limit):
    e = Env(limit)
    out = []
    for N in range(6, limit + 1, 2):
        cap = N // 3
        B, rows = set(), {}
        for p in range(3, cap + 1, 2):
            if not e.isp[p] or N % p == 0:
                continue
            v = N - p
            if v < 2 or e.isp[v]:
                continue
            B.add(p)
            rows[p] = pf(v, e.spf)
        changed = True
        while changed:
            changed = False
            for p in list(B):
```

```
            for q in rows[p]:
                if N % q and q not in B:
                    B.discard(p)
                    changed = True
                    break
        if B:
            out.append((N, sorted(B)))
    return {"limit": limit, "nonempty_count": len(out),
            "nonempty": [(N, B) for N, B in out]}


# ============================ EXP8: ablacja obserwabli p_0 vs p_1
def exp8_observables(limit):
    e = Env(limit)
    acc = {k: {n: [] for n in ("depth", "world", "esc", "Omega", "ret", "kap")}
           | {"fail": 0, "n": 0, "pp": []} for k in (0, 1)}
    for N in range(50, limit + 1, 2):
        x = N // 2
        r0 = e.p0(N)
        r1 = e.nextp(r0)
        for k, r in ((0, r0), (1, r1)):
            if r >= N or N - r <= 1:
                continue
            m = N - r
            ok, dep, W, esc = e.world(N, r)
            a = acc[k]
            a["n"] += 1
            if not ok:
                a["fail"] += 1
            if dep is not None:
                a["depth"].append(dep)
            a["world"].append(W)
            a["esc"].append(esc)
            a["Omega"].append(bigomega(m, e.spf))
            Q = pf(m, e.spf)
            Qs = set(Q)
            a["pp"].append(max(Q) / m)
            a["ret"].append(sum(1 for q in Q for z in pf(N - q, e.spf) if z in Qs))
            hh = r - x
            tt = (hh - 1) // 2
            a["kap"].append(sum(1 for j in range(1, tt + 1) if e.isp[r - 2 * j]))

    def st(v):
        v = sorted(v)
        return {"mean": round(sum(v) / len(v), 4), "median": v[len(v) // 2],
                "max": v[-1]}
    return {"limit": limit,
            **{f"p{k}": {n: st(acc[k][n]) for n in
                          ("depth", "world", "esc", "Omega", "ret", "kap")}
               | {"fail": acc[k]["fail"], "n": acc[k]["n"],
                  "Pplus_over_m_mean": round(sum(acc[k]["pp"]) / len(acc[k]["pp"]), 5),
                  "Omega_hist": {str(a): b for a, b in
                                 sorted(Counter(acc[k]["Omega"]).items())},
                  "depth_hist": {str(a): b for a, b in
                                 sorted(Counter(acc[k]["depth"]).items())}}
               for k in (0, 1)}}


# ==================================================== pojedynczy swiadkowie
def witnesses(limit=1_100_000):
    """Wszyscy pojedynczy swiadkowie cytowani w dossier."""
    e = Env(limit)

    def fac(n):
        c = Counter(pf(n, e.spf))
        d = Counter()
        m = n
        for q in c:
            while m % q == 0:
                m //= q
                d[q] += 1
        return dict(sorted(d.items()))

    w = {}
    for N in (124, 242, 240, 2200, 103344, 364844, 14198, 11822):
```

```
        P = e.p0(N)
        Pm = e.prevp(P)
        h, G, m0 = P - N // 2, P - Pm, N - P
        ok, dep, W, esc = e.world(N, P)
        t = (h - 1) // 2
        w[str(N)] = {"p0": P, "h": h, "G": G, "prev_prime": Pm, "m0": m0,
                     "m0_fac": {str(a): b for a, b in fac(m0).items()},
                     "type": "diag" if h == 0 else
                             ("I" if 2 * h < G else
                              ("II" if 2 * h == G else "III")),
                     "kappa_gen": sum(1 for j in range(1, t + 1)
                                      if e.isp[P - 2 * j]),
                     "rho0": sum(1 for j in range(1, t + 1) if e.isp[m0 + 2 * j]),
                     "verdict": "SUCCESS" if ok else "FAILURE",
                     "depth": dep, "world": W, "escapes": esc}
    w["2200_core"] = {"2200-3": 2200 - 3, "13^3": 13 ** 3,
                      "2200-13": 2200 - 13, "3^7": 3 ** 7,
                      "shift": 3 ** 7 - 3 == 13 ** 3 - 13 == 2184}
    N = 1_000_000
    Q, R = (61, 67, 73), (113, 179, 599)
    w["K33"] = {"N": N, "Q": Q, "R": R,
                "all_entries_prime": all(e.isp[N - q * r] for q in Q for r in R),
                "all_factor_vertices_bad": all(not e.isp[N - v] for v in Q + R)}
    w["cells"] = [{"P": P, "G": P - e.prevp(P),
                   "all_h_have_p0_eq_P": all(e.p0(2 * (P - h)) == P
                                             for h in range(0, P - e.prevp(P)))}
                  for P in (67, 127, 1009, 104729)]
    return w


# ================================================================ driver
def main():
    ap = argparse.ArgumentParser()
    ap.add_argument("--scan", type=int, default=1_000_000)
    ap.add_argument("--ablation", type=int, default=200_000)
    ap.add_argument("--census", type=int, default=20_000)
    ap.add_argument("--margin", type=int, default=200_000)
    ap.add_argument("--phase", type=int, default=200_000)
    ap.add_argument("--filtration", type=int, default=20_000)
    ap.add_argument("--fixedpoint", type=int, default=20_000)
    ap.add_argument("--observables", type=int, default=200_000)
    ap.add_argument("--json", default="dane_p0.json")
    a = ap.parse_args()
    out = {
        "EXP1_scan": exp1_scan(a.scan),
        "EXP2_ablation": exp2_ablation(a.ablation),
        "EXP3_upper_census": exp3_upper_census(a.census),
        "EXP4_margin": exp4_margin(a.margin),
        "EXP5_phase": exp5_phase(a.phase),
        "EXP6_filtration": exp6_filtration(a.filtration),
        "EXP7_fixedpoint": exp7_fixedpoint(a.fixedpoint),
        "EXP8_observables": exp8_observables(a.observables),
        "witnesses": witnesses(),
    }
    with open(a.json, "w") as f:
        json.dump(out, f, indent=1)
    print(json.dumps({k: (v if k == "witnesses" else
                          {kk: vv for kk, vv in v.items()
                           if not isinstance(vv, (list, dict)) or len(str(vv)) < 400})
                      for k, v in out.items()}, indent=1)[:6000])


if __name__ == "__main__":
    main()
```

A.2 The Euler-defect check

This script produced the refutations of normalisations (N1) and (N2) and the zero-violation results for (N3) and (N4) in §5.7. It is new in this file.

Listing A.2: `euler_check.py` — tests the four candidate normalisations of the layered Euler defect on all canonical failed worlds.

```
#!/usr/bin/env python3
"""Test the four candidate normalisations of the layered Euler defect
on every canonical failed world with 50 <= N <= LIMIT."""
from collections import deque

LIM = 200001
spf = list(range(LIM))
i = 2
while i * i < LIM:
    if spf[i] == i:
        for j in range(i * i, LIM, i):
            if spf[j] == j:
                spf[j] = i
    i += 1


def fac(n):
    f = {}
    while n > 1:
        p = spf[n]
        e = 0
        while n % p == 0:
            n //= p
            e += 1
        f[p] = e
    return f


def isp(n):
    return n >= 2 and spf[n] == n


def world(N):
    """Canonical stopped traversal, world mode.
    Returns (R, rows, depth, success, root) or None when the root wins
    immediately (diagonal case or N - p0 prime)."""
    x = N // 2
    r = x if isp(x) else None
    if r is None:
        p = x + 1
        while not isp(p):
            p += 1
        r = p
    if 2 * r == N:
        return None
    if isp(N - r):
        return None
    q = deque([(r, 0)])
    seen = {r}
    R, rows, depth = set(), {}, {}
    ok = False
    while q:
        p, d = q.popleft()
        if p in R:
            continue
        if isp(N - p):
            ok = True
            continue
        R.add(p)
        rows[p] = fac(N - p)
        depth[p] = d
        for c in sorted(rows[p]):
            if N % c == 0:
                continue
            if isp(N - c):
                ok = True
                continue
            if c not in R and c not in seen:
                seen.add(c)
                q.append((c, d + 1))
    return (R, rows, depth, ok, r)


```

```python
bad = [0, 0, 0, 0]
first = [None, None, None, None]
tested = 0
for N in range(50, 60001, 2):
    w = world(N)
    if w is None:
        continue
    R, rows, depth, ok, r = w
    if not R:
        continue
    tested += 1
    nu = len(set(depth.values()))
    # (N1) row-excess form
    eta1 = sum(sum(rows[p].values()) - len([c for c in rows[p] if c in R])
               for p in R)
    # (N2) valuation / reuse split
    d = {}
    kap = {}
    for p in R:
        for q_, e_ in rows[p].items():
            if N % q_ == 0:
                continue
            d[q_] = d.get(q_, 0) + e_
            kap[q_] = kap.get(q_, 0) + 1
    eta2 = sum(d[q_] - kap[q_] for q_ in d)
    nu2 = sum(kap[q_] - 1 for q_ in d)
    # (N3) active-mass form ; (N4) trivial form
    mass = sum(d.values())
    eta4 = sum(sum(rows[p].values()) - 1 for p in R)
    target = len(R) + 1
    for k, val in enumerate((eta1 + nu, eta2 + nu2, mass, eta4 + nu)):
        if val < target:
            bad[k] += 1
            if first[k] is None:
                first[k] = (N, len(R), val, target)
print("worlds tested:", tested)
for k, name in enumerate(("N1", "N2", "N3", "N4")):
    print(name, "violations:", bad[k], "first:", first[k])
```

A.3 Auxiliary scripts not embedded here

The project repository also contains the following, which are *not* embedded because no number in this file depends on them.

File	Role
`p0_algorithm.py`	Listing 3.1: the self-contained reference algorithm (embedded, §3.4).
`p0_experiments.py`	Listing A.1: the data generator (embedded above).
`p0_core.py`	Extended engine: both traversal modes, canonical geometry, reflected block, genealogical rank, D_q/M_q, FRESH/SMOOTH split, $B_{\mathrm{abs}}/B_{\mathrm{re}}$, CLI.
`p0_certificates.py`	The 34 hostile certificates of §11.4: regressions 11 822 and 3 542 948, counterexamples 242, 128, 322, 858, 4736, 6142, 14198, 20, 1718, 51 032, the Exact-D models, the $K_{3,3}$ grid and the 2-core.
`p0_bigscan.py`, `p0_phase.py`, `p0_figures.py`, `p0_upper_root_census.py`, `p0_escape_margin.py`	Supporting scripts for individual experiments; all of their outputs are reproducible from `p0_experiments.py`.

Appendix B

Exact Post-v4.81 Raw Checkpoint Transcript

This appendix is the lossless canonical payload of the September 5–7C continuation. It contains all 6420 source lines in original order. The standalone archive's 5108-line payload is an exact prefix of this transcript, so it is not duplicated separately. Status labels and later-overrides-earlier semantics are source data, not editorial promotion by this merge.

```
GOLDBACH RESEARCH CONTINUATION - 2026-09-05
Single raw checkpoint. Work in progress; no Goldbach proof asserted.
NOVELTY of new statements: NOT YET AUDITED unless explicitly stated otherwise.

INPUT PROVENANCE
Latest checkpoint: upload/astra_checkpoint.txt, mathematical sections 1-10.
Recent reports: upload/astra_run_1.md (singleton-excess construction),
upload/astra_run_2.md (PR-MQ repair and global counting attempt).
Navigation: upload/goldbach_master_for_future_ai(5).tex.
Historical sources are being searched selectively, not certified wholesale.
Later corrections override earlier PROVED labels.

RESULT-ID: AUDIT-PREFIX-TRANSFER-20260905
STATUS: PROVED (independent proof audit of the supplied checkpoint, not a new claim).
EXACT STATEMENT:
Let 3=P_0<P_1<... be the odd primes, X>=4 an integer, I the first index
with P_I>=X, K>=0 an integer, b_j=(P_j-1)/2, L=b_(I+K).
For x>=4 let i satisfy P_(i-1)<x<=P_i, p_k(2x)=P_(i+k),
m_k(2x)=2x-p_k(2x). For any E of positive odd integers <=X, B=|E|,
T=sum_(x=4..X) sum_(k=0..K) 1_E(m_k(2x)) satisfies
T<=6(K+1) B (1+log^+(L/B)) when B>0, and T=0 when B=0.
HYPOTHESES: No failure hypothesis; E may depend on X and K. Nonpositive
complements do not belong to E. The claim counts incidences, not just inputs.
PROOF:
Fix selected endpoint j. The disjoint source cells for ranks 0..K unite,
after restricting to x>=4, into the portion of
P_max(0,j-K-1)<x<=P_j that actually occurs. Dropping x<=X enlarges it.
In output coordinate t=(m-1)/2 the containing interval is
J_j=(2a_j-b_j,b_j] intersect Z, a_j=b_max(0,j-K-1).
Its core C_j=(a_j,b_j] has |J_j|=2|C_j|. Zero-length cores are absent.
Each integer belongs to at most K+1 cores, since the cores are unions of
at most K+1 consecutive disjoint elementary gaps (b_(i-1),b_i]. All are
inside {1,...,L}. With A={(m-1)/2:m in E}, use the uncentred integer-interval
maximal function M_A(t)=sup_(J containing t) |A intersect J|/|J|.
A longest-first disjoint selection of witnessing intervals for M_A>u,
0<u<=1, covers their union after tripling the retained intervals. Integer
intervals of cardinality n can be tripled to cardinality 3n-2; hence
|{M_A>u}|<=3B/u. Only finitely many witnessing intervals exist: their
length is <B/u and each intersects the finite A.
Layer cake gives, on any set U of cardinality at most L,
sum_U M_A <= integral_0^1 min(L,3B/u)du
<=3B(1+log^+(L/B)).
For every t in C_j, |A intersect J_j|<=2|C_j|M_A(t).
Average over C_j, sum j, and use core multiplicity K+1. This proves T.
HOSTILE CHECK:
The source cell is closed at its upper endpoint, so x prime is included.
Endpoint j=0 never occurs for x>=4. Initial cells are truncated using P_0=3.
Truncating the last cell and excluding nonpositive complements only decrease T.
Grouping by endpoint is essential: summing separate k bounds would lose
another K+1. No prime-gap distribution enters the finite inequality.
DEPENDENCIES: Elementary maximal-interval covering only.
IMPLICATION:
The supplied density-zero transfer, fixed-rank normal-order deduction, and
growing direct-prefix no-go follow, subject to their stated standard
number-theoretic inputs. The squarefree-density proof also needs finite
residue equidistribution; density-zero transfer alone is insufficient.
RELATION TO GOLDBACH: No positive representation count. No inference about
factorization-graph traversal from the direct-prefix failure estimate.
RELATION TO p0 EXCEPTIONALITY: Shared by all fixed nearby roots.
WHAT REMAINS OPEN: Nonlocal rooted reachability and every-point positivity.
```

```

AUDIT NOTE: The supplied Scott-Bozek branch is not used as a classification.
No conclusion about its completeness is promoted without rechecking primary
sources. The global/rooted failure distinction is maintained throughout.

FAIL-ID: AUDIT-FBTU-PARTIAL-VERSUS-COMPLETE-20260905
STATUS: SCOPE-CORRECTED; the blanket bounded-depth BFS no-go is NOT ESTABLISHED
by the supplied CRT argument. This is not a proof of a bounded-depth closer.
CLAIM / ROUTE:
Master theorem FBTU and its following finite-local wall (around lines
4320-4333), citing the post-CRD dossier, use programming of finite BAD
subtrees to rule out bounded-depth failure-dynamics arguments.
The dossier explicitly calls the object a BAD-subtree. It supplies no proof
that every unspecified prime factor of every expanded row is also BAD.
The p0 dossier marks the imported claims PENDING INDEPENDENT AUDIT.
EXACT FAILURE:
Prescribed divisibilities force selected edges, not an entire factor row.
A factor not named in a CRT transcript may be a GOOD child. A selected BAD
path or BAD cycle is an existential statement. Failure of the full BFS
through depth d is a universal statement about ALL reachable vertices at
depths <=d. CRT compatibility alone does not justify that quantifier change.
COUNTEREXAMPLE / OBSTRUCTION:
N=92, canonical P=47, h=1 (46 composite, 47 prime).
92-47=45=3^2*5; 92-3=89 is prime.
92-5=87=3*29; 92-29=63=3^2*7; 92-7=85=5*17.
Thus 47->5->29->7 is a selected all-BAD path and 5->29->7->5
is an all-BAD cycle, while the complete canonical traversal succeeds at
depth 1 via 3. Arbitrarily long BAD walks can be unrolled in this one world;
they do not give arbitrarily large shortest GOOD depth.
This example refutes the implication from a selected BAD transcript to a
full BAD traversal, not the existence of prescribed partial trees.
WHAT SURVIVES:
The finite congruence/valuation programming theorem with its actual
compatibility hypothesis. The historical FAIL-P0-FINITE-VALUATION-SIGNATURE
and reflection obstruction survive on those restricted domains.
DO NOT REPEAT:
Do not use FAIL-P0-FINITE-BAD-TRAVERSAL or the master finite-local wall to
claim that ALL complete finite BFS neighbourhoods can be reproduced by CRT.
Do not reverse this correction into a claimed p0 proof or into a claim that
arbitrarily deep complete BAD neighbourhoods do not exist.
POSSIBLE REPLACEMENT:
An exact theorem about complete neighbourhoods, controlling the residual
cofactors, or a correctly restricted theorem about selected edges only.

RESULT-ID: CRT-TRANSCRIPT-EXACT-SCOPE-20260905
STATUS: PROVED (precise repair of FACP, not a novelty claim).
EXACT STATEMENT / HYPOTHESES:
Fix finitely many integer affine forms L_j(P)=a_j P+b_j and finitely many
requirements v_l(L_j(P))=e or v_l(L_j(P))>=e. Assume all conditions admit
one residue class P=A mod M with gcd(A,M)=1; for an exact valuation include
l^(e+1) in M. Assume every form required to be composite has a specified
prime divisor and is positive and tends to infinity along the progression.
Then infinitely many primes P realize these conditions; every required
composite form is composite for all sufficiently large such P. These P
are canonical roots of N=2(P-1), h=1.
PROOF:
The constraints are unions of residue classes for finite prime-power
moduli. Choose the assumed common reduced class A mod M. Dirichlet's
theorem supplies infinitely many primes in it. Exact valuations are decided
mod l^(e+1), lower valuations mod l^e. A growing positive form divisible
by a fixed prime eventually exceeds that prime. Finally P-1 is an even
composite integer for P>=5, proving canonicity. No assertion about the
unprescribed cofactors occurs in the proof.
HOSTILE CHECK:
Conditions involving the same prime are NOT independently programmable.
For a fixed-label edge a->b in an h=1 world one needs
2P=a+2 mod b; if b also divides P-2, compatibility forces a=2 mod b.
If b divides a+2, the edge may instead force P=0 mod b, violating the
reduced-class hypothesis. Compatibility must be checked, not waved away.
DEPENDENCIES: CRT and Dirichlet's theorem on reduced progressions.
RELATION TO GOLDBACH: No global failure hypothesis or conclusion.
RELATION TO p0 EXCEPTIONALITY: Shows a precise class of partial local
constraints is compatible with p0; does not cover complete BFS closure.
WHAT REMAINS OPEN: Uniform control of all unprescribed child factors.

EXPERIMENT-ID: EXACT-WITNESS-N92
STATUS: VERIFIED COMPUTATION, with displayed arithmetic independently sufficient.
DOMAIN: one even N=92, active vertices 47,3,5,29,7,17; unit rows excluded.
METRIC: selected BAD path/cycle versus shortest path to a GOOD vertex.
METHOD: trial division through the integer square root, exact arithmetic.
RESULT: displayed rows above; additionally 92-17=75=3*5^2.
No exhaustive-range or asymptotic claim is attached to this witness.

EXTERNAL-INPUT: BS-1996
STATUS: PRIMARY STATEMENT VERIFIED.
F. Beukers and H. P. Schlickewei, The equation x+y=1 in finitely generated
groups, Acta Arithmetica 78 (1996), 189-199, Theorem 1.1.
Author-hosted full text: https://webspace.science.uu.nl/~beuke106/s-units.pdf
(The available author PDF has a January 8, 2007 typesetting date.)
Exact input: in the Q-closure of a finitely generated subgroup of
(C^*)^2 of rank r, x+y=1 has at most 2^(8r+8) solutions. In particular the
same bound holds in the subgroup itself. No height bound is asserted.
Only this input theorem is external; applications and rank bookkeeping follow.

RESULT-ID: COPRIME-SUPPORTED-AFFINE-COUNT-20260905
STATUS: PROVED.
EXACT STATEMENT:
Let S be a finite set of rational primes, s=|S|, and let A,B,C be fixed
nonzero rational numbers. The number of ordered pairs of positive coprime
S-smooth integers (u,v) satisfying A*u+B*v=C is at most 2^(9s+16).
Here 1 is allowed as an S-smooth integer. This bound is uniform in S,A,B,C.
PROOF:
Since gcd(u,v)=1, choose a partition S=S_1 disjoint-union S_2 such that
PF(u) is contained in S_1 and PF(v) in S_2. An unused prime can be assigned
to either side. There are at most 2^s partitions covering all possibilities.
For a fixed partition take the subgroup of (Q^*)^2 generated by
```

```
(l,1) for l in S_1, (1,l) for l in S_2, and the single vector (A/C,B/C).
Its rank is at most s+1, independently of all prime divisors of A,B,C.
The injective map (u,v)->(A*u/C,B*v/C) lands in this group and satisfies
x+y=1. BS-1996 gives at most 2^(8s+16) solutions in this partition.
Summing over 2^s partitions proves the assertion.
HOSTILE CHECK:
The coefficient vector adds ONE group generator, not one generator per
prime dividing the coefficients. Signs are permitted in Q^*; the group need
not consist of positive vectors. The Q-closure theorem bounds its subgroup.
Overcounting unused-prime assignments only enlarges the upper bound.
The coprimality hypothesis is essential to the s+1 rank accounting.
DEPENDENCIES: BS-1996. No Goldbach or primality-distribution assumption.
IMPLICATION: A usable uniform COUNT of coprime supported affine pairs,
with a common alphabet; it is not a bound on the size of a solution.
RELATION TO GOLDBACH: Auxiliary counting lemma; no positive prime-pair count.
RELATION TO p0 EXCEPTIONALITY: None by itself.
WHAT REMAINS OPEN: A consumer forcing more than this many pairs in a shared
small alphabet, or controlling moving labels more strongly.
NOVELTY: NOT YET AUDITED; an elementary application of an established theorem.

RESULT-ID: COMPLETE-H1-TWO-ROW-FINITENESS-20260905
STATUS: PROVED.
EXACT STATEMENT:
Fix a finite set S of odd primes, s=|S|. There are at most
s * 2^(9s+16) even integers N with P=N/2+1 prime, m=N-P=P-2>1,
PF(m) subset S, and PF(N-q) subset S for q=min PF(m).
No assertion of complete traversal failure is assumed; N-q need not be BAD.
PROOF:
For each such N the selected q belongs to S. Write v=N-q=2m+2-q.
Any common prime divisor of m and v divides q-2 and is at least q by
minimality of q. Since 0<q-2<q, this is impossible. Hence gcd(m,v)=1.
Apply COPRIME-SUPPORTED-AFFINE-COUNT with (u,v)=(m,N-q) and
A=2, B=-1, C=q-2. The equation 2m-v=q-2 has nonzero right side.
For each of the s choices of q there are at most 2^(9s+16) possible pairs.
Each m determines N=2m+2 uniquely. Summing proves the bound.
HOSTILE CHECK:
Uses the COMPLETE supports of m AND N-q. Specifying only selected divisors
of these rows does not satisfy the hypotheses. The theorem allows m prime,
so restricting it to BAD roots only reduces the set. N-q is positive for
all admitted N, because q<=m<N. q is odd and q-2 is never zero.
DEPENDENCIES: Elementary h=1 minimum-factor orthogonality (reproved above)
and COPRIME-SUPPORTED-AFFINE-COUNT.
IMPLICATION:
A fixed finite alphabet cannot carry complete root and next-generation rows
for infinitely many canonical h=1 inputs, even when all exponents vary.
This gives a quantitative obstruction to the strong/full-support reading of
FBTU, beyond the elementary observation that fixing all factors AND all
exponents fixes N. Partial CRT programming survives.
RELATION TO HISTORICAL FAILURES:
The corpus already warns that fixed-S finiteness is not a moving-S proof.
We do NOT use this as a Goldbach closer. Its consumer here is a correction
to an infinite full-transcript assertion with genuinely FIXED labels.
RELATION TO GOLDBACH: Does not rule out a single failed world.
RELATION TO p0 EXCEPTIONALITY: A real arithmetic restriction in the h=1
subfamily, but not a distinction between all p0 and all nearby roots.
WHAT REMAINS OPEN: Complete BAD neighbourhoods with moving prime labels;
existence of arbitrarily large shortest GOOD depth for canonical roots.
NOVELTY: NOT YET AUDITED.

RESULT-ID: CLOSED-ALPHABET-COUNT-20260905
STATUS: PROVED.
EXACT STATEMENT:
Fix a finite set S of odd primes, s=|S|. Let C_S be the set of even N for
which there is a nonempty set C subset S of active primes (p does not divide
N), all N-p composite, with PF(N-p) subset C for every p in C.
Then |C_S| <= binomial(s,2) * 2^(9s+16), over ALL positive N.
PROOF:
A singleton C={a} is impossible: closure would give N-a=a^e and hence a|N,
contrary to activity. Let a<b be the two smallest elements of C.
The two complete rows U=N-a and V=N-b are coprime. Indeed, a common prime
l belongs to C and divides U-V=b-a. It cannot be a because a does not
divide U by activity. Every other prime of C is at least b>b-a, also
impossible. Thus gcd(U,V)=1. Both rows are S-smooth and satisfy
U-V=b-a. For each of binomial(s,2) choices of a,b, apply
COPRIME-SUPPORTED-AFFINE-COUNT with A=1, B=-1, C=b-a.
Each U determines N=U+a. The union bound proves the claim.
HOSTILE CHECK:
No SCC minimality is needed; full factor closure and activity are needed.
The set C and all its valuations may vary with N. They all lie inside ONE
fixed S. Dropping composite-row or prime-root conditions in the counting
lemma is harmless because the count becomes larger. Singleton inactivity
is why trivial p|N components are excluded.
DEPENDENCIES: BS-1996; two-smallest-row coprimality, already in the corpus
and reproved here. No Scott/Bozek two-vertex classification is used.
IMPLICATION / NEW CONSUMER:
With S={odd primes <=y}, s=pi(y)-1, the total number of N admitting ANY
closed active BAD set supported below y is at most
binomial(s,2)*2^(9s+16), with no upper bound on N required.
Consequently, for y(X)=c log X log log X and fixed 0<c<1/(9 log 2),
#{N<=X: such a closed set exists with max C<=y(X)}
<= X^(9c log 2+o(1))=o(X).
For each X, all moving supports in this event are contained in the SAME
alphabet S_y(X), so taking their union is justified. The prime number theorem
gives pi(y(X))=(c+o(1))log X. No union over arbitrary unbounded labels is made.
RELATION TO HISTORICAL FIXED-S WALL:
This does not repeat the invalid within-one-world capacity argument.
It counts DISTINCT N across a family sharing an explicitly bounded envelope.
It still gives no every-N exclusion and no height bound for exceptional N.
RELATION TO GOLDBACH:
Applies also to ordinary successful N containing bad components. Under global
Goldbach failure a closed BAD set exists, but nothing proved here forces
its labels below y(X). Hence this is not a Goldbach proof or a new estimate
for the full Goldbach exceptional set.
RELATION TO p0 EXCEPTIONALITY: Global graph restriction; no selector dependence.
WHAT REMAINS OPEN: Large or moving support, and canonical access to it.
NOVELTY: NOT YET AUDITED; new application within this checkpoint, not a claim
```

```
that bounded-prime smooth-pair counting is new to number theory.

RESULT-ID: SMALL-BAD-CHILDREN-PREFIX-20260905
STATUS: PROVED.
EXACT STATEMENT:
Fix 0<alpha<1/2 and 0<epsilon<1/2. Put z=(log X)^alpha,
s=#{odd primes q<=z}, mu=sum_(3<=q<=z, q prime) 1/q.
Let K=K(X)>=0 be an integer with K+1=o(mu), equivalently it suffices to
assume K+1=o(log log log X). For all but o(X) integers 4<=x<=X,
every upper root p_k(2x), 0<=k<=K, is BAD and has at least
(1-epsilon)mu distinct active BAD children q<=z.
In particular the lower bound is at least
(1-2epsilon)log log log X for all sufficiently large X when 0<epsilon<1/2.
This concerns actual vertices on the COMPLETE first frontier, not merely
an arbitrarily prescribed partial tree. It does NOT say that all children
are BAD or that shortest GOOD depth tends to infinity.

HYPOTHESES / CONVENTIONS:
BAD here means a positive composite complement, excluding unit rows.
All quantifiers over k are simultaneous on one set of x of density 1.
Logs are natural and X tends to infinity; no useful small-X rate is claimed.

PROOF:
1. Uniform arithmetic progression counts.
Let H=X-3 and J be the number of source prime cells meeting [4,X], so
J=O(X/log X). For ANY k>=0 and ANY odd d>=1, as x runs across a source
cell, m_k(2x) is an arithmetic progression with difference 2. Therefore
  A_d(k):=#{4<=x<=X: d divides m_k(2x)} = H/d + O(J),
with absolute error at most J, uniform in d and k. Negative complements
can provisionally be counted here as integers; we remove them in step 4.
The final partial cell also has discrepancy at most 1.

2. Mean and variance on actual root complements.
Define f_k(x)=sum_(3<=q<=z, prime) 1_(q divides m_k(2x)). Then
  sum_x f_k = H mu + O(J s),
  sum_x f_k^2 = H(mu^2+mu-sum_q 1/q^2) + O(J s^2).
The latter follows by expanding the square and using A_(qr) for distinct
q,r. It follows, without assuming independence in x, that
  sum_x (f_k-mu)^2 <= H mu + O(J(s^2+mu s)).
Consequently
  #{x: f_k(x)<(1-epsilon)mu}
  <= H/(epsilon^2 mu)
     + O(J(s^2+mu s)/(epsilon^2 mu^2)).
All constants are independent of k. Taking a union over 0..K is allowed.
The elementary bound s<=z and mu<=sum_(n<=z)1/n=O(log z) shows that
the second error, divided by X and multiplied by K+1=o(mu), tends to 0:
J/X=O(1/log X) and z^2=(log X)^(2alpha) with 2alpha<1.
The first error tends to 0 because K+1=o(mu).

3. Every sufficiently small prime vertex is BAD on almost all inputs.
For each odd prime q<=z, the condition 2x-q prime holds for at most pi(2X)
values of x. Thus the number of x having ANY GOOD q<=z is at most
  s*pi(2X)=O(X*z/log X)=o(X).
This is an ordinary counting upper bound, not an independence heuristic.
For x>=sqrt X all N-q>1 and exceed q, so rows avoiding the prime case are
composite. Also all roots exceed z.

4. Positivity, activity and root status.
The prime number theorem gives at least c*x/log x primes in [x,2x) for all
sufficiently large x. Uniformly for sqrt X<=x<=X this number exceeds K+1,
since K+1=o(mu)=O(log z). Hence p_k(2x)<2x and m_k(2x)>0 for all k<=K
in this range. Removing x<sqrt X costs o(X).
Outside the exceptional set in step 2, f_k>1 eventually, so m_k is composite
and p_k is not diagonal. Because x<=p_k<2x, a nondiagonal prime p_k cannot
divide 2x. For q|m_k, q|2x would imply q|p_k and thus q=p_k, impossible
since m_k<p_k. All counted child primes are therefore active. Step 3 makes
their complete complements composite. This proves the assertion.

5. Growth of mu; no normal-order theorem is needed here.
The Euler product over primes <=z contains all reciprocals 1/n for n<=z,
so product_(q<=z)(1-1/q)^(-1) >= sum_(n<=z)1/n >= log z.
Taking logarithms and using sum_q sum_(a>=2)1/(a*q^a)=O(1) gives
sum_(q<=z)1/q >= log log z-O(1). Removing q=2 changes a constant.
Thus mu>=log log log X+log alpha-O(1), proving the asserted lower bound.
The standard sharper Mertens estimate is unnecessary for this lower bound.

HOSTILE CHECK:
The square expansion involves q*r, never prime powers q^2 for the diagonal
term. Divisibility discrepancy is counted by SOURCE cells, whose number is
independent of k; no unjustified prime-residue distribution is assumed.
Goodness is tested in N-q, not in the root complement. Step 3 is a union
bound over all small q, so no conditional-primality assumption is hidden.
The full frontier can still have a large GOOD child and succeed at depth 1.

DEPENDENCIES:
Exact source-cell geometry; elementary second moments and Euler product;
pi(t)=O(t/log t) and the prime number theorem pi(t)~t/log t. The latter is
used only for uniform positivity of the first slowly growing prefix.

ACTUAL TRAVERSAL CONSUMERS:
(a) In world/exhaustion mode, which expands all reachable BAD vertices and
stops only each GOOD branch, |R_N(p_k)|>=1+(1-epsilon)mu simultaneously.
Each small BAD child of a BAD root is necessarily in that world even when
some other child is GOOD.
(b) In the reference algorithm the root's children are tested in increasing
prime order. Before finding a GOOD child or exhausting the root row, it
must perform at least (1-epsilon)mu DISTINCT unsuccessful child-goodness
tests, because every child <=z precedes every possible GOOD child.
The same conclusion holds for a lazy FIFO BFS with children enqueued in
increasing order, counting tested vertices before first GOOD.
(c) This is NOT a lower bound of mu on expanded BAD rows before first escape
in an eager-child implementation: it may detect a large GOOD child while
still processing the root. The distinction is material.

RELATION TO GOLDBACH:
An algorithmic lower bound on a specific search, not a lack of prime pairs.
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RELATION TO p0 EXCEPTIONALITY:
A nontrivial traversal statistic, but it is shared simultaneously by p0 and
a growing prefix of nearby roots. Slow early testing / a growing BAD first
front cannot by itself explain unique eventual success of p0.
WHAT REMAINS OPEN:
Control of LARGE child exits, complete BAD depth, and multigeneration
correlations. Nothing here forces either success or closed failure.
NOVELTY: NOT YET AUDITED.

SOURCE-CORRECTION ADDENDUM TO AUDIT-FBTU:
The older Biblia independently records the precise obstruction already:
iteration 108, immediately before theorem iteration108-support (around
lines 28288-28310 of upload/goldbach_biblia(1).tex), reports that CRT retained
up to eight prescribed BAD-path edges but additional prime factors let BFS
escape. It explicitly distinguishes retaining existing edges from preventing
new factors. Therefore the present audit repairs a later scope regression,
not the first discovery of cofactor leakage. The small N=92 certificate and
the complete-two-row finiteness bound provide additional exact witnesses.

HISTORICAL LEMMA RECOVERED FOR FURTHER IMPROVEMENT:
Biblia theorem iteration108-support (around lines 28317-28373) uses Matveev
to obtain a support-height bound, but assumes at least two nonbranching
rows, via |R|>=b+2. The text correctly notes this excludes highly branched
worlds. The current continuation is checking whether that restriction can
be removed by choosing two arbitrary distinct rows and using max C itself
as the row-gap bound. This is a genuine changed hypothesis/consumer, not
an attempt to reuse generic fixed-S finiteness as a full Goldbach proof.

EXTERNAL-INPUT: MATVEEV-IN-BMS-9.4
STATUS: EXACT STATEMENT AND DERIVATION CHECKED IN PRIMARY RESEARCH SOURCE.
Y. Bugeaud, M. Mignotte, S. Siksek, Classical and modular approaches to
exponential Diophantine equations I. Fibonacci and Lucas perfect powers,
Annals of Mathematics 163 (2006), 969-1018, Theorem 9.4, real-field clause.
Author's full text, printed pp.16-17:
https://samirsiksek.github.io/siksek.github.io/papers/fibannalsfinal.pdf
The source derives this product-minus-one form from Matveev's Corollary 2.3.
For positive rational alpha_1,...,alpha_t and integer b_i, B=max|b_i|>=1,
Lambda=product alpha_i^b_i-1 !=0, and
A_i>=max(h(alpha_i), |log alpha_i|, 0.16), its D=1 real case gives
log|Lambda| > -1.4*30^(t+3)*t^(9/2)*(1+log B)*product A_i.
Here h(a/b)=log max(|a|,|b|) for coprime integers a,b, b>0.
We use this verified form, not an unverified stronger interpretation.

RESULT-ID: UNRESTRICTED-CLOSED-SUPPORT-HEIGHT-20260905
STATUS: PROVED WITH REPAIR / GENERALIZATION of Biblia iteration108-support.
EXACT STATEMENT:
Let N>=8 be even and let C be a nonempty set of active prime labels <N,
with every N-p composite and PF(N-p) subset C for every p in C.
Put s=|C|, M=max C, and
  A(C)=1.4*30^(s+3)*s^(9/2)*product_(q in C) log q.
Then s>=2, all q in C are odd, and
  log((N-M)/M)
  < A(C) * (1+log(log N/log 3)).                    (H)
No assumption on the number of nonbranching rows, SCC minimality,
canonicity, or global Goldbach failure is made.

PROOF:
Activity excludes 2 because N is even. The singleton case is excluded as
in CLOSED-ALPHABET-COUNT. Choose any a<b in C; one may choose the two smallest.
Write U=N-a and V=N-b. Closure implies U,V are C-smooth and U>V>0.
Let e_q=v_q(U)-v_q(V). Since U,V<N and q>=3,
  |e_q|<=log N/log q<=log N/log 3.
Omit zero coefficients if necessary. At least one remains, because U!=V.
For Lambda=U/V-1=(b-a)/(N-b)>0, use the preceding Matveev form with
alpha_i=q_i, A_i=log q_i (h(q_i)=log q_i>0.16), D=1.
All logarithms of odd primes exceed 1. Thus adding back omitted primes
only enlarges the product; the factor 1.4*30^(t+3)*t^(9/2) increases with t.
The result is
  log Lambda > -A(C)*(1+log(log N/log 3)).
On the other hand b-a<M and N-b>=N-M, so
  Lambda < M/(N-M).
Compare and negate to obtain (H).

HOSTILE CHECK:
Lambda is PRODUCT-minus-one, precisely the verified external theorem's form.
No confusion with the logarithmic form is needed. The inequality remains
valid if M>N/2, but its left side then gives no information. This is an
explicit limitation, not an omitted hypothesis. No coprimality of U,V is
needed here, and no assumption about two small pure-power rows is used.
For M<=sqrt N, (H) immediately yields the familiar positive order-log N
lower bound on A(C), without any branching restriction.

DEPENDENCIES: Exact full factor closure; the verified real rational Matveev
bound. No Scott classification, prime-gap bound, RH, or abc assumption.

RESULT-ID: EFFECTIVE-BOUNDED-CORE-THRESHOLD-20260905
STATUS: PROVED.
EXACT STATEMENT:
For real y>=5, put s_y=#{odd primes <=y},
  A_y=1.4*30^(s_y+3)*s_y^(9/2)*product_(3<=q<=y, prime) log q,
  D_y=A_y+log(2y),
  F(y)=exp(2 D_y log(2 D_y)).
If an even N admits ANY nonempty closed active BAD set C with max C<=y,
then N<F(y). Thus N>=F(y) excludes all such sets, regardless of their
branching and cardinality. F is explicit and effectively computable.

PROOF:
When N<=2y the bound is immediate because D_y>log(2y) and F(y)>2y.
When N>2y, use (H), A(C)<=A_y, M<=y, and t=log N>1 to get
  t-log(2y) < A_y*(1+log t).
Indeed log((N-M)/M)>=log((N-y)/y)>t-log(2y), while log3>1
implies log(t/log3)<log t. Consequently t<D_y*(1+log t).
For any D>=1, the inequality t<D(1+log t), t>1, implies
  t<2D log(2D).
To verify it, put t_0=2D log(2D), u=log(2D)>0. Then
 t_0/D-(1+log t_0)=u-1-log u>=0.
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The function t/D-1-log t is strictly increasing for t>D, and t_0>D.
Thus t>=t_0 is impossible. Apply this with D=D_y and exponentiate.

HOSTILE CHECK:
A_y is a uniform envelope for ALL choices of C inside primes <=y.
The bound is enormous and is not a practical census endpoint even for small y.
It bounds N for a specified maximum label y; it is NOT a Goldbach N_0.
It supplies the height control that BS counting alone does not provide.

RELATION TO HISTORICAL FAILURES:
The old Matveev support lemma required |R|>=b+2 to obtain two rows with
labels <=sqrt N. This proof removes that condition: it uses arbitrary
distinct rows and max C as their gap bound. If max C is already large,
the desired small-support exclusion is automatic. The old warning about
huge constants and moving/unbounded support still applies.

RESULT-ID: POINTWISE-CORE-LABEL-GROWTH-20260905
STATUS: PROVED (asymptotic consequence, no assertion of novelty).
EXACT STATEMENT:
Uniformly along any sequence N->infinity possessing a nonempty closed
active BAD set C, its largest prime label M obeys
  M >= (1-o(1)) *
       (log log N)(log log log N)/(log log log log N).       (LG)
The iterated logarithms are used only beyond their positivity thresholds.

PROOF:
The effective theorem first implies M->infinity along such a sequence.
The prime number theorem gives s_y=(1+o(1))y/log y. Therefore
  log A_y
  <= s_y*(log 30+log log y)+O(log y)
  <= (1+o(1))*y log log y/log y.
Since D_y=A_y+log(2y), the same main upper bound holds for log D_y,
and also for log(2 D_y log(2 D_y)), the extra term log log D_y being
lower order. Apply N<F(M), take log twice, and conclude
  log log N <= (1+o(1))*M log log M/log M.
The function g(u)=u log log u/log u is increasing for sufficiently large u.
Its asymptotic inverse is v log v/log log v, since direct substitution
shows g(v log v/log log v)~v. This proves (LG).

IMPLICATION / SCOPE FOR THE THREE HEIGHT RESULTS:
These constrain every genuine closed BAD set, including one arising from
rooted failure in an otherwise Goldbach-successful N. A full failed rooted
world has a terminal SCC; all its vertices are factors of composite odd
rows and hence <N/3. Apply the same results to this terminal core, so the
bound need not be vacuously satisfied just by the large starting root.
The existence of a terminal SCC follows from finite graph closure; it has
at least two vertices by activity. No classification of such SCCs is used.
RELATION TO GOLDBACH:
Global Goldbach failure forces a closed BAD component, but its labels can
satisfy (LG). The result is a necessary condition, not a contradiction.
RELATION TO p0 EXCEPTIONALITY:
Shared by all failed traversals. No canonical asymmetry is proved.
WHAT REMAINS OPEN:
An upper restriction incompatible with this lower growth, or a direct
arithmetic escape theorem for the moving support. We have neither.
NOVELTY: NOT YET AUDITED. Classical logarithmic-form methods are the source;
the advance over the supplied lemma is the removal of its branching hypothesis
and an explicit common-envelope threshold with correct scope.

RESULT-ID: H1-TWO-GENERATION-ENVELOPE-ESCAPE-20260905
STATUS: PROVED.
EXACT STATEMENT:
For y>=5 let s_y=#{odd primes <=y} and define
 A_y^+=1.4*30^(s_y+4)*(s_y+1)^(9/2)*log 2*
       product_(3<=l<=y, prime) log l,
 D_y^+=2*A_y^++log(2y),
 F_2(y)=exp(2*D_y^+*log(2*D_y^+)).
Let N=2(P-1), P>=5 prime, with m=P-2 composite and PF(m) contained in
{odd primes <=y}. Put q=min PF(m). If N>=F_2(y), then
  P^+(N-q)>y.                                              (E2)
Thus either q itself is GOOD, giving success at depth 1, or the stopped
traversal from canonical P reaches a prime r>y at depth 2.
No global failure or full rooted failure is assumed.

PROOF:
Suppose instead that both complete rows m and v=N-q are supported on odd
primes <=y. q<=y and v=2m+2-q, so
 Lambda=2m/v-1=(q-2)/v>0.
For N>2y, Lambda<2y/N. (N>=F_2(y) implies N>2y.)
The expression 2m/v has a nonzero exponent 1 on base 2, and its other bases
are odd primes <=y. Their exponent differences have absolute value at most
log N/log 3; all exponents are bounded by 2 log N.
Apply the verified rational real Matveev form to these bases. Base 2 has
height log2>0.16 and is always present. Omitted odd bases may be padded since
their logarithms exceed 1. Hence, with t=log N,
 log Lambda > -A_y^+*(2+log t),
where 1+log(2 log N)<2+log t has been used.
Combining bounds gives
 t<log(2y)+A_y^+*(2+log t) <= D_y^+*(1+log t).
The inversion proved above implies N<F_2(y), a contradiction. This proves E2.
If q is BAD, every prime r|N-q is an actual child and is active: r|N would
force r|q, hence r=q and q|N, already impossible because q|m and
N=2m+2. Consequently a factor r>y is reached at depth 2. If q is GOOD, the
stopped traversal stops that branch at q and has already succeeded.

HOSTILE CHECK:
The large r is a factor of the COMPLETE row N-q, not a residual quotient
obtained by gcd removal. It is genuinely outside the WHOLE prescribed
prime envelope, not just a surplus valuation of an old prime. A GOOD q is
handled before claiming that its children are traversed. A large reached
prime r need not be GOOD; no Goldbach conclusion follows in the BAD-q branch.
The finite h=1 transcript can still be prescribed by CRT: this theorem
forces some of its previously uncontrolled cofactors to carry larger primes.

DEPENDENCIES: Matveev-in-BMS-9.4, exact h=1 arithmetic, elementary inversion.
RELATION TO HISTORICAL FAILURES:
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This is an actual escape consumer, so the fixed-alphabet finiteness warning
does not invalidate it. It still cannot iterate with a fixed useful bound,
since the new alphabet may grow and its threshold changes. It does not
revive the failed collision-to-new-support argument PR-MQ.
RELATION TO GOLDBACH: Success OR a provable enlarged reachable alphabet.
RELATION TO p0 EXCEPTIONALITY:
A concrete reachability theorem in the canonical h=1 subfamily. We have not
shown a corresponding property fails at p1 for the same N, nor covered all
canonical gaps. It therefore is not an explanation of universal p0 success.
WHAT REMAINS OPEN: A mechanism that turns repeated necessary envelope escape
into GOOD reachability, with control that survives the changing envelope.
NOVELTY: NOT YET AUDITED.

EXTERNAL CLASSICAL INPUT CHECKS
PNT: pi(t)~t/log t, Hadamard and de la Vallee Poussin (1896).
Statement and a complete proof checked in D. Zagier, Newman's Short Proof
of the Prime Number Theorem, American Mathematical Monthly 104 (1997),
705-708: https://sites.math.rutgers.edu/~zeilberg/EM18/zagier.pdf
The upper bound pi(t)=O(t/log t) and the interval consequence
pi(2t)-pi(t)~t/log t follow directly. No error term or AP uniformity is used.
Dirichlet (1837): if gcd(a,m)=1, m>=1, infinitely many primes are
a mod m. The exact statement is independently given in Paul Pollack,
Hypothesis H and an impossibility theorem, Rendiconti del Seminario
Matematico Universita Politecnico Torino 68 (2010), starting at p.183:
https://seminariomatematico.polito.it/rendiconti/68-2/183.pdf
No theorem about several simultaneously prime affine forms is used.

EXPERIMENT-ID: FINITE-CELL-AND-CLOSED-WORLD-STRESS-20260905
STATUS: VERIFIED COMPUTATION.
DOMAIN:
All integer 4<=x<=200000 for the residue and moment checks; ranks
k=0,1,2,5,20; small-prime cutoffs z=7,31,101 (15 configurations).
These z are diagnostics of the finite identities; they are not an assertion
that the asymptotic scale z=(log X)^alpha is numerically informative here.
The residue/moment identities include negative complements. The graph
consumer excludes nonpositive complements, x below sqrt(X), and inputs
with a GOOD prime <=z. Unit rows contribute zero to small factor counts.
For graph closure, exactly the six previously supplied N were reconstructed:
128,1718,1862,1928,2200,6142; this is NOT a scan for all N<=6142 or beyond.
METRICS:
Exact residue discrepancy for every q and pair-product q*r; second moment;
actual BAD-child count; maximal low closed set by deletion to a fixed point;
terminal SCCs; smallest-row gcd; pure-row count; and complete stopped BFS
from ranks 0,1,2. Pure row means exactly one distinct prime factor.
CHECKS:
For all residue counts, the integer certificate |d*A_d-H|<=d*J passed.
The explicit second-moment upper bound
H*mu+J*(s^2+2*mu*s) passed in all 15 cases. On every eligible input
the actual number of small BAD children equalled the small-divisor count.
Every reconstructed terminal component was closed, active, entirely BAD,
and its two smallest rows were coprime. All assertions passed.
EXTREMAL / RELEVANT WITNESSES:
N=128: terminal size=8, pure rows=2, max label=41; C=[3, 5, 7, 11, 13, 23, 29, 41].
N=1718: terminal size=28, pure rows=1, max label=571; C=[3, 5, 7, 11, 13, 17, 29, 31, 37, 41, 43, 53, 59, 67, 79, 89, 127, 137, 149, 181, 211, 239, 241, 383, 523, 563, 569, 571].
N=1862: terminal size=21, pure rows=1, max label=619; C=[3, 5, 11, 13, 17, 41, 43, 47, 53, 67, 83, 107, 113, 167, 179, 251, 359, 593, 607, 617, 619].
N=1928: terminal size=14, pure rows=0, max label=641; C=[3, 5, 7, 11, 13, 17, 53, 71, 73, 103, 113, 383, 619, 641].
N=2200: terminal size=2, pure rows=2, max label=13; C=[3, 13].
N=6142: terminal size=10, pure rows=0, max label=877; C=[3, 5, 7, 13, 17, 19, 157, 227, 409, 877].
The components at N=1928 and 6142 have ZERO pure rows. They show the
branching hypothesis removed from the old Matveev lemma is a real domain
extension. They satisfy the new necessary height inequality, with enormous
slack; they do not refute any global-Goldbach-failure theorem.
Maximum finite residue discrepancy among the recorded configurations: 5335.66666667; source-cell upper bound J=17983.
Maximum recorded small BAD-child count: 5.
All asymptotic and all-N assertions rely on the written proofs.
Floating-point logarithmic comparisons in the code illustrate the size
of the Matveev margin and do not constitute a rigorous numerical theorem.

REPRODUCIBLE CODE (single-file Python, requires NumPy)
Save the following code separately only when reproducing the diagnostics.
BEGIN DIAGNOSTIC CODE
from math import isqrt, gcd, log
from collections import deque
import json
from pathlib import Path
import numpy as np

X=200000
limit=2*X+2000
isp=np.ones(limit+1,dtype=bool);isp[:2]=False
for p in range(2,isqrt(limit)+1):
    if isp[p]:isp[p*p::p]=False
ps=np.flatnonzero(isp)
spf=np.arange(limit+1,dtype=np.int64)
for p in ps[ps<=isqrt(limit)]:
    sl=spf[int(p)*int(p)::int(p)]
    mask=sl==np.arange(int(p)*int(p),limit+1,int(p))
    sl[mask]=p

def fac(n):
    out={}
    while n>1:
        p=int(spf[n]);out[p]=out.get(p,0)+1;n//=p
    return out

def world(N,r):
    if 2*r==N:return {'depth':0,'R':[],'good':[r]}
    todo=deque([(r,0)]);seen={r};R=[];good=[];first=None
    while todo:
        p,d=todo.popleft()
        if isp[N-p]:
            good.append(p)
            if first is None:first=d
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            continue
        assert N-p>1
        R.append(p)
        for q in fac(N-p):
            assert N%q!=0
            if q not in seen:seen.add(q);todo.append((q,d+1))
    return {'depth':first,'R':sorted(R),'good':sorted(good)}

def tarjan(adj):
    index={};low={};stack=[];active=set();comps=[]
    def go(v):
        index[v]=low[v]=len(index);stack.append(v);active.add(v)
        for w in adj[v]:
            if w not in index:go(w);low[v]=min(low[v],low[w])
            elif w in active:low[v]=min(low[v],index[w])
        if low[v]==index[v]:
            comp=[]
            while True:
                w=stack.pop();active.remove(w);comp.append(w)
                if w==v:break
            comps.append(sorted(comp))
    for v in adj:
        if v not in index:go(v)
    return comps

xs=np.arange(4,X+1,dtype=np.int64);ii=np.searchsorted(ps,xs)
H=len(xs);J=len(np.unique(ii));res=[]
for k in [0,1,2,5,20]:
    ms=2*xs-ps[ii+k]
    for z in [7,31,101]:
        small=[int(q) for q in ps if 3<=q<=z]
        s=len(small);mu=sum(1/q for q in small)
        f=np.zeros(H,dtype=np.int64)
        maxdisc=0
        for j,q in enumerate(small):
            f+=(ms%q==0)
            ds=[q]+[q*r for r in small[:j]]
            for d in ds:
                count=int((ms%d==0).sum())
                # Exact integer discrepancy certificate, no floating point.
                assert abs(d*count-H)<=d*J
                maxdisc=max(maxdisc,abs(count-H/d))
        variance=float(((f-mu)**2).sum())
        variance_bound=H*mu+J*(s*s+2*mu*s)
        assert variance<=variance_bound+1e-7
        pos=ms>0
        smallgood=np.zeros(H,dtype=bool)
        smallbad_count=np.zeros(H,dtype=np.int64)
        for q in small:
            good=isp[2*xs-q] if 2*xs[0]-q>=0 else np.array([2*int(x)-q>=2 and bool(isp[2*int(x)-q]) for x in xs])
            smallgood|=good
            smallbad_count+=(pos & (ms%q==0) & (2*xs>2*q) & ~good)
        eligible=pos & (xs>=isqrt(X)+1) & ~smallgood
        assert np.all(smallbad_count[eligible]==f[eligible])
        res.append({'k':k,'z':z,'nonpositive':int((~pos).sum()),
                    'mu':mu,'actual_mean':float(f.mean()),'actual_variance':variance/H,
                    'max_residue_discrepancy':maxdisc,'source_cell_bound':J,
                    'any_small_good_inputs':int(smallgood.sum()),
                    'eligible_graph_bridge_inputs':int(eligible.sum()),
                    'max_small_bad_children':int(smallbad_count.max())})

cores=[]
for N in [128,1718,1862,1928,2200,6142]:
    B={int(p) for p in ps if 3<=p<N/3 and N%int(p)!=0 and not isp[N-int(p)]}
    while True:
        nxt={p for p in B if set(fac(N-p))<=B}
        if nxt==B:break
        B=nxt
    adj={p:set(fac(N-p)) for p in B}
    comps=tarjan(adj)
    terminal=[c for c in comps if all(adj[p]<=set(c) for p in c)]
    out=[]
    for C in terminal:
        a,b=C[:2]
        assert gcd(N-a,N-b)==1
        rows={p:fac(N-p) for p in C}
        assert all(set(row)<=set(C) and sum(row.values())>=2 for row in rows.values())
        pure=sum(len(row)==1 for row in rows.values())
        M=max(C);s=len(C)
        logA=log(1.4)+(s+3)*log(30)+4.5*log(s)+sum(log(log(q)) for q in C)
        lhs=log((N-M)/M)
        # Logarithmic comparison avoids overflows and is diagnostic only.
        assert log(lhs)<logA+log(1+log(log(N)/log(3)))
        out.append({'C':C,'size':s,'pure_rows':pure,'M':M,'two_min_rows_gcd':gcd(N-a,N-b),
                    'height_lhs':lhs,'log10_A':logA/log(10),'rows':rows})
    root_index=int(np.searchsorted(ps,N//2))
    roots=[]
    for k in range(3):
        p=int(ps[root_index+k]);w=world(N,p)
        roots.append({'k':k,'p':p,'m_factors':fac(N-p),'depth':w['depth'],'R_size':len(w['R'])})
    cores.append({'N':N,'B_size':len(B),'terminal':out,'roots':roots})

n92=world(92,47)
assert n92['depth']==1
for a,b in [(47,5),(5,29),(29,7),(7,5)]:
    assert (92-a)%b==0 and not isp[92-a] and not isp[92-b]
output={'domain':{'x_min':4,'x_max':X,'ranks':[0,1,2,5,20],
                  'z_values':[7,31,101],'scope':'exhaustive residue/moment checks at these finite parameters; not an asymptotic census'},
        'moment_checks':res,'known_closed_worlds':cores,
        'N92':{'root':47,'shortest_good_depth':n92['depth'],'R':n92['R'],'good':n92['good']},
        'checks':'all assertions passed'}
Path('/workspace/scratch/db0c19a2d4d9/checkpoint_diagnostics.json').write_text(json.dumps(output,indent=2)+'\n')
print(json.dumps({'checks':output['checks'],'moment_cases':len(res),
                 'known_worlds':[{'N':c['N'],'B_size':c['B_size'],'terminal':[{key:t[key] for key in ['size','pure_rows','M','log10_A']} for t in
                ↪  c['terminal']],'roots':c['roots']} for c in cores],
```

```
                'N92':output['N92']},indent=2))

END DIAGNOSTIC CODE

COMPLETE DIAGNOSTIC OUTPUT
```


```
{
  "domain": {
    "x_min": 4,
    "x_max": 200000,
    "ranks": [
      0,
      1,
      2,
      5,
      20
    ],
    "z_values": [
      7,
      31,
      101
    ],
    "scope": "exhaustive residue/moment checks at these finite parameters; not an asymptotic census"
  },
  "moment_checks": [
    {
      "k": 0,
      "z": 7,
      "nonpositive": 0,
      "mu": 0.6761904761904762,
      "actual_mean": 0.6255343830157453,
      "actual_variance": 0.5041111680167265,
      "max_residue_discrepancy": 5335.6666666666715,
      "source_cell_bound": 17983,
      "any_small_good_inputs": 90156,
      "eligible_graph_bridge_inputs": 109748,
      "max_small_bad_children": 3
    },
    {
      "k": 0,
      "z": 31,
      "nonpositive": 0,
      "mu": 1.065696836388161,
      "actual_mean": 0.9840997614964224,
      "actual_variance": 0.900648678697622,
      "max_residue_discrepancy": 5335.6666666666715,
      "source_cell_bound": 17983,
      "any_small_good_inputs": 171998,
      "eligible_graph_bridge_inputs": 27998,
      "max_small_bad_children": 5
    },
    {
      "k": 0,
      "z": 101,
      "nonpositive": 0,
      "mu": 1.312718191147881,
      "actual_mean": 1.2083131246968704,
      "actual_variance": 1.1748641703513216,
      "max_residue_discrepancy": 5335.6666666666715,
      "source_cell_bound": 17983,
      "any_small_good_inputs": 197552,
      "eligible_graph_bridge_inputs": 2445,
      "max_small_bad_children": 5
    },
    {
      "k": 1,
      "z": 7,
      "nonpositive": 0,
      "mu": 0.6761904761904762,
      "actual_mean": 0.6853552803292049,
      "actual_variance": 0.5027517190535636,
      "max_residue_discrepancy": 752.3333333333285,
      "source_cell_bound": 17983,
      "any_small_good_inputs": 90156,
      "eligible_graph_bridge_inputs": 109748,
      "max_small_bad_children": 3
    },
    {
      "k": 1,
      "z": 31,
      "nonpositive": 0,
      "mu": 1.065696836388161,
      "actual_mean": 1.0837412561188418,
      "actual_variance": 0.8788562862563026,
      "max_residue_discrepancy": 752.3333333333285,
      "source_cell_bound": 17983,
      "any_small_good_inputs": 171998,
      "eligible_graph_bridge_inputs": 27998,
      "max_small_bad_children": 5
    },
    {
      "k": 1,
      "z": 101,
      "nonpositive": 0,
      "mu": 1.312718191147881,
      "actual_mean": 1.330524957874368,
      "actual_variance": 1.1190787532897006,
      "max_residue_discrepancy": 752.3333333333285,
      "source_cell_bound": 17983,
      "any_small_good_inputs": 197552,
      "eligible_graph_bridge_inputs": 2445,
      "max_small_bad_children": 5
    },
    {
      "k": 2,
      "z": 7,
      "nonpositive": 4,
```

```
      "mu": 0.6761904761904762,
      "actual_mean": 0.6757051355770337,
      "actual_variance": 0.4963820996997494,
      "max_residue_discrepancy": 505.66666666666606,
      "source_cell_bound": 17983,
      "any_small_good_inputs": 90156,
      "eligible_graph_bridge_inputs": 109748,
      "max_small_bad_children": 3
    },
    {
      "k": 2,
      "z": 31,
      "nonpositive": 4,
      "mu": 1.065696836388161,
      "actual_mean": 1.0690610359155388,
      "actual_variance": 0.865234941052481,
      "max_residue_discrepancy": 505.66666666666606,
      "source_cell_bound": 17983,
      "any_small_good_inputs": 171998,
      "eligible_graph_bridge_inputs": 27998,
      "max_small_bad_children": 5
    },
    {
      "k": 2,
      "z": 101,
      "nonpositive": 4,
      "mu": 1.312718191147881,
      "actual_mean": 1.318204773071596,
      "actual_variance": 1.1091539806479949,
      "max_residue_discrepancy": 505.66666666666606,
      "source_cell_bound": 17983,
      "any_small_good_inputs": 197552,
      "eligible_graph_bridge_inputs": 2445,
      "max_small_bad_children": 5
    },
    {
      "k": 5,
      "z": 7,
      "nonpositive": 18,
      "mu": 0.6761904761904762,
      "actual_mean": 0.6767401511022665,
      "actual_variance": 0.5068375422456733,
      "max_residue_discrepancy": 130.80000000000018,
      "source_cell_bound": 17983,
      "any_small_good_inputs": 90156,
      "eligible_graph_bridge_inputs": 109748,
      "max_small_bad_children": 3
    },
    {
      "k": 5,
      "z": 31,
      "nonpositive": 18,
      "mu": 1.065696836388161,
      "actual_mean": 1.0651409771146567,
      "actual_variance": 0.8621199575250112,
      "max_residue_discrepancy": 161.30909090909108,
      "source_cell_bound": 17983,
      "any_small_good_inputs": 171998,
      "eligible_graph_bridge_inputs": 27998,
      "max_small_bad_children": 5
    },
    {
      "k": 5,
      "z": 101,
      "nonpositive": 18,
      "mu": 1.312718191147881,
      "actual_mean": 1.3149047235708535,
      "actual_variance": 1.1085079110185263,
      "max_residue_discrepancy": 161.30909090909108,
      "source_cell_bound": 17983,
      "any_small_good_inputs": 197552,
      "eligible_graph_bridge_inputs": 2445,
      "max_small_bad_children": 5
    },
    {
      "k": 20,
      "z": 7,
      "nonpositive": 111,
      "mu": 0.6761904761904762,
      "actual_mean": 0.6762601439021585,
      "actual_variance": 0.5039366415893064,
      "max_residue_discrepancy": 100.66666666666606,
      "source_cell_bound": 17983,
      "any_small_good_inputs": 90156,
      "eligible_graph_bridge_inputs": 109748,
      "max_small_bad_children": 3
    },
    {
      "k": 20,
      "z": 31,
      "nonpositive": 111,
      "mu": 1.065696836388161,
      "actual_mean": 1.064990974864623,
      "actual_variance": 0.8678897531228617,
      "max_residue_discrepancy": 157.5294117647063,
      "source_cell_bound": 17983,
      "any_small_good_inputs": 171998,
      "eligible_graph_bridge_inputs": 27998,
      "max_small_bad_children": 5
    },
    {
      "k": 20,
      "z": 101,
      "nonpositive": 111,
      "mu": 1.312718191147881,
      "actual_mean": 1.3126396895953438,
```

```
      "actual_variance": 1.1086295262486547,
      "max_residue_discrepancy": 157.5294117647063,
      "source_cell_bound": 17983,
      "any_small_good_inputs": 197552,
      "eligible_graph_bridge_inputs": 2445,
      "max_small_bad_children": 5
    }
  ],
  "known_closed_worlds": [
    {
      "N": 128,
      "B_size": 10,
      "terminal": [
        {
          "C": [
            3,
            5,
            7,
            11,
            13,
            23,
            29,
            41
          ],
          "size": 8,
          "pure_rows": 2,
          "M": 41,
          "two_min_rows_gcd": 1,
          "height_lhs": 0.752336051950276,
          "log10_A": 23.377296381473453,
          "rows": {
            "3": {
              "5": 3
            },
            "5": {
              "3": 1,
              "41": 1
            },
            "7": {
              "11": 2
            },
            "11": {
              "3": 2,
              "13": 1
            },
            "13": {
              "5": 1,
              "23": 1
            },
            "23": {
              "3": 1,
              "5": 1,
              "7": 1
            },
            "29": {
              "3": 2,
              "11": 1
            },
            "41": {
              "3": 1,
              "29": 1
            }
          }
        }
      ],
      "roots": [
        {
          "k": 0,
          "p": 67,
          "m_factors": {
            "61": 1
          },
          "depth": 0,
          "R_size": 0
        },
        {
          "k": 1,
          "p": 71,
          "m_factors": {
            "3": 1,
            "19": 1
          },
          "depth": 1,
          "R_size": 9
        },
        {
          "k": 2,
          "p": 73,
          "m_factors": {
            "5": 1,
            "11": 1
          },
          "depth": null,
          "R_size": 9
        }
      ]
    },
    {
      "N": 1718,
      "B_size": 55,
      "terminal": [
        {
          "C": [
            3,
            5,
            7,
```

```
        11,
        13,
        17,
        29,
        31,
        37,
        41,
        43,
        53,
        59,
        67,
        79,
        89,
        127,
        137,
        149,
        181,
        211,
        239,
        241,
        383,
        523,
        563,
        569,
        571
      ],
      "size": 28,
      "pure_rows": 1,
      "M": 571,
      "two_min_rows_gcd": 1,
      "height_lhs": 0.6975159074733605,
      "log10_A": 69.09427702086347,
      "rows": {
        "3": {
          "5": 1,
          "7": 3
        },
        "5": {
          "3": 1,
          "571": 1
        },
        "7": {
          "29": 1,
          "59": 1
        },
        "11": {
          "3": 1,
          "569": 1
        },
        "13": {
          "5": 1,
          "11": 1,
          "31": 1
        },
        "17": {
          "3": 5,
          "7": 1
        },
        "29": {
          "3": 1,
          "563": 1
        },
        "31": {
          "7": 1,
          "241": 1
        },
        "37": {
          "41": 2
        },
        "41": {
          "3": 1,
          "13": 1,
          "43": 1
        },
        "43": {
          "5": 2,
          "67": 1
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        "53": {
          "3": 2,
          "5": 1,
          "37": 1
        },
        "59": {
          "3": 1,
          "7": 1,
          "79": 1
        },
        "67": {
          "13": 1,
          "127": 1
        },
        "79": {
          "11": 1,
          "149": 1
        },
        "89": {
          "3": 2,
          "181": 1
        },
        "127": {
          "37": 1,
          "43": 1
        },
        "137": {
          "3": 1,
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```
              "17": 1,
              "31": 1
            },
            "149": {
              "3": 1,
              "523": 1
            },
            "181": {
              "29": 1,
              "53": 1
            },
            "211": {
              "11": 1,
              "137": 1
            },
            "239": {
              "3": 1,
              "17": 1,
              "29": 1
            },
            "241": {
              "7": 1,
              "211": 1
            },
            "383": {
              "3": 1,
              "5": 1,
              "89": 1
            },
            "523": {
              "5": 1,
              "239": 1
            },
            "563": {
              "3": 1,
              "5": 1,
              "7": 1,
              "11": 1
            },
            "569": {
              "3": 1,
              "383": 1
            },
            "571": {
              "31": 1,
              "37": 1
            }
          }
        }
      ],
      "roots": [
        {
          "k": 0,
          "p": 859,
          "m_factors": {
            "859": 1
          },
          "depth": 0,
          "R_size": 0
        },
        {
          "k": 1,
          "p": 863,
          "m_factors": {
            "3": 2,
            "5": 1,
            "19": 1
          },
          "depth": 1,
          "R_size": 29
        },
        {
          "k": 2,
          "p": 877,
          "m_factors": {
            "29": 2
          },
          "depth": null,
          "R_size": 29
        }
      ]
    },
    {
      "N": 1862,
      "B_size": 55,
      "terminal": [
        {
          "C": [
            3,
            5,
            11,
            13,
            17,
            41,
            43,
            47,
            53,
            67,
            83,
            107,
            113,
            167,
            179,
            251,
            359,
            593,
```

```
          607,
          617,
          619
        ],
        "size": 21,
        "pure_rows": 1,
        "M": 619,
        "two_min_rows_gcd": 1,
        "height_lhs": 0.697177818826115,
        "log10_A": 54.142312309949745,
        "rows": {
          "3": {
            "11": 1,
            "13": 2
          },
          "5": {
            "3": 1,
            "619": 1
          },
          "11": {
            "3": 1,
            "617": 1
          },
          "13": {
            "43": 2
          },
          "17": {
            "3": 2,
            "5": 1,
            "41": 1
          },
          "41": {
            "3": 1,
            "607": 1
          },
          "43": {
            "17": 1,
            "107": 1
          },
          "47": {
            "3": 1,
            "5": 1,
            "11": 2
          },
          "53": {
            "3": 3,
            "67": 1
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          "67": {
            "5": 1,
            "359": 1
          },
          "83": {
            "3": 1,
            "593": 1
          },
          "107": {
            "3": 3,
            "5": 1,
            "13": 1
          },
          "113": {
            "3": 1,
            "11": 1,
            "53": 1
          },
          "167": {
            "3": 1,
            "5": 1,
            "113": 1
          },
          "179": {
            "3": 2,
            "11": 1,
            "17": 1
          },
          "251": {
            "3": 2,
            "179": 1
          },
          "359": {
            "3": 2,
            "167": 1
          },
          "593": {
            "3": 3,
            "47": 1
          },
          "607": {
            "5": 1,
            "251": 1
          },
          "617": {
            "3": 1,
            "5": 1,
            "83": 1
          },
          "619": {
            "11": 1,
            "113": 1
          }
        }
      }
    ],
    "roots": [
      {
```

```
          "k": 0,
          "p": 937,
          "m_factors": {
            "5": 2,
            "37": 1
          },
          "depth": 2,
          "R_size": 23
        },
        {
          "k": 1,
          "p": 941,
          "m_factors": {
            "3": 1,
            "307": 1
          },
          "depth": null,
          "R_size": 24
        },
        {
          "k": 2,
          "p": 947,
          "m_factors": {
            "3": 1,
            "5": 1,
            "61": 1
          },
          "depth": 1,
          "R_size": 22
        }
      ]
    },
    {
      "N": 1928,
      "B_size": 41,
      "terminal": [
        {
          "C": [
            3,
            5,
            7,
            11,
            13,
            17,
            53,
            71,
            73,
            103,
            113,
            383,
            619,
            641
          ],
          "size": 14,
          "pure_rows": 0,
          "M": 641,
          "two_min_rows_gcd": 1,
          "height_lhs": 0.6970397506754565,
          "log10_A": 37.78714243994619,
          "rows": {
            "3": {
              "5": 2,
              "7": 1,
              "11": 1
            },
            "5": {
              "3": 1,
              "641": 1
            },
            "7": {
              "17": 1,
              "113": 1
            },
            "11": {
              "3": 3,
              "71": 1
            },
            "13": {
              "5": 1,
              "383": 1
            },
            "17": {
              "3": 1,
              "7": 2,
              "13": 1
            },
            "53": {
              "3": 1,
              "5": 4
            },
            "71": {
              "3": 1,
              "619": 1
            },
            "73": {
              "5": 1,
              "7": 1,
              "53": 1
            },
            "103": {
              "5": 2,
              "73": 1
            },
            "113": {
              "3": 1,
              "5": 1,
```

```
                "11": 2
              },
              "383": {
                "3": 1,
                "5": 1,
                "103": 1
              },
              "619": {
                "7": 1,
                "11": 1,
                "17": 1
              },
              "641": {
                "3": 2,
                "11": 1,
                "13": 1
              }
            }
          }
        ],
        "roots": [
          {
            "k": 0,
            "p": 967,
            "m_factors": {
              "31": 2
            },
            "depth": 2,
            "R_size": 16
          },
          {
            "k": 1,
            "p": 971,
            "m_factors": {
              "3": 1,
              "11": 1,
              "29": 1
            },
            "depth": null,
            "R_size": 18
          },
          {
            "k": 2,
            "p": 977,
            "m_factors": {
              "3": 1,
              "317": 1
            },
            "depth": null,
            "R_size": 17
          }
        ]
      },
      {
        "N": 2200,
        "B_size": 2,
        "terminal": [
          {
            "C": [
              3,
              13
            ],
            "size": 2,
            "pure_rows": 2,
            "M": 13,
            "two_min_rows_gcd": 1,
            "height_lhs": 5.125336663215231,
            "log10_A": 9.33629253712542,
            "rows": {
              "3": {
                "13": 3
              },
              "13": {
                "3": 7
              }
            }
          }
        ],
        "roots": [
          {
            "k": 0,
            "p": 1103,
            "m_factors": {
              "1097": 1
            },
            "depth": 0,
            "R_size": 0
          },
          {
            "k": 1,
            "p": 1109,
            "m_factors": {
              "1091": 1
            },
            "depth": 0,
            "R_size": 0
          },
          {
            "k": 2,
            "p": 1117,
            "m_factors": {
              "3": 1,
              "19": 2
            },
            "depth": 3,
            "R_size": 5
```

```
        }
      ]
    },
    {
      "N": 6142,
      "B_size": 66,
      "terminal": [
        {
          "C": [
            3,
            5,
            7,
            13,
            17,
            19,
            157,
            227,
            409,
            877
          ],
          "size": 10,
          "pure_rows": 0,
          "M": 877,
          "two_min_rows_gcd": 1,
          "height_lhs": 1.792329432195893,
          "log10_A": 28.764079106274437,
          "rows": {
            "3": {
              "7": 1,
              "877": 1
            },
            "5": {
              "17": 1,
              "19": 2
            },
            "7": {
              "3": 1,
              "5": 1,
              "409": 1
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            "13": {
              "3": 3,
              "227": 1
            },
            "17": {
              "5": 3,
              "7": 2
            },
            "19": {
              "3": 1,
              "13": 1,
              "157": 1
            },
            "157": {
              "3": 2,
              "5": 1,
              "7": 1,
              "19": 1
            },
            "227": {
              "5": 1,
              "7": 1,
              "13": 2
            },
            "409": {
              "3": 2,
              "7": 2,
              "13": 1
            },
            "877": {
              "3": 4,
              "5": 1,
              "13": 1
            }
          }
        }
      ],
      "roots": [
        {
          "k": 0,
          "p": 3079,
          "m_factors": {
            "3": 1,
            "1021": 1
          },
          "depth": 2,
          "R_size": 12
        },
        {
          "k": 1,
          "p": 3083,
          "m_factors": {
            "7": 1,
            "19": 1,
            "23": 1
          },
          "depth": 2,
          "R_size": 13
        },
        {
          "k": 2,
          "p": 3089,
          "m_factors": {
            "43": 1,
            "71": 1
          },
```

```
          "depth": null,
          "R_size": 15
        }
      ]
    }
  ],
  "N92": {
    "root": 47,
    "shortest_good_depth": 1,
    "R": [
      5,
      7,
      17,
      29,
      47
    ],
    "good": [
      3
    ]
  },
  "checks": "all assertions passed"
}
```


```
RESULT-ID: H1-ALL-BRANCH-INJECTIVE-ESCAPE-20260905
STATUS: PROVED (strict strengthening of the preceding two-generation theorem).
EXACT STATEMENT:
Keep the hypotheses and explicit F_2(y) of H1-TWO-GENERATION-ENVELOPE-ESCAPE:
N=2(P-1), P prime, m=P-2 composite, Q=PF(m) subset {odd primes <=y},
and N>=F_2(y). Then EVERY q in Q satisfies P^+(N-q)>y.
Moreover the sets
  L_q={r>y: r prime and r divides N-q}, q in Q,
are all nonempty and pairwise disjoint.
Consequently either the canonical traversal succeeds at depth <=1, or its
actual depth-2 frontier contains at least |Q| DISTINCT prime labels >y.
If any of those new labels is GOOD, the traversal succeeds at depth <=2.

PROOF:
Inspect the previous full proof. Its use of q=min PF(m) was unnecessary:
all it needed was q in Q, 3<=q<=y, and v=N-q=2m+2-q.
The same nonzero form Lambda=2m/v-1=(q-2)/v has the same uniform bound
for every q in Q. Thus every L_q is nonempty.
If a prime r belonged to L_q and L_q' for q!=q', it would divide
(N-q)-(N-q')=q'-q. But 0<|q'-q|<y<r, impossible.
If every q in Q is BAD, the stopped traversal expands each such child in
world mode, and each selected r_q in L_q is reached. Their disjointness
makes them |Q| distinct new prime vertices, none in Q. Also each N-q
is odd and composite, so each factor r_q< N/3<P; none is the original
root. Their shortest root distance is therefore exactly 2. They are active
by the activity proof in the preceding result. If some q in Q is GOOD, success
has already occurred at depth 1. This proves every alternative.

HOSTILE CHECK:
This is NOT a claim that every residual gcd quotient is free of Q. Old
prime-power excess may coexist with the forced large factor in each row.
The injection uses r>y and the ACTUAL row difference; it is not Hall
expansion inferred from a generic collision average. In an implementation
that stops as soon as any GOOD is found, the entire frontier need not be
materialized; the alternative is success or the complete stopped graph's
stated expansion. No expansion is asserted at later generations.

DEPENDENCIES:
Exact h=1 relation; verified Matveev bound; elementary row-gap divisibility.
No PR-MQ, Scott classification, global failure, or conjectural prime tuples.
RELATION TO HISTORICAL FAILURES:
The positive consumer escaping the previous collision-budget obstruction is
an INDEPENDENT height bound on complete y-smooth rows. It is precisely the
piece absent from attempts to infer new support merely from singleton mass.
This does not erase those counterexamples: for each fixed y the result
applies only beyond the enormous explicit F_2(y).
RELATION TO GOLDBACH:
Success OR an injective collection of actual new reachable primes. Still no
control of the complete rows at those new primes, so no proof of termination.
RELATION TO p0 EXCEPTIONALITY:
A nontrivial dynamical theorem for canonical h=1 roots with bounded initial
prime envelope relative to N. It is not yet a theorem distinguishing p0 from
p1: no same-N failure of this property for p1 has been established.
WHAT REMAINS OPEN:
Control after this forced exit. The new labels may be as large as N/3, at
which point F_2 of the enlarged envelope greatly exceeds N; automatic
iteration is invalid.
NOVELTY: NOT YET AUDITED.

FINAL HOSTILE AUDIT / LIMITATIONS
1. H1-ALL-BRANCH-INJECTIVE-ESCAPE is a conditional arithmetic theorem with a
   very severe N-versus-y hypothesis. No explicit example with N>=F_2(y),
   and no infinite prime family satisfying that small-envelope hypothesis,
   is supplied in this run. Its logical consumer is correct; its practical
   nonvacuity at that scale and usefulness for a Goldbach architecture are
   not established. Do not silently report it as a typical-input statement.
2. The support-height theorem applies to a terminal core without assuming
   two pure rows. The finite cores with no pure rows verify this is an actual
   extension of the historical domain, but the numeric bound is very weak.
3. BS-1996 counts solutions; it does not bound their heights. The height
   bound uses a separate, explicitly verified Matveev input. These dependencies
   must not be interchanged.
4. The new bounded-alphabet counts do not bound the union over arbitrary
   moving alphabets. Their density corollary explicitly confines all supports
   to the primes <=y(X) before applying the bound.
5. The small-child theorem produces MANY actual BAD children, not an ALL-BAD
   first frontier. Large GOOD children are its exact remaining obstruction.
   Runtime consequences specify the sorting and early-stopping conventions.
6. The strongest growing-direct-prefix estimate from the INPUT checkpoint
   remains valid after the audit. It controls direct Goldbach pairs only;
   no theorem here promotes its failure to graph-traversal failure.
7. FBTU is not declared false as a theorem about admissible partial trees.
   The inferred blanket theorem about complete BFS neighbourhoods is not
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   established by the supplied proof. Older Biblia iteration 108 already
   documents the residual-factor obstruction. No opposite complete-depth
   theorem has been proved in this continuation.
8. The Scott-Bozek even-exponent branch was neither eliminated nor used as
   a dependency. No published theorem is declared false in this report.
9. No source correction, necessary condition, finite computation, or
   density-one statement is called a proof of binary Goldbach.
10. Proofs and targeted arithmetic checks were performed in this continuation.
    They have not received a separate external adversarial review. No claim
    of literature novelty is made for any new application.

FAIL-ID: ENVELOPE-ESCAPE-AUTOMATIC-ITERATION-20260905
STATUS: KILLED AS AN INFERENCE; the two-generation theorem remains PROVED.
CLAIM / ROUTE: Iterate H1-ALL-BRANCH-INJECTIVE-ESCAPE until a finite graph
cannot contain more vertices, thereby obtaining a Goldbach pair.
WHY IT LOOKED PROMISING: The theorem gives an injective set of actual new
large primes using an independent height bound, rather than a scalar identity.
EXACT FAILURE: At the next generation the relation N=2m+2 is no longer a
relation with the NEW row anchor. Also the larger alphabet envelope may be
of order N, while the premise N>=F_2(y_new) then fails spectacularly.
The new labels may enter a closed large core. Neither injectivity nor the
height premise persists automatically.
COUNTEREXAMPLE / OBSTRUCTION: The explicit changing-envelope dependence of
F_2 and loss of the h=1 row identity are sufficient obstructions; this is
not a constructed canonical failure.
WHAT SURVIVES: One fully justified two-generation expansion, and the global
support-height constraint on any subsequently closed core.
DO NOT REPEAT: Do not replace a fixed y in the proved premise by the newly
produced, larger y without verifying the premise again.
POSSIBLE REPLACEMENT: A different invariant or a height-sensitive argument
that survives changing alphabets and is forced by actual closed failure.

REFERENCE BOOK
Yann Bugeaud, Linear Forms in Logarithms and Applications, EMS, 2018.
Publisher: https://ems.press/content/book-files/22003
For ordinary prime-counting and probabilistic number theory, the input
checkpoint's G. Tenenbaum, Introduction to Analytic and Probabilistic
Number Theory, 3rd ed., AMS, 2015, remains an appropriate background book.

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STRONGEST NEW RESULTS
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1. UNRESTRICTED-CLOSED-SUPPORT-HEIGHT: removes the historical two-pure-row
   assumption. EFFECTIVE-BOUNDED-CORE-THRESHOLD gives a completely explicit
   F(y) excluding every closed active BAD core supported below y once
   N>=F(y). POINTWISE-CORE-LABEL-GROWTH gives the resulting iterated-log
   lower bound on the largest core prime, uniformly over arbitrary branching.
2. H1-ALL-BRANCH-INJECTIVE-ESCAPE: beyond explicit F_2(y), a canonical h=1
   root with y-smooth composite complement either succeeds at depth 1 or
   reaches at least |PF(m)| distinct primes >y at depth 2. Severe premise;
   automatic iteration and large-scale nonvacuity are not established.
3. CLOSED-ALPHABET-COUNT: at most binomial(s,2)*2^(9s+16) distinct even N
   support a closed active BAD set inside any one fixed alphabet of s primes.
   A common slowly growing prime envelope gives a rigorous density corollary.
4. SMALL-BAD-CHILDREN-PREFIX: simultaneous actual first-front BAD growth for
   a slowly growing prefix of roots, with exact search-cost consumers and
   no inference about shortest GOOD depth. This is shared by nearby roots.
5. AUDIT-FBTU: corrects the later partial-tree/full-BFS scope regression;
   N=92 is a short exact certificate, and complete h=1 two-row finiteness
   disproves an infinite fixed-alphabet/full-row interpretation.
The original prefix-transfer estimate survived the audit; it is inherited
work, not counted as a new result of this continuation.

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CURRENT FRONTIER
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No proof of binary Goldbach. No unconditional Goldbach threshold N_0.
A globally failed N supplies a closed active BAD core. A failed canonical
traversal also supplies a reachable closed BAD core, but does not imply that
all other vertices of the graph are BAD. The new results constrain each such
core's height and support, and exclude a quantitatively specified small-label
regime. They do not rule out moving supports above that regime.
The h=1 expansion theorem ends at new prime labels, not at GOOD vertices.
No proved invariant controls their next complete rows strongly enough to
force another expansion, and no contradictory upper support bound was found.
A pointwise positive prime-pair count or a contradiction to full composite
closure remains missing. Existing density-one conclusions cannot remove
every exceptional N.

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p0 STATUS
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Found a genuine restricted h=1 arithmetic/reachability theorem with an
explicit size hypothesis. Did NOT find a property proved for every canonical
p0 whose corresponding claim fails at a nearby root for the same N.
The density-transfer and small-BAD-child results are shared by nearby roots.
The claimed broad finite-local no-go needs the scope correction above; partial
CRT programming is not a theorem about complete bounded-depth failure.
There is still no convincing mathematical explanation of universal canonical
traversal success, and that universal success remains unproved.

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EXACT NEXT QUESTION
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For every integer d>=1, does there exist a prime P>=5 with N=2(P-1),
P-2 composite, such that EVERY vertex at directed distance at most d from
canonical P in the factorization graph is BAD (positive composite complement)?
Equivalently, are the minimum distances from canonical h=1 roots to GOOD
vertices unbounded, allowing infinity for a failed traversal?
A proof must control the COMPLETE factor frontiers, including every residual
cofactor; programming a selected BAD path is not an answer. Resolving this
would settle the strongest presently unsupported bounded-depth no-go in the
corpus, or provide a uniform finite-depth success theorem on the h=1 family.

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CONTINUATION 2026-09-06
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Previous sections remain an archive of the 2026-09-05 run; later entries
below override any conflicting frontier statement. The present run starts
from the complete-support / moving-support obstruction. No assertion of a
Goldbach proof or of universal canonical traversal success is assumed.

RESULT-ID: TWO-SMALLEST-ROWS-HEIGHT-20260906
STATUS: PROVED; refinement of UNRESTRICTED-CLOSED-SUPPORT-HEIGHT.
EXACT STATEMENT:
Let N>=8 be even, and C a nonempty finite set of primes p<N with p not
 dividing N, N-p composite, and PF(N-p) subset C. Write a<b for the two
smallest labels in C, U=N-a, V=N-b, S=PF(UV), t=|S|, and
 A(S)=1.4*30^(t+3)*t^(9/2)*product_(q in S) log q.
Then:
 (i) a<b<sqrt(N), and gcd(U,V)=1;
 (ii) log((N-b)/(b-a))
       < A(S)*(1+log(log N/log 3));
 (iii) log(sqrt(N)-1)
       < A(S)*(1+log(log N/log 3)).
In particular the full support of just these two rows, even inside an
arbitrarily branching core with large other labels, must satisfy
 product_(q in S) log q >
 log(sqrt(N)-1) /
 [1.4*30^(t+3)*t^(9/2)*(1+log(log N/log 3))].

PROOF:
A singleton C={a} would give N-a=a^e, hence a|N, contradicting activity.
All labels are odd. The least prime divisor r of the odd composite U is
at most sqrt(U)<sqrt(N). Closure gives r in C, and activity gives r!=a,
because a|(N-a) would imply a|N. Hence b<=r<sqrt(N), proving (i)'s
size assertion, without any assumption about pure-power rows.
If a prime ell divided U and V, then ell belongs to C and divides b-a.
It cannot equal a, by activity of a, and every other element of C is at
least b>b-a. Thus no such ell exists and gcd(U,V)=1.
Now U/V-1=(b-a)/(N-b)>0. Its prime bases are exactly S, with nonzero
signed exponents e_q=v_q(U)-v_q(V) and |e_q|<=log N/log 3. Apply the
rational real-field product-minus-one Matveev bound recorded above, with
A_q=log q, to obtain (ii). Finally b-a<b<sqrt(N) and
N-b>N-sqrt(N), whence (N-b)/(b-a)>sqrt(N)-1. This gives (iii).

HOSTILE CHECK:
The fragile elementary step is excluding a as a factor of its own row;
that requires activity, not merely compositeness. If inactive vertices
were allowed, singleton examples N=a+a^2 invalidate the claim. The bound
uses the actual union of two row supports, not necessarily all of C.
Nothing requires that either row be a prime power. The older result that
a terminal BAD SCC has a cycle below sqrt(N) is already stated in the
Biblia (late synthesis, near line 49182 in the supplied copy); the small
cycle itself is not claimed as new. Here the consumer is the quantitative
height comparison of the two least rows. This improves the previous
M/(N-M) comparison, but remains only a necessary condition on a failed
world. It does not put an upper bound on the moving support.

DEPENDENCIES:
Full factor closure; activity; Matveev-in-BMS-9.4. The external input is
being rechecked in the author's PDF in this run. No Scott classification.
RELATION TO GOLDBACH: Applies to any closed failed core, whether supplied
by global failure or by one failed traversal. Does not exclude all cores.
RELATION TO p0: Shared by all roots whose traversals fail. No asymmetry.
WHAT REMAINS OPEN: A contradictory upper support bound or another genuine
consumer of the close pair of coprime rows.
NOVELTY: NOT YET AUDITED.


SOURCE-AUDIT 2026-09-06:
The Matveev input was rechecked in Bugeaud--Mignotte--Siksek, Theorem 9.4,
author PDF printed pp.16--17. It is indeed the product-minus-one bound,
with real-field coefficient 1.4*30^(n+3)*n^4.5*(1+log B) for D=1.
The preceding two-smallest-row refinement uses exactly that statement.

EXTERNAL-INPUT: FKL-2010-LEMMA-5.1
STATUS: EXACT STATEMENT CHECKED in primary research, including uniformity.
Kevin Ford, Sergei V. Konyagin, Florian Luca, Prime chains and Pratt trees,
Geom. Funct. Anal. 20 (2010), 1231--1258, Lemma 5.1, printed pp.12--13.
https://arxiv.org/pdf/0904.0473
For k primitive integer linear forms a_i n+b_i, a_i>0, let nu(ell) count
their roots modulo ell, B=sum_ell (k-nu(ell))*log(ell)/ell, and
 S=product_ell (1-nu(ell)/ell)*(1-1/ell)^(-k).
There is an absolute delta>0 such that, if x>10,
k<=delta*log(x)/loglog(x), B<=delta*log(x), the number of positive n<=x
for which all forms are prime and exceed k is at most
 2^k*k!*x*S/(log x)^k *
 exp(O((k*B+k^2*loglog x)/log x)).
The displayed O constant is absolute. We only use k=2,3,4. All pairwise
determinants are checked nonzero below; nonprimitive forms and prime
values <=k are removed before application. This is an upper-bound sieve,
not a prime-tuples lower bound.

RESULT-ID: UNIFORM-FEW-FORMS-SIEVE-CONSUMER-20260906
STATUS: PROVED from FKL-2010-LEMMA-5.1.
EXACT STATEMENT:
Fix r in {2,3,4}, epsilon>0, C>0. Let primitive linear forms
 L_i(n)=a_i*n+b_i, a_i>0, have nonzero pairwise determinants and
 1<=Delta=abs(product_i a_i * product_(i<j)(a_i*b_j-a_j*b_i))<=X^C.
Suppose X^epsilon<=z<=X^C. Uniformly in these forms, the number of
positive n<=z with every L_i(n) prime and >r is
 O_(epsilon,C)(z*(loglog X)^r/(log z)^r).
If nu(ell)>=1 for every prime ell, improve the exponent r on loglog X
to r-1. The same improvement holds when the only possible exception is
one odd prime q>=3; its extra local factor is bounded by (3/2)^r.
All assertions are for sufficiently large X; changing constants covers
bounded X in any use where the logarithms are replaced by positive logs.

PROOF:
For ell not dividing Delta, nu(ell)=r, since all leading coefficients are
invertible and the r roots are distinct. Always 0<=nu(ell)<=r. Thus
 B<=r*sum_(ell|Delta) log(ell)/ell = O_C(loglog X).
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Here is an elementary justification of the uniform bound: put
T=max(3,log(3*Delta)). For ell<=T the sum is O(log T), by partial
summation from theta(t)=sum_(ell<=t)log ell=O(t). For ell>T it is at most
(log Delta)/T<=1. Chebyshev's theta bound is sufficient (and also follows
from the PNT already used in the checkpoint).
Similarly
 product_(ell|Delta)(1-1/ell)^(-1)=O(loglog(3*Delta)).
For completeness, the factors ell>T have bounded product because their
logarithms are O(sum_(ell|Delta,ell>T)1/ell)=O(1). For ell<=T, put
sigma=1+1/log T. Comparing Euler factors at 1 and sigma costs O(1) in
logarithm: the first-power difference is at most
(sigma-1)*sum_(ell<=T)log(ell)/ell=O(1), and higher powers have a
bounded sum. The product at sigma is <=zeta(sigma)<=1+1/(sigma-1),
by the integral comparison for sum n^(-sigma). Thus the product is O(log T).
For ell not dividing Delta, the local singular-series factor is <=1,
by 1-r/ell<=(1-1/ell)^r; primes ell<=r outside Delta cannot occur
unless local admissibility fails. At exceptional primes use the bound
(1-1/ell)^(-r). If nu(ell)>=1, use the sharper exponent -(r-1).
This proves S=O_C((loglog X)^r), or the stated improvement. If S=0,
a prime ell<=r divides at least one form for every n, so no tuple of
prime values >r occurs. Finally B=O_C(loglog X) and log z>=epsilon log X
verify both FKL size hypotheses for large X; its exponential factor is
bounded. This proves the claimed uniform bounds.

HOSTILE CHECK:
A fixed-coefficient asymptotic sieve result would be insufficient here;
FKL supplies the needed uniform B condition. Common leading primes may
have nu=0 even for primitive forms, so the exponent r-1 is NOT used
without the explicitly stated local check. A zero pair determinant
would destroy the r-dimensional sieve gain; such cases are checked
separately in each consumer. This lemma is an application, not a new
claim about the underlying sieve theorem.
DEPENDENCIES: FKL Lemma 5.1; elementary Euler products and Chebyshev bound.
NOVELTY: NOT YET AUDITED.

RESULT-ID: COMPLETE-H1-FIRST-FRONT-RARE-SUCCESS-20260906
STATUS: PROVED.
EXACT STATEMENT:
For X sufficiently large, the number of primes P in [X,2X] for which
N=2(P-1), root P, has a GOOD vertex at directed distance 0 or 1 is
 O(X*(loglog X)^2/(log X)^2).
Thus, with relative density one among primes P, P and EVERY prime
factor q of P-2 are BAD for this same N. This is a statement about the
COMPLETE radius-one frontier, including all large prime factors.
It does not say the traversal fails at later generations.

PROOF:
P is canonical because N/2=P-1 and P-1 is an even integer >2.
A GOOD root requires P-2 prime. The two-form sieve for n,n-2 counts
these primes by O(X/(log X)^2).
Otherwise m=P-2 is an odd composite. If a first-generation vertex q is
GOOD, write m=a*q, where q is an odd prime and a>=3 is odd. Then
 P=a*q+2,             N-q=(2a-1)*q+2
are both prime. Count such pairs (a,q); overcounting several GOOD
children of the same P is harmless.
For q<=sqrt X, fix q and sieve in a<=2X/q the two forms
 q*a+2,              2q*a+2-q.
They are primitive: gcd(q,2)=1 and gcd(2q,2-q)=1. Their determinant is
-q*(q+2), nonzero, and Delta=2q^3*(q+2)<=O(X^2). Except at ell=q
there is at least one root of their product modulo every prime ell
(the first form suffices unless ell=q). The uniform two-form consumer
therefore gives
 O(X*loglog X/(q*(log X)^2)).
Summing over prime q<=sqrt X costs O(loglog X), by the Euler-product
bound proved above, giving O(X*(loglog X)^2/(log X)^2).
For q>sqrt X, fix odd a>=3. Now a<=2sqrt X and q<=2X/a. Sieve in q
using the THREE forms
 q,                  a*q+2,                (2a-1)*q+2.
All are primitive. Their pair determinants are 2, 2, 2(1-a), nonzero.
The product Delta is polynomially bounded in X. There is at least one
root modulo every prime, because the first form is q itself. Hence
the count for each a is
 O(X*(loglog X)^2/(a*(log X)^3)).
Sum over a<=2sqrt X; the harmonic sum is O(log X). The resulting bound
is again O(X*(loglog X)^2/(log X)^2). All prime values in the actual
count exceed the sieve's r, since q>sqrt X and P,N-q have size at least
X up to an absolute constant. The sieve variable in both cases has
upper limit >=sqrt X, verifying the uniformity premise. This proves
the asserted count.
PNT gives #{P in [X,2X]}~X/log X; dividing proves relative density one.
When the root is BAD, each q|m is active: a prime dividing q and N
would divide P, impossible since q<P. Also q<=m/3, so N-q>1. Removing
the prime case makes N-q positive composite, not a unit or inactive row.

HOSTILE CHECK:
The identity in the sieve is N-q=2a*q+2-q, not 2a*q+4-q.
The large-q range is NOT discarded as negligible and is NOT treated
with a short progression estimate. Switching to the small cofactor a
creates three distinct prime forms and controls that entire range.
This is the step absent from a small-child-only argument. The result
is relative to the thin family N/2+1 prime, not density one among all
even N. The union bound counts all first-front GOOD children; no
independence of primality and factorization is assumed.
DEPENDENCIES: The uniform two-/three-form sieve consumer; PNT only for
the relative-density interpretation. No conjectural prime tuples.
RELATION TO GOLDBACH: A rigorous limitation on shallow traversal in an
infinite canonical family. It neither supplies nor denies a Goldbach
representation using deeper reached vertices or other primes.
RELATION TO p0: Actual complete-frontier statement for canonical h=1.
Nearby roots for the SAME N are tested in the next result.
WHAT REMAINS OPEN: Complete radius-two control, and every fixed depth.
NOVELTY: NOT YET AUDITED.

RESULT-ID: SAME-N-GROWING-ROOTS-COMPLETE-FIRST-FRONT-20260906
STATUS: PROVED.
EXACT STATEMENT:
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Let P run over primes in [X,2X], let N=2(P-1), and let p_k(N) be the
(k+1)-st prime at or above N/2; in particular p_0=P. Let K=K(X)>=1 be
integer-valued with
 K=o(log X/(loglog X)^3).
For all but o(X/log X) such primes P, EVERY root p_k, 0<=k<=K, is BAD
and EVERY prime divisor q of N-p_k is also BAD. Thus none of these
roots has a GOOD vertex at distance <=1 in its complete stopped graph.
This compares all the roots for the SAME N, and includes all their
large as well as small first-front factors.
For 1<=K<=log X/(loglog X)^3, a quantitative bound for the exceptional
proportion (relative to X/log X) is
 O((loglog X)^2/log X + sqrt(K*(loglog X)^3/log X)).            (R1)
For example K=floor(log X/(loglog X)^5) gives proportion O(1/loglog X).
These are asymptotic statements; they do not assert this proportion at
currently accessible numerical sizes.

PROOF:
Write L=log X and l=loglog X. First isolate p_0: the preceding theorem
counts its exceptional P by O(X*l^2/L^2).
For any later prime Q=p_k, put h=Q-P+1. Then h is an odd integer >=3,
 P=Q-h+1,       N=2(Q-h),       m=N-Q=Q-2h.
We first count all such pairs (P,Q) with 3<=h<=H, without imposing any
rank condition on Q. Assume 2<=H<=sqrt(X)/10, P in [X,2X], Q prime.
Then Q<=3X for large X. For EACH h the number of these pairs with a
GOOD root or a GOOD first-front child is
 O(X*l^3/L^3), uniformly in h.                              (R2)
We prove (R2) carefully.
If m is prime, the three forms in Q are
 Q,                    Q-h+1,                    Q-2h.
They are primitive, distinct since h>1, with nonzero shift differences
h-1, 2h, h+1. Their Delta is polynomially bounded in X and nu(ell)>=1.
The three-form consumer bounds the count by O(X*l^2/L^3).
Otherwise m is odd composite. For a GOOD child q write m=a*q, a>=3 odd.
The quantities
 Q=a*q+2h,
 P=a*q+h+1,
 g=N-q=(2a-1)*q+2h
are prime. Split into q<=sqrt X and q>sqrt X.
For small prime q, sieve in a<=3X/q the THREE forms
 q*a+2h,             q*a+h+1,              2q*a+2h-q.
If q divides h or h+1, one of Q,P would be divisible by q while larger
than q, so this tuple contributes nothing. For every remaining q these
three forms are primitive, including the last since its constant is
odd and coprime to q. Their pair determinants are
 q*(1-h),           -q*(q+2h),           -q*(q+2),
all nonzero. Delta is bounded by X to an absolute constant power.
Except possibly at ell=q there is a root modulo every prime. Hence
the uniform three-form count is O(X*l^2/(q*L^3)). Summing 1/q over
prime q<=sqrt X gives O(l), and this range contributes O(X*l^3/L^3).
For large q>sqrt X, fix a<=3sqrt X and sieve in q<=3X/a the FOUR forms
 q,       a*q+2h,       a*q+h+1,       (2a-1)*q+2h.
If any of the last three forms is not primitive, every value is a
multiple of a fixed integer d>1 with d<=2h+1<X. But its actual prime
value (Q, P, or g) is >=X for large X. Hence that a has no solutions.
For the remaining a all forms are primitive. The determinants between
the first form and the others are 2h, h+1, 2h. The other three are
 a*(1-h),           2h*(1-a),           h+1-2a.
The first two do not vanish. If the last vanished, then h+1=2a and
gcd(a,h+1)=a>1, a case just excluded. Thus there is no hidden
coincidence of two prime forms. Delta is polynomially bounded in X,
and nu(ell)>=1 because the first form is the variable q. The four-form
bound is O(X*l^3/(a*L^4)). Summing 1/a up to 3sqrt X costs O(L),
giving O(X*l^3/L^3). All relevant prime values exceed four. Both
sieve-variable upper limits are >=sqrt X, so the uniform constants
are absolute, independently of h<=H. This proves (R2).
Summing (R2) over h<=H bounds the number of exceptional P with ANY
such nearby Q by O(H*X*l^3/L^3). No extra factor K is required: every
relevant pair (P,Q) is already in this count.
It remains to show that the first K successors usually lie within
Q-P<=H-1. For the indicated range of K, PNT ensures that p_K<=3X for
every P in [X,2X] and sufficiently large X. Write each p_K-P as the
sum of its K consecutive prime gaps. Summing over starting primes P
in [X,2X], every gap is used at most K times and all gaps lie between
primes in [X,3X]. Their total length is <=2X. Therefore
 sum_(P in [X,2X] prime)(p_K-P)<=2KX,
so at most 2KX/(H-1) of these P have p_K-P>H-1.
Combining all three exceptional sets and dividing by X/L yields
 O(l^2/L + K*L/(H-1) + H*l^3/L^2).
Take H to be an integer within O(1) of sqrt(K*L^3/l^3). Uniformly in
1<=K<=L/l^3, H tends to infinity and is <sqrt X/10 for large X.
This proves (R1). If K=o(L/l^3), both terms in (R1) tend to zero.
For every remaining P, all roots have positive odd complements; their
proper prime factors are active by gcd(N,Q)=1 and q<Q. Indeed Q>P
is close to P, Q>N/2 and Q<N, so it cannot divide the even number N
(the only possible quotient would be 1). A root is BAD after excluding
the prime case, and each child complement exceeds 1 and is composite
after excluding all the counted GOOD children. This completes the
full stopped-graph conclusion.

HOSTILE CHECK:
1. The rank condition was used only in the unweighted prime-gap sum.
   No claim about typical polylogarithmic gaps for uniformly sampled
   even N is required. Sampling is over prime P, not over gap-weighted x.
2. The extra primality P=Q-h+1 is essential. Without it the sieve loses
   one logarithm and summing over h in the required range does not give
   this result. For h=1 these forms coincide; p_0 was handled separately
   by the two-/three-form theorem, not by a false four-prime count.
3. The potentially vanishing determinant h+1-2a forces an imprimitive
   form and has no actual prime solutions in the stated large-X range.
4. The conclusion is FULL radius-one BAD, not a selected path, not a
   statement that only many children are BAD, and not failure of BFS.
5. The property is explicitly SHARED by nearby roots at the same N.
   It is not evidence of an advantage peculiar to p_0. It rules out
   shallow-success explanations only in the precise asymptotic family
   and prefix stated here, leaving all deeper behavior open.
6. The count remains true if the algorithm has arbitrary child order:
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   it is a graph-theoretic distance assertion, not a runtime statistic.

DEPENDENCIES: Uniform FKL sieve consumer, elementary harmonic sums,
PNT for the prime-sample density and availability of K primes below 3X.
RELATION TO GOLDBACH: No representation is ruled out outside the tested
one-step factor neighborhoods. Many steps or other roots may succeed.
RELATION TO p0: A rigorous SAME-N asymptotic no-go for distinguishing
p_0 from a growing nearby prefix by existence of a depth-0/1 GOOD node.
WHAT REMAINS OPEN: Control the complete second frontier; prove or refute
unbounded shortest GOOD distance in the h=1 family.
NOVELTY: NOT YET AUDITED.

RESULT-ID: SAME-N-NEARBY-ROOT-FIRST-FRONT-SIEVE-20260906
STATUS: PROVED.
EXACT STATEMENT:
Let X be sufficiently large and 3<=H<=sqrt(X)/10. Count pairs of odd
primes P,Q with X<=P<=2X, P<Q<=3X, and h=Q-P+1<=H. Set N=2(P-1).
The number of these pairs for which the stopped traversal from Q has
a GOOD vertex at distance 0 or 1 is
 O(H*X*(loglog X)^3/(log X)^3),
with an absolute implied constant. No consecutiveness assumption on
P,Q is needed. All first-front prime factors are covered.

PROOF:
Fix h. Since P,Q are odd and Q>P, h is odd and h>=3. The identities are
 P=Q-h+1,   N=2Q-2h,   m=N-Q=Q-2h.
Because h<=sqrt X/10 and Q>=X, m>1, and all relevant rows below have
size bounded below by a constant times X. A GOOD root Q requires the
THREE forms Q, Q-h+1, Q-2h to be prime. Their nonzero pair differences
are h-1, 2h, h+1. Leading coefficients are 1, so they are primitive
and nu(ell)>=1. Uniform three-form sieving in Q<=3X bounds the count
by O(X*(loglog X)^2/(log X)^3).
Otherwise m is odd composite. Write m=a*q, with a>=3 odd and q an odd
prime. A GOOD child q requires primality of
 Q=a*q+2h,       P=a*q+h+1,       g=N-q=(2a-1)*q+2h.             (S)
For q<=sqrt X fix q and sieve in a<=3X/q the three forms
 q*a+2h,        q*a+h+1,         2q*a+2h-q.
If q divides 2h, Q would be a multiple of q smaller than Q, impossible.
If q divides h+1, the same holds for P. Thus parameters violating
these coprimality conditions contribute no actual pair. In the remaining
case all three forms are primitive (their odd constant in the third
form also excludes a common factor 2). Their pair determinants are
 q*(1-h),       -q*(q+2h),       -q*(q+2),
all nonzero. Delta is polynomially bounded in X, uniformly in h<=H.
For ell!=q, the first form has a root modulo ell; only the one odd
prime q may have nu(q)=0. The improved r=3 sieve bound gives
 O(X*(loglog X)^2/(q*(log X)^3)).
The sum over prime q<=sqrt X is therefore
 O(X*(loglog X)^3/(log X)^3).
For q>sqrt X, fix a<=3sqrt X and sieve in q<=3X/a using FOUR forms:
 q,             a*q+2h,        a*q+h+1,        (2a-1)*q+2h.
If gcd(a,2h)>1 or gcd(a,h+1)>1 or gcd(2a-1,2h)>1, one of Q,P,g
has a fixed divisor between 2 and 2h, whereas the relevant prime value
is >=cX>2h. Hence there are no actual pairs for that parameter.
For the remaining parameters the four forms are primitive. Determinants
involving the first form are 2h,h+1,2h. The other three are
 a*(1-h),       2h*(1-a),       h+1-2a.
The first two are nonzero because h>=3 and a>=3. If h+1-2a=0, then
P=a*(q+2) is composite; equivalently gcd(a,h+1)=a>1 already excluded
this case. Thus no zero-determinant case is being passed to the sieve.
The first form ensures nu(ell)>=1 at every prime. Delta remains bounded
by a fixed power of X. The r=4 bound is
 O(X*(loglog X)^3/(a*(log X)^4)).
Summing a<=3sqrt X costs O(log X). This proves the same bound as in the
small-q range, for each fixed h. All actual prime values exceed 4 in
this range, and both sieve variable limits are >=sqrt X. Adding the
root-GOOD bound and summing over h<=H proves the statement.

HOSTILE CHECK:
The extra prime P=N/2+1 is essential for the extra sieve dimension.
Ignoring its primality would lose one log X and would not prove the
nearby-root prefix result below. It is available because all roots are
being compared for the SAME N in the h=1 family. For h=1, P and Q
coincide and this dimension gain disappears; that case is handled by
COMPLETE-H1-FIRST-FRONT-RARE-SUCCESS instead. The other possible
proportional pair at h+1=2a is not ignored: it has no prime solutions.
No typical prime-gap estimate is assumed in this counting lemma.
DEPENDENCIES: Uniform few-form sieve consumer, elementary identities.
RELATION TO GOLDBACH: Counts successful radius-one traversals only.
RELATION TO p0: Enables an actual same-N comparison with nearby roots.
WHAT REMAINS OPEN: Extend complete-front estimates to deeper generations.
NOVELTY: NOT YET AUDITED.

RESULT-ID: COMPLETE-FIRST-FRONT-NO-GO-GROWING-SAME-N-PREFIX-20260906
STATUS: PROVED.
EXACT STATEMENT:
Let K=K(X)>=1 be integers with
 K=o(log X/(loglog X)^3).
For all but o(X/log X) primes P in [X,2X], set N=2(P-1). Let Q_0=P,
Q_1,Q_2,... be the successive primes starting at P. Simultaneously for
EVERY 0<=k<=K, Q_k is BAD and EVERY prime factor of N-Q_k is BAD.
Equivalently none of these complete stopped traversals succeeds at
radius 0 or 1. Each Q_k is exactly p_k(N); these roots refer to the SAME N.
This is a relative-density-one theorem in the prime-parametrized h=1
family, not a density-one theorem among all even N.
One admissible concrete prefix is floor(log X/(loglog X)^5).
A quantitative upper bound for the exceptional proportion, up to an
absolute constant for sufficiently large X, is
 (loglog X)^2/log X + sqrt(K*(loglog X)^3/log X).

PROOF:
For K in the stated range, PNT ensures that there are more than K primes
between 2X and 3X for large X. Thus Q_K<=3X for every P in [X,2X].
Summing Q_K-P over these starting primes counts each consecutive prime
gap inside [X,3X] at most K times. The sum of all such gaps is at most
2X. Consequently
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 #{P in [X,2X] prime: Q_K-P>H-1} <=2KX/(H-1).                (G)
This uses only the sum of gaps, not a bound on individual gaps or a
second moment. If Q_K-P<=H-1, every k>=1 has h_k=Q_k-P+1<=H and the
pair (P,Q_k) lies in SAME-N-NEARBY-ROOT-FIRST-FRONT-SIEVE. Hence the
number of starting P admitting ANY successful radius-one traversal
among k=1,...,K with Q_K-P<=H-1 is at most
 O(H*X*(loglog X)^3/(log X)^3).
There is no extra K factor: that lemma already counts all eligible
pairs P,Q and all offsets h<=H, and a union is bounded by that count.
The k=0 exception is O(X*(loglog X)^2/(log X)^2), by the preceding
canonical theorem. Divide the sum of these three bounds by
#{P in [X,2X]}~X/log X. The exceptional proportion is
 O((loglog X)^2/log X + K*log X/H
                        + H*(loglog X)^3/(log X)^2).
Choose H equal to an integer within 1 of
 sqrt(K*(log X)^3/(loglog X)^3).
It tends to infinity and lies below sqrt X/10 for large X; the rounding
and H-1 in (G) change only the constant. The last two terms are both
of the asserted square-root order, and tend to zero by the assumption
on K. This proves the quantitative and qualitative assertions.
For good starting P the complements are positive: Q_k=P+O(H), so
N-Q_k=P-2-O(H)>1. No root divides N, since each root is prime in
(N/2,N), strictly above the diagonal. If a child q divided N then
q would divide its root Q_k, impossible because q<Q_k. Thus the BAD
conclusions have the active, positive-composite meaning, as required.

HOSTILE CHECK:
This gives COMPLETE first-front BADness, not merely a selected CRT
subtree or many small BAD children. It therefore repairs one limited
part of the old full-BFS no-go with a different, analytic proof.
It proves neither arbitrary depth nor failure of the complete traversal.
The relative family restriction is indispensable in this proof: the
additional prime P=N/2+1 supplies a sieve dimension for nearby roots.
Markov's estimate in (G) is over prime starts P, not over all integer
midpoints weighted by their containing prime gap. Confusing those two
sampling measures would be an invalid extension to all even N.
The theorem does not assert that the factor supports or deeper dynamics
of Q_0 and Q_1 agree; it asserts a shared entire radius-one failure event.

DEPENDENCIES:
FKL Lemma 5.1 via the fully checked consumers above; PNT; exact gap sum.
No Goldbach-failure hypothesis, no lower-bound prime-tuples conjecture,
no Scott--Bozek classification, no iteration of an envelope theorem.
RELATION TO GOLDBACH:
An unconditional obstruction to using a growing nearby-root prefix with
one factorization generation as a typical solver in this infinite family.
It does not obstruct multigeneration traversal or general Goldbach pairs.
RELATION TO p0:
A substantive same-N negative comparison: even an entire complete first
front is asymptotically BAD for p0 and a growing family of its neighbours.
The empirical exceptional behavior must, if real, be attributed to
something stronger than ubiquitous shallow success. No universal
p0-specific reachability advantage is proved here.
WHAT REMAINS OPEN:
In particular, no corresponding complete radius-two theorem is proved.
The original question about every fixed complete BAD BFS depth remains
open beyond radius one.
NOVELTY: NOT YET AUDITED.

RESULT-ID: COMPLETE-H1-RADIUS-TWO-RARE-SUCCESS-20260906
STATUS: PROVED (new full-generation extension; proof audited case by case).
EXACT STATEMENT:
Let P range over primes in [X,2X], N=2(P-1), canonical root P. The
proportion of these P for which ANY vertex at directed distance <=2
from P is GOOD tends to zero as X tends to infinity. In particular
there are infinitely many canonical h=1 roots whose COMPLETE radius-two
neighbourhood consists of BAD vertices.
More quantitatively, with L=log X and l=loglog X, the number of exceptional
P is O((X/L)*(l^3/L)^(1/5)). This bound is asymptotic with an absolute
constant; it is not an asserted numerical proportion at small X.
No conclusion is made about distance three or eventual traversal failure.

PROOF:
Success at distance <=1 is already counted by O(X*l^2/L^2). Consider a
shortest successful path P -> q -> r with q BAD and r GOOD. Put
 a=(P-2)/q,       b=(N-q)/r,       g=N-r.
Then q,r,P,g are odd primes, a,b are odd integers >=3, q!=r, and
 P=a*q+2,
 b*r=(2a-1)*q+2,
 g=2a*q+2-r.                                                (T)
We have q<(2/3)X, r<(4/3)X, P>=X, and g>2N/3>=cX.
Fix 0<eta<1/8, initially independent of X. Remove all P for which
P-2 has a prime divisor q in
 [X^(1/2-eta), X^(1/2+eta)].                                (BAND)
For a fixed prime q here, the one-form FKL bound applied to q*a+2,
a<=2X/q, gives O(X/(q*L)), uniformly for eta<1/8. Indeed its singular
series is (1-1/q)^(-1)<=3/2, its B is log(q)/q, and log(2X/q)>=3L/8.
Partial summation using pi(t)=O(t/log t) gives
 sum_(q in BAND prime)1/q=O(eta+1/L).
Consequently at most O((eta+1/L)*X/L) starting primes are removed.
This is a count of a factor occurrence, not an independence assertion.

Every remaining path has either q<X^(1/2-eta) or q>X^(1/2+eta).
Split also at r=sqrt X. In each of the following four cases we obtain
an arithmetic progression with at least a power of X available for
sieving. The product of the fixed parameters is <=8X^(1-eta).
All stated sieve-variable upper limits are therefore >=cX^eta.

CASE LL: q<X^(1/2-eta), r<=sqrt X. Fix the two distinct primes q,r.
The conditions are P=2 modulo q and 2P=2+q modulo r. They specify one
class A modulo q*r. Choose 1<=A<=q*r and write P=A+q*r*t. Since
q*r<X and P>=X, t>=1 and t<=2X/(q*r).
Sieve the TWO forms P and g=2P-2-r. Their determinant is
 -(q*r)*(r+2), nonzero. If either is imprimitive, its fixed divisor is
at most 2q*r=o(X), smaller than its actual prime value, so that class
has no solutions and can be discarded. Their common leading primes
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are only q,r; away from those primes there is a root of the first
form (with the harmless factor 2 handled by its odd leading coefficient).
Thus their singular series is O(1), since at most two odd primes can
have nu=0, each contributing a bounded local factor. FKL bounds this
case for each q,r by
 O(X*l/(q*r*(eta*L)^2)).
Summing reciprocals of q,r over primes <=2X gives O(l^2). The total is
 O(X*l^3/(eta^2*L^2)).

CASE LC: q<X^(1/2-eta), r>sqrt X. Fix the prime q and the odd integer
b=(N-q)/r<=4sqrt X. The congruences are P=2 modulo q and
2P=2+q modulo b. They are compatible only if gcd(q,b)=1: otherwise
q|b would imply q|2. Because b is odd, the compatible case gives one
class A modulo q*b. Write P=A+q*b*t, 1<=A<=q*b, so t>=1 and
t<=2X/(q*b). The THREE prime forms are
 P=A+q*b*t,
 r=(2A-2-q)/b+2q*t,
 g=2P-2-r.
All coefficients are positive. Their leading coefficients are q*b,
2q, 2q*(b-1). For large X each is smaller than its actual prime value:
q*b<=4X^(1-eta), P,g>=cX, and 2q<sqrt X<r. Therefore an imprimitive
form has no actual prime solutions and is discarded.
To check the determinants, regard P=(b*r+q+2)/2 and
g=(b-1)*r+q as affine forms in r. The three relevant nonzero constants
are q+2, q, and q-2(b-1). The last cannot vanish: q is odd and
2(b-1) even. Changing to the t parameter preserves nonvanishing.
The leading coefficient 2q of r gives a root at every odd prime except
q; at 2, the leading coefficient q*b of P is odd. Hence S=O(l^2).
The contribution for fixed q,b is
 O(X*l^2/(q*b*(eta*L)^3)).
The reciprocal prime sum costs O(l), and the reciprocal b sum O(L).
This case contributes O(X*l^3/(eta^3*L^2)).

CASE CL: q>X^(1/2+eta), r<=sqrt X. Fix odd a=(P-2)/q<=2X^(1/2-eta)
and the prime r. The condition r|((2a-1)q+2) forces gcd(r,2a-1)=1;
otherwise the odd prime r would divide 2. Thus q=A+r*t is a unique
progression with 1<=A<=r, t>=1, t<=2X/(a*r).
Sieve the THREE forms q, P=a*q+2, g=2a*q+2-r. Before substituting
q=A+r*t, their pair determinants are 2, 2-r, -a*(r+2), all nonzero.
Their t-coefficients are r,a*r,2a*r. They are smaller than their actual
prime values because q>sqrt X>=r and a*r<=2X^(1-eta). Discard any
imprimitive forms, which therefore have no solutions. The first form
has a root at every prime except possibly r, so S=O(l^2).
The count for each a,r is O(X*l^2/(a*r*(eta*L)^3)). Sum over a and
prime r to get O(X*l^3/(eta^3*L^2)).

CASE CC: q>X^(1/2+eta), r>sqrt X. Fix odd a<=2X^(1/2-eta) and
b<=4sqrt X. The relation b*r=(2a-1)q+2 forces gcd(b,2a-1)=1, since
this odd gcd divides 2. Thus q=A+b*t for a unique 1<=A<=b. For large
X, q>X^(1/2+eta)>b, so t>=1, and t<=2X/(a*b). Let
 R=((2a-1)*A+2)/b, an integer. The FOUR prime forms in t are
 q=A+b*t,
 r=R+(2a-1)*t,
 P=a*A+2+a*b*t,
 g=2a*A+2-R+[2a*b-2a+1]*t.
All coefficients are positive. The coefficients of q,r are <their
actual prime values since b<q and 2a-1<r for large X. The coefficients
of P,g are <=16X^(1-eta)=o(X), also smaller than their actual values.
Therefore nonprimitive forms again give no solutions. Their six
pairwise determinants, in the order (q,P),(q,r),(q,g),(P,r),(P,g),(r,g),
are
 2b, 2, 2(b-1), 2(1-a), -2[a(b-1)+1], -2.
They are all nonzero for a,b>=3. Moreover gcd(b,2a-1)=1, so at least
one of the first two forms has invertible leading coefficient modulo
every prime. Hence nu(ell)>=1 for all ell and S=O(l^3).
The count is O(X*l^3/(a*b*(eta*L)^4)). Two harmonic sums over a,b
cost O(L^2), giving O(X*l^3/(eta^4*L^2)).

In all four cases, the representatives, coefficients and determinants
are bounded by fixed powers of X; B_FKL=O(1) uniformly. For fixed eta
the sieve exponential is bounded, and adding the four cases gives
 O(X*l^3/(eta^4*L^2)). Adding BAND and radius-one exceptions, and
dividing by X/L, gives
 O(eta+1/L+l^2/L+l^3/(eta^4*L)).                         (R2)
For each fixed eta the last three terms tend to zero. Letting eta
decrease to zero proves the qualitative relative-density-one claim.
For the quantitative claim choose eta=(l^3/L)^(1/5). Then eta<1/8
for large X, eta*L tends to infinity much faster than l, and all the
coefficient-vs-prime comparisons above remain valid. The FKL hypotheses
and its bounded exponential factor hold uniformly, since every sieve
variable limit is >=cX^eta and B_FKL=O(1)=o(eta*L). Factors eta^(-r)
from the denominator were already retained explicitly. Thus (R2)
is O((l^3/L)^(1/5)), as claimed.
After excluding success at distances 0,1,2, every vertex in this complete
radius-two neighbourhood is active with positive composite complement:
activity propagates along edges, and children of BAD rows are <N/3.
This is exactly the asserted stopped-graph conclusion.

HOSTILE CHECK:
The proof counts a shortest successful path, but bounds the UNION of
all such paths. That is sufficient to exclude every GOOD vertex in the
complete radius-two frontier. It is not a selected-path construction.
The exceptional BAND covers every possible first child in that size
range, even if it is not on a shortest path. Removing it is an upper
bound, hence safe. Its estimate uses a one-form prime progression bound,
not an assumption that shifted-prime factors behave independently.
The four switches cover all remaining q,r, and none silently applies
an asymptotic in a progression with only O(1) terms. The positive t>=1
checks avoid adding one unsieved term for every parameter pair.
The equations are for h=1. Replacing 2 by 2h in (T) can introduce
nontrivial common divisors in the progression moduli; this proof is
NOT automatically a theorem for neighbouring roots or general h.
A third generation is not obtained by repeating this four-case argument:
three selected smaller parameters can already have product >X on a
region of nonvanishing size. That requires a new counting input.
DEPENDENCIES: FKL Lemma 5.1, including its k=1 case; Chebyshev upper
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prime count; PNT for relative prime density. No lower prime-tuple bound.
RELATION TO GOLDBACH: Rules out a universal depth-two canonical solver
on the h=1 family, and shows depth-two success is rare there. It leaves
eventual success and all ordinary Goldbach representations untouched.
RELATION TO p0: A negative theorem about complete depth two at p0. The
same-N nearby-prefix theorem currently remains proved only at depth one.
WHAT REMAINS OPEN: Complete radius three, arbitrary fixed radius, and
any positive restriction forbidding a closed BAD reachable set.
NOVELTY: NOT YET AUDITED.

RESULT-ID: MULTIGENERATION-SUCCESS-PATH-PRODUCT-BARRIER-20260906
STATUS: PROVED.
EXACT STATEMENT:
Fix epsilon,delta in (0,1). Let
 D(X)=floor((1-delta)*loglog X/logloglog X).
For all but o(X/log X) primes P in [X,2X], N=2(P-1), every successful
stopped path P -> q_1 -> ... -> q_d with 1<=d<=D(X) satisfies
 product_(i=1..d) q_i > X^(1-epsilon),
after deleting cycles if necessary. Equivalently no simple successful
path of those lengths has product <=X^(1-epsilon). The root is also
BAD outside o(X/log X) exceptions.
Consequently, if success occurs by depth D, any simple successful path
of length d<=D contains some label >X^((1-epsilon)/d).
This is a growing-depth restriction on actual successful paths with
moving prime labels, not on an alphabet held fixed through an iteration.

PROOF:
A successful stopped path has only BAD vertices before its GOOD endpoint.
Deleting a cycle shortens it and decreases the product of its labels,
so it is enough to count simple paths. The labels q_1,...,q_d are then
distinct odd primes <N/3<2X. Put M=product q_i. The edge conditions are
 P=2 modulo q_1,
 2P=2+q_(i-1) modulo q_i, i=2,...,d.
By CRT they specify exactly one residue class A modulo M. Choose
1<=A<=M and write P=A+M*t. Since M<=X^(1-epsilon)<X, t>=1 and
 t<=2X/M. The final GOOD condition says that
 g=2P-2-q_d
is prime. Sieve the two forms A+M*t and 2A-2-q_d+2M*t.
They have determinant -M*(2+q_d), which is nonzero. If the first
form is imprimitive, any common prime divisor belongs to {q_i};
then a prime P in this class would equal that q_i, impossible since
all path labels are <N/3<P. If the second form is imprimitive, its
odd fixed prime divisor also belongs to {q_i}; but g>2N/3 is larger
than all q_i, also impossible. Thus every contributing class has two
primitive forms. Their Delta is 2M^3*(q_d+2), bounded by a fixed power
of X. Apply the general r=2 sieve consumer, retaining its l^2 singular-
series bound, with z=2X/M>=2X^epsilon. Uniformly in d and the labels
this counts at most
 O_epsilon(X*(loglog X)^2/(M*(log X)^2)).
There is no dependence on d in that constant: there are still only TWO
prime forms, and Delta and the FKL B parameter have uniform bounds.
Let H_X=sum_(q<=2X prime)1/q. Dropping restrictions on distinctness and
on the product only increases the reciprocal sum, so for a given d
 sum_(eligible q_1,...,q_d)1/(q_1*...*q_d)<=H_X^d.
The elementary Euler-product bound proved above gives
 H_X<=loglog X+O(1). Summing d=1,...,D costs at most 2H_X^D for large X.
The total exception bound, relative to X/log X, is therefore
 O_epsilon((loglog X)^2*H_X^D/log X).
Now H_X^D<=(log X)^(1-delta+o(1)), so this tends to zero.
The prime root-complement case adds O(X/(log X)^2) exceptions, already
proved above. This establishes the result and its maximum-label corollary.

HOSTILE CHECK:
No claim is made that a typical complete BFS remains within this product
budget. The theorem constrains successful paths only; large-product
paths remain possible, and are the precise missing class in this count.
The product excludes the initial prime P, which would otherwise make
the condition vacuous. It includes the final GOOD label, and simple
paths are required to use pairwise coprime prime moduli. Cycle deletion
justifies that reduction for the existence question.
This is not an iteration of H1-ALL-BRANCH-INJECTIVE-ESCAPE. It replaces
that mechanism with one simultaneous CRT modulus and a prime-pair
upper bound; it has its own explicit size restriction M<X^(1-epsilon).
At radius two the preceding four-switch theorem also treats large-
product paths except for a negligible band. No analogous full treatment
has been proved here for growing D or even for radius three.
DEPENDENCIES: CRT; the uniform two-form sieve consumer; PNT only to
interpret density among prime P. No prime-tuple lower bound.
RELATION TO GOLDBACH: Gives a quantitative necessary size condition on
shallow successful paths, not an obstruction to all successful paths.
RELATION TO p0: Nonlocal path constraint in the h=1 family. It is not
proved to distinguish p0 from p1 for the same N.
WHAT REMAINS OPEN: Count paths whose label product is large, especially
at three or more generations.
NOVELTY: NOT YET AUDITED.

FAIL-ID: AUTOMATIC-THIRD-GENERATION-SIEVE-SWITCH-20260906
STATUS: KILLED AS AN INFERENCE; no third-generation theorem is refuted.
CLAIM / ROUTE: Repeat the four-switch proof of the radius-two theorem
without a new argument to obtain every fixed complete BAD radius.
WHY IT LOOKED PROMISING: Each edge offers a choice between fixing its
prime label and fixing its cofactor, and at depth two the smaller
choices give a long enough progression outside one negligible band.
EXACT FAILURE:
At depth three, all three labels may have size about X^0.4, with their
row cofactors about X^0.6. The product of even the smaller choices then
has size about X^1.2. The progression parameter need not have a growing
interval to sieve. A whole interval of exponent choices has this
problem; it is not confined to a shrinking band around sqrt X that
can be discarded as in the radius-two proof.
COUNTEREXAMPLE / OBSTRUCTION:
This is a parameter-range obstruction to the proposed proof, not an
assertion that prime paths in that range have been constructed in
positive density. Those paths must be bounded arithmetically; size
constraints alone do not eliminate them. Common factors in some moduli
could help particular patterns but have not been controlled uniformly.
```

```
WHAT SURVIVES: The complete radius-two theorem and the growing-depth
path-product barrier. Neither requires a fixed factor alphabet.
DO NOT REPEAT: Do not promote the heuristic loglog(N) branching factor
and a 1/log N endpoint-prime probability to an independence theorem;
do not apply a prime-tuple asymptotic with a bounded sieve variable.
POSSIBLE REPLACEMENT: A simultaneous estimate for the three-generation
bilinear congruences, or a different parametrization that resolves the
large-product region with all degeneracies and remainder terms controlled.

AUDIT / PROVENANCE NOTES 2026-09-06
The result named SAME-N-GROWING-ROOTS-COMPLETE-FIRST-FRONT and the later
COMPLETE-FIRST-FRONT-NO-GO-GROWING-SAME-N-PREFIX are the same theorem,
with the latter splitting off SAME-N-NEARBY-ROOT-FIRST-FRONT-SIEVE as a
separate lemma. They must not be counted as two independent advances.
Both versions are retained as incremental proof records.
The older failure-log statement about a prime-tuple/codegree wall is not
contradicted: the present proofs only need prime-tuple UPPER bounds to
show rare shallow success. They provide no pointwise lower prime count.
The old automatic envelope-iteration failure likewise remains valid;
the new radius-two proof uses four explicit arithmetic parametrizations,
not iteration with an enlarged smoothness envelope.
A selective search in the master, Biblia and post-CRD failure log found
no earlier complete-radius density theorem supplying this consumer.
That search is not a literature novelty audit.

EXPERIMENT-ID: COMPLETE-FRONT-AND-RADIUS-TWO-AUDIT-20260906
STATUS: VERIFIED COMPUTATION; exact exhaustive finite scans.
METHOD:
A full smallest-prime-factor sieve through 4,000,100; integer-only
factor extraction; exact prime table lookups. No probable-prime tests.
Scan A: all prime starts in each of [100,1000), [1000,10000),
[10000,100000), [100000,1000000), [1000000,2000000). For SAME N=2(P-1),
compare roots of ranks 0,1,2,5. Classify success at depth 0, at depth 1,
or absence of success through depth 1. The last category does NOT
assert success at depth 2.
Scan B: all prime starts in [1000,10000), [10000,100000),
[100000,200000), [1000000,2000000). Classify actual shortest success
at depth 0,1,2, or no success through depth 2. Enumerate all successful
length-two paths after excluding earlier success, and check (T), the
three necessary gcd conditions, and all six CC determinants exactly.
There are 43,150 such paths in these domains; every check passed.
No claim of a global-depth BFS census is made. All four switching regimes
occur in the finite scan, so omitting any of them would omit real paths.
The finite prime interval [1000000,2000000) has 70,435 starts. Of these,
14,498 have the complete radius-two BAD property. This is about 20.6%,
not close to the limiting density one; the proven bounds are asymptotic
and are not being calibrated to this numerical range.
The first complete radius-two BAD certificate in the scanned domain
starting at P=1000 is P=1091, N=2180:
 N-P=1089=3^2*11^2;
 N-3=2177=7*311;
 N-11=2169=3^2*241;
 N-7=2173=41*53;
 N-241=1939=7*277;
 N-311=1869=3*7*89.
Every row at distances 0,1,2 is composite. The repeated label 3 is
already at distance 1; it is not counted as a new distance-2 vertex.
The certificate concerns finite depth only, not a closed BAD set.
For the six historical N=128,1718,1862,1928,2200,6142, the maximal low
closed BAD sets were also reconstructed exactly, and their two least
labels satisfy b^2<N and coprime rows. This is a targeted audit of the
height refinement, not a new census of autonomous components.
Full reproducible diagnostic code and exact output follow. These are
finite verification records, not the proof of any asymptotic theorem.
import numpy as np
from math import isqrt, gcd
from collections import Counter
import json
L=4000100
spf=np.zeros(L+1,dtype=np.int32)
for p in range(2,isqrt(L)+1):
    if spf[p]==0:
        z=spf[p*p::p]
        z[z==0]=p
primes=np.flatnonzero(spf[2:]==0)+2
isprime=np.zeros(L+1,dtype=np.bool_);isprime[primes]=True

def pf(n):
    out=[]
    while n>1:
        p=int(spf[n]) or n
        out.append(p)
        while n%p==0:n//=p
    return out

def first(N,Q):
    m=N-Q
    if isprime[m]:return 0,[]
    qs=pf(m)
    good=[q for q in qs if isprime[N-q]]
    return (1 if good else 2),good
out={}
for lo,hi in [(100,1000),(1000,10000),(10000,100000),(100000,1000000),(1000000,2000000)]:
    inds=np.flatnonzero((primes>=lo)&(primes<hi))
    hist=Counter(); examples={}
    for idx in inds:
        P=int(primes[idx]); N=2*(P-1)
        v=tuple(first(N,int(primes[idx+k]))[0] for k in [0,1,2,5])
        hist[v]+=1
        examples.setdefault(v,N)
    out[f'{lo}<=P<{hi}']={'prime_count':len(inds),'root_depth_capped_2_counts':[[sum(c for v,c in hist.items() if v[j]==d) for d in [0,1,2]] for j in range(4)],'all_four_no_success_depth_0_or_1':sum(c for v,c in hist.items() if min(v)==2),'first_witness_all_four':examples.get((2,2,2,2))}
checks={}
for N in [128,1718,1862,1928,2200,6142]:
```

```
    C=set(int(p) for p in primes if p<N/3 and N%p and not isprime[N-p])
    while True:
        D={p for p in C if set(pf(N-p))<=C}
        if D==C:break
        C=D
    a,b=sorted(C)[:2]
    assert b*b<N and gcd(N-a,N-b)==1
    checks[N]={'closed_size':len(C),'a':a,'b':b,'b_squared':b*b,'row_supports':[pf(N-a),pf(N-b)]}
out['closed_pair_checks']=checks
out['conventions']='Exact exhaustive sieve and trial extraction via SPF. Depth 2 means no success at depth 0 or 1, and is not an actual success-depth assertion. Roots k=0,1,2,5 refer to SAME N=2(P-1).'
print(json.dumps(out,indent=2))
```


```
{
  "100<=P<1000": {
    "prime_count": 143,
    "root_depth_capped_2_counts": [
      [
        27,
        79,
        37
      ],
      [
        59,
        50,
        34
      ],
      [
        44,
        65,
        34
      ],
      [
        42,
        57,
        44
      ]
    ],
    "all_four_no_success_depth_0_or_1": 4,
    "first_witness_all_four": 692
  },
  "1000<=P<10000": {
    "prime_count": 1061,
    "root_depth_capped_2_counts": [
      [
        170,
        517,
        374
      ],
      [
        306,
        394,
        361
      ],
      [
        293,
        398,
        370
      ],
      [
        288,
        409,
        364
      ]
    ],
    "all_four_no_success_depth_0_or_1": 52,
    "first_witness_all_four": 2024
  },
  "10000<=P<100000": {
    "prime_count": 8363,
    "root_depth_capped_2_counts": [
      [
        1019,
        3662,
        3682
      ],
      [
        1777,
        2937,
        3649
      ],
      [
        1754,
        2915,
        3694
      ],
      [
        1774,
        2930,
        3659
      ]
    ],
    "all_four_no_success_depth_0_or_1": 704,
    "first_witness_all_four": 20276
  },
  "100000<=P<1000000": {
    "prime_count": 68906,
    "root_depth_capped_2_counts": [
      [
        6945,
        27567,
        34394
      ],
      [
        12469,
        22233,
```

```
          34204
        ],
        [
          11973,
          22621,
          34312
        ],
        [
          12174,
          22872,
          33860
        ]
      ],
      "all_four_no_success_depth_0_or_1": 8124,
      "first_witness_all_four": 200096
    },
    "1000000<=P<2000000": {
      "prime_count": 70435,
      "root_depth_capped_2_counts": [
        [
          6702,
          26908,
          36825
        ],
        [
          11667,
          21869,
          36899
        ],
        [
          11365,
          22330,
          36740
        ],
        [
          11264,
          22327,
          36844
        ]
      ],
      "all_four_no_success_depth_0_or_1": 9390,
      "first_witness_all_four": 2000064
    },
    "closed_pair_checks": {
      "128": {
        "closed_size": 10,
        "a": 3,
        "b": 5,
        "b_squared": 25,
        "row_supports": [
          [
            5
          ],
          [
            3,
            41
          ]
        ]
      },
      "1718": {
        "closed_size": 55,
        "a": 3,
        "b": 5,
        "b_squared": 25,
        "row_supports": [
          [
            5,
            7
          ],
          [
            3,
            571
          ]
        ]
      },
      "1862": {
        "closed_size": 55,
        "a": 3,
        "b": 5,
        "b_squared": 25,
        "row_supports": [
          [
            11,
            13
          ],
          [
            3,
            619
          ]
        ]
      },
      "1928": {
        "closed_size": 41,
        "a": 3,
        "b": 5,
        "b_squared": 25,
        "row_supports": [
          [
            5,
            7,
            11
          ],
          [
            3,
            641
          ]
```

```
      ]
    },
    "2200": {
      "closed_size": 2,
      "a": 3,
      "b": 13,
      "b_squared": 169,
      "row_supports": [
        [
          13
        ],
        [
          3
        ]
      ]
    },
    "6142": {
      "closed_size": 66,
      "a": 3,
      "b": 5,
      "b_squared": 25,
      "row_supports": [
        [
          7,
          877
        ],
        [
          17,
          19
        ]
      ]
    }
  },
  "conventions": "Exact exhaustive sieve and trial extraction via SPF. Depth 2 means no success at depth 0 or 1, and is not an actual success-depth assertion. Roots k=0,1,2,5 refer to SAME N=2(P-1)."
}
```


```
exec(open('diagnostic_20260906.py').read().split('out={}')[0])
from fractions import Fraction

def radius2(N,P):
    if isprime[N-P]:return 0,0,[]
    qs=pf(N-P)
    if any(isprime[N-q] for q in qs):return 1,0,[]
    successes=[]
    cases=Counter()
    for q in qs:
        a=(P-2)//q
        for r in pf(N-q):
            if not isprime[N-r]:continue
            b=(N-q)//r
            assert q!=r and a>=3 and b>=3 and a%2==b%2==1
            assert b*r==(2*a-1)*q+2 and P==a*q+2
            g=N-r
            successes.append((q,r,a,b,g))
            assert gcd(q,b)==1
            assert gcd(r,2*a-1)==1
            assert gcd(b,2*a-1)==1
            A=(-2*pow(2*a-1,-1,b))%b
            if A==0:A=b
            R=((2*a-1)*A+2)//b
            forms=[(b,A),(2*a-1,R),(a*b,a*A+2),(2*a*b-2*a+1,2*a*A+2-R)]
            pairs=[(0,2),(0,1),(0,3),(2,1),(2,3),(1,3)]
            expected=[2*b,2,2*(b-1),2*(1-a),-2*(a*(b-1)+1),-2]
            actual=[forms[i][0]*forms[j][1]-forms[j][0]*forms[i][1] for i,j in pairs]
            assert actual==expected and all(actual)
    return (2 if successes else 3),len(successes),successes

out={}; cert=None; totalpaths=0
for lo,hi in [(1000,10000),(10000,100000),(100000,200000),(1000000,2000000)]:
    ps=primes[(primes>=lo)&(primes<hi)]
    c=Counter(); pathcount=0; regimes=Counter(); smallest={}
    for pp in ps:
        P=int(pp);N=2*(P-1)
        d,n,rows=radius2(N,P);c[d]+=1;pathcount+=n
        smallest.setdefault(d,N)
        for q,r,a,b,g in rows:
            regime=('L' if q*q<=lo else 'C')+('L' if r*r<=lo else 'C')
            regimes[regime]+=1
        if d==3 and cert is None:
            frontier1=pf(P-2)
            frontier2=sorted(set(r for q in frontier1 for r in pf(N-q)))
            cert={'P':P,'N':N,'root_row':pf(P-2),'first_rows':{q:pf(N-q) for q in frontier1},'second_rows':{r:pf(N-r) for r in frontier2}}
            assert all(not isprime[N-v] and N-v>1 for v in [P]+frontier1+frontier2)
    out[f'{lo}<=P<{hi}']={'prime_count':len(ps),'exact_success_depth_0_1_2_or_no_success_through_2':[c[j] for j in range(4)],'number_of_depth2_successful_paths_after_excluding_depth0_1':pathcount,'parameter_regimes_at_sqrt_lo':dict(regimes),'first_N_in_each_depth_category':smallest}
    totalpaths+=pathcount
out['certificate_complete_BAD_radius_two']=cert
out['all_radius2_identity_and_six_CC_determinant_checks']=f'passed for {totalpaths} actual successful length-two paths'
print(json.dumps(out,indent=2))
```


```
{
  "1000<=P<10000": {
    "prime_count": 1061,
    "exact_success_depth_0_1_2_or_no_success_through_2": [
      170,
      517,
      251,
      123
    ],
    "number_of_depth2_successful_paths_after_excluding_depth0_1": 397,
    "parameter_regimes_at_sqrt_lo": {
      "CL": 85,
      "CC": 70,
      "LL": 109,
      "LC": 133
    },
```

```
    "first_N_in_each_depth_category": {
      "1": 2016,
      "2": 2024,
      "0": 2040,
      "3": 2180
    }
  },
  "10000<=P<100000": {
    "prime_count": 8363,
    "exact_success_depth_0_1_2_or_no_success_through_2": [
      1019,
      3662,
      2363,
      1319
    ],
    "number_of_depth2_successful_paths_after_excluding_depth0_1": 3770,
    "parameter_regimes_at_sqrt_lo": {
      "LL": 1321,
      "LC": 1123,
      "CL": 746,
      "CC": 580
    },
    "first_N_in_each_depth_category": {
      "2": 20012,
      "0": 20016,
      "1": 20120,
      "3": 20324
    }
  },
  "100000<=P<200000": {
    "prime_count": 8392,
    "exact_success_depth_0_1_2_or_no_success_through_2": [
      936,
      3481,
      2543,
      1432
    ],
    "number_of_depth2_successful_paths_after_excluding_depth0_1": 4103,
    "parameter_regimes_at_sqrt_lo": {
      "LL": 1835,
      "CC": 426,
      "LC": 1056,
      "CL": 786
    },
    "first_N_in_each_depth_category": {
      "1": 200004,
      "3": 200084,
      "2": 200096,
      "0": 200304
    }
  },
  "1000000<=P<2000000": {
    "prime_count": 70435,
    "exact_success_depth_0_1_2_or_no_success_through_2": [
      6702,
      26908,
      22327,
      14498
    ],
    "number_of_depth2_successful_paths_after_excluding_depth0_1": 34880,
    "parameter_regimes_at_sqrt_lo": {
      "LL": 16192,
      "CL": 6598,
      "LC": 8591,
      "CC": 3499
    },
    "first_N_in_each_depth_category": {
      "1": 2000004,
      "3": 2000064,
      "0": 2000076,
      "2": 2000160
    }
  },
  "certificate_complete_BAD_radius_two": {
    "P": 1091,
    "N": 2180,
    "root_row": [
      3,
      11
    ],
    "first_rows": {
      "3": [
        7,
        311
      ],
      "11": [
        3,
        241
      ]
    },
    "second_rows": {
      "3": [
        7,
        311
      ],
      "7": [
        41,
        53
      ],
      "241": [
        7,
        277
      ],
      "311": [
        3,
        7,
```

```
        89
      ]
    }
  },
  "all_radius2_identity_and_six_CC_determinant_checks": "passed for 43150 actual successful length-two paths"
}
```


```
EXTERNAL-INPUT: PINTZ-GOLDBACH-EXCEPTIONAL-SET
STATUS: Exact theorem statement and hypotheses checked in primary paper.
Janos Pintz, A new explicit formula in the additive theory of primes with
applications II. The exceptional set in Goldbach's problem, Theorem 1,
arXiv:1804.09084v2 (30 April 2018), printed p.2.
https://arxiv.org/pdf/1804.09084
Writing E(Y) for the number of even n<=Y not expressible as two primes,
the theorem states E(Y)<Y^0.72 for Y above an ineffective constant.
Only the consequence E(4X)=o(X/log X) is used below. This run checked
the statement, not the entire 45-page proof and its prerequisite papers.
The exponent is an inherited external theorem, not improved in this run.

RESULT-ID: GOLDBACH-TRUE-WITH-COMPLETE-SHALLOW-BAD-20260906
STATUS: PROVED, using the preceding external exceptional-set theorem.
EXACT STATEMENT:
For relative density one of primes P in [X,2X], N=2(P-1) simultaneously:
 (a) has an ordinary binary Goldbach representation;
 (b) has no GOOD vertex through depth two from its canonical root P;
 (c) has no GOOD vertex through depth one from any of the first K+1
     roots, whenever K=o(log X/(loglog X)^3).
For fixed epsilon,delta in (0,1), the growing-depth path-product barrier
may also be imposed simultaneously. In particular there are infinitely
many genuine Goldbach numbers with all these shallow BAD properties.
PROOF:
The map P -> 2(P-1) is injective and its values lie below 4X. Hence the
number of starting primes violating (a) is at most E(4X)=O(X^0.72),
which is o(X/log X). The exceptions to (b), (c), and the optional
path-product assertion are each o(X/log X) by the proved results above.
The union of these finitely many exceptional sets remains o(X/log X),
whereas PNT gives ~X/log X prime starts. This proves the statement.
HOSTILE CHECK:
This does not claim that their complete canonical traversals eventually
succeed: the Goldbach pair in (a) might be outside the reachable set.
It does prove that a complete shallow BAD neighbourhood is compatible
with ordinary Goldbach truth on an asymptotically full part of this
prime-parametrized family. No theorem requiring global failure is being
applied to such ordinary successful N.
DEPENDENCIES: The new shallow-front theorems plus the stated external
exceptional-set bound. No pointwise Goldbach assertion is smuggled in.
RELATION TO p0: Reinforces the distinction between generic shallow BADness
and whatever deeper canonical-root phenomenon the computations suggest.
WHAT REMAINS OPEN: Eventual canonical reachability, and binary Goldbach
for every even N. NOVELTY: NOT YET AUDITED.

FINAL HOSTILE AUDIT 2026-09-06
1. The radius-two proof's weakest step was availability of a long sieve
   variable. All four parametrizations explicitly have fixed-parameter
   product <=8X^(1-eta). The only removed first-factor range has mass
   O(eta+1/log X) in the prime-start sample. eta*log X grows for the
   quantitative choice, so all variable-length and coefficient-size
   comparisons used by the sieve remain valid.
2. Primitive forms and nonzero determinants were checked separately in
   every use, including the nearby-root degeneracy h+1=2a. An upper
   prime-tuples bound was never silently used for coincident forms.
3. Source uniformity comes from FKL's explicit B condition. A generic
   theorem with constants depending arbitrarily on coefficients would
   not justify the switching proofs or their growing prefixes.
4. The sample throughout the analytic continuation is P prime, with
   N=2(P-1). It is not the uniform-even-N sample of the earlier transfer
   theorem. No unproved change of sampling measure has been made.
5. Shallow BADness is a union-over-all-successful-paths conclusion,
   hence controls the complete BFS radius in question. It does not
   control its remaining radii or the final closed reachable set.
6. The h=1 radius-two switching result is not automatically shared by
   p1 for the same N; no p0 advantage may be inferred from the absence
   of that extension. The full first-front theorem IS explicitly shared
   by a growing same-N nearby prefix.
7. Numerical scans verify exact identities and finite classifications;
   they do not verify the limiting densities. At the scanned scales,
   many depth-two successes still occur. The upper bounds were not
   fitted to the data and no finite threshold was claimed.
8. The height refinement is a necessary condition shared by any closed
   active BAD set. It neither excludes arbitrary moving support nor
   repairs the killed automatic envelope iteration.
9. Literature novelty has not been audited. The mathematical proofs
   have been locally stress-tested in this run, but have not received
   a separate external adversarial review.
Reproduction note: in the diagnostic section, save the first new Python
block as diagnostic_20260906.py and the second as audit_radius2_20260906.py;
the intervening JSON blocks are their exact outputs. Both use numpy.
No additional diagnostic files are required as permanent deliverables.

========================
STRONGEST NEW RESULTS
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1. COMPLETE-H1-RADIUS-TWO-RARE-SUCCESS: almost every prime start P in
   the h=1 family has its entire radius-two neighbourhood BAD, with
   exceptional proportion O(((loglog X)^3/log X)^(1/5)). Four explicit
   sieve switches control both generations, including large factors.
2. SAME-N-GROWING-ROOTS-COMPLETE-FIRST-FRONT: for almost all such P,
   every root of rank <=K=o(log X/(loglog X)^3) has its entire first
   front BAD. This is a same-N comparison, not a selected CRT tree.
3. MULTIGENERATION-SUCCESS-PATH-PRODUCT-BARRIER: up to depth
   (1-delta)loglog X/logloglog X, successful simple paths must have
   product of nonroot labels >X^(1-epsilon), outside o(X/log X) starts.
4. TWO-SMALLEST-ROWS-HEIGHT: every active closed BAD set has two least
   labels below sqrt N with coprime rows, yielding a height constraint
   on their actual joint support without a pure-row assumption.
The first three statements coexist with ordinary Goldbach truth for
```

relative density one of the h=1 prime starts, by a known exceptional-set bound. That external Goldbach bound has not been improved here.

```
========================
CURRENT FRONTIER
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```

No proof of binary Goldbach, no pointwise Goldbach threshold, and no proof of universal canonical traversal success. The new analytic results settle full shallow BADness through radius two in the h=1 family in a relative-density-one sense. They do not yet settle every fixed radius. The radius-three large-product parameter region is not controlled by the current one-variable sieve switching. Closed BAD reachable sets with moving large support remain unexcluded. No upper support bound contradicts the new or inherited necessary height bounds.

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========================
p0 STATUS
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```

No nontrivial property proved for every p0 with a demonstrated same-N failure at p1. A substantial SAME-N negative theorem was proved: complete first-front BADness is shared by p0 and a growing nearby prefix in the h=1 family. Complete radius-two BADness was proved asymptotically for p0 there, but no conclusion comparing the radius-two laws of p0 and p1 is justified yet. The empirical eventual-traversal exceptionality of p0 remains unexplained.

```
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EXACT NEXT QUESTION
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```

Is the number of primes P in [X,2X] for which the canonical root of N=2(P-1) reaches a GOOD vertex within THREE edges o(X/log X)? A proof must control the complete third frontier, especially successful paths with large product of labels for which all naive choices of fixed labels or cofactors leave too short a sieve variable.

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CONTINUATION 2026-09-06B
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The previous run's end sections are archival. This continuation audits its radius-two theorem and investigates a deeper SAME-N comparison. No pointwise Goldbach or eventual p0-reachability theorem is assumed.

RESULT-ID: AUDIT-RADIUS-TWO-INPUT-20260906B
STATUS: AUDITED; previous theorem retained as PROVED.
The four parametrizations in COMPLETE-H1-RADIUS-TWO-RARE-SUCCESS were rechecked directly. The first and second cofactors are odd and >=3; q!=r follows from activity; after removing the first-factor band each fixed-parameter product is <=8X^(1-eta). Prime values exceed their corresponding positive leading coefficients for sufficiently large X, including eta=((loglog X)^3/log X)^(1/5). Therefore discarded imprimitive forms truly have no actual prime solutions. The six CC determinants agree with the written formulas. No fixed-depth/global-failure scope change is needed.
The exact FKL Lemma 5.1 was reopened in the primary author paper, https://arxiv.org/pdf/0904.0473, printed pp.12--13. Its rank/size hypotheses and explicit B term also permit FIVE fixed linear forms; no new external theorem is required for that extension.
The potential new obstacle at later roots is real: the equations now contain 2h rather than 2, so gcd(b,2a-1) can be a nontrivial divisor of h. The following lemma controls its total contribution instead of assuming it away.

RESULT-ID: WEIGHTED-COMPATIBLE-GCD-AVERAGE-20260906B
STATUS: PROVED.
EXACT STATEMENT:
For A,B>=2 and an odd positive integer h, define

```
 W_h(A,B)=sum_(1<=a<=A,1<=b<=B; gcd(b,2a-1)|h)
                    gcd(b,2a-1)/(a*b).
```

Then, with an absolute constant C,

```
 W_h(A,B)<=C*log(2A)*log(2B)*sigma_{-1}(h),
```

where sigma_{-1}(h)=sum_(e|h)1/e. Consequently, for H>=1,

```
 sum_(1<=h<=H,h odd) W_h(A,B)
 <=C*zeta(2)*H*log(2A)*log(2B).
```

Restricting a,b to odd integers >=3 preserves these upper bounds.
Also, for any odd prime r,

```
 sum_(1<=a<=A; 2a=1 mod r)1/a <=C*log(2A)/r,
```

and

```
 sum_(h<=H)sum_(r|h, r prime)1/r <=H*sum_r 1/r^2 <=H*zeta(2).
```

PROOF:
For d=gcd(b,2a-1), the elementary identity d=sum_(e|d)phi(e) holds. If d|h, all divisors e of d divide h. Dropping the condition d|h after expanding gives

```
 W_h <=sum_(e|h)phi(e)
            [sum_(b<=B,e|b)1/b]
            [sum_(a<=A,2a=1 mod e)1/a].
```

All e here are odd. The first bracket is <=(1+log B)/e. For e>=3, the least positive solution of 2a=1 mod e is (e+1)/2. The subsequent solutions are spaced e apart. Hence its reciprocal sum is at most 2/e+(1/e)*log(2A), by comparison with the integral of 1/(e*t+(e+1)/2). For e=1 the ordinary harmonic bound gives the same assertion with an absolute constant. Thus the second bracket is <=C*log(2A)/e. Since phi(e)<=e, this proves the first bound. Finally

```
 sum_(h<=H)sigma_{-1}(h)
 =sum_(e<=H)(1/e)*floor(H/e)<=H*sum_(e>=1)1/e^2=H*zeta(2).
```

The prime-r variants follow from the same progression estimate and by summing floor(H/r)/r. All quantities are nonnegative, so the additional parity and lower-end restrictions can be dropped safely.

HOSTILE CHECK:
The hypothesis gcd(b,2a-1)|h is essential to the stated per-h estimate; there is no claimed uniform O(log A log B) bound for the unrestricted gcd sum. The proof does not replace a moving divisor by a fixed one. It sums every compatible divisor and retains its exact multiplicative weight before averaging h. No prime distribution or Goldbach premise is used in this lemma.

```
DEPENDENCIES: gcd=sum phi, elementary harmonic estimates.
IMPLICATION: Compensates for the d-fold increase in progression length
when b*r=(2a-1)*q+2h has gcd(b,2a-1)=d>1. This is a real consumer
of the compatibility relation d|h, not a collision-to-new-support claim.
RELATION TO GOLDBACH: Counting tool only, not a positivity theorem.
RELATION TO p0: Makes deeper same-N nearby-root comparisons possible.
WHAT REMAINS OPEN: Verify every accompanying prime-form and exceptional
progression case in that comparison. NOVELTY: NOT YET AUDITED.

RESULT-ID: DISTINCT-PRIME-FORMS-DEGENERACY-FILTER-20260906B
STATUS: PROVED; elementary lemma used to audit five-form switches.
EXACT STATEMENT:
Suppose a parametrized arithmetic configuration requires k integer
linear forms L_i(t)=u_i*t+v_i, with u_i>0, to take pairwise DISTINCT
prime values, each greater than u_i. If one form is imprimitive, or
if a pair of primitive forms has zero determinant u_i*v_j-u_j*v_i,
then the configuration has no solutions.
PROOF:
If d=gcd(u_i,v_i)>1, it divides every value and satisfies d<=u_i<L_i(t);
no such value is prime. If two primitive coefficient vectors are
proportional with positive ratio lambda, Bezout for each vector shows
both lambda and 1/lambda are positive integers. Hence lambda=1 and
the forms are identical. Their values cannot be distinct. This proves
both exclusions. A zero constant term causes no exception to Bezout.
HOSTILE CHECK: The coefficient-versus-value comparison and DISTINCT
prime-value requirement must be proved from the actual configuration.
Neither is automatic for arbitrary linear forms. The filter excludes
parameter choices with no actual solutions; it does not count their
solutions by an incorrectly high-dimensional sieve.
DEPENDENCIES: Bezout and primality. NOVELTY: NOT YET AUDITED.

SIEVE EXTENSION 2026-09-06B:
UNIFORM-FEW-FORMS-SIEVE-CONSUMER applies unchanged to k=5 (as well as
k=2,3,4): the primary FKL lemma permits every fixed k. Write
L=log X, l=loglog X. If coefficients and nonzero determinants are
bounded by fixed powers of X, log z>=c*eta*L, z<=X^C, and
L^(-1/4)<=eta<=1/8, then B_FKL=O(1)=o(eta*L). Thus the sieve count
is O(z*S/(eta*L)^k), with an absolute constant when k<=5 and c,C
are fixed. The earlier Euler-product proof gives S=O(l^(k-1)) when
nu(ell)>=1 at every prime. The same bound survives at most TWO
exceptional odd primes where nu may vanish, since each extra local
factor is bounded by a fixed power of 3/2. No fixed-coefficient-only
asymptotic is substituted for the uniform FKL statement.

RESULT-ID: SAME-N-RADIUS-TWO-PAIR-COUNT-20260906B
STATUS: PROVED.
EXACT STATEMENT:
Let L=log X and l=loglog X. Assume X sufficiently large,
 3<=H<=L^2,        L^(-1/4)<=eta<=1/8.
Let T_2(X,H) be the number of pairs of primes (P,Q) with
 X<=P<=2X,        P<Q,        h=Q-P+1<=H,
such that in the graph for N=2(P-1), the stopped traversal rooted at Q
has a GOOD vertex at distance <=2. Then, with an absolute constant,
 T_2(X,H) << H*X*[l*(eta+1/L)/L^2 + l^4/(eta^5*L^3)].       (PAIR2)
In particular, taking eta=l^(1/2)/L^(1/6),
 T_2(X,H) << H*X*l^(3/2)/L^(13/6).                          (PAIR2*)
These count pairs (P,Q), not paths with multiplicity. No rank restriction
or consecutiveness assumption on Q is imposed.

PROOF:
All offsets h in the count are odd and >=3. Also Q<=2X+H<=3X. The
known same-N first-front pair sieve counts success at distances 0 or 1
by O(H*X*l^3/L^3), which is absorbed by (PAIR2). Hence consider a
shortest successful path Q -> q -> r with Q and q BAD and r GOOD.
Write
 m=N-Q=Q-2h=a*q,
 N-q=b*r,
 g=N-r.
Then a,b are odd integers >=3, and
 Q=a*q+2h,         P=a*q+h+1,
 b*r=(2a-1)*q+2h,           g=2a*q+2h-r.                    (E)
The prime values q,r,P,Q,g are pairwise distinct: q!=r by activity,
q,r<N/3; P=(N+2)/2; Q=P+h-1=P+o(X); and g>2N/3>Q for large X.
Thus g>Q>P>max(q,r), apart from the already separately ordered q,r.
In particular every tuple of variable prime forms below really must
have distinct values.

BAND REMOVAL:
For each h remove any pair whose root complement has a prime divisor
 q in [X^(1/2-eta),X^(1/2+eta)].
For fixed q, the two forms q*a+2h and q*a+h+1 must be prime.
Here q>2H for large X, so both forms are primitive. Their determinant
q*(1-h) is nonzero. Except at ell=q there is a root modulo every prime;
the two-form singular series is O(l). Sieving a<=3X/q, whose logarithm
is at least 3L/8, gives O(X*l/(q*L^2)). By partial summation from the
Chebyshev bound pi(t)<<t/log t, the reciprocal sum over this band is
O(eta+1/L). Sum h<=H to bound the removed PAIRS by
 O(H*X*l*(eta+1/L)/L^2).                                    (B2)
Removing all such pairs only enlarges the exceptional set; the divisor
need not actually belong to the successful path being counted.

For a remaining pair choose any shortest successful length-two path.
Its q is either below X^(1/2-eta) or above X^(1/2+eta). Split also at
r=sqrt X. Use the following four counts to bound the existence of such
a path. Parameters outside the indicated arithmetic compatibility
conditions have no actual solutions.

CASE LL: q<X^(1/2-eta), r<=sqrt X. Fix h and distinct odd primes q,r.
The edge conditions give
 Q=2h modulo q,        2Q=2h+q modulo r.
They specify one class A modulo q*r. Choose 1<=A<=q*r and write
 Q=A+q*r*t,         P=Q-h+1,        g=2Q-2h-r.
Here t>=1 and t<=3X/(q*r), since q*r<X and Q>=X. These are THREE
integer prime forms with positive leading coefficients q*r,q*r,2q*r.
All coefficients are o(X), uniformly for eta>=L^(-1/4), and smaller
than their corresponding actual prime values. By the degeneracy filter,
```

```
nonprimitive forms or zero pair determinants imply no actual solution
and are discarded. (Directly the determinants are proportional to
h-1, 2h+r, and r+2, all nonzero.) Every prime outside {q,r} has an
invertible leading coefficient; at 2, q*r is odd. Thus S=O(l^2).
The uniform three-form bound is
 O(X*l^2/(q*r*(eta*L)^3)).
Summing reciprocals of the two prime parameters costs O(l^2), giving
 O(X*l^4/(eta^3*L^3)) for each h.

CASE LC: q<X^(1/2-eta), r>sqrt X. Fix h, prime q, and odd
b=(N-q)/r<=4sqrt X. The congruences are
 Q=2h modulo q,       2Q=2h+q modulo b.
They require gcd(q,b)=1. Indeed q|b would imply q|2h by (E), and
then q|Q=a*q+2h, impossible since Q is prime and q<Q.
Thus there is a unique class A modulo q*b. Write
 Q=A+q*b*t,   P=Q-h+1,
 r=(2A-2h-q)/b+2q*t,   g=2Q-2h-r.
The FOUR positive leading coefficients are q*b,q*b,2q,2q*(b-1).
We have q*b<=4X^(1-eta), and 2q<sqrt X<r for large X. Thus every
coefficient is smaller than its actual prime value. The degeneracy
filter removes all nonprimitive or coincident forms without losing a
possible solution. The coefficient 2q of r is invertible at every
odd prime except q, while Q has odd leading coefficient at 2.
Consequently S=O(l^3). The variable satisfies
 1<=t<=3X/(q*b),       log(3X/(q*b))>=eta*L-O(1).
The four-form count is O(X*l^3/(q*b*(eta*L)^4)). Sum the prime q
reciprocals and integer b reciprocals to obtain
 O(X*l^4/(eta^4*L^3)) for each h.

CASE CL: q>X^(1/2+eta), r<=sqrt X. Fix h, odd
 a=(Q-2h)/q<=3X^(1/2-eta),
and the prime r. If gcd(r,2a-1)=1, the equation (E) gives one class
q=A+r*t, 1<=A<=r, with t>=1 and t<=3X/(a*r). Sieve the FOUR forms
 q,       Q=a*q+2h,       P=a*q+h+1,       g=2a*q+2h-r.
Their t-coefficients are r,a*r,a*r,2a*r, smaller than the actual prime
values because a*r<=3X^(1-eta) and q>sqrt X>=r. The degeneracy filter
again applies. Outside ell=r the first form has an invertible leading
coefficient, giving S=O(l^3). Summing the bound
 O(X*l^3/(a*r*(eta*L)^4))
over integer a and prime r gives O(X*l^4/(eta^4*L^3)) for each h.

CL EXCEPTION (must not be omitted): If r|2a-1, compatibility forces
r|2h and hence r|h. The progression condition on q then degenerates.
For these pairs it suffices to DROP q,a and count the necessary prime
conditions directly: P, Q=P+h-1, and g=2P-2-r are all prime for some
odd prime r|h. These three forms are primitive and their determinants
are nonzero (h-1, -(r+2), -(2h+r)). At every prime there is a root of
the form P, so S=O(l^2). Since P<=2X, the count for each h,r is
O(X*l^2/L^3). Sum over h<=H and r|h:
 sum_(h<=H)omega(h)<=sum_(r<=H prime)floor(H/r)<<H*loglog(3H).
Hence this exceptional branch contributes
 O(H*X*l^2*loglog(3H)/L^3)=O(H*X*l^3/L^3).
This branch counts PAIRS directly, not all paths with multiplicity.
That is sufficient for the stated union/existence problem.

CASE CC: q>X^(1/2+eta), r>sqrt X. Fix h, odd
 a<=3X^(1/2-eta),       b<=4sqrt X.
Put c=2a-1 and d=gcd(b,c). By (E), d|2h, hence d|h because d is odd.
No assumption d=1 is made. Set b'=b/d, c'=c/d, h'=h/d, so
 gcd(b',c')=1,       b'*r=c'*q+2h'.
There is exactly one class q=A+b'*t, with 1<=A<=b'. Put
 R=(c'*A+2h')/b', an integer. Then the FIVE prime forms in t are
 q=A+b'*t,
 r=R+c'*t,
 Q=a*A+2h+a*b'*t,
 P=a*A+h+1+a*b'*t,
 g=2a*A+2h-R+(2a*b'-c')*t.
The last leading coefficient is positive since b>=3 and
2a*b'-c'=[2a*(b-1)+1]/d>0. Moreover q>b>=b' and r>c>=c' for
large X; the Q,P,g coefficients are O(a*b)=O(X^(1-eta)) and are
smaller than their actual prime values. Thus the degeneracy filter
rigorously excludes every imprimitive or proportional case. For all
remaining parameter choices Delta is nonzero and polynomially bounded
in X. The coprime coefficients b',c' guarantee nu(ell)>=1 at every
prime. Therefore S=O(l^4).
Here t>=1 and
 t<=3X/(a*b')=3dX/(a*b),
whose logarithm is >=eta*L-O(1). The five-form bound for fixed h,a,b is
 O(d*X*l^4/(a*b*(eta*L)^5)).
Crucially, the d in this count is retained. Summing all compatible a,b
using WEIGHTED-COMPATIBLE-GCD-AVERAGE gives at most
 O(X*l^4*sigma_{-1}(h)/(eta^5*L^3))
for this h, because the two logarithmic harmonic sums are O(L^2).
Sum h<=H. The identity
 sum_(h<=H)sigma_{-1}(h)<=zeta(2)*H
makes the TOTAL CC bound
 O(H*X*l^4/(eta^5*L^3)),
with no growing extra divisor-count factor.

In every case the representatives, positive leading coefficients and
constant terms have absolute values bounded by fixed powers of X.
After nondegenerate primitive filtering, Delta has the same property.
The variable logarithms are >=c*eta*L, while B_FKL=O(l)=o(eta*L),
uniformly in the stipulated eta interval. Actual prime values in the
sieve exceed the number of forms (fixed q,r are not sieved as variables).
Thus all FKL applications are uniform, including k=5. Add LL, LC, CL,
the CL exception, CC, BAND, and earlier-depth successes. Since eta<=1,
the resulting total is (PAIR2).
For eta=l^(1/2)/L^(1/6), the hypotheses hold for sufficiently large X,
and both main terms in the brackets are O(l^(3/2)/L^(13/6)). The
1/L part of BAND is smaller. This proves (PAIR2*).

HOSTILE CHECK:
1. gcd(q,b)=1 in LC follows from ROOT primality; it is not an independent
   CRT-programming assumption. The different gcd in CC can be >1 and
   is handled by reducing the modulus and then averaging its full cost.
```

```
2. The CL zero-coefficient congruence branch r|h is treated separately.
   Simply writing an inverse of 2a-1 modulo r would omit real cases.
3. Primitive/proportional filtering uses actual distinct prime values
   and coefficient-size bounds, verified above in all four cases. A
   generic promise that forms are 'different' would not suffice.
4. The count concerns existence of SOME shortest successful path and
   therefore controls the COMPLETE radius-two frontier. It does not
   prescribe a path or claim the neighborhoods themselves are identical.
5. The prime P=N/2+1 is an additional sieve condition for the later root
   Q. Without it every case loses a logarithm; this proof cannot be
   exported unchanged to the uniform-even-N sample.
6. This is an explicit repair of the changing-offset/gcd obstruction
   mentioned in the prior radius-two theorem, not its automatic reuse.
DEPENDENCIES: FKL Lemma 5.1, k<=5; compatible-gcd harmonic average;
Chebyshev prime upper bound; the audited radius-one pair count.
RELATION TO GOLDBACH: Upper bound on shallow successful traversal pairs.
RELATION TO p0: Supplies a deeper SAME-N comparison with nearby roots.
WHAT REMAINS OPEN: Radius three and eventual rooted reachability.
NOVELTY: NOT YET AUDITED.

RESULT-ID: SAME-N-GROWING-ROOTS-COMPLETE-RADIUS-TWO-20260906B
STATUS: PROVED.
EXACT STATEMENT:
Write L=log X and l=loglog X. Let K=K(X) be a positive integer with
 K=o(L^(1/6)/l^(3/2)).
For all but o(X/L) primes P in [X,2X], set N=2(P-1) and let Q_k be the
(k+1)-st prime at or above N/2. Then EVERY root Q_k, 0<=k<=K, has a
COMPLETE stopped radius-two neighborhood consisting of BAD vertices.
An upper bound for the exceptional proportion among primes P is
 O((l^3/L)^(1/5) + sqrt(K*l^(3/2)/L^(1/6))).               (PREFIX2)
All implied constants are absolute, for X sufficiently large along
any K satisfying the displayed hypothesis. A concrete admissible
choice is K=floor((log X)^(1/8)).

HYPOTHESES / SCOPE:
The sample is primes P, with N=2(P-1); it is not all even N. BADness is
asserted for every vertex reachable within at most TWO edges, including
the root itself, with expansion stopped at GOOD vertices. The growing
prefix is for the SAME N. There is no assertion of eventual failure,
closedness, or a p0-specific invariant.

PROOF:
For large P the integer P-1 is composite and there is no integer between
P-1 and P, so Q_0=P. The previous audited canonical radius-two theorem
bounds starts with a GOOD vertex within two edges from Q_0 by
 O((X/L)*(l^3/L)^(1/5)).                                    (C0)

We use an elementary aggregate gap bound. List primes as s_i. For each
P=s_i in [X,2X], Q_K=s_(i+K). The prime number theorem, together with
K=o(L^(1/6)/l^(3/2)), ensures s_(i+K)<=3X for every such i when X is
large, since pi(3X)-pi(2X)>>X/L>K. Expand
 Q_K-P=sum_(j=i)^(i+K-1)(s_(j+1)-s_j).
After summing over the eligible i, each consecutive prime gap occurs
at most K times. All participating gaps lie in [X,3X], up to endpoints
already lying inside that interval; their total length is at most 2X.
Consequently
 sum_(P prime in [X,2X])(Q_K-P)<=2KX.
For H>=3, the number of starts with Q_K-P>H-1 is therefore at most
 2KX/(H-1)=O(KX/H).                                         (GAP)
No assertion about a uniform short interval containing K primes has
been made; the long-gap starts have been counted explicitly.

For every remaining start, all 1<=k<=K satisfy
 Q_k>P,       Q_k-P+1<=H.
If any such root has success within two edges, the pair (P,Q_k) is
included in SAME-N-RADIUS-TWO-PAIR-COUNT. Counting all pairs only
increases the number of exceptional starts. Thus (PAIR2*) gives
 O(H*X*l^(3/2)/L^(13/6))                                    (LATER)
provided H<=L^2. There is NO further factor K in (LATER), since that
pair count already includes every prime Q at the allowed offset.

Take H to be an integer within 1 of
 H_* = sqrt(K)*L^(13/12)/l^(3/4).
Then H_* tends to infinity, and the bound on K gives
 H_* = o(L^(7/6)/l^(3/2)) < L^2,
so the pair theorem applies. Divide (C0), (GAP), and (LATER) by
pi(2X)-pi(X) asymptotic to X/L. The last two resulting bounds are
 O(K*L/H + H*l^(3/2)/L^(7/6))
 =O(sqrt(K*l^(3/2)/L^(1/6))).
This proves (PREFIX2), which tends to zero under the stated condition.
For K=floor(L^(1/8)), the required ratio is at most
 l^(3/2)/L^(1/24), and tends to zero. The proof is complete.

HOSTILE CHECK:
1. This does not carry the earlier, larger radius-one prefix
   K=o(L/l^3) to radius two. The new sufficient range is shorter.
2. The extra primality condition P=N/2+1 in the pair count is retained;
   that count is not being applied to a generic root at a generic N.
3. Both a root being directly GOOD and success at the first frontier
   are included in the exceptional set. Therefore outside it the whole
   radius-two BFS, rather than selected BAD paths, is BAD.
4. The gap bound is a double count of actual consecutive prime gaps;
   it does not replace a mean gap by a pointwise gap estimate.
5. H is chosen only after the uniform pair estimate has been proved
   for the full range 3<=H<=L^2. Hence no hidden dependence of a
   sieve constant on an arbitrarily growing H is introduced.
6. The theorem compares shallow BADness only. It gives no coupling of
   later BFS generations and no equality of eventual success rates.
DEPENDENCIES: SAME-N-RADIUS-TWO-PAIR-COUNT-20260906B; audited
COMPLETE-H1-RADIUS-TWO-RARE-SUCCESS; PNT and elementary gap summation.
IMPLICATION: A deeper SAME-N negative result for explanations based on
universally shallow success of p0: asymptotically almost every member
of this family has complete radius-two BADness simultaneously at p0
and a growing nearby prefix.
RELATION TO GOLDBACH: This is compatible with Goldbach being true for
all the starts. It is not a lower-bound argument for prime pairs.
```

RELATION TO p0: A property previously established at radius two only for p0 is now proved to be shared by nearby roots at the SAME N.
WHAT REMAINS OPEN: Radius three, large-product successful paths, eventual reachability, and any pointwise p0-specific distinction.
NOVELTY: NOT YET AUDITED.

COROLLARY-ID: GOLDBACH-TRUE-WITH-SAME-N-PREFIX-RADIUS-TWO-BAD-20260906B
STATUS: PROVED, using the previously source-checked global exceptional set theorem, not an assumption of rooted success.
For all but o(X/log X) primes P in [X,2X], N=2(P-1) has a Goldbach representation while every root of the prefix in the preceding theorem has a complete radius-two BAD neighborhood.
PROOF: Pintz's checked bound E(Y)<Y^0.72 for sufficiently large Y gives at most E(4X)=o(X/log X) possible Goldbach counterexamples in this injectively parametrized family. Unite that exceptional set with the one from (PREFIX2). No root reachability conclusion follows merely from existence of a Goldbach pair elsewhere in the graph.
SOURCE: J. Pintz, A new explicit formula in the additive theory of primes with applications II, primary https://arxiv.org/pdf/1804.09084, Theorem 1 (statement previously checked; its long proof is not reproved in this checkpoint). Ineffective threshold is harmless here.

RESULT-ID: SAME-N-RADIUS-TWO-ADVERSARIAL-SCAN-20260906B
STATUS: VERIFIED COMPUTATION; finite evidence, not an asymptotic test.
EXACT DOMAIN:
All primes P in [1000,2000), [10000,20000), [100000,200000), and [500000,1000000): respectively 135, 1033, 8392, 36960 starts. For each start set N=2(P-1), and examine the SAME-N roots of ranks 0,1,2,5. No starts excluded. All distinct prime factors of every expanded row are included. The stopped search checks success at distances 0,1,2. Category 3 means no success by distance 2, not success at distance 3. Exact smallest-prime-factor sieve through 4,000,100 suffices for every integer factored or tested. Arithmetic tests below concern successful length-two paths from ranks 1,2,5 only, after excluding roots with ANY success at distance 0 or 1. Thus they are actual shortest paths.

MAIN DIAGNOSTICS:
70,836 shortest successful length-two paths were reconstructed from the reduced five-form parametrization. Every affine identity and every compatibility d|h held; gcd(q,b)=1 held for all of them. Of these, 15,761 had d=gcd(b,2a-1)>1. There were 12,256 paths with r|(2a-1), and in all such cases r|h, as required. At the illustrative split eta=0.1, 1,180 paths with d>1 were in the CC size regime, and 2,996 r|(2a-1) paths were in the CL size regime. This finite eta is for case diagnostics only; these small X need not meet the theorem's asymptotic eta bounds.
For 16,752 reconstructions all five actual prime values exceeded their positive leading coefficients; in every such reconstruction all five forms were primitive and all ten determinants nonzero. The remaining reconstructions were NOT declared counterexamples: the size hypothesis of the degeneracy filter was not being imposed outside the large-factor regime.

REAL NONCOPRIME WITNESS:
P=1423, N=2844, p1=Q=1427, h=5, q=109, r=547, a=13, b=5, g=2297. The path Q -> 109 -> 547 succeeds at depth two. Here d=gcd(5,25)=5, not 1. The reduced variable t=108 gives
 q=t+1, r=5t+7, Q=13t+23, P=13t+19, g=21t+29.
This certifies why reusing the canonical gcd=1 proof would miss an actual nearby-root configuration. The compatible-gcd averaging lemma has a necessary arithmetic consumer.

REAL DEGENERATE-CONGRUENCE WITNESS:
P=1277, N=2552, p5=Q=1297, h=21, q=251, r=3, a=5, b=767, g=2549. Then 2a-1=9 is divisible by r=3 and h=21 is divisible by r. The congruence for q modulo r does not specify a unique class. The separate CL-exception sieve includes this case.

SAME-N POINTWISE NON-DOMINANCE AT RADIUS TWO:
At N=2180, the previously verified p0=1091 certificate has its complete radius-two neighborhood BAD; p1=1093 is directly GOOD since 1087 is prime. In the other direction, N=2024 has p0=1013 with a successful length-two path 1013 -> 337 -> 7 (2024-7=2017 prime), while p1=1019 has complete radius-two BADness. Its full certificate is:
 2024-1019=3*5*67;
 2024-3=43*47, 2024-5=3*673, 2024-67=19*103;
 2024-43=7*283, 2024-47=3*659, 2024-673=7*193,
 2024-19=5*401, 2024-103=17*113.
Every displayed factor is prime, and every radius<=2 row is composite. Thus neither radius-two BADness event includes the other pointwise for p0 and p1. This is stronger evidence of a precise scope limit than simply comparing average percentages. No eventual-reachability claim is made for either example.

At the largest interval, category counts [depth0,depth1,depth2,BAD<=2] were p0=[3604,14500,11540,7316], p1=[6406,11779,11426,7349], p2=[6242,11976,11366,7376], p5=[6297,12160,11217,7286].
722 starts had complete radius-two BADness at all four selected roots. The finite proportions are far from the asymptotic density-one limit; no numerical convergence rate or eventual advantage is inferred. The complete diagnostic source and output follow, preserving all metrics and first witnesses without a second report artifact.

BEGIN DIAGNOSTIC SOURCE audit_same_n_radius2_20260906b.py
```
import numpy as np
from math import isqrt, gcd
from collections import Counter
from itertools import combinations
import json
LIMIT=4000100
spf=np.zeros(LIMIT+1,dtype=np.int32)
for p in range(2,isqrt(LIMIT)+1):
    if spf[p]==0:
        v=spf[p*p::p]
        v[v==0]=p
primes=np.flatnonzero(spf[2:]==0)+2
```

```
isprime=np.zeros(LIMIT+1,dtype=np.bool_)
isprime[primes]=True

def pf(n):
    out=[]
    while n>1:
        p=int(spf[n]) or n
        out.append(p)
        while n%p==0:n//=p
    return out

def depth2(N,Q):
    m=N-Q
    assert m>1 and N%Q
    if isprime[m]:return 0,[],[]
    qs=pf(m)
    if any(isprime[N-q] for q in qs):return 1,qs,[]
    paths=[]
    for q in qs:
        for r in pf(N-q):
            if isprime[N-r]:paths.append((q,r))
    return (2 if paths else 3),qs,paths

out={'domain':'Exhaustive primes P in each stated half-open dyadic interval; N=2(P-1); same-N roots k=0,1,2,5. Category 3 means no GOOD within distance 2, not actual success at distance 3. All prime factors included, no size cutoffs. Exact SPF sieve through 4,000,100.', 'intervals':{},'audit':{}}
audit=Counter();witness={}
ks=[0,1,2,5]
for X in [1000,10000,100000,500000]:
    inds=np.flatnonzero((primes>=X)&(primes<2*X))
    hist=Counter(); firstexamples={}
    for idx in inds:
        P=int(primes[idx]);N=2*(P-1);cats=[]
        for k in ks:
            Q=int(primes[idx+k]);h=Q-P+1
            depth,qs,paths=depth2(N,Q);cats.append(depth)
            if k==0:continue
            for q,r in paths:
                a=(N-Q)//q;b=(N-q)//r;c=2*a-1;d=gcd(b,c);g=N-r
                audit['shortest_success_length2_paths']+=1
                assert Q==a*q+2*h and P==a*q+h+1
                assert b*r==c*q+2*h and g==2*a*q+2*h-r
                assert gcd(q,b)==1 and h%d==0
                assert len({q,r,P,Q,g})==5
                audit['identities_activity_compatibility_checked']+=1
                bp=b//d;cp=c//d;hp=h//d
                A=((-2*hp)*pow(cp,-1,bp))%bp if bp>1 else 0
                if A==0:A=bp
                R=(cp*A+2*hp)//bp
                assert (cp*A+2*hp)%bp==0
                assert (q-A)%bp==0
                t=(q-A)//bp
                forms=[(bp,A),(cp,R),(a*bp,a*A+2*h),(a*bp,a*A+h+1),(2*a*bp-cp,2*a*A+2*h-R)]
                vals=[u*t+v for u,v in forms]
                assert vals==[q,r,Q,P,g]
                assert all(u>0 for u,v in forms)
                audit['reduced_five_form_reconstruction_checked']+=1
                if all(value>u for value,(u,v) in zip(vals,forms)):
                    assert all(gcd(u,v)==1 for u,v in forms)
                    assert all(u*y-w*v!=0 for (u,v),(w,y) in combinations(forms,2))
                    audit['full_filter_hypotheses_and_conclusion_checked']+=1
                rec={'X':X,'P':P,'N':N,'k':k,'Q':Q,'h':h,'q':q,'r':r,'a':a,'b':b,'g':g,'d':d,'reduced_forms':forms,'t':t}
                if d>1:
                    audit['d_greater_than_one']+=1
                    witness.setdefault('noncoprime_CC_equation',rec)
                    if q>X**0.6 and r>isqrt(X):
                        audit['noncoprime_CC_regime_eta_point1']+=1
                        witness.setdefault('noncoprime_CC_regime_eta_point1',rec)
                if c%r==0:
                    assert h%r==0
                    audit['degenerate_r_divides_2a_minus1']+=1
                    witness.setdefault('CL_exception_equation',rec)
                    if q>X**0.6 and r<=isqrt(X):
                        audit['CL_exception_regime_eta_point1']+=1
                        witness.setdefault('CL_exception_regime_eta_point1',rec)
        key=tuple(cats);hist[key]+=1;firstexamples.setdefault(key,P)
    out['intervals'][str(X)]={
        'prime_count':len(inds),
        'root_categories_0_1_2_bad_radius2':{str(k):[sum(n for v,n in hist.items() if v[j]==d) for d in range(4)] for j,k in enumerate(ks)},
        'all_four_radius2_BAD':sum(n for v,n in hist.items() if min(v)==3),
        'first_P_all_four_radius2_BAD':firstexamples.get((3,3,3,3)),
        'p0_bad_radius2_p1_success_within2':sum(n for v,n in hist.items() if v[0]==3 and v[1]<3),
        'p1_bad_radius2_p0_success_within2':sum(n for v,n in hist.items() if v[1]==3 and v[0]<3)
    }
out['audit']=dict(audit);out['witnesses']=witness
print(json.dumps(out,indent=2))
END DIAGNOSTIC SOURCE
BEGIN DIAGNOSTIC OUTPUT
```


```
{
  "domain": "Exhaustive primes P in each stated half-open dyadic interval; N=2(P-1); same-N roots k=0,1,2,5. Category 3 means no GOOD within distance 2, not actual success at distance 3. All prime factors included, no size cutoffs. Exact SPF sieve through 4,000,100.",
  "intervals": {
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          15
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          34,
```

```
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      "p0_bad_radius2_p1_success_within2": 13,
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    "10000": {
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    "100000": {
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      "first_P_all_four_radius2_BAD": 100931,
      "p0_bad_radius2_p1_success_within2": 898,
      "p1_bad_radius2_p0_success_within2": 964
    },
    "500000": {
      "prime_count": 36960,
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```

```
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```
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          1277
        ],
        [
          7661,
          2549
        ]
      ],
      "t": 0
    },
    "noncoprime_CC_regime_eta_point1": {
      "X": 1000,
      "P": 1423,
      "N": 2844,
      "k": 1,
      "Q": 1427,
      "h": 5,
      "q": 109,
      "r": 547,
      "a": 13,
      "b": 5,
      "g": 2297,
      "d": 5,
      "reduced_forms": [
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          1
        ],
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          7
        ],
        [
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          23
        ],
        [
          13,
          19
        ],
        [
          21,
          29
        ]
      ],
      "t": 108
    }
  }
}
```


```
END DIAGNOSTIC OUTPUT

HISTORICAL CHECK 2026-09-06B:
The original post-CRD failure log was searched for common-divisor,
local-offset, and support-height routes. Its Tail Hard-Edge Universality
entry and All-root collision identity do not imply the present theorem:
they establish identities without a BFS consumer. Here the compatibility
weight is consumed by a uniform prime-form count, and complete success
is bounded by a union over ALL possible first two edges. No assertion
that a gcd collision forces new support is used. The log's h=1
root-factor descent still has its stated scope; the present proof does
not presume that descent persists after an offset or support change.

RESULT-ID: COFACTOR-CHAIN-EXACT-MODULUS-20260906B
STATUS: PROVED; affine identity archival in substance, exact modulus
and averaging consumer recorded explicitly for the new sieve use.
EXACT STATEMENT:
Let x,h be positive integers, q_0=x+h, and let a_1,...,a_d be odd
integers >=3. Put A_0=C_0=1 and, for 1<=j<=d,
 A_j=a_1*...*a_j,       C_j=2*A_(j-1)-C_(j-1).
The recurrence a_j*q_j=2x-q_(j-1) is equivalent to
 A_j*q_j=C_j*x+(-1)^j*h.                                  (CF)
All q_j are integers if and only if
 C_d*x=(-1)^(d+1)*h modulo A_d.                            (CM)
Writing e=gcd(C_d,A_d), this condition is soluble precisely when e|h;
if soluble, it specifies exactly one residue class of x modulo A_d/e.
Every q_j is an integer affine form in the resulting progression
variable, with positive leading coefficient C_j*A_d/(A_j*e).
In particular for h=1 the compatible modulus is exactly A_d, since
e=1 is necessary. This is not assumed for later same-N roots.

PROOF:
Induction in the recurrence gives (CF), with the initial identity
q_0=x+h. Since every a_j>=3, C_1=1, and induction gives
 0<C_j<=2*A_(j-1) for all j>=1.
Indeed the subtraction term C_(j-1) is positive and, for j>=2,
C_(j-1)<=2*A_(j-2)<2*A_(j-1). Thus all coefficients are positive.
```

For i<j, reduction modulo A_i in the recurrence for C shows
 C_j=(-1)^(j-i)*C_i modulo A_i,
because A_(j-1) is divisible by A_i whenever j>i. Consequently (CM)
implies C_i*x+(-1)^i*h=0 modulo A_i for EVERY i<=d. This proves
that final integrality implies all earlier integrality; the converse
is immediate. The solvability and modulus assertions are the elementary
linear-congruence theorem after dividing by e.
For x=x_0+(A_d/e)*t, divisibility of each earlier numerator for all
integer t implies A_i divides C_i*A_d/e. Alternatively C_i agrees
up to sign with C_d modulo A_i and C_d*A_d/e is divisible by A_i.
Hence every stated leading coefficient is a positive integer. The
constants are integers by the choice of x_0. This proves everything.

HOSTILE CHECK:
One cannot simply divide every earlier modulus A_j by the final gcd e:
e may not divide A_j. What is integral is C_j*A_d/(A_j*e), by the
congruence just proved. The reduced modulus need not equal the product
of separately reduced edge moduli. At h=1 the coprimality conclusion
comes from the numerator +/-1, not from hypothetical independence of
cofactors. No prime or BAD hypothesis is needed for the algebra.
HISTORICAL RELATION: The master identifies
G-CRD-PATH-AFFINE-COFACTOR-NORMAL-FORM as a surviving exact transport
lemma. FAIL-V48-DIRECT-REFLECTED-TAG-INHERITANCE-001 warns against
preserving an old divisibility tag along a path; (CF) transports the
changing term instead. FAIL-V51-ALTERNATE-PATH-CONGRUENCE-AMPLIFIES-
DEEP-POWER-001 does not apply: no extra valuation is inferred from an
invertible common factor. The consumer below counts prime solutions.
DEPENDENCIES: Elementary induction and linear congruences.
NOVELTY: NOT YET AUDITED; do not claim novelty of affine transport.

RESULT-ID: ORDERED-COFACTOR-HARMONIC-SIMPLEX-20260906B
STATUS: PROVED; elementary counting lemma.
EXACT STATEMENT:
For every integer d>=1 and Y>=1,
 sum_(a_1,...,a_d>=3; product a_j<=Y) 1/(product a_j)
 <= (max(log Y-d*log 2,0))^d/d! <= (log Y)^d/d!
(the last expression used for Y>=1). Restricting the a_j to odd
integers preserves the bound. The tuples are ORDERED; their factorial
saving is not an unjustified division by permutations.
PROOF:
For each integer a>=3,
 1/a <= integral_(a-1)^a dt/t.
The boxes product_j[a_j-1,a_j] have disjoint interiors. If product a_j
is at most Y, every point of its box has all coordinates >=2 and
product coordinates <=Y. Integrate product_j dt_j/t_j over the union
of boxes, then over this larger region. In variables u_j=log t_j-log2
it is the simplex u_j>=0, sum u_j<=log Y-d log2. Its volume is exactly
the claimed expression. A negative right endpoint gives an empty
region. This proves the bound with no symmetry assumption.
IMPLICATION: Cancels the factorial cost of sieving d+O(1) prime forms
when summing over all short-product cofactor sequences.
NOVELTY: NOT YET AUDITED; standard volume method.

RESULT-ID: GROWING-FORMS-UNIFORM-SIEVE-CONSUMER-20260906B
STATUS: PROVED from the same primary FKL Lemma 5.1.
EXACT STATEMENT USED:
Fix epsilon>0. Let L=log X, l=loglog X, and
 2<=k<=2*l/log l,   X^epsilon<=z<=3X.
Consider k primitive positive-leading-coefficient integer linear forms
with coefficients and constants of absolute value <=10X^2, and all
pairwise determinants nonzero. If their root count nu(ell)>=1 at every
prime ell, then the number of positive t<=z with every form prime and
>k is at most
 2^(k+1)*k!*z*(C*l)^(k-1)/(epsilon*L)^k   (GF)
for an absolute C and sufficiently large X depending on epsilon.
For any fixed k the same assertion holds with (C*l)^k without the
nu>=1 assumption. The constants are uniform in the varying forms.

PROOF:
Let Delta be the absolute product of all leading coefficients and all
pairwise determinants. It is a nonzero integer, with
 log(3*Delta)=O(k^2*L).
Outside primes dividing Delta, the roots of the k forms are distinct,
so nu(ell)=k. As in the earlier Euler-product proof,
 sum_(ell|Delta) log ell/ell = O(loglog(3*Delta))=O(l),
 prod_(ell|Delta)(1-1/ell)^(-1) <= C*loglog(3*Delta)<=C'*l.
The first estimate follows by splitting at log(3*Delta), using the
Chebyshev bound below that cutoff, and summing log ell over divisors
above it. The second is the already proved elementary Euler bound.
Thus B_FKL<=k*sum_(ell|Delta)log ell/ell=O(k*l).
If the singular series S is zero there are no such prime values >k.
Otherwise its factors outside Delta are at most 1 by
1-k/ell<=(1-1/ell)^k, and at primes dividing Delta the condition nu>=1
gives factors <=(1-1/ell)^(-(k-1)). Therefore S<=(C*l)^(k-1).
For fixed k without nu>=1 the exponent k gives a valid upper bound.
Both FKL hypotheses hold uniformly: k=O(l/log l)=o(L/l) and
B=O(k*l)=o(L), while log z>=epsilon*L. Its explicit exponential
error is exp(O(k^2*l/(epsilon*L)))=1+o(1), hence <=2 for large X.
Substitution gives (GF). No unspecified fixed-k constant has been
allowed to grow with the number of generations.

SOURCE-AUDIT:
The exact growing-k statement and its error term were reopened in
Ford--Konyagin--Luca, Prime chains and Pratt trees, Lemma 5.1, primary
https://arxiv.org/pdf/0904.0473, printed pp.12--13. Their following
Lemma 5.2 confirms the same Euler-product method, but the argument
above checks it directly for the present forms. This does not invoke
their more specialized averaging theorem for Pratt-chain forms.
HOSTILE CHECK: The k! remains explicit and will only be cancelled by
the ordered-simplex bound; ignoring it without that bound would reduce
the provable depth. All determinants are required nonzero before B
is estimated; repeating a form could make B infinite.

RESULT-ID: MULTIGENERATION-COFACTOR-PRODUCT-BARRIER-20260906B
STATUS: PROVED.
EXACT STATEMENT:

```
Fix epsilon in (0,1) and c in (0,1). Write L=log X, l=loglog X, and
 D=floor(c*l/log l).
For all but o(X/L) primes P in [X,2X], N=2(P-1), the root P is BAD
and every simple successful stopped path
 P=q_0 -> q_1 -> ... -> q_d,       1<=d<=D,
has
 product_(j=1..d) a_j > X^(1-epsilon),
where a_j=(N-q_(j-1))/q_j is the integer cofactor of the actual edge.
In fact the exceptional count is <=X*L^(-2+c+o(1)).
This is complementary to, not a restatement of, the earlier barrier
on product q_j: a large prime child has a small cofactor.

The following uniform pair version is part of the theorem. For
3<=H<=L^2, the number of pairs of primes (P,Q) with
 X<=P<=2X, P<Q, h=Q-P+1<=H,
for which the Q-rooted stopped traversal at N=2(P-1) has a simple
successful path of length <=D with cofactor product <=X^(1-epsilon)
(or Q is directly GOOD), is
 <=H*X*L^(-3+c+o(1)).                                    (CFPAIR)
The o(1) is uniform over the stated H range, depending only on fixed
epsilon,c. No fixed alphabet, support-size bound, or closed-core
hypothesis is used.

PROOF:
Put x=P-1, so N=2x and Q=x+h. For a successful stopped path of positive
length, all earlier vertices are BAD. Activity propagates to all
children: if a prime q divides both N and N-u, it also divides u;
for prime active u this is impossible. All labels are odd, and proper
prime factors of a composite odd row are <N/3. Thus the path labels
q_1,...,q_d are <N/3, while g=N-q_d>2N/3>Q for large X. For a simple
path, the required prime values q_1,...,q_d,P,Q,g are all distinct,
except that Q=P when h=1; in that case we include that form only once.
The cofactors are odd integers >=3.

Fix d and an ordered tuple of cofactors a_1,...,a_d with
 A=A_d=product a_j<=X^(1-epsilon).
Use C_j from COFACTOR-CHAIN-EXACT-MODULUS. If e=gcd(C_d,A) does not
divide h there are no integer paths. Otherwise let x_0 in [1,A/e]
represent its unique compatible class, and set
 M=A/e,       x=x_0+M*t.
All q_j=(C_j*x+(-1)^j*h)/A_j are integer affine forms in t, with
positive leading coefficients C_j*M/A_j. In particular the last
coefficient is C_d/e, coprime to M. Also g=2x-q_d has positive leading
coefficient (2A-C_d)/e: indeed C_d<=2A_(d-1)<=2A/3.
Since x>=X-1, M<=A<=X^(1-epsilon), and h<=L^2, the progression variable
satisfies 1<=t<=2X/M=2eX/A for large X. Every required prime value is
larger than its leading coefficient. For q_j, subtracting that
coefficient gives
 [C_j*(x-M)+(-1)^j*h]/A_j>0,
because C_j>=1 and x-M>h. For g the same assertion follows from
(2-C_d/A)*(x-M)-(-1)^d*h/A>0. For P,Q it is immediate.
All q_j also exceed X/(2A)>=X^epsilon/2: the first row x-h is >=X/2
and all subsequent rows are >2N/3; divide by a_j<=A. Hence all variable
prime values exceed k=d+2 or d+3 when X is sufficiently large.

Discard any imprimitive or zero-determinant form tuple. It has no
actual solution by DISTINCT-PRIME-FORMS-DEGENERACY-FILTER, whose
coefficient and distinctness hypotheses were just verified. For all
remaining tuples, the two coprime leading coefficients M and C_d/e
ensure nu(ell)>=1 at every prime. The coefficients and constants of
P,Q,q_j,g are bounded in absolute value by 10X^2, uniformly in j,d,h:
C_j<=2A_(j-1), x_0<=A/e, and h<=L^2 suffice. Thus the growing-form
sieve applies, with log(2eX/A)>=epsilon*L.

CANONICAL CASE h=1:
Compatibility forces e=1. There are k=d+2 forms: P, all q_j, and g.
For each cofactor tuple the sieve gives at most
 2^(d+3)*(d+2)!*(2X/A)*(C*l)^(d+1)/(epsilon*L)^(d+2).
Sum 1/A over all ordered cofactor tuples of product at most
X^(1-epsilon). The harmonic-simplex lemma bounds this by L^d/d!.
Therefore the total at depth d is at most
 C_epsilon*(X/L^2)*(d+3)^3*(C_epsilon*l)^(d+2).            (CF0d)
This bound is deliberately loose in a fixed power of l; the key
factorial ratio (d+2)!/d! is only quadratic in d.

LATER-ROOT PAIRS:
For h>=3 there are k=d+3 forms, including the extra prime P=x+1 and
Q=x+h separately. The fixed tuple, fixed compatible h bound has the
same shape with d+3 in place of d+2, and the progression length carries
its full factor e:
 2^(d+4)*(d+3)!*(2eX/A)*(C*l)^(d+2)/(epsilon*L)^(d+3).
For the chosen cofactor tuple e is fixed and depends on that tuple,
not on h. Sum over all compatible odd 3<=h<=H. Crucially,
 sum_(h<=H; e|h) e <= e*floor(H/e)<=H.
This includes every compatible gcd with no growing divisor loss.
Now sum 1/A using the ordered-simplex bound. The depth-d pair count is
at most
 C_epsilon*(H*X/L^3)*(d+3)^3*(C_epsilon*l)^(d+2).          (CFPd)
It was essential to average the actual reduced modulus; setting e=1
for later roots would have omitted valid paths.

SUM OVER DEPTH:
For D=floor(c*l/log l),
 log[sum_(d=1..D)(d+3)^3*(C_epsilon*l)^(d+2)]
 <=(D+2)*log(C_epsilon*l)+O(log(D+3))=(c+o(1))*l.
Thus that sum is L^(c+o(1)). Summing (CF0d) and (CFPd) proves the
claimed bounds for positive path lengths. Direct canonical success
has the previously proved O(X/L^2) count. Direct later-root success
requires P, P+h-1, P-h-1 prime; these are three primitive forms with
nonzero determinants for h>=3 and nu>=1. The same fixed-three-form
sieve gives O(H*X*l^2/L^3), absorbed by (CFPAIR).
Finally X*L^(-2+c+o(1))=o(X/L) because c<1. This proves the theorem.

HOSTILE CHECK:
1. The bound controls all cofactor sequences of short product, not just
   a chosen transcript. It remains a restriction on successful paths,
```

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   not a proof of full BFS failure.
2. A simple path is used to ensure distinct variable prime forms. The
   proof does not sieve repeated forms as independent prime conditions.
3. The growing factorial in FKL is cancelled by an explicitly proved
   volume bound for ORDERED tuples, not by an illicit d! symmetry.
4. For later roots, reduced earlier coefficients are integers because
   of final-to-earlier congruence compatibility, not because e divides
   every partial product A_j.
5. The e factor is averaged before the cofactor tuples are summed.
   No uncontrolled average of a moving gcd is hidden in the estimate.
6. Short cofactor product gives a polynomially long progression. When
   the product is larger, the proof makes no estimate for those paths.
7. This uses the full explicit growing-k FKL bound; a fixed-k big-O
   sieve assertion alone would not justify D tending to infinity.
DEPENDENCIES: Exact cofactor modulus; harmonic-simplex bound; growing
prime-form sieve; activity; no prime-pair lower bound or global failure.
RELATION TO GOLDBACH: A restriction on shallow successful routes. A
closed BAD core has no successful path, so this theorem alone cannot
contradict its existence.
RELATION TO p0: The pair theorem prepares a same-N extension. The h=1
coprimality is arithmetically special but not a success mechanism.
WHAT REMAINS OPEN: Paths with both large label product and large
cofactor product; complete radius three; eventual reachability.
NOVELTY: NOT YET AUDITED.

RESULT-ID: MULTIGENERATION-LABEL-PRODUCT-PAIR-COUNT-20260906B
STATUS: PROVED; same-N extension of the previous label-product barrier.
EXACT STATEMENT:
For fixed epsilon,c in (0,1), D=floor(c*l/log l), 3<=H<=L^2,
the number of prime pairs P<Q, P in [X,2X], h=Q-P+1<=H, such that
Q is GOOD or a simple successful path from Q of length d<=D has
M=product_(j=1..d)q_j<=X^(1-epsilon), is
 <=H*X*L^(-3+c+o(1)),
uniformly over H. The analogous canonical count is
 <=X*L^(-2+c+o(1)), as in the previously audited theorem.

PROOF OF THE PAIR EXTENSION:
Fix h and a tuple of distinct odd prime labels q_1,...,q_d with the
specified product bound. Put x=P-1. The actual edges impose
 x=h modulo q_1,
 2x=q_(j-1) modulo q_j for j>=2.
CRT gives one class x=x_0+M*t, 1<=x_0<=M, and
1<=t<=2X/M for actual starts. The three necessary prime forms are
 P=x+1,       Q=x+h,       g=2x-q_d.
Their positive leading coefficients are M,M,2M and are smaller than
actual prime values. The determinants, up to sign, are
 M*(h-1),       M*(q_d+2),       M*(q_d+2h),
all nonzero. Imprimitive forms have no actual prime solutions since
the values exceed the leading coefficients. Their Delta is bounded by
a fixed power of X uniformly in d,h: only three forms are present.
The fixed-three-form sieve with the general singular-series bound
O(l^3) gives O_epsilon(X*l^3/(M*L^3)). All path primes are <2X.
Sum ordered tuples, dropping distinctness and the product cutoff only
in the positive reciprocal sum. It is at most
 (sum_(q<=2X prime)1/q)^d <=(l+O(1))^d.
Sum d<=D and h<=H. The result is at most
 O_epsilon(H*X*l^3/L^3)*(l+O(1))^D
 =H*X*L^(-3+c+o(1)).
Direct GOOD roots are included by the three-form count already proved
in the cofactor theorem. For h=1 P=Q, so the canonical proof correctly
uses only TWO prime forms and has the stated one-logarithm weaker count.
HOSTILE CHECK: No extra prime condition is counted at h=1. The varying
path length does not enter the Delta bound because the modulus M itself
is bounded; no infinite-rank sieve is invoked here.
NOVELTY: NOT YET AUDITED.

RESULT-ID: SAME-N-GROWING-DEPTH-DUAL-PRODUCT-BARRIER-20260906B
STATUS: PROVED.
EXACT STATEMENT:
Fix epsilon in (0,1), c in (0,1), and kappa>=0 with c+kappa<1. Put
 L=log X, l=loglog X,
 D=floor(c*l/log l),       K=floor(L^kappa).
For all but o(X/L) primes P in [X,2X], let N=2(P-1). Then every
SAME-N root p_k(N), 0<=k<=K, is BAD, and every simple successful
stopped path from any one of those roots, of length 1<=d<=D, satisfies
 BOTH
  product_(j=1..d) q_j > X^(1-epsilon),
  product_(j=1..d) [(N-q_(j-1))/q_j] > X^(1-epsilon).        (DUAL)
The exceptional proportion among primes P is at most
 L^(-(1-c-kappa)/2+o(1)).
Here q_0 is the selected root, and the products exclude q_0 as a label
but include the cofactor of its first edge. The result is simultaneous
over the entire root prefix and over every eligible simple path.

PROOF:
The canonical union of the two small-product exceptions and direct
success has size <=X*L^(-2+c+o(1)). For later roots, the union of the
two pair exceptions just proved has size <=H*X*L^(-3+c+o(1)), uniformly
for 3<=H<=L^2. These are counts of all possible later-root pairs, so
there will be no extra factor K when using them for a root prefix.
Take H to be an integer within 1 of
 L^((3-c+kappa)/2).
The inequality c+kappa<1 guarantees H<L^2 for large X. The same
consecutive-gap summation proved in SAME-N-GROWING-ROOTS-COMPLETE-
RADIUS-TWO gives at most O(KX/H) starts with p_K-P>H-1. The use of PNT
to place all p_K below 3X remains valid because K is a fixed power of
log X. Outside these long-gap starts, every later prefix root has
h<=H, so all its exceptions are counted by the pair union.
Divide the total exceptional count by pi(2X)-pi(X) asymptotic to X/L.
It is bounded by
 L^(-1+c+o(1)) + O(K*L/H) + H*L^(-2+c+o(1))
 <= L^(-(1-c-kappa)/2+o(1)).
The exponent is negative. Every root and every successful simple path
outside this exceptional set has (DUAL), proving the assertion.

HOSTILE CHECK:
```

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1. This is a restriction on successful paths, not complete BADness
   through the growing depth D. Large products remain possible.
2. Both products are for the ACTUAL same path. Their simultaneous
   lower bounds are obtained by removing the union of the two failure
   sets; no independence is assumed.
3. The depth-prefix tradeoff c+kappa<1 is explicit. It is not legitimate
   to let both parameters approach 1 with this proof.
4. The cofactor proof had to handle the changing gcd of a moving
   product; the label proof had to retain the extra prime P. Neither
   is an automatic h=1-to-h>1 substitution.
5. A shortest successful path is simple, so the theorem supplies a
   necessary condition for success by D. A closed BAD core has no
   successful path and therefore satisfies the conclusion vacuously.
6. The proof does not claim that the scalar data C_j or A_j distinguish
   p0. The SAME-N conclusion proves the asymptotic obstruction is shared.
DEPENDENCIES: Both product pair bounds; PNT; prime-gap double count.
RELATION TO GOLDBACH: Constrains increasingly long routes but supplies
no contradiction to arbitrary closed support or to large-product success.
RELATION TO p0: A multigeneration property previously available only
for p0 is now shared by a growing SAME-N nearby prefix, with a second,
complementary arithmetic product constraint added.
WHAT REMAINS OPEN: The middle range where both products are large,
complete radius three, and eventual escape from a closed BAD set.
NOVELTY: NOT YET AUDITED.

COROLLARY / EXACT RESIDUAL WINDOW:
At any successful path the identity
 (product q_j)*(product a_j)=product_(j=1..d)(N-q_(j-1))
is exact. Since N<4X, (DUAL) implies, on the same typical starts,
 X^(1-epsilon)<product q_j<4^d*X^(d-1+epsilon),
and the same upper and lower bounds for product a_j. At depth three,
the unexcluded label-product range is therefore between approximately
X^(1-epsilon) and X^(2+epsilon). It still includes the region with
all three labels of size X^0.4. Nothing here resolves the existing
FAIL-ID AUTOMATIC-THIRD-GENERATION-SIEVE-SWITCH-20260906.

FAIL-ID: SUCCESS-PRODUCT-BARRIER-TO-CLOSED-CORE-HEIGHT-20260906B
STATUS: KILLED AS AN INFERENCE; both input theorems survive.
CLAIM / ROUTE:
Combine the growing-depth product barriers with TWO-SMALLEST-ROWS-
HEIGHT to rule out a closed active BAD core reached from p0.
WHY IT LOOKED PROMISING:
One input controls products along moving paths; the other forces
complexity in a core's two least rows. Both refer to large arithmetic
objects, so it is tempting to regard their bounds as opposing capacities.
EXACT FAILURE:
The product barriers quantify over SUCCESSFUL paths, which a closed
BAD core has none of. Applying them inside such a core yields only a
vacuously true statement. The height theorem is a lower complexity
bound, not a conflicting upper bound. The new cofactor theorem does
not alter this logical domain mismatch.
In its sieve proof the final prime g=N-q_d is a crucial EXTRA prime
condition. Dropping it to count paths ending at BAD vertices loses a
factor log X: for canonical starts k changes from d+2 to d+1. The
resulting crude bound is of size (X/L)*L^(c+o(1)), not o(X/L). Thus
the written proof itself identifies the missing consumer, rather than
merely failing to state one.
WHAT SURVIVES:
Both complete radius-two density theorems, both path-product barriers,
and the pointwise two-smallest-row height restriction. None assumes a
fixed prime alphabet through the path.
DO NOT REPEAT:
Do not apply a successful-endpoint sieve to a failed closed core;
do not treat two lower bounds as a lower-versus-upper contradiction.
POSSIBLE REPLACEMENT:
A bound for BAD path endpoints or moving support that supplies an
independent upper restriction in a genuinely closed failed world.

RESULT-ID: MULTIGENERATION-COFACTOR-ADVERSARIAL-SCAN-20260906B
STATUS: VERIFIED COMPUTATION; no asymptotic conclusion inferred.
DOMAIN AND METHOD:
Exhaustive primes P in [1000,2000) and [10000,20000), respectively
135 and 1033 starts. For each, N=2(P-1), same-N roots k=0,1,2,5.
Enumerate every SIMPLE stopped successful path of lengths 1 through 4;
these need not be shortest paths if another branch succeeds earlier.
Stop expansion at GOOD vertices, and exclude repeated labels within
one path. Exact SPF sieve through 1,000,100. No factor-size exclusions.
The diagnostic's depth 4 is fixed independently of asymptotic D(X).
The small-product diagnostic uses the exact integer test product^10
<=X^9, i.e. epsilon=1/10, so floating-point comparisons do not decide
membership.

CHECKS AND RESULTS:
22,562 actual successful paths passed every affine-identity, exact-
modulus, reduced-coefficient-integrality, and gcd-compatibility check.
For all canonical paths the final gcd e was exactly 1. For 2,320 paths
with small cofactor product, the progression variable was >=1, all
actual prime values exceeded their leading coefficients, every form
was primitive, and every pair determinant was nonzero.
There were 4,767 depth-three-or-four paths where e failed to divide
at least one earlier partial product A_j. Thus the proof of integer
reduced coefficients is indispensable; division of each A_j by e
would be an actual error.
Maximum observed e was 85, at P=10343, N=20684, Q=p5=10427, h=85,
path Q -> 13 -> 7 -> 29 -> 3. Its cofactor product is
11,437,574,202,585. This large-product path is outside the cofactor
sieve domain, and its progression variable is t=0. It is evidence of
the exact large-modulus obstruction, not a counterexample to the bound.

NONREDUNDANCY WITNESS:
P=1279, N=2556, Q=p5=1301, h=23 admits
 1301 -> 251 -> 461 -> 419,       N-419=2137 prime.
All three edge cofactors equal 5. Hence A=125 while the label product
is 48,482,909. The new cofactor-product count includes this type of
configuration even though the earlier label-product count does not.
Its exact forms at t=10 are
```

```
 P=125t+29, Q=125t+51, q1=25t+1, q2=45t+11,
 q3=41t+9, g=209t+47.
```

An actual finite example inside an asymptotically sparse region is expected and does not contradict the theorem.

At X=10000, among all depth-three-or-four successful paths, 14,200 had both products large, 1046 had only the label product small, and 58 had only the cofactor product small. These finite multiplicity counts identify regions tested by the two arguments; they are not counts of exceptional starts, convergence evidence, or proof that the residual region has positive asymptotic density.

```
BEGIN DIAGNOSTIC SOURCE audit_cofactor_paths_20260906b.py
import numpy as np
from math import isqrt,gcd,prod
from collections import Counter
from itertools import combinations
import json
LIMIT=1000100
spf=np.zeros(LIMIT+1,dtype=np.int32)
for p in range(2,isqrt(LIMIT)+1):
 if spf[p]==0:
  v=spf[p*p::p];v[v==0]=p
primes=np.flatnonzero(spf[2:]==0)+2
ip=np.zeros(LIMIT+1,dtype=np.bool_);ip[primes]=True

def pf(n):
 out=[]
 while n>1:
  p=int(spf[n]) or n;out.append(p)
  while n%p==0:n//=p
 return out

stats=Counter();domains={};witness={};max_e=(0,None)
for X in [1000,10000]:
 inds=np.flatnonzero((primes>=X)&(primes<2*X))
 local=Counter()
 for idx in inds:
  P=int(primes[idx]);x=P-1;N=2*x
  for k in [0,1,2,5]:
   Q=int(primes[idx+k]);h=Q-x
   def visit(path,aa):
    global max_e
    u=path[-1];d=len(aa)
    if ip[N-u]:
     if d==0:
      local['direct_GOOD_roots']+=1;return
     local[f'success_paths_depth_{d}']+=1
     AP=[1];CP=[1]
     for a in aa:
      CP.append(2*AP[-1]-CP[-1]);AP.append(AP[-1]*a)
     A=AP[-1];C=CP[-1];e=gcd(A,C);mod=A//e
     assert h%e==0
     if k==0:assert e==1
     residue=((-1)**(d+1)*(h//e)*pow(C//e,-1,mod))%mod if mod>1 else 0
     if residue==0:residue=mod
     assert (x-residue)%mod==0
     t=(x-residue)//mod
     forms=[(mod,residue+1)]
     values=[P]
     if k:
      forms.append((mod,residue+h));values.append(Q)
     for j in range(1,d+1):
      assert AP[j]*path[j]==CP[j]*x+(-1)**j*h
      assert CP[j]*mod%AP[j]==0
      assert (CP[j]*residue+(-1)**j*h)%AP[j]==0
      forms.append((CP[j]*mod//AP[j],(CP[j]*residue+(-1)**j*h)//AP[j]))
      values.append(path[j])
     last_u,last_v=forms[-1]
     forms.append((2*mod-last_u,2*residue-last_v));values.append(N-u)
     assert all(z>0 for z,b in forms)
     assert [z*t+b for z,b in forms]==values
     assert gcd(mod,C//e)==1 and len(set(values))==len(values)
     stats['all_affine_and_modulus_checks']+=1
     M=prod(path[1:]);small_A=A**10<=X**9;small_M=M**10<=X**9
     if small_A:
      assert t>=1 and all(v>z for v,(z,b) in zip(values,forms))
      assert all(gcd(z,b)==1 for z,b in forms)
      assert all(z*d-w*b!=0 for (z,b),(w,d) in combinations(forms,2))
      stats['short_cofactor_product_full_filter_checks']+=1
     rec={'X':X,'P':P,'N':N,'k':k,'Q':Q,'h':h,'path':path,'cofactors':aa,'label_product':M,'cofactor_product':A,'C_d':C,'e':e,'modulus':mod,'residue':residue,'t':t,'prime_forms':forms,'prime_values':values}
     if e>max_e[0]:max_e=(e,rec)
     if d>=3:
      local[f'depth3or4_small_label_{small_M}_small_cofactor_{small_A}']+=1
      if small_A and not small_M:witness.setdefault('large_labels_small_cofactors_depth3or4',rec)
      if e>1 and any(z%e for z in AP[1:-1]):
       stats['e_does_not_divide_all_earlier_partial_products_depth3or4']+=1
       witness.setdefault('reduced_earlier_coefficients_require_proof',rec)
     return
    if d==4:return
    for q in pf(N-u):
     if q in path:continue
     visit(path+[q],aa+[(N-u)//q])
   visit([Q],[])
 domains[str(X)]={'prime_starts':len(inds),'counts':dict(local)}
print(json.dumps({'domain':'Exhaustive primes in [X,2X), X=1000,10000; roots k=0,1,2,5; all SIMPLE stopped successful paths through length 4, not only shortest paths. No size exclusions. Exact SPF through 1,000,100. Small-product flag is exact product^10<=X^9, i.e. epsilon=1/10. This diagnostic depth is fixed independently of the asymptotic D(X).','intervals':domains,'audit':dict(stats),'witnesses':witness,'maximum_e':max_e},indent=2))
END DIAGNOSTIC SOURCE
BEGIN DIAGNOSTIC OUTPUT
```


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        7,
        29,
        3,
        20681
      ]
    }
  ]
}
```


```
END DIAGNOSTIC OUTPUT

FINAL HOSTILE AUDIT 2026-09-06B:
The radius-two pair proof retains the two independent degeneracies:
gcd(b,2a-1)|h in CC and r|(2a-1), r|h in CL. Both occur in actual
successful paths. The cofactor-chain proof uses final integrality to
justify all earlier affine forms, rather than assuming reductions
commute with partial products. The growing-k sieve constants, the
factorial sum, the h-average of the exact gcd, and the prime-gap prefix
consumer are written explicitly. Parameters epsilon,c,kappa in the
dual theorem are FIXED; no uniformity as they approach their endpoints
is claimed. The only asymptotic variable in that theorem is X.
No independent external referee audit or literature novelty audit has
been performed. Computational checks test algebra and scope, not the
asymptotic sieve theorem. All original corpus files are unchanged.

========================
STRONGEST NEW RESULTS
========================
1. SAME-N-GROWING-DEPTH-DUAL-PRODUCT-BARRIER: for fixed c+kappa<1,
   almost every prime P in [X,2X] has, simultaneously for the same-N
   roots k<=floor((log X)^kappa), no successful simple path of length
   <=floor(c*loglog X/logloglog X) whose LABEL product or COFACTOR
   product is <=X^(1-epsilon). The new cofactor argument uses an exact
   moving modulus, growing-form sieve, and an ordered-simplex bound.
2. SAME-N-GROWING-ROOTS-COMPLETE-RADIUS-TWO: almost every such P has
   COMPLETE radius-two BADness simultaneously for all same-N roots
   k<=K=o((log X)^(1/6)/(loglog X)^(3/2)). This extends the previous
   canonical radius-two theorem to nearby roots, with all gcd cases.
3. Exact same-N non-dominance: at N=2180 p0 is BAD through radius two
   while p1 is directly GOOD; at N=2024 p0 succeeds at depth two while
   p1 has complete radius-two BADness. This refutes either universal
   inclusion of the two shallow BADness events, not eventual success.
All novelty claims remain NOT YET AUDITED.

========================
CURRENT FRONTIER
========================
No proof of binary Goldbach, no universal p0 success theorem, and no
exclusion of moving closed BAD support. Complete radius three remains
uncontrolled. The new count removes another class of large-label paths,
namely those with small cofactor product, but the region where BOTH
products are large remains. The successful-path barriers have no
closed-core consumer: inside a BAD closed core their hypotheses never
occur. The two-smallest-row height restriction remains a compatible
lower bound, with no contradictory upper bound. The Scott--Bozek
classification issue was not used or advanced in this continuation.

========================
p0 STATUS
========================
A stronger SAME-N no-go was obtained: complete radius-two BADness and
the growing-depth dual product restriction are shared by nearby roots.
No nontrivial property holding for every p0 and failing at a nearby
same-N root with an eventual-reachability consequence was proved.
Finite shallow success can favor either p0 or p1. The empirical
exceptionality of eventual p0 traversal remains unexplained.

========================
EXACT NEXT QUESTION
========================
Is the number of primes P in [X,2X] admitting the following system
o(X/log X)?
 P=a*q+2,
 b*r=(2a-1)*q+2,
 c*s=2*a*q+2-r,
 g=2*a*q+2-s prime,
where q,r,s are distinct odd primes, a,b,c are odd integers >=3,
and BOTH q*r*s>X^(9/10) and a*b*c>X^(9/10).
Count distinct starts P, not paths with multiplicity. This is the
explicit remaining region for canonical stopped success at depth three.
Together with the proved small-product barriers and the existing
radius-two estimate, an affirmative answer would establish complete
canonical radius-three BADness for almost all h=1 starts. It would
still be an asymptotic traversal theorem, not a Goldbach proof.


====================================================================
CONTINUATION 2026-09-07: RETURN FROM ARTICLE PREPARATION TO RESEARCH
====================================================================
The 2026-09-06B frontier above remains the starting point. No recent
asymptotic result is being used critically in the following exact arguments.
Research resumes in this same raw TXT file. The article is a separate,
completed deliverable and is not the research checkpoint.

RESULT-ID: SAME-N-SUPPORT-WITNESS-RANK-CORRECTION-20260907
STATUS: SCOPE-CORRECTED / VERIFIED COMPUTATION.
EXACT STATEMENT:
For N=212, the primes at or above N/2 begin 107,109,113. Consequently
113 is p2, not p1. A temporary p1 label during article preparation was
incorrect. The support comparison at N=212 is valid as p0 versus p2:
 PF(212-107)={3,5,7}, PF(212-113)={3,11}, PF(212-3)={11,19}.
```

```
A genuine same-N p0 versus p1 replacement is N=344:
 p0=173, p1=179,
 344-173=171=3^2*19,
 344-179=165=3*5*11,
 344-3=341=11*31.
Thus the least first factor 3 has a second row disjoint from the p0
factor set {3,19}, but meeting the p1 factor set {3,5,11} at 11.
Both searches have first-success depth 2 and inclusive work W=5.
The common successful tail is 3 ->31, with 344-31=313 prime.
For p0 the FIFO test order is 173,3,19,11,31; for p1 it is
179,3,5,11,31. All indicated primalities and all root ranks were checked
by exact trial division, including the absence of intervening primes.
DEPENDENCIES / PROVENANCE:
The underlying least-factor escape is already PCRD-H1-1 in the master,
not a newly discovered general theorem. The new point is the exact
same-N witness and correction of the previously mislabelled index.
HOSTILE CHECK:
This distinguishes initial-support retention, not eventual success.
Both roots here even have the SAME first-success depth and work.
Neither this witness nor the minimum-factor lemma implies universal p0
success. The same-N radius-two witnesses N=2024 and N=2180 retain their
previously stated p0/p1 ranks after explicit enumeration.
NOVELTY: NOT YET AUDITED.

RESULT-ID: EXACT-SCAN-WORK-CONVENTION-AUDIT-20260907
STATUS: VERIFIED COMPUTATION / SOURCE-CORRECTION.
DOMAIN:
Exact full FIFO BFS, ascending distinct prime factors, duplicate suppression
on discovery; every even 4<=N<=10^7. No work/depth cutoff, random sampling,
probable-prime test or incomplete factorization.
RESULT:
4,999,999 canonical inputs; zero failures. Inclusive work W counts the
successful vertex. Mean W=7.052697210539442, maximum W=151 first at
N=3542948; maximum success depth 10 first at N=11822.
Through N<=10^6: mean W=5.804359608719217, max W=133 at N=976904.
The old article's displayed step statistics use W, but its pseudocode
counted expanded composite rows E. On success W=E+1; on failure W=E.
This is a counting-convention repair, not a change to success verdicts.
Root-rank ablation k=0,...,6 through N<=200000 reproduces failures
[0,2,4,5,3,2,1], with eligible composite-complement denominators
[73394,81448,81043,80909,80788,80980,80829].
The complete upper-half composite-root census through N<=20000 has
4,041,839 eligible roots, 301 failures at the six known hosts, and
2261 excluded unit-complement rows. Host multiplicities are
128:8, 1718:68, 1862:63, 1928:57, 2200:4, 6142:101.
Independent trial division verifies all 999 canonical inputs through
2000, all three named record inputs, and ten exact example roots with
explicit prime ranks. This is NOT a second independent full-range scan.
SOURCE:
Reproduction source and raw compressed records are in the completed
p0_zenodo_release archive. These regenerated records do not replace
proof and do not extend the original canonical numerical bound.

RESULT-ID: PRIME-LIFT-FULL-BFS-CONJUGACY-20260907
STATUS: PROVED.
EXACT STATEMENT:
Let N>=4 be even, d a prime divisor of N, x=N/d, and m=x-1>=2.
Suppose R=N-m=(d-1)x+1 is prime. Use full factor BFS from any admissible
prime, including the inactive start d; remove duplicate visits, use FIFO
and enqueue all distinct prime factors in increasing order.
Then R>N/2, R<=N-2, and:
(A) If m is prime, R succeeds immediately with (depth,W)=(0,1), while
    d succeeds via d ->m with (depth,W)=(1,2).
(B) If m is composite, the two searches have identical verdicts. If they
    succeed, their first-success depths and inclusive work counts agree.
    In fact their entire ordered test transcripts agree after replacing
    the initial d by R. All complete stopped frontiers after the initial
    root are the same. On failure the reachable sets differ only by
    replacing d with R.
At graph level in (B), deletion of the harmless self-loop d->d and
renaming that root gives the same complete reachable stopped graph.

PROOF:
Since d>=2, R=N-N/d+1>=N/2+1. The assumption m>=2 gives R<=N-2.
Also R-d=(d-1)(x-1)=(d-1)m>0. If d|m, then d|N-m=R, contradicting
primality of R>d. Hence d does not divide m.
If a prime q divides both m=x-1 and N=dx, it must divide d, since
q cannot divide x and x-1. This would mean q=d, already excluded.
Therefore every prime factor of m is active, i.e. coprime to N.
We have the EXACT row relation
 N-d=d*m,            N-R=m,
so PF(N-d)={d} disjoint-union PF(m).
The self-loop at d is ignored by duplicate suppression. Thus, after
processing the first BAD row, both queues contain exactly the same
ordered set PF(m), each at distance 1.
Activity is preserved: if p is an active prime and q|N-p also divides
N, then q|p, so q=p, contradicting p not dividing N. Consequently no
later vertex can equal the inactive prime d.
Nor can a later prime parent p have an edge to R in the composite-m
case. Since R>N/2, a positive N-p<N divisible by R must equal R;
this would give p=N-R=m, which is composite. Thus the root R is never
encountered again from any prime vertex, including any terminal row.
The two discovered sets therefore differ only in their initial root;
this difference cannot affect any later duplicate test. Inductively,
FIFO removes the same next prime at the same distance, tests the same
row, and enqueues the same new factors. This proves (B), including
complete-frontier equality and the graph statement. No assumption of
success is used in the induction.
If m is prime, the R-root test is immediately successful. The d-root
row has factors d and m, with m!=d as established above. After ignoring
d's self-loop, m is tested at depth 1 and is GOOD because N-m=R is
prime. This proves (A).

HOSTILE CHECK:
The initial d is INACTIVE. The active-root theorem cannot be invoked
at that initial vertex; the proof explicitly separates its self-loop
```

```
and proves that all other children are active. The conjugacy is for
composite m; applying its depth equality to prime m would be false.
The absence of an incoming edge to R is proved, rather than inferred
from a selected BAD path. Entire BFS queues are compared, not one path.
The theorem does not assert that either search succeeds.
DEPENDENCIES: Elementary divisibility, primality and finite full BFS only.
RELATION TO GOLDBACH:
An exact equivalence of two algorithmic reachability questions, not
an independent source of a Goldbach representation.
RELATION TO p0:
Taking d=2 gives x=P-1, N=2(P-1), R=P. If P>=5 is prime, P is the
canonical root. Therefore every non-direct canonical h=1 traversal
is, for ALL generations, exactly the traversal from prime 2 after the
stated initial-root replacement. An invariant of the resulting stopped
transition graph which ignores that replacement and its self-loop
cannot distinguish these two starts.
This is a restricted h=1 theorem; root 2 is not a nearby upper root.
It does not claim equivalence between p0 and pk for k>=1.
WHAT REMAINS OPEN:
Whether the prime-lift condition with d=2 excludes eventual failure.
The same question for all prime divisors d is FALSE, as the following
exact example shows. Thus lift primality alone is not a general closer.
NOVELTY: NOT YET AUDITED. Targeted corpus search found the h=1
least-factor/shift-2 lemmas but no prior full-BFS conjugacy entry.

FAIL-ID: PRIME-LIFT-PRIMALITY-FORCES-SUCCESS-20260907
STATUS: KILLED.
CLAIM / ROUTE:
If the affine lift R=(d-1)N/d+1 of a prime divisor d|N is prime, then
full factor search from d (equivalently its lift R) must succeed.
WHY IT LOOKED PROMISING:
Lift primality automatically removes inactive factors from PF(N/d-1),
and for d=2 gives exactly the canonical h=1 initial condition.
EXACT COUNTEREXAMPLE:
N=1862, d=7, x=266, m=265=5*53, R=1597 prime.
Both complete BFS runs fail, each after W=22 tests. Their common active
nonroot reachable set U has the 21 labels listed by the rows below.
For every p in U, N-p is composite and has all factors in U. The root
1597 also has all its factors in U. This is a COMPLETE closure
certificate, independent of any claim about terminal-SCC classification.
 1862-3 = 1859 = 11 * 13^2
 1862-5 = 1857 = 3 * 619
 1862-11 = 1851 = 3 * 617
 1862-13 = 1849 = 43^2
 1862-17 = 1845 = 3^2 * 5 * 41
 1862-41 = 1821 = 3 * 607
 1862-43 = 1819 = 17 * 107
 1862-47 = 1815 = 3 * 5 * 11^2
 1862-53 = 1809 = 3^3 * 67
 1862-67 = 1795 = 5 * 359
 1862-83 = 1779 = 3 * 593
 1862-107 = 1755 = 3^3 * 5 * 13
 1862-113 = 1749 = 3 * 11 * 53
 1862-167 = 1695 = 3 * 5 * 113
 1862-179 = 1683 = 3^2 * 11 * 17
 1862-251 = 1611 = 3^2 * 179
 1862-359 = 1503 = 3^2 * 167
 1862-593 = 1269 = 3^3 * 47
 1862-607 = 1255 = 5 * 251
 1862-617 = 1245 = 3 * 5 * 83
 1862-619 = 1243 = 11 * 113
 1862-1597 = 265 = 5 * 53

The inactive root has 1862-7=1855=7*5*53, so only its own duplicate
self-loop is added. Primality and all rows were independently checked
by trial division. The prime-divisor/lift enumeration was exhaustive
only over divisors of the six known hosts, not over all N: this was the
only admissible prime lift in that small diagnostic.
For the SAME N, p0=937 succeeds along 937 ->37 ->73, since
1862-937=5^2*37, 1862-37=5^2*73, and 1862-73=1789 prime.
Thus a successful canonical search can have a separate branch into U;
reaching a BAD closed component does not imply full rooted failure.
WHAT SURVIVES:
The exact conjugacy and the special d=2 question. The counterexample
has d=7 and does not contradict canonical h=1 success.
DO NOT REPEAT:
Do not infer success merely from a prime affine lift or the absence of
additional inactive factors. Those conditions hold in this example.
POSSIBLE REPLACEMENT:
A restriction using d=2 essentially, or a pointwise theorem controlling
access from PF(N/2-1) to moving closed active support.


RESULT-ID: LOW-GAP-COMPLETE-THREE-LABEL-ALTERNATIVE-20260907
STATUS: PROVED.
EXACT STATEMENT:
Let N>=4 be even, R>=N/2 an admissible prime root, m=N-R composite,
h=R-N/2, and q=min PF(m). Suppose h<q. In the COMPLETE stopped factor
search from R, either:
 (i) a GOOD vertex occurs at depth at most 2, or
 (ii) at least THREE DISTINCT nonroot primes occur by depth 3.
In particular, if the full rooted traversal fails, its nonroot reachable
active set has at least three elements. No active closed BAD alphabet
of size at most two can contain the whole first factor set PF(m).
This is a pointwise statement, with no asymptotic or fixed-envelope premise.

PROOF:
Since m is composite and R>=N/2 is prime, R>N/2, R is odd and active,
m is odd, q>=3, and h is a positive integer. Every subsequent prime is
active, by the activity lemma. A child t of an odd composite BAD row
N-p has t< N/3: its cofactor is an odd integer at least 3.
Assume there is no GOOD vertex at depth <=2. Put S=PF(m).
If |S|>=3, the conclusion already holds at depth 1.
If |S|=2, consider its least element q. For any r in S,
 r|N-q implies r|(2h-q), since N=2m+2h.
But h<q implies 0<|2h-q|<q<=r; the zero case is excluded because q
```

```
is odd. Thus PF(N-q) is disjoint from S. The row N-q is composite
under our assumption and has at least one prime factor, reached at
depth 2 outside S. Together with S this gives three distinct labels.
It remains to treat S={q}. Then m=q^u for an integer u>=2. The BAD
row N-q has no q factor, by activity. If it has two distinct factors,
we already have at least three nonroot labels at depth <=2.
Otherwise N-q=b^v for a prime b!=q and v>=2. The vertex b occurs
at depth 2, is BAD by assumption, and b<N/3. It cannot divide its
own row. If no third label occurred by depth 3, the entire factor set
of N-b would consequently be {q}, so N-b=q^w with w>=2.
If w<=u, then b=N-q^w>=N-q^u=R>N/2, contradicting b<N/3.
If w>u, then q^w>=q*m. But
 q*m-N=(q-2)m-2h >= m-2h >= q^2-2h >0,
since q>=3 and h<q. This contradicts q^w=N-b<N.
Thus N-b must introduce a third distinct nonroot prime by depth 3.
Every case has been covered.
For the closed-alphabet corollary, a set containing PF(m) and closed
under all BAD rows contains every nonroot descendant. If it had at
most two elements, the proved alternative would force a GOOD vertex,
contrary to BAD closure. No terminal-core assumption is used.

HOSTILE CHECK:
The assertion is about COMPLETE rows, not a selected-factor path.
In the singleton case the strict inequality q*m>N is the fragile
arithmetic step, and is explicitly justified by h<q and u>=2.
Dropping the small-gap premise is false: at N=2200, R=1471 prime,
 m=729=3^6, h=371, q=3,
 2200-3=2197=13^3, 2200-13=2187=3^7.
The full search is BAD and has exactly the two nonroot labels {3,13}.
This uses the existing N=2200 two-core certificate, not a new EAC.
Crucially the conclusion does NOT exclude a two-vertex TERMINAL core
reached after additional transient labels; it only excludes a whole
nonroot reachable alphabet of size <=2. This avoids the open
Scott--Bozek classification issue and does not repair that citation.

DEPENDENCIES / HISTORICAL AUDIT:
The |S|>=2 step is an explicit consumer of the existing minimum-factor
escape corollary (master cor:minfactor, PCRD-H1-1 at h=1). The singleton
step uses a gap between consecutive powers of q and the upper-root
geometry. The historical C-MONOMIAL-TWO-EDGE-RIGIDITY and its novelty
corrections concern a square-map/high-floor mechanism; no claim is made
that pure-power path arguments as such are new. No Matveev estimate,
height threshold, growing sieve or global Goldbach failure is assumed.
RELATION TO GOLDBACH:
A small, unconditional restriction on a possible failed traversal.
It does not rule out larger alphabets or prove a prime-pair existence.
RELATION TO p0:
For every canonical h=1 root with composite first complement it applies
automatically, since q>=3. It ALSO applies to every nearby or other
upper root satisfying its stated h<q premise. Thus it is not a universal
p0-only invariant and supplies no explanation of eventual p0 reliability.
WHAT REMAINS OPEN:
A quantitative lower bound growing with N on the complete moving
reachable support, or any new arithmetic mechanism excluding its closure.
The old automatic envelope-iteration obstruction remains in force.
NOVELTY: NOT YET AUDITED.


RESULT-ID: PRIME-LIFT-AND-LOW-GAP-EXHAUSTIVE-AUDIT-20260907
STATUS: VERIFIED COMPUTATION.
DOMAIN / METHOD:
Exact smallest-prime-factor sieve; FIFO queue; ascending distinct factors;
duplicates suppressed on enqueue; no work or depth cutoff for full BFS.
For every even N from 4 through 1,000,000 (499,999 inputs), run BFS from 2.
For every prime divisor d|N with m=N/d-1>=2 and R=N-m prime, compare the
full queue, depths, verdict and number of composite rows in the two starts
d and R, including the separate direct-complement case. This is exhaustive
in that domain, not a random sample or a scan of selected paths.
COUNTS:
41,536 admissible h=1 inputs, of which 36,971 have composite P-2.
265,847 admissible prime-divisor lifts: 200,602 composite-complement lifts
and 65,245 direct-complement lifts. All claimed conjugacies and direct-case
corrections passed. Exactly one lift failed: (N,d,R)=(1862,7,1597), the
complete certificate recorded above. No h=1 input failed in this range.
The ONLY root-2 failures in this finite range are
 N=2^a+2, 2<=a<=19, each W=1,
and N=128 (W=9), N=1718 (W=29).
The power-of-two family is rigorously a self-loop: N-2=2^a. For a=1,
N=4 succeeds directly; thus that endpoint is deliberately excluded.
The statement that the list is exhaustive beyond the scanned range is NOT
claimed. In particular a finite list of known active cores was not assumed
complete by the scan. At N=6142, root 2 actually succeeds along
 2 ->307 ->389 ->11, with final complement 6131 prime:
 6142-2=2^2*5*307, 6142-307=3*5*389,
 6142-389=11*523. Inclusive BFS work W=12.
Thus the mere existence of a known closed BAD component at the SAME N does
not imply failure from 2 or from any other specified root.

SECOND DOMAIN / METRIC:
Every even 4<=N<=200000 and each admissible upper rank k=0,...,6;
retain exactly composite m=N-p_k with h=p_k-N/2<min PF(m).
If no GOOD vertex occurs at depths 0,1,2, expand ALL these rows and count
the distinct nonroot labels discovered at depths 1,2,3, without expanding
depth-3 rows for the count. Early exit when a GOOD vertex at <=2 is found
is permitted only for deciding that the input is outside the BAD2 class.
Counts (rank: eligible low-gap inputs, complete BAD through depth 2):
 0: 38715, 7135
 1: 22044, 4503
 2: 14356, 3136
 3: 11229, 2495
 4: 9285, 2096
 5: 7915, 1808
 6: 6790, 1516
Every input passed the proved lower bound of three distinct nonroot labels.
The observed minimum was FIVE labels, first at (N,R,k)=(4422,2213,0).
This stronger minimum is EMPIRICAL ONLY. Exact complete first rows:
```

```
 4422-2213=47^2;
 4422-47=5^4*7;
 4422-5=7*631;
 4422-7=5*883.
Hence depths are {47}:1, {5,7}:2, {631,883}:3. The last vertex 883 is
GOOD (4422-883=3539 prime); 631 is BAD (3791=17*223).
Thus this is complete BADness THROUGH depth 2, not through depth 3.
Full BFS succeeds at depth 3 after W=6 tests.

INDEPENDENT HOSTILE CHECK:
Trial division independently verified the N=4422 complete rows and ranks,
the two nontrivial root-2 failure reachable sets, the N=6142 successful
path, and the N=1862 lift closure certificate. This is an independent
certificate audit, not a second exhaustive million-input scan.
The three-label theorem is not made stronger by the observed five-label
minimum. The existing master already treats three-vertex closed cores as
an open coupled prime-power/support system; empirical absence is not a
classification theorem. No Scott--Bozek uniqueness claim is used.
REPRODUCTION:
The retained C++ source audit_prime_lifts_20260907.cpp uses an SPF array
through 1,001,000, enough for all roots and rows in the stated ranges.
Its raw result is audit_prime_lifts_20260907.out. All counts, domains,
exceptions and the direction-changing witnesses are preserved here so
this checkpoint remains usable without those scratch diagnostics.
RELATION TO p0 / GOLDBACH:
Confirms exact finite behaviour and the theorem's breadth across nearby
roots; proves neither universal canonical success nor binary Goldbach.
NOVELTY: NOT YET AUDITED.


RESULT-ID: CANONICAL-FIRST-SUPPORT-CYCLE-REALIZATION-20260907
STATUS: PROVED; an explicit consumer of the surviving partial-CRT method.
EXACT STATEMENT:
For every ordered list q_1,...,q_l of distinct odd primes, l>=2, there
exists a FIXED positive integer h and infinitely many even integers N
such that, with P=p0(N)=N/2+h and m=N-P:
 (a) each q_i divides m, so every listed label lies at depth 1;
 (b) every q_i is active and BAD;
 (c) q_1 -> q_2 -> ... -> q_l -> q_1 is a directed cycle of the actual
     factorization graph induced on the complete first support PF(m).
There may be additional edges, factors and GOOD children. No assertion
is made that PF(m) equals the prescribed list, that the induced cycle
is a terminal component, or that the COMPLETE BFS fails.
Consequently canonical roots have first-factor induced subgraphs with
arbitrarily long directed BAD cycles and arbitrarily large strongly
connected subgraphs. The strong component may have additional vertices
and edges, and need not be closed in the full factorization graph.

HYPOTHESES / QUANTIFIERS:
The list is fixed before constructing h or the arithmetic progression.
The infinitude is in N for each such fixed list and fixed h. It is not
uniform in growing l, not a density assertion, and not a statement about
the h=1 subfamily. The construction does NOT prescribe all cofactors.

PROOF:
Set M=product q_i and write indices cyclically, q_0=q_l. Choose h by
 2h = q_{i-1} (mod q_i), for i=1,...,l.
The Chinese remainder theorem applies because the odd primes q_i are
distinct and 2 is invertible modulo each q_i. It determines one residue
class h modulo M. None of its residues is zero: q_{i-1} is a distinct
prime, hence is not divisible by q_i. In particular gcd(2h,M)=1.
Choose any fixed positive representative h of that class.

We will choose a large odd prime P with P=2h (mod M), and set
 N=2(P-h), m=P-2h. To enforce canonicity, for EVERY even integer j
with 2<=j<=h choose a new prime ell_j>h, outside {q_1,...,q_l}, with
all these new primes distinct, and impose P=j (mod ell_j). Add P=1
(mod 2). These congruences have pairwise coprime moduli. Their common
class A modulo L=2M*product ell_j is reduced: its residues are nonzero
modulo every q_i, every ell_j (because 0<j<ell_j), and 2.
Dirichlet's theorem therefore supplies infinitely many primes P=A
(mod L). Discard finitely many small P so that P>2h+M, P-h>2,
P-j>ell_j for every chosen j, and all inequalities below are strict.
For odd 1<=j<=h, P-j is even and >2. For even j, ell_j is a proper
factor of P-j. Thus EVERY integer in [P-h,P) is composite. It follows
that P is exactly the least prime at or above N/2=P-h: P=p0(N).

By P=2h (mod M), all q_i divide m=P-2h. The root complement is positive
and divisible by M with l>=2, hence composite; each q_i is reached in
the complete first factor set. Modulo q_i, N=2P-2h=2h=q_{i-1}, which
is nonzero; therefore all q_i are active. For each i,
 N-q_i = 2h-q_i =0 (mod q_{i+1}).
For sufficiently large P, N-q_i>q_{i+1}; so this row is composite and
its COMPLETE factorization includes q_{i+1}. This gives every claimed
BAD cycle edge, including the closing edge. Increasing P changes N,
so infinitely many distinct N are obtained. Every label of the cycle
is in PF(m); therefore the cycle is genuinely in that induced subgraph,
not merely somewhere deeper in the reachable graph. This proves the
statement without a prime-tuple conjecture or a bounded-BFS assertion.

EXTERNAL INPUT AUDIT:
Only Dirichlet's theorem for one FIXED reduced arithmetic progression
is used: if gcd(A,L)=1, there are infinitely many primes P=A (mod L).
Verified in Andrew V. Sutherland, MIT 18.785, Fall 2021, Lecture 18,
Theorem 18.1, page 1:
https://math.mit.edu/classes/18.785/2021fa/LectureNotes18.pdf
No simultaneous primality of different linear forms is requested.

HOSTILE CHECK / NECESSARY SMALL-GAP BARRIER:
For ANY upper root R=N/2+h with m=N-R and S=PF(m), an internal edge
u->v with u,v in S obeys v|(2h-u), the old shadow identity. Since R
is prime and m<R, no v in S divides 2h; in particular internal self-
loops are impossible. If a is the least label on a directed cycle and
b is its next label, then b>a. A nonzero multiple of b equal to 2h-a
cannot be negative: if 2h<a its absolute value is <a<b. Consequently
2h-a>=b and h> a. Thus a first-support cycle necessarily has a label
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below h. In particular the existing h=1 DAG theorem is untouched.
The CRT construction is compatible with this necessary bound: its h
cannot lie below every prescribed cycle label, because its congruences
would then give a strictly decreasing edge at every step of the cycle.
The fragile step in canonical CRT programming is primitivity of the
P progression; it was checked for EACH old and new prime modulus above.
Using the q_i themselves as arbitrary prime-gap blockers without that
check would be unjustified. Proper-factor thresholds were also imposed,
so a small shifted prime equal to its assigned divisor is excluded.

TWO-CYCLE SPECIALIZATION:
For distinct odd primes a,b in S, the two edges a->b and b->a occur
if and only if 2h=a+b (mod ab), equivalently
 h=(a+b)/2 (mod ab).
Indeed each edge supplies one of the two congruences, and their CRT
combination is 2h=a+b (mod ab); division by 2 is legitimate modulo odd
ab. The smallest positive representative is (a+b)/2<ab. All positive
solutions are h=(a+b)/2+t*ab with integers t>=0. For a=3,b=11 one can
therefore choose the small fixed value h=7. The general construction
then gives infinitely many canonical first-support 3<->11 cycles.

HISTORICAL RELATION:
This does not revive the killed reading of FBTU as complete-BFS
universality. Cite the existing FAIL-P0-FINITE-BAD-TRAVERSAL together
with its binding correction AUDIT-FBTU-PARTIAL-VERSUS-COMPLETE-20260905
and the Biblia iteration-108 cofactor-leakage warning. The construction
is a concrete, fully audited partial-factor consumer of what survived.
It adds an explicit FIRST-SUPPORT cyclic obstruction to a proposed
acyclicity explanation; it is not a new general finite-local no-go.
RELATION TO GOLDBACH:
No Goldbach failure follows. Extra branches can and do reach GOOD.
RELATION TO p0:
An all-p0 acyclicity or uniform cycle-size bound is false. The h=1
acyclicity property and its full-BFS conjugacy with root 2 remain valid,
but neither property extends automatically to the whole canonical family.
WHAT REMAINS OPEN:
Any obstruction to complete BAD closure, with all moving factors retained.
NOVELTY: NOT YET AUDITED; no novelty claim for CRT programming itself.


RESULT-ID: SAME-N-FIRST-SUPPORT-CYCLE-NON-DOMINANCE-20260907
STATUS: VERIFIED COMPUTATION with exact hand-checkable certificates.
EXACT STATEMENT:
Whether the first-support induced factorization subgraph is acyclic can
favor p0 or p1 at the SAME N. It is not a universal separator of the roots.
EXAMPLE A (p0 acyclic, p1 cyclic):
 N=344, p0=173 (h=1), p1=179 (h=7).
 S0=PF(171)={3,19}; its induced graph has no edges:
 344-3=11*31, 344-19=5^2*13.
 S1=PF(165)={3,5,11}; its induced edges are
 3->11, 5->3, 11->3, because
 344-3=11*31, 344-5=3*113, 344-11=3^2*37.
 Thus 3<->11 is a BAD cycle inside S1. As already recorded above,
 BOTH full searches succeed at depth 2 with W=5. The cycle does not
 force even a larger work count here.
EXAMPLE B (p0 cyclic, p1 acyclic):
 N=1268, p0=641 (h=7), p1=643 (h=9).
 The integers 634,...,640 are composite, and 641 and 643 are primes;
 hence the ranks are exact. The first complements are
 1268-641=627=3*11*19; 1268-643=625=5^4.
 In S0={3,11,19}, the BAD vertices 3 and 11 form a two-cycle:
 1268-3=1265=5*11*23; 1268-11=1257=3*419.
 The remaining child 19 is GOOD, since 1268-19=1249 prime. Therefore
 the COMPLETE stopped induced subgraph on S0 has exactly edges
 3->11 and 11->3. The p0 search succeeds at depth 1, W=4, via 19.
 In S1={5}, there is no internal edge: 1268-5=1263=3*421.
 Thus p1's first-support induced graph is acyclic while p0's is cyclic.

HOSTILE CHECK:
All roots, prime factors, prime endpoints and root ranks were verified
by trial division. The induced graph uses all internal edges; GOOD
vertices contribute no outgoing edges in the stopped graph. In B the
cycle labels are BAD even though another first child is GOOD. This is
precisely why a prescribed BAD cycle must not be confused with complete
first-front BADness or full BFS failure. These are exact examples,
not an exhaustive claim about first occurrences.
DEPENDENCIES: Elementary arithmetic and the fixed graph definition.
RELATION TO p0 / GOLDBACH:
Same-N non-dominance of this structural diagnostic. Neither example is
a failed canonical search or a Goldbach counterexample. The earlier
performance non-dominance examples N=2024 and N=2180 remain separate.
NOVELTY: NOT YET AUDITED.


RESULT-ID: FIXED-GAP-CANONICAL-VALUATION-EMBEDDING-20260907
STATUS: PROVED; precise extension/consumer of CRT-TRANSCRIPT-EXACT-SCOPE.
EXACT STATEMENT:
Let N0>=4 be even and let R0>=N0/2 be an admissible prime upper root
with COMPOSITE m0=N0-R0. Set h0=R0-N0/2 and S=PF(m0).
There exist infinitely many even N with canonical prime root
 P=p0(N)=N/2+h0, m=N-P composite, such that for every q in S,
  v_q(m)=v_q(m0),
and for every p,q in S,
  v_q(N-p)=v_q(N0-p).
All primes of S occur at depth 1 in the COMPLETE search from P.
Every such N sufficiently large has additional first factors outside S.

GRAPH FORM OF THE CONCLUSION:
Construct the first-support graph on PF(N-R), labelling each vertex q
by its prime value and by v_q(N-R), and labelling every internal edge
p->q by v_q(N-p)>0. This agrees with the internal edges of the stopped
factorization graph: an internal first-support edge cannot come from
a prime complement. On the original subset S, the new canonical graph
has EXACTLY the original labelled vertices, edge valuations and NONEDGES.
The numerical gap h0 is also preserved. Additional vertices outside S
and edges incident to them remain unrestricted. GOOD/BAD flags at a
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vertex with no internal edges are not asserted to be preserved.

PROOF:
Because m0 is composite, R0>N0/2, h0 is a positive integer, R0 is odd, and m0 is odd. Moreover R0>m0, so R0 is distinct from every prime in S. For each q in S set
 t_q = 1 + max(v_q(m0), max_{p in S} v_q(N0-p)),
and put L=product_{q in S} q^(t_q). These numbers are finite, L is odd, and gcd(R0,L)=1. Require a new prime P to satisfy P=R0 (mod L). Keep h0 FIXED and define N=2(P-h0), m=P-2h0. Then
 m=m0 (mod L),   N=N0 (mod L),
since m0=R0-2h0 and N0=2(R0-h0). If a positive integer n is congruent to a modulo q^(v_q(a)+1), then v_q(n)=v_q(a): division by q^v leaves the same nonzero residue modulo q. By the definition of t_q this gives all the asserted root and row valuations, including their zero values.

To make P canonical with the very SAME h0, add P=1 (mod 2). For every even j in [2,h0], choose a distinct new prime ell_j>h0 not dividing L, and add P=j (mod ell_j). The moduli are pairwise coprime. The common class is reduced: R0 is coprime to L, P is odd, and 0<j<ell_j at every new modulus. By Dirichlet's theorem there are infinitely many primes P in this class. Take P>R0 and large enough that P-h0>2 and P-j>ell_j for every even j used. Each odd j in [1,h0] makes P-j even and >2; each even j gives a proper factor ell_j. The entire interval [P-h0,P) is therefore composite. Hence P=p0(2(P-h0)), with gap h0. This verifies canonicity by actual compositeness certificates rather than silently treating an arbitrary upper root as canonical.

The exact positive valuations of m at every q in S place all S in PF(m). Since m>m0>=4 and m is divisible by the original m0, the new root is BAD and every label in S is indeed reached at depth 1 in the complete search. For any original p,q in S, p<=m0<R0 and q<=m0, so
 N0-p>=N0-m0=R0>q.
Thus if q divides N0-p it is a proper divisor, and the corresponding row is BAD. For the larger N the same properness holds. Positive row valuations therefore give actual stopped-graph edges on S in both worlds, with identical weights; zero valuations give identical nonedges. This proves the asserted induced-graph equality, including the case of an originally GOOD vertex whose complement prime is outside S. Finally, all exponents at primes of S in m equal those in m0. If there were no extra factor outside S, unique factorization would give m=m0, contrary to P>R0. So additional support is NECESSARY in every retained new world. Distinct primes P give distinct even N, proving infinitude.

HOSTILE CHECK:
Primitivity uses R0>m0. The diagonal/direct case m0=R0 is NOT allowed: then the needed congruence P=R0 (mod R0) would not be reduced. The hypothesis m0 composite removes this problem and fixes the relevant BAD-root domain. A single-pair Goldbach success among first children is not required to survive: preserving finite valuations is not the same as preserving complement primality. No factor is declared absent outside S, and no complete BFS transcript is copied. The proof itself shows that new first factors MUST appear. Consequently it cannot conflict with complete fixed-alphabet finiteness or supply an iteration with an unchanged alphabet.

DEPENDENCIES / HISTORICAL AUDIT:
CRT, elementary exact valuations, and Dirichlet's fixed reduced-progression theorem verified above in MIT Lecture 18, Theorem 18.1. This extends the checkpoint's CRT-TRANSCRIPT-EXACT-SCOPE-20260905 from already compatible h=1 partial constraints to a concrete compatible table obtained from ANY actual upper BAD root, with its own h0 held fixed. The additional step enforces the prime-free canonical interval. It is a precise consumer of what survived FAIL-P0-FINITE-VALUATION-SIGNATURE and FAIL-P0-FINITE-BAD-TRAVERSAL, subject to the later AUDIT-FBTU-PARTIAL-VERSUS-COMPLETE-20260905 correction. No duplicate claim of complete-BFS universality is introduced.
RELATION TO GOLDBACH:
No root failure is copied, and no prime-pair existence theorem follows.
RELATION TO p0:
A rigorous limit on what the root's finite internal factor arithmetic can distinguish. The useful consumer is the hereditary-property no-go below; the statement is not itself an eventual-success mechanism.
WHAT REMAINS OPEN:
Properties using the COMPLETE alphabet, external row factors, GOOD status, absolute N or root location, or later reachability can escape this embedding theorem. No conclusion about them is inferred here.
NOVELTY: NOT YET AUDITED; no novelty claim for the underlying CRT method.

RESULT-ID: HEREDITARY-FIRST-SUPPORT-P0-NO-GO-20260907
STATUS: PROVED, as an exact consumer of the preceding embedding.
EXACT STATEMENT:
Consider a property Q(h,G) of finite directed graphs G whose vertices carry prime labels q and root exponents, and whose edges carry row valuations, as in FIXED-GAP-CANONICAL-VALUATION-EMBEDDING. Suppose Q is hereditary under induced subgraphs at fixed h: whenever Q(h,G) holds, it holds on every induced subgraph, retaining vertex/edge labels and exponents. Q may use the numerical labels and h, but has no extra unrecorded dependence on the absolute N, root R, or outside factors. Then the following two statements are equivalent:
 (A) Q holds for the first-support graph of EVERY canonical BAD root p0.
 (B) Q holds for the first-support graph of EVERY admissible upper BAD prime root R>=N/2, including all nearby roots that are BAD.
Moreover, if (B) fails, (A) fails for infinitely many N with one fixed h.

PROOF:
(B) implies (A) because canonical roots are upper roots. Conversely, assume (A) and take an arbitrary upper BAD root (N0,R0). The preceding embedding supplies a canonical BAD-root graph G' at the same h0 that contains its full original graph G0 as an induced subgraph with all stated labels and weights unchanged. By (A), Q(h0,G') holds. Heredity then implies Q(h0,G0). The original upper BAD root was arbitrary, so (B) follows. If Q(h0,G0) fails for one such root, every canonical embedding G' fails Q by the contrapositive of heredity. There are infinitely many distinct N carrying those embeddings. This proves

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the strengthened failure conclusion as well.

HOSTILE CHECK / EXACT CLASS OF THE NO-GO:
Heredity is essential and is not silently assumed for an arbitrary
invariant. Examples inside the class include acyclicity, forbidding a
fixed labelled finite cycle/pattern, a uniform upper bound on directed
cycle length or strong-component size, and hereditary restrictions on
internal row valuations. These properties cannot universally hold for
p0 while failing for a nearby BAD root. The theorem does not say their
frequencies are equal, or compare p0 and p1 at every same N.
OUTSIDE its scope are complete BADness (GOOD flags can change), equality
of the WHOLE prime alphabet with a fixed set, completeness of all row
factorizations inside that alphabet, a lower support bound, full BFS
closure, or full multigeneration success. Such properties need not be
hereditary under the stated restriction, and external factors are the
precise information deliberately not preserved by the construction.
This is therefore a mathematically specified subclass of a finite-local
no-go, NOT the old overbroad claim that any finite BFS invariant fails.

RELATION TO p0 / GOLDBACH:
A broad, rigorous negative answer for these graph-and-valuation
properties; it does not answer whether the complete canonical traversal
always reaches GOOD. Same-N witnesses above illustrate both directions
for acyclicity; this theorem supplies the universal logical limitation.
WHAT REMAINS OPEN:
A non-hereditary complete-support or multigeneration invariant that
actually forbids a closed BAD canonical reachable world.
NOVELTY: NOT YET AUDITED; theorem consumer and scope are explicitly new
entries here, not a claim of priority over the literature.


RESULT-ID: CANONICAL-VALUATION-EMBEDDING-EXACT-CERTIFICATE-20260907
STATUS: VERIFIED COMPUTATION.
DOMAIN:
One constructed example, using the noncanonical source (N0,R0)=(344,179)
with h0=7 and S={3,5,11}. No exhaustive-range claim is made.
CONSTRUCTION / VALUES:
The proof gives t_3=3,t_5=2,t_11=2, hence L=81675.
Use the progression P=179 (mod L), P=1 (mod 2),
 P=2 (mod 13), P=4 (mod 17), P=6 (mod 19).
Its common class is P=499197779 (mod 685906650).
A prime member is P=9415984229, yielding N=18831968444 and h=7.
The interval [P-7,P) is composite: odd shifts are even >2; shifts
2,4,6 have the proper divisors 13,17,19 respectively. Trial division
also checked primality of P. Thus P=p0(N) exactly, with no reliance on
a probable-prime certificate or an assumed preceding prime gap.
COMPLETE ROWS NEEDED FOR THE VALUATION CERTIFICATE:
 m=P-14=3*5*11*229*249199;
 N-3 = 11*13*131692087;
 N-5 = 3*89*6709*10513;
 N-11 = 3^2*17*101*239*5099.
All displayed factors are prime. On S these reproduce the original
root exponents (1,1,1), and exactly the internal weighted edges
 3->11 (weight 1), 5->3 (weight 1), 11->3 (weight 2).
The extra root multiplier is 57066571=229*249199, coprime to 3*5*11.
The full first support is therefore strictly larger than S, as the
proof requires. Both extra first children are BAD:
 N-229=5*53*653*108827;
 N-249199=5*37*199*511523.
The COMPLETE first frontier is BAD in this particular example, but
that extra fact was verified here and does not follow from the embedding.
The full FIFO search nevertheless succeeds at depth 2 after W=11 tests:
 P ->5 ->10513, and N-10513=18831957931 is prime.
Primality, all displayed factors and this BFS were checked using exact
trial division. The finite example illustrates preservation of the
internal cycle and valuations while an unprescribed factor supplies a
GOOD escape. It is not a failed canonical world.
REPRODUCTION:
CRT was solved with integer modular inverses, starting with class 179
modulo 81675 and then combining moduli 2,13,17,19. The first retained
prime in the resulting nonnegative progression is the displayed P;
that order is diagnostic only, not a minimality theorem about all N.
The complete certificate is retained in this TXT; the auxiliary scratch
record is canonical_valuation_embedding_20260907.json.
NOVELTY: NOT YET AUDITED.


AUDIT ADDENDUM: NECESSARY BOUNDARIES OF THE EMBEDDING NO-GO
STATUS: VERIFIED COMPUTATION / SCOPE-CORRECTED.
1. GOOD flags really are not preserved. A small exact example is
 (N0,R0)=(20,11), h0=1, m0=9=3^2, S={3}.
The original child 3 is GOOD because 20-3=17 prime. Take P=173,
N=344 and h=1. Then P=11 (mod 27), m=171=3^2*19 has the same
3-adic root exponent, and 344-3=341=11*31 has the same 3-adic row
valuation zero as 17. The entire induced graph on the labelled subset
{3} (one vertex of root weight 2, no internal edge) is unchanged,
but that vertex is now BAD. Thus adding GOOD/BAD flags to the theorem
without new hypotheses is false. This is a concrete boundary of the
retained theorem, not a counterexample to it.
2. Arbitrary incompatible edge data are still impossible. On labels
{3,5,7}, simultaneous edges 5->3 and 7->3 would require N=2 and N=1
modulo 3. No N realizes them. The fixed-gap theorem starts from an
actual upper-root graph, so its compatibility is guaranteed and its
primitivity is then proved. The cycle-realization theorem prescribes
one incoming predecessor per target and permits extra edges; it does
not assert universality of every directed graph on arbitrary labels.
3. The low-gap support LOWER bound proved earlier is not hereditary
under deleting vertices. It therefore escapes the hereditary no-go,
as do the complete-support height restrictions. This logical escape
alone does not turn them into a contradiction for large moving support.
4. None of these arguments assumes global Goldbach failure. The
failure counterexample (1862,1597) is a rooted failure in an ordinary
Goldbach-successful N. Its use kills only the unconditional prime-lift
success claim, not a theorem restricted to a global failed world.
5. No recent asymptotic estimate was needed in this continuation.
The checkpoint's earlier growing-root/growing-depth sieve claims have
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not been strengthened or reclassified by the elementary results here.

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STRONGST NEW RESULTS
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1. FIXED-GAP-CANONICAL-VALUATION-EMBEDDING and its HEREDITARY-FIRST-
   SUPPORT-P0-NO-GO: exact preservation of the original induced first-
   support graph, root exponents and internal row valuations at the
   SAME numerical h, inside infinitely many canonical worlds. This
   rules out a precisely specified class of universal p0 separators.
   Complete support and eventual reachability are explicitly outside it.
2. PRIME-LIFT-FULL-BFS-CONJUGACY: for N=d*x and prime lift
   R=(d-1)*x+1 with composite x-1, the full BFS from inactive d and
   from R are identical after renaming their initial root. At d=2
   this is an exact all-generation equivalence for canonical h=1.
   The tempting general success consequence is FALSE at
   (N,d,R)=(1862,7,1597), with a complete 22-vertex closure certificate.
3. LOW-GAP-COMPLETE-THREE-LABEL-ALTERNATIVE: if h<min PF(N-R), either
   GOOD occurs by depth 2 or at least three distinct nonroot labels
   occur by depth 3. A whole closed BAD nonroot alphabet of size <=2
   is excluded in that domain. This also applies to nearby roots.
4. Canonical first-support BAD cycles can have arbitrary prescribed
   length. Same-N acyclicity reverses between p0 and p1 at N=344 and
   N=1268. Exact root-rank corrections, million-input lift checks and
   the explicit canonical valuation-embedding example are recorded above.

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CURRENT FRONTIER
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Binary Goldbach is not proved, and no universal p0 success theorem was
obtained. The new embedding result explains why inherited finite graph
patterns and their internal valuations cannot provide certain universal
p0 distinctions; it does NOT control the extra factors whose presence
prevents the transplant from remaining closed. The exact conjugacy makes
root 2 an equivalent starting point in the h=1 subfamily, but gives no
mechanism forcing that root to succeed when P=N/2+1 is prime.
A possible large, moving, completely closed active BAD reachable set is
still unexcluded. Earlier support-height bounds remain lower constraints,
not contradictory upper bounds. The earlier asymptotic frontier is also
unchanged: the large-label-product AND large-cofactor-product region at
radius three is not controlled here. Shallow BADness is compatible with
later success, and none of it is a global Goldbach counterexample.

========================
p0 STATUS
========================
Found a nontrivial all-generation equivalence for canonical h=1, but its
partner is root 2, not a nearby upper root, and it does not cover all p0.
Found a precise hereditary graph/valuation no-go and additional same-N
structural non-dominance witnesses. The small-gap support restriction is
shared by nearby roots satisfying the same premise. Canonical first-
support acyclicity holds at h=1 but fails for other h, even infinitely
often with arbitrarily long cycles. No convincing explanation of the
empirical eventual reliability of p0 has been established.

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EXACT NEXT QUESTION
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Can complete factor search from 2 fail at N=2(P-1) with P>=5 prime
and P-2 composite? Equivalently, can there exist such a P and a finite
set C of odd primes below 2(P-1)/3 such that
 PF(P-2) is contained in C,
 and for EVERY q in C the integer 2(P-1)-q is composite with its ENTIRE
 prime support contained in C?
This is the exact d=2 obstruction left by the proved conjugacy. A
solution must use complete support or another feature not preserved
by the canonical valuation embedding. Showing selected BAD cycles,
reproducing finite internal valuations, or merely increasing a support-
height lower bound does not answer it. A counterexample would refute
canonical h=1 traversal success, not by itself binary Goldbach.


====================================================================
CONTINUATION IN LOCAL WORKSPACE -- 2026-09-07
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SOURCE AND SCOPE:
The preceding checkpoint was copied intact from the supplied Downloads
file goldbach_research_checkpoint_2026-09-05 (3).txt. Its latest entries
already extend through 2026-09-07 and supersede the earlier master where
they explicitly correct it. This continuation uses the documents as
research data, not as instructions. All new research files are in nowe.
Only one independent subagent was used. No PDF or TeX output is produced.
The immediate focus is complete-support restrictions, rather than another
transplant of selected edges or an unsupported all-N conclusion.

FAIL-ID: LOW-GAP-FIVE-LABEL-OVERGENERALIZATION-20260907C
STATUS: REFUTED by an exact counterexample; the earlier three-label
theorem and the stated rank-0-through-6 finite experiment are unaffected.
CLAIM / ROUTE:
Strengthen LOW-GAP-COMPLETE-THREE-LABEL-ALTERNATIVE by replacing THREE
with FIVE for every upper prime root with h<min PF(N-r).
WHY IT LOOKED PROMISING:
The old experiment found no fewer than five nonroot labels by depth 3
under complete BADness through depth 2, at ranks 0,...,6 and N<=200000.
EXACT COUNTEREXAMPLE:
 N=44670, r=22469, m=N-r=22201=149^2, h=134<149.
 N-149=44521=211^2.
 N-211=44459=23*1933.
 N-23=44647 prime; N-1933=42737 prime.
All displayed prime labels, the root and the two prime complements are
prime by independent trial division through the integer square root.
Thus complete BADness holds at depths 0,1,2. The distinct nonroot
labels discovered through depth 3 are exactly {149,211,23,1933}:
depth 1 is {149}, depth 2 is {211}, depth 3 is {23,1933}.
The complete FIFO search, ascending factors and testing on dequeue,
succeeds at depth 3 after four tests r,149,211,23. No branch is omitted.
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The exact upper rank is 12, since the primes at or above N/2=22335
through r are
 22343,22349,22367,22369,22381,22391,22397,
 22409,22433,22441,22447,22453,22469.
This explains why the earlier rank-restricted experiment missed it.
WHAT SURVIVES:
The proved bound THREE. The possibility of a universal FOUR under the
same low-gap hypotheses remains under investigation. FIVE for p0 itself,
or for ranks at most six outside the tested range, is not decided by
this noncanonical example. No five-label theorem for those roots is
claimed. The observed five-label minimum was already explicitly empirical.
RELATION TO p0 / GOLDBACH:
An exact boundary of a candidate strengthening, not a failed canonical
traversal. This N has many Goldbach representations, including those above.
REPRODUCTION:
Discovered by the single independent agent in a targeted singleton-
first-support scan, with scratch source agent_scan_singles.cjs. The
displayed arithmetic is a self-contained certificate; the domain and
results of any larger scan are recorded separately after verification.
NOVELTY: no literature-priority claim.


RESULT-ID: LOW-GAP-SHARP-TWO-STEP-RETURN-BARRIER-20260907C
STATUS: PROVED, elementary pointwise statement.
EXACT STATEMENT:
Let N be even and r>=N/2 a prime upper root with composite complement
m=N-r. Put h=r-N/2 and q=min PF(m), and assume 0<h<q.
Suppose q and an odd prime b are BAD and form an actual two-cycle:
 b divides N-q, and q divides N-b.
Then
 b=2h+(2j+1)q for an integer j>=1;
 in particular b>=3q+2h.
The numerical lower bound is sharp, both for canonical roots and for
noncanonical upper roots. It bounds a two-cycle through the minimum
first child; it is not a no-cycle theorem for the whole reached graph.

PROOF:
The BAD root is odd, m is odd, r>N/2, and q>=3. Every child of an
active prime is active: if t divides both N-p and N then t divides p,
forcing t=p and contradicting activity of p. Since r is an upper
nondiagonal prime, it is active; so q,b are active and b!=q.
Because q divides m, N=2m+2h is congruent to 2h modulo q.
The return edge b->q gives b=2h+kq for an integer k. The primes b,q
are odd, so k is odd. Since 0<2h<2q and b>0, k<=-3 is impossible.
If k=-1, then b=2h-q<q and the other edge q->b gives
 b divides N-q=2m+b.
Thus b divides m, contradicting the minimality of q.
If k=1, then b=q+2h, and
 N-q=2m+2h-q=2(m+2h)-b=2r-b.
Hence b divides 2r, so b=r because b is odd and r is prime.
This is impossible: a prime factor b of the odd composite row N-q
satisfies b< N/3<r. Thus k>=3, proving the assertion.

SHARPNESS / EXACT CERTIFICATES:
Canonical case: N=80, r=p0=41, h=1, m=39=3*13, q=3.
 N-3=77=7*11; N-11=69=3*23.
Thus b=11=3q+2h is an actual BAD two-cycle partner.
The child 13 is GOOD (80-13=67 prime), so this cycle does not cause
full BFS failure or even complete first-front BADness.
Noncanonical case: N=1638, r=p1=823, p0=821, h=4,
 m=815=5*163, q=5.
 N-5=1633=23*71; N-23=1615=5*17*19.
Here b=23=3q+2h again. The root ranks are exact: 819 is composite,
820 and 822 are even, and 821 and 823 are prime.
All displayed factors, roots and prime complements were checked by
integer trial division / exact sieve lookup.

HISTORICAL AUDIT:
The underlying odd parent lattice p=2h+ell*q is already present in
the Biblia, proposition prop:v73-parent-lattice (near line 45279 in
the supplied file). It is not claimed as new. This consumer removes
the two closest positive candidates in the small-gap setting:
ell=-1 by complete first-support minimality, ell=1 by primality of
the upper root. An exact-string search for 3q+2h and its reordered
forms in the master, Biblia, core, failure log and input checkpoint
found no earlier version. This is a selective corpus check, not a
literature priority audit.
RELATION TO p0:
The bound applies automatically at h=1 but is shared by every upper
prime root satisfying h<q. Its sharp p1 example prevents interpreting
its sharpness as a p0-only signature. It is an actual two-step return
restriction, but it does not force a GOOD child or prevent longer cycles.
RELATION TO GOLDBACH: no prime-pair existence conclusion.
NOVELTY: not externally audited; no priority claim.


RESULT-ID: LOW-GAP-SQUARE-ROW-NO-RETURN-20260907C
STATUS: PROVED; independent elementary lemma supplied by the one agent
and checked by the root agent.
EXACT STATEMENT:
Under the same root, gap and minimum-factor hypotheses, suppose
 N-q=b^2 for a prime b.
Then q does not divide N-b, and
 (b-1)^2 < N-b < b^2.
PROOF:
Since m is composite and q is its least prime factor, m>=q^2.
Consequently b^2=2m+2h-q>q^2, hence b>q.
If q divided N-b, then b=N=2h (mod q). The equation b^2=N-q
also gives b^2=N=b (mod q). Since b!=q is prime, this implies
b=1 (mod q) and 2h=1 (mod q). The only even integer strictly
between 0 and 2q in this residue class is q+1. Thus 2h=q+1.
But then b^2=2m+1=3 (mod 4), impossible because m is odd.
Finally N-b=b^2+q-b<b^2, and subtraction of (b-1)^2 leaves
q+b-1>0. This proves the strict interval and the no-return assertion.

COMPLETE-SUPPORT CONSEQUENCE:
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If PF(m)={q}, N-q=b^2, and no GOOD vertex occurs through depth 2,
then either at least FOUR distinct nonroot primes are discovered by
depth 3, or the next COMPLETE row is
 N-b=c^w with a new prime c!=q,b and an ODD exponent w>=3.
Indeed q is excluded by the lemma, b by activity. Two distinct factors
of N-b introduce two new labels. Otherwise it is a prime power; its
exponent is at least two by BADness and cannot be even because N-b
lies strictly between consecutive integer squares.
Thus a three-label square bottleneck would satisfy the exact system
 2q^u+2h=q+b^2=b+c^w,
 u>=2, w>=3 odd, 0<h<q, q^u+2h prime.
This system has NOT been excluded. It is an obstruction to completing
a universal FOUR-label proof, not a claimed counterexample to FOUR.
The child c need not be BAD because its row is at depth 3.
RELATION TO p0: shared under the stated gap condition.
NOVELTY: no literature-priority claim.


RESULT-ID: LOW-GAP-MONOMIAL-RETURN-EXPONENT-DROP-20260907C
STATUS: PROVED, as a consumer of the preceding two elementary lemmas.
EXACT STATEMENT:
Assume m=q^u, u>=2, 0<h<q, r=q^u+2h prime, N=2q^u+2h,
and N-q=b^v for a prime b and integer v>=2.
If q divides N-b (a BAD return q->b->q), necessarily
 3<=v<=u-1, hence u>=4.
More precisely (3q+2h)^v <= 2q^u+2h-q.
In particular neither m=q^2 nor m=q^3 can have such a return when
the complete row N-q has only one distinct prime factor.
PROOF:
q is the minimum first factor automatically. Since N-q is composite,
q is BAD. The condition q|N-b also makes b BAD: b<N/3 and N-b>q,
so q is a proper factor. Apply the sharp return bound b>=3q+2h.
If v>=u, then
 b^v >= (3q+2h)^u > (3q)^u >= 9q^u
 > 2q^u+2h-q=N-q,
where 2h-q<q<=q^u. This is impossible. Thus v<=u-1.
The square-row lemma excludes v=2. The remaining inequalities follow.
HOSTILE CHECK:
The conclusion concerns a direct return to q, not absence of all
later returns or full BAD closure. The assumption that N-q is a
COMPLETE prime power is essential; choosing one divisor of a
multifactor row does not satisfy it. At N=80 a squarefree multifactor
row has a return at the sharp size threshold, as recorded above.
No root minimality within the upper primes was used.
WHAT REMAINS OPEN:
Longer return paths, branching rows, and singleton-support roots with
u>=4 remain possible. Even an acyclic initial segment need not succeed.
NOVELTY: no literature-priority claim.

ADDITIONAL HOSTILE BOUNDARY OF THE RETURN LEMMA:
The sharpened lattice applies to RECIPROCAL edges. It is not a correction
asserting that all parents of q exceed 3q+2h. At N=344, r=173, h=1,
m=171=3^2*19, q=3, the label b=5=q+2h is a genuine parent of q,
since 344-5=3*113, but q->b is absent since 344-3=11*31.
The prime-root contradiction needs the second divisibility b|(N-q).


RESULT-ID: H1-SINGLETON-SQUARE-ROW-RESIDUE-FILTER-20260907C
STATUS: PROVED; necessary conditions only.
EXACT STATEMENT:
If r=q^u+2 is prime, u>=2, N=2q^u+2, and N-q=b^2 with b prime,
then u is odd and at least three, and q=23 (mod 24).
PROOF:
If q=3, then N-q=2*3^u-1=2 (mod 3), which cannot be a square.
For q!=3, even u gives r=q^u+2=0 (mod 3), with r>3, contradicting
primality. Thus u is odd and u>=3. The primality of r now excludes
q=1 (mod 3), so q=2 (mod 3). Since an odd integer to an odd power
is congruent to itself modulo 8, b^2=2q^u+2-q=q+2 (mod 8).
Odd squares are 1 modulo 8; hence q=7 (mod 8), giving q=23 (mod 24).
SCOPE:
This uses h=1 and complete pure-power rows. It is not asserted for all
p0 or all h, and supplies neither existence nor impossibility of this
sparse configuration. It merely reduces the unresolved square bottleneck.


EXPERIMENT-ID: LOW-GAP-RETURN-AND-COMPLETE-SUPPORT-AUDIT-20260907C
STATUS: VERIFIED COMPUTATION; exact finite exhaustive domain.
DOMAIN:
Every even 4<=N<=200000, each upper rank k=0,...,12, retaining exactly
composite m=N-p_k and 0<h=p_k-N/2<q=min PF(m).
METHOD:
An independently written exact linear smallest-prime-factor sieve through
201000. All rows have arguments within that range. Enumerate every factor
b of the COMPLETE BAD row N-q and test whether q divides N-b; check the
sharp return barrier. Check all prime-square rows and the consecutive-
square interval. For singleton monomial returns check exponent drop.
For support counts, expand complete depths 0,1,2 only when these depths
are all BAD; count all distinct nonroot labels discovered through depth 3.
Suppress repeated labels. Do not expand newly discovered depth-3 rows.
COUNTS:
134898 eligible (N,k) inputs; 28289 completely BAD through depth 2.
10249 reciprocal q->b->q incidences, all satisfying b>=3q+2h.
1360 incidences attain equality, so the sharp boundary is repeatedly
realized in the finite domain. There are 97 eligible prime-square rows;
every no-return and interval check passed.
There are ZERO singleton monomial returns in this finite domain; thus
the exponent-drop condition received no nonvacuous returned-path test
here. Its status rests on the proof, not that vacuous check.
Per rank: k, eligible, complete BAD through 2, least label count through 3:
 0   38715 7135 5
 1   22044 4503 5
 2   14356 3136 6
 3   11229 2495 6
 4    9285 2096 6
 5    7915 1808 6
 6    6790 1516 6
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 7    5824 1318 6
 8    4930 1144 6
 9    4191  965 6
10    3670  836 6
11    3186  698 6
12    2763  639 4
The unique four-label case in THIS DOMAIN is (N,k)=(44670,12), whose
complete certificate was recorded above. No three-label case occurred.
The rank-0-through-6 eligible/BAD2 counts independently reproduce all
seven counts from the supplied checkpoint. This is a check on definitions
and implementation, not a promotion of any finite minimum to a theorem.
REPRODUCTION: nowe/return_barrier_audit.cjs; no random or probable-prime tests.

SEPARATE AGENT DIAGNOSTIC, RETAINED HERE IN FULL NUMERICAL SCOPE:
All odd-prime powers m=q^u, u>=2, and ALL integers 1<=h<q such that
N=2m+2h<=10000000 and r=m+2h is prime, with no root-rank restriction.
The exact SPF scan in agent_scan_singles.cjs found 47490 eligible roots,
25413 completely BAD through depth 2, including 11 with a singleton
factor set for N-q. The minimum count through depth 3 is FOUR and
occurs on exactly three inputs:
 (N,r,q,u,h)=(44670,22469,149,2,134),
 (241428,121019,347,2,305),
 (2037338,1019257,1009,2,588).
For the latter two, the complete first rows are
 241428-121019=347^2;
 241428-347=491^2;
 241428-491=479*503;
 2037338-1019257=1009^2;
 2037338-1009=1427^2;
 2037338-1427=3*678637.
The root agent independently verified all displayed prime factors and
roots by trial division; it did not repeat the full ten-million-range
singleton scan. That scan covers ONLY singleton initial support and is
not an all-root or complete closed-core census.


RESULT-ID: CLOSED-SUPPORT-MULTIPLICATIVE-SUBGROUP-TEST-20260907C
STATUS: PROVED elementary necessary condition; investigated consumer
does not yield a general p0 distinction.
EXACT STATEMENT:
Let C be a nonempty set of active primes with every N-p composite and
PF(N-p) contained in C. In the unit group modulo N let H_C be the
subgroup generated by the residue classes of all primes in C.
Then -1 belongs to H_C. If an upper prime root r has PF(N-r) contained
in C and N-r>1, then r belongs to H_C as well.
PROOF:
Choose p in C. Complete factor closure makes N-p an element of H_C
modulo N. Since p is a unit in H_C, (N-p)*p^(-1)=-1 modulo N belongs
to H_C. If m=N-r has its full support in C, then m belongs to H_C,
so r=-m belongs to H_C. This proves both claims.
CAUTION:
Excluding r from H_C excludes failure contained in THIS C only. To
deduce actual success by this test one must use an envelope containing
every possible failed nonroot vertex, for example the maximal closed
low BAD set. Checking only one terminal component is insufficient.

FAIL-ID: SUBGROUP-SEPARATION-OF-P0-AS-GENERAL-EXPLANATION-20260907C
STATUS: NO GENERAL SEPARATION; necessary condition and occasional use survive.
ROUTE:
Test whether the group generated by all failed low primes is a proper
subgroup that excludes p0, thereby explaining the known exceptional hosts.
EXACT DIAGNOSTIC:
For EACH of the six historical hosts reconstruct C by starting with
all active BAD primes p<N/3 and repeatedly deleting rows having a prime
factor outside the current set, until stabilization. This is complete
factor pruning and supplies the maximal closed low BAD set at that N.
Generate H_C by finite multiplication modulo N. Counts are:
 N       |C|   |H_C|   phi(N)    index
 128      10      64       64       1
 1718     55     858      858       1
 1862     55     756      756       1
 1928     41     960      960       1
 2200      2      40      800      20
 6142     66    2952     2952       1
At five of the six hosts the group is the ENTIRE unit group; consequently
no multiplicative character modulo N that is trivial on C can separate
p0, or any other active root, there.
At N=2200, C={3,13}, the group has order 40 and does exclude p0=1103.
All upper prime members of H_C with complement>1 are
 1471,1523,1693,2083,2161.
Exactly 1471,1693,2083,2161 fail. The remaining member 1523 succeeds
DIRECTLY, since 2200-1523=677 is prime. Thus even in the proper-group
case membership is NOT sufficient for failure.
The full group at N=2200 is
 {1,3,9,13,27,39,81,117,163,169,243,351,489,507,677,679,729,733,
  959,1053,1147,1241,1467,1471,1521,1523,1693,1711,1849,1957,
  2031,2037,2083,2119,2161,2173,2187,2191,2197,2199} modulo 2200.
Additional diagnostic: of the 301 failed upper roots at these SIX N,
the only roots satisfying h<min PF(N-r) are
 (N,r)=(1718,877) and (6142,3089).
They are noncanonical. This confirms that the low-gap premise itself
does not force eventual success. For N=1718 the midpoint 859 is prime
and is p0; its h=0 diagonal success must not be mislabeled h=1.
REPRODUCTION: nowe/structural_probe.py. These are six targeted cases,
not an exhaustive search for all hosts or a classification of bad cores.
HISTORICAL AUDIT:
The Biblia already contains the stronger valuation-matrix relation
product_j p_j^(E_ij-delta_ij)=-1 (mod N) and its determinant/resonance
consumer near lines 35135 onward. The subgroup lemma is an elementary
consequence of that known row relation; it is NOT claimed as new theory.
The new work here is the direct test and its negative explanatory result.
WHAT SURVIVES:
A sufficient exclusion test outside a verified full BAD-envelope group.
Smaller moduli dividing N and their characters cannot recover information
lost when H_C is already the entire unit group modulo N. More structured
valuation or additive constraints were not tested by this argument.
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No inference is made that all possible residue methods are impossible.


RESULT-ID: LOW-GAP-THREE-LABEL-EXACT-OBSTRUCTION-SPLIT-20260907C
STATUS: PROVED finite-support reduction; the resulting Diophantine
systems are unresolved, so this is NOT a four-label theorem.
EXACT STATEMENT:
Fix a prime upper root r with composite m=N-r, 0<h<q=min PF(m),
and no GOOD vertex through depth 2. If the complete nonroot support
discovered through depth 3 has at most three labels, then it has
exactly three, and at least one of the following holds:

(A) There is a closed active BAD set C of exactly three primes that
    contains PF(m); the whole nonroot reachable set is C.

(B) There are distinct odd primes q,b,c and integers u,v>=2,
    a>=0,w>=1,a+w>=2 satisfying the COMPLETE row equations
      m=q^u,
      N-q=b^v,
      N-b=q^a*c^w,
      N=2q^u+2h, 0<h<q, r=q^u+2h prime.
    The row N-c is not constrained; c is first discovered at depth 3.
    If a>0, then 3<=v<=u-1. If v=2, then a=0 and w is odd>=3.

Conversely, (A) in the stated low-gap root domain yields no GOOD and
exactly three discovered nonroot labels by depth 3; and a system (B)
with the listed hypotheses yields no GOOD through depth 2 and exactly
three labels through depth 3. Cases (A) and (B) may overlap.

PROOF OF EXHAUSTIVENESS:
Let S=PF(m). Minimum-factor escape gives PF(N-q) disjoint from S:
any common prime t would divide 2h-q, a nonzero integer of absolute
value <q<=t. The old LOW-GAP-COMPLETE-THREE-LABEL-ALTERNATIVE proves
that at least three labels occur. If |S|>=3, expansion of the BAD
row at q adds an outside label by depth 2, giving at least four.
Therefore |S|<=2 in a three-label case.
If |S|=2, PF(N-q) must be a singleton {b}, or at least four labels
are already present. The three labels S union {b} are all reached
by depth 2. Since no GOOD occurs there, all three rows are expanded,
and no child outside this set is discovered by depth 3. Thus it is
a complete closed active BAD set C, giving (A).
If |S|=1, write m=q^u, u>=2, and T=PF(N-q). If |T|>=3, at least
four labels occur. If |T|=2, then {q} union T consists of three
BAD labels reached by depth 2; expanding all three again proves (A).
Finally |T|=1 gives N-q=b^v, v>=2. Activity gives b!=q. Under the
three-label count the entire support of N-b lies in {q,c}, where c
is the sole new label. The earlier three-label theorem ensures c
really occurs. Hence N-b=q^a*c^w, w>=1, with a+w>=2 by BADness.
The sharp monomial return lemma applies when a>0. If v=2, the
square-row lemma gives a=0 and excludes an even w. This proves (B).
For the converse in (A), closure and BADness prevent any GOOD or
fourth label, while the old three-label theorem forces three labels
by depth 3. In (B), the complete rows show exactly the successive
fronts {q}, {b}, {c}, with a possible repeated q in the last row.
The root and q,b are BAD; the status of c is immaterial through depth 2.

HOSTILE CHECK:
Case (A) is a statement about the entire reachable set, not just a
terminal SCC of size three. No classification of three-vertex cores
or of two-vertex terminal cores is assumed. In (B), requiring N-c
to be BAD or supported on {q,b,c} would be an extra, unjustified
condition. The reduction explicitly retains the full cofactors in
both expanded rows. A selected path alone would not establish it.
RELATION TO p0:
The reduction is shared by all low-gap roots; in the h=1 square
subcase one can add u odd>=3 and q=23 (mod 24). It does not prove
that even one of the reduced configurations exists for p0, nor that
all are impossible. The five-label counterexample with r=p12 has
FOUR labels and therefore lies outside this three-label obstruction.
NOVELTY: no literature priority claim; this is a precise consumer of
the existing three-label theorem and the new reciprocal-row lemmas.


======================================================================
END-OF-RUN CHECKPOINT -- 2026-09-07C
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STRONGEST RESULTS OF THIS LOCAL CONTINUATION:
1. Sharp reciprocal-edge bound b>=3q+2h for a low-gap return
   q->b->q through the minimum first factor. Complete proof and sharp
   p0 and p1 examples are recorded. The root's primality excludes
   the candidate b=q+2h; minimum-factor arithmetic excludes b=2h-q.
2. For m=q^u and complete prime-power row N-q=b^v, a return forces
   3<=v<=u-1. Thus u=2 and u=3 prohibit this immediate monomial return.
   The separate square-row lemma forces the next row between two
   consecutive squares and coprime to q.
3. The unrestricted low-gap FIVE-label strengthening is false at
   N=44670, r=p12=22469. Both depth-3 endpoints are GOOD. The old
   rank-0-through-6 experiment remains correct within its exact domain.
4. A precise reduction of the open FOUR-label strengthening to a full
   three-label closed world or an explicit complete prime-power system.
5. A modular-subgroup explanation was tested and found uninformative
   at five of the six known hosts, where the failed support generates
   the whole unit group. The necessary subgroup lemma is retained.

VALIDATION:
One subagent supplied an independent counterexample search and proof
audit; no additional agents were created. The root checked the exact
certificates independently and ran the new return/support audit across
134898 eligible rank-0-through-12 inputs. All 10249 reciprocal-edge
checks passed; 1360 attain equality. Proofs are elementary and use no
new external sieve, prime-tuple, height or classification theorem.
The supplied TXT's complete previous contents remain above this run.

CURRENT LIMIT:
Neither binary Goldbach nor universal p0 reachability is proved.
The new return restrictions use the primality of an upper root and a
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small numerical gap; they are not properties unique to the canonical selector. No general bound of FOUR, let alone FIVE, on the depth-3 nonroot support has been proved. The earlier asymptotic radius-three large-product obstruction and moving closed-support obstruction are unchanged. Finite counts are not all-N theorems or literature novelty.

EXACT NEXT QUESTIONS:
Local: can either configuration in LOW-GAP-THREE-LABEL-EXACT-OBSTRUCTION-SPLIT occur? In its singleton square branch the remaining equations are 2q^u+2h=q+b^2=b+c^w with w odd>=3; in h=1 also u is odd>=3 and q=23 (mod 24). These equations still need an actual Diophantine exclusion or a checked example.
Global: the h=1 full-BFS question from the previous checkpoint remains the same: can a complete closed active BAD set contain PF(P-2) when N=2(P-1) and P is prime? Excluding only immediate two-cycles, or only three-label worlds, does not settle this moving-support problem.

FINAL INDEPENDENT AUDIT:
The same single subagent checked the strengthened reciprocal bound, the exponent-drop consumer, and the exhaustive three-label split, including both converses and the unconstrained status of the depth-3 prime c. No correction was needed. This is an internal independent proof check, not external peer review or a novelty audit.

Bibliography

[1] J. Nagura, *On the interval containing at least one prime number*, Proceedings of the Japan Academy 28(4) (1952), 177–181. https://doi.org/10.3792/pja/1195570997.

[2] E. M. Matveev, *An explicit lower bound for a homogeneous rational linear form in logarithms of algebraic numbers*, Izvestiya: Mathematics 62(4) (1998), 723–772. https://doi.org/10.1070/IM1998v062n04ABEH000190.

[3] D. Ridout, *The p-adic generalization of the Thue–Siegel–Roth theorem*, Mathematika 5(1) (1958), 40–48.

[4] Y. Bugeaud, M. Mignotte, S. Siksek, *Classical and modular approaches to exponential Diophantine equations II. The Lebesgue–Nagell equation*, Compositio Mathematica 142 (2006), 31–62; arXiv:math/0405220. https://arxiv.org/abs/math/0405220.

[5] M. A. Bennett, S. Siksek, *Rational points on Erdős–Selfridge superelliptic curves and the Lebesgue–Nagell equation with S-unit right-hand side*, arXiv:2109.09128 (2021). https://arxiv.org/abs/2109.09128.

[6] J. Hančl, T. P. Nguyen, *On a Theorem of Legendre on Diophantine Approximation*, arXiv:2407.11166 (2024). https://arxiv.org/abs/2407.11166.

[7] J. Dodziuk, *Finite-Difference Approach to the Hodge Theory of Harmonic Forms*, American Journal of Mathematics 98 (1976), 79–104. https://doi.org/10.2307/2373615.

[8] A. Bożek, *Exceptional autonomous components of Goldbach factorization graphs*, arXiv:1909.09900 (2019). https://arxiv.org/abs/1909.09900.

[9] D. Holzmüller, *Efficient Neighbor-Finding on Space-Filling Curves*, arXiv:1710.06384, version 3 (2019). https://arxiv.org/abs/1710.06384.

[10] OEIS Foundation Inc., *A306746: Goldbug numbers*. https://oeis.org/A306746.

[11] R. Scott, *On the equations $p^x - b^y = c$ and $a^x + b^y = c^z$*, Journal of Number Theory 44(2) (1993), 153–165. https://doi.org/10.1006/jnth.1993.1041.

[12] T. Oliveira e Silva, S. Herzog, S. Pardi, *Empirical verification of the even Goldbach conjecture and computation of prime gaps up to $4 \cdot 10^{18}$*, Mathematics of Computation 83(288) (2014), 2033–2060. https://doi.org/10.1090/S0025-5718-2013-02787-1.

[13] R. C. Baker, G. Harman, J. Pintz, *The Difference Between Consecutive Primes, II*, Proceedings of the London Mathematical Society 83(3) (2001), 532–562. https://doi.org/10.1112/plms/83.3.532.

[14] R. Li, *The number of primes in short intervals and numerical calculations for Harman's sieve*, arXiv:2308.04458 (2023; version checked 24 VIII 2026). https://arxiv.org/abs/2308.04458.

[15] J. Li, J. Liu, *Theorem* $(1 + 1.9)$ *on the Goldbach Conjecture*, arXiv:2606.05224 (2026). https://arxiv.org/abs/2606.05224.

[16] H. A. Helfgott, *Minor arcs for Goldbach's problem*, arXiv:1205.5252 (2012). https://arxiv.org/abs/1205.5252.

[17] H. A. Helfgott, *Major arcs for Goldbach's theorem*, arXiv:1305.2897 (2013). https://arxiv.org/abs/1305.2897.

[18] J.-H. Evertse, H. P. Schlickewei, W. M. Schmidt, *Linear equations in variables which lie in a multiplicative group*, Annals of Mathematics 155 (2002), 807–836; arXiv:math/0409604. https://arxiv.org/abs/math/0409604.

[19] M. A. Bennett, S. Siksek, *Differences between perfect powers: prime power gaps*, arXiv:2110.05553 (2021). https://arxiv.org/abs/2110.05553.

[20] Y. Bugeaud, *On the difference between perfect powers and integral S-units*, arXiv:2604.27490 (2026). https://arxiv.org/abs/2604.27490.

[21] S. Siksek, M. Stoll, *Partial Descent on Hyperelliptic Curves and the Generalized Fermat Equation $x^3 + y^4 + z^5 = 0$*, arXiv:1103.1979 (2011). https://arxiv.org/abs/1103.1979.

[22] F. Costello, P. Watts, *A computational upper bound on Jacobsthal's function*, arXiv:1208.5342 (2012). https://arxiv.org/abs/1208.5342.

[23] J. Merikoski, *Large prime factors on short intervals*, Mathematical Proceedings of the Cambridge Philosophical Society 170(1) (2021), 1–50. https://doi.org/10.1017/S0305004119000306.

[24] R. Li, *On the largest prime factor of integers in short intervals III*, arXiv:2601.00910 (2026). https://arxiv.org/abs/2601.00910.

[25] J.-J. Huang, H. Li, *On the connection between the Goldbach conjecture and the Elliott–Halberstam conjecture*, arXiv:2005.03811 (2020). https://arxiv.org/abs/2005.03811.

[26] J. D. Lichtman, *Primes in arithmetic progressions to large moduli, and Goldbach beyond the square-root barrier*, arXiv:2309.08522 (2023). https://arxiv.org/abs/2309.08522.

[27] R. Williamson, *An elementary conjecture which implies the Goldbach conjecture*, arXiv:1802.02428 (2018). https://arxiv.org/abs/1802.02428.

[28] E. Assing, V. Blomer, J. Li, *Uniform Titchmarsh divisor problems*, arXiv:2005.13915 (2020). https://arxiv.org/abs/2005.13915.

[29] O. Gorodetsky, L. Grimmelt, *On an Erdős–Kac-type conjecture of Elliott*, arXiv:2307.13585 (2023). https://arxiv.org/abs/2307.13585.

[30] K. Ford, *Poisson approximation of prime divisors of shifted primes*, arXiv:2408.03803 (2024). https://arxiv.org/abs/2408.03803.

[31] K. Ford, *Sieving very thin sets of primes, and Pratt trees with missing primes*, International Mathematics Research Notices (2014). https://doi.org/10.1093/imrn/rnt026.

[32] P. D. T. A. Elliott, *Products of shifted primes: multiplicative analogues of Goldbach's problem*, Acta Arithmetica 88(1) (1999), 31–50. https://doi.org/10.4064/AA-88-1-31-50.

[33] J. H. Loxton, *Some problems involving powers of integers*, Acta Arithmetica 46 (1986), 113–123.

[34] D. J. Bernstein, *Detecting perfect powers in essentially linear time*, Mathematics of Computation 67 (1998), 1253–1283. https://doi.org/10.1090/S0025-5718-98-00952-1.

[35] C. L. Stewart, *On heights of multiplicatively dependent algebraic numbers*, Acta Arithmetica 133 (2008), 97–108.

[36] G. Bhowmik, L. Grimmelt, *The exceptional set of the Goldbach problem*, arXiv:2607.27282, version 2 (2026). https://arxiv.org/abs/2607.27282.

[37] G. Zhao, *The exceptional set of Goldbach problem and Linnik's constant*, arXiv:2511.05631, version 2 (2026). https://arxiv.org/abs/2511.05631.

[38] J. Friedlander, H. Iwaniec, *Asymptotic sieve for primes*, Annals of Mathematics 148 (1998), 1041–1065; arXiv:math/9811186. https://arxiv.org/abs/math/9811186.

[39] D. H. J. Polymath, *Variants of the Selberg sieve, and bounded intervals containing many primes*, arXiv:1407.4897 (2014). https://arxiv.org/abs/1407.4897.

[40] B. Green, T. Tao, *Restriction theory of the Selberg sieve, with applications*, arXiv:math/0405581 (2004). https://arxiv.org/abs/math/0405581.

[41] P. X. Gallagher, *A larger sieve*, Acta Arithmetica 18 (1971), 77–81. https://doi.org/10.4064/aa-18-1-77-81.

[42] B. Green, A. J. Harper, *Inverse questions for the large sieve*, arXiv:1311.6176 (2013). https://arxiv.org/abs/1311.6176.

[43] R. Li, *Largest square divisors of shifted primes*, arXiv:2505.23779 (2025). https://arxiv.org/abs/2505.23779.

[44] P.-Y. Bienvenu, X. Shao, J. Teräväinen, *A transference principle for systems of linear equations, and applications to almost twin primes*, arXiv:2106.09001 (2021). https://arxiv.org/abs/2106.09001.

[45] L. Grimmelt, J. Teräväinen, *The Exceptional Set in Goldbach's Problem with two Chen Primes*, arXiv:2508.16400, version 2 (2026). https://arxiv.org/abs/2508.16400.

[46] K. Halupczok, *Goldbach's problem with primes in arithmetic progressions and in short intervals*, arXiv:1212.4406 (2012). https://arxiv.org/abs/1212.4406.

[47] K. Ford, *A Hardy–Ramanujan type inequality for shifted primes and sifted sets*, arXiv:2101.03440 (2021). https://arxiv.org/abs/2101.03440.

[48] K. Matomäki, M. Radziwiłł, *Multiplicative functions in short intervals*, Annals of Mathematics 183 (2016), 1015–1056; arXiv:1501.04585. https://arxiv.org/abs/1501.04585.

[49] J. D. Vaaler, *Heights on Groups and Small Multiplicative Dependencies*, arXiv:1211.5066 (2012). https://arxiv.org/abs/1211.5066.

[50] F. Amoroso, M. Sombra, U. Zannier, *Unlikely intersections and multiple roots of sparse polynomials*, arXiv:1412.8059 (2014). https://arxiv.org/abs/1412.8059.

[51] R. Bellman, *On a routing problem*, Quarterly of Applied Mathematics 16 (1958), 87–90. https://doi.org/10.1090/qam/102435.

[52] R. M. Karp, *A characterization of the minimum cycle mean in a digraph*, Discrete Mathematics 23 (1978), 309–311. https://doi.org/10.1016/0012-365X(78)90011-0.

[53] S. Prajna, A. Jadbabaie, *Safety Verification of Hybrid Systems Using Barrier Certificates*, Hybrid Systems: Computation and Control, LNCS 2993 (2004), 477–492. https://doi.org/10.1007/978-3-540-24743-2_32.

[54] D. Angeli, M. Philippe, N. Athanasopoulos, R. M. Jungers, *Path-Complete Graphs and Common Lyapunov Functions*, arXiv:1612.03983 (2016). https://arxiv.org/abs/1612.03983.

[55] J. Friedman, J.-P. Tillich, *Calculus on Graphs*, arXiv:cs/0408028 (2004). https://arxiv.org/abs/cs/0408028.

[56] X. Jiang, L.-H. Lim, Y. Yao, Y. Ye, *Statistical ranking and combinatorial Hodge theory*, Mathematical Programming 127 (2011), 203–244; arXiv:0811.1067. https://arxiv.org/abs/0811.1067.

[57] P. De Leenheer, *An elementary proof of a matrix tree theorem for directed graphs*, arXiv:1904.12221 (2019). https://arxiv.org/abs/1904.12221.

[58] H. Bass, *The Ihara–Selberg zeta function of a tree lattice*, International Journal of Mathematics 3(6) (1992), 717–797. https://doi.org/10.1142/S0129167X92000357.

[59] B. Rangarajan, *A Combinatorial Proof of Ihara–Bass's Formula for the Zeta Function of Regular Graphs*, FSTTCS 2017, LIPIcs 93 (2018), 46:1–46:13. https://doi.org/10.4230/LIPIcs.FSTTCS.2017.46.

[60] B. Chen, V. Gupta, Y. C. Li, *Large Value Estimates for Dirichlet Polynomials with Characters and Zero Density of Dirichlet L-Functions*, arXiv:2507.08296, version 2 (2026). https://arxiv.org/abs/2507.08296.

[61] L. Guth, J. Maynard, *New large value estimates for Dirichlet polynomials*, arXiv:2405.20552, version 2 (2026). https://arxiv.org/abs/2405.20552.

[62] T. Tao, *Almost all orbits of the Collatz map attain almost bounded values*, arXiv:1909.03562, version checked 2026. https://arxiv.org/abs/1909.03562.

[63] C. Elsholtz, T. Tao, *Counting the number of solutions to the Erdős–Straus equation on unit fractions*, arXiv:1107.1010 (2011). https://arxiv.org/abs/1107.1010.

[64] M. Bright, D. Loughran, *Brauer–Manin obstruction for Erdős–Straus surfaces*, arXiv:1908.02526 (2019). https://arxiv.org/abs/1908.02526.

[65] J. Maynard, *Small gaps between primes*, Annals of Mathematics 181 (2015), 383–413; arXiv:1311.4600. https://arxiv.org/abs/1311.4600.

[66] A. Wiles, *Modular elliptic curves and Fermat's Last Theorem*, Annals of Mathematics 141 (1995), 443–551. https://annals.math.princeton.edu/1995/141-3/p01.

[67] G. Perelman, *The entropy formula for the Ricci flow and its geometric applications*, arXiv:math/0211159 (2002). https://arxiv.org/abs/math/0211159.

[68] M. Baker, S. Norine, *Riemann–Roch and Abel–Jacobi theory on a finite graph*, Advances in Mathematics 215 (2007), 766–788; arXiv:math/0608360. https://arxiv.org/abs/math/0608360.

[69] J. Hansen, R. Ghrist, *Toward a Spectral Theory of Cellular Sheaves*, Journal of Applied and Computational Topology 3 (2019), 315–358; arXiv:1808.01513. https://arxiv.org/abs/1808.01513.

[70] B. Romera-Paredes et al., *Mathematical discoveries from program search with large language models*, Nature 625 (2024), 468–475. https://www.nature.com/articles/s41586-023-06924-6.

[71] A. Novikov et al., *AlphaEvolve: A coding agent for scientific and algorithmic discovery*, arXiv:2506.13131 (2025). https://arxiv.org/abs/2506.13131.

[72] OpenAI, *Planar Point Sets with Many Unit Distances*, preprint (2026). https://cdn.openai.com/pdf/74c24085-19b0-4534-9c90-465b8e29ad73/unit-distance-proof.pdf.

[73] OpenAI, *Ten advances in mathematics and theoretical computer science*, manuscripts and formal certificates (2026). https://openai.com/index/ten-advances-in-mathematics/.

[74] S. S. Pillai, *On the equation* $2^x - 3^y = 2^X + 3^Y$, Bulletin of the Calcutta Mathematical Society 37 (1945), 15–20.

[75] C. L. Siegel, *Über einige Anwendungen diophantischer Approximationen*, Abhandlungen der Preussischen Akademie der Wissenschaften, Physikalisch-mathematische Klasse (1929), Nr. 1.

[76] A. Baker, *Bounds for the solutions of the hyperelliptic equation*, Mathematical Proceedings of the Cambridge Philosophical Society 65 (1969), 439–444. https://doi.org/10.1017/S0305004100044418.

[77] P. Mihăilescu, *Primary cyclotomic units and a proof of Catalan's conjecture*, Journal für die reine und angewandte Mathematik 572 (2004), 167–195. https://doi.org/10.1515/crll.2004.048.

[78] D. Ridout, *Rational approximations to algebraic numbers*, Mathematika 4 (1957), 125–131. https://doi.org/10.1112/S0025579300001182.

[79] E. M. Matveev, *An explicit lower bound for a homogeneous rational linear form in logarithms of algebraic numbers. II*, Izvestiya: Mathematics 64 (2000), 1217–1269. https://doi.org/10.1070/IM2000v064n06ABEH000314.

[80] K. H. Monfared, S. Mallik, *An analog of Matrix Tree Theorem for signless Laplacians*, Linear Algebra and its Applications 560 (2019), 43–55.

[81] R. H. Villarreal, *Rees algebras of edge ideals*, Communications in Algebra 23(9) (1995), 3513–3524, DOI: https://doi.org/10.1080/00927879508825412.

[82] J. Martinez-Bernal, R. H. Villarreal, *Toric ideals generated by circuits*, arXiv:0812.0195 (2008).

[83] A. Geroldinger, Q. Zhong, *Factorization theory in commutative monoids*, Semigroup Forum 100 (2020), 22–51, DOI: https://doi.org/10.1007/s00233-019-10079-0.

[84] R. Bacher, *Euclid meets Popeye: The Euclidean Algorithm for* 2×2 *Matrices*, Comptes Rendus Mathematique 361 (2023), 889–895, DOI: https://doi.org/10.5802/crmath.451.

[85] M. B. Nathanson, *Free monoids and forests of rational numbers*, Discrete Applied Mathematics 216 (2017), 662–669, DOI: https://doi.org/10.1016/j.dam.2015.07.011.

[86] J. H. Conway, H. S. M. Coxeter, *Triangulated polygons and frieze patterns*, Parts I–II, The Mathematical Gazette 57 (1973), DOI: https://doi.org/10.2307/3615344 and https://doi.org/10.2307/3615561.

[87] C. Bessenrodt, T. Holm, P. Jorgensen, *All* SL_2*-tilings come from infinite triangulations*, Advances in Mathematics 315 (2017), DOI: https://doi.org/10.1016/j.aim.2017.05.019.

[88] F. Beukers, H. P. Schlickewei, *The equation* $x + y = 1$ *in finitely generated groups*, Acta Arithmetica 78 (1996), 189–199, DOI: https://doi.org/10.4064/aa-78-2-189-199.

[89] F. Santos, B. Sturmfels, *Higher Lawrence configurations*, Journal of Combinatorial Theory, Series A 103 (2003), 151–164, DOI: https://doi.org/10.1016/S0097-3165(03)00092-X.

[90] J. C. Lagarias, *Cyclic systems of simultaneous congruences*, International Journal of Number Theory 6 (2010), 219–245, DOI: https://doi.org/10.1142/S1793042110002892.

[91] P. Leonetti, S. Tringali, *On a system of equations with primes*, Journal de Theorie des Nombres de Bordeaux 26 (2014), 399–413, DOI: https://doi.org/10.5802/jtnb.873.

[92] P. T. Bateman, E. Grosswald, *On a theorem of Erdős and Szekeres*, Illinois Journal of Mathematics 2(1) (1958), 88–98. http://projecteuclid.org/euclid.ijm/1255380836.

[93] A. Pascadi, *On the exponents of distribution of primes and smooth numbers*, arXiv:2505.00653 (2025). https://arxiv.org/abs/2505.00653.

[94] R. Li, *On Chen's theorem, Goldbach's conjecture and almost prime twins II*, Mathematical Reports **28** (2026).

[95] A. P. Mangerel, *On a Goldbach-type problem for the Liouville function*, International Mathematics Research Notices (2024).

[96] Y. Bugeaud, M. Mignotte, S. Siksek, *Classical and modular approaches to exponential Diophantine equations I. Fibonacci and Lucas perfect powers*, Annals of Mathematics 163 (2006), 969–1018. https://samirsiksek.github.io/siksek.github.io/papers/fibannalsfinal.pdf.

[97] K. Ford, S. V. Konyagin, F. Luca, *Prime chains and Pratt trees*, Geometric and Functional Analysis 20 (2010), 1231–1258. https://arxiv.org/abs/0904.0473.

[98] J. Pintz, *A new explicit formula in the additive theory of primes with applications II. The exceptional set in Goldbach's problem*, arXiv:1804.09084v2 (2018). https://arxiv.org/abs/1804.09084.